Configuration Options Reference

Subsequent sections contain the list of available ESP-IDF options automatically generated from Kconfig files. Note that due to dependencies between options, some options listed here may not be visible by default in `menuconfig`.

By convention, all option names are upper-case letters with underscores. When Kconfig generates `sdkconfig` and `sdkconfig.h` files, option names are prefixed with `CONFIG_`. So if an option `ENABLE_FOO` is defined in a Kconfig file and selected in `menuconfig`, then the `sdkconfig` and `sdkconfig.h` files will have `CONFIG_ENABLE_FOO` defined. In the following sections, option names are also prefixed with `CONFIG_`, same as in the source code.

Build type

Contains:

- CONFIG_APP_COMPATIBLE_PRE_V3_1_BOOTLOADERS
- CONFIG_APP_COMPATIBLE_PRE_V2_1_BOOTLOADERS
- CONFIG_APP_BUILD_TYPE
- CONFIG_APP_BUILD_TYPE_PURE_RAM_APP
- CONFIG_APP_REPRODUCIBLE_BUILD
- CONFIG_APP_NO_BLOBS

CONFIG_APP_BUILD_TYPE

Application build type

Found in: Build type

Select the way the application is built.

By default, the application is built as a binary file in a format compatible with the ESP-IDF bootloader. In addition to this application, 2nd stage bootloader is also built. Application and bootloader binaries can be written into flash and loaded/executed from there.

Another option, useful for only very small and limited applications, is to only link the .elf file of the application, such that it can be loaded directly into RAM over JTAG or UART. Note that since IRAM and DRAM sizes are very limited, it is not possible to build any complex application this way. However for some kinds of testing and debugging, this option may provide faster iterations, since the application does not need to be written into flash.

Note: when APP_BUILD_TYPE_RAM is selected and loaded with JTAG, ESP-IDF does not contain all the startup code required to initialize the CPUs and ROM memory (data/bss). Therefore it is necessary to execute a bit of ROM code prior to executing the application. A gdbinit file may look as follows (for ESP32):

> # Connect to a running instance of OpenOCD target remote :3333 # Reset and halt the target mon reset halt # Run to a specific point in ROM code, # where most of initialization is complete. thb *0x40007d54 c # Load the application into RAM load # Run till app_main tb app_main c

Execute this gdbinit file as follows:

> xtensa-esp32-elf-gdb build/app-name.elf -x gdbinit

Example gdbinit files for other targets can be found in tools/test_apps/system/gdb_loadable_elf/

When loading the BIN with UART, the ROM will jump to ram and run the app after finishing the ROM startup code, so there's no additional startup initialization required. You can use the *load-ram* in esptool to load the generated .bin file into ram and execute.

Example:

> esptool --chip {chip} -p {port} -b {baud} --no-stub load-ram {app.bin}

Recommended sdkconfig.defaults for building loadable ELF files is as follows. CONFIG_APP_BUILD_TYPE_RAM is required, other options help reduce application memory footprint.

```
CONFIG_APP_BUILD_TYPE_RAM=y CONFIG_VFS_SUPPORT_TERMIOS=
CONFIG_LIBC_NEWLIB_NANO_FORMAT=y
CONFIG_ESP_SYSTEM_PANIC_PRINT_HALT=y
CONFIG_ESP_DEBUG_STUBS_ENABLE= CONFIG_ESP_ERR_TO_NAME_LOOKUP=
```

Available options:

- Default (binary application + 2nd stage bootloader) (CONFIG_APP_BUILD_TYPE_APP_2NDBOOT)
- Build app runs entirely in RAM (EXPERIMENTAL) (CONFIG_APP_BUILD_TYPE_RAM)

CONFIG_APP_BUILD_TYPE_PURE_RAM_APP

Build app without SPI_FLASH/PSRAM support (saves ram)

Found in: Build type

If this option is enabled, external memory and related peripherals, such as Cache, MMU, Flash and PSRAM, won't be initialized. Corresponding drivers won't be introduced either. Components that depend on the spi_flash component will also be unavailable, such as app_update, etc. When this option is enabled, about 26KB of RAM space can be saved.

CONFIG_APP_REPRODUCIBLE_BUILD

Enable reproducible build

Found in: Build type

If enabled, all date, time, and path information would be eliminated. A .gdbinit file would be create automatically. (or will be append if you have one already)

Default value:

- No (disabled)

CONFIG_APP_NO_BLOBS

No Binary Blobs

Found in: Build type

If enabled, this disables the linking of binary libraries in the application build. Note that after enabling this Wi-Fi/Bluetooth will not work.

Default value:

- No (disabled)

CONFIG_APP_COMPATIBLE_PRE_V2_1_BOOTLOADERS

App compatible with bootloaders before ESP-IDF v2.1

Found in: Build type

Bootloaders before ESP-IDF v2.1 did less initialisation of the system clock. This setting needs to be enabled to build an app which can be booted by these older bootloaders.

If this setting is enabled, the app can be booted by any bootloader from IDF v1.0 up to the current version.

If this setting is disabled, the app can only be booted by bootloaders from IDF v2.1 or newer.

Enabling this setting adds approximately 1KB to the app's IRAM usage.

Default value:

- No (disabled)

CONFIG_APP_COMPATIBLE_PRE_V3_1_BOOTLOADERS

App compatible with bootloader and partition table before ESP-IDF v3.1

Found in: Build type

Partition tables before ESP-IDF V3.1 do not contain an MD5 checksum field, and the bootloader before ESP-IDF v3.1 cannot read a partition table that contains an MD5 checksum field.

Enable this option only if your app needs to boot on a bootloader and/or partition table that was generated from a version *before* ESP-IDF v3.1.

If this option and Flash Encryption are enabled at the same time, and any data partitions in the partition table are marked Encrypted, then the partition encrypted flag should be manually verified in the app before accessing the partition (see CVE-2021-27926).

Default value:

- No (disabled)

App Update config

Contains:

- CONFIG_SECURE_SIGNED_DATA_PARTITION

CONFIG_SECURE_SIGNED_DATA_PARTITION

Require signed Data partition images

Found in: App Update config

If set, the Data partition images will be verified during OTA updates. Only partitions with subtype ESP_PARTITION_SUBTYPE_DATA_UNDEFINED will be verified.

Default value:

- No (disabled) if CONFIG_SECURE_SIGNED_ON_UPDATE_NO_SECURE_BOOT

Bootloader config

Contains:

- Application Rollback

- Bootloader manager
- CONFIG_BOOTLOADER_COMPILER_OPTIMIZATION
- CONFIG_BOOTLOADER_REGION_PROTECTION_ENABLE
- CONFIG_BOOTLOADER_APP_TEST
- CONFIG_BOOTLOADER_FACTORY_RESET
- CONFIG_BOOTLOADER_HOLD_TIME_GPIO
- Log
- Recovery Bootloader and Rollback
- CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC
- Serial Flash Configurations
- CONFIG_BOOTLOADER_SKIP_VALIDATE_ALWAYS
- CONFIG_BOOTLOADER_SKIP_VALIDATE_ON_POWER_ON
- CONFIG_BOOTLOADER_SKIP_VALIDATE_IN_DEEP_SLEEP
- CONFIG_BOOTLOADER_WDT_ENABLE
- CONFIG_BOOTLOADER_VDDSDIO_BOOST

Bootloader manager

Contains:

- CONFIG_BOOTLOADER_PROJECT_VER
- CONFIG_BOOTLOADER_COMPILE_TIME_DATE

CONFIG_BOOTLOADER_COMPILE_TIME_DATE

Use time/date stamp for bootloader

Found in: Bootloader config > Bootloader manager

If set, then the bootloader will be built with the current time/date stamp. It is stored in the bootloader description structure. If not set, time/date stamp will be excluded from bootloader image. This can be useful for getting the same binary image files made from the same source, but at different times.

CONFIG_BOOTLOADER_PROJECT_VER

Project version

Found in: Bootloader config > Bootloader manager

Project version. It is placed in "version" field of the esp_bootloader_desc structure. The type of this field is "uint32_t".

Range:

- from 0 to 4294967295

Default value:

- 1

Application Rollback

Contains:

- CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE

CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE

Enable app rollback support

Found in: Bootloader config > Application Rollback

After updating the app, the bootloader runs a new app with the "ESP_OTA_IMG_PENDING_VERIFY" state set. This state prevents the re-run of this app. After the first boot of the new app in the user code, the function should be called to confirm the operability of the app or vice versa about its non-operability. If the app is working, then it is marked as valid. Otherwise, it is marked as not valid and rolls back to the previous working app. A reboot is performed, and the app is booted before the software update. Note: If during the first boot a new app the power goes out or the WDT works, then roll back will happen. Rollback is possible only between the apps with the same security versions.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

Enable app anti-rollback support

Found in: Bootloader config > Application Rollback > CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE

This option prevents rollback to previous firmware/application image with lower security version.

Default value:

- No (disabled) if CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE

CONFIG_BOOTLOADER_APP_SECURE_VERSION

eFuse secure version of app

Found in: Bootloader config > Application Rollback > CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE > CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

The secure version is the sequence number stored in the header of each firmware. The security version is set in the bootloader, version is recorded in the eFuse field as the number of set ones. The allocated number of bits in the efuse field for storing the security version is limited (see BOOTLOADER_APP_SEC_VER_SIZE_EFUSE_FIELD option).

Bootloader: When bootloader selects an app to boot, an app is selected that has a security version greater or equal that recorded in eFuse field. The app is booted with a higher (or equal) secure version.

The security version is worth increasing if in previous versions there is a significant vulnerability and their use is not acceptable.

Your partition table should has a scheme with ota_0 + ota_1 (without factory).

Default value:

- 0 if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

CONFIG_BOOTLOADER_APP_SEC_VER_SIZE_EFUSE_FIELD

Size of the efuse secure version field

Found in: Bootloader config > Application Rollback > CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE > CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

The size of the efuse secure version field. Its length is limited to 32 bits for ESP32, 4 bits for ESP32-C2, 9 bits for ESP32-C5, and 16 bits for other targets. This determines how many times the security version can be increased.

Range:

- from 1 to 32 if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK
- from 1 to 16 if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

Default value:

- 32 if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK
- 16 if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

CONFIG_BOOTLOADER_EFUSE_SECURE_VERSION_EMULATE

Emulate operations with efuse secure version(only test)

Found in: Bootloader config > Application Rollback > CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE > CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

This option allows to emulate read/write operations with all eFuses and efuse secure version. It allows to test anti-rollback implementation without permanent write eFuse bits. There should be an entry in partition table with following details: *emul_efuse*, *data*, *efuse*, , *0x2000*.

This option enables: EFUSE_VIRTUAL and EFUSE_VIRTUAL_KEEP_IN_FLASH.

Default value:

- No (disabled) if CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK

Recovery Bootloader and Rollback

Contains:

- CONFIG_BOOTLOADER_RECOVERY_ENABLE

CONFIG_BOOTLOADER_RECOVERY_ENABLE

Enable Recovery Bootloader

Found in: Bootloader config > Recovery Bootloader and Rollback

The recovery bootloader feature is implemented in the ROM bootloader. It is required for safe OTA updates of the bootloader. The feature is activated when the eFuse field (ESP_EFUSE_RECOVERY_BOOTLOADER_FLASH_SECTOR) is set, which defines the flash address of the recovery bootloader. If activated and the primary bootloader fails to load, the ROM bootloader will attempt to load the recovery bootloader from the address specified in eFuse.

Default value:

- No (disabled) if SOC_RECOVERY_BOOTLOADER_SUPPORTED

CONFIG_BOOTLOADER_RECOVERY_OFFSET

Recovery Bootloader Flash Offset

Found in: Bootloader config > Recovery Bootloader and Rollback > CONFIG_BOOTLOADER_RECOVERY_ENABLE

Flash address where the recovery bootloader is stored. This value must be written to the eFuse field (ESP_EFUSE_RECOVERY_BOOTLOADER_FLASH_SECTOR) to activate the recovery bootloader in the ROM bootloader. The eFuse can be programmed using espefuse or in the user application with the API esp_efuse_set_recovery_bootloader_offset(). Setting this value in the config allows parttool.py to verify that it does not overlap with existing partitions in the partition table.

The address must be a multiple of the flash sector size (0x1000 bytes). The eFuse field stores the offset in sectors. If the feature is no longer needed or unused, you can burn the 0xFFF value to disable this feature in the ROM bootloader.

Range:

- from 0x0 to 0xFFE000 if CONFIG_BOOTLOADER_RECOVERY_ENABLE

Default value:

- "0x3F0000" if CONFIG_BOOTLOADER_RECOVERY_ENABLE

CONFIG_BOOTLOADER_ANTI_ROLLBACK_ENABLE

Enable bootloader rollback support

Found in: Bootloader config > Recovery Bootloader and Rollback > CONFIG_BOOTLOADER_RECOVERY_ENABLE

This option prevents rollback to previous bootloader image with lower security version.

Default value:

- No (disabled) if CONFIG_BOOTLOADER_RECOVERY_ENABLE

CONFIG_BOOTLOADER_SECURE_VERSION

Secure version of bootloader

Found in: Bootloader config > Recovery Bootloader and Rollback > CONFIG_BOOTLOADER_RECOVERY_ENABLE > CONFIG_BOOTLOADER_ANTI_ROLLBACK_ENABLE

The secure version is the sequence number stored in the header of each bootloader.

The ROM Bootloader which runs the 2nd stage bootloader (PRIMARY or RECOVERY) checks that the security version is greater or equal that recorded in the eFuse field. Bootloaders that have a secure version in the image < secure version in efuse will not boot.

The security version is worth increasing if in previous versions there is a significant vulnerability and their use is not acceptable.

Range:

- from 0 to 4 if CONFIG_BOOTLOADER_ANTI_ROLLBACK_ENABLE

Default value:

- 0 if CONFIG_BOOTLOADER_ANTI_ROLLBACK_ENABLE

CONFIG_BOOTLOADER_COMPILER_OPTIMIZATION

Bootloader optimization Level

Found in: Bootloader config

This option sets compiler optimization level (gcc -O argument) for the bootloader.

- The default "Size" setting will add the -Os (-Oz with clang) flag to CFLAGS.
- The "Debug" setting will add the -Og flag to CFLAGS.
- The "Performance" setting will add the -O2 flag to CFLAGS.

Note that custom optimization levels may be unsupported.

Available options:

- Size (-Os with GCC, -Oz with Clang) (CONFIG_BOOTLOADER_COMPILER_OPTIMIZATION_SIZE)
- Debug (-Og) (CONFIG_BOOTLOADER_COMPILER_OPTIMIZATION_DEBUG)
- Optimize for performance (-O2) (CONFIG_BOOTLOADER_COMPILER_OPTIMIZATION_PERF)

Log

Contains:

- CONFIG_BOOTLOADER_LOG_LEVEL
- Format
- CONFIG_BOOTLOADER_LOG_VERSION
- Settings

CONFIG_BOOTLOADER_LOG_VERSION

Log version

Found in: Bootloader config > Log

Select the log version to be used by the ESP log component. The app log version (CONFIG_LOG_VERSION) controls the version used in the bootloader, preventing the selection of different versions. For description of V1 and V2 see CONFIG_LOG_VERSION.

Available options:

- V1 (CONFIG_BOOTLOADER_LOG_VERSION_1)
- V2 (CONFIG_BOOTLOADER_LOG_VERSION_2)

CONFIG_BOOTLOADER_LOG_LEVEL

Bootloader log verbosity

Found in: Bootloader config > Log

Specify how much output to see in bootloader logs.

Available options:

- No output (CONFIG_BOOTLOADER_LOG_LEVEL_NONE)
- Error (CONFIG_BOOTLOADER_LOG_LEVEL_ERROR)
- Warning (CONFIG_BOOTLOADER_LOG_LEVEL_WARN)
- Info (CONFIG_BOOTLOADER_LOG_LEVEL_INFO)
- Debug (CONFIG_BOOTLOADER_LOG_LEVEL_DEBUG)
- Verbose (CONFIG_BOOTLOADER_LOG_LEVEL_VERBOSE)

Format

Contains:

- CONFIG_BOOTLOADER_LOG_COLORS_SUPPORT
- CONFIG_BOOTLOADER_LOG_TIMESTAMP_SUPPORT
- CONFIG_BOOTLOADER_LOG_COLORS
- CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE

CONFIG_BOOTLOADER_LOG_COLORS

Color

Found in: Bootloader config > Log > Format

Enable ANSI terminal color codes. Logs (info, errors, warnings) will contain color codes. In order to view these, your terminal program must support ANSI color codes.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_LOG_COLORS_SUPPORT

Allow enabling color output at run time

Found in: Bootloader config > Log > Format

Enables support for color codes in the esp_log() function. If CONFIG_LOG_COLORS is enabled, this option is always active. If CONFIG_LOG_COLORS is disabled, this option allows you to still handle color codes in specific files by defining ESP_LOG_COLOR_DISABLED as 0 before including esp_log.h.

Note that enabling this option may slightly increase RAM/FLASH usage due to additional color handling functionality. It provides flexibility to manage color output even when CONFIG_LOG_COLORS is turned off.

Default value:

- No (disabled) if CONFIG_BOOTLOADER_LOG_VERSION_2

CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE

Timestamp

Found in: Bootloader config > Log > Format

Choose what sort of timestamp is displayed in the log output:

- "None" - The log will only contain the actual log messages themselves without any time-related information. Avoiding timestamps can help conserve processing power and memory. It might useful when you perform log analysis or debugging, sometimes it's more straightforward to work with logs that lack timestamps, especially if the time of occurrence is not critical for understanding the issues. "I log_test: info message"
- "Milliseconds since boot" is calculated from the RTOS tick count multiplied by the tick period. This time will reset after a software reboot. "I (112500) log_test: info message"

Available options:

- None (CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE_NONE)
- Milliseconds Since Boot (CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE_CPU_TICKS)

CONFIG_BOOTLOADER_LOG_TIMESTAMP_SUPPORT

Allow enabling timestamp output at run time

Found in: Bootloader config > Log > Format

Enables support for timestamp in the esp_log() function. If CONFIG_LOG_TIMESTAMP_SOURCE_NONE, this option allows you to still handle timestamp in specific files by defining ESP_LOG_TIMESTAMP_DISABLED as 0 before including esp_log.h.

Note that enabling this option may slightly increase RAM/FLASH usage due to additional timestamp handling functionality. It provides flexibility to manage timestamp output even when CONFIG_LOG_TIMESTAMP_SOURCE_NONE.

Default value:

- Yes (enabled) if CONFIG_BOOTLOADER_LOG_VERSION_2

Settings

Contains:

- CONFIG_BOOTLOADER_LOG_MODE

CONFIG_BOOTLOADER_LOG_MODE

Log Mode

Found in: Bootloader config > Log > Settings

Available options:

- Text Log Mode (CONFIG_BOOTLOADER_LOG_MODE_TEXT)

  Enables text-based logging, where log messages are stored in a human-readable format. This mode is useful for development and debugging, as it allows logs to be easily read and interpreted without additional processing.

- Binary Log Mode (CONFIG_BOOTLOADER_LOG_MODE_BINARY)

  Enables binary logging with host-side format string expansion. In this mode, the format argument of ESP_LOGx, ESP_EARLY_LOG, and ESP_DRAM_LOG macros is stored in a NOLOAD section, not included in the final binary file. This reduces flash usage by approximately 10% - 35%. The esp_log() function uses the binary log handler to output messages. Instead of sending the full log string, the chip transmits only the addresses of these strings (if present in the ELF file). If the format string cannot be found in the ELF file, the chip sends the entire string. The host-side monitor tool, which has access to the ELF file, reconstructs the log message using the format string. This reduces firmware size by eliminating format strings from flash memory and removing the usage of printf-like functions, potentially freeing up a few kilobytes of space. To further reduce firmware size, wrap string data with ESP_LOG_ATTR_STR.

Serial Flash Configurations

Contains:

- CONFIG_BOOTLOADER_FLASH_DC_AWARE
- CONFIG_BOOTLOADER_SPI_WP_PIN
- CONFIG_BOOTLOADER_CACHE_32BIT_ADDR_QUAD_FLASH
- CONFIG_BOOTLOADER_FLASH_XMC_SUPPORT
- CONFIG_BOOTLOADER_SPI_CUSTOM_WP_PIN

CONFIG_BOOTLOADER_SPI_CUSTOM_WP_PIN

Use custom SPI Flash WP Pin when flash pins set in eFuse (read help)

Found in: Bootloader config > Serial Flash Configurations

This setting is only used if the SPI flash pins have been overridden by setting the eFuses SPI_PAD_CONFIG_xxx, and the SPI flash mode is QIO or QOUT.

When this is the case, the eFuse config only defines 3 of the 4 Quad I/O data pins. The WP pin (aka ESP32 pin "SD_DATA_3" or SPI flash pin "IO2") is not specified in eFuse. The same pin is also used for external SPIRAM if it is enabled.

If this config item is set to N (default), the correct WP pin will be automatically used for any Espressif chip or module with integrated flash. If a custom setting is needed, set this config item to Y and specify the GPIO number connected to the WP.

Default value:

- No (disabled) if CONFIG_ESPTOOLPY_FLASHMODE_QIO || CONFIG_ESPTOOLPY_FLASHMODE_QOUT

CONFIG_BOOTLOADER_SPI_WP_PIN

Custom SPI Flash WP Pin

Found in: Bootloader config > Serial Flash Configurations

The option "Use custom SPI Flash WP Pin" must be set or this value is ignored

If burning a customized set of SPI flash pins in eFuse and using QIO or QOUT mode for flash, set this value to the GPIO number of the SPI flash WP pin.

Range:

- from 0 to 33 if CONFIG_ESPTOOLPY_FLASHMODE_QIO || CONFIG_ESPTOOLPY_FLASHMODE_QOUT

Default value:

- 7 if CONFIG_ESPTOOLPY_FLASHMODE_QIO || CONFIG_ESPTOOLPY_FLASHMODE_QOUT

CONFIG_BOOTLOADER_FLASH_DC_AWARE

Allow app adjust Dummy Cycle bits in SPI Flash for higher frequency (READ HELP FIRST)

Found in: Bootloader config > Serial Flash Configurations

This will force 2nd bootloader to be loaded by DOUT mode, and will restore Dummy Cycle setting by resetting the Flash

CONFIG_BOOTLOADER_FLASH_XMC_SUPPORT

Enable the support for flash chips of XMC (READ DOCS FIRST)

Found in: Bootloader config > Serial Flash Configurations

Perform the startup flow recommended by XMC. Please consult XMC for the details of this flow. XMC chips will be forbidden to be used, when this option is disabled.

DON'T DISABLE THIS UNLESS YOU KNOW WHAT YOU ARE DOING.

comment "Features below require specific hardware (READ DOCS FIRST!)"

Default value:

- Yes (enabled)

CONFIG_BOOTLOADER_CACHE_32BIT_ADDR_QUAD_FLASH

Enable cache access to 32-bit-address (over 16MB) range of SPI Flash (READ DOCS FIRST)

Found in: Bootloader config > Serial Flash Configurations

Enabling this option allows the CPU to access 32-bit-address flash beyond 16M range. 1. This option only valid for 4-line flash. Octal flash doesn't need this. 2. This option is experimental, which means it can’t use on all flash chips stable, for more information, please contact Espressif Business support.

CONFIG_BOOTLOADER_VDDSDIO_BOOST

VDDSDIO LDO voltage

Found in: Bootloader config

If this option is enabled, and VDDSDIO LDO is set to 1.8V (using eFuse or MTDI bootstrapping pin), bootloader will change LDO settings to output 1.9V instead. This helps prevent flash chip from browning out during flash programming operations.

This option has no effect if VDDSDIO is set to 3.3V, or if the internal VDDSDIO regulator is disabled via eFuse.

Available options:

- 1.8V (CONFIG_BOOTLOADER_VDDSDIO_BOOST_1_8V)
- 1.9V (CONFIG_BOOTLOADER_VDDSDIO_BOOST_1_9V)

CONFIG_BOOTLOADER_FACTORY_RESET

GPIO triggers factory reset

Found in: Bootloader config

Allows to reset the device to factory settings: - clear one or more data partitions; - boot from "factory" partition. The factory reset will occur if there is a GPIO input held at the configured level while device starts up. See settings below.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_NUM_PIN_FACTORY_RESET

Number of the GPIO input for factory reset

Found in: Bootloader config > CONFIG_BOOTLOADER_FACTORY_RESET

The selected GPIO will be configured as an input with internal pull-up enabled. To trigger a factory reset, this GPIO must be held high or low (as configured) on startup.

Note that on some SoCs not all pins have an internal pull-up and certain pins are already used by ROM bootloader as bootstrapping. Refer to the technical reference manual for further details on the selected SoC.

Range:

- from 0 to 39 if CONFIG_BOOTLOADER_FACTORY_RESET

Default value:

- 4 if CONFIG_BOOTLOADER_FACTORY_RESET

CONFIG_BOOTLOADER_FACTORY_RESET_PIN_LEVEL

Factory reset GPIO level

Found in: Bootloader config > CONFIG_BOOTLOADER_FACTORY_RESET

Pin level for factory reset, can be triggered on low or high.

Available options:

- Reset on GPIO low (CONFIG_BOOTLOADER_FACTORY_RESET_PIN_LOW)
- Reset on GPIO high (CONFIG_BOOTLOADER_FACTORY_RESET_PIN_HIGH)

CONFIG_BOOTLOADER_OTA_DATA_ERASE

Clear OTA data on factory reset (select factory partition)

Found in: Bootloader config > CONFIG_BOOTLOADER_FACTORY_RESET

The device will boot from "factory" partition (or OTA slot 0 if no factory partition is present) after a factory reset.

CONFIG_BOOTLOADER_DATA_FACTORY_RESET

Comma-separated names of partitions to clear on factory reset

Found in: Bootloader config > CONFIG_BOOTLOADER_FACTORY_RESET

Allows customers to select which data partitions will be erased while factory reset.

Specify the names of partitions as a comma-delimited with optional spaces for readability. (Like this: "nvs, phy_init, ...") Make sure that the name specified in the partition table and here are the same. Partitions of type "app" cannot be specified here.

Default value:

- "nvs" if CONFIG_BOOTLOADER_FACTORY_RESET

CONFIG_BOOTLOADER_APP_TEST

GPIO triggers boot from test app partition

Found in: Bootloader config

Allows to run the test app from "TEST" partition. A boot from "test" partition will occur if there is a GPIO input pulled low while device starts up. See settings below.

CONFIG_BOOTLOADER_NUM_PIN_APP_TEST

Number of the GPIO input to boot TEST partition

Found in: Bootloader config > CONFIG_BOOTLOADER_APP_TEST

The selected GPIO will be configured as an input with internal pull-up enabled. To trigger a test app, this GPIO must be pulled low on reset. After the GPIO input is deactivated and the device reboots, the old application will boot. (factory or OTA[x]).

Note that on some SoCs not all pins have an internal pull-up and certain pins are already used by ROM bootloader as bootstrapping. Refer to the technical reference manual for further details on the selected SoC.

Range:

- from 0 to 39 if CONFIG_BOOTLOADER_APP_TEST

Default value:

- 18 if CONFIG_BOOTLOADER_APP_TEST

CONFIG_BOOTLOADER_APP_TEST_PIN_LEVEL

App test GPIO level

Found in: Bootloader config > CONFIG_BOOTLOADER_APP_TEST

Pin level for app test, can be triggered on low or high.

Available options:

- Enter test app on GPIO low (CONFIG_BOOTLOADER_APP_TEST_PIN_LOW)
- Enter test app on GPIO high (CONFIG_BOOTLOADER_APP_TEST_PIN_HIGH)

CONFIG_BOOTLOADER_HOLD_TIME_GPIO

Hold time of GPIO for reset/test mode (seconds)

Found in: Bootloader config

The GPIO must be held low continuously for this period of time after reset before a factory reset or test partition boot (as applicable) is performed.

Default value:

- 5 if CONFIG_BOOTLOADER_FACTORY_RESET || CONFIG_BOOTLOADER_APP_TEST

CONFIG_BOOTLOADER_REGION_PROTECTION_ENABLE

Enable protection for unmapped memory regions

Found in: Bootloader config

Protects the unmapped memory regions of the entire address space from unintended accesses. This will ensure that an exception will be triggered whenever the CPU performs a memory operation on unmapped regions of the address space. NOTE: Disabling this config on some targets (ESP32-C6, ESP32-H2, ESP32-C5) would not generate an exception when reading from or writing to 0x0.

Default value:

- Yes (enabled)

CONFIG_BOOTLOADER_WDT_ENABLE

Use RTC watchdog in start code

Found in: Bootloader config

Tracks the execution time of startup code. If the execution time is exceeded, the RTC_WDT will restart system. It is also useful to prevent a lock up in start code caused by an unstable power source. NOTE: Tracks the execution time starts from the bootloader code - re-set timeout, while selecting the source for slow_clk - and ends calling app_main. Re-set timeout is needed due to WDT uses a SLOW_CLK clock source. After changing a frequency slow_clk a time of WDT needs to re-set for new frequency. slow_clk depends on RTC_CLK_SRC (INTERNAL_RC or EXTERNAL_CRYSTAL).

Default value:

- Yes (enabled)

CONFIG_BOOTLOADER_WDT_DISABLE_IN_USER_CODE

Allows RTC watchdog disable in user code

Found in: Bootloader config > CONFIG_BOOTLOADER_WDT_ENABLE

If this option is set, the ESP-IDF app must explicitly reset, feed, or disable the rtc_wdt in the app's own code. If this option is not set (default), then rtc_wdt will be disabled by ESP-IDF before calling the app_main() function.

Use function wdt_hal_feed() for resetting counter of RTC_WDT. For esp32/s2 you can also use rtc_wdt_feed().

Use function wdt_hal_disable() for disabling RTC_WDT. For esp32/s2 you can also use rtc_wdt_disable().

Default value:

- No (disabled)

CONFIG_BOOTLOADER_WDT_TIME_MS

Timeout for RTC watchdog (ms)

Found in: Bootloader config > CONFIG_BOOTLOADER_WDT_ENABLE

Verify that this parameter is correct and more then the execution time. Pay attention to options such as reset to factory, trigger test partition and encryption on boot - these options can increase the execution time. Note: RTC_WDT will reset while encryption operations will be performed.

Range:

- from 0 to 120000

Default value:

- 9000

CONFIG_BOOTLOADER_SKIP_VALIDATE_IN_DEEP_SLEEP

Skip image validation when exiting deep sleep

Found in: Bootloader config

This option disables the normal validation of an image coming out of deep sleep (checksums, SHA256, and signature). This is a trade-off between wakeup performance from deep sleep, and image integrity checks.

Only enable this if you know what you are doing. It should not be used in conjunction with using deep_sleep() entry and changing the active OTA partition as this would skip the validation upon first load of the new OTA partition.

It is possible to enable this option with Secure Boot if "allow insecure options" is enabled, however it's strongly recommended to NOT enable it as it may allow a Secure Boot bypass.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT && CONFIG_SECURE_BOOT_INSECURE

CONFIG_BOOTLOADER_SKIP_VALIDATE_ON_POWER_ON

Skip image validation from power on reset (READ HELP FIRST)

Found in: Bootloader config

Some applications need to boot very quickly from power on. By default, the entire app binary is read from flash and verified which takes up a significant portion of the boot time.

Enabling this option will skip validation of the app when the SoC boots from power on. Note that in this case it's not possible for the bootloader to detect if an app image is corrupted in the flash, therefore it's not possible to safely fall back to a different app partition. Flash corruption of this kind is unlikely but can happen if there is a serious firmware bug or physical damage.

Following other reset types, the bootloader will still validate the app image. This increases the chances that flash corruption resulting in a crash can be detected following soft reset, and the bootloader will fall back to a valid app image. To increase the chances of successfully recovering from a flash corruption event, keep the option BOOTLOADER_WDT_ENABLE enabled and consider also enabling BOOTLOADER_WDT_DISABLE_IN_USER_CODE - then manually disable the RTC Watchdog once the app is running. In addition, enable both the Task and Interrupt watchdog timers with reset options set.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_SKIP_VALIDATE_ALWAYS

Skip image validation always (READ HELP FIRST)

Found in: Bootloader config

Selecting this option prevents the bootloader from ever validating the app image before booting it. Any flash corruption of the selected app partition will make the entire SoC unbootable.

Although flash corruption is a very rare case, it is not recommended to select this option. Consider selecting "Skip image validation from power on reset" instead. However, if boot time is the only important factor then it can be enabled.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC

Reserve RTC FAST memory for custom purposes

Found in: Bootloader config

This option allows the customer to place data in the RTC FAST memory, this area remains valid when rebooted, except for power loss. This memory is located at a fixed address and is available for both the bootloader and the application. (The application and bootloader must be compiled with the same option). The RTC FAST memory has access only through PRO_CPU.

Default value:

- No (disabled)

CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC_IN_CRC

Include custom memory in the CRC calculation

Found in: Bootloader config > CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC

This option allows the customer to use the legacy bootloader behavior when the RTC FAST memory CRC calculation takes place. When this option is enabled, the allocated user custom data will be taken into account in the CRC calculation. This means that any change to the custom data would need a CRC update to prevent the bootloader from marking this data as corrupted. If this option is disabled, the custom data will not be taken into account when calculating the RTC FAST memory CRC. The user custom data can be changed freely, without the need to update the CRC. THIS OPTION MUST BE THE SAME FOR BOTH THE BOOTLOADER AND THE APPLICATION BUILDS.

Default value:

- No (disabled) if CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC

CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC_SIZE

Size in bytes for custom purposes

Found in: Bootloader config > CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC

This option reserves in RTC FAST memory the area for custom purposes. If you want to create your own bootloader and save more information in this area of memory, you can increase it. It must be a multiple of 4 bytes. This area (rtc_retain_mem_t) is reserved and has access from the bootloader and an application.

Default value:

- 0 if CONFIG_BOOTLOADER_CUSTOM_RESERVE_RTC

Security features

Contains:

- CONFIG_SECURE_BOOT_INSECURE
- CONFIG_SECURE_SIGNED_APPS_SCHEME
- CONFIG_SECURE_SIGNED_ON_BOOT_NO_SECURE_BOOT
- CONFIG_SECURE_FLASH_CHECK_ENC_EN_IN_APP
- CONFIG_SECURE_BOOT_V2_ALLOW_EFUSE_RD_DIS
- CONFIG_SECURE_BOOT_ECDSA_KEY_LEN_SIZE
- CONFIG_SECURE_BOOT_ENABLE_AGGRESSIVE_KEY_REVOKE
- CONFIG_SECURE_FLASH_ENC_ENABLED
- CONFIG_SECURE_BOOT
- CONFIG_SECURE_FLASH_ENCRYPT_ONLY_IMAGE_LEN_IN_APP_PART
- CONFIG_SECURE_BOOT_FLASH_BOOTLOADER_DEFAULT
- CONFIG_SECURE_BOOTLOADER_KEY_ENCODING
- CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC
- Potentially insecure options
- CONFIG_SECURE_SIGNED_APPS_NO_SECURE_BOOT
- CONFIG_SECURE_BOOT_VERIFICATION_KEY
- CONFIG_SECURE_BOOTLOADER_MODE
- CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES
- CONFIG_SECURE_UART_ROM_DL_MODE
- CONFIG_SECURE_SIGNED_ON_UPDATE_NO_SECURE_BOOT

CONFIG_SECURE_SIGNED_APPS_NO_SECURE_BOOT

Require signed app images

Found in: Security features

Require apps to be signed to verify their integrity.

This option uses the same app signature scheme as hardware secure boot, but unlike hardware secure boot it does not prevent the bootloader from being physically updated. This means that the device can be secured against remote network access, but not physical access. Compared to using hardware Secure Boot this option is much simpler to implement.

CONFIG_SECURE_SIGNED_APPS_SCHEME

App Signing Scheme

Found in: Security features

Select the Secure App signing scheme. Depends on the Chip Revision. There are two secure boot versions:

1. **Secure boot V1**
    - Legacy custom secure boot scheme. Supported in ESP32 SoC.
2. **Secure boot V2**
    - RSA based secure boot scheme. Supported in ESP32-ECO3 (ESP32 Chip Revision 3 onwards), ESP32-S2, ESP32-C3, ESP32-S3 SoCs.
    - ECDSA based secure boot scheme. Supported in ESP32-C2 SoC.

Available options:

- ECDSA (CONFIG_SECURE_SIGNED_APPS_ECDSA_SCHEME)

  Embeds the ECDSA public key in the bootloader and signs the application with an ECDSA key. Refer to the documentation before enabling.

- RSA (CONFIG_SECURE_SIGNED_APPS_RSA_SCHEME)

  Appends the RSA-3072 based Signature block to the application. Refer to <Secure Boot Version 2 documentation link> before enabling.

- ECDSA (V2) (CONFIG_SECURE_SIGNED_APPS_ECDSA_V2_SCHEME)

  For Secure boot V2 (e.g., ESP32-C2 SoC), appends ECDSA based signature block to the application. Refer to documentation before enabling.

CONFIG_SECURE_BOOT_ECDSA_KEY_LEN_SIZE

ECDSA key size

Found in: Security features

Select the ECDSA key size. Two key sizes are supported depending on the target:

- 256 bit key using NISTP256 curve (Recommended)
- 384 bit key using NISTP384 curve (Recommended)

At present, both key sizes are practically implausible to bruteforce.

Available options:

- Using ECC curve NISTP256 (Recommended) (CONFIG_SECURE_BOOT_ECDSA_KEY_LEN_256_BITS)
- Using ECC curve NISTP384 (Recommended) (CONFIG_SECURE_BOOT_ECDSA_KEY_LEN_384_BITS)

CONFIG_SECURE_SIGNED_ON_BOOT_NO_SECURE_BOOT

Bootloader verifies app signatures

Found in: Security features

If this option is set, the bootloader will be compiled with code to verify that an app is signed before booting it.

If hardware secure boot is enabled, this option is always enabled and cannot be disabled. If hardware secure boot is not enabled, this option doesn't add significant security by itself so most users will want to leave it disabled.

Default value:

- No (disabled) if CONFIG_SECURE_SIGNED_APPS_NO_SECURE_BOOT && CONFIG_SECURE_SIGNED_APPS_ECDSA_SCHEME

CONFIG_SECURE_SIGNED_ON_UPDATE_NO_SECURE_BOOT

Verify app signature on update

Found in: Security features

If this option is set, any OTA updated apps will have the signature verified before being considered valid.

When enabled, the signature is automatically checked whenever the esp_ota_ops.h APIs are used for OTA updates, or esp_image_format.h APIs are used to verify apps.

If hardware secure boot is enabled, this option is always enabled and cannot be disabled. If hardware secure boot is not enabled, this option still adds significant security against network-based attackers by preventing spoofing of OTA updates.

Default value:

- Yes (enabled) if CONFIG_SECURE_SIGNED_APPS_NO_SECURE_BOOT

CONFIG_SECURE_BOOT

Enable hardware Secure Boot in bootloader (READ DOCS FIRST)

Found in: Security features

Build a bootloader which enables Secure Boot on first boot.

Once enabled, Secure Boot will not boot a modified bootloader. The bootloader will only load a partition table or boot an app if the data has a verified digital signature. There are implications for reflashing updated apps once secure boot is enabled.

When enabling secure boot, JTAG and ROM BASIC Interpreter are permanently disabled by default.

Default value:

- No (disabled)

CONFIG_SECURE_BOOT_VERSION

Select secure boot version

Found in: Security features > CONFIG_SECURE_BOOT

Select the Secure Boot Version. Depends on the Chip Revision. Secure Boot V2 is the new RSA / ECDSA based secure boot scheme.

- RSA based scheme is supported in ESP32 (Revision 3 onwards), ESP32-S2, ESP32-C3 (ECO3), ESP32-S3.
- ECDSA based scheme is supported in ESP32-C2 SoC.

Please note that, RSA or ECDSA secure boot is property of specific SoC based on its HW design, supported crypto accelerators, die-size, cost and similar parameters. Please note that RSA scheme has requirement for bigger key sizes but at the same time it is comparatively faster than ECDSA verification.

Secure Boot V1 is the AES based (custom) secure boot scheme supported in ESP32 SoC.

Available options:

- Enable Secure Boot version 1 (CONFIG_SECURE_BOOT_V1_ENABLED)

  Build a bootloader which enables secure boot version 1 on first boot. Refer to the Secure Boot section of the ESP-IDF Programmer's Guide for this version before enabling.

- Enable Secure Boot version 2 (CONFIG_SECURE_BOOT_V2_ENABLED)

  Build a bootloader which enables Secure Boot version 2 on first boot. Refer to Secure Boot V2 section of the ESP-IDF Programmer's Guide for this version before enabling.

CONFIG_SECURE_BOOTLOADER_MODE

Secure bootloader mode

Found in: Security features

Available options:

- One-time flash (CONFIG_SECURE_BOOTLOADER_ONE_TIME_FLASH)

  On first boot, the bootloader will generate a key which is not readable externally or by software. A digest is generated from the bootloader image itself. This digest will be verified on each subsequent boot.

  Enabling this option means that the bootloader cannot be changed after the first time it is booted.

- Reflashable (CONFIG_SECURE_BOOTLOADER_REFLASHABLE)

  Generate a reusable secure bootloader key, derived (via SHA-256) from the secure boot signing key.

  This allows the secure bootloader to be re-flashed by anyone with access to the secure boot signing key.

  This option is less secure than one-time flash, because a leak of the digest key from one device allows reflashing of any device that uses it.

CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES

Sign binaries during build

Found in: Security features

Once secure boot or signed app requirement is enabled, app images are required to be signed.

If enabled (default), these binary files are signed as part of the build process. The file named in "Secure boot private signing key" will be used to sign the image.

If disabled, unsigned app/partition data will be built. They must be signed manually using espsecure. Version 1 to enable ECDSA Based Secure Boot and Version 2 to enable RSA based Secure Boot. (for example, on a remote signing server.)

CONFIG_SECURE_BOOT_SIGNING_KEY

Secure boot private signing key

Found in: Security features > CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES

Path to the key file used to sign app images.

Key file is an ECDSA private key (NIST256p curve) in PEM format for Secure Boot V1. Key file is an RSA private key in PEM format for Secure Boot V2 (RSA scheme). Key file is an ECDSA private key (256p or 384p) in PEM format for Secure Boot V2 (ECDSA scheme).

Path is evaluated relative to the project directory.

You can generate a new signing key by running the following command: espsecure generate-signing-key secure_boot_signing_key.pem

See the Secure Boot section of the ESP-IDF Programmer's Guide for this version for details.

Default value:

- "secure_boot_signing_key.pem" if CONFIG_SECURE_BOOT_BUILD_SIGNED_BINARIES

CONFIG_SECURE_BOOT_VERIFICATION_KEY

Secure boot public signature verification key

Found in: Security features

Path to a public key file used to verify signed images. Secure Boot V1: This ECDSA public key is compiled into the bootloader and/or app, to verify app images.

Key file is in raw binary format, and can be extracted from a PEM formatted private key using the espsecure extract-public-key command.

Refer to the Secure Boot section of the ESP-IDF Programmer's Guide for this version before enabling.

CONFIG_SECURE_BOOT_ENABLE_AGGRESSIVE_KEY_REVOKE

Enable Aggressive key revoke strategy

Found in: Security features

If this option is set, ROM bootloader will revoke the public key digest burned in efuse block if it fails to verify the signature of software bootloader with it. Revocation of keys does not happen when enabling secure boot. Once secure boot is enabled, key revocation checks will be done on subsequent boot-up, while verifying the software bootloader

This feature provides a strong resistance against physical attacks on the device.

NOTE: Once a digest slot is revoked, it can never be used again to verify an image This can lead to permanent bricking of the device, in case all keys are revoked because of signature verification failure.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT && SOC_SUPPORT_SECURE_BOOT_REVOKE_KEY

CONFIG_SECURE_BOOT_V2_ALLOW_EFUSE_RD_DIS

Do not disable the ability to further read protect eFuses

Found in: Security features

If not set (default, recommended), on first boot the bootloader will burn the WR_DIS_RD_DIS efuse when Secure Boot is enabled. This prevents any more efuses from being read protected.

If this option is set, it will remain possible to write the EFUSE_RD_DIS efuse field after Secure Boot is enabled. This may allow an attacker to read-protect the BLK2 efuse (for ESP32) and BLOCK4-BLOCK10 (i.e. BLOCK_KEY0-BLOCK_KEY5)(for other chips) holding the secure boot public key digest, causing an immediate denial of service and possibly allowing an additional fault injection attack to bypass the signature protection.

The option must be set when you need to program any read-protected key type into the efuses, e.g., HMAC, ECDSA etc. after secure boot has already been enabled on the device. Please refer to secure boot V2 documentation guide for more details.

NOTE: Once a BLOCK is read-protected, the application will read all zeros from that block

NOTE: If "UART ROM download mode (Permanently disabled (recommended))" or "UART ROM download mode (Permanently switch to Secure mode (recommended))" is set, then it is __NOT__ possible to read/write efuses using espefuse utility. However, efuse can be read/written from the application

Please refer to the Secure Boot V2 documentation guide for more information.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT_V2_ENABLED

CONFIG_SECURE_BOOT_FLASH_BOOTLOADER_DEFAULT

Flash bootloader along with other artifacts when using the default flash command

Found in: Security features

When Secure Boot V2 is enabled, by default the bootloader is not flashed along with other artifacts like the application and the partition table images, i.e. bootloader has to be separately flashed using the command *idf.py bootloader flash*, whereas, the application and partition table can be flashed using the command *idf.py flash* itself. Enabling this option allows flashing the bootloader along with the other artifacts by invocation of the command *idf.py flash*.

If this option is enabled make sure that even the bootloader is signed using the correct secure boot key, otherwise the bootloader signature verification would fail, as hash of the public key which is present in the bootloader signature would not match with the digest stored into the efuses and thus the device will not be able to boot up.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT_V2_ENABLED

CONFIG_SECURE_BOOTLOADER_KEY_ENCODING

Hardware Key Encoding

Found in: Security features

In reflashable secure bootloader mode, a hardware key is derived from the signing key (with SHA-256) and can be written to eFuse with espefuse.

Normally this is a 256-bit key, but if 3/4 Coding Scheme is used on the device then the eFuse key is truncated to 192 bits.

This configuration item doesn't change any firmware code, it only changes the size of key binary which is generated at build time.

Available options:

- No encoding (256 bit key) (CONFIG_SECURE_BOOTLOADER_KEY_ENCODING_256BIT)
- 3/4 encoding (192 bit key) (CONFIG_SECURE_BOOTLOADER_KEY_ENCODING_192BIT)

CONFIG_SECURE_BOOT_INSECURE

Allow potentially insecure options

Found in: Security features

You can disable some of the default protections offered by secure boot, in order to enable testing or a custom combination of security features.

Only enable these options if you are very sure.

Refer to the Secure Boot section of the ESP-IDF Programmer's Guide for this version before enabling.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT

CONFIG_SECURE_FLASH_ENC_ENABLED

Enable flash encryption on boot (READ DOCS FIRST)

Found in: Security features

If this option is set, flash contents will be encrypted by the bootloader on first boot.

Note: After first boot, the system will be permanently encrypted. Re-flashing an encrypted system is complicated and not always possible.

Read Flash Encryption before enabling.

Default value:

- No (disabled)

CONFIG_SECURE_FLASH_ENCRYPTION_KEY_SOURCE

Flash Encryption Key Source

Found in: Security features > CONFIG_SECURE_FLASH_ENC_ENABLED

Specify the key source for the Flash Encryption Key

Available options:

- eFuse Key Block (CONFIG_SECURE_FLASH_ENCRYPTION_KEY_SOURCE_EFUSES)

  Use a key that is stored in the eFuses key blocks.

- Key Manager (CONFIG_SECURE_FLASH_ENCRYPTION_KEY_SOURCE_KEY_MGR)

  Use a key that is deployed using the Key Manager

CONFIG_SECURE_FLASH_ENCRYPTION_KEYSIZE

Size of generated XTS-AES key

Found in: Security features > CONFIG_SECURE_FLASH_ENC_ENABLED

Size of generated XTS-AES key.

- AES-128 uses a 256-bit key (32 bytes) derived from 128 bits (16 bytes) burned in half Efuse key block. Internally, it calculates SHA256(128 bits)
- AES-128 uses a 256-bit key (32 bytes) which occupies one Efuse key block.
- AES-256 uses a 512-bit key (64 bytes) which occupies two Efuse key blocks.

This setting is ignored if either type of key is already burned to Efuse before the first boot. In this case, the pre-burned key is used and no new key is generated.

Available options:

- AES-128 key derived from 128 bits (SHA256(128 bits)) (CONFIG_SECURE_FLASH_ENCRYPTION_AES128_DERIVED)
- AES-128 (256-bit key) (CONFIG_SECURE_FLASH_ENCRYPTION_AES128)
- AES-256 (512-bit key) (CONFIG_SECURE_FLASH_ENCRYPTION_AES256)

CONFIG_SECURE_FLASH_ENCRYPTION_MODE

Enable usage mode

Found in: Security features > CONFIG_SECURE_FLASH_ENC_ENABLED

By default Development mode is enabled which allows ROM download mode to perform flash encryption operations (plaintext is sent to the device, and it encrypts it internally and writes ciphertext to flash.) This mode is not secure, it's possible for an attacker to write their own chosen plaintext to flash.

Release mode should always be selected for production or manufacturing. Once enabled it's no longer possible for the device in ROM Download Mode to use the flash encryption hardware.

When EFUSE_VIRTUAL is enabled, SECURE_FLASH_ENCRYPTION_MODE_RELEASE is not available. For CI tests we use IDF_CI_BUILD to bypass it ("export IDF_CI_BUILD=1"). We do not recommend bypassing it for other purposes.

Refer to the Flash Encryption section of the ESP-IDF Programmer's Guide for details.

Available options:

- Development (NOT SECURE) (CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT)
- Release (CONFIG_SECURE_FLASH_ENCRYPTION_MODE_RELEASE)

Potentially insecure options

Contains:

- CONFIG_SECURE_BOOT_ALLOW_SHORT_APP_PARTITION
- CONFIG_SECURE_BOOT_ALLOW_JTAG
- CONFIG_SECURE_BOOT_V2_FORCE_ENABLE_ECDSA
- CONFIG_SECURE_BOOT_ALLOW_ROM_BASIC
- CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_DEC
- CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_ENC
- CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_CACHE
- CONFIG_SECURE_BOOT_ALLOW_UNUSED_DIGEST_SLOTS
- CONFIG_SECURE_FLASH_REQUIRE_ALREADY_ENABLED
- CONFIG_SECURE_FLASH_SKIP_WRITE_PROTECTION_CACHE
- CONFIG_SECURE_BOOT_SKIP_WRITE_PROTECTION_SCA

CONFIG_SECURE_BOOT_V2_FORCE_ENABLE_ECDSA

Force enable ECDSA based Secure Boot V2

Found in: Security features > Potentially insecure options

ECDSA based Secure Boot V2 is not functional for certain input vectors on this SoC and is therefore not offered by default. Refer to the hardware errata document for details.

Setting this option re-enables the ECDSA based Secure Boot V2 signing scheme despite the known vulnerability. Only set this option if you fully understand the risk. RSA based Secure Boot V2 is the recommended scheme on SoCs that support it.

CONFIG_SECURE_BOOT_ALLOW_ROM_BASIC

Leave ROM BASIC Interpreter available on reset

Found in: Security features > Potentially insecure options

By default, the BASIC ROM Console starts on reset if no valid bootloader is read from the flash.

When either flash encryption or secure boot are enabled, the default is to disable this BASIC fallback mode permanently via eFuse.

If this option is set, this eFuse is not burned and the BASIC ROM Console may remain accessible. Only set this option in testing environments.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT_INSECURE || CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_BOOT_ALLOW_JTAG

Allow JTAG Debugging

Found in: Security features > Potentially insecure options

If not set (default), the bootloader will permanently disable JTAG (across entire chip) on first boot when either secure boot or flash encryption is enabled.

Setting this option leaves JTAG on for debugging, which negates all protections of flash encryption and some of the protections of secure boot.

Only set this option in testing environments.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT_INSECURE || CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_BOOT_ALLOW_SHORT_APP_PARTITION

Allow app partition length not 64KB aligned

Found in: Security features > Potentially insecure options

If not set (default), app partition size must be a multiple of 64KB. App images are padded to 64KB length, and the bootloader checks any trailing bytes after the signature (before the next 64KB boundary) have not been written. This is because flash cache maps entire 64KB pages into the address space. This prevents an attacker from appending unverified data after the app image in the flash, causing it to be mapped into the address space.

Setting this option allows the app partition length to be unaligned, and disables padding of the app image to this length. It is generally not recommended to set this option, unless you have a legacy partitioning scheme which doesn't support 64KB aligned partition lengths.

CONFIG_SECURE_BOOT_ALLOW_UNUSED_DIGEST_SLOTS

Leave unused digest slots available (not revoke)

Found in: Security features > Potentially insecure options

If not set (default), during startup in the app all unused digest slots will be revoked. To revoke unused slot will be called esp_efuse_set_digest_revoke(num_digest) for each digest. Revoking unused digest slots makes ensures that no trusted keys can be added later by an attacker. If set, it means that you have a plan to use unused digests slots later.

Note that if you plan to enable secure boot during the first boot up, the bootloader will intentionally revoke the unused digest slots while enabling secure boot, even if the above config is enabled because keeping the unused key slots un-revoked would a security hazard. In case for any development workflow if you need to avoid this revocation, you should enable secure boot externally (host based mechanism) rather than enabling it during the boot up, so that the bootloader would not need to enable secure boot and thus you could avoid its revocation strategy.

Default value:

- No (disabled) if CONFIG_SECURE_BOOT_INSECURE && SOC_EFUSE_REVOKE_BOOT_KEY_DIGESTS

CONFIG_SECURE_BOOT_SKIP_WRITE_PROTECTION_SCA

Skip write-protection of SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH

Found in: Security features > Potentially insecure options

If not set (default, recommended), on the first boot when Secure Boot is enabled for targets that support Secure Boot using ECDSA-P384, the bootloader will burn the write-protection bit of of SECURE_BOOT_SHA384_EN that could be shared by multiple other efuse bits like SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH / XTS_DPA_PSEUDO_LEVEL.

Once this efuse bit is write-protected you cannot update the values of the shared efuses, for example, the security strength value of XTS_DPA_PSEUDO_LEVEL or setting ECC_FORCE_CONST_TIME.

List of eFuses with the same write protection bit:

ESP32-C5: XTS_DPA_PSEUDO_LEVEL and ECC_FORCE_CONST_TIME

Default value:

- No (disabled) if SOC_ECDSA_SUPPORT_CURVE_P384 && SOC_FLASH_ENCRYPTION_XTS_AES_SUPPORT_PSEUDO_ROUND

CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_ENC

Leave UART bootloader encryption enabled

Found in: Security features > Potentially insecure options

If not set (default), the bootloader will permanently disable UART bootloader encryption access on first boot. If set, the UART bootloader will still be able to access hardware encryption.

It is recommended to only set this option in testing environments.

Default value:

- No (disabled) if CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_DEC

Leave UART bootloader decryption enabled

Found in: Security features > Potentially insecure options

If not set (default), the bootloader will permanently disable UART bootloader decryption access on first boot. If set, the UART bootloader will still be able to access hardware decryption.

Only set this option in testing environments. Setting this option allows complete bypass of flash encryption.

Default value:

- No (disabled) if CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_CACHE

Leave UART bootloader flash cache enabled

Found in: Security features > Potentially insecure options

If not set (default), the bootloader will permanently disable UART bootloader flash cache access on first boot. If set, the UART bootloader will still be able to access the flash cache.

Only set this option in testing environments.

Default value:

- No (disabled) if CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_FLASH_REQUIRE_ALREADY_ENABLED

Require flash encryption to be already enabled

Found in: Security features > Potentially insecure options

If not set (default), and flash encryption is not yet enabled in eFuses, the 2nd stage bootloader will enable flash encryption: generate the flash encryption key and program eFuses. If this option is set, and flash encryption is not yet enabled, the bootloader will error out and reboot. If flash encryption is enabled in eFuses, this option does not change the bootloader behavior.

Only use this option in testing environments, to avoid accidentally enabling flash encryption on the wrong device. The device needs to have flash encryption already enabled using espefuse.

Default value:

- No (disabled) if CONFIG_SECURE_FLASH_ENCRYPTION_MODE_DEVELOPMENT

CONFIG_SECURE_FLASH_SKIP_WRITE_PROTECTION_CACHE

Skip write-protection of DIS_CACHE (DIS_ICACHE, DIS_DCACHE)

Found in: Security features > Potentially insecure options

If not set (default, recommended), on the first boot the bootloader will burn the write-protection of DIS_CACHE(for ESP32) or DIS_ICACHE/DIS_DCACHE(for other chips) eFuse when Flash Encryption is enabled. Write protection for cache disable efuse prevents the chip from being blocked if it is set by accident. App and bootloader use cache so disabling it makes the chip useless for IDF. Due to other eFuses are linked with the same write protection bit (see the list below) then write-protection will not be done if these SECURE_FLASH_UART_BOOTLOADER_ALLOW_ENC, SECURE_BOOT_ALLOW_JTAG or SECURE_FLASH_UART_BOOTLOADER_ALLOW_CACHE options are selected to give a chance to turn on the chip into the release mode later.

List of eFuses with the same write protection bit: ESP32: MAC, MAC_CRC, DISABLE_APP_CPU, DISABLE_BT, DIS_CACHE, VOL_LEVEL_HP_INV.

ESP32-C3: DIS_ICACHE, DIS_USB_JTAG, DIS_DOWNLOAD_ICACHE, DIS_USB_SERIAL_JTAG, DIS_FORCE_DOWNLOAD, DIS_TWAI, JTAG_SEL_ENABLE, DIS_PAD_JTAG, DIS_DOWNLOAD_MANUAL_ENCRYPT.

ESP32-C6: SWAP_UART_SDIO_EN, DIS_ICACHE, DIS_USB_JTAG, DIS_DOWNLOAD_ICACHE, DIS_USB_SERIAL_JTAG, DIS_FORCE_DOWNLOAD, DIS_TWAI, JTAG_SEL_ENABLE, DIS_PAD_JTAG, DIS_DOWNLOAD_MANUAL_ENCRYPT.

ESP32-H2 & ESP32H21: DIS_ICACHE, DIS_ICACHE, DIS_USB_JTAG, POWERGLITCH_EN, DIS_FORCE_DOWNLOAD, SPI_DOWNLOAD_MSPI_DIS, DIS_TWAI, JTAG_SEL_ENABLE, DIS_PAD_JTAG, DIS_DOWNLOAD_MANUAL_ENCRYPT, DIS_USB_SERIAL_JTAG

ESP32-S2: DIS_ICACHE, DIS_DCACHE, DIS_DOWNLOAD_ICACHE, DIS_DOWNLOAD_DCACHE, DIS_FORCE_DOWNLOAD, DIS_USB, DIS_TWAI, DIS_BOOT_REMAP, SOFT_DIS_JTAG, HARD_DIS_JTAG, DIS_DOWNLOAD_MANUAL_ENCRYPT.

ESP32-S3: DIS_ICACHE, DIS_DCACHE, DIS_DOWNLOAD_ICACHE, DIS_DOWNLOAD_DCACHE, DIS_FORCE_DOWNLOAD, DIS_USB_OTG, DIS_TWAI, DIS_APP_CPU, DIS_PAD_JTAG, DIS_DOWNLOAD_MANUAL_ENCRYPT, DIS_USB_JTAG, DIS_USB_SERIAL_JTAG, STRAP_JTAG_SEL, USB_PHY_SEL.

CONFIG_SECURE_FLASH_ENCRYPT_ONLY_IMAGE_LEN_IN_APP_PART

Encrypt contents upto app image length in app partition

Found in: Security features

If set (default), optimise encryption time for the partition of type APP, by only encrypting the app image that is present in the partition, instead of the whole partition. The image length used for encryption is derived from the image metadata, which includes the size of the app image, checksum, hash and also the signature sector when secure boot is enabled.

If not set, the whole partition of type APP would be encrypted, which increases the encryption time but might be useful if there is any custom data appended to the firmware image.

CONFIG_SECURE_FLASH_CHECK_ENC_EN_IN_APP

Check Flash Encryption enabled on app startup

Found in: Security features

If set (default), in an app during startup code, there is a check of the flash encryption eFuse bit is on (as the bootloader should already have set it). The app requires this bit is on to continue work otherwise abort.

If not set, the app does not care if the flash encryption eFuse bit is set or not.

Default value:

- Yes (enabled) if CONFIG_SECURE_FLASH_ENC_ENABLED

CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC

Permanently enable XTS-AES's pseudo rounds function

Found in: Security features

If set (default), the bootloader will permanently enable the XTS-AES peripheral's pseudo rounds function. Note: Enabling this config would burn an efuse.

Default value:

- Yes (enabled) if CONFIG_SECURE_FLASH_ENCRYPTION_MODE_RELEASE && CONFIG_SECURE_FLASH_ENC_ENABLED && SOC_FLASH_ENCRYPTION_XTS_AES_SUPPORT_PSEUDO_ROUND
- No (disabled) if CONFIG_SECURE_FLASH_ENC_ENABLED && SOC_FLASH_ENCRYPTION_XTS_AES_SUPPORT_PSEUDO_ROUND

CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH

Strength of the pseudo rounds function

Found in: Security features > CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC

The strength of the pseudo rounds functions can be configured to low, medium and high, each denoting the values that would be stored in the efuses field. By default the value to set to low.

It is recommended that the required strength of the pseudo rounds function should be set during the first boot itself. If your workflow needs to update the function's strength after the first boot, you should enable CONFIG_SECURE_BOOT_SKIP_WRITE_PROTECTION_SCA to avoid write protecting this bit during the boot up for targets that support Secure Boot using ECDSA-P384.

You can configure the strength of the pseudo rounds functions according to your use cases, for example, increasing the strength would provide higher security but would slow down the flash encryption/decryption operations. For more info regarding the performance impact, please checkout the pseudo round function section of the security guide documentation.

Available options:

- Low (CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH_LOW)
- Medium (CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH_MEDIUM)
- High (CONFIG_SECURE_FLASH_PSEUDO_ROUND_FUNC_STRENGTH_HIGH)

CONFIG_SECURE_UART_ROM_DL_MODE

UART ROM download mode

Found in: Security features

Available options:

- UART ROM download mode (Permanently disabled (recommended)) (CONFIG_SECURE_DISABLE_ROM_DL_MODE)

  If set, during startup the app will burn an eFuse bit to permanently disable the UART ROM Download Mode. This prevents any future use of esptool, espefuse and similar tools.

  Once disabled, if the SoC is booted with strapping pins set for ROM Download Mode then an error is printed instead.

  It is recommended to enable this option in any production application where Flash Encryption and/or Secure Boot is enabled and access to Download Mode is not required.

  It is also possible to permanently disable Download Mode by calling esp_efuse_disable_rom_download_mode() at runtime.

- UART ROM download mode (Permanently switch to Secure mode (recommended)) (CONFIG_SECURE_ENABLE_SECURE_ROM_DL_MODE)

  If set, during startup the app will burn an eFuse bit to permanently switch the UART ROM Download Mode into a separate Secure Download mode. This option can only work if Download Mode is not already disabled by eFuse.

  Secure Download mode limits the use of Download Mode functions to update SPI config, changing baud rate, basic flash write and a command to return a summary of currently enabled security features (*get-security-info*).

  Secure Download mode is not compatible with the esptool flasher stub feature, espefuse, read/writing memory or registers, encrypted download, or any other features that interact with unsupported Download Mode commands.

  Secure Download mode should be enabled in any application where Flash Encryption and/or Secure Boot is enabled. Disabling this option does not immediately cancel the benefits of the security features, but it increases the potential "attack surface" for an attacker to try and bypass them with a successful physical attack.

  It is also possible to enable secure download mode at runtime by calling esp_efuse_enable_rom_secure_download_mode()

  Note: Secure Download mode is not available for ESP32.

- UART ROM download mode (Enabled (not recommended)) (CONFIG_SECURE_INSECURE_ALLOW_DL_MODE)

  This is a potentially insecure option. Enabling this option will allow the full UART download mode to stay enabled. This option SHOULD NOT BE ENABLED for production use cases.

Application manager

Contains:

- CONFIG_APP_EXCLUDE_PROJECT_NAME_VAR
- CONFIG_APP_EXCLUDE_PROJECT_VER_VAR
- CONFIG_APP_PROJECT_VER_FROM_CONFIG
- CONFIG_APP_RETRIEVE_LEN_ELF_SHA
- CONFIG_APP_COMPILE_TIME_DATE

CONFIG_APP_COMPILE_TIME_DATE

Use time/date stamp for app

Found in: Application manager

If set, then the app will be built with the current time/date stamp. It is stored in the app description structure. If not set, time/date stamp will be excluded from app image. This can be useful for getting the same binary image files made from the same source, but at different times.

CONFIG_APP_EXCLUDE_PROJECT_VER_VAR

Exclude PROJECT_VER from firmware image

Found in: Application manager

The PROJECT_VER variable from the build system will not affect the firmware image. This value will not be contained in the esp_app_desc structure.

Default value:

- No (disabled)

CONFIG_APP_EXCLUDE_PROJECT_NAME_VAR

Exclude PROJECT_NAME from firmware image

Found in: Application manager

The PROJECT_NAME variable from the build system will not affect the firmware image. This value will not be contained in the esp_app_desc structure.

Default value:

- No (disabled)

CONFIG_APP_PROJECT_VER_FROM_CONFIG

Get the project version from Kconfig

Found in: Application manager

If this is enabled, then config item APP_PROJECT_VER will be used for the variable PROJECT_VER. Other ways to set PROJECT_VER will be ignored.

Default value:

- No (disabled)

CONFIG_APP_PROJECT_VER

Project version

Found in: Application manager > CONFIG_APP_PROJECT_VER_FROM_CONFIG

Project version

Default value:

- 1 if CONFIG_APP_PROJECT_VER_FROM_CONFIG

CONFIG_APP_RETRIEVE_LEN_ELF_SHA

The length of APP ELF SHA is stored in RAM(chars)

Found in: Application manager

At startup, the app will read the embedded APP ELF SHA-256 hash value from flash and convert it into a string and store it in a RAM buffer. This ensures the panic handler and core dump will be able to print this string even when cache is disabled. The size of the buffer is APP_RETRIEVE_LEN_ELF_SHA plus the null terminator. Changing this value will change the size of this buffer, in bytes.

Range:

- from 8 to 64

Default value:

- 9

Serial flasher config

Contains:

- CONFIG_ESPTOOLPY_AFTER
- CONFIG_ESPTOOLPY_BEFORE
- CONFIG_ESPTOOLPY_FLASH_MODE_AUTO_DETECT
- CONFIG_ESPTOOLPY_HEADER_FLASHSIZE_UPDATE
- CONFIG_ESPTOOLPY_NO_STUB
- CONFIG_ESPTOOLPY_OCT_FLASH
- CONFIG_ESPTOOLPY_FLASH_SAMPLE_MODE
- CONFIG_ESPTOOLPY_FLASHSIZE
- CONFIG_ESPTOOLPY_FLASHMODE
- CONFIG_ESPTOOLPY_FLASHFREQ

CONFIG_ESPTOOLPY_NO_STUB

Disable download stub

Found in: Serial flasher config

The flasher tool sends a precompiled download stub first by default. That stub allows things like compressed downloads and more. Usually you should not need to disable that feature

CONFIG_ESPTOOLPY_OCT_FLASH

Enable Octal Flash

Found in: Serial flasher config

CONFIG_ESPTOOLPY_FLASH_MODE_AUTO_DETECT

Choose flash mode automatically (please read help)

Found in: Serial flasher config

This config option helps decide whether flash is Quad or Octal, but please note some limitations:

1. If the flash chip is an Octal one, even if one of "QIO", "QOUT", "DIO", "DOUT" options is selected in *ESPTOOLPY_FLASHMODE*, our code will automatically change the mode to "OPI" and the sample mode will be STR.
2. If the flash chip is a Quad one, even if "OPI" is selected in *ESPTOOLPY_FLASHMODE*, our code will automatically change the mode to "DIO".
3. This option is mainly to improve the out-of-box experience of developers. It doesn't guarantee the feature-complete. Some code still rely on *ESPTOOLPY_OCT_FLASH*. Please do not rely on this option when you are pretty sure that you are using Octal flash. In this case, please enable *ESPTOOLPY_OCT_FLASH* option, then you can choose *DTR* sample mode in *ESPTOOLPY_FLASH_SAMPLE_MODE*. Otherwise, only *STR* mode is available.
4. Enabling this feature reduces available internal RAM size (around 900 bytes). If your IRAM space is insufficient and you're aware of your flash type, disable this option and select corresponding flash type options.

CONFIG_ESPTOOLPY_FLASHMODE

Flash SPI mode

Found in: Serial flasher config

Mode the flash chip is flashed in, as well as the default mode for the binary to run in.

Available options:

- QIO (CONFIG_ESPTOOLPY_FLASHMODE_QIO)
- QOUT (CONFIG_ESPTOOLPY_FLASHMODE_QOUT)
- DIO (CONFIG_ESPTOOLPY_FLASHMODE_DIO)
- DOUT (CONFIG_ESPTOOLPY_FLASHMODE_DOUT)
- OPI (CONFIG_ESPTOOLPY_FLASHMODE_OPI)

CONFIG_ESPTOOLPY_FLASH_SAMPLE_MODE

Flash Sampling Mode

Found in: Serial flasher config

Available options:

- STR Mode (CONFIG_ESPTOOLPY_FLASH_SAMPLE_MODE_STR)
- DTR Mode (CONFIG_ESPTOOLPY_FLASH_SAMPLE_MODE_DTR)

CONFIG_ESPTOOLPY_FLASHFREQ

Flash SPI speed

Found in: Serial flasher config

Available options:

- 80 MHz (CONFIG_ESPTOOLPY_FLASHFREQ_80M)
- 40 MHz (CONFIG_ESPTOOLPY_FLASHFREQ_40M)
- 26 MHz (CONFIG_ESPTOOLPY_FLASHFREQ_26M)
- 20 MHz (CONFIG_ESPTOOLPY_FLASHFREQ_20M)

CONFIG_ESPTOOLPY_FLASHSIZE

Flash size

Found in: Serial flasher config

SPI flash size, in megabytes

Available options:

- 1 MB (CONFIG_ESPTOOLPY_FLASHSIZE_1MB)
- 2 MB (CONFIG_ESPTOOLPY_FLASHSIZE_2MB)
- 4 MB (CONFIG_ESPTOOLPY_FLASHSIZE_4MB)
- 8 MB (CONFIG_ESPTOOLPY_FLASHSIZE_8MB)
- 16 MB (CONFIG_ESPTOOLPY_FLASHSIZE_16MB)
- 32 MB (CONFIG_ESPTOOLPY_FLASHSIZE_32MB)
- 64 MB (CONFIG_ESPTOOLPY_FLASHSIZE_64MB)
- 128 MB (CONFIG_ESPTOOLPY_FLASHSIZE_128MB)

CONFIG_ESPTOOLPY_HEADER_FLASHSIZE_UPDATE

Detect flash size when flashing bootloader

Found in: Serial flasher config

If this option is set, flashing the project will automatically detect the flash size of the target chip and update the bootloader image before it is flashed.

Enabling this option turns off the image protection against corruption by a SHA256 digest. Updating the bootloader image before flashing would invalidate the digest.

CONFIG_ESPTOOLPY_BEFORE

Before flashing

Found in: Serial flasher config

Configure whether esptool should reset the ESP32 before flashing.

Automatic resetting depends on the RTS & DTR signals being wired from the serial port to the ESP32. Most USB development boards do this internally.

Available options:

- Reset to bootloader (CONFIG_ESPTOOLPY_BEFORE_RESET)
- No reset (CONFIG_ESPTOOLPY_BEFORE_NORESET)

CONFIG_ESPTOOLPY_AFTER

After flashing

Found in: Serial flasher config

Configure whether esptool should reset the ESP32 after flashing.

Automatic resetting depends on the RTS & DTR signals being wired from the serial port to the ESP32. Most USB development boards do this internally.

Available options:

- Reset after flashing (CONFIG_ESPTOOLPY_AFTER_RESET)
- Stay in bootloader (CONFIG_ESPTOOLPY_AFTER_NORESET)

Partition Table

Contains:

- CONFIG_PARTITION_TABLE_CUSTOM_FILENAME
- CONFIG_PARTITION_TABLE_MD5
- CONFIG_PARTITION_TABLE_OFFSET
- CONFIG_PARTITION_TABLE_TYPE

CONFIG_PARTITION_TABLE_TYPE

Partition Table

Found in: Partition Table

The partition table to flash to the ESP32. The partition table determines where apps, data and other resources are expected to be found.

The predefined partition table CSV descriptions can be found in the components/partition_table directory. These are mostly intended for example and development use, it's expect that for production use you will copy one of these CSV files and create a custom partition CSV for your application.

Available options:

- Single factory app, no OTA (CONFIG_PARTITION_TABLE_SINGLE_APP)

  This is the default partition table, designed to fit into a 2MB or larger flash with a single 1MB app partition.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_singleapp.csv

  This partition table is not suitable for an app that needs OTA (over the air update) capability.

- Single factory app (large), no OTA (CONFIG_PARTITION_TABLE_SINGLE_APP_LARGE)

  This is a variation of the default partition table, that expands the 1MB app partition size to 1.5MB to fit more code.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_singleapp_large.csv

  This partition table is not suitable for an app that needs OTA (over the air update) capability.

- Factory app, two OTA definitions (CONFIG_PARTITION_TABLE_TWO_OTA)

  This is a basic OTA-enabled partition table with a factory app partition plus two OTA app partitions. All are 1MB, so this partition table requires 4MB or larger flash size.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_two_ota.csv

- Two large size OTA partitions (CONFIG_PARTITION_TABLE_TWO_OTA_LARGE)

  This is a basic OTA-enabled partition table with two OTA app partitions. Both app partition sizes are 1700K, so this partition table requires 4MB or larger flash size.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_two_ota_large.csv

- Custom partition table CSV (CONFIG_PARTITION_TABLE_CUSTOM)

  Specify the path to the partition table CSV to use for your project.

  Consult the Partition Table section in the ESP-IDF Programmers Guide for more information.

- Single factory app, no OTA, encrypted NVS (CONFIG_PARTITION_TABLE_SINGLE_APP_ENCRYPTED_NVS)

  This is a variation of the default "Single factory app, no OTA" partition table that supports encrypted NVS when using flash encryption. See the Flash Encryption section in the ESP-IDF Programmers Guide for more information.

The corresponding CSV file in the IDF directory is components/partition_table/partitions_singleapp_encr_nvs.csv

- Single factory app (large), no OTA, encrypted NVS (CONFIG_PARTITION_TABLE_SINGLE_APP_LARGE_ENC_NVS)

  This is a variation of the "Single factory app (large), no OTA" partition table that supports encrypted NVS when using flash encryption. See the Flash Encryption section in the ESP-IDF Programmers Guide for more information.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_singleapp_large_encr_nvs.csv

- Factory app, two OTA definitions, encrypted NVS (CONFIG_PARTITION_TABLE_TWO_OTA_ENCRYPTED_NVS)

  This is a variation of the "Factory app, two OTA definitions" partition table that supports encrypted NVS when using flash encryption. See the Flash Encryption section in the ESP-IDF Programmers Guide for more information.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_two_ota_encr_nvs.csv

- Single factory app, no OTA, TEE (CONFIG_PARTITION_TABLE_SINGLE_APP_TEE)

  This is a variation of the default "Single factory app, no OTA" partition table that supports the ESP-TEE framework. See the Trusted Execution Environment (TEE) section in the ESP-IDF Programmers Guide for more information.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_singleapp_tee.csv

- Two OTA definitions, TEE (CONFIG_PARTITION_TABLE_TWO_OTA_TEE)

  This is a basic OTA-enabled partition table with two OTA app partitions each for the TEE and the user (REE) application. The user app partition sizes are 1536K, so this partition table requires 4MB or larger flash size. See the Trusted Execution Environment (TEE) section in the ESP-IDF Programmers Guide for more information.

  The corresponding CSV file in the IDF directory is components/partition_table/partitions_two_ota_tee.csv

CONFIG_PARTITION_TABLE_CUSTOM_FILENAME

Custom partition CSV file

Found in: Partition Table

Name of the custom partition CSV filename. This path is evaluated relative to the project root directory by default. However, if the absolute path for the CSV file is provided, then the absolute path is configured.

Default value:

- "partitions.csv"

CONFIG_PARTITION_TABLE_OFFSET

Offset of partition table

Found in: Partition Table

The offset address of the partition table is, by default, 0x8000, and it starts right after the bootloader. This offset can be changed if more space is required for the bootloader. If you change this offset, both the bootloader and the app must be compiled with the same PARTITION_TABLE_OFFSET value.

This value must be a multiple of 0x1000.

If you change the partition table offset, make sure to update the partition offsets in the partition table CSV file. To let the build system automatically adjust partition offsets based on the configured partition table offset, leave the offset fields blank in the CSV file.

Note: If you specify fixed offsets in the partition table CSV (i.e., the offset column is not blank), the first partition must start at PARTITION_TABLE_OFFSET + 0x1000, since the partition table occupies 0x1000 bytes of flash space.

Default value:

- "0x8000"

CONFIG_PARTITION_TABLE_MD5

Generate an MD5 checksum for the partition table

Found in: Partition Table

Generate an MD5 checksum for the partition table for protecting the integrity of the table. The generation should be turned off for legacy bootloaders which cannot recognize the MD5 checksum in the partition table.

Compiler options

Contains:

- CONFIG_COMPILER_OPTIMIZATION_ASSERTION_LEVEL
- CONFIG_COMPILER_FLOAT_LIB_FROM
- CONFIG_COMPILER_RT_LIB
- CONFIG_COMPILER_CXX_GLIBCXX_CONSTEXPR
- CONFIG_COMPILER_DISABLE_DEFAULT_ERRORS
- CONFIG_COMPILER_NO_MERGE_CONSTANTS
- CONFIG_COMPILER_OPTIMIZATION_CHECKS_SILENT
- CONFIG_COMPILER_DISABLE_GCC12_WARNINGS
- CONFIG_COMPILER_DISABLE_GCC13_WARNINGS
- CONFIG_COMPILER_DISABLE_GCC14_WARNINGS
- CONFIG_COMPILER_DISABLE_GCC15_WARNINGS
- CONFIG_COMPILER_DUMP_RTL_FILES
- CONFIG_COMPILER_WARN_WRITE_STRINGS
- CONFIG_COMPILER_CXX_EXCEPTIONS
- CONFIG_COMPILER_CXX_RTTI
- CONFIG_COMPILER_STATIC_ANALYZER
- CONFIG_COMPILER_ENABLE_RISCV_ZCMP
- CONFIG_COMPILER_ASSERT_NDEBUG_EVALUATE
- CONFIG_COMPILER_OPTIMIZATION
- CONFIG_COMPILER_ORPHAN_SECTIONS
- CONFIG_COMPILER_HIDE_PATHS_MACROS
- CONFIG_COMPILER_STACK_CHECK_MODE

CONFIG_COMPILER_OPTIMIZATION

Optimization Level

Found in: Compiler options

This option sets compiler optimization level (gcc -O argument) for the app.

- The "Debug" setting will add the -Og flag to CFLAGS.
- The "Size" setting will add the -Os flag to CFLAGS (-Oz with Clang).
- The "Performance" setting will add the -O2 flag to CFLAGS.
- The "None" setting will add the -O0 flag to CFLAGS.

The "Size" setting cause the compiled code to be smaller and faster, but may lead to difficulties of correlating code addresses to source file lines when debugging.

The "Performance" setting causes the compiled code to be larger and faster, but will be easier to correlated code addresses to source file lines.

"None" with -O0 produces compiled code without optimization.

Note that custom optimization levels may be unsupported.

Compiler optimization for the IDF bootloader is set separately, see the BOOTLOADER_COMPILER_OPTIMIZATION setting.

Available options:

- Debug (-Og) (CONFIG_COMPILER_OPTIMIZATION_DEBUG)
- Optimize for size (-Os with GCC, -Oz with Clang) (CONFIG_COMPILER_OPTIMIZATION_SIZE)
- Optimize for performance (-O2) (CONFIG_COMPILER_OPTIMIZATION_PERF)
- Debug without optimization (-O0) (CONFIG_COMPILER_OPTIMIZATION_NONE)

CONFIG_COMPILER_ENABLE_RISCV_ZCMP

Enable RISCV ZCMP extension

Found in: Compiler options

Enable the RISC-V ZCMP (Compressed Macro) extension to reduce binary size by optimizing function prologue and epilogue sequences.

Note: Due to a hardware issue on some ESP32 chips (e.g., ESP32C5, ESP32C61, ESP32H4), executing "cm.push" may re-enable interrupts even when global interrupts are disabled (mstatus.mie = 0). This can cause unexpected interrupts during CPU retention or within critical sections.

Workarounds are implemented in the IDF codebase. However, if user code directly disables interrupts, additional actions may be required. Refer to code examples under the SOC_CPU_ZCMP_WORKAROUND macro, or disable the ZCMP extension for source files that contain functions which may execute while mstatus.mie = 0.

Default value:

- No (disabled) if SOC_CPU_ZCMP_WORKAROUND

CONFIG_COMPILER_OPTIMIZATION_ASSERTION_LEVEL

Assertion level

Found in: Compiler options

Assertions can be:

- Enabled. Failure will print verbose assertion details. This is the default.
- Set to "silent" to save code size (failed assertions will abort() but user needs to use the aborting address to find the line number with the failed assertion.)
- Disabled entirely (not recommended for most configurations.) -DNDEBUG is added to CPPFLAGS in this case.

Available options:

- Enabled (CONFIG_COMPILER_OPTIMIZATION_ASSERTIONS_ENABLE)

  Enable assertions. Assertion content and line number will be printed on failure.

- Silent (saves code size)
  (CONFIG_COMPILER_OPTIMIZATION_ASSERTIONS_SILENT)

  Enable silent assertions. Failed assertions will abort(), user needs to use the aborting address to find the line number with the failed assertion.

- Disabled (sets -DNDEBUG)
  (CONFIG_COMPILER_OPTIMIZATION_ASSERTIONS_DISABLE)

  If assertions are disabled, -DNDEBUG is added to CPPFLAGS.

CONFIG_COMPILER_ASSERT_NDEBUG_EVALUATE

Enable the evaluation of the expression inside assert(X) when NDEBUG is set

Found in: Compiler options

When NDEBUG is set, assert(X) will not cause code to trigger an assertion. With this option set, assert(X) will still evaluate the expression X, though the result will never cause an assertion. This means that if X is a function then the function will be called.

This is not according to the standard, which states that the assert(X) should be replaced with ((void)0) if NDEBUG is defined.

Default value:

- No (disabled)

CONFIG_COMPILER_FLOAT_LIB_FROM

Compiler float lib source

Found in: Compiler options

In the soft-fp part of libgcc, riscv version is written in C, and handles all edge cases in IEEE754, which makes it larger and performance is slow.

RVfplib is an optimized RISC-V library for FP arithmetic on 32-bit integer processors, for single and double-precision FP. RVfplib is "fast", but it has a few exceptions from IEEE 754 compliance.

Available options:

- libgcc (CONFIG_COMPILER_FLOAT_LIB_FROM_GCCLIB)
- librvfp (CONFIG_COMPILER_FLOAT_LIB_FROM_RVFPLIB)

CONFIG_COMPILER_OPTIMIZATION_CHECKS_SILENT

Disable messages in ESP_RETURN_ON_* and ESP_EXIT_ON_* macros

Found in: Compiler options

If enabled, the error messages will be discarded in following check macros: - ESP_RETURN_ON_ERROR - ESP_EXIT_ON_ERROR - ESP_RETURN_ON_FALSE - ESP_EXIT_ON_FALSE

Default value:

- No (disabled)

CONFIG_COMPILER_HIDE_PATHS_MACROS

Replace ESP-IDF and project paths in binaries

Found in: Compiler options

When expanding the __FILE__ and __BASE_FILE__ macros, replace paths inside ESP-IDF with paths relative to the placeholder string "IDF", and convert paths inside the project directory to relative paths.

This allows building the project with assertions or other code that embeds file paths, without the binary containing the exact path to the IDF or project directories.

This option passes -fmacro-prefix-map options to the GCC command line. To replace additional paths in your binaries, modify the project CMakeLists.txt file to pass custom -fmacro-prefix-map or -ffile-prefix-map arguments.

Default value:

- Yes (enabled)

CONFIG_COMPILER_CXX_EXCEPTIONS

Enable C++ exceptions

Found in: Compiler options

Enabling this option compiles all IDF C++ files with exception support enabled.

Disabling this option disables C++ exception support in all compiled files, and any libstdc++ code which throws an exception will abort instead.

Enabling this option currently adds an additional ~500 bytes of heap overhead when an exception is thrown in user code for the first time.

Default value:

- No (disabled)

Contains:

- CONFIG_COMPILER_CXX_EXCEPTIONS_EMG_POOL_SIZE

CONFIG_COMPILER_CXX_EXCEPTIONS_EMG_POOL_SIZE

Emergency Pool Size

Found in: Compiler options > CONFIG_COMPILER_CXX_EXCEPTIONS

Size (in bytes) of the emergency memory pool for C++ exceptions. This pool will be used to allocate memory for thrown exceptions when there is not enough memory on the heap.

Default value:

- 0 if CONFIG_COMPILER_CXX_EXCEPTIONS

CONFIG_COMPILER_CXX_RTTI

Enable C++ run-time type info (RTTI)

Found in: Compiler options

Enabling this option compiles all C++ files with RTTI support enabled. This increases binary size (typically by tens of kB) but allows using dynamic_cast conversion and typeid operator.

Default value:

- No (disabled)

CONFIG_COMPILER_STACK_CHECK_MODE

Stack smashing protection mode

Found in: Compiler options

Stack smashing protection mode. Emit extra code to check for buffer overflows, such as stack smashing attacks. This is done by adding a guard variable to functions with vulnerable objects. The guards are initialized when a function is entered and then checked when the function exits. If a guard check fails, program is halted. Protection has the following modes:

- In NORMAL mode (GCC flag: -fstack-protector) only functions that call alloca, and functions with buffers larger than 8 bytes are protected.
- STRONG mode (GCC flag: -fstack-protector-strong) is like NORMAL, but includes additional functions to be protected -- those that have local array definitions, or have references to local frame addresses.
- In OVERALL mode (GCC flag: -fstack-protector-all) all functions are protected.

Modes have the following impact on code performance and coverage:

- performance: NORMAL > STRONG > OVERALL
- coverage: NORMAL < STRONG < OVERALL

The performance impact includes increasing the amount of stack memory required for each task.

Available options:

- None (CONFIG_COMPILER_STACK_CHECK_MODE_NONE)
- Normal (CONFIG_COMPILER_STACK_CHECK_MODE_NORM)
- Strong (CONFIG_COMPILER_STACK_CHECK_MODE_STRONG)
- Overall (CONFIG_COMPILER_STACK_CHECK_MODE_ALL)

CONFIG_COMPILER_NO_MERGE_CONSTANTS

Disable merging const sections

Found in: Compiler options

Disable merging identical constants (string/floating-point) across compilation units. This helps in better size analysis of the application binary as the rodata section distribution is more uniform across libraries. On downside, it may increase the binary size and hence should be used during development phase only.

CONFIG_COMPILER_WARN_WRITE_STRINGS

Enable -Wwrite-strings warning flag

Found in: Compiler options

Adds -Wwrite-strings flag for the C/C++ compilers.

For C, this gives string constants the type `const char[]` so that copying the address of one into a non-const `char \*` pointer produces a warning. This warning helps to find at compile time code that tries to write into a string constant.

For C++, this warns about the deprecated conversion from string literals to `char \*`.

Default value:

- No (disabled)

CONFIG_COMPILER_DISABLE_DEFAULT_ERRORS

Disable errors for default warnings

Found in: Compiler options

Enable this option if you do not want default warnings to be considered as errors, especially when updating IDF.

This is a temporary flag that could help to allow upgrade while having some time to address the warnings raised by those default warnings. Alternatives are: 1) fix code (preferred), 2) remove specific warnings, 3) do not consider specific warnings as error.

Default value:

- No (disabled)

CONFIG_COMPILER_DISABLE_GCC12_WARNINGS

Disable new warnings introduced in GCC 12

Found in: Compiler options

Enable this option if use GCC 12 or newer, and want to disable warnings which don't appear with GCC 11.

Default value:

- No (disabled)

CONFIG_COMPILER_DISABLE_GCC13_WARNINGS

Disable new warnings introduced in GCC 13

Found in: Compiler options

Enable this option if use GCC 13 or newer, and want to disable warnings which don't appear with GCC 12.

Default value:

- No (disabled)

CONFIG_COMPILER_DISABLE_GCC14_WARNINGS

Disable new warnings introduced in GCC 14

Found in: Compiler options

Enable this option if use GCC 14 or newer, and want to disable warnings which don't appear with GCC 13.

Default value:

- No (disabled)

CONFIG_COMPILER_DISABLE_GCC15_WARNINGS

Disable new warnings introduced in GCC 15

Found in: Compiler options

Enable this option if use GCC 15 or newer, and want to disable warnings which don't appear with GCC 14.

Default value:

- No (disabled)

CONFIG_COMPILER_DUMP_RTL_FILES

Dump RTL files during compilation

Found in: Compiler options

If enabled, RTL files will be produced during compilation. These files can be used by other tools, for example to calculate call graphs.

CONFIG_COMPILER_RT_LIB

Compiler runtime library

Found in: Compiler options

Select runtime library to be used by compiler. - GCC toolchain supports libgcc only. - Clang allows to choose between libgcc or libclang_rt. - For host builds ("linux" target), uses the default library.

Available options:

- libgcc (CONFIG_COMPILER_RT_LIB_GCCLIB)
- libclang_rt (CONFIG_COMPILER_RT_LIB_CLANGRT)
- Host (CONFIG_COMPILER_RT_LIB_HOST)

CONFIG_COMPILER_ORPHAN_SECTIONS

Orphan sections handling

Found in: Compiler options

If the linker finds orphan sections, it attempts to place orphan sections after sections of the same attribute such as code vs data, loadable vs non-loadable, etc. That means that orphan sections could placed between sections defined in IDF linker scripts. This could lead to corruption of the binary image. Configure the linker action here.

Available options:

- Fail if orphan sections found (CONFIG_COMPILER_ORPHAN_SECTIONS_ERROR)

  Fails the link step with an error if orphan sections are detected.

- Place with warning (CONFIG_COMPILER_ORPHAN_SECTIONS_WARNING)

  Places orphan sections with a warning message.

- Place silently (CONFIG_COMPILER_ORPHAN_SECTIONS_PLACE)

  Places orphan sections without a warning/error message.

CONFIG_COMPILER_STATIC_ANALYZER

Enable compiler static analyzer

Found in: Compiler options

Enable compiler static analyzer. This may produce false-positive results and increases compile time.

CONFIG_COMPILER_CXX_GLIBCXX_CONSTEXPR

Define _GLIBCXX_CONSTEXPR

Found in: Compiler options

Modify libstdc++ _GLIBCXX20_CONSTEXPR and _GLIBCXX23_CONSTEXPR definitions to provide size optimizations. The total size optimization depends on the application's structure. There is no robust way to determine which option would be better in a particular case. Please try all available options to find the best size optimization.

Available options:

- No change (CONFIG_COMPILER_CXX_GLIBCXX_CONSTEXPR_NO_CHANGE)

  Use default _GLIBCXX20_CONSTEXPR and _GLIBCXX23_CONSTEXPR defined in libstdc++

- _GLIBCXX2X_CONSTEXPR=__attribute__((cold)) constexpr (CONFIG_COMPILER_CXX_GLIBCXX_CONSTEXPR_COLD_CONSTEXPR)

  Define _GLIBCXX20_CONSTEXPR=__attribute__((cold)) constexpr Define _GLIBCXX23_CONSTEXPR=__attribute__((cold)) constexpr

- _GLIBCXX2X_CONSTEXPR=__attribute__((cold)) (CONFIG_COMPILER_CXX_GLIBCXX_CONSTEXPR_COLD)

  Define _GLIBCXX20_CONSTEXPR=__attribute__((cold)). Define _GLIBCXX23_CONSTEXPR=__attribute__((cold)).

Component config

Contains:

- 2D-DMA Configurations
- ADC and ADC Calibration
- Bluetooth
- Common ESP-related
- Console Library
- Core dump
- DW_GDMA Configurations
- eFuse Bit Manager
- CONFIG_BLE_MESH
- ESP HID
- ESP HTTP client
- ESP HTTPS OTA
- ESP HTTPS server

- ESP NETIF Adapter
- ESP PSRAM
- ESP Ringbuf
- ESP Security Specific
- ESP System Settings
- ESP Timer (High Resolution Timer)
- ESP Trace Configuration
- ESP-Driver:Analog Comparator Configurations
- ESP-Driver:BitScrambler Configurations
- ESP-Driver:Camera Controller Configurations
- ESP-Driver:DAC Configurations
- ESP-Driver:GPIO Configurations
- ESP-Driver:GPTimer Configurations
- ESP-Driver:I2C Configurations
- ESP-Driver:I2S Configurations
- ESP-Driver:I3C Master Configurations
- ESP-Driver:ISP Configurations
- ESP-Driver:JPEG-Codec Configurations
- ESP-Driver:LCD Controller Configurations
- ESP-Driver:LEDC Configurations
- ESP-Driver:MCPWM Configurations
- ESP-Driver:Parallel IO Configurations
- ESP-Driver:PCNT Configurations
- ESP-Driver:RMT Configurations
- ESP-Driver:SD Host SDMMC Controller Configurations
- ESP-Driver:Sigma Delta Modulator Configurations
- ESP-Driver:SPI Configurations
- ESP-Driver:Temperature Sensor Configurations
- ESP-Driver:Touch Sensor Configurations
- ESP-Driver:TWAI Configurations
- ESP-Driver:UART Configurations
- ESP-Driver:UHCI Configurations
- ESP-Driver:USB Serial/JTAG Configuration
- ESP-MM: Memory Management Configurations
- ESP-ROM
- ESP-STDIO
- ESP-TLS
- Ethernet
- Event Loop Library
- FAT Filesystem support
- FreeRTOS
- GDB Stub
- GDMA Configurations
- Hardware Abstraction Layer (HAL) and Low Level (LL)
- Hardware Settings

- Heap memory debugging
- HTTP Server
- IEEE 802.15.4
- IPC (Inter-Processor Call)
- Legacy Driver Configurations
- LibC
- Log
- LWIP
- Main Flash configuration
- mbedTLS
- NVS
- NVS Security Provider
- OpenThread
- Partition API Configuration
- PHY
- Power Management
- Protocomm
- PThreads
- SD Protocol Layer Configuration
- SoC Settings
- SPI Flash driver
- SPIFFS Configuration
- TCP Transport
- Ultra Low Power (ULP) Co-processor
- Unity unit testing library
- Virtual file system
- Wear Levelling
- Wi-Fi
- Wireless Coexistence

Bluetooth

Contains:

- Bluedroid Options
- CONFIG_BT_ENABLED
- Common Options
- Controller Options
- CONFIG_BT_HCI_LOG_DEBUG_EN
- NimBLE Options
- CONFIG_BT_RELEASE_IRAM

CONFIG_BT_ENABLED

Bluetooth

Found in: Component config > Bluetooth

Select this option to enable Bluetooth and show the submenu with Bluetooth configuration choices.

CONFIG_BT_HOST

Host

Found in: Component config > Bluetooth > CONFIG_BT_ENABLED

This helps to choose Bluetooth host stack

Available options:

- Bluedroid - Dual-mode (CONFIG_BT_BLUEDROID_ENABLED)

  This option is recommended for classic Bluetooth or for dual-mode usecases

- NimBLE - BLE only (CONFIG_BT_NIMBLE_ENABLED)

  This option is recommended for BLE only usecases to save on memory

- Disabled (CONFIG_BT_CONTROLLER_ONLY)

  This option is recommended when you want to communicate directly with the controller (without any host) or when you are using any other host stack not supported by Espressif (not mentioned here).

CONFIG_BT_CONTROLLER

Controller

Found in: Component config > Bluetooth > CONFIG_BT_ENABLED

This helps to choose Bluetooth controller stack

Available options:

- Enabled (CONFIG_BT_CONTROLLER_ENABLED)

  This option is recommended for Bluetooth controller usecases

- Disabled (CONFIG_BT_CONTROLLER_DISABLED)

  This option is recommended for Bluetooth Host only usecases

Common Options

Contains:

- BT Logs
- CONFIG_BT_LE_USED_MEM_STATISTICS_ENABLED
- CONFIG_BT_ALARM_MAX_NUM
- CONFIG_BT_SMP_CRYPTO_STACK

CONFIG_BT_ALARM_MAX_NUM

Maximum number of Bluetooth alarms

Found in: Component config > Bluetooth > Common Options

This option decides the maximum number of alarms which could be used by Bluetooth host.

Default value:

- 50 if CONFIG_BT_BLUEDROID_ENABLED || CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_SMP_CRYPTO_STACK

SMP cryptographic stack

Found in: Component config > Bluetooth > Common Options

Select the cryptographic library to use for SMP operations (AES, AES-CMAC, ECDH P-256).

Available options:

- TinyCrypt (CONFIG_BT_SMP_CRYPTO_STACK_TINYCRYPT)

  Use TinyCrypt library for cryptographic operations. TinyCrypt is a lightweight cryptographic library designed for constrained devices. This can reduce code size compared to the native implementation. This is the default option.

- mbedTLS (CONFIG_BT_SMP_CRYPTO_STACK_MBEDTLS)

  Use mbedTLS library for cryptographic operations. This can provide hardware acceleration on supported platforms and reduce code size by sharing crypto implementations with other components.

BT Logs

Contains:

- CONFIG_BT_LOG_CRITICAL_ONLY
- CONFIG_BLE_LOG_ENABLED
- Legacy SPI Log Output (Deprecated - use BT Log Async Output instead)
- Log Sources

Log Sources

Contains:

- CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Enable Controller logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources

Default value:

- No (disabled) if SOC_ESP_NIMBLE_CONTROLLER

Contains:

- CONFIG_BT_LE_CONTROLLER_LOG_DUMP_ONLY
- CONFIG_BT_LE_CONTROLLER_LOG_MOD_OUTPUT_SWITCH
- CONFIG_BT_LE_CONTROLLER_LOG_OUTPUT_LEVEL
- CONFIG_BT_LE_CONTROLLER_LOG_OUTPUT_MODE
- CONFIG_BT_LE_CONTROLLER_LOG_HCI_ENABLED
- CONFIG_BT_LE_CONTROLLER_LOG_CTRL_ENABLED
- CONFIG_BT_LE_CONTROLLER_LOG_TASK_WDT_USER_HANDLER_ENABLE
- CONFIG_BT_LE_CONTROLLER_LOG_WRAP_PANIC_HANDLER_ENABLE
- CONFIG_BT_LE_LOG_CTRL_BUF1_SIZE
- CONFIG_BT_LE_LOG_HCI_BUF_SIZE
- CONFIG_BT_LE_CONTROLLER_LOG_SPI_OUT_ENABLED
- CONFIG_BT_LE_LOG_CTRL_BUF2_SIZE
- CONFIG_BT_LE_CONTROLLER_LOG_STORAGE_ENABLE

CONFIG_BT_LE_CONTROLLER_LOG_OUTPUT_MODE

Controller log output mode

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Available options:

- BLE Log v2 mode (CONFIG_BT_LE_CONTROLLER_LOG_MODE_BLE_LOG_V2)

  Utilize BLE Log v2 for controller log

- Legacy log mode (Deprecated) (CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED)

CONFIG_BT_LE_CONTROLLER_LOG_CTRL_ENABLED

Enable controller link layer logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Default value:

- Yes (enabled) if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_HCI_ENABLED

Enable controller HCI logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Default value:

- Yes (enabled) if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_DUMP_ONLY

Controller log dump mode only

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Only operate in dump mode. Logs are cached internally only, not output asynchronously.

Default value:

- Yes (enabled) if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_SPI_OUT_ENABLED

Output controller logs to SPI bus (Experimental)

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Output ble controller logs to SPI bus

CONFIG_BT_LE_CONTROLLER_LOG_STORAGE_ENABLE

Store controller logs to flash (Experimental)

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Store ble controller logs to flash memory.

CONFIG_BT_LE_CONTROLLER_LOG_PARTITION_SIZE

Controller log partition size (multiples of 4K)

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED > CONFIG_BT_LE_CONTROLLER_LOG_STORAGE_ENABLE

The size of ble controller log partition shall be a multiples of 4K. The name of log partition shall be "bt_ctrl_log". The partition type shall be ESP_PARTITION_TYPE_DATA. The partition sub_type shall be ESP_PARTITION_SUBTYPE_ANY.

Default value:

- 65536 if CONFIG_BT_LE_CONTROLLER_LOG_STORAGE_ENABLE && CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_LOG_CTRL_BUF1_SIZE

First controller log buffer size

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Configure the size of the first BLE controller LOG buffer.

Default value:

- 4096 if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_LOG_CTRL_BUF2_SIZE

Second controller log buffer size

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Configure the size of the second BLE controller LOG buffer.

Default value:

- 1024 if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_LOG_HCI_BUF_SIZE

HCI log buffer size

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Configure the size of the BLE HCI LOG buffer.

Default value:

- 4096 if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED && CONFIG_BT_LE_CTRL_LEGACY_LOG_MODE_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_OUTPUT_LEVEL

Controller log output level

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

The output level of controller log.

Range:

- from 0 to 5 if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Default value:

- 2 if CONFIG_BT_LOG_CRITICAL_ONLY_CTRL && CONFIG_BT_LE_CONTROLLER_LOG_ENABLED
- 1 if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_MOD_OUTPUT_SWITCH

Controller log module output switch

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Bitmask to enable/disable logging for individual controller modules. 0xFFFFFFFF enables all modules.

Range:

- from 0 to 0xFFFFFFFF if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Default value:

- "0xFFFFFFFF" if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_WRAP_PANIC_HANDLER_ENABLE

Enable wrap panic handler

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Wrap esp_panic_handler to get controller logs when PC pointer exception crashes.

Default value:

- No (disabled) if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

CONFIG_BT_LE_CONTROLLER_LOG_TASK_WDT_USER_HANDLER_ENABLE

Enable esp_task_wdt_isr_user_handler implementation

Found in: Component config > Bluetooth > Common Options > BT Logs > Log Sources > CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

Implement esp_task_wdt_isr_user_handler to get controller logs when task wdt issue is triggered.

Default value:

- No (disabled) if CONFIG_BT_LE_CONTROLLER_LOG_ENABLED

CONFIG_BT_LOG_CRITICAL_ONLY

Enable bandwidth-optimized log mode (critical logs only)

Found in: Component config > Bluetooth > Common Options > BT Logs

Enable bandwidth-optimized logging for the BLE Log Async Output system. When enabled, only high-severity logs are captured and log encoding is applied to reduce UART/SPI DMA bandwidth usage.

Each stack component below can be independently enabled. Requires a DMA-backed output peripheral — configure in: BLE Log Module → BLE Log peripheral choice.

Contains:

- CONFIG_BT_LOG_CRITICAL_ONLY_CTRL
- CONFIG_BT_LOG_CRITICAL_ONLY_HOST
- CONFIG_BT_LOG_CRITICAL_ONLY_MESH

CONFIG_BT_LOG_CRITICAL_ONLY_CTRL

Controller: bandwidth-optimized logging

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BT_LOG_CRITICAL_ONLY

Enable controller log output via the async transport with a reduced output level for bandwidth optimization. The controller log level defaults to 2 when active.

Default value:

- Yes (enabled) if (SOC_ESP_NIMBLE_CONTROLLER || CONFIG_BLE_LOG_IS_ESP_LEGACY_CONTROLLER) && CONFIG_BT_LOG_CRITICAL_ONLY

CONFIG_BT_LOG_CRITICAL_ONLY_HOST

Host: bandwidth-optimized logging

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BT_LOG_CRITICAL_ONLY

Enable host stack log encoding via the async transport. For Bluedroid, the per-level compression options control which severity levels are encoded — configure in: Settings of BLE Log Compression → BLE Host log compression.

Default value:

- Yes (enabled) if CONFIG_BT_LOG_CRITICAL_ONLY

CONFIG_BT_LOG_CRITICAL_ONLY_MESH

Mesh: bandwidth-optimized logging

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BT_LOG_CRITICAL_ONLY

Enable mesh log encoding via the async transport. The per-level compression options are configured in: Settings of BLE Log Compression → BLE Mesh log compression.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH && CONFIG_BT_LOG_CRITICAL_ONLY

CONFIG_BLE_LOG_ENABLED

Enable BT Log Async Output (Dev Only)

Found in: Component config > Bluetooth > Common Options > BT Logs

Enable BT Log Async Output

Default value:

- No (disabled)

CONFIG_BLE_LOG_TASK_STACK_SIZE

Stack size for BLE Log Task

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Stack size for BLE Log Task

Default value:

- 2048 if CONFIG_BLE_LOG_ENABLED
- 1024 if CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LBM_TRANS_BUF_SIZE

Total buffer memory per common LBM (bytes)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Total buffer memory allocated for each common pool log buffer manager (LBM). This memory is divided equally among internal transport buffers. Must be a multiple of BLE_LOG_TRANS_BUF_CNT (currently 4).

The common pool contains: - BLE_LOG_LBM_ATOMIC_LOCK_TASK_CNT atomic LBMs (task context) - BLE_LOG_LBM_ATOMIC_LOCK_ISR_CNT atomic LBMs (ISR context) - 2 spinlock-protected LBMs (one for task, one for ISR fallback)

Total common pool memory: (ATOMIC_TASK_CNT + ATOMIC_ISR_CNT + 2) * BLE_LOG_LBM_TRANS_BUF_SIZE

Default value:

- 512 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_LOG_ENABLED
- 1024 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LBM_ATOMIC_LOCK_TASK_CNT

Count of log buffer managers with atomic lock protection for task context

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

BLE Log module will search for an LBM with atomic lock protection first; if all LBMs with atomic lock protection are unavailable, BLE Log module will try to use the LBM with spin lock protection. So the more LBMs with atomic lock protection are created, the better the logging performance will be.

Default value:

- 2 if CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LBM_ATOMIC_LOCK_ISR_CNT

Count of log buffer managers with atomic lock protection for ISR context

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

BLE Log module will search for an LBM with atomic lock protection first; if all LBMs with atomic lock protection are unavailable, BLE Log module will try to use the LBM with spin lock protection. So the more LBMs with atomic lock protection are created, the more ISRs can nest.

Default value:

- 1 if CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_IS_ESP_CONTROLLER

Current BLE Controller is ESP BLE Controller

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Current BLE Controller is ESP BLE Controller

Default value:

- Yes (enabled) if CONFIG_BT_CONTROLLER_ENABLED && SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_IS_ESP_LEGACY_CONTROLLER

Current BLE Controller is ESP BLE Legacy Controller

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Current BLE Controller is ESP BLE Legacy Controller

CONFIG_BLE_LOG_LL_ENABLED

Enable BLE Log for Link Layer

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Enable BLE Log for Link Layer

Default value:

- Yes (enabled) if CONFIG_BLE_LOG_IS_ESP_LEGACY_CONTROLLER && CONFIG_BT_CONTROLLER_ENABLED && CONFIG_BLE_LOG_ENABLED
- No (disabled) if CONFIG_BT_CONTROLLER_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LBM_LL_TRANS_BUF_SIZE

Total buffer memory per Link Layer LBM (bytes)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_LL_ENABLED

Total buffer memory allocated for each Link Layer dedicated log buffer manager (LBM). This memory is divided equally among internal transport buffers. Must be a multiple of BLE_LOG_TRANS_BUF_CNT (currently 4).

There are 2 Link Layer LBMs without lock protection (each is accessed from a single context only): - LL task LBM (Link Layer task context logs) - LL HCI LBM (Link Layer HCI context logs)

Total LL pool memory: 2 * BLE_LOG_LBM_LL_TRANS_BUF_SIZE

Default value:

- 2048 if CONFIG_BLE_LOG_LL_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LL_HCI_LOG_PAYLOAD_LEN_LIMIT_ENABLED

Enable LL HCI Log Payload Length Limit

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_LL_ENABLED

Enable length limit for LL HCI Log payload (addr_append). When enabled, if len_append exceeds the configured limit, it will be truncated to the maximum length.

Default value:

- No (disabled) if CONFIG_BLE_LOG_LL_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_LL_HCI_LOG_PAYLOAD_LEN_LIMIT

LL HCI Log Payload Length Limit (bytes)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_LL_ENABLED > CONFIG_BLE_LOG_LL_HCI_LOG_PAYLOAD_LEN_LIMIT_ENABLED

Maximum length for LL HCI Log payload (len_append). When the feature is enabled and len_append exceeds this value, it will be truncated.

Default value:

- 32 if CONFIG_BLE_LOG_LL_HCI_LOG_PAYLOAD_LEN_LIMIT_ENABLED && CONFIG_BLE_LOG_LL_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_HOST_LOG

Enable BLE Log for Host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Enable BLE Log for the Host stack (Bluedroid or NimBLE).

When enabled, host-side log statements are routed through the BLE Log async output transport instead of (or in addition to) the standard UART logging path. Disable this option to keep BLE Log enabled for the controller/link layer only.

Default value:

- Yes (enabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_HOST_SIDE_HCI_LOG_ENABLED

Enable BLE Host side HCI Logging

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Enable HCI packet logging captured on the Host side (Bluedroid / NimBLE HCI HAL). Useful for correlating host events with controller-side HCI traces.

Default value:

- Yes (enabled) if CONFIG_BLE_LOG_IS_ESP_LEGACY_CONTROLLER && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_TS_ENABLED

Enable BLE Log Timestamp Synchronization (TS)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Enable BLE Log TS with external logging module

Default value:

- No (disabled) if CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_SYNC_IO_NUM

GPIO number for Timestamp Synchronization (TS) toggle output

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_TS_ENABLED

GPIO number for TS toggle output

Default value:

- 0 if CONFIG_BLE_LOG_TS_ENABLED && CONFIG_BLE_LOG_TS_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_TS_TRIGGER_TIMEOUT_MS

Timeout (ms) for Timestamp Synchronization toggle

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_TS_ENABLED

Timeout (ms) for Timestamp Synchronization toggle

Default value:

- 1000 if CONFIG_BLE_LOG_TS_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_TS_TRIGGER_CHOICE

BLE Log Timestamp Synchronization trigger choice

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_TS_ENABLED

Choose BLE Log Timestamp Synchronization trigger

Available options:

- BLE Log Timestamp Synchronization trigger - ESP Timer (CONFIG_BLE_LOG_TS_TRIGGER_ESP_TIMER)

  ESP Timer based periodic TS trigger

- BLE Log Timestamp Synchronization trigger - Task Event (CONFIG_BLE_LOG_TS_TRIGGER_TASK_EVENT)

  Task Event based TS trigger (Light Sleep Test Compatibility)

CONFIG_BLE_LOG_TS_TRIGGER_ESP_TIMER_ISR_DISPATCH_METHOD

Utilize ISR dispatch method for ESP Timer as Timestamp Synchronization trigger

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > CONFIG_BLE_LOG_TS_ENABLED

Utilize ISR dispatch method for ESP Timer as Timestamp Synchronization trigger

Default value:

- No (disabled) if CONFIG_BLE_LOG_TS_TRIGGER_ESP_TIMER && CONFIG_BLE_LOG_TS_ENABLED && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_CHOICE

BLE Log peripheral choice

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Choose BLE Log peripheral

Available options:

- Dummy transport (CONFIG_BLE_LOG_PRPH_DUMMY)

  Dummy transport (dump only)

- Utilize SPI master DMA driver as transport (CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA)

  Utilize SPI master DMA driver as transport

- Utilize UART DMA driver as transport (CONFIG_BLE_LOG_PRPH_UART_DMA)

  Utilize UART DMA driver as transport

CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_MOSI_IO_NUM

GPIO number of MOSI port for SPI master DMA transport

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

GPIO number of MOSI port for SPI master DMA transport

Default value:

- 0 if CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_SCLK_IO_NUM

GPIO number of SCLK port for SPI master DMA transport

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

GPIO number of SCLK port for SPI master DMA transport

Default value:

- 0 if CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_CS_IO_NUM

GPIO number of CS port for SPI master DMA transport

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

GPIO number of CS port for SPI master DMA transport

Default value:

- 0 if CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_UART_DMA_PORT

UART port number for UART DMA transport

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

UART port number for UART DMA

Default value:

- 0 if CONFIG_BLE_LOG_PRPH_UART_DMA && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_UART_DMA_BAUD_RATE

Baud rate of UART port for UART DMA transport

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

Determine the baud rate of UART port

Default value:

- 3000000 if CONFIG_BLE_LOG_PRPH_UART_DMA && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_LOG_PRPH_UART_DMA_TX_IO_NUM

GPIO number for UART TX

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED

GPIO number for UART TX

Default value:

- 0 if CONFIG_BLE_LOG_PRPH_UART_DMA && CONFIG_BLE_LOG_ENABLED

Settings of BLE Log Compression

Contains:

- CONFIG_BLE_COMPRESSED_LOG_ENABLE

CONFIG_BLE_COMPRESSED_LOG_ENABLE

Enable BLE log compression(Preview, Please read help information)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression

Compress BLE logs during application build to reduce flash usage and improve output speed. When enabled, log data from Bluetooth Low Energy components will be compressed before storage, optimizing both memory footprint and transmission efficiency.

Note: This library depends on additional Python packages. It will function correctly only after these dependencies are installed; refer to: "components/bt/common/ble_log/log_compression/README.en.md" for installation instructions.

Default value:

- No (disabled) if CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE

Enable BLE Mesh log compression(Preview)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE

Apply compression to ESP-BLE-MESH protocol stack logs. Requires base BLE compression to be enabled. Specifically optimizes log storage and transmission for ble mesh.

Note: This library depends on additional Python packages. It will function correctly only after these dependencies are installed; refer to: "components/bt/common/ble_log/log_compression/README.en.md" for installation instructions.

If the required packages are not installed, the log-compression mechanism will remain disabled even when this Config is enabled.

Default value:

- No (disabled) if CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_MESH && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Contains:

- CONFIG_BLE_MESH_COMPRESSED_LOG_BUFFER_LEN
- Select the net buf log tag to be compressed
- Select the stack log tag to be compressed

CONFIG_BLE_MESH_COMPRESSED_LOG_BUFFER_LEN

BLE Mesh log buffer length

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE

Maximum output length for a single log

Default value:

- 400 if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the stack log tag to be compressed

Contains:

- CONFIG_BLE_MESH_STACK_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_MESH_STACK_ERR_LOG_COMPRESSION
- CONFIG_BLE_MESH_STACK_INFO_LOG_COMPRESSION
- CONFIG_BLE_MESH_STACK_WARN_LOG_COMPRESSION

CONFIG_BLE_MESH_STACK_ERR_LOG_COMPRESSION

Compress ERROR log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed

The error log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_ERR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed > CONFIG_BLE_MESH_STACK_ERR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_STACK_ERR_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_WARN_LOG_COMPRESSION

Compress warn log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed

The warn log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_WARN_LOG_PRESERVE

Keep the original warn log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed > CONFIG_BLE_MESH_STACK_WARN_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_STACK_ERR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_STACK_WARN_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_INFO_LOG_COMPRESSION

Compress info log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed

The info log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_INFO_LOG_PRESERVE

Keep the original info log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed > CONFIG_BLE_MESH_STACK_INFO_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_STACK_ERR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_MESH_STACK_INFO_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_DEBUG_LOG_COMPRESSION

Compress debug log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed

The debug log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_STACK_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the stack log tag to be compressed > CONFIG_BLE_MESH_STACK_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_STACK_ERR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_MESH_STACK_DEBUG_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the net buf log tag to be compressed

Contains:

- CONFIG_BLE_MESH_NET_BUF_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_MESH_NET_BUF_ERR_LOG_COMPRESSION
- CONFIG_BLE_MESH_NET_BUF_INFO_LOG_COMPRESSION
- CONFIG_BLE_MESH_NET_BUF_WARN_LOG_COMPRESSION

CONFIG_BLE_MESH_NET_BUF_ERR_LOG_COMPRESSION

Compress ERROR log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed

The error log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_ERR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed > CONFIG_BLE_MESH_NET_BUF_ERR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_NET_BUF_ERR_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_WARN_LOG_COMPRESSION

Compress warn log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed

The warn log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_WARN_LOG_PRESERVE

Keep the original warn log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed > CONFIG_BLE_MESH_NET_BUF_WARN_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_NET_BUF_ERR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_NET_BUF_WARN_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_INFO_LOG_COMPRESSION

Compress info log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed

The info log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_INFO_LOG_PRESERVE

Keep the original info log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed > CONFIG_BLE_MESH_NET_BUF_INFO_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_NET_BUF_ERR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_MESH_NET_BUF_INFO_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_DEBUG_LOG_COMPRESSION

Compress debug log of ESP-BLE-MESH

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed

The debug log in the BLE-MESH component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_MESH_NET_BUF_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE > Select the net buf log tag to be compressed > CONFIG_BLE_MESH_NET_BUF_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_MESH_NET_BUF_ERR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_MESH_NET_BUF_DEBUG_LOG_COMPRESSION && CONFIG_BLE_MESH_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE

Enable BLE Host log compression(Preview)

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE

Apply compression to host logs. Requires base BLE compression to be enabled. Specifically optimizes log storage and transmission.

Note: This library depends on additional Python packages. It will function correctly only after these dependencies are installed; refer to: "components/bt/common/ble_log/log_compression/README.en.md" for installation instructions.

Default value:

- No (disabled) if CONFIG_BLE_COMPRESSED_LOG_ENABLE && (CONFIG_BT_BLUEDROID_ENABLED || CONFIG_BT_NIMBLE_ENABLED) && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Contains:

- CONFIG_BLE_HOST_COMPRESSED_LOG_BUFFER_LEN
- Select the APPL layer log tag to be compressed
- Select the BTM layer log tag to be compressed
- Select the GAP layer log tag to be compressed
- Select the GATT layer log tag to be compressed
- Select the LA2CAP layer log tag to be compressed

- Select the SMP layer log tag to be compressed

CONFIG_BLE_HOST_COMPRESSED_LOG_BUFFER_LEN

Host log buffer length

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE

Maximum output length for a single log

Default value:

- 300 if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the BTM layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_BTM_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_BTM_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_BTM_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_BTM_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_BTM_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_BTM_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_BTM_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_BTM_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_BTM_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_BTM_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_BTM_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_BTM_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_BTM_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the BTM layer log tag to be compressed > CONFIG_BLE_BLUEDROID_BTM_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_BTM_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_BTM_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the LA2CAP layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_L2CAP_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_L2CAP_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_L2CAP_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_L2CAP_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_L2CAP_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_L2CAP_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_L2CAP_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_L2CAP_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_L2CAP_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_L2CAP_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_L2CAP_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_L2CAP_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_L2CAP_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the LA2CAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_L2CAP_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_L2CAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_L2CAP_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the GAP layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_GAP_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GAP_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GAP_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GAP_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GAP_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GAP_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_GAP_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_GAP_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_GAP_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GAP_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GAP_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GAP_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GAP_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GAP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GAP_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GAP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GAP_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the GATT layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_GATT_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GATT_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GATT_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GATT_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GATT_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_GATT_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_GATT_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_GATT_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_GATT_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GATT_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GATT_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GATT_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_GATT_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the GATT layer log tag to be compressed > CONFIG_BLE_BLUEDROID_GATT_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_GATT_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_GATT_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the SMP layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_SMP_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_SMP_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_SMP_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_SMP_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_SMP_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_SMP_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_SMP_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_SMP_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_SMP_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_SMP_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_SMP_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_SMP_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_SMP_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the SMP layer log tag to be compressed > CONFIG_BLE_BLUEDROID_SMP_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_SMP_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_SMP_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Select the APPL layer log tag to be compressed

Contains:

- CONFIG_BLE_BLUEDROID_APPL_API_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_APPL_DEBUG_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_APPL_ERROR_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_APPL_EVENT_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_APPL_VERBOSE_LOG_COMPRESSION
- CONFIG_BLE_BLUEDROID_APPL_WARNING_LOG_COMPRESSION

CONFIG_BLE_BLUEDROID_APPL_ERROR_LOG_COMPRESSION

Compress error log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The error log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Keep the original error log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_ERROR_LOG_COMPRESSION

When this option is enabled, the log data will be output through both the compressed log interface and the original UART interface at the same time, meaning that the log statements will appear on both paths. However, please note that this dual-output approach introduces additional code and string constants, which will increase the size of the firmware binary file. When this option is disabled, the logs will no longer be printed through the original UART output path; instead, they will only be output through the compressed log interface. As the code and strings related to the original UART output are omitted, the size of the firmware binary file can be effectively reduced.

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_APPL_ERROR_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_WARNING_LOG_COMPRESSION

Compress warning log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The warning log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_WARNING_LOG_PRESERVE

Keep the original warning log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_WARNING_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Default value:

- Yes (enabled) if CONFIG_BLE_BLUEDROID_APPL_WARNING_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_API_LOG_COMPRESSION

Compress api log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The api log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_API_LOG_PRESERVE

Keep the original api log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_API_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_APPL_API_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_EVENT_LOG_COMPRESSION

Compress event log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The event log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_EVENT_LOG_PRESERVE

Keep the original event log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_EVENT_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_APPL_EVENT_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_DEBUG_LOG_COMPRESSION

Compress debug log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The debug log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_DEBUG_LOG_PRESERVE

Keep the original debug log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_DEBUG_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_APPL_DEBUG_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_VERBOSE_LOG_COMPRESSION

Compress verbose log of Bluedroid host

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed

The verbose log in the Bluedroid host component will be compressed

Default value:

- Yes (enabled) if CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

CONFIG_BLE_BLUEDROID_APPL_VERBOSE_LOG_PRESERVE

Keep the original verbose log statement

Found in: Component config > Bluetooth > Common Options > BT Logs > CONFIG_BLE_LOG_ENABLED > Settings of BLE Log Compression > CONFIG_BLE_COMPRESSED_LOG_ENABLE > CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE > Select the APPL layer log tag to be compressed > CONFIG_BLE_BLUEDROID_APPL_VERBOSE_LOG_COMPRESSION

Please refer to the help information in BLE_BLUEDROID_APPL_ERROR_LOG_PRESERVE

Default value:

- No (disabled) if CONFIG_BLE_BLUEDROID_APPL_VERBOSE_LOG_COMPRESSION && CONFIG_BLE_HOST_COMPRESSED_LOG_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BLE_COMPRESSED_LOG_ENABLE && CONFIG_BLE_LOG_ENABLED

Legacy SPI Log Output (Deprecated - use BT Log Async Output instead)

Contains:

- CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Output ble logs to SPI bus (Experimental)

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead)

Output ble logs to SPI bus

CONFIG_BT_BLE_LOG_SPI_OUT_UL_TASK_BUF_SIZE

SPI transaction buffer size for upper layer task logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

SPI transaction buffer size for upper layer task logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 512 if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HCI_ENABLED

Enable HCI log output to SPI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable logging of HCI packets to the SPI bus when BLE SPI log output is enabled.

Default value:

- No (disabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HCI_BUF_SIZE

SPI transaction buffer size for HCI logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_HCI_ENABLED

SPI transaction buffer size for HCI logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 1024 if CONFIG_BT_BLE_LOG_SPI_OUT_HCI_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HCI_TASK_CNT

HCI task count

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_HCI_ENABLED

HCI task count

Default value:

- 1 if CONFIG_BT_BLE_LOG_SPI_OUT_HCI_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HOST_ENABLED

Enable Host log output to SPI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

This configuration applies to the logs of both Bluedroid Host and NimBLE Host. When BLE SPI log output is enabled, this option allows host logs to be transmitted via SPI.

Default value:

- No (disabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HOST_BUF_SIZE

SPI transaction buffer size for host logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_HOST_ENABLED

SPI transaction buffer size for host logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 1024 if CONFIG_BT_BLE_LOG_SPI_OUT_HOST_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_HOST_TASK_CNT

Host task count

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_HOST_ENABLED

Host task count.

Default value:

- 2 if CONFIG_BT_BLE_LOG_SPI_OUT_HOST_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED

Enable Controller log output to SPI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable controller log output to SPI bus.

Default value:

- No (disabled) if CONFIG_BT_LE_CONTROLLER_LOG_SPI_OUT_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LL_TASK_BUF_SIZE

SPI transaction buffer size for lower layer task logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED

SPI transaction buffer size for lower layer task logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 1024 if CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LL_ISR_BUF_SIZE

SPI transaction buffer size for lower layer ISR logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED

SPI transaction buffer size for lower layer ISR logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 512 if CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LL_HCI_BUF_SIZE

SPI transaction buffer size for lower layer HCI logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED

SPI transaction buffer size for upper layer HCI logs. There will be 2 SPI DMA buffers with the same size

Default value:

- 512 if CONFIG_BT_BLE_LOG_SPI_OUT_LL_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_MOSI_IO_NUM

GPIO number of SPI MOSI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

GPIO number of SPI MOSI

Default value:

- 0 if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_SCLK_IO_NUM

GPIO number of SPI SCLK

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

GPIO number of SPI SCLK

Default value:

- 1 if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_CS_IO_NUM

GPIO number of SPI CS

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

GPIO number of SPI CS

Default value:

- 2 if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_TS_SYNC_ENABLED

Enable ble log & logic analyzer log time sync

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable ble log & logic analyzer log time sync

Default value:

- Yes (enabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_SYNC_IO_NUM

GPIO number of SYNC IO

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_TS_SYNC_ENABLED

GPIO number of SYNC IO

Default value:

- 3 if CONFIG_BT_BLE_LOG_SPI_OUT_TS_SYNC_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_FLUSH_TIMER_ENABLED

Enable periodic buffer flush out

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable periodic buffer flush out Not recommended when SPI receiver is unavailable

Default value:

- No (disabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_FLUSH_TIMEOUT

Buffer flush out period in unit of ms

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_FLUSH_TIMER_ENABLED

Buffer flush out period in unit of ms

Default value:

- 1000 if CONFIG_BT_BLE_LOG_SPI_OUT_FLUSH_TIMER_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_ENABLED

Enable LE Audio log output to SPI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable LE Audio log output to SPI

Default value:

- No (disabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_BUF_SIZE

SPI transaction buffer size for LE Audio logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_ENABLED

SPI transaction buffer size for LE Audio logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 1024 if CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_TASK_CNT

LE audio task count

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_ENABLED

LE audio task count

Default value:

- 1 if CONFIG_BT_BLE_LOG_SPI_OUT_LE_AUDIO_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_MESH_ENABLED

Enable BLE mesh log output to SPI

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

Enable BLE mesh log output to SPI

Default value:

- No (disabled) if CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_MESH_BUF_SIZE

SPI transaction buffer size for BLE mesh logs

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_MESH_ENABLED

SPI transaction buffer size for BLE mesh logs. There will be 2 SPI DMA buffers with the same size.

Default value:

- 1024 if CONFIG_BT_BLE_LOG_SPI_OUT_MESH_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_BLE_LOG_SPI_OUT_MESH_TASK_CNT

Mesh task count

Found in: Component config > Bluetooth > Common Options > BT Logs > Legacy SPI Log Output (Deprecated - use BT Log Async Output instead) > CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED > CONFIG_BT_BLE_LOG_SPI_OUT_MESH_ENABLED

Mesh task count

Default value:

- 3 if CONFIG_BT_BLE_LOG_SPI_OUT_MESH_ENABLED && CONFIG_BT_BLE_LOG_SPI_OUT_ENABLED

CONFIG_BT_LE_USED_MEM_STATISTICS_ENABLED

Enable used memory statistics

Found in: Component config > Bluetooth > Common Options

Used in internal tests only. Enable used memory statistics.

Default value:

- No (disabled)

Bluedroid Options

Contains:

- CONFIG_BT_ABORT_WHEN_ALLOCATION_FAILS
- CONFIG_BT_BLE_HOST_ALLOW_SUB_SPEC_MIN_CONN_INT
- Bluedroid debug option
- CONFIG_BT_BTU_TASK_STACK_SIZE
- CONFIG_BT_BTC_TASK_STACK_SIZE
- CONFIG_BT_BLE_ENABLED
- BT DEBUG LOG LEVEL
- CONFIG_BT_ACL_CONNECTIONS
- CONFIG_BT_SMP_MAX_BONDS
- CONFIG_BT_ALLOCATION_FROM_SPIRAM_FIRST
- CONFIG_BT_CLASSIC_ENABLED
- CONFIG_BT_STACK_NO_LOG
- CONFIG_BT_BLE_FEAT_ADV_CODING_SELECTION
- CONFIG_BT_BLE_42_FEATURES_SUPPORTED
- CONFIG_BT_BLE_50_FEATURES_SUPPORTED
- CONFIG_BT_BLE_FEAT_ISO_EN
- CONFIG_BT_BLE_FEAT_ADV_MONITOR
- CONFIG_BT_BLE_FEAT_CHANNEL_SOUNDING
- CONFIG_BT_BLE_FEAT_CONN_SUBRATING
- CONFIG_BT_BLE_FEAT_CTE_EN
- CONFIG_BT_BLE_HIGH_DUTY_ADV_INTERVAL
- CONFIG_BT_MULTI_CONNECTION_ENBALE
- CONFIG_BT_BLE_FEAT_PERIODIC_ADV_SYNC_TRANSFER
- CONFIG_BT_BLE_FEAT_POWER_CONTROL
- CONFIG_BT_BLE_VENDOR_HCI_EN
- CONFIG_BT_BLE_FEAT_CREATE_SYNC_ENH
- CONFIG_BT_BLUEDROID_ESP_COEX_VSC
- CONFIG_BT_BLE_FEAT_PERIODIC_ADV_ENH
- CONFIG_BT_BLE_FEAT_PAWR_EN
- CONFIG_BT_MAX_DEVICE_NAME_LEN
- CONFIG_BT_BLE_FEAT_CREATE_SYNC_RETRY_MAX
- CONFIG_BT_BLE_ACT_SCAN_REP_ADV_SCAN
- CONFIG_BT_BLUEDROID_PINNED_TO_CORE_CHOICE
- CONFIG_BT_BLE_RPA_TIMEOUT
- CONFIG_BT_BLE_DYNAMIC_ENV_MEMORY

CONFIG_BT_BTC_TASK_STACK_SIZE

Bluetooth event (callback to application) task stack size

Found in: Component config > Bluetooth > Bluedroid Options

This select btc task stack size

Default value:

- 3072 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUEDROID_PINNED_TO_CORE_CHOICE

The cpu core which Bluedroid run

Found in: Component config > Bluetooth > Bluedroid Options

Which the cpu core to run Bluedroid. Can choose core0 and core1. Can not specify no-affinity.

Available options:

- Core 0 (PRO CPU) (CONFIG_BT_BLUEDROID_PINNED_TO_CORE_0)
- Core 1 (APP CPU) (CONFIG_BT_BLUEDROID_PINNED_TO_CORE_1)

CONFIG_BT_BTU_TASK_STACK_SIZE

Bluetooth Bluedroid Host Stack task stack size

Found in: Component config > Bluetooth > Bluedroid Options

This select btu task stack size

Default value:

- 4352 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUEDROID_ESP_COEX_VSC

Enable Espressif Vendor-specific HCI commands for coexist status configuration

Found in: Component config > Bluetooth > Bluedroid Options

Enable Espressif Vendor-specific HCI commands for coexist status configuration

Default value:

- Yes (enabled) if (CONFIG_ESP_COEX_SW_COEXIST_ENABLE || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED
- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_CLASSIC_ENABLED

Classic Bluetooth

Found in: Component config > Bluetooth > Bluedroid Options

For now this option needs "SMP_ENABLE" to be set to yes

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && (CONFIG_BT_CONTROLLER_ENABLED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_CLASSIC_ENABLE_POWER_CTRL_VSC

Enable Espressif Vendor-specific HCI commands for power control of Classic Bluetooth

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

Enable Espressif Vendor-specific HCI commands for power control of Classic Bluetooth

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_ENC_KEY_SIZE_CTRL_ENABLED

configure encryption key size

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This chooses the support status of configuring encryption key size

Available options:

- Supported by standard HCI command (CONFIG_BT_ENC_KEY_SIZE_CTRL_STD)
- Supported by Vendor-specific HCI command (CONFIG_BT_ENC_KEY_SIZE_CTRL_VSC)
- Not supported (CONFIG_BT_ENC_KEY_SIZE_CTRL_NONE)

CONFIG_BT_CLASSIC_BQB_ENABLED

Host Qualitifcation support for Classic Bluetooth

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables functionalities of Host qualification for Classic Bluetooth.

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_A2DP_ENABLE

A2DP

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

Advanced Audio Distribution Profile

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_A2DP_USE_EXTERNAL_CODEC

Use External Codec for A2DP

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_A2DP_ENABLE

If enable, user shall register audio codec capability to A2DP and encode/decode audio data in application layer. The internal codec in A2DP will be remove in the future, it is recommend to use external codec for new design.

Default value:

- No (disabled) if CONFIG_BT_A2DP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

AVRCP Features

Contains:

- CONFIG_BT_AVRCP_CT_COVER_ART_ENABLED

CONFIG_BT_AVRCP_CT_COVER_ART_ENABLED

AVRCP CT Cover Art

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_A2DP_ENABLE > AVRCP Features

This enable Cover Art feature for AVRCP CT role, Cover Art requires AVRCP version 1.6 or higher. Therefore, if the Cover Art feature is enabled, AVRCP version will set to 1.6, otherwise, it will remain at 1.5

CONFIG_BT_SPP_ENABLED

SPP

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables the Serial Port Profile

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_L2CAP_ENABLED

BT L2CAP

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables the Logical Link Control and Adaptation Layer Protocol. Only supported classic bluetooth.

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_SDP_COMMON_ENABLED

BT SDP COMMON

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables common SDP operation, such as SDP record creation and deletion.

Default value:

- Yes (enabled) if CONFIG_BT_L2CAP_ENABLED && CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED
- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_SDP_PAD_LEN

One or more BT SDP attributes total allocated length (bytes)

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This is the total size of all SDP attributes allowed. Any attributes that exceed this size are truncated. The default value is 300.

Range:

- from to 1024 if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 300 if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_SDP_ATTR_LEN

Single BT SDP attribute allocated length (bytes)

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This is the maximum allowed size for a single SDP attribute. Any attributes that exceed this size are truncated. The default value is 300.

Range:

- from 300 to 1024 if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 300 if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HFP_ENABLE

Hands Free/Handset Profile

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

Hands Free Unit and Audio Gateway can be included simultaneously but they cannot run simultaneously due to internal limitations.

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_HFP_AG_ENABLE
- CONFIG_BT_HFP_AUDIO_DATA_PATH
- CONFIG_BT_HFP_CLIENT_ENABLE
- CONFIG_BT_HFP_USE_EXTERNAL_CODEC
- CONFIG_BT_HFP_WBS_ENABLE

CONFIG_BT_HFP_CLIENT_ENABLE

Hands Free Unit

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HFP_ENABLE

Default value:

- Yes (enabled) if CONFIG_BT_HFP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HFP_AG_ENABLE

Audio Gateway

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HFP_ENABLE

Default value:

- Yes (enabled) if CONFIG_BT_HFP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HFP_AUDIO_DATA_PATH

audio(SCO) data path

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HFP_ENABLE

SCO data path, i.e. HCI or PCM. This option is set using API "esp_bredr_sco_datapath_set" in Bluetooth host. Default SCO data path can also be set in Bluetooth Controller.

Available options:

- PCM (CONFIG_BT_HFP_AUDIO_DATA_PATH_PCM)
- HCI (CONFIG_BT_HFP_AUDIO_DATA_PATH_HCI)

CONFIG_BT_HFP_USE_EXTERNAL_CODEC

Use External Codec for HFP

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HFP_ENABLE

If enable, user shall encode/decode audio data in application layer. The internal codec in HFP will be remove in the future, it is recommend to use external codec for new design.

Default value:

- No (disabled) if CONFIG_BT_HFP_ENABLE && CONFIG_BT_HFP_AUDIO_DATA_PATH_HCI && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HFP_WBS_ENABLE

Wide Band Speech

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HFP_ENABLE

This enables Wide Band Speech. Should disable it when SCO data path is PCM. Otherwise there will be no data transmitted via GPIOs.

Default value:

- Yes (enabled) if CONFIG_BT_HFP_ENABLE && CONFIG_BT_HFP_AUDIO_DATA_PATH_HCI && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HID_ENABLED

Classic BT HID

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables the BT HID functionalities

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_HID_DEVICE_ENABLED
- CONFIG_BT_HID_HOST_ENABLED
- CONFIG_BT_HID_REMOVE_DEVICE_BONDING_ENABLED

CONFIG_BT_HID_HOST_ENABLED

Classic BT HID Host

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HID_ENABLED

This enables the BT HID Host

Default value:

- No (disabled) if CONFIG_BT_HID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_HID_DEVICE_ENABLED

Classic BT HID Device

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HID_ENABLED

This enables the BT HID Device

CONFIG_BT_HID_REMOVE_DEVICE_BONDING_ENABLED

Remove Device Bonding Information when HID Virtual Cable Unplugging

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_HID_ENABLED

This enables the BT HID to remove device bonding information when virtual cable unplugging, removing device bonding information is optional in HID 1.0 but mandatory in HID 1.1

Default value:

- Yes (enabled) if CONFIG_BT_HID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_PBAC_ENABLED

PBAP Client

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED

This enables the Phone Book Access Profile Client

Default value:

- No (disabled) if CONFIG_BT_CLASSIC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_PBAC_PREFERRED_MTU
- CONFIG_BT_PBAC_SUPPORTED_FEAT

CONFIG_BT_PBAC_SUPPORTED_FEAT

PBAP Client Supported Features

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_PBAC_ENABLED

Set the supported features of PBAP Client, the default value is supported all features

Default value:

- "0x000003FF" if CONFIG_BT_PBAC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_PBAC_PREFERRED_MTU

PBAP Client Preferred MTU

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_CLASSIC_ENABLED > CONFIG_BT_PBAC_ENABLED

MTU is limited by the max MTU of transport layer, and should not be smaller than 255, but can be set to zero to use a default MTU of transport layer

Default value:

- 0 if CONFIG_BT_PBAC_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_ENABLED

Bluetooth Low Energy

Found in: Component config > Bluetooth > Bluedroid Options

This enables Bluetooth Low Energy

Default value:

- Yes (enabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_ENABLE

Include GATT server module(GATTS)

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED

This option can be disabled when the app work only on gatt client mode

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_GATTS_APPEARANCE_WRITABLE
- CONFIG_BT_GATTS_DEVICE_NAME_WRITABLE
- CONFIG_BT_GATTS_KEY_MATERIAL_CHAR
- CONFIG_BT_GATTS_SECURITY_LEVELS_CHAR
- CONFIG_BT_GATTS_PPCP_CHAR_GAP
- CONFIG_BT_GATTS_ROBUST_CACHING_ENABLED
- CONFIG_BT_GATTS_SEND_SERVICE_CHANGE_MODE
- CONFIG_BT_BLE_BLUFI_ENABLE
- CONFIG_BT_GATT_MAX_SR_PROFILES
- CONFIG_BT_GATT_MAX_SR_ATTRIBUTES

CONFIG_BT_GATTS_PPCP_CHAR_GAP

Enable Peripheral Preferred Connection Parameters characteristic in GAP service

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

This enables "Peripheral Preferred Connection Parameters" characteristic (UUID: 0x2A04) in GAP service that has connection parameters like min/max connection interval, slave latency and supervision timeout multiplier

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_BLUFI_ENABLE

Include blufi function

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

This option can be close when the app does not require blufi function.

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUFI_BLE_SMP_ENABLE

Enable BLE SMP support for BluFi

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE > CONFIG_BT_BLE_BLUFI_ENABLE

Enable BLE Security Manager Protocol (SMP) for BluFi. When enabled, BluFi will support BLE pairing and encryption before Wi-Fi provisioning, providing a more secure provisioning process. This feature is only supported with the Bluedroid host.

Default value:

- No (disabled) if CONFIG_BT_BLE_BLUFI_ENABLE && CONFIG_BT_BLE_SMP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATT_MAX_SR_PROFILES

Max GATT Server Profiles

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Maximum GATT Server Profiles Count

Range:

- from 1 to 32 if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 8 if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATT_MAX_SR_ATTRIBUTES

Max GATT Service Attributes

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Maximum GATT Service Attributes Count

Range:

- from 1 to 500 if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 100 if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_SEND_SERVICE_CHANGE_MODE

GATTS Service Change Mode

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Service change indication mode for GATT Server.

Available options:

- GATTS manually send service change indication (CONFIG_BT_GATTS_SEND_SERVICE_CHANGE_MANUAL)

  Manually send service change indication through API esp_ble_gatts_send_service_change_indication()

- GATTS automatically send service change indication (CONFIG_BT_GATTS_SEND_SERVICE_CHANGE_AUTO)

  Let Bluedroid handle the service change indication internally

CONFIG_BT_GATTS_ROBUST_CACHING_ENABLED

Enable Robust Caching on Server Side

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

This option enables the GATT robust caching feature on the server. if turned on, the Client Supported Features characteristic, Database Hash characteristic, and Server Supported Features characteristic will be included in the GAP SERVICE.

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_DEVICE_NAME_WRITABLE

Allow to write device name by GATT clients

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Enabling this option allows remote GATT clients to write device name

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_APPEARANCE_WRITABLE

Allow to write appearance by GATT clients

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Enabling this option allows remote GATT clients to write appearance

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_SECURITY_LEVELS_CHAR

Enable LE GATT Security Levels Characteristic

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Enable LE GATT Security Levels Characteristic

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTS_KEY_MATERIAL_CHAR

Enable Encrypted Data Key Material Characteristic

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTS_ENABLE

Enable the Encrypted Data Key Material characteristic in GAP service. This characteristic allows advertising data to be decrypted and authenticated using the key material (session key + IV) as defined in Bluetooth Core Specification Version 5.4. The characteristic requires encrypted link to read.

Default value:

- No (disabled) if CONFIG_BT_GATTS_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTC_ENABLE

Include GATT client module(GATTC)

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED

This option can be close when the app work only on gatt server mode

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_GATTC_MAX_CACHE_CHAR
- CONFIG_BT_GATTC_NOTIF_REG_MAX
- CONFIG_BT_GATTC_CACHE_NVS_FLASH
- CONFIG_BT_GATTC_CONNECT_RETRY_COUNT
- CONFIG_BT_BLE_ESTAB_LINK_CONN_TOUT

CONFIG_BT_GATTC_MAX_CACHE_CHAR

Max gattc cache characteristic for discover

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTC_ENABLE

Maximum GATTC cache characteristic count

Range:

- from 1 to 500 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 40 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTC_NOTIF_REG_MAX

Max gattc notify(indication) register number

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTC_ENABLE

Maximum GATTC notify(indication) register number

Range:

- from 1 to 255 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 5 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTC_CACHE_NVS_FLASH

Save gattc cache data to nvs flash

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTC_ENABLE

This select can save gattc cache data to nvs flash

Default value:

- No (disabled) if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_GATTC_CONNECT_RETRY_COUNT

The number of attempts to reconnect if the connection establishment failed

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTC_ENABLE

The number of attempts to reconnect if the connection establishment failed

Range:

- from 0 to 255 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 3 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_ESTAB_LINK_CONN_TOUT

Timeout of BLE connection establishment

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_GATTC_ENABLE

Bluetooth Connection establishment maximum time, if connection time exceeds this value, the connection establishment fails, ESP_GATTC_OPEN_EVT or ESP_GATTS_OPEN_EVT is triggered.

Range:

- from 1 to 60 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 30 if CONFIG_BT_GATTC_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_SMP_ENABLE

Include BLE security module(SMP)

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED

This option can be close when the app not used the ble security connect.

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_BLE_SMP_ID_RESET_ENABLE
- CONFIG_BT_BLE_SMP_BOND_NVS_FLASH
- CONFIG_BT_SMP_SLAVE_CON_PARAMS_UPD_ENABLE
- CONFIG_BT_BLE_RPA_SUPPORTED

CONFIG_BT_SMP_SLAVE_CON_PARAMS_UPD_ENABLE

Slave enable connection parameters update during pairing

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_BLE_SMP_ENABLE

In order to reduce the pairing time, slave actively initiates connection parameters update during pairing.

Default value:

- No (disabled) if CONFIG_BT_BLE_SMP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_SMP_ID_RESET_ENABLE

Reset device identity when all bonding records are deleted

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_BLE_SMP_ENABLE

There are tracking risks associated with using a fixed or static IRK. If enabled this option, Bluedroid will assign a new randomly-generated IRK when all pairing and bonding records are deleted. This would decrease the ability of a previously paired peer to be used to determine whether a device with which it previously shared an IRK is within range.

Default value:

- No (disabled) if CONFIG_BT_BLE_SMP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_SMP_BOND_NVS_FLASH

Save SMP bonding keys to nvs flash

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_BLE_SMP_ENABLE

This select can save SMP bonding keys to nvs flash

Default value:

- Yes (enabled) if CONFIG_BT_BLE_SMP_ENABLE && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_RPA_SUPPORTED

Update RPA to Controller

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_ENABLED > CONFIG_BT_BLE_SMP_ENABLE

This enables controller RPA list function. For ESP32, ESP32 only support network privacy mode. If this option is enabled, ESP32 will only accept advertising packets from peer devices that contain private address, HW will not receive the advertising packets contain identity address after IRK changed. If this option is disabled, address resolution will be performed in the host, so the functions that require controller to resolve address in the white list cannot be used. This option is disabled by default on ESP32, please enable or disable this option according to your own needs.

For other BLE chips, devices support network privacy mode and device privacy mode, users can switch the two modes according to their own needs. So this option is enabled by default.

Default value:

- No (disabled) if CONFIG_BT_CONTROLLER_ENABLED && CONFIG_BT_BLE_SMP_ENABLE && (CONFIG_BT_CONTROLLER_ENABLED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED
- Yes (enabled) if CONFIG_BT_CONTROLLER_DISABLED && CONFIG_BT_BLE_SMP_ENABLE && (CONFIG_BT_CONTROLLER_ENABLED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

Bluedroid debug option

Contains:

- CONFIG_BT_BLUEDROID_MEM_DEBUG
- CONFIG_BT_BLUEDROID_MEM_STATS
- CONFIG_BT_BLUEDROID_THREAD_DEBUG

CONFIG_BT_BLUEDROID_MEM_DEBUG

Bluedroid memory debug

Found in: Component config > Bluetooth > Bluedroid Options > Bluedroid debug option

Bluedroid memory debug

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUEDROID_MEM_STATS

Bluedroid memory statistics

Found in: Component config > Bluetooth > Bluedroid Options > Bluedroid debug option

Enable Bluedroid memory usage statistics

CONFIG_BT_BLUEDROID_THREAD_DEBUG

Bluedroid thread debug

Found in: Component config > Bluetooth > Bluedroid Options > Bluedroid debug option

Enable Bluedroid thread debug mode. Used to debug whether the thread is blocked and dump information about the thread's related work queue.

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUEDROID_THREAD_BLOCK_TIME

OSI thread block time (in ms)

Found in: Component config > Bluetooth > Bluedroid Options > Bluedroid debug option > CONFIG_BT_BLUEDROID_THREAD_DEBUG

Indicates how long it takes for the thread to execute an item before it is considered blocked.

Default value:

- 1000 if CONFIG_BT_BLUEDROID_THREAD_DEBUG && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLUEDROID_THREAD_BLOCK_MSG

OSI thread block message count

Found in: Component config > Bluetooth > Bluedroid Options > Bluedroid debug option > CONFIG_BT_BLUEDROID_THREAD_DEBUG

Indicates how many messages are added to the queue while the threadis executing an item before it is considered blocked.

Default value:

- 50 if CONFIG_BT_BLUEDROID_THREAD_DEBUG && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_STACK_NO_LOG

Disable BT debug logs (minimize bin size)

Found in: Component config > Bluetooth > Bluedroid Options

This select can save the rodata code size

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

BT DEBUG LOG LEVEL

Contains:

- CONFIG_BT_LOG_A2D_TRACE_LEVEL

- CONFIG_BT_LOG_APPL_TRACE_LEVEL
- CONFIG_BT_LOG_AVCT_TRACE_LEVEL
- CONFIG_BT_LOG_AVDT_TRACE_LEVEL
- CONFIG_BT_LOG_AVRC_TRACE_LEVEL
- CONFIG_BT_LOG_BLUFI_TRACE_LEVEL
- CONFIG_BT_LOG_BNEP_TRACE_LEVEL
- CONFIG_BT_LOG_BTC_TRACE_LEVEL
- CONFIG_BT_LOG_BTIF_TRACE_LEVEL
- CONFIG_BT_LOG_BTM_TRACE_LEVEL
- CONFIG_BT_LOG_GAP_TRACE_LEVEL
- CONFIG_BT_LOG_GATT_TRACE_LEVEL
- CONFIG_BT_LOG_HCI_TRACE_LEVEL
- CONFIG_BT_LOG_HID_TRACE_LEVEL
- CONFIG_BT_LOG_L2CAP_TRACE_LEVEL
- CONFIG_BT_LOG_MCA_TRACE_LEVEL
- CONFIG_BT_LOG_OSI_TRACE_LEVEL
- CONFIG_BT_LOG_PAN_TRACE_LEVEL
- CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL
- CONFIG_BT_LOG_SDP_TRACE_LEVEL
- CONFIG_BT_LOG_SMP_TRACE_LEVEL

CONFIG_BT_LOG_HCI_TRACE_LEVEL

HCI layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for HCI layer

Available options:

- NONE (CONFIG_BT_LOG_HCI_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_HCI_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_HCI_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_HCI_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_HCI_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_HCI_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_HCI_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_BTM_TRACE_LEVEL

BTM layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for BTM layer

Available options:

- NONE (CONFIG_BT_LOG_BTM_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_BTM_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_BTM_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_BTM_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_BTM_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_BTM_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_BTM_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_L2CAP_TRACE_LEVEL

L2CAP layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for L2CAP layer

Available options:

- NONE (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL

RFCOMM layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for RFCOMM layer

Available options:

- NONE (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_RFCOMM_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_SDP_TRACE_LEVEL

SDP layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for SDP layer

Available options:

- NONE (CONFIG_BT_LOG_SDP_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_SDP_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_SDP_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_SDP_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_SDP_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_SDP_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_SDP_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_GAP_TRACE_LEVEL

GAP layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for GAP layer

Available options:

- NONE (CONFIG_BT_LOG_GAP_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_GAP_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_GAP_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_GAP_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_GAP_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_GAP_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_GAP_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_BNEP_TRACE_LEVEL

BNEP layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for BNEP layer

Available options:

- NONE (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_BNEP_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_PAN_TRACE_LEVEL

PAN layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for PAN layer

Available options:

- NONE (CONFIG_BT_LOG_PAN_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_PAN_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_PAN_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_PAN_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_PAN_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_PAN_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_PAN_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_A2D_TRACE_LEVEL

A2D layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for A2D layer

Available options:

- NONE (CONFIG_BT_LOG_A2D_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_A2D_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_A2D_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_A2D_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_A2D_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_A2D_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_A2D_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_AVDT_TRACE_LEVEL

AVDT layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for AVDT layer

Available options:

- NONE (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_AVDT_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_AVCT_TRACE_LEVEL

AVCT layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for AVCT layer

Available options:

- NONE (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_AVCT_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_AVRC_TRACE_LEVEL

AVRC layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for AVRC layer

Available options:

- NONE (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_AVRC_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_MCA_TRACE_LEVEL

MCA layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for MCA layer

Available options:

- NONE (CONFIG_BT_LOG_MCA_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_MCA_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_MCA_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_MCA_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_MCA_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_MCA_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_MCA_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_HID_TRACE_LEVEL

HID layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for HID layer

Available options:

- NONE (CONFIG_BT_LOG_HID_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_HID_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_HID_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_HID_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_HID_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_HID_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_HID_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_APPL_TRACE_LEVEL

APPL layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for APPL layer

Available options:

- NONE (CONFIG_BT_LOG_APPL_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_APPL_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_APPL_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_APPL_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_APPL_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_APPL_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_APPL_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_GATT_TRACE_LEVEL

GATT layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for GATT layer

Available options:

- NONE (CONFIG_BT_LOG_GATT_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_GATT_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_GATT_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_GATT_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_GATT_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_GATT_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_GATT_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_SMP_TRACE_LEVEL

SMP layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for SMP layer

Available options:

- NONE (CONFIG_BT_LOG_SMP_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_SMP_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_SMP_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_SMP_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_SMP_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_SMP_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_SMP_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_BTIF_TRACE_LEVEL

BTIF layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for BTIF layer

Available options:

- NONE (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_BTIF_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_BTC_TRACE_LEVEL

BTC layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for BTC layer

Available options:

- NONE (CONFIG_BT_LOG_BTC_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_BTC_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_BTC_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_BTC_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_BTC_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_BTC_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_BTC_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_OSI_TRACE_LEVEL

OSI layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for OSI layer

Available options:

- NONE (CONFIG_BT_LOG_OSI_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_OSI_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_OSI_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_OSI_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_OSI_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_OSI_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_OSI_TRACE_LEVEL_VERBOSE)

CONFIG_BT_LOG_BLUFI_TRACE_LEVEL

BLUFI layer

Found in: Component config > Bluetooth > Bluedroid Options > BT DEBUG LOG LEVEL

Define BT trace level for BLUFI layer

Available options:

- NONE (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_WARNING)
- API (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_API)
- EVENT (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_EVENT)
- DEBUG (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL_VERBOSE)

CONFIG_BT_ACL_CONNECTIONS

BT/BLE MAX ACL CONNECTIONS

Found in: Component config > Bluetooth > Bluedroid Options

Maximum BT/BLE connection count. The ESP32-C3/S3 chip supports a maximum of 10 instances, including ADV, SCAN and connections. The ESP32-C3/S3 chip can connect up to 9 devices if ADV or SCAN uses only one. If ADV and SCAN are both used, The ESP32-C3/S3 chip is connected to a maximum of 8 devices. Because Bluetooth cannot reclaim used instances once ADV or SCAN is used.

Range:

- from 1 to 9 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED
- from 1 to 50 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 4 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_MULTI_CONNECTION_ENBALE

Enable BLE multi-connections

Found in: Component config > Bluetooth > Bluedroid Options

Enable this option if there are multiple connections

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_ALLOCATION_FROM_SPIRAM_FIRST

BT/BLE will first malloc the memory from the PSRAM

Found in: Component config > Bluetooth > Bluedroid Options

This select can save the internal RAM if there have the PSRAM

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_DYNAMIC_ENV_MEMORY

Use dynamic memory allocation in BT/BLE stack

Found in: Component config > Bluetooth > Bluedroid Options

This select can make the allocation of memory will become more flexible

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_SMP_MAX_BONDS

BT/BLE maximum bond device count

Found in: Component config > Bluetooth > Bluedroid Options

The number of security records for peer devices.

CONFIG_BT_BLE_ACT_SCAN_REP_ADV_SCAN

Report adv data and scan response individually when BLE active scan

Found in: Component config > Bluetooth > Bluedroid Options

Originally, when doing BLE active scan, Bluedroid will not report adv to application layer until receive scan response. This option is used to disable the behavior. When enable this option, Bluedroid will report adv data or scan response to application layer immediately.

Memory reserved at start of DRAM for Bluetooth stack

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_MAX_DEVICE_NAME_LEN

length of bluetooth device name

Found in: Component config > Bluetooth > Bluedroid Options

Bluetooth Device name length shall be no larger than 248 octets, If the broadcast data cannot contain the complete device name, then only the shortname will be displayed, the rest parts that can't fit in will be truncated.

Range:

- from 32 to 248 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 32 if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_RPA_TIMEOUT

Timeout of resolvable private address

Found in: Component config > Bluetooth > Bluedroid Options

This set RPA timeout of Controller and Host. Default is 900 s (15 minutes). Range is 1 s to 1 hour (3600 s).

Range:

- from 1 to 3600 if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 900 if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_HOST_ALLOW_SUB_SPEC_MIN_CONN_INT

Allow BLE connection interval below Bluetooth Core Spec minimum (disable host check)

Found in: Component config > Bluetooth > Bluedroid Options

When enabled, the Bluedroid host skips the minimum BLE connection interval validation (Bluetooth Core Spec minimum is 0x0006 / 7.5 ms) and accepts any non-zero interval value from the application. The BLE controller then enforces what is actually allowed; how small the connection interval may be depends on controller capability and its own configuration, not on this host option text.

End users should NOT set this option directly. In typical IDF builds it follows the active Controller integration when that Controller supports this mode; use the Controller's own configuration (menu entries and symbol names differ by chip) instead of toggling this host symbol manually.

Default value:

- No (disabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_50_FEATURES_SUPPORTED

Enable BLE 5.0 and above features(please disable BLE 4.2 if enable BLE 5.0)

Found in: Component config > Bluetooth > Bluedroid Options

Enabling this option activates BLE 5.0 features. This option is universally supported in chips that support BLE, except for ESP32. BLE 4.2 and BLE 5.0 cannot be used simultaneously.

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_50_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_BLE_50_DTM_TEST_EN
- CONFIG_BT_BLE_50_EXTEND_ADV_EN
- CONFIG_BT_BLE_50_EXTEND_SCAN_EN

CONFIG_BT_BLE_50_EXTEND_ADV_EN

Enable BLE extend advertising

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_50_FEATURES_SUPPORTED

This enables BLE extend advertising

Default value:

- Yes (enabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_50_PERIODIC_ADV_EN

Enable BLE periodic advertising

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_50_FEATURES_SUPPORTED > CONFIG_BT_BLE_50_EXTEND_ADV_EN

This enables BLE periodic advertising

Default value:

- Yes (enabled) if CONFIG_BT_BLE_50_EXTEND_ADV_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_50_EXTEND_SCAN_EN

Enable BLE extend scan

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_50_FEATURES_SUPPORTED

This enables BLE extend scan

Default value:

- Yes (enabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_50_EXTEND_SYNC_EN

Enable BLE periodic advertising sync

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_50_FEATURES_SUPPORTED > CONFIG_BT_BLE_50_EXTEND_SCAN_EN

This enables BLE periodic advertising sync

Default value:

- Yes (enabled) if CONFIG_BT_BLE_50_EXTEND_SCAN_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_50_DTM_TEST_EN

Enable BLE 5.0 DTM test

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_50_FEATURES_SUPPORTED

This enables BLE 5.0 direct test mode

Default value:

- Yes (enabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_PERIODIC_ADV_SYNC_TRANSFER

Enable BLE periodic advertising sync transfer feature

Found in: Component config > Bluetooth > Bluedroid Options

This enables BLE periodic advertising sync transfer feature

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLE_50_PERIODIC_ADV_EN && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_ESP_NIMBLE_CONTROLLER) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_PERIODIC_ADV_ENH

Enable periodic adv enhancements(adi support)

Found in: Component config > Bluetooth > Bluedroid Options

Enable the periodic advertising enhancements

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLE_50_PERIODIC_ADV_EN && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_ESP_NIMBLE_CONTROLLER) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_CREATE_SYNC_ENH

Enable create sync enhancements(reporting disable and duplicate filtering enable support)

Found in: Component config > Bluetooth > Bluedroid Options

Enable the create sync enhancements

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLE_50_EXTEND_SYNC_EN && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_ESP_NIMBLE_CONTROLLER) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_CREATE_SYNC_RETRY_MAX

Maximum retry count for periodic advertising create sync

Found in: Component config > Bluetooth > Bluedroid Options

Set the maximum retry count when periodic advertising create sync fails with error code 0x3E (Connection Failed to be Established). Set to 0 to disable retry. Default is 3.

Range:

- from 0 to 16 if CONFIG_BT_BLE_50_EXTEND_SYNC_EN && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 3 if CONFIG_BT_BLE_50_EXTEND_SYNC_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 iso feature

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE 5.2 iso

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_AUDIO_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_BLE_FEAT_ISO_BIG_BROCASTER
- CONFIG_BT_BLE_FEAT_ISO_BIG_SYNCER
- CONFIG_BT_BLE_FEAT_ISO_CIG_CENTRAL
- CONFIG_BT_BLE_FEAT_ISO_CIG_PERIPHERAL
- CONFIG_BT_BLE_FEAT_ISO_60_EN
- CONFIG_BT_BLE_ISO_BIS_MAX_COUNT
- CONFIG_BT_BLE_ISO_CIS_MAX_COUNT
- CONFIG_BT_BLE_ISO_FLOW_CONTROL

CONFIG_BT_BLE_FEAT_ISO_60_EN

Enable ISO v6.0 feature

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable iso 6.0 feature

Default value:

- No (disabled) if CONFIG_BT_BLE_FEAT_ISO_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ISO_BIG_BROCASTER

Enable BLE iso BIG broadcaster

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 BIG broadcaster

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_ISO_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ISO_BIG_SYNCER

Enable BLE iso BIG syncer

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 BIG syncer

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_ISO_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_ISO_BIS_MAX_COUNT

Maximum bis count

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 CIG peripheral

Range:

- from 1 to 31 if CONFIG_BT_BLE_FEAT_ISO_EN && (CONFIG_BT_BLE_FEAT_ISO_BIG_BROCASTER || CONFIG_BT_BLE_FEAT_ISO_BIG_SYNCER) && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 2 if CONFIG_BT_BLE_FEAT_ISO_EN && (CONFIG_BT_BLE_FEAT_ISO_BIG_BROCASTER || CONFIG_BT_BLE_FEAT_ISO_BIG_SYNCER) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ISO_CIG_CENTRAL

Enable BLE iso CIG central

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 CIG central

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_ISO_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ISO_CIG_PERIPHERAL

Enable BLE iso CIG peripheral

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 CIG peripheral

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_ISO_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_ISO_CIS_MAX_COUNT

Maximum cis count

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Enable BLE 5.2 CIG peripheral

Range:

- from 1 to 31 if CONFIG_BT_BLE_FEAT_ISO_EN && (CONFIG_BT_BLE_FEAT_ISO_CIG_CENTRAL || CONFIG_BT_BLE_FEAT_ISO_CIG_PERIPHERAL) && CONFIG_BT_BLUEDROID_ENABLED

Default value:

- 2 if CONFIG_BT_BLE_FEAT_ISO_EN && (CONFIG_BT_BLE_FEAT_ISO_CIG_CENTRAL || CONFIG_BT_BLE_FEAT_ISO_CIG_PERIPHERAL) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_ISO_FLOW_CONTROL

Select ISO flow control type

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_ISO_EN

Select ISO flow control type

Available options:

- ISO standard flow control (CONFIG_BT_BLE_ISO_STD_FLOW_CTRL)

  Enable ISO standard flow control
- ISO non-standard flow control (CONFIG_BT_BLE_ISO_NON_STD_FLOW_CTRL)

  Enable ISO non-standard flow control

CONFIG_BT_BLE_FEAT_CTE_EN

Enable BLE CTE feature

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE 5.1 CTE

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_CTE_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_BLE_FEAT_CTE_CONNECTION_EN
- CONFIG_BT_BLE_FEAT_CTE_CONNECTIONLESS_EN

CONFIG_BT_BLE_FEAT_CTE_CONNECTIONLESS_EN

Enable BLE CTE connectionless feature

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_CTE_EN

Transmission of CTE in periodic advertising

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_CTE_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_CTE_CONNECTION_EN

Enable BLE CTE connection feature

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_FEAT_CTE_EN

Transmission of CTE by ACL connection

Default value:

- Yes (enabled) if CONFIG_BT_BLE_FEAT_CTE_EN && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_POWER_CONTROL

Enable BLE power control feature

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE power control feature

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_POWER_CONTROL_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_CONN_SUBRATING

Enable BLE connection subrating feature

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE connection subrating feature

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_SUBRATE_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_PAWR_EN

Enable Periodic Advertisement with Response(PAwR)

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE Periodic Advertisement with Response(PAwR) feature

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_PERIODIC_ADV_WITH_RESPONSE) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ADV_CODING_SELECTION

Enable Advertising Coding Selection

Found in: Component config > Bluetooth > Bluedroid Options

Enable Advertising Coding Selection

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_ADV_CODING_SELECT_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_CHANNEL_SOUNDING

Enable BLE channel sounding

Found in: Component config > Bluetooth > Bluedroid Options

Enable BLE channel sounding

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_CHANNEL_SOUNDING_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_FEAT_ADV_MONITOR

Enable BLE Advertising Monitor (LE Monitor Advertisement)

Found in: Component config > Bluetooth > Bluedroid Options

Enable LE Monitor Advertisement feature (Bluetooth Core 6.0). Allows the host to add devices to a monitor list and receive reports when the controller detects advertising from those devices (e.g. RSSI threshold).

Default value:

- No (disabled) if CONFIG_BT_BLE_50_FEATURES_SUPPORTED && ((CONFIG_BT_CONTROLLER_ENABLED && SOC_BLE_ADV_MONITOR_SUPPORTED) || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_42_FEATURES_SUPPORTED

Enable BLE 4.2 features(please disable BLE 5.0 if enable BLE 4.2)

Found in: Component config > Bluetooth > Bluedroid Options

This enables BLE 4.2 features. This option is universally supported by all ESP chips with BLE capabilities. BLE 4.2 and BLE 5.0 cannot be used simultaneously.

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED
- No (disabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

Contains:

- CONFIG_BT_BLE_42_ADV_EN
- CONFIG_BT_BLE_42_DTM_TEST_EN
- CONFIG_BT_BLE_42_SCAN_EN

CONFIG_BT_BLE_42_DTM_TEST_EN

Enable BLE 4.2 DTM test

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_42_FEATURES_SUPPORTED

This enables BLE 4.2 direct test mode

Default value:

- Yes (enabled) if CONFIG_BT_BLE_42_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_42_ADV_EN

Enable BLE 4.2 advertising

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_42_FEATURES_SUPPORTED

This enables BLE v4.2 advertising

Default value:

- Yes (enabled) if CONFIG_BT_BLE_42_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_42_SCAN_EN

Enable BLE 4.2 scan

Found in: Component config > Bluetooth > Bluedroid Options > CONFIG_BT_BLE_42_FEATURES_SUPPORTED

This enables BLE v4.2 scan

Default value:

- Yes (enabled) if CONFIG_BT_BLE_42_FEATURES_SUPPORTED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_VENDOR_HCI_EN

Enable BLE Vendor HCI command and event

Found in: Component config > Bluetooth > Bluedroid Options

This enables BLE vendor HCI command and event

Default value:

- Yes (enabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_BLE_HIGH_DUTY_ADV_INTERVAL

Enable BLE high duty advertising interval feature

Found in: Component config > Bluetooth > Bluedroid Options

This enable BLE high duty advertising interval feature

Default value:

- No (disabled) if CONFIG_BT_BLE_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

CONFIG_BT_ABORT_WHEN_ALLOCATION_FAILS

Abort when memory allocation fails in BT/BLE stack

Found in: Component config > Bluetooth > Bluedroid Options

This enables abort when memory allocation fails

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED && CONFIG_BT_BLUEDROID_ENABLED

NimBLE Options

Contains:

- BLE 5.x Features
- BLE 6.x Features
- Debugging/Testing
- Extra Features
- GAP
- GATT / ATT
- General
- Helper Utils
- Host-controller Transport
- L2CAP
- Memory Settings
- NimBLE Mesh
- Roles and Profiles
- Security (SMP)
- Services
- Vendor / Optimization

General

Contains:

- CONFIG_BT_NIMBLE_MEM_ALLOC_MODE
- CONFIG_BT_NIMBLE_HOST_TASK_STACK_SIZE
- CONFIG_BT_NIMBLE_PINNED_TO_CORE_CHOICE

CONFIG_BT_NIMBLE_MEM_ALLOC_MODE

Memory allocation strategy

Found in: Component config > Bluetooth > NimBLE Options > General

Allocation strategy for NimBLE host stack. Provides ability to allocate all required dynamic allocations from:

- Internal DRAM memory only
- External SPIRAM memory only
- Either internal or external memory based on default malloc() behavior in ESP-IDF
- Internal IRAM memory wherever applicable else internal DRAM

Available options:

- Internal memory (CONFIG_BT_NIMBLE_MEM_ALLOC_MODE_INTERNAL)
- External SPIRAM (CONFIG_BT_NIMBLE_MEM_ALLOC_MODE_EXTERNAL)
- Default alloc mode (CONFIG_BT_NIMBLE_MEM_ALLOC_MODE_DEFAULT)
- Internal IRAM (CONFIG_BT_NIMBLE_MEM_ALLOC_MODE_IRAM_8BIT)

  Allows use of IRAM memory region as 8-bit accessible region.

  Every unaligned (8bit or 16bit) access will result in an exception and incur penalty of certain clock cycles per unaligned read/write.

CONFIG_BT_NIMBLE_PINNED_TO_CORE_CHOICE

The CPU core on which NimBLE host will run

Found in: Component config > Bluetooth > NimBLE Options > General

The CPU core on which NimBLE host will run. You can choose Core 0 or Core 1. Cannot specify no-affinity

Available options:

- Core 0 (PRO CPU) (CONFIG_BT_NIMBLE_PINNED_TO_CORE_0)
- Core 1 (APP CPU) (CONFIG_BT_NIMBLE_PINNED_TO_CORE_1)

CONFIG_BT_NIMBLE_HOST_TASK_STACK_SIZE

NimBLE Host task stack size

Found in: Component config > Bluetooth > NimBLE Options > General

This configures stack size of NimBLE host task

Default value:

- 5120 if CONFIG_BLE_MESH && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED
- 4096 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Roles and Profiles

Contains:

- CONFIG_BT_NIMBLE_ROLE_BROADCASTER
- CONFIG_BT_NIMBLE_ROLE_CENTRAL
- CONFIG_BT_NIMBLE_GATT_CLIENT
- CONFIG_BT_NIMBLE_GATT_SERVER
- CONFIG_BT_NIMBLE_ROLE_OBSERVER
- CONFIG_BT_NIMBLE_ROLE_PERIPHERAL

CONFIG_BT_NIMBLE_ROLE_CENTRAL

Enable BLE Central role

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enables BLE Central role

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ROLE_PERIPHERAL

Enable BLE Peripheral role

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enable BLE Peripheral role

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ROLE_BROADCASTER

Enable BLE Broadcaster role

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enables BLE broadcaster role

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ROLE_OBSERVER

Enable BLE Observer role

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enables BLE Observer role

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CLIENT

Enable BLE GATT Client support

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enables support for GATT Client

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ROLE_CENTRAL && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_SERVER

Enable BLE GATT Server support

Found in: Component config > Bluetooth > NimBLE Options > Roles and Profiles

Enables support for GATT Server

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ROLE_PERIPHERAL && CONFIG_BT_NIMBLE_ENABLED

Security (SMP)

Contains:

- CONFIG_BT_NIMBLE_SECURITY_ENABLE

CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable BLE SM feature

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP)

Enable BLE sm feature

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_ENCRYPTION

- CONFIG_BT_NIMBLE_SM_SC_ONLY
- CONFIG_BT_NIMBLE_HANDLE_REPEAT_PAIRING_DELETION
- CONFIG_BT_NIMBLE_MAX_BONDS
- CONFIG_BT_NIMBLE_NVS_PERSIST
- CONFIG_BT_NIMBLE_SMP_ID_RESET
- CONFIG_BT_NIMBLE_SM_LVL
- CONFIG_BT_NIMBLE_SM_LEGACY
- CONFIG_BT_NIMBLE_SM_SC

CONFIG_BT_NIMBLE_SM_LEGACY

Security manager legacy pairing

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable security manager legacy pairing

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SM_SC

Security manager secure connections (4.2)

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable security manager secure connections

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SM_SC_DEBUG_KEYS

Use predefined public-private key pair

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE > CONFIG_BT_NIMBLE_SM_SC

If this option is enabled, SM uses predefined DH key pair as described in Core Specification, Vol. 3, Part H, 2.3.5.6.1. This allows to decrypt air traffic easily and thus should only be used for debugging.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SM_SC && CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_ENCRYPTION

Enable LE encryption

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable encryption connection

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SM_LVL

Security level

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

LE Security Mode 1 Levels: 1(0). No Security 2(1). Unauthenticated pairing with encryption 3(2). Authenticated pairing with encryption 4(3). Authenticated LE Secure Connections pairing with encryption using a 128-bit strength encryption key.

Default value:

- 0 if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SM_SC_ONLY

Enable Secure Connections Only Mode

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable Secure Connections Only Mode

Default value:

- 0 if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SMP_ID_RESET

Reset device identity when all bonding records are deleted

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

There are tracking risks associated with using a fixed or static IRK. If enabled this option, NimBLE will assign a new randomly-generated IRK when all pairing and bonding records are deleted. This would decrease the ability of a previously paired peer to be used to determine whether a device with which it previously shared an IRK is within range.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_NVS_PERSIST

Persist the BLE Bonding keys in NVS

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Enable this flag to make bonding persistent across device reboots

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_BONDS

Maximum number of bonds to save across reboots

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Defines maximum number of bonds to save for peer security and our security

Default value:

- 3 if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HANDLE_REPEAT_PAIRING_DELETION

Enable stack handling of repeat pairing

Found in: Component config > Bluetooth > NimBLE Options > Security (SMP) > CONFIG_BT_NIMBLE_SECURITY_ENABLE

Use this option to let stack internally handle the request for repeat pairing. Enabling this option will delete the pairing of the device and stack will NOT post any event to application. If this option is disabled, application will get BLE_GAP_EVENT_REPEAT_PAIRING event.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

GAP

Contains:

- CONFIG_BT_NIMBLE_HOST_ALLOW_CONNECT_WITH_SCAN
- CONFIG_BT_NIMBLE_HS_STOP_TIMEOUT_MS
- CONFIG_BT_NIMBLE_HOST_QUEUE_CONG_CHECK
- CONFIG_BT_NIMBLE_WHITELIST_SIZE
- CONFIG_BT_NIMBLE_ENABLE_CONN_REATTEMPT
- CONFIG_BT_NIMBLE_USE_ESP_TIMER
- CONFIG_BT_NIMBLE_HS_FLOW_CTRL
- CONFIG_BT_NIMBLE_HS_PVCY
- CONFIG_BT_NIMBLE_MAX_CCCDS
- CONFIG_BT_NIMBLE_MAX_CONNECTIONS
- CONFIG_BT_NIMBLE_CRYPTO_STACK_MBEDTLS
- CONFIG_BT_NIMBLE_RPA_TIMEOUT

CONFIG_BT_NIMBLE_RPA_TIMEOUT

RPA timeout in seconds

Found in: Component config > Bluetooth > NimBLE Options > GAP

Time interval between RPA address change.

Range:

- from 1 to 41400 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 900 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_WHITELIST_SIZE

BLE white list size

Found in: Component config > Bluetooth > NimBLE Options > GAP

BLE Whitelist size

Range:

- from 1 to 31 if SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED
- from 1 to 15 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 12 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ENABLE_CONN_REATTEMPT

Enable connection reattempts on connection establishment error

Found in: Component config > Bluetooth > NimBLE Options > GAP

Enable to make the NimBLE host to reattempt GAP connection attempt in case of connection establishment failure due to reason 0x3E.

Default value:

- Yes (enabled) if SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLED
- No (disabled) if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_CONN_REATTEMPT

Maximum number connection reattempts

Found in: Component config > Bluetooth > NimBLE Options > GAP > CONFIG_BT_NIMBLE_ENABLE_CONN_REATTEMPT

Defines maximum number of connection reattempts.

Range:

- from 1 to 255 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLE_CONN_REATTEMPT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 3 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLE_CONN_REATTEMPT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HS_PVCY

Enable privacy related APIs

Found in: Component config > Bluetooth > NimBLE Options > GAP

Enable this option to include host-side APIs related to BLE privacy. Disable it if your application does not require privacy features (such as RPA generation, resolving, or privacy-based pairing). Disabling this option excludes privacy-related code from the host to reduce memory usage.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HOST_BASED_PRIVACY

Enable host based privacy for random address.

Found in: Component config > Bluetooth > NimBLE Options > GAP > CONFIG_BT_NIMBLE_HS_PVCY

Use this option to do host based Random Private Address resolution. This option is valid only for ESP32

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_HS_PVCY && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HOST_ALLOW_CONNECT_WITH_SCAN

Allow Connections with scanning in progress

Found in: Component config > Bluetooth > NimBLE Options > GAP

This enables support for user to initiate a new connection with scan in progress

CONFIG_BT_NIMBLE_HOST_QUEUE_CONG_CHECK

BLE queue congestion check

Found in: Component config > Bluetooth > NimBLE Options > GAP

When scanning and scan duplicate is not enabled, if there are a lot of adv packets around or application layer handling adv packets is slow, it will cause the controller memory to run out. if enabled, adv packets will be lost when host queue is congested.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_CONNECTIONS

Maximum number of concurrent connections

Found in: Component config > Bluetooth > NimBLE Options > GAP

Defines maximum number of concurrent BLE connections. For ESP32, user is expected to configure BTDM_CTRL_BLE_MAX_CONN from controller menu along with this option. Similarly for ESP32-C3 or ESP32-S3, user is expected to configure BT_CTRL_BLE_MAX_ACT from controller menu. For ESP32C2, ESP32C6 and ESP32H2, each connection will take about 1k DRAM.

Range:

- from 1 to 70 if SOC_ESP_NIMBLE_CONTROLLER && (CONFIG_BT_NIMBLE_ROLE_CENTRAL || CONFIG_BT_NIMBLE_ROLE_PERIPHERAL) && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED
- from 1 to 9 if (CONFIG_BT_NIMBLE_ROLE_CENTRAL || CONFIG_BT_NIMBLE_ROLE_PERIPHERAL) && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 3 if (CONFIG_BT_NIMBLE_ROLE_CENTRAL || CONFIG_BT_NIMBLE_ROLE_PERIPHERAL) && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_CCCDS

Maximum number of CCC descriptors to save across reboots

Found in: Component config > Bluetooth > NimBLE Options > GAP

Defines maximum number of CCC descriptors to save

Default value:

- 8 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_CRYPTO_STACK_MBEDTLS

Override TinyCrypt with mbedTLS for crypto computations

Found in: Component config > Bluetooth > NimBLE Options > GAP

Enable this option to choose mbedTLS instead of TinyCrypt for crypto computations.

Default value:

- No (disabled) if CONFIG_BT_SMP_CRYPTO_STACK_TINYCRYPT && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HS_STOP_TIMEOUT_MS

BLE host stop timeout in msec

Found in: Component config > Bluetooth > NimBLE Options > GAP

BLE Host stop procedure timeout in milliseconds.

Default value:

- 2000 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_USE_ESP_TIMER

Enable Esp Timer for Nimble

Found in: Component config > Bluetooth > NimBLE Options > GAP

Set this option to use Esp Timer which has higher priority timer instead of FreeRTOS timer

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HS_FLOW_CTRL

Enable Host Flow control

Found in: Component config > Bluetooth > NimBLE Options > GAP

Enable Host Flow control

CONFIG_BT_NIMBLE_HS_FLOW_CTRL_ITVL

Host Flow control interval

Found in: Component config > Bluetooth > NimBLE Options > GAP > CONFIG_BT_NIMBLE_HS_FLOW_CTRL

Host flow control interval in msecs

Default value:

- 1000 if CONFIG_BT_NIMBLE_HS_FLOW_CTRL && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HS_FLOW_CTRL_THRESH

Host Flow control threshold

Found in: Component config > Bluetooth > NimBLE Options > GAP > CONFIG_BT_NIMBLE_HS_FLOW_CTRL

Host flow control threshold, if the number of free buffers are at or below this threshold, send an immediate number-of-completed-packets event

Default value:

- 2 if CONFIG_BT_NIMBLE_HS_FLOW_CTRL && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HS_FLOW_CTRL_TX_ON_DISCONNECT

Host Flow control on disconnect

Found in: Component config > Bluetooth > NimBLE Options > GAP > CONFIG_BT_NIMBLE_HS_FLOW_CTRL

Enable this option to send number-of-completed-packets event to controller after disconnection

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_HS_FLOW_CTRL && CONFIG_BT_NIMBLE_ENABLED

GATT / ATT

Contains:

- CONFIG_BT_NIMBLE_BLE_GATT_BLOB_TRANSFER
- CONFIG_BT_NIMBLE_GATT_CACHING_DISABLE_AUTO
- CONFIG_BT_NIMBLE_GATT_CACHING_ASSOC_ENABLE
- CONFIG_BT_NIMBLE_GATT_CACHING
- CONFIG_BT_NIMBLE_INCL_SVC_DISCOVERY
- CONFIG_BT_NIMBLE_ATT_MAX_PREP_ENTRIES
- CONFIG_BT_NIMBLE_GATT_CACHING_MAX_CONNS
- CONFIG_BT_NIMBLE_GATT_CACHING_MAX_CHRS
- CONFIG_BT_NIMBLE_GATT_CACHING_MAX_DSCS
- CONFIG_BT_NIMBLE_GATT_MAX_PROCS
- CONFIG_BT_NIMBLE_GATT_CACHING_MAX_INCL_SVCS
- CONFIG_BT_NIMBLE_GATT_CACHING_MAX_SVCS
- CONFIG_BT_NIMBLE_ATT_PREFERRED_MTU

CONFIG_BT_NIMBLE_ATT_PREFERRED_MTU

Preferred MTU size in octets

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

This is the default value of ATT MTU indicated by the device during an ATT MTU exchange. This value can be changed using API ble_att_set_preferred_mtu()

Default value:

- 256 if (CONFIG_BT_NIMBLE_GATT_CLIENT || CONFIG_BT_NIMBLE_GATT_SERVER) && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ATT_MAX_PREP_ENTRIES

Max Prepare write entries

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

This is the default value of ATT Maximum prepare entries

Default value:

- 64 if (CONFIG_BT_NIMBLE_GATT_CLIENT || CONFIG_BT_NIMBLE_GATT_SERVER) && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_MAX_PROCS

Maximum number of GATT client procedures

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Maximum number of GATT client procedures that can be executed.

Default value:

- 4 if (CONFIG_BT_NIMBLE_GATT_CLIENT || CONFIG_BT_NIMBLE_GATT_SERVER) && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_BLE_GATT_BLOB_TRANSFER

Blob transfer

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

This option is used when data to be sent is more than 512 bytes. For peripheral role, BT_NIMBLE_MSYS_1_BLOCK_COUNT needs to be increased according to the need.

CONFIG_BT_NIMBLE_GATT_CACHING

Enable GATT caching

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Enable GATT caching

Contains:

- CONFIG_BT_NIMBLE_GATT_CACHING_INCLUDE_SERVICES

CONFIG_BT_NIMBLE_GATT_CACHING_INCLUDE_SERVICES

Include services in GATT caching

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT > CONFIG_BT_NIMBLE_GATT_CACHING

Enable this option to include *included services* (e.g., services referenced by other services) in the GATT database cache. Disabling this will skip caching of included service entries.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_INCL_SVC_DISCOVERY

Enable Included service discovery

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Enable this option to start discovery for included service.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_GATT_CACHING_INCLUDE_SERVICES && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_MAX_CONNS

Maximum connections to be cached

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Set this option to set the upper limit on number of connections to be cached.

Default value:

- if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_MAX_SVCS

Maximum number of services per connection

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Set this option to set the upper limit on number of services per connection to be cached.

Default value:

- 64 if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_MAX_INCL_SVCS

Maximum number of included services per connection

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Set this option to set the upper limit on number of included services per connection to be cached.

Default value:

- 64 if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_MAX_CHRS

Maximum number of characteristics per connection

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Set this option to set the upper limit on number of characteristics per connection to be cached.

Default value:

- 64 if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_MAX_DSCS

Maximum number of descriptors per connection

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Set this option to set the upper limit on number of descriptors per connection to be cached.

Default value:

- 64 if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_DISABLE_AUTO

Do not start discovery procedure automatically upon receiving Out of Sync

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

When client receives ATT out-of-sync error message, it will not automatically start the discovery procedure to correct the invalid cache.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATT_CACHING_ASSOC_ENABLE

Enable association-based GATT caching

Found in: Component config > Bluetooth > NimBLE Options > GATT / ATT

Enable this option to use associated address caching instead of performing service discovery.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GATT_CACHING && CONFIG_BT_NIMBLE_ENABLED

L2CAP

Contains:

- CONFIG_BT_NIMBLE_L2CAP_ENHANCED_COC
- CONFIG_BT_NIMBLE_L2CAP_COC_MAX_NUM

CONFIG_BT_NIMBLE_L2CAP_COC_MAX_NUM

Maximum number of connection oriented channels

Found in: Component config > Bluetooth > NimBLE Options > L2CAP

Defines maximum number of BLE Connection Oriented Channels. When set to (0), BLE COC is not compiled in

Range:

- from 0 to 9 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 0 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_L2CAP_ENHANCED_COC

L2CAP Enhanced Connection Oriented Channel

Found in: Component config > Bluetooth > NimBLE Options > L2CAP

Enable Enhanced Credit Based Flow Control Mode

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_L2CAP_COC_MAX_NUM >= 1 && CONFIG_BT_NIMBLE_ENABLED

Memory Settings

Contains:

- CONFIG_BT_NIMBLE_TRANSPORT_ACL_FROM_LL_COUNT
- CONFIG_BT_NIMBLE_TRANSPORT_EVT_DISCARD_COUNT
- CONFIG_BT_NIMBLE_MSYS_BUF_FROM_HEAP
- CONFIG_BT_NIMBLE_L2CAP_COC_SDU_BUFF_COUNT
- CONFIG_BT_NIMBLE_MSYS_1_BLOCK_COUNT
- CONFIG_BT_NIMBLE_MSYS_1_BLOCK_SIZE
- CONFIG_BT_NIMBLE_MSYS_2_BLOCK_COUNT
- CONFIG_BT_NIMBLE_MSYS_2_BLOCK_SIZE
- CONFIG_BT_NIMBLE_MEMPOOL_RUNTIME_ALLOC
- CONFIG_BT_NIMBLE_TRANSPORT_ACL_SIZE
- CONFIG_BT_NIMBLE_TRANSPORT_EVT_COUNT

- CONFIG_BT_NIMBLE_TRANSPORT_EVT_SIZE

CONFIG_BT_NIMBLE_MSYS_1_BLOCK_COUNT

MSYS_1 Block Count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

MSYS is a system level mbuf registry. For prepare write & prepare responses MBUFs are allocated out of msys_1 pool. For NIMBLE_MESH enabled cases, this block count is increased by 8 than user defined count.

Default value:

- 24 if SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MSYS_1_BLOCK_SIZE

MSYS_1 Block Size

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

Dynamic memory size of block 1

Default value:

- 128 if SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MSYS_2_BLOCK_COUNT

MSYS_2 Block Count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

Dynamic memory count for block 2

Default value:

- 24 if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MSYS_2_BLOCK_SIZE

MSYS_2 Block Size

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

Dynamic memory size of block 2

Default value:

- 320 if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MSYS_BUF_FROM_HEAP

Get Msys Mbuf from heap

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This option sets the source of the shared msys mbuf memory between the Host and the Controller. Allocate the memory from the heap if this option is sets, from the mempool otherwise.

Default value:

- Yes (enabled) if BT_LE_MSYS_INIT_IN_CONTROLLER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_ACL_FROM_LL_COUNT

ACL Buffer count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

The number of ACL data buffers allocated for host.

Default value:

- 24 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_ACL_SIZE

Transport ACL Buffer size

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This is the maximum size of the data portion of HCI ACL data packets. It does not include the HCI data header (of 4 bytes)

Default value:

- 255 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_EVT_SIZE

Transport Event Buffer size

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This is the size of each HCI event buffer in bytes. In case of extended advertising, packets can be fragmented. 257 bytes is the maximum size of a packet.

Default value:

- 257 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED
- 70 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_EVT_COUNT

Transport Event Buffer count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This is the high priority HCI events' buffer size. High-priority event buffers are for everything except advertising reports. If there are no free high-priority event buffers then host will try to allocate a low-priority buffer instead

Default value:

- 30 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_EVT_DISCARD_COUNT

Discardable Transport Event Buffer count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This is the low priority HCI events' buffer size. Low-priority event buffers are only used for advertising reports. If there are no free low-priority event buffers, then an incoming advertising report will get dropped

Default value:

- 8 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_L2CAP_COC_SDU_BUFF_COUNT

L2cap coc Service Data Unit Buffer count

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

This is the service data unit buffer count for l2cap coc.

Default value:

- 1 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MEMPOOL_RUNTIME_ALLOC

Support on-demand runtime memory allocation for mempool

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings

When this option is enabled, mempool does not require pre-allocating memory. Instead, memory for each block will be dynamically allocated and released during mempool usage. This can significantly reduce memory consumption after mempool initialization, but may have some impact on performance.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_STATIC_TO_DYNAMIC && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MEMPOOL_BLOCK_REUSED

Support block reuse for mempool runtime memory allocation

Found in: Component config > Bluetooth > NimBLE Options > Memory Settings > CONFIG_BT_NIMBLE_MEMPOOL_RUNTIME_ALLOC

When this option is enabled, dynamically allocated blocks will not be freed but will be reused instead. This ensures virtually no impact on performance while reducing the memory consumption of the mempool.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MEMPOOL_RUNTIME_ALLOC && CONFIG_BT_NIMBLE_ENABLED

BLE 5.x Features

Contains:

- CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable BLE 5 feature

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features

Enable BLE 5 feature

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && SOC_BLE_50_SUPPORTED && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_AOA_AOD
- CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_2M_PHY
- CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_CODED_PHY
- CONFIG_BT_NIMBLE_EXT_ADV
- CONFIG_BT_NIMBLE_EXT_SCAN
- CONFIG_BT_NIMBLE_BLE_POWER_CONTROL
- CONFIG_BT_NIMBLE_ISO

CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_2M_PHY

Enable 2M Phy

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable 2M-PHY

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_LL_CFG_FEAT_LE_CODED_PHY

Enable coded Phy

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable coded-PHY

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EXT_ADV

Enable extended advertising

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable this option to do extended advertising. Extended advertising will be supported from BLE 5.0 onwards.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EXT_ADV_V2

Enable support for extended adv v2

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV

Enable this option to use Extended Adv V2 command instead of V1.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_EXT_ADV_INSTANCES

Maximum number of extended advertising instances.

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV

Change this option to set maximum number of extended advertising instances. Minimum there is always one instance of advertising. Enter how many more advertising instances you want. For ESP32C2, ESP32C6 and ESP32H2, each extended advertising instance will take about 0.5k DRAM.

Range:

- from 0 to 4 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 1 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED
- 0 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EXT_ADV_MAX_SIZE

Maximum length of the advertising data.

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV

Defines the length of the extended adv data. The value should not exceed 1650.

Range:

- from 0 to 1650 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 1650 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED
- 0 if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV

Enable periodic advertisement.

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV

Enable this option to start periodic advertisement.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_PERIODIC_ADV_ENH

Periodic adv enhancements(adi support)

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV > CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV

Enable the periodic advertising enhancements

CONFIG_BT_NIMBLE_PERIODIC_ADV_SYNC_TRANSFER

Enable Transfer Sync Events

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV > CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV

This enables controller transfer periodic sync events to host

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_PERIODIC_ADV_WITH_RESPONSES

Enable Periodic Advertisement with Response (EXPERIMENTAL)

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_ADV > CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV

This enables controller PAwR (Periodic Advertisement with Response).

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV && SOC_BLE_PERIODIC_ADV_WITH_RESPONSE && CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EXT_SCAN

Enable extended scanning

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable this option to do extended scanning.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_EXT_ADV && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC

Enable periodic sync

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_SCAN

Enable this option to receive periodic advertisement.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_EXT_SCAN && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_PERIODIC_SYNCS

Maximum number of periodic advertising syncs

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_SCAN > CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC

Set this option to set the upper limit for number of periodic sync connections. This should be less than maximum connections allowed by controller.

Range:

- from 0 to 8 if CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 1 if CONFIG_BT_NIMBLE_ENABLE_PERIODIC_ADV && CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED
- 0 if CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_PERIODIC_ADVERTISER_LIST

Maximum number of periodic advertiser list

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_EXT_SCAN > CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC

Set this option to set the upper limit for number of periodic advertiser list.

Range:

- from 1 to 5 if SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 5 if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && SOC_ESP_NIMBLE_CONTROLLER && CONFIG_BT_NIMBLE_ENABLE_PERIODIC_SYNC && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_BLE_POWER_CONTROL

Enable support for BLE Power Control

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Set this option to enable the Power Control feature

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && SOC_BLE_POWER_CONTROL_SUPPORTED && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_AOA_AOD

Direction Finding

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable support for Connectionless and Connection Oriented Direction Finding

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && SOC_BLE_CTE_SUPPORTED && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO

Isochronous channels

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT

Enable BLE Isochronous functionality.

CONFIG_BT_NIMBLE_ISO_FLOW_CONTROL

ISO flow control mode

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

ISO flow control mode

Available options:

- Standard (CONFIG_BT_NIMBLE_ISO_STD_FLOW_CTRL)

  ISO standard flow control
- Non-standard (CONFIG_BT_NIMBLE_ISO_NON_STD_FLOW_CTRL)

  ISO non-standard flow control

CONFIG_BT_NIMBLE_ISO_TEST

ISO Test mode

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Enable BLE Isochronous Test functionality.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_BIG

Maximum number of BIG

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of BIG.

Range:

- from 1 to 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 1 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_BIS

Maximum number of BIS

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of BIS.

Range:

- from 1 to 3 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_BIS_PER_BIG

Maximum number of BIS per BIG

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of BIS per BIG.

Range:

- from 1 to 3 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_CIG

Maximum number of CIG

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of CIG.

Range:

- from 1 to 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 1 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_CIS

Maximum number of CIS

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of CIS.

Range:

- from 1 to 3 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_CIS_PER_CIG

Maximum number of CIS per CIG

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Maximum number of CIS per CIG.

Range:

- from 1 to 3 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

Default value:

- 2 if CONFIG_BT_NIMBLE_ISO && CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ISO_CIS_ESTAB_V2

LE CIS Established event [v2]

Found in: Component config > Bluetooth > NimBLE Options > BLE 5.x Features > CONFIG_BT_NIMBLE_50_FEATURE_SUPPORT > CONFIG_BT_NIMBLE_ISO

Enable this to support LE CIS Established event [v2].

BLE 6.x Features

Contains:

- CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT

CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT

Enable BLE 6 feature

Found in: Component config > Bluetooth > NimBLE Options > BLE 6.x Features

Enable BLE 6.x feature support

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && (SOC_BLE_60_SUPPORTED || CONFIG_BT_CONTROLLER_DISABLED) && (SOC_BLE_60_SUPPORTED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_CHANNEL_SOUNDING
- CONFIG_BT_NIMBLE_MONITOR_ADV

CONFIG_BT_NIMBLE_CHANNEL_SOUNDING

ble channel sounding feature

Found in: Component config > Bluetooth > NimBLE Options > BLE 6.x Features > CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT

Used to enable/disable the channel sounding feature

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT && (SOC_BLE_60_SUPPORTED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MONITOR_ADV

Enable Monitor Advertising

Found in: Component config > Bluetooth > NimBLE Options > BLE 6.x Features > CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT

Enable support for Monitor Advertisers

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_60_FEATURE_SUPPORT && (SOC_BLE_60_SUPPORTED || CONFIG_BT_CONTROLLER_DISABLED) && CONFIG_BT_NIMBLE_ENABLED

Services

Contains:

- CONFIG_BT_NIMBLE_ANS_SERVICE
- CONFIG_BT_NIMBLE_BAS_SERVICE
- CONFIG_BT_NIMBLE_CTS_SERVICE
- CONFIG_BT_NIMBLE_DIS_SERVICE
- CONFIG_BT_NIMBLE_GAP_SERVICE
- CONFIG_BT_NIMBLE_HTP_SERVICE
- CONFIG_BT_NIMBLE_HR_SERVICE

- CONFIG_BT_NIMBLE_HID_SERVICE
- CONFIG_BT_NIMBLE_IAS_SERVICE
- CONFIG_BT_NIMBLE_IPSS_SERVICE
- CONFIG_BT_NIMBLE_LLS_SERVICE
- CONFIG_BT_NIMBLE_PROX_SERVICE
- CONFIG_BT_NIMBLE_SPS_SERVICE
- CONFIG_BT_NIMBLE_TPS_SERVICE

CONFIG_BT_NIMBLE_PROX_SERVICE

Proximity service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Proximity Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ANS_SERVICE

Alert Notification service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Alert Notification Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_CTS_SERVICE

Current Time Service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Current Time Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HTP_SERVICE

Health Thermometer service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Health Thermometer Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_IPSS_SERVICE

Internet Protocol Support service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Internet Protocol Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TPS_SERVICE

Tx Power service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Tx Power Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_IAS_SERVICE

Immediate Alert service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Immediate Alert Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_LLS_SERVICE

Link Loss service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Link Loss Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SPS_SERVICE

Serial Port service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Serial Port Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HR_SERVICE

Heart Rate service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable HeartRate Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HID_SERVICE

Human Interface Device service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable HID service support

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_SVC_HID_MAX_RPTS
- CONFIG_BT_NIMBLE_SVC_HID_MAX_INSTANCES

CONFIG_BT_NIMBLE_SVC_HID_MAX_INSTANCES

Maximum HID service instances

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_HID_SERVICE

Defines maximum number of HID service instances

Default value:

- 2 if CONFIG_BT_NIMBLE_HID_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_HID_MAX_RPTS

Maximum HID Report characteristics per service instance

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_HID_SERVICE

Defines maximum number of report characteristics per service instance

Default value:

- 3 if CONFIG_BT_NIMBLE_HID_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_BAS_SERVICE

Battery service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Battery service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_SVC_BAS_BATTERY_LEVEL_NOTIFY

CONFIG_BT_NIMBLE_SVC_BAS_BATTERY_LEVEL_NOTIFY

BAS Battery Level NOTIFY permission

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_BAS_SERVICE

Enable/Disable notifications on BAS Battery Level Characteristic

CONFIG_BT_NIMBLE_DIS_SERVICE

Device Information Service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable Device Information service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_SVC_DIS_INCLUDED
- CONFIG_BT_NIMBLE_SVC_DIS_FIRMWARE_REVISION
- CONFIG_BT_NIMBLE_SVC_DIS_HARDWARE_REVISION
- CONFIG_BT_NIMBLE_SVC_DIS_MANUFACTURER_NAME
- CONFIG_BT_NIMBLE_SVC_DIS_PNP_ID
- CONFIG_BT_NIMBLE_SVC_DIS_SERIAL_NUMBER
- CONFIG_BT_NIMBLE_SVC_DIS_SOFTWARE_REVISION
- CONFIG_BT_NIMBLE_SVC_DIS_SYSTEM_ID

CONFIG_BT_NIMBLE_SVC_DIS_MANUFACTURER_NAME

Manufacturer Name

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS characteristic Manufacturer Name String characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_SERIAL_NUMBER

Serial Number

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS Serial Number characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_HARDWARE_REVISION

Hardware Revision

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS Hardware Revision characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_FIRMWARE_REVISION

Firmware Revision

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS Firmware Revision characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_SOFTWARE_REVISION

Software Revision

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS Software Revision characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_SYSTEM_ID

System ID

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS System ID characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_PNP_ID

PnP ID

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Enable the DIS PnP ID characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_DIS_INCLUDED

DIS as an Included Service

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_DIS_SERVICE

Use DIS as an included service

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_DIS_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GAP_SERVICE

GAP Service

Found in: Component config > Bluetooth > NimBLE Options > Services

Enable GAP Service support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_SVC_GAP_DEVICE_NAME
- CONFIG_BT_NIMBLE_SVC_GAP_APPEARANCE
- GAP Appearance write permissions
- CONFIG_BT_NIMBLE_SVC_GAP_CENT_ADDR_RESOLUTION
- GAP device name write permissions
- CONFIG_BT_NIMBLE_SVC_GAP_GATT_SECURITY_LEVEL
- CONFIG_BT_NIMBLE_GAP_DEVICE_NAME_MAX_LEN
- Peripheral Preferred Connection Parameters (PPCP) settings
- CONFIG_BT_NIMBLE_SVC_GAP_RPA_ONLY

CONFIG_BT_NIMBLE_SVC_GAP_DEVICE_NAME

BLE GAP default device name

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

The Device Name characteristic shall contain the name of the device as an UTF-8 string. This name can be changed by using API ble_svc_gap_device_name_set()

Default value:

- "nimble" if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GAP_DEVICE_NAME_MAX_LEN

Maximum length of BLE device name in octets

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

Device Name characteristic value shall be 0 to 248 octets in length

Default value:

- 31 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_APPEARANCE

External appearance of the device

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

Standard BLE GAP Appearance value in HEX format e.g. 0x02C0

Default value:

- 0 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_GATT_SECURITY_LEVEL

LE GATT Security Level Characteristic

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

Enable the LE GATT Security Level Characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_RPA_ONLY

Resolvable Private Address Only characteristic

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

Enable the Resolvable Private Address Only characteristic

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_CENT_ADDR_RESOLUTION

GAP Characteristic - Central Address Resolution

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE

Weather or not Central Address Resolution characteristic is supported on the device, and if supported, weather or not Central Address Resolution is supported.

- Central Address Resolution characteristic not supported
- Central Address Resolution not supported
- Central Address Resolution supported

Available options:

- Characteristic not supported (CONFIG_BT_NIMBLE_SVC_GAP_CAR_CHAR_NOT_SUPP)
- Central Address Resolution not supported (CONFIG_BT_NIMBLE_SVC_GAP_CAR_NOT_SUPP)
- Central Address Resolution supported (CONFIG_BT_NIMBLE_SVC_GAP_CAR_SUPP)

GAP Appearance write permissions

Contains:

- CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE

CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE

Write

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP Appearance write permissions

Enable write permission (BLE_GATT_CHR_F_WRITE)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE_ENC

Write with encryption

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP Appearance write permissions > CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE

Enable write with encryption permission (BLE_GATT_CHR_F_WRITE_ENC)

CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE_AUTHEN

Write with authentication

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP Appearance write permissions > CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE

Enable write with authentication permission (BLE_GATT_CHR_F_WRITE_AUTHEN)

CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE_AUTHOR

Write with authorisation

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP Appearance write permissions > CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE

Enable write with authorisation permission (BLE_GATT_CHR_F_WRITE_AUTHOR)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SVC_GAP_APPEAR_WRITE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

GAP device name write permissions

Contains:

- CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE

CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE

Write

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP device name write permissions

Enable write permission (BLE_GATT_CHR_F_WRITE)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE_ENC

Write with encryption

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP device name write permissions > CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE

Enable write with encryption permission (BLE_GATT_CHR_F_WRITE_ENC)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE_AUTHEN

Write with authentication

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP device name write permissions > CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE

Enable write with authentication permission (BLE_GATT_CHR_F_WRITE_AUTHEN)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE_AUTHOR

Write with authorisation

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > GAP device name write permissions > CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE

Enable write with authorisation permission (BLE_GATT_CHR_F_WRITE_AUTHOR)

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_SVC_GAP_NAME_WRITE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Peripheral Preferred Connection Parameters (PPCP) settings

Contains:

- CONFIG_BT_NIMBLE_SVC_GAP_PPCP_MAX_CONN_INTERVAL
- CONFIG_BT_NIMBLE_SVC_GAP_PPCP_MIN_CONN_INTERVAL
- CONFIG_BT_NIMBLE_SVC_GAP_PPCP_SLAVE_LATENCY
- CONFIG_BT_NIMBLE_SVC_GAP_PPCP_SUPERVISION_TMO

CONFIG_BT_NIMBLE_SVC_GAP_PPCP_MAX_CONN_INTERVAL

PPCP Connection Interval Max (Unit: 1.25 ms)

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > Peripheral Preferred Connection Parameters (PPCP) settings

Peripheral Preferred Connection Parameter: Connection Interval maximum value Interval Max = value * 1.25 ms

Default value:

- 0 if CONFIG_BT_NIMBLE_ROLE_PERIPHERAL && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_PPCP_MIN_CONN_INTERVAL

PPCP Connection Interval Min (Unit: 1.25 ms)

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > Peripheral Preferred Connection Parameters (PPCP) settings

Peripheral Preferred Connection Parameter: Connection Interval minimum value Interval Min = value * 1.25 ms

Default value:

- 0 if CONFIG_BT_NIMBLE_ROLE_PERIPHERAL && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_PPCP_SLAVE_LATENCY

PPCP Slave Latency

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > Peripheral Preferred Connection Parameters (PPCP) settings

Peripheral Preferred Connection Parameter: Slave Latency

Default value:

- 0 if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SVC_GAP_PPCP_SUPERVISION_TMO

PPCP Supervision Timeout (Uint: 10 ms)

Found in: Component config > Bluetooth > NimBLE Options > Services > CONFIG_BT_NIMBLE_GAP_SERVICE > Peripheral Preferred Connection Parameters (PPCP) settings

Peripheral Preferred Connection Parameter: Supervision Timeout Timeout = Value * 10 ms

Default value:

- 0 if CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GAP_SERVICE && CONFIG_BT_NIMBLE_GATT_SERVER && CONFIG_BT_NIMBLE_ENABLED

Extra Features

Contains:

- CONFIG_BT_NIMBLE_GATTC_AUTO_PAIR
- CONFIG_BT_NIMBLE_ADV_UUID_CONCAT
- CONFIG_BT_NIMBLE_BLUFI_ENABLE
- CONFIG_BT_NIMBLE_CPFD_CAFD
- CONFIG_BT_NIMBLE_DTM_MODE_TEST
- CONFIG_BT_NIMBLE_DYNAMIC_SERVICE
- CONFIG_BT_NIMBLE_MEM_OPTIMIZATION
- CONFIG_BT_NIMBLE_SM_SIGN_CNT
- CONFIG_BT_NIMBLE_SUBRATE
- CONFIG_BT_NIMBLE_STATIC_PASSKEY
- CONFIG_BT_NIMBLE_ENC_ADV_DATA
- CONFIG_BT_NIMBLE_GATTC_PROC_PREEMPTION_PROTECT
- CONFIG_BT_NIMBLE_MAX_EADS
- CONFIG_BT_NIMBLE_EATT_CHAN_NUM
- CONFIG_BT_NIMBLE_RECONFIG_MTU
- CONFIG_BT_NIMBLE_LOW_SPEED_MODE
- CONFIG_BT_NIMBLE_STATIC_TO_DYNAMIC

CONFIG_BT_NIMBLE_DYNAMIC_SERVICE

Enable dynamic services

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

This enables user to add/remove Gatt services at runtime

CONFIG_BT_NIMBLE_BLUFI_ENABLE

Enable blufi functionality

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Set this option to enable blufi functionality.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_ENC_ADV_DATA

Encrypted Advertising Data

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

This option is used to enable encrypted advertising data.

CONFIG_BT_NIMBLE_ADV_UUID_CONCAT

concatenate uuids while parsing advertising data

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enables concatenation of multiple UUIDs of the same type while parsing advertising data on the central device. When disabled, only the last parsed UUID of a given type is retained.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MAX_EADS

Maximum number of EAD devices to save across reboots

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Defines maximum number of encrypted advertising data key material to save

Default value:

- 10 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENC_ADV_DATA && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATTC_PROC_PREEMPTION_PROTECT

Gatt-proc preemption protect check

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

When BLE and Wireless protocol/IEEE 802.15.4 operate in coexistence, BLE preemption can disrupt the GATT context,causing the service discovery callback to not be invoked. A temporary list is maintained to preserve the GATT context and use it in case of preemption.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_GATTC_AUTO_PAIR

Automatically pair upon receiving service request failure

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

If enabled, when a service request (e.g. read, write) to a server fails, and the ATT error suggests insufficient security, then the central will initiate pairing and retry the service request.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EATT_CHAN_NUM

Maximum number of EATT channels

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Defines the number of channels EATT bearers can use

Default value:

- 0 if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SUBRATE

Enable Subrate Change

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable connection subrate change feature

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_STATIC_PASSKEY

Enable support for Static Passkey

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable support for user defined static passkey for SMP

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_SECURITY_ENABLE && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_DTM_MODE_TEST

Enable DTM related test procedure support

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable DTM related test procedure support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MEM_OPTIMIZATION

Enable memory optimization

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

This option enables memory optimizations in the NimBLE stack.

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_STATIC_TO_DYNAMIC

Use Dynamic Memory allocations instead of static

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

This option enables use of dynamic memory allocation instead of static allocation for variable global / local variable. This helps in improving memory footprint

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_SM_SIGN_CNT

Enable Sign counter operations

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable Sign counter operations

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_CPFD_CAFD

Enable CPFD/CAPD support

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable CPFD and CAPD gatt descriptors support

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_RECONFIG_MTU

Reconfig L2CAP MTU support

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Enable Support to reconfigure L2CAP MTU / MPS

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_LOW_SPEED_MODE

Reduce DRAM/IRAM usage by moving GATT/GAP/NPL functions to FLASH

Found in: Component config > Bluetooth > NimBLE Options > Extra Features

Disabling IRAM-optimized paths will place functions in FLASH (.irom.text) instead of DRAM (.iram.text/.text in internal RAM). Suitable for low-speed GAP/GATT operations to reduce RAM usage.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED

NimBLE Mesh

Contains:

- CONFIG_BT_NIMBLE_MESH

CONFIG_BT_NIMBLE_MESH

Enable BLE mesh functionality

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh

Enable BLE Mesh example present in upstream mynewt-nimble and not maintained by Espressif.

IDF maintains ESP-BLE-MESH as the official Mesh solution. Please refer to ESP-BLE-MESH guide at: *:doc:../esp32/api-guides/esp-ble-mesh/ble-mesh-index``*

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Contains:

- CONFIG_BT_NIMBLE_MESH_PROVISIONER
- CONFIG_BT_NIMBLE_MESH_PROV
- CONFIG_BT_NIMBLE_MESH_GATT_PROXY
- CONFIG_BT_NIMBLE_MESH_FRIEND
- CONFIG_BT_NIMBLE_MESH_LOW_POWER
- CONFIG_BT_NIMBLE_MESH_PROXY
- CONFIG_BT_NIMBLE_MESH_RELAY
- CONFIG_BT_NIMBLE_MESH_DEVICE_NAME
- CONFIG_BT_NIMBLE_MESH_NODE_COUNT

CONFIG_BT_NIMBLE_MESH_PROXY

Enable mesh proxy functionality

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Enable proxy. This is automatically set whenever NIMBLE_MESH_PB_GATT or NIMBLE_MESH_GATT_PROXY is set

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_PROV

Enable BLE mesh provisioning

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Enable mesh provisioning

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_PB_ADV

Enable mesh provisioning over advertising bearer

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH > CONFIG_BT_NIMBLE_MESH_PROV

Enable this option to allow the device to be provisioned over the advertising bearer

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_MESH_PROV && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_PB_GATT

Enable mesh provisioning over GATT bearer

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH > CONFIG_BT_NIMBLE_MESH_PROV

Enable this option to allow the device to be provisioned over the GATT bearer

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_MESH_PROV && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_GATT_PROXY

Enable GATT Proxy functionality

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

This option enables support for the Mesh GATT Proxy Service, i.e. the ability to act as a proxy between a Mesh GATT Client and a Mesh network

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_RELAY

Enable mesh relay functionality

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Support for acting as a Mesh Relay Node

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_LOW_POWER

Enable mesh low power mode

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Enable this option to be able to act as a Low Power Node

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_FRIEND

Enable mesh friend functionality

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Enable this option to be able to act as a Friend Node

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_DEVICE_NAME

Set mesh device name

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

This value defines Bluetooth Mesh device/node name

Default value:

- "nimble-mesh-node" if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_NODE_COUNT

Set mesh node count

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Defines mesh node count.

Default value:

- 1 if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_MESH_PROVISIONER

Enable BLE mesh provisioner

Found in: Component config > Bluetooth > NimBLE Options > NimBLE Mesh > CONFIG_BT_NIMBLE_MESH

Enable mesh provisioner.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_MESH && CONFIG_BT_NIMBLE_ENABLED

Host-controller Transport

Contains:

- CONFIG_BT_NIMBLE_TRANSPORT_UART
- CONFIG_BT_NIMBLE_HCI_UART_CTS_PIN
- CONFIG_BT_NIMBLE_USE_HCI_UART_FLOW_CTRL
- CONFIG_BT_NIMBLE_HCI_UART_RTS_PIN

CONFIG_BT_NIMBLE_TRANSPORT_UART

Enable Uart Transport

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport

Use UART transport

Default value:

- Yes (enabled) if CONFIG_BT_CONTROLLER_DISABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TRANSPORT_UART_PORT

Uart port

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport > CONFIG_BT_NIMBLE_TRANSPORT_UART

Uart port

Default value:

- 1 if CONFIG_BT_CONTROLLER_DISABLED && CONFIG_BT_NIMBLE_TRANSPORT_UART && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HCI_USE_UART_BAUDRATE

Uart Hci Baud Rate

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport > CONFIG_BT_NIMBLE_TRANSPORT_UART

Uart Baud Rate

Available options:

- 115200 (CONFIG_UART_BAUDRATE_115200)
- 230400 (CONFIG_UART_BAUDRATE_230400)
- 460800 (CONFIG_UART_BAUDRATE_460800)
- 921600 (CONFIG_UART_BAUDRATE_921600)

CONFIG_BT_NIMBLE_USE_HCI_UART_PARITY

Uart PARITY

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport > CONFIG_BT_NIMBLE_TRANSPORT_UART

Uart Parity

Available options:

- None (CONFIG_UART_PARITY_NONE)
- Odd (CONFIG_UART_PARITY_ODD)
- Even (CONFIG_UART_PARITY_EVEN)

CONFIG_BT_NIMBLE_UART_RX_PIN

UART Rx pin

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport > CONFIG_BT_NIMBLE_TRANSPORT_UART

Rx pin for Nimble Transport

Default value:

- 5 if CONFIG_BT_CONTROLLER_DISABLED && CONFIG_BT_NIMBLE_TRANSPORT_UART && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_UART_TX_PIN

UART Tx pin

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport > CONFIG_BT_NIMBLE_TRANSPORT_UART

Tx pin for Nimble Transport

Default value:

- 4 if CONFIG_BT_CONTROLLER_DISABLED && CONFIG_BT_NIMBLE_TRANSPORT_UART && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_USE_HCI_UART_FLOW_CTRL

Uart Flow Control

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport

Uart Flow Control

Available options:

- Disable (CONFIG_UART_HW_FLOWCTRL_DISABLE)
- Enable hardware flow control (CONFIG_UART_HW_FLOWCTRL_CTS_RTS)

CONFIG_BT_NIMBLE_HCI_UART_RTS_PIN

UART Rts Pin

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport

UART HCI RTS pin

Default value:

- 19 if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HCI_UART_CTS_PIN

UART Cts Pin

Found in: Component config > Bluetooth > NimBLE Options > Host-controller Transport

UART HCI CTS pin

Default value:

- 23 if CONFIG_BT_NIMBLE_ENABLED

Debugging/Testing

Contains:

- CONFIG_BT_NIMBLE_DEBUG
- CONFIG_BT_NIMBLE_PRINT_ERR_NAME
- CONFIG_BT_NIMBLE_LOG_LEVEL
- CONFIG_BT_NIMBLE_MEM_DEBUG
- CONFIG_BT_NIMBLE_TEST_THROUGHPUT_TEST

CONFIG_BT_NIMBLE_MEM_DEBUG

NimBLE memory debug

Found in: Component config > Bluetooth > NimBLE Options > Debugging/Testing

Enable NimBLE memory debug to track memory allocations and deallocations. This feature records the function name, line number, and size of each allocation, and provides APIs to display memory usage statistics, find memory leaks, and monitor peak memory usage.

CONFIG_BT_NIMBLE_LOG_LEVEL

NimBLE Host log verbosity

Found in: Component config > Bluetooth > NimBLE Options > Debugging/Testing

Select NimBLE log level. Note that the selected NimBLE log verbosity can not exceed the level set in "Component config --> Log output --> Default log verbosity".

Available options:

- No logs (CONFIG_BT_NIMBLE_LOG_LEVEL_NONE)
- Error logs (CONFIG_BT_NIMBLE_LOG_LEVEL_ERROR)
- Warning logs (CONFIG_BT_NIMBLE_LOG_LEVEL_WARNING)
- Info logs (CONFIG_BT_NIMBLE_LOG_LEVEL_INFO)
- Debug logs (CONFIG_BT_NIMBLE_LOG_LEVEL_DEBUG)

CONFIG_BT_NIMBLE_PRINT_ERR_NAME

Enable feature to print Error description

Found in: Component config > Bluetooth > NimBLE Options > Debugging/Testing

Enable feature to give useful explanation for HCI errors

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_DEBUG

Enable extra runtime asserts and host debugging

Found in: Component config > Bluetooth > NimBLE Options > Debugging/Testing

This enables extra runtime asserts and host debugging

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_TEST_THROUGHPUT_TEST

Throughput Test Mode enable

Found in: Component config > Bluetooth > NimBLE Options > Debugging/Testing

Enable the throughput test mode

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED

Vendor / Optimization

Contains:

- CONFIG_BT_NIMBLE_ADV_SEND_CONSTANT_DID
- CONFIG_BT_NIMBLE_SCAN_ALLOW_ENH_ADI_FILTER
- CONFIG_BT_NIMBLE_HIGH_DUTY_ADV_ITVL
- CONFIG_BT_NIMBLE_VS_SUPPORT
- CONFIG_BT_NIMBLE_OPTIMIZE_MULTI_CONN

CONFIG_BT_NIMBLE_VS_SUPPORT

Enable support for VSC and VSE

Found in: Component config > Bluetooth > NimBLE Options > Vendor / Optimization

This option is used to enable support for sending Vendor Specific HCI commands and handling Vendor Specific HCI Events.

CONFIG_BT_NIMBLE_OPTIMIZE_MULTI_CONN

Enable the optimization of multi-connection

Found in: Component config > Bluetooth > NimBLE Options > Vendor / Optimization

This option enables the use of vendor-specific APIs for multi-connections, which can greatly enhance the stability of coexistence between numerous central and peripheral devices. It will prohibit the usage of standard APIs.

Default value:

- No (disabled) if CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_HIGH_DUTY_ADV_ITVL

Enable BLE high duty advertising interval feature

Found in: Component config > Bluetooth > NimBLE Options > Vendor / Optimization

This enable BLE high duty advertising interval feature

CONFIG_BT_NIMBLE_ADV_SEND_CONSTANT_DID

Enable BLE Ext Adv to Send with Specific DID

Found in: Component config > Bluetooth > NimBLE Options > Vendor / Optimization

This enable vendor-specific APIs to send specific DID for different advertising data in extended advertising.

CONFIG_BT_NIMBLE_SCAN_ALLOW_ENH_ADI_FILTER

Enable BLE Ext Scan to Receive Packet with Specific ADI

Found in: Component config > Bluetooth > NimBLE Options > Vendor / Optimization

This enables vendor-specific APIs to only receive specific groups of ADIs, and implements DIDs duplicate filtering for extended scanning. Enabling this can reduce the unnecessary reception of secondary channel PDUs carrying non-targeted ADIs.

Helper Utils

Contains:

- CONFIG_BT_NIMBLE_EXTRA_ADV_FIELDS
- CONFIG_BT_NIMBLE_UTIL_API
- CONFIG_BT_NIMBLE_CHK_HOST_STATUS

CONFIG_BT_NIMBLE_CHK_HOST_STATUS

Enable host status query support

Found in: Component config > Bluetooth > NimBLE Options > Helper Utils

Enable support to query host status

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_UTIL_API

Enable helper APIs

Found in: Component config > Bluetooth > NimBLE Options > Helper Utils

Enable support for misc APIs like set data addr change, read local resolv addr and set host feature

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_NIMBLE_EXTRA_ADV_FIELDS

Enable extra Adv report fields parsing

Found in: Component config > Bluetooth > NimBLE Options > Helper Utils

Enable support for parsing of more adv fields as per spec

Default value:

- Yes (enabled) if CONFIG_BT_NIMBLE_ENABLED && CONFIG_BT_NIMBLE_ENABLED

Controller Options

Contains:

- CONFIG_BTDM_CTRL_AUTO_LATENCY
- CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP
- BLE disconnects when Instant Passed (0x28) occurs
- CONFIG_BTDM_CTRL_FULL_SCAN_SUPPORTED
- CONFIG_BTDM_CTRL_BLE_MAX_CONN
- CONFIG_BTDM_BLE_SCAN_DUPL
- CONFIG_BTDM_BLE_SLEEP_CLOCK_ACCURACY
- CONFIG_BTDM_CTRL_MODE
- CONFIG_BTDM_CTRL_BR_EDR_MAX_ACL_CONN
- CONFIG_BTDM_CTRL_BR_EDR_MIN_ENC_KEY_SZ_DFT
- CONFIG_BTDM_CTRL_BR_EDR_SCO_DATA_PATH
- CONFIG_BTDM_CTRL_BR_EDR_MAX_SYNC_CONN
- CONFIG_BTDM_CTRL_SCAN_BACKOFF_UPPERLIMITMAX
- CONFIG_BTDM_CTRL_CONTROLLER_DEBUG_MODE_1
- CONFIG_BTDM_BLE_CHAN_ASS_EN

- CONFIG_BTDM_CTRL_CHECK_CONNECT_IND_ACCESS_ADDRESS
- CONFIG_BTDM_BLE_PING_EN
- CONFIG_BTDM_CTRL_HCI_MODE_CHOICE
- HCI UART(H4) Options
- CONFIG_BTDM_CTRL_HLI
- CONFIG_BTDM_CTRL_LEGACY_AUTH_VENDOR_EVT
- MODEM SLEEP Options
- CONFIG_BTDM_CTRL_PCM_ROLE_EDGE_CONFIG
- CONFIG_BTDM_CTRL_PINNED_TO_CORE_CHOICE

CONFIG_BTDM_CTRL_MODE

Bluetooth controller mode (BR/EDR/BLE/DUALMODE)

Found in: Component config > Bluetooth > Controller Options

Specify the bluetooth controller mode (BR/EDR, BLE or dual mode).

Available options:

- BLE Only (CONFIG_BTDM_CTRL_MODE_BLE_ONLY)
- BR/EDR Only (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY)
- Bluetooth Dual Mode (CONFIG_BTDM_CTRL_MODE_BTDM)

CONFIG_BTDM_CTRL_BLE_MAX_CONN

BLE Max Connections

Found in: Component config > Bluetooth > Controller Options

BLE maximum connections of bluetooth controller. Each connection uses 1KB static DRAM whenever the BT controller is enabled.

Range:

- from 1 to 9 if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 3 if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_BR_EDR_MIN_ENC_KEY_SZ_DFT

BR/EDR default minimum size of encryption key

Found in: Component config > Bluetooth > Controller Options

BR/EDR default minimum size of encryption key when start encryption.

Range:

- from 7 to 16 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 7 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_BR_EDR_MAX_ACL_CONN

BR/EDR ACL Max Connections

Found in: Component config > Bluetooth > Controller Options

BR/EDR ACL maximum connections of bluetooth controller. Each connection uses 1.2 KB DRAM whenever the BT controller is enabled.

Range:

- from 1 to 7 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 2 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_BR_EDR_MAX_SYNC_CONN

BR/EDR Sync(SCO/eSCO) Max Connections

Found in: Component config > Bluetooth > Controller Options

BR/EDR Synchronize maximum connections of bluetooth controller. Each connection uses 2 KB DRAM whenever the BT controller is enabled.

Range:

- from 0 to 3 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 0 if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_BR_EDR_SCO_DATA_PATH

BR/EDR Sync(SCO/eSCO) default data path

Found in: Component config > Bluetooth > Controller Options

SCO data path, i.e. HCI or PCM. SCO data can be sent/received through HCI synchronous packets, or the data can be routed to on-chip PCM module on ESP32. PCM input/output signals can be "matrixed" to GPIOs. The default data path can also be set using API "esp_bredr_sco_datapath_set"

Available options:

- HCI (CONFIG_BTDM_CTRL_BR_EDR_SCO_DATA_PATH_HCI)
- PCM (CONFIG_BTDM_CTRL_BR_EDR_SCO_DATA_PATH_PCM)

CONFIG_BTDM_CTRL_PCM_ROLE_EDGE_CONFIG

PCM Signal Configurations: Role, Polar and Channel Mode(Stereo/Mono)

Found in: Component config > Bluetooth > Controller Options

Default value:

- Yes (enabled) if CONFIG_BTDM_CTRL_BR_EDR_SCO_DATA_PATH_PCM && CONFIG_BT_CONTROLLER_ENABLED

Contains:

- CONFIG_BTDM_CTRL_PCM_FSYNCSHP
- CONFIG_BTDM_CTRL_PCM_POLAR
- CONFIG_BTDM_CTRL_PCM_ROLE

CONFIG_BTDM_CTRL_PCM_ROLE

PCM Role

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_CTRL_PCM_ROLE_EDGE_CONFIG

PCM role can be configured as PCM master or PCM slave

Available options:

- PCM Master (CONFIG_BTDM_CTRL_PCM_ROLE_MASTER)
- PCM Slave (CONFIG_BTDM_CTRL_PCM_ROLE_SLAVE)

CONFIG_BTDM_CTRL_PCM_POLAR

PCM Polar

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_CTRL_PCM_ROLE_EDGE_CONFIG

PCM polarity can be configured as Falling Edge or Rising Edge

Available options:

- Falling Edge (CONFIG_BTDM_CTRL_PCM_POLAR_FALLING_EDGE)
- Rising Edge (CONFIG_BTDM_CTRL_PCM_POLAR_RISING_EDGE)

CONFIG_BTDM_CTRL_PCM_FSYNCSHP

Channel Mode(Stereo/Mono)

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_CTRL_PCM_ROLE_EDGE_CONFIG

PCM frame synchronization signal shape can be configured as Stereo Mode or Mono Mode. (There are detailed instructions under the path examples/bluetooth/bluedroid/classic_bt/hfp_ag/README.md)

Available options:

- Stereo Mode (CONFIG_BTDM_CTRL_PCM_FSYNCSHP_STEREO_MODE)

  Stereo Mode(Dual channel): FSYNC and DOUT signals both change simultaneously on the edge of CLK. The FSYNC signal continues until the end of the current channel-data transmission. (There is a waveform graph under the path examples/bluetooth/bluedroid/classic_bt/hfp_ag/image)

- Mono Mode 1 (CONFIG_BTDM_CTRL_PCM_FSYNCSHP_MONO_MODE_LF)

  Mono Mode 1(Single channel): FSYNC signal starts to change a CLK clock cycle earlier than the DOUT signal. The FSYNC signal continues for one extra CLK clock cycle. (There is a waveform graph under the path examples/bluetooth/bluedroid/classic_bt/hfp_ag/image)

- Mono Mode 2 (CONFIG_BTDM_CTRL_PCM_FSYNCSHP_MONO_MODE_FF)

  Mono Mode 2(Single channel): FSYNC and DOUT signals both change simultaneously on the edge of CLK. The FSYNC signal continues for one extra CLK clock cycle. (There is a waveform graph under the path examples/bluetooth/bluedroid/classic_bt/hfp_ag/image)

CONFIG_BTDM_CTRL_AUTO_LATENCY

Auto latency

Found in: Component config > Bluetooth > Controller Options

BLE auto latency, used to enhance classic BT performance while classic BT and BLE are enabled at the same time.

Default value:

- No (disabled) if CONFIG_BTDM_CTRL_MODE_BTDM && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_LEGACY_AUTH_VENDOR_EVT

Legacy Authentication Vendor Specific Event Enable

Found in: Component config > Bluetooth > Controller Options

To protect from BIAS attack during Legacy authentication, Legacy authentication Vendor specific event should be enabled

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BR_EDR_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_PINNED_TO_CORE_CHOICE

The cpu core which bluetooth controller run

Found in: Component config > Bluetooth > Controller Options

Specify the cpu core to run bluetooth controller. Can not specify no-affinity.

Available options:

- Core 0 (PRO CPU) (CONFIG_BTDM_CTRL_PINNED_TO_CORE_0)
- Core 1 (APP CPU) (CONFIG_BTDM_CTRL_PINNED_TO_CORE_1)

CONFIG_BTDM_CTRL_HCI_MODE_CHOICE

HCI mode

Found in: Component config > Bluetooth > Controller Options

Specify HCI mode as VHCI or UART(H4)

Available options:

- VHCI (CONFIG_BTDM_CTRL_HCI_MODE_VHCI)

  Normal option. Mostly, choose this VHCI when bluetooth host run on ESP32, too.

- UART(H4) (CONFIG_BTDM_CTRL_HCI_MODE_UART_H4)

  If use external bluetooth host which run on other hardware and use UART as the HCI interface, choose this option.

HCI UART(H4) Options

Contains:

- CONFIG_BTDM_CTRL_HCI_UART_FLOW_CTRL_EN
- CONFIG_BTDM_CTRL_HCI_UART_BAUDRATE
- CONFIG_BTDM_CTRL_HCI_UART_NO

CONFIG_BTDM_CTRL_HCI_UART_NO

UART Number for HCI

Found in: Component config > Bluetooth > Controller Options > HCI UART(H4) Options

Uart number for HCI. The available uart is UART1 and UART2.

Range:

- from 1 to 2 if CONFIG_BTDM_CTRL_HCI_MODE_UART_H4 && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 1 if CONFIG_BTDM_CTRL_HCI_MODE_UART_H4 && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_HCI_UART_BAUDRATE

UART Baudrate for HCI

Found in: Component config > Bluetooth > Controller Options > HCI UART(H4) Options

UART Baudrate for HCI. Please use standard baudrate.

Range:

- from 115200 to 921600 if CONFIG_BTDM_CTRL_HCI_MODE_UART_H4 && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 921600 if CONFIG_BTDM_CTRL_HCI_MODE_UART_H4 && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_HCI_UART_FLOW_CTRL_EN

Enable UART flow control

Found in: Component config > Bluetooth > Controller Options > HCI UART(H4) Options

Default value:

- Yes (enabled) if CONFIG_BTDM_CTRL_HCI_MODE_UART_H4 && CONFIG_BT_CONTROLLER_ENABLED

MODEM SLEEP Options

Contains:

- CONFIG_BTDM_CTRL_LOW_POWER_CLOCK
- CONFIG_BTDM_CTRL_MODEM_SLEEP

CONFIG_BTDM_CTRL_MODEM_SLEEP

Bluetooth modem sleep

Found in: Component config > Bluetooth > Controller Options > MODEM SLEEP Options

Enable/disable bluetooth controller low power mode.

Default value:

- Yes (enabled) if CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_MODEM_SLEEP_MODE

Bluetooth Modem sleep mode

Found in: Component config > Bluetooth > Controller Options > MODEM SLEEP Options > CONFIG_BTDM_CTRL_MODEM_SLEEP

To select which strategy to use for modem sleep

Available options:

- ORIG Mode(sleep with low power clock) (CONFIG_BTDM_CTRL_MODEM_SLEEP_MODE_ORIG)

  ORIG mode is a bluetooth sleep mode that can be used for dual mode controller. In this mode, bluetooth controller sleeps between BR/EDR frames and BLE events. A low power clock is used to maintain bluetooth reference clock.

- EVED Mode(For internal test only) (CONFIG_BTDM_CTRL_MODEM_SLEEP_MODE_EVED)

  EVED mode is for BLE only and is only for internal test. Do not use it for production. this mode is not compatible with DFS nor light sleep

CONFIG_BTDM_CTRL_LOW_POWER_CLOCK

Bluetooth low power clock

Found in: Component config > Bluetooth > Controller Options > MODEM SLEEP Options

Select the low power clock source for bluetooth controller. Bluetooth low power clock is the clock source to maintain time in sleep mode.

- "Main crystal" option provides good accuracy and can support Dynamic Frequency Scaling to be used with Bluetooth modem sleep. Light sleep is not supported.
- "External 32kHz crystal" option allows user to use a 32.768kHz crystal as Bluetooth low power clock. This option is allowed as long as External 32kHz crystal is configured as the system RTC clock source. This option provides good accuracy and supports Bluetooth modem sleep to be used alongside Dynamic Frequency Scaling or light sleep.

Available options:

- Main crystal (CONFIG_BTDM_CTRL_LPCLK_SEL_MAIN_XTAL)

  Main crystal can be used as low power clock for bluetooth modem sleep. If this option is selected, bluetooth modem sleep can work under Dynamic Frequency Scaling(DFS) enabled, but cannot work when light sleep is enabled. Main crystal has a good performance in accuracy as the bluetooth low power clock source.

- External 32kHz crystal/oscillator (CONFIG_BTDM_CTRL_LPCLK_SEL_EXT_32K_XTAL)

  External 32kHz crystal/oscillator has a nominal frequency of 32.768kHz and provides good frequency stability. If used as Bluetooth low power clock, External 32kHz can support Bluetooth modem sleep to be used with both DFS and light sleep.

CONFIG_BTDM_BLE_SLEEP_CLOCK_ACCURACY

BLE Sleep Clock Accuracy

Found in: Component config > Bluetooth > Controller Options

BLE Sleep Clock Accuracy(SCA) for the local device is used to estimate window widening in BLE connection events. With a lower level of clock accuracy(e.g. 500ppm over 250ppm), the slave needs a larger RX window to synchronize with master in each anchor point, thus resulting in an increase of power consumption but a higher level of robustness in keeping connected. According to the requirements of Bluetooth Core specification 4.2, the worst-case accuracy of Classic Bluetooth Low Power Oscillator (LPO) is +/-250ppm in STANDBY and in low power modes such as sniff. For BLE the worst-case SCA is +/-500ppm.

- "151ppm to 250ppm" option is the default value for Bluetooth Dual mode
- **"251ppm to 500ppm" option can be used in BLE only mode when using external 32kHz crystal as**

  low power clock. This option is provided in case that BLE sleep clock has a lower level of accuracy, or other error sources contribute to the inaccurate timing during sleep.

Available options:

- 251ppm to 500ppm (CONFIG_BTDM_BLE_DEFAULT_SCA_500PPM)
- 151ppm to 250ppm (CONFIG_BTDM_BLE_DEFAULT_SCA_250PPM)

CONFIG_BTDM_BLE_SCAN_DUPL

BLE Scan Duplicate Options

Found in: Component config > Bluetooth > Controller Options

This select enables parameters setting of BLE scan duplicate.

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BTDM || CONFIG_BTDM_CTRL_MODE_BLE_ONLY) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_SCAN_DUPL_TYPE

Scan Duplicate Type

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_SCAN_DUPL

Scan duplicate have three ways. one is "Scan Duplicate By Device Address", This way is to use advertiser address filtering. The adv packet of the same address is only allowed to be reported once. Another way is "Scan Duplicate By Device Address And Advertising Data". This way is to use advertising data and device address filtering. All different adv packets with the same address are allowed to be reported. The last way is "Scan Duplicate By Advertising Data". This way is to use advertising data filtering. All same advertising data only allow to be reported once even though they are from different devices.

Available options:

- Scan Duplicate By Device Address (CONFIG_BTDM_SCAN_DUPL_TYPE_DEVICE)

  This way is to use advertiser address filtering. The adv packet of the same address is only allowed to be reported once

- Scan Duplicate By Advertising Data (CONFIG_BTDM_SCAN_DUPL_TYPE_DATA)

  This way is to use advertising data filtering. All same advertising data only allow to be reported once even though they are from different devices.

- Scan Duplicate By Device Address And Advertising Data (CONFIG_BTDM_SCAN_DUPL_TYPE_DATA_DEVICE)

  This way is to use advertising data and device address filtering. All different adv packets with the same address are allowed to be reported.

CONFIG_BTDM_SCAN_DUPL_CACHE_SIZE

Maximum number of devices in scan duplicate filter

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_SCAN_DUPL

Maximum number of devices which can be recorded in scan duplicate filter. When the maximum amount of device in the filter is reached, the oldest device will be refreshed.

Range:

- from 10 to 1000 if CONFIG_BTDM_BLE_SCAN_DUPL && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 100 if CONFIG_BTDM_BLE_SCAN_DUPL && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_SCAN_DUPL_CACHE_REFRESH_PERIOD

Duplicate scan list refresh period (seconds)

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_SCAN_DUPL

If the period value is non-zero, the controller will periodically clear the device information stored in the scan duuplicate filter. If it is 0, the scan duuplicate filter will not be cleared until the scanning is disabled. Duplicate advertisements for this period should not be sent to the Host in advertising report events. There are two scenarios where the ADV packet will be repeatedly reported: 1. The duplicate scan cache is full, the controller will delete the oldest device information and add new device information. 2. When the refresh period is up, the controller will clear all device information and start filtering again.

Range:

- from 0 to 1000 if CONFIG_BTDM_BLE_SCAN_DUPL && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 0 if CONFIG_BTDM_BLE_SCAN_DUPL && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_MESH_SCAN_DUPL_EN

Special duplicate scan mechanism for BLE Mesh scan

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_SCAN_DUPL

This enables the BLE scan duplicate for special BLE Mesh scan.

Default value:

- No (disabled) if CONFIG_BTDM_BLE_SCAN_DUPL && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_MESH_DUPL_SCAN_CACHE_SIZE

Maximum number of Mesh adv packets in scan duplicate filter

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_SCAN_DUPL > CONFIG_BTDM_BLE_MESH_SCAN_DUPL_EN

Maximum number of adv packets which can be recorded in duplicate scan cache for BLE Mesh. When the maximum amount of device in the filter is reached, the cache will be refreshed.

Range:

- from 10 to 1000 if CONFIG_BTDM_BLE_MESH_SCAN_DUPL_EN && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 100 if CONFIG_BTDM_BLE_MESH_SCAN_DUPL_EN && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_FULL_SCAN_SUPPORTED

BLE full scan feature supported

Found in: Component config > Bluetooth > Controller Options

The full scan function is mainly used to provide BLE scan performance. This is required for scenes with high scan performance requirements, such as BLE Mesh scenes.

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_SCAN_BACKOFF_UPPERLIMITMAX

Disable active scan backoff

Found in: Component config > Bluetooth > Controller Options

Disable active scan backoff. The bluetooth spec requires that scanners should run a backoff procedure to minimize collision of scan request PDUs from multiple scanners. If scan backoff is disabled, in active scanning, scan request PDU will be sent every time when HW receives scannable ADV PDU.

Default value:

- No (disabled) if CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_CHECK_CONNECT_IND_ACCESS_ADDRESS

Enable enhanced Access Address check in CONNECT_IND

Found in: Component config > Bluetooth > Controller Options

Enabling this option will add stricter verification of the Access Address in the CONNECT_IND PDU. This improves security by ensuring that only connection requests with valid Access Addresses are accepted. If disabled, only basic checks are applied, improving compatibility.

Default value:

- No (disabled) if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP

BLE adv report flow control supported

Found in: Component config > Bluetooth > Controller Options

The function is mainly used to enable flow control for advertising reports. When it is enabled, advertising reports will be discarded by the controller if the number of unprocessed advertising reports exceeds the size of BLE adv report flow control.

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BTDM || CONFIG_BTDM_CTRL_MODE_BLE_ONLY) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_NUM

BLE adv report flow control number

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP

The number of unprocessed advertising report that Bluedroid can save.If you set *BTDM_BLE_ADV_REPORT_FLOW_CTRL_NUM* to a small value, this may cause adv packets lost. If you set *BTDM_BLE_ADV_REPORT_FLOW_CTRL_NUM* to a large value, Bluedroid may cache a lot of adv packets and this may cause system memory run out. For example, if you set it to 50, the maximum memory consumed by host is 35 * 50 bytes. Please set *BTDM_BLE_ADV_REPORT_FLOW_CTRL_NUM* according to your system free memory and handle adv packets as fast as possible, otherwise it will cause adv packets lost.

Range:

- from 50 to 1000 if CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 100 if CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_ADV_REPORT_DISCARD_THRSHOLD

BLE adv lost event threshold value

Found in: Component config > Bluetooth > Controller Options > CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP

When adv report flow control is enabled, The ADV lost event will be generated when the number of ADV packets lost in the controller reaches this threshold. It is better to set a larger value. If you set *BTDM_BLE_ADV_REPORT_DISCARD_THRSHOLD* to a small value or printf every adv lost event, it may cause adv packets lost more.

Range:

- from 1 to 1000 if CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP && CONFIG_BT_CONTROLLER_ENABLED

Default value:

- 20 if CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP && CONFIG_BT_CONTROLLER_ENABLED

BLE disconnects when Instant Passed (0x28) occurs

Contains:

- CONFIG_BTDM_BLE_LLCP_CHAN_MAP_UPDATE
- CONFIG_BTDM_BLE_LLCP_CONN_UPDATE

CONFIG_BTDM_BLE_LLCP_CONN_UPDATE

BLE ACL connection update procedure

Found in: Component config > Bluetooth > Controller Options > BLE disconnects when Instant Passed (0x28) occurs

If this option is enabled, Controller will terminate the connection when Instant Passed (0x28) error occurs during connection update procedure.

Default value:

- No (disabled) if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_LLCP_CHAN_MAP_UPDATE

BLE ACL channel map update procedure

Found in: Component config > Bluetooth > Controller Options > BLE disconnects when Instant Passed (0x28) occurs

If this option is enabled, Controller will terminate the connection when Instant Passed (0x28) error occurs in channel map update procedure.

Default value:

- No (disabled) if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_CHAN_ASS_EN

Enable channel assessment

Found in: Component config > Bluetooth > Controller Options

If this option is enabled, The Controller will records the communication quality for each channel and then start a timer to check and update the channel map every 4 seconds.

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BLE_ONLY || CONFIG_BTDM_CTRL_MODE_BTDM) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_BLE_PING_EN

Enable LE Ping procedure

Found in: Component config > Bluetooth > Controller Options

If this option is disabled, The Controller will not start the LE authenticated payload timer. This option is used for some compatibility problems related to LE ping procedure.

Default value:

- Yes (enabled) if (CONFIG_BTDM_CTRL_MODE_BTDM || CONFIG_BTDM_CTRL_MODE_BLE_ONLY) && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_CONTROLLER_DEBUG_MODE_1

Enable Bluetooth controller debugging mode 1 (for internal use only)

Found in: Component config > Bluetooth > Controller Options

Enables specific debugging features for the Bluetooth controller. This option is strictly for internal debugging purposes and should not be enabled in production environments, as it may impact performance and stability.

Default value:

- No (disabled) if CONFIG_BT_ENABLED && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BTDM_CTRL_HLI

High level interrupt

Found in: Component config > Bluetooth > Controller Options

Using Level 4 interrupt for Bluetooth.

Default value:

- Yes (enabled) if CONFIG_BT_ENABLED && CONFIG_BT_CONTROLLER_ENABLED

CONFIG_BT_RELEASE_IRAM

Release Bluetooth text (READ DOCS FIRST)

Found in: Component config > Bluetooth

This option release Bluetooth text section and merge Bluetooth data, bss & text into a large free heap region when esp_bt_mem_release is called, total saving ~21kB or more of IRAM. ESP32-C2 only 3 configurable PMP entries available, rest of them are hard-coded. We cannot split the memory into 3 different regions (IRAM, BLE-IRAM, DRAM). So this option will disable the memory protection scheme (ESP_SYSTEM_MEMPROT)

Default value:

- No (disabled) if CONFIG_BT_ENABLED && BT_LE_RELEASE_IRAM_SUPPORTED

CONFIG_BT_HCI_LOG_DEBUG_EN

Enable Bluetooth HCI debug mode

Found in: Component config > Bluetooth

This option is used to enable bluetooth debug mode, which saves the hci layer data stream.

Default value:

- No (disabled) if CONFIG_BT_BLUEDROID_ENABLED || CONFIG_BT_NIMBLE_ENABLED

CONFIG_BT_HCI_LOG_DATA_BUFFER_SIZE

Size of the cache used for HCI data in Bluetooth HCI debug mode (N*1024 bytes)

Found in: Component config > Bluetooth > CONFIG_BT_HCI_LOG_DEBUG_EN

This option is to configure the buffer size of the hci data steam cache in hci debug mode. This is a ring buffer, the new data will overwrite the oldest data if the buffer is full.

Range:

- from 1 to 100 if CONFIG_BT_HCI_LOG_DEBUG_EN

Default value:

- 5 if CONFIG_BT_HCI_LOG_DEBUG_EN

CONFIG_BT_HCI_LOG_ADV_BUFFER_SIZE

Size of the cache used for adv report in Bluetooth HCI debug mode (N*1024 bytes)

Found in: Component config > Bluetooth > CONFIG_BT_HCI_LOG_DEBUG_EN

This option is to configure the buffer size of the hci adv report cache in hci debug mode. This is a ring buffer, the new data will overwrite the oldest data if the buffer is full.

Range:

- from 1 to 100 if CONFIG_BT_HCI_LOG_DEBUG_EN

Default value:

- 8 if CONFIG_BT_HCI_LOG_DEBUG_EN

CONFIG_BLE_MESH

ESP BLE Mesh Support

Found in: Component config

This option enables ESP BLE Mesh support. The specific features that are available may depend on other features that have been enabled in the stack, such as Bluetooth Support, Bluedroid Support & GATT support.

Contains:

- BLE Mesh and BLE coexistence support
- CONFIG_BLE_MESH_GATT_PROXY_CLIENT
- CONFIG_BLE_MESH_GATT_PROXY_SERVER
- BLE Mesh NET BUF DEBUG LOG LEVEL
- CONFIG_BLE_MESH_PROV
- CONFIG_BLE_MESH_PROXY
- BLE Mesh specific test option
- BLE Mesh STACK DEBUG LOG LEVEL
- CONFIG_BLE_MESH_NO_LOG
- CONFIG_BLE_MESH_IVU_DIVIDER
- CONFIG_BLE_MESH_FAST_PROV
- CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC
- CONFIG_BLE_MESH_EXPERIMENTAL
- CONFIG_BLE_MESH_CRPL
- CONFIG_BLE_MESH_RX_SDU_MAX
- CONFIG_BLE_MESH_MODEL_KEY_COUNT
- CONFIG_BLE_MESH_APP_KEY_COUNT
- CONFIG_BLE_MESH_MODEL_GROUP_COUNT
- CONFIG_BLE_MESH_LABEL_COUNT
- CONFIG_BLE_MESH_SUBNET_COUNT
- CONFIG_BLE_MESH_TX_SEG_MAX
- CONFIG_BLE_MESH_RX_SEG_MSG_COUNT
- CONFIG_BLE_MESH_TX_SEG_MSG_COUNT
- CONFIG_BLE_MESH_MEM_ALLOC_MODE
- CONFIG_BLE_MESH_MSG_CACHE_SIZE
- CONFIG_BLE_MESH_NOT_RELAY_REPLAY_MSG
- CONFIG_BLE_MESH_ADV_BUF_COUNT
- CONFIG_BLE_MESH_PB_GATT
- CONFIG_BLE_MESH_PB_ADV
- CONFIG_BLE_MESH_IVU_RECOVERY_IVI
- CONFIG_BLE_MESH_RELAY
- CONFIG_BLE_MESH_SAR_ENHANCEMENT
- CONFIG_BLE_MESH_SETTINGS

- CONFIG_BLE_MESH_ACTIVE_SCAN
- CONFIG_BLE_MESH_DEINIT
- CONFIG_BLE_MESH_USE_DUPLICATE_SCAN
- CONFIG_BLE_MESH_V11_SUPPORT
- Support for BLE Mesh Client/Server models
- Support for BLE Mesh Foundation models
- CONFIG_BLE_MESH_NODE
- CONFIG_BLE_MESH_PROVISIONER
- CONFIG_BLE_MESH_FRIEND
- CONFIG_BLE_MESH_LOW_POWER
- CONFIG_BLE_MESH_PROXY_CLI_SRV_COEXIST
- CONFIG_BLE_MESH_HCI_5_0
- CONFIG_BLE_MESH_USE_BLE_50
- CONFIG_BLE_MESH_RANDOM_ADV_INTERVAL
- CONFIG_BLE_MESH_IV_UPDATE_TEST
- CONFIG_BLE_MESH_CLIENT_MSG_TIMEOUT
- CONFIG_BLE_MESH_USE_UNIFIED_CRYPTO

CONFIG_BLE_MESH_HCI_5_0

Support sending 20ms non-connectable adv packets

Found in: Component config > CONFIG_BLE_MESH

It is a temporary solution and needs further modifications.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_V11_SUPPORT

Support ESP BLE Mesh v1.1 features (Preview)

Found in: Component config > CONFIG_BLE_MESH

Support BLE Mesh v1.1 features

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_RANDOM_ADV_INTERVAL

Support using random adv interval for mesh packets

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow using random advertising interval for mesh packets. And this could help avoid collision of advertising packets.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_USE_UNIFIED_CRYPTO

Use the unified BLE tinycrypt implementation(Deprecated)

Found in: Component config > CONFIG_BLE_MESH

This configuration path is deprecated and will be removed in a future version. Please use the corresponding Kconfig options under $IDF_PATH/components/bt/common/Kconfig.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_USE_BLE_50

Support using BLE 5.0 APIs for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH

This option to enable BLE Mesh using some BLE 5.0 APIs.

Default value:

- No (disabled) if CONFIG_BLE_MESH_EXPERIMENTAL && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_EXT_ADV
- CONFIG_BLE_MESH_ADV_INST_ID
- CONFIG_BLE_MESH_SUPPORT_MULTI_ADV

CONFIG_BLE_MESH_EXT_ADV

Enable extended advertising for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50

Enable broadcasting of BLE Mesh messages using BLE 5.0 extended advertising. This allows control of extended advertising parameters (e.g., PHY selection) while maintaining standard BLE Mesh packet length.

Default value:

- No (disabled) if CONFIG_BLE_MESH_USE_BLE_50 && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_LONG_PACKET
- CONFIG_BLE_MESH_EXT_ADV_BUF_COUNT
- CONFIG_BLE_MESH_EXT_RELAY_ADV_BUF_COUNT

CONFIG_BLE_MESH_EXT_ADV_BUF_COUNT

Number of extended advertising buffers

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV

Number of buffer slots allocated for extended advertising packets.

Range:

- from 6 to 256 if CONFIG_BLE_MESH_EXT_ADV && CONFIG_BLE_MESH

Default value:

- 60 if CONFIG_BLE_MESH_EXT_ADV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_EXT_RELAY_ADV_BUF_COUNT

Number of extended relay advertising buffers

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV

Number of buffer slots allocated for extended advertising packets used in message relay.

Range:

- from 0 to 256 if CONFIG_BLE_MESH_EXT_ADV && CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

Default value:

- 60 if CONFIG_BLE_MESH_EXT_ADV && CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LONG_PACKET

Enable non-standard long packet mode for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV

Enable extended-length advertising packets for BLE Mesh using BLE 5.0 extended advertising. This overrides the standard BLE Mesh packet length limitations.

Default value:

- No (disabled) if CONFIG_BLE_MESH_EXT_ADV && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_LONG_PACKET_ADV_BUF_COUNT
- CONFIG_BLE_MESH_LONG_PACKET_RELAY_ADV_BUF_COUNT
- CONFIG_BLE_MESH_LONG_PACKET_ADV_LEN
- CONFIG_BLE_MESH_LONG_PACKET_RX_SEG_CNT
- CONFIG_BLE_MESH_LONG_PACKET_TX_SEG_CNT

CONFIG_BLE_MESH_LONG_PACKET_ADV_LEN

Maximum advertising payload length

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV > CONFIG_BLE_MESH_LONG_PACKET

Maximum payload length for extended advertising packets (bytes). Range: 30-249 bytes. Default: 105. Note: Maximum access payload = (LENGTH - 17) × SEGMENT_COUNT - Transmission: Uses BLE_MESH_LONG_PACKET_TX_SEG_CNT - Reception: Uses BLE_MESH_LONG_PACKET_RX_SEG_CNT

Range:

- from 30 to 249 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

Default value:

- 105 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LONG_PACKET_ADV_BUF_COUNT

Long packet advertising buffer count

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV > CONFIG_BLE_MESH_LONG_PACKET

Number of advertising buffers allocated for long packet transmissions.

Default value:

- 20 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LONG_PACKET_RELAY_ADV_BUF_COUNT

Long packet relay buffer count

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV > CONFIG_BLE_MESH_LONG_PACKET

Number of advertising buffers allocated for relay long packets.

Default value:

- 20 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LONG_PACKET_TX_SEG_CNT

Maximum transmission segments per message

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV > CONFIG_BLE_MESH_LONG_PACKET

Maximum number of segments for outgoing long packet messages. Upper bound: min(BLE_MESH_TX_SEG_MAX, BLE_MESH_LONG_PACKET_ADV_BUF_COUNT)

Range:

- from 1 to if CONFIG_BLE_MESH_TX_SEG_MAX < CONFIG_BLE_MESH_LONG_PACKET_ADV_BUF_COUNT && CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH
- from 1 to if CONFIG_BLE_MESH_TX_SEG_MAX >= CONFIG_BLE_MESH_LONG_PACKET_ADV_BUF_COUNT && CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

Default value:

- 20 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LONG_PACKET_RX_SEG_CNT

Maximum reception segments per message

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_EXT_ADV > CONFIG_BLE_MESH_LONG_PACKET

Maximum number of segments supported for receiving long packet messages.

Range:

- from 1 to if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

Default value:

- 20 if CONFIG_BLE_MESH_LONG_PACKET && CONFIG_BLE_MESH

CONFIG_BLE_MESH_ADV_INST_ID

Extended adv instance for Mesh normal packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50

Extended ADV instance used by Mesh normal advertising packets.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_USE_BLE_50 && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_USE_BLE_50 && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SUPPORT_MULTI_ADV

Support using multiple adv instance for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50

Enable this option to support using multiple adv instance while running BLE Mesh.

Default value:

- No (disabled) if CONFIG_BLE_MESH_USE_BLE_50 && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_PROXY_ADV_INST_ID
- CONFIG_BLE_MESH_SEPARATE_BLE_ADV_INSTANCE
- CONFIG_BLE_MESH_SEPARATE_RELAY_ADV_INSTANCE

CONFIG_BLE_MESH_PROXY_ADV_INST_ID

Extended adv instance for Mesh proxy packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_SUPPORT_MULTI_ADV

Extended ADV instance used by Mesh proxy advertising packets.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_PROXY && (CONFIG_BLE_MESH_PB_GATT || CONFIG_BLE_MESH_GATT_PROXY_SERVER) && CONFIG_BLE_MESH_SUPPORT_MULTI_ADV && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_PROXY && (CONFIG_BLE_MESH_PB_GATT || CONFIG_BLE_MESH_GATT_PROXY_SERVER) && CONFIG_BLE_MESH_SUPPORT_MULTI_ADV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SEPARATE_RELAY_ADV_INSTANCE

Use separate extended adv instance for Mesh relay packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_SUPPORT_MULTI_ADV

Enable this option to support using a separate extended ADV instance for Mesh relay packets.

Default value:

- No (disabled) if CONFIG_BLE_MESH_SUPPORT_MULTI_ADV && CONFIG_BLE_MESH_RELAY_ADV_BUF && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_RELAY_ADV_INST_ID

CONFIG_BLE_MESH_RELAY_ADV_INST_ID

Extended adv instance for Mesh relay packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_SUPPORT_MULTI_ADV > CONFIG_BLE_MESH_SEPARATE_RELAY_ADV_INSTANCE

Extended ADV instance used by Mesh relay advertising packets.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_SEPARATE_RELAY_ADV_INSTANCE && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_SEPARATE_RELAY_ADV_INSTANCE && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SEPARATE_BLE_ADV_INSTANCE

Use separate extended adv instance for BLE normal packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_SUPPORT_MULTI_ADV

Enable this option to support using a separate extended ADV instance for normal BLE advertising packets.

Default value:

- No (disabled) if CONFIG_BLE_MESH_SUPPORT_MULTI_ADV && CONFIG_BLE_MESH_SUPPORT_BLE_ADV && CONFIG_BLE_MESH

Contains:

- CONFIG_BLE_MESH_BLE_ADV_INST_ID

CONFIG_BLE_MESH_BLE_ADV_INST_ID

Extended adv instance for normal BLE packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_USE_BLE_50 > CONFIG_BLE_MESH_SUPPORT_MULTI_ADV > CONFIG_BLE_MESH_SEPARATE_BLE_ADV_INSTANCE

Extended ADV instance used by normal BLE advertising packets.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_SEPARATE_BLE_ADV_INSTANCE && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_SEPARATE_BLE_ADV_INSTANCE && CONFIG_BLE_MESH

CONFIG_BLE_MESH_USE_DUPLICATE_SCAN

Support Duplicate Scan in BLE Mesh

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow using specific duplicate scan filter in BLE Mesh, and Scan Duplicate Type must be set by choosing the option in the Bluetooth Controller section in menuconfig, which is "Scan Duplicate By Device Address and Advertising Data".

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_ACTIVE_SCAN

Support Active Scan in BLE Mesh

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow using BLE Active Scan for BLE Mesh.

CONFIG_BLE_MESH_MEM_ALLOC_MODE

Memory allocation strategy

Found in: Component config > CONFIG_BLE_MESH

Allocation strategy for BLE Mesh stack, essentially provides ability to allocate all required dynamic allocations from,

- Internal DRAM memory only
- External SPIRAM memory only
- Either internal or external memory based on default malloc() behavior in ESP-IDF
- Internal IRAM memory wherever applicable else internal DRAM

Recommended mode here is always internal (*), since that is most preferred from security perspective. But if application requirement does not allow sufficient free internal memory then alternate mode can be selected.

(*) In case of ESP32-S2/ESP32-S3, hardware allows encryption of external SPIRAM contents provided hardware flash encryption feature is enabled. In that case, using external SPIRAM allocation strategy is also safe choice from security perspective.

Available options:

- Internal DRAM (CONFIG_BLE_MESH_MEM_ALLOC_MODE_INTERNAL)
- External SPIRAM (CONFIG_BLE_MESH_MEM_ALLOC_MODE_EXTERNAL)
- Default alloc mode (CONFIG_BLE_MESH_MEM_ALLOC_MODE_DEFAULT)

  Enable this option to use the default memory allocation strategy when external SPIRAM is enabled. See the SPIRAM options for more details.

- Internal IRAM (CONFIG_BLE_MESH_MEM_ALLOC_MODE_IRAM_8BIT)

  Allows to use IRAM memory region as 8bit accessible region. Every unaligned (8bit or 16bit) access will result in an exception and incur penalty of certain clock cycles per unaligned read/write.

CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC

Enable FreeRTOS static allocation

Found in: Component config > CONFIG_BLE_MESH

Enable this option to use FreeRTOS static allocation APIs for BLE Mesh, which provides the ability to use different dynamic memory (i.e. SPIRAM or IRAM) for FreeRTOS objects. If this option is disabled, the FreeRTOS static allocation APIs will not be used, and internal DRAM will be allocated for FreeRTOS objects.

Default value:

- No (disabled) if (CONFIG_SPIRAM || CONFIG_ESP32_IRAM_AS_8BIT_ACCESSIBLE_MEMORY) && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC_MODE

Memory allocation for FreeRTOS objects

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC

Choose the memory to be used for FreeRTOS objects.

Available options:

- External SPIRAM (CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC_EXTERNAL)

  If enabled, BLE Mesh allocates dynamic memory from external SPIRAM for FreeRTOS objects, i.e. mutex, queue, and task stack. External SPIRAM can only be used for task stack when SPIRAM_ALLOW_STACK_EXTERNAL_MEMORY is enabled. See the SPIRAM options for more details.

- Internal IRAM (CONFIG_BLE_MESH_FREERTOS_STATIC_ALLOC_IRAM_8BIT)

  If enabled, BLE Mesh allocates dynamic memory from internal IRAM for FreeRTOS objects, i.e. mutex, queue. Note: IRAM region cannot be used as task stack.

CONFIG_BLE_MESH_DEINIT

Support de-initialize BLE Mesh stack

Found in: Component config > CONFIG_BLE_MESH

If enabled, users can use the function esp_ble_mesh_deinit() to de-initialize the whole BLE Mesh stack.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

BLE Mesh and BLE coexistence support

Contains:

- CONFIG_BLE_MESH_SUPPORT_BLE_SCAN
- CONFIG_BLE_MESH_SUPPORT_BLE_ADV

CONFIG_BLE_MESH_SUPPORT_BLE_ADV

Support sending normal BLE advertising packets

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh and BLE coexistence support

When selected, users can send normal BLE advertising packets with specific API.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_BLE_ADV_BUF_COUNT

Number of advertising buffers for BLE advertising packets

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh and BLE coexistence support > CONFIG_BLE_MESH_SUPPORT_BLE_ADV

Number of advertising buffers for BLE packets available.

Range:

- from 1 to 255 if CONFIG_BLE_MESH_SUPPORT_BLE_ADV && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_SUPPORT_BLE_ADV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SUPPORT_BLE_SCAN

Support scanning normal BLE advertising packets

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh and BLE coexistence support

When selected, users can register a callback and receive normal BLE advertising packets in the application layer.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_FAST_PROV

Enable BLE Mesh Fast Provisioning

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow BLE Mesh fast provisioning solution to be used. When there are multiple unprovisioned devices around, fast provisioning can greatly reduce the time consumption of the whole provisioning process. When this option is enabled, and after an unprovisioned device is provisioned into a node successfully, it can be changed to a temporary Provisioner.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_NODE

Support for BLE Mesh Node

Found in: Component config > CONFIG_BLE_MESH

Enable the device to be provisioned into a node. This option should be enabled when an unprovisioned device is going to be provisioned into a node and communicate with other nodes in the BLE Mesh network.

CONFIG_BLE_MESH_PROVISIONER

Support for BLE Mesh Provisioner

Found in: Component config > CONFIG_BLE_MESH

Enable the device to be a Provisioner. The option should be enabled when a device is going to act as a Provisioner and provision unprovisioned devices into the BLE Mesh network.

CONFIG_BLE_MESH_WAIT_FOR_PROV_MAX_DEV_NUM

Maximum number of unprovisioned devices that can be added to device queue

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many unprovisioned devices can be added to device queue for provisioning. Users can use this option to define the size of the queue in the bottom layer which is used to store unprovisioned device information (e.g. Device UUID, address).

Range:

- from 1 to 100 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_PROV_NODES

Maximum number of devices that can be provisioned by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many devices can be provisioned by a Provisioner. This value indicates the maximum number of unprovisioned devices which can be provisioned by a Provisioner. For instance, if the value is 6, it means the Provisioner can provision up to 6 unprovisioned devices. Theoretically a Provisioner without the limitation of its memory can provision up to 32766 unprovisioned devices, here we limit the maximum number to 100 just to limit the memory used by a Provisioner. The bigger the value is, the more memory it will cost by a Provisioner to store the information of nodes.

Range:

- from 1 to 1000 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PBA_SAME_TIME

Maximum number of PB-ADV running at the same time by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many devices can be provisioned at the same time using PB-ADV. For examples, if the value is 2, it means a Provisioner can provision two unprovisioned devices with PB-ADV at the same time.

Range:

- from 1 to 10 if CONFIG_BLE_MESH_PB_ADV && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_PB_ADV && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PBG_SAME_TIME

Maximum number of PB-GATT running at the same time by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many devices can be provisioned at the same time using PB-GATT. For example, if the value is 2, it means a Provisioner can provision two unprovisioned devices with PB-GATT at the same time.

Range:

- from 1 to 5 if CONFIG_BLE_MESH_PB_GATT && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_PB_GATT && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROVISIONER_SUBNET_COUNT

Maximum number of mesh subnets that can be created by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many subnets per network a Provisioner can create. Indeed, this value decides the number of network keys which can be added by a Provisioner.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROVISIONER_APP_KEY_COUNT

Maximum number of application keys that can be owned by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

This option specifies how many application keys the Provisioner can have. Indeed, this value decides the number of the application keys which can be added by a Provisioner.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROVISIONER_RECV_HB

Support receiving Heartbeat messages

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER

When this option is enabled, Provisioner can call specific functions to enable or disable receiving Heartbeat messages and notify them to the application layer.

Default value:

- No (disabled) if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROVISIONER_RECV_HB_FILTER_SIZE

Maximum number of filter entries for receiving Heartbeat messages

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROVISIONER > CONFIG_BLE_MESH_PROVISIONER_RECV_HB

This option specifies how many heartbeat filter entries Provisioner supports. The heartbeat filter (acceptlist or rejectlist) entries are used to store a list of SRC and DST which can be used to decide if a heartbeat message will be processed and notified to the application layer by Provisioner. Note: The filter is an empty rejectlist by default.

Range:

- from 1 to 1000 if CONFIG_BLE_MESH_PROVISIONER_RECV_HB && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_PROVISIONER_RECV_HB && CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROV

BLE Mesh Provisioning support

Found in: Component config > CONFIG_BLE_MESH

Enable this option to support BLE Mesh Provisioning functionality. For BLE Mesh, this option should be always enabled.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROV_EPA

BLE Mesh enhanced provisioning authentication

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROV

Enable this option to support BLE Mesh enhanced provisioning authentication functionality. This option can increase the security level of provisioning. It is recommended to enable this option.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_PROV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_CERT_BASED_PROV

Support Certificate-based provisioning

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROV

Enable this option to support BLE Mesh Certificate-Based Provisioning.

Default value:

- No (disabled) if CONFIG_BLE_MESH_PROV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RECORD_FRAG_MAX_SIZE

Maximum size of the provisioning record fragment that Provisioner can receive

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PROV > CONFIG_BLE_MESH_CERT_BASED_PROV

This option sets the maximum size of the provisioning record fragment that the Provisioner can receive. The range depends on provisioning bearer.

Range:

- from 1 to 57 if CONFIG_BLE_MESH_CERT_BASED_PROV && CONFIG_BLE_MESH

Default value:

- 56 if CONFIG_BLE_MESH_CERT_BASED_PROV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PB_ADV

Provisioning support using the advertising bearer (PB-ADV)

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow the device to be provisioned over the advertising bearer. This option should be enabled if PB-ADV is going to be used during provisioning procedure.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_UNPROVISIONED_BEACON_INTERVAL

Interval between two consecutive Unprovisioned Device Beacon

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_PB_ADV

This option specifies the interval of sending two consecutive unprovisioned device beacon, users can use this option to change the frequency of sending unprovisioned device beacon. For example, if the value is 5, it means the unprovisioned device beacon will send every 5 seconds. When the option of BLE_MESH_FAST_PROV is selected, the value is better to be 3 seconds, or less.

Range:

- from 1 to 100 if CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH_PB_ADV && CONFIG_BLE_MESH

Default value:

- 5 if CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH_PB_ADV && CONFIG_BLE_MESH
- 3 if CONFIG_BLE_MESH_FAST_PROV && CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH_PB_ADV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PB_GATT

Provisioning support using GATT (PB-GATT)

Found in: Component config > CONFIG_BLE_MESH

Enable this option to allow the device to be provisioned over GATT. This option should be enabled if PB-GATT is going to be used during provisioning procedure.

Virtual option enabled whenever any Proxy protocol is needed

CONFIG_BLE_MESH_PROXY

BLE Mesh Proxy protocol support

Found in: Component config > CONFIG_BLE_MESH

Enable this option to support BLE Mesh Proxy protocol used by PB-GATT and other proxy pdu transmission.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_GATT_PROXY_SERVER

BLE Mesh GATT Proxy Server

Found in: Component config > CONFIG_BLE_MESH

This option enables support for Mesh GATT Proxy Service, i.e. the ability to act as a proxy between a Mesh GATT Client and a Mesh network. This option should be enabled if a node is going to be a Proxy Server.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH

CONFIG_BLE_MESH_NODE_ID_TIMEOUT

Node Identity advertising timeout

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_SERVER

This option determines for how long the local node advertises using Node Identity. The given value is in seconds. The specification limits this to 60 seconds and lists it as the recommended value as well. So leaving the default value is the safest option. When an unprovisioned device is provisioned successfully and becomes a node, it will start to advertise using Node Identity during the time set by this option. And after that, Network ID will be advertised.

Range:

- from 1 to 60 if CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH

Default value:

- 60 if CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROXY_FILTER_SIZE

Maximum number of filter entries per Proxy Client

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_SERVER

This option specifies how many Proxy Filter entries the local node supports. The entries of Proxy filter (whitelist or blacklist) are used to store a list of addresses which can be used to decide which messages will be forwarded to the Proxy Client by the Proxy Server.

Range:

- from 1 to 32767 if CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH

Default value:

- 4 if CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROXY_PRIVACY

Support Proxy Privacy

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_SERVER

The Proxy Privacy parameter controls the privacy of the Proxy Server over the connection. The value of the Proxy Privacy parameter is controlled by the type of proxy connection, which is dependent on the bearer used by the proxy connection.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_PRB_SRV && CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROXY_SOLIC_PDU_RX

Support receiving Proxy Solicitation PDU

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_SERVER

Enable this option to support receiving Proxy Solicitation PDU.

CONFIG_BLE_MESH_PROXY_SOLIC_RX_CRPL

Maximum capacity of solicitation replay protection list

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_SERVER > CONFIG_BLE_MESH_PROXY_SOLIC_PDU_RX

This option specifies the maximum capacity of the solicitation replay protection list. The solicitation replay protection list is used to reject Solicitation PDUs that were already processed by a node, which will store the solicitation src and solicitation sequence number of the received Solicitation PDU message.

Range:

- from 1 to 65536 if CONFIG_BLE_MESH_PROXY_SOLIC_PDU_RX && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_PROXY_SOLIC_PDU_RX && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROXY_CLI_SRV_COEXIST

Support Proxy Client and Proxy Server coexistence

Found in: Component config > CONFIG_BLE_MESH

Enable this option to support the coexistence of proxy client and proxy server.

Default value:

- No (disabled) if CONFIG_BLE_MESH_EXPERIMENTAL && CONFIG_BLE_MESH

CONFIG_BLE_MESH_GATT_PROXY_CLIENT

BLE Mesh GATT Proxy Client

Found in: Component config > CONFIG_BLE_MESH

This option enables support for Mesh GATT Proxy Client. The Proxy Client can use the GATT bearer to send mesh messages to a node that supports the advertising bearer.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_PROXY_SOLIC_PDU_TX

Support sending Proxy Solicitation PDU

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_CLIENT

Enable this option to support sending Proxy Solicitation PDU.

CONFIG_BLE_MESH_PROXY_SOLIC_TX_SRC_COUNT

Maximum number of SSRC that can be used by Proxy Client

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_GATT_PROXY_CLIENT > CONFIG_BLE_MESH_PROXY_SOLIC_PDU_TX

This option specifies the maximum number of Solicitation Source (SSRC) that can be used by Proxy Client for sending a Solicitation PDU. A Proxy Client may use the primary address or any of the secondary addresses as the SSRC for a Solicitation PDU. So for a Proxy Client, it's better to choose the value based on its own element count.

Range:

- from 1 to 16 if CONFIG_BLE_MESH_PROXY_SOLIC_PDU_TX && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_PROXY_SOLIC_PDU_TX && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SETTINGS

Store BLE Mesh configuration persistently

Found in: Component config > CONFIG_BLE_MESH

When selected, the BLE Mesh stack will take care of storing/restoring the BLE Mesh configuration persistently in flash. If the device is a BLE Mesh node, when this option is enabled, the configuration of the device will be stored persistently, including unicast address, NetKey, AppKey, etc. And if the device is a BLE Mesh Provisioner, the information of the device will be stored persistently, including the information of provisioned nodes, NetKey, AppKey, etc.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_STORE_TIMEOUT

Delay (in seconds) before storing anything persistently

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

This value defines in seconds how soon any pending changes are actually written into persistent storage (flash) after a change occurs. The option allows nodes to delay a certain period of time to save proper information to flash. The default value is 0, which means information will be stored immediately once there are updates.

Range:

- from 0 to 1000000 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SEQ_STORE_RATE

How often the sequence number gets updated in storage

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

This value defines how often the local sequence number gets updated in persistent storage (i.e. flash). e.g. a value of 100 means that the sequence number will be stored to flash on every 100th increment. If the node sends messages very frequently a higher value makes more sense, whereas if the node sends infrequently a value as low as 0 (update storage for every increment) can make sense. When the stack gets initialized it will add sequence number to the last stored one, so that it starts off with a value that's guaranteed to be larger than the last one used before power off.

Range:

- from 0 to 1000000 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RPL_STORE_TIMEOUT

Minimum frequency that the RPL gets updated in storage

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

This value defines in seconds how soon the RPL (Replay Protection List) gets written to persistent storage after a change occurs. If the node receives messages frequently, then a large value is recommended. If the node receives messages rarely, then the value can be as low as 0 (which means the RPL is written into the storage immediately). Note that if the node operates in a security-sensitive case, and there is a risk of sudden power-off, then a value of 0 is strongly recommended. Otherwise, a power loss before RPL being written into the storage may introduce message replay attacks and system security will be in a vulnerable state.

Range:

- from 0 to 1000000 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SETTINGS_BACKWARD_COMPATIBILITY

A specific option for settings backward compatibility

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

This option is created to solve the issue of failure in recovering node information after mesh stack updates. In the old version mesh stack, there is no key of "mesh/role" in nvs. In the new version mesh stack, key of "mesh/role" is added in nvs, recovering node information needs to check "mesh/role" key in nvs and implements selective recovery of mesh node information. Therefore, there may be failure in recovering node information during node restarting after OTA.

The new version mesh stack adds the option of "mesh/role" because we have added the support of storing Provisioner information, while the old version only supports storing node information.

If users are updating their nodes from old version to new version, we recommend enabling this option, so that system could set the flag in advance before recovering node information and make sure the node information recovering could work as expected.

Default value:

- No (disabled) if CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SPECIFIC_PARTITION

Use a specific NVS partition for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

When selected, the mesh stack will use a specified NVS partition instead of default NVS partition. Note that the specified partition must be registered with NVS using nvs_flash_init_partition() API, and the partition must exists in the csv file. When Provisioner needs to store a large amount of nodes' information in the flash (e.g. more than 20), this option is recommended to be enabled.

Default value:

- No (disabled) if CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PARTITION_NAME

Name of the NVS partition for BLE Mesh

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS > CONFIG_BLE_MESH_SPECIFIC_PARTITION

This value defines the name of the specified NVS partition used by the mesh stack.

Default value:

- "ble_mesh" if CONFIG_BLE_MESH_SPECIFIC_PARTITION && CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_USE_MULTIPLE_NAMESPACE

Support using multiple NVS namespaces by Provisioner

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS

When selected, Provisioner can use different NVS namespaces to store different instances of mesh information. For example, if in the first room, Provisioner uses NetKey A, AppKey A and provisions three devices, these information will be treated as mesh information instance A. When the Provisioner moves to the second room, it uses NetKey B, AppKey B and provisions two devices, then the information will be treated as mesh information instance B. Here instance A and instance B will be stored in different namespaces. With this option enabled, Provisioner needs to use specific functions to open the corresponding NVS namespace, restore the mesh information, release the mesh information or erase the mesh information.

Default value:

- No (disabled) if CONFIG_BLE_MESH_PROVISIONER && CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_NVS_NAMESPACE

Maximum number of NVS namespaces

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_SETTINGS > CONFIG_BLE_MESH_USE_MULTIPLE_NAMESPACE

This option specifies the maximum NVS namespaces supported by Provisioner.

Range:

- from 1 to 255 if CONFIG_BLE_MESH_USE_MULTIPLE_NAMESPACE && CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_USE_MULTIPLE_NAMESPACE && CONFIG_BLE_MESH_SETTINGS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SUBNET_COUNT

Maximum number of mesh subnets per network

Found in: Component config > CONFIG_BLE_MESH

This option specifies how many subnets a Mesh network can have at the same time. Indeed, this value decides the number of the network keys which can be owned by a node.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_APP_KEY_COUNT

Maximum number of application keys per network

Found in: Component config > CONFIG_BLE_MESH

This option specifies how many application keys the device can store per network. Indeed, this value decides the number of the application keys which can be owned by a node.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_MODEL_KEY_COUNT

Maximum number of application keys per model

Found in: Component config > CONFIG_BLE_MESH

This option specifies the maximum number of application keys to which each model can be bound.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_MODEL_GROUP_COUNT

Maximum number of group address subscriptions per model

Found in: Component config > CONFIG_BLE_MESH

This option specifies the maximum number of addresses to which each model can be subscribed.

Range:

- from 1 to 4096 if CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_LABEL_COUNT

Maximum number of Label UUIDs used for Virtual Addresses

Found in: Component config > CONFIG_BLE_MESH

This option specifies how many Label UUIDs can be stored. Indeed, this value decides the number of the Virtual Addresses can be supported by a node.

Range:

- from 0 to 4096 if CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_CRPL

Maximum capacity of the replay protection list

Found in: Component config > CONFIG_BLE_MESH

This option specifies the maximum capacity of the replay protection list. It is similar to Network message cache size, but has a different purpose. The replay protection list is used to prevent a node from replay attack, which will store the source address and sequence number of the received mesh messages. For Provisioner, the replay protection list size should not be smaller than the maximum number of nodes whose information can be stored. And the element number of each node should also be taken into consideration. For example, if Provisioner can provision up to 20 nodes and each node contains two elements, then the replay protection list size of Provisioner should be at least 40.

Range:

- from 2 to 65535 if CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_NOT_RELAY_REPLAY_MSG

Not relay replayed messages in a mesh network

Found in: Component config > CONFIG_BLE_MESH

There may be many expired messages in a complex mesh network that would be considered replayed messages. Enable this option will refuse to relay such messages, which could help to reduce invalid packets in the mesh network. However, it should be noted that enabling this option may result in packet loss in certain environments. Therefore, users need to decide whether to enable this option according to the actual usage situation.

Default value:

- No (disabled) if CONFIG_BLE_MESH_EXPERIMENTAL && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MSG_CACHE_SIZE

Network message cache size

Found in: Component config > CONFIG_BLE_MESH

Number of messages that are cached for the network. This helps prevent unnecessary decryption operations and unnecessary relays. This option is similar to Replay protection list, but has a different purpose. A node is not required to cache the entire Network PDU and may cache only part of it for tracking, such as values for SRC/SEQ or others.

Range:

- from 2 to 65535 if CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_ADV_BUF_COUNT

Number of advertising buffers

Found in: Component config > CONFIG_BLE_MESH

Number of advertising buffers available. The transport layer reserves ADV_BUF_COUNT - 3 buffers for outgoing segments. The maximum outgoing SDU size is 12 times this value (out of which 4 or 8 bytes are used for the Transport Layer MIC). For example, 5 segments means the maximum SDU size is 60 bytes, which leaves 56 bytes for application layer data using a 4-byte MIC, or 52 bytes using an 8-byte MIC.

Range:

- from 6 to 256 if CONFIG_BLE_MESH

Default value:

- 60 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_IVU_DIVIDER

Divider for IV Update state refresh timer

Found in: Component config > CONFIG_BLE_MESH

When the IV Update state enters Normal operation or IV Update in Progress, we need to keep track of how many hours has passed in the state, since the specification requires us to remain in the state at least for 96 hours (Update in Progress has an additional upper limit of 144 hours).

In order to fulfill the above requirement, even if the node might be powered off once in a while, we need to store persistently how many hours the node has been in the state. This doesn't necessarily need to happen every hour (thanks to the flexible duration range). The exact cadence will depend a lot on the ways that the node will be used and what kind of power source it has.

Since there is no single optimal answer, this configuration option allows specifying a divider, i.e. how many intervals the 96 hour minimum gets split into. After each interval the duration that the node has been in the current state gets stored to flash. E.g. the default value of 4 means that the state is saved every 24 hours (96 / 4).

Range:

- from 2 to 96 if CONFIG_BLE_MESH

Default value:

- 4 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_IVU_RECOVERY_IVI

Recovery the IV index when the latest whole IV update procedure is missed

Found in: Component config > CONFIG_BLE_MESH

According to Section 3.10.5 of Mesh Specification v1.0.1. If a node in Normal Operation receives a Secure Network beacon with an IV index equal to the last known IV index+1 and the IV Update Flag set to 0, the node may update its IV without going to the IV Update in Progress state, or it may initiate an IV Index Recovery procedure (Section 3.10.6), or it may ignore the Secure Network beacon. The node makes the choice depending on the time since last IV update and the likelihood that the node has missed the Secure Network beacons with the IV update Flag. When the above situation is encountered, this option can be used to decide whether to perform the IV index recovery procedure.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_SAR_ENHANCEMENT

Segmentation and reassembly enhancement

Found in: Component config > CONFIG_BLE_MESH

Enable this option to use the enhanced segmentation and reassembly mechanism introduced in Bluetooth Mesh Protocol 1.1.

Default value:

- No (disabled) if CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_TX_SEG_MSG_COUNT

Maximum number of simultaneous outgoing segmented messages

Found in: Component config > CONFIG_BLE_MESH

Maximum number of simultaneous outgoing multi-segment and/or reliable messages. The default value is 1, which means the device can only send one segmented message at a time. And if another segmented message is going to be sent, it should wait for the completion of the previous one. If users are going to send multiple segmented messages at the same time, this value should be configured properly.

Range:

- from 1 to if CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_RX_SEG_MSG_COUNT

Maximum number of simultaneous incoming segmented messages

Found in: Component config > CONFIG_BLE_MESH

Maximum number of simultaneous incoming multi-segment and/or reliable messages. The default value is 1, which means the device can only receive one segmented message at a time. And if another segmented message is going to be received, it should wait for the completion of the previous one. If users are going to receive multiple segmented messages at the same time, this value should be configured properly.

Range:

- from 1 to 255 if CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_RX_SDU_MAX

Maximum incoming Upper Transport Access PDU length

Found in: Component config > CONFIG_BLE_MESH

Maximum incoming Upper Transport Access PDU length. Leave this to the default value, unless you really need to optimize memory usage.

Range:

- from 36 to 384 if CONFIG_BLE_MESH

Default value:

- 384 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_TX_SEG_MAX

Maximum number of segments in outgoing messages

Found in: Component config > CONFIG_BLE_MESH

Maximum number of segments supported for outgoing messages. This value should typically be fine-tuned based on what models the local node supports, i.e. what's the largest message payload that the node needs to be able to send. This value affects memory and call stack consumption, which is why the default is lower than the maximum that the specification would allow (32 segments).

The maximum outgoing SDU size is 12 times this number (out of which 4 or 8 bytes is used for the Transport Layer MIC). For example, 5 segments means the maximum SDU size is 60 bytes, which leaves 56 bytes for application layer data using a 4-byte MIC and 52 bytes using an 8-byte MIC.

Be sure to specify a sufficient number of advertising buffers when setting this option to a higher value. There must be at least three more advertising buffers (BLE_MESH_ADV_BUF_COUNT) as there are outgoing segments.

Range:

- from 2 to 32 if CONFIG_BLE_MESH

Default value:

- 32 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_RELAY

Relay support

Found in: Component config > CONFIG_BLE_MESH

Support for acting as a Mesh Relay Node. Enabling this option will allow a node to support the Relay feature, and the Relay feature can still be enabled or disabled by proper configuration messages. Disabling this option will let a node not support the Relay feature.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_NODE && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RELAY_ADV_BUF

Use separate advertising buffers for relay packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_RELAY

When selected, self-send packets will be put in a high-priority queue and relay packets will be put in a low-priority queue.

Default value:

- No (disabled) if CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RELAY_ADV_BUF_COUNT

Number of advertising buffers for relay packets

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_RELAY > CONFIG_BLE_MESH_RELAY_ADV_BUF

Number of advertising buffers for relay packets available.

Range:

- from 6 to 256 if CONFIG_BLE_MESH_RELAY_ADV_BUF && CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

Default value:

- 60 if CONFIG_BLE_MESH_RELAY_ADV_BUF && CONFIG_BLE_MESH_RELAY && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LOW_POWER

Support for Low Power features

Found in: Component config > CONFIG_BLE_MESH

Enable this option to operate as a Low Power Node. If low power consumption is required by a node, this option should be enabled. And once the node enters the mesh network, it will try to find a Friend node and establish a friendship.

CONFIG_BLE_MESH_LPN_ESTABLISHMENT

Perform Friendship establishment using low power

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Perform the Friendship establishment using low power with the help of a reduced scan duty cycle. The downside of this is that the node may miss out on messages intended for it until it has successfully set up Friendship with a Friend node. When this option is enabled, the node will stop scanning for a period of time after a Friend Request or Friend Poll is sent, so as to reduce more power consumption.

Default value:

- No (disabled) if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_AUTO

Automatically start looking for Friend nodes once provisioned

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Once provisioned, automatically enable LPN functionality and start looking for Friend nodes. If this option is disabled LPN mode needs to be manually enabled by calling bt_mesh_lpn_set(true). When an unprovisioned device is provisioned successfully and becomes a node, enabling this option will trigger the node starts to send Friend Request at a certain period until it finds a proper Friend node.

Default value:

- No (disabled) if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_AUTO_TIMEOUT

Time from last received message before going to LPN mode

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER > CONFIG_BLE_MESH_LPN_AUTO

Time in seconds from the last received message, that the node waits out before starting to look for Friend nodes.

Range:

- from 0 to 3600 if CONFIG_BLE_MESH_LPN_AUTO && CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 15 if CONFIG_BLE_MESH_LPN_AUTO && CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_RETRY_TIMEOUT

Retry timeout for Friend requests

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Time in seconds between Friend Requests, if a previous Friend Request did not yield any acceptable Friend Offers.

Range:

- from 1 to 3600 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 6 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_RSSI_FACTOR

RSSIFactor, used in Friend Offer Delay calculation

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

The contribution of the RSSI, measured by the Friend node, used in Friend Offer Delay calculations. 0 = 1, 1 = 1.5, 2 = 2, 3 = 2.5. RSSIFactor, one of the parameters carried by Friend Request sent by Low Power node, which is used to calculate the Friend Offer Delay.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_RECV_WIN_FACTOR

ReceiveWindowFactor, used in Friend Offer Delay calculation

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

The contribution of the supported Receive Window used in Friend Offer Delay calculations. 0 = 1, 1 = 1.5, 2 = 2, 3 = 2.5. ReceiveWindowFactor, one of the parameters carried by Friend Request sent by Low Power node, which is used to calculate the Friend Offer Delay.

Range:

- from 0 to 3 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_MIN_QUEUE_SIZE

Minimum size of the acceptable friend queue (MinQueueSizeLog)

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

The MinQueueSizeLog field is defined as log_2(N), where N is the minimum number of maximum size Lower Transport PDUs that the Friend node can store in its Friend Queue. As an example, MinQueueSizeLog value 1 gives N = 2, and value 7 gives N = 128.

Range:

- from 1 to 7 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_RECV_DELAY

Receive delay requested by the local node

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

The ReceiveDelay is the time between the Low Power node sending a request and listening for a response. This delay allows the Friend node time to prepare the response. The value is in units of milliseconds.

Range:

- from 10 to 255 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 100 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_POLL_TIMEOUT

The value of the PollTimeout timer

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

PollTimeout timer is used to measure time between two consecutive requests sent by a Low Power node. If no requests are received the Friend node before the PollTimeout timer expires, then the friendship is considered terminated. The value is in units of 100 milliseconds, so e.g. a value of 300 means 30 seconds. The smaller the value, the faster the Low Power node tries to get messages from corresponding Friend node and vice versa.

Range:

- from 10 to 244735 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 300 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_INIT_POLL_TIMEOUT

The starting value of the PollTimeout timer

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

The initial value of the PollTimeout timer when Friendship is to be established for the first time. After this, the timeout gradually grows toward the actual PollTimeout, doubling in value for each iteration. The value is in units of 100 milliseconds, so e.g. a value of 300 means 30 seconds.

Range:

- from 10 to if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_SCAN_LATENCY

Latency for enabling scanning

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Latency (in milliseconds) is the time it takes to enable scanning. In practice, it means how much time in advance of the Receive Window, the request to enable scanning is made.

Range:

- from 0 to 50 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_GROUPS

Number of groups the LPN can subscribe to

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Maximum number of groups to which the LPN can subscribe.

Range:

- from 0 to 16384 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

Default value:

- 8 if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_LPN_SUB_ALL_NODES_ADDR

Automatically subscribe all nodes address

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_LOW_POWER

Automatically subscribe all nodes address when friendship established.

Default value:

- No (disabled) if CONFIG_BLE_MESH_LOW_POWER && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FRIEND

Support for Friend feature

Found in: Component config > CONFIG_BLE_MESH

Enable this option to be able to act as a Friend Node.

CONFIG_BLE_MESH_FRIEND_RECV_WIN

Friend Receive Window

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FRIEND

Receive Window in milliseconds supported by the Friend node.

Range:

- from 1 to 255 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

Default value:

- 255 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FRIEND_QUEUE_SIZE

Minimum number of buffers supported per Friend Queue

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FRIEND

Minimum number of buffers available to be stored for each local Friend Queue. This option decides the size of each buffer which can be used by a Friend node to store messages for each Low Power node.

Range:

- from 2 to 65536 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

Default value:

- 16 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FRIEND_SUB_LIST_SIZE

Friend Subscription List Size

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FRIEND

Size of the Subscription List that can be supported by a Friend node for a Low Power node. And Low Power node can send Friend Subscription List Add or Friend Subscription List Remove messages to the Friend node to add or remove subscription addresses.

Range:

- from 0 to 1023 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

Default value:

- 3 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FRIEND_LPN_COUNT

Number of supported LPN nodes

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FRIEND

Number of Low Power Nodes with which a Friend can have Friendship simultaneously. A Friend node can have friendship with multiple Low Power nodes at the same time, while a Low Power node can only establish friendship with only one Friend node at the same time.

Range:

- from 1 to 1000 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

CONFIG_BLE_MESH_FRIEND_SEG_RX

Number of incomplete segment lists per LPN

Found in: Component config > CONFIG_BLE_MESH > CONFIG_BLE_MESH_FRIEND

Number of incomplete segment lists tracked for each Friends' LPN. In other words, this determines from how many elements can segmented messages destined for the Friend queue be received simultaneously.

Range:

- from 1 to 1000 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_FRIEND && CONFIG_BLE_MESH

CONFIG_BLE_MESH_NO_LOG

Disable BLE Mesh debug logs (minimize bin size)

Found in: Component config > CONFIG_BLE_MESH

Select this to save the BLE Mesh related rodata code size. Enabling this option will disable the output of BLE Mesh debug log.

Default value:

- No (disabled) if CONFIG_BLE_MESH && CONFIG_BLE_MESH

BLE Mesh STACK DEBUG LOG LEVEL

Contains:

- CONFIG_BLE_MESH_STACK_TRACE_LEVEL

CONFIG_BLE_MESH_STACK_TRACE_LEVEL

BLE_MESH_STACK

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh STACK DEBUG LOG LEVEL

Define BLE Mesh trace level for BLE Mesh stack.

Available options:

- NONE (CONFIG_BLE_MESH_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BLE_MESH_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BLE_MESH_TRACE_LEVEL_WARNING)
- INFO (CONFIG_BLE_MESH_TRACE_LEVEL_INFO)
- DEBUG (CONFIG_BLE_MESH_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BLE_MESH_TRACE_LEVEL_VERBOSE)

BLE Mesh NET BUF DEBUG LOG LEVEL

Contains:

- CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL

CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL

BLE_MESH_NET_BUF

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh NET BUF DEBUG LOG LEVEL

Define BLE Mesh trace level for BLE Mesh net buffer.

Available options:

- NONE (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_NONE)
- ERROR (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_ERROR)
- WARNING (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_WARNING)
- INFO (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_INFO)
- DEBUG (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_DEBUG)
- VERBOSE (CONFIG_BLE_MESH_NET_BUF_TRACE_LEVEL_VERBOSE)

CONFIG_BLE_MESH_CLIENT_MSG_TIMEOUT

Timeout(ms) for client message response

Found in: Component config > CONFIG_BLE_MESH

Timeout value used by the node to get response of the acknowledged message which is sent by the client model. This value indicates the maximum time that a client model waits for the response of the sent acknowledged messages. If a client model uses 0 as the timeout value when sending acknowledged messages, then the default value will be used which is four seconds.

Range:

- from 100 to 1200000 if CONFIG_BLE_MESH

Default value:

- 4000 if CONFIG_BLE_MESH

Support for BLE Mesh Foundation models

Contains:

- CONFIG_BLE_MESH_BRC_CLI
- CONFIG_BLE_MESH_BRC_SRV
- CONFIG_BLE_MESH_CFG_CLI
- CONFIG_BLE_MESH_DF_CLI
- CONFIG_BLE_MESH_DF_SRV
- CONFIG_BLE_MESH_HEALTH_CLI
- CONFIG_BLE_MESH_HEALTH_SRV
- CONFIG_BLE_MESH_LCD_CLI
- CONFIG_BLE_MESH_LCD_SRV
- CONFIG_BLE_MESH_PRB_CLI
- CONFIG_BLE_MESH_PRB_SRV
- CONFIG_BLE_MESH_ODP_CLI
- CONFIG_BLE_MESH_ODP_SRV
- CONFIG_BLE_MESH_AGG_CLI
- CONFIG_BLE_MESH_AGG_SRV
- CONFIG_BLE_MESH_RPR_CLI
- CONFIG_BLE_MESH_RPR_SRV
- CONFIG_BLE_MESH_SAR_CLI
- CONFIG_BLE_MESH_SAR_SRV
- CONFIG_BLE_MESH_SRPL_CLI
- CONFIG_BLE_MESH_SRPL_SRV
- CONFIG_BLE_MESH_COMP_DATA_1
- CONFIG_BLE_MESH_COMP_DATA_128
- CONFIG_BLE_MESH_MODELS_METADATA_0

CONFIG_BLE_MESH_CFG_CLI

Configuration Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Configuration Client model.

CONFIG_BLE_MESH_HEALTH_CLI

Health Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Health Client model.

CONFIG_BLE_MESH_HEALTH_SRV

Health Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Health Server model.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_BRC_CLI

Bridge Configuration Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Bridge Configuration Client model.

CONFIG_BLE_MESH_BRC_SRV

Bridge Configuration Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Bridge Configuration Server model.

Default value:

- No (disabled) if CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_BRIDGING_TABLE_ENTRY_COUNT

Maximum number of Bridging Table entries

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_BRC_SRV

Maximum number of Bridging Table entries that the Bridge Configuration Server can support.

Range:

- from 16 to 65535 if CONFIG_BLE_MESH_BRC_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 16 if CONFIG_BLE_MESH_BRC_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_BRIDGE_CRPL

Maximum capacity of bridge replay protection list

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_BRC_SRV

This option specifies the maximum capacity of the bridge replay protection list. The bridge replay protection list is used to prevent a bridged subnet from replay attack, which will store the source address and sequence number of the received bridge messages.

Range:

- from 1 to 255 if CONFIG_BLE_MESH_BRC_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 5 if CONFIG_BLE_MESH_BRC_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PRB_CLI

Mesh Private Beacon Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Mesh Private Beacon Client model.

CONFIG_BLE_MESH_PRB_SRV

Mesh Private Beacon Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Mesh Private Beacon Server model.

CONFIG_BLE_MESH_ODP_CLI

On-Demand Private Proxy Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for On-Demand Private Proxy Client model.

CONFIG_BLE_MESH_ODP_SRV

On-Demand Private Proxy Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for On-Demand Private Proxy Server model.

CONFIG_BLE_MESH_SRPL_CLI

Solicitation PDU RPL Configuration Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Solicitation PDU RPL Configuration Client model.

CONFIG_BLE_MESH_SRPL_SRV

Solicitation PDU RPL Configuration Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Solicitation PDU RPL Configuration Server model. Note: This option depends on the functionality of receiving Solicitation PDU. If the device doesn't support receiving Solicitation PDU, then there is no need to enable this server model.

CONFIG_BLE_MESH_AGG_CLI

Opcodes Aggregator Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Opcodes Aggregator Client model.

CONFIG_BLE_MESH_AGG_SRV

Opcodes Aggregator Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Opcodes Aggregator Server model.

CONFIG_BLE_MESH_SAR_CLI

SAR Configuration Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for SAR Configuration Client model.

CONFIG_BLE_MESH_SAR_SRV

SAR Configuration Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for SAR Configuration Server model.

CONFIG_BLE_MESH_COMP_DATA_1

Support Composition Data Page 1

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Composition Data Page 1 contains information about the relationships among models. Each model either can be a root model or can extend other models.

CONFIG_BLE_MESH_COMP_DATA_128

Support Composition Data Page 128

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Composition Data Page 128 is used to indicate the structure of elements, features, and models of a node after the successful execution of the Node Address Refresh procedure or the Node Composition Refresh procedure, or after the execution of the Node Removal procedure followed by the provisioning process. Composition Data Page 128 shall be present if the node supports the Remote Provisioning Server model; otherwise it is optional.

CONFIG_BLE_MESH_MODELS_METADATA_0

Support Models Metadata Page 0

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

The Models Metadata state contains metadata of a node's models. The Models Metadata state is composed of a number of pages of information. Models Metadata Page 0 shall be present if the node supports the Large Composition Data Server model.

CONFIG_BLE_MESH_MODELS_METADATA_128

Support Models Metadata Page 128

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_MODELS_METADATA_0

The Models Metadata state contains metadata of a node's models. The Models Metadata state is composed of a number of pages of information. Models Metadata Page 128 contains metadata for the node's models after the successful execution of the Node Address Refresh procedure or the Node Composition Refresh procedure, or after the execution of the Node Removal procedure followed by the provisioning process. Models Metadata Page 128 shall be present if the node supports the Remote Provisioning Server model and the node supports the Large Composition Data Server model.

CONFIG_BLE_MESH_LCD_CLI

Large Composition Data Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Large Composition Data Client model.

CONFIG_BLE_MESH_LCD_SRV

Large Composition Data Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Large Composition Data Server model.

CONFIG_BLE_MESH_RPR_CLI

Remote Provisioning Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Remote Provisioning Client model

CONFIG_BLE_MESH_RPR_CLI_PROV_SAME_TIME

Maximum number of PB-Remote running at the same time by Provisioner

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_RPR_CLI

This option specifies how many devices can be provisioned at the same time using PB-REMOTE. For example, if the value is 2, it means a Provisioner can provision two unprovisioned devices with PB-REMOTE at the same time.

Range:

- from 1 to 5 if CONFIG_BLE_MESH_RPR_CLI && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_RPR_CLI && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RPR_SRV

Remote Provisioning Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Remote Provisioning Server model

CONFIG_BLE_MESH_RPR_SRV_MAX_SCANNED_ITEMS

Maximum number of device information can be scanned

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_RPR_SRV

This option specifies how many device information can a Remote Provisioning Server store each time while scanning.

Range:

- from 4 to 255 if CONFIG_BLE_MESH_RPR_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH_RPR_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RPR_SRV_ACTIVE_SCAN

Support Active Scan for remote provisioning

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_RPR_SRV

Enable this option to support Active Scan for remote provisioning.

CONFIG_BLE_MESH_RPR_SRV_MAX_EXT_SCAN

Maximum number of extended scan procedures

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_RPR_SRV

This option specifies how many extended scan procedures can be started by the Remote Provisioning Server.

Range:

- from 1 to 10 if CONFIG_BLE_MESH_RPR_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_RPR_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DF_CLI

Directed Forwarding Configuration Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Directed Forwarding Configuration Client model.

CONFIG_BLE_MESH_DF_SRV

Directed Forwarding Configuration Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models

Enable support for Directed Forwarding Configuration Server model.

CONFIG_BLE_MESH_MAX_DISC_TABLE_ENTRY_COUNT

Maximum number of discovery table entries in a given subnet

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_DF_SRV

Maximum number of Discovery Table entries supported by the node in a given subnet.

Range:

- from 2 to 255 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_FORWARD_TABLE_ENTRY_COUNT

Maximum number of forward table entries in a given subnet

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_DF_SRV

Maximum number of Forward Table entries supported by the node in a given subnet.

Range:

- from 2 to 64 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_DEPS_NODES_PER_PATH

Maximum number of dependent nodes per path

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_DF_SRV

Maximum size of dependent nodes list supported by each forward table entry.

Range:

- from 2 to 64 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_PATH_MONITOR_TEST

Enable Path Monitoring test mode

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_DF_SRV

The option only removes the Path Use timer; all other behavior of the device is not changed. If Path Monitoring test mode is going to be used, this option should be enabled.

Default value:

- No (disabled) if CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SUPPORT_DIRECTED_PROXY

Enable Directed Proxy functionality

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Foundation models > CONFIG_BLE_MESH_DF_SRV

Support Directed Proxy functionality.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_GATT_PROXY_SERVER && CONFIG_BLE_MESH_DF_SRV && CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Support for BLE Mesh Client/Server models

Contains:

- Binary Larger Object Transfer model
- CONFIG_BLE_MESH_MBT_CLI
- CONFIG_BLE_MESH_MBT_SRV
- Device Firmware Update model
- CONFIG_BLE_MESH_GENERIC_BATTERY_CLI
- CONFIG_BLE_MESH_GENERIC_DEF_TRANS_TIME_CLI
- CONFIG_BLE_MESH_GENERIC_LEVEL_CLI
- CONFIG_BLE_MESH_GENERIC_LOCATION_CLI
- CONFIG_BLE_MESH_GENERIC_ONOFF_CLI
- CONFIG_BLE_MESH_GENERIC_POWER_LEVEL_CLI
- CONFIG_BLE_MESH_GENERIC_POWER_ONOFF_CLI
- CONFIG_BLE_MESH_GENERIC_PROPERTY_CLI
- CONFIG_BLE_MESH_GENERIC_SERVER
- CONFIG_BLE_MESH_LIGHT_CTL_CLI
- CONFIG_BLE_MESH_LIGHT_HSL_CLI
- CONFIG_BLE_MESH_LIGHT_LC_CLI
- CONFIG_BLE_MESH_LIGHT_LIGHTNESS_CLI
- CONFIG_BLE_MESH_LIGHT_XYL_CLI
- CONFIG_BLE_MESH_LIGHTING_SERVER
- CONFIG_BLE_MESH_SCENE_CLI
- CONFIG_BLE_MESH_SCHEDULER_CLI
- CONFIG_BLE_MESH_SENSOR_CLI
- CONFIG_BLE_MESH_SENSOR_SERVER
- CONFIG_BLE_MESH_TIME_SCENE_SERVER
- CONFIG_BLE_MESH_TIME_CLI

CONFIG_BLE_MESH_GENERIC_ONOFF_CLI

Generic OnOff Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic OnOff Client model.

CONFIG_BLE_MESH_GENERIC_LEVEL_CLI

Generic Level Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Level Client model.

CONFIG_BLE_MESH_GENERIC_DEF_TRANS_TIME_CLI

Generic Default Transition Time Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Default Transition Time Client model.

CONFIG_BLE_MESH_GENERIC_POWER_ONOFF_CLI

Generic Power OnOff Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Power OnOff Client model.

CONFIG_BLE_MESH_GENERIC_POWER_LEVEL_CLI

Generic Power Level Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Power Level Client model.

CONFIG_BLE_MESH_GENERIC_BATTERY_CLI

Generic Battery Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Battery Client model.

CONFIG_BLE_MESH_GENERIC_LOCATION_CLI

Generic Location Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Location Client model.

CONFIG_BLE_MESH_GENERIC_PROPERTY_CLI

Generic Property Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic Property Client model.

CONFIG_BLE_MESH_SENSOR_CLI

Sensor Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Sensor Client model.

CONFIG_BLE_MESH_TIME_CLI

Time Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Time Client model.

CONFIG_BLE_MESH_SCENE_CLI

Scene Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Scene Client model.

CONFIG_BLE_MESH_SCHEDULER_CLI

Scheduler Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Scheduler Client model.

CONFIG_BLE_MESH_LIGHT_LIGHTNESS_CLI

Light Lightness Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Light Lightness Client model.

CONFIG_BLE_MESH_LIGHT_CTL_CLI

Light CTL Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Light CTL Client model.

CONFIG_BLE_MESH_LIGHT_HSL_CLI

Light HSL Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Light HSL Client model.

CONFIG_BLE_MESH_LIGHT_XYL_CLI

Light XYL Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Light XYL Client model.

CONFIG_BLE_MESH_LIGHT_LC_CLI

Light LC Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Light LC Client model.

CONFIG_BLE_MESH_GENERIC_SERVER

Generic server models

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Generic server models.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_SENSOR_SERVER

Sensor server models

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Sensor server models.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_TIME_SCENE_SERVER

Time and Scenes server models

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Time and Scenes server models.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_LIGHTING_SERVER

Lighting server models

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for Lighting server models.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_MBT_CLI

BLOB Transfer Client model(Deprecated)

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for BLOB Transfer Client model. Warn: This version of the Mesh Binary Large Object Transfer Model will be deprecated, and a new version will be released in the future.

Default value:

- No (disabled) if CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MAX_BLOB_RECEIVERS

Maximum number of simultaneous blob receivers

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > CONFIG_BLE_MESH_MBT_CLI

Maximum number of BLOB Transfer Server models that can participating in the BLOB transfer with a BLOB Transfer Client model.

Range:

- from 1 to 255 if CONFIG_BLE_MESH_MBT_CLI && CONFIG_BLE_MESH

Default value:

- 2 if CONFIG_BLE_MESH_MBT_CLI && CONFIG_BLE_MESH

CONFIG_BLE_MESH_MBT_SRV

BLOB Transfer Server model(Deprecated)

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models

Enable support for BLOB Transfer Server model. Warn: This version of the Mesh Binary Large Object Transfer Model will be deprecated, and a new version will be released in the future.

Default value:

- No (disabled) if CONFIG_BLE_MESH_V11_SUPPORT && CONFIG_BLE_MESH

Binary Larger Object Transfer model

Contains:

- BLOB models common configuration
- CONFIG_BLE_MESH_BLOB_CLI
- CONFIG_BLE_MESH_BLOB_SRV

CONFIG_BLE_MESH_BLOB_SRV

Support for BLOB Transfer Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model

Enable the Binary Large Object (BLOB) Transfer Server model.

CONFIG_BLE_MESH_BLOB_SRV_PULL_REQ_COUNT

Number of chunks to request for each pull

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

In Pull mode (Pull BLOB Transfer Mode), the BLOB Transfer Server requests a fixed number of chunks from the Client. Use this option to control the chunk count in the request. If the BLOB Transfer Server is instantiated on a Low Power node, the pull request count will be trimmed to not overflow the Friend queue.

Range:

- from 1 to 16 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

Default value:

- 4 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_BLOB_SIZE_MAX

Largest BLOB size in bytes

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

The maximum object size a BLOB Transfer Server can receive.

Range:

- from 1 to 3257617792 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

Default value:

- 524288 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_BLOB_BLOCK_SIZE_MIN

Minimum block size

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

Minimum acceptable block size in a BLOB transfer. The transfer block size will be some number that is a power of two, and is between block size min and block size max. If no such number exists, a compile time warning will be issued.

Range:

- from 64 to 1048576 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

Default value:

- 4096 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_BLOB_BLOCK_SIZE_MAX

Maximum block size

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

Maximum acceptable block size in a BLOB transfer. The transfer block size will be some number that is a power of two, and is between block size min and block size max. If no such number exists, a compile time warning will be issued.

Range:

- from to 1048576 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

Default value:

- 4096 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_BLOB_REPORT_TIMEOUT

Partial Block Report interval in Pull mode

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

The timer value that Pull BLOB Transfer Server uses to report missed chunks.

Range:

- from 1 to 31 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

Default value:

- 10 if CONFIG_BLE_MESH_BLOB_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_RX_BLOB_CHUNK_SIZE

BLOB Server chunk size

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_SRV

Set the chunk size for the BLOB Server. The actual maximum chunk size depends on how many segments are possible and will be clamped to the max possible if set above. see also: BLE_MESH_RX_SEG_MAX, and the maximum SDU a node can receive.

CONFIG_BLE_MESH_BLOB_CLI

Support for BLOB Transfer Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model

Enable the Binary Large Object (BLOB) Transfer Client model.

CONFIG_BLE_MESH_BLOB_CLI_BLOCK_RETRIES

Number of retries per block

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_CLI

Controls the number of times the client will attempt to resend missing chunks to the BLOB receivers for every block.

Default value:

- 5 if CONFIG_BLE_MESH_BLOB_CLI && CONFIG_BLE_MESH

CONFIG_BLE_MESH_TX_BLOB_CHUNK_SIZE

BLOB Client chunk size

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_CLI

Set the chunk size for the BLOB Client. The actual maximum chunk size depends on how many segments are possible and will be clamped to the max possible if set above. see also: BLE_MESH_TX_SEG_MAX, and the maximum SDU a node can receive.

CONFIG_BLE_MESH_TX_BLOB_CHUNK_SEND_INTERVAL

BLOB Client chunk send interval

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > CONFIG_BLE_MESH_BLOB_CLI

Set the interval in milliseconds in which chunks are sent during the BLOB transfer. Note: Without a delay between each sent chunk, the network might become too busy with the BLOB transfer for other communications to succeed. Note: Timing restrictions, like the timeout base, should be considered or changed accordingly when setting this interval. Otherwise, the interval might be too big for the timeout settings and cause timeouts.

Range:

- from 0 to 2147483647 if CONFIG_BLE_MESH_BLOB_CLI && CONFIG_BLE_MESH

Default value:

- 0 if CONFIG_BLE_MESH_BLOB_CLI && CONFIG_BLE_MESH

BLOB models common configuration

Contains:

- CONFIG_BLE_MESH_ALIGN_CHUNK_SIZE_TO_MAX_SEGMENT
- CONFIG_BLE_MESH_BLOB_CHUNK_COUNT_MAX

CONFIG_BLE_MESH_BLOB_CHUNK_COUNT_MAX

Maximum chunk count per block

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > BLOB models common configuration

A BLOB transfer contains several blocks. Each block is made up of several chunks. This option controls the maximum chunk count per block.

Range:

- from 1 to 2992 if CONFIG_BLE_MESH

Default value:

- 256 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_ALIGN_CHUNK_SIZE_TO_MAX_SEGMENT

Align chunk size to max segmented message size

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Binary Larger Object Transfer model > BLOB models common configuration

Default value:

- Yes (enabled) if CONFIG_BLE_MESH

Device Firmware Update model

Contains:

- CONFIG_BLE_MESH_DFU_SLOTS
- Firmware Update model configuration
- CONFIG_BLE_MESH_DFD_CLI
- CONFIG_BLE_MESH_DFD_SRV
- CONFIG_BLE_MESH_DFU_CLI

- CONFIG_BLE_MESH_DFU_SRV

CONFIG_BLE_MESH_DFU_SRV

Support for Firmware Update Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model

Enable the Firmware Update Server model.

CONFIG_BLE_MESH_DFU_CLI

Support for Firmware Update Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model

Enable the Firmware Update Client model.

Firmware Update model configuration

Contains:

- CONFIG_BLE_MESH_DFU_FWID_MAXLEN
- CONFIG_BLE_MESH_DFU_METADATA_MAXLEN
- CONFIG_BLE_MESH_DFU_URI_MAXLEN
- CONFIG_BLE_MESH_DFU_METADATA

CONFIG_BLE_MESH_DFU_FWID_MAXLEN

DFU FWID max length

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > Firmware Update model configuration

This value defines the maximum length of an image's firmware ID.

Range:

- from 0 to 106 if CONFIG_BLE_MESH

Default value:

- 16 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFU_METADATA_MAXLEN

DFU metadata max length

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > Firmware Update model configuration

This value defines the maximum length of an image's metadata.

Range:

- from 18 to 255 if CONFIG_BLE_MESH_DFU_METADATA && CONFIG_BLE_MESH
- from 0 to 255 if CONFIG_BLE_MESH

Default value:

- 32 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFU_METADATA

Support for the default metadata format

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > Firmware Update model configuration

This option adds a set of functions that can be used to encode and decode a firmware metadata using the format defined in the Bluetooth mesh DFU subsystem.

CONFIG_BLE_MESH_DFU_URI_MAXLEN

DFU URI max length

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > Firmware Update model configuration

This value defines the maximum length of an image's URI, not including a string terminator.

Range:

- from 0 to 255 if CONFIG_BLE_MESH

Default value:

- 32 if CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFU_SLOTS

Firmware image slot manager

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model

Enable the DFU image slot manager, for managing firmware distribution slots for the Firmware Update Client model.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_DFU_CLI && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFU_SLOT_CNT

Number of firmware image slots

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > CONFIG_BLE_MESH_DFU_SLOTS

This value defines the number of firmware slots the DFU image slot manager can keep simultaneously.

Range:

- from 1 to 32767 if CONFIG_BLE_MESH_DFU_SLOTS && CONFIG_BLE_MESH

Default value:

- 1 if CONFIG_BLE_MESH_DFU_SLOTS && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFD_CLI

Support for Device Distribution Client model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model

Enable the Device Distribution Client model

CONFIG_BLE_MESH_DFD_SRV

Support for Firmware Distribution Server model

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model

Enable the Firmware Distribution Server model.

CONFIG_BLE_MESH_DFD_SRV_SLOT_MAX_SIZE

Largest DFU image that can be stored

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > CONFIG_BLE_MESH_DFD_SRV

This value defines the largest DFU image a single slot can store.

Range:

- from 0 to if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

Default value:

- if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFD_SRV_SLOT_SPACE

Total DFU image storage space

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > CONFIG_BLE_MESH_DFD_SRV

This value defines the total storage space dedicated to storing DFU images on the Firmware Distribution Server.

Range:

- from 0 to 4294967295 if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

Default value:

- if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFD_SRV_TARGETS_MAX

Maximum Target node count

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > CONFIG_BLE_MESH_DFD_SRV

This value defines the maximum number of Target nodes the Firmware Distribution Server can target simultaneously.

Range:

- from 1 to 65535 if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

Default value:

- 8 if CONFIG_BLE_MESH_DFD_SRV && CONFIG_BLE_MESH

CONFIG_BLE_MESH_DFD_SRV_OOB_UPLOAD

Support for DFU image OOB upload

Found in: Component config > CONFIG_BLE_MESH > Support for BLE Mesh Client/Server models > Device Firmware Update model > CONFIG_BLE_MESH_DFD_SRV

This enables support for OOB upload of firmware images for distribution. This makes several callbacks and use of the init macro BLE_MESH_DFD_SRV_INIT_OOB mandatory. See the API documentation for bt_mesh_dfd_srv_cb for details about the mandatory callbacks.

CONFIG_BLE_MESH_IV_UPDATE_TEST

Test the IV Update Procedure

Found in: Component config > CONFIG_BLE_MESH

This option removes the 96 hour limit of the IV Update Procedure and lets the state to be changed at any time. If IV Update test mode is going to be used, this option should be enabled.

Default value:

- No (disabled) if CONFIG_BLE_MESH

BLE Mesh specific test option

Contains:

- CONFIG_BLE_MESH_DEBUG
- CONFIG_BLE_MESH_SHELL
- CONFIG_BLE_MESH_BQB_TEST
- CONFIG_BLE_MESH_SELF_TEST
- CONFIG_BLE_MESH_TEST_AUTO_ENTER_NETWORK
- CONFIG_BLE_MESH_TEST_USE_WHITE_LIST

CONFIG_BLE_MESH_SELF_TEST

Perform BLE Mesh self-tests

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

This option adds extra self-tests which are run every time BLE Mesh networking is initialized.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_BQB_TEST

Enable BLE Mesh specific internal test

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

This option is used to enable some internal functions for auto-pts test.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_TEST_AUTO_ENTER_NETWORK

Unprovisioned device enters mesh network automatically

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

With this option enabled, an unprovisioned device can automatically enters mesh network using a specific test function without the pro- visioning procedure. And on the Provisioner side, a test function needs to be invoked to add the node information into the mesh stack.

Default value:

- Yes (enabled) if CONFIG_BLE_MESH_SELF_TEST && CONFIG_BLE_MESH

CONFIG_BLE_MESH_TEST_USE_WHITE_LIST

Use white list to filter mesh advertising packets

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

With this option enabled, users can use white list to filter mesh advertising packets while scanning.

Default value:

- No (disabled) if CONFIG_BLE_MESH_SELF_TEST && CONFIG_BLE_MESH

CONFIG_BLE_MESH_SHELL

Enable BLE Mesh shell

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

Activate shell module that provides BLE Mesh commands to the console.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_DEBUG

Enable BLE Mesh debug logs

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option

Enable debug logs for the BLE Mesh functionality.

Default value:

- No (disabled) if CONFIG_BLE_MESH

CONFIG_BLE_MESH_DEBUG_NET

Network layer debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Network layer debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_TRANS

Transport layer debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Transport layer debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_BEACON

Beacon debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Beacon-related debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_CRYPTO

Crypto debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable cryptographic debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_PROV

Provisioning debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Provisioning debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_ACCESS

Access layer debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Access layer debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_MODEL

Foundation model debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Foundation Models debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_ADV

Advertising debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable advertising debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_LOW_POWER

Low Power debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Low Power debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_FRIEND

Friend debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Friend debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_DEBUG_PROXY

Proxy debug

Found in: Component config > CONFIG_BLE_MESH > BLE Mesh specific test option > CONFIG_BLE_MESH_DEBUG

Enable Proxy protocol debug logs for the BLE Mesh functionality.

CONFIG_BLE_MESH_EXPERIMENTAL

Make BLE Mesh experimental features visible

Found in: Component config > CONFIG_BLE_MESH

Make BLE Mesh Experimental features visible. Experimental features list: - CONFIG_BLE_MESH_NOT_RELAY_REPLAY_MSG - CONFIG_BLE_MESH_USE_BLE_50 - CONFIG_BLE_MESH_PROXY_CLI_SRV_COEXIST

Default value:

- No (disabled) if CONFIG_BLE_MESH

Console Library

Contains:

- CONFIG_CONSOLE_SORTED_HELP

CONFIG_CONSOLE_SORTED_HELP

Enable sorted help

Found in: Component config > Console Library

Instead of listing the commands in the order of registration, the help command lists the available commands in sorted order, if this option is enabled.

Default value:

- No (disabled)

Legacy Driver Configurations

Contains:

- Legacy I2C Driver Configurations
- Legacy Touch Sensor Driver Configurations
- Legacy TWAI Driver Configurations

Legacy TWAI Driver Configurations

Contains:

- CONFIG_TWAI_ISR_IN_IRAM_LEGACY

- CONFIG_TWAI_SKIP_LEGACY_CONFLICT_CHECK
- CONFIG_TWAI_SUPPRESS_DEPRECATE_WARN

CONFIG_TWAI_ISR_IN_IRAM_LEGACY

Place TWAI ISR function in IRAM

Found in: Component config > Legacy Driver Configurations > Legacy TWAI Driver Configurations

Place the TWAI ISR in to IRAM to reduce latency and increase performance

Default value:

- No (disabled)

CONFIG_TWAI_SUPPRESS_DEPRECATE_WARN

Suppress legacy driver deprecated warning

Found in: Component config > Legacy Driver Configurations > Legacy TWAI Driver Configurations

whether to suppress the deprecation warnings when using legacy twai driver (driver/twai.h). If you want to continue using the legacy driver, and don't want to see related deprecation warnings, you can enable this option.

Default value:

- No (disabled)

CONFIG_TWAI_SKIP_LEGACY_CONFLICT_CHECK

Skip legacy driver conflict check

Found in: Component config > Legacy Driver Configurations > Legacy TWAI Driver Configurations

This configuration option used to bypass the conflict check mechanism with legacy code.

Default value:

- No (disabled)

Legacy I2C Driver Configurations

Contains:

- CONFIG_I2C_SKIP_LEGACY_CONFLICT_CHECK
- CONFIG_I2C_SUPPRESS_DEPRECATE_WARN

CONFIG_I2C_SUPPRESS_DEPRECATE_WARN

Suppress legacy driver deprecated warning

Found in: Component config > Legacy Driver Configurations > Legacy I2C Driver Configurations

whether to suppress the deprecation warnings when using legacy i2c driver (driver/i2c.h). If you want to continue using the legacy driver, and don't want to see related deprecation warnings, you can enable this option.

Default value:

- No (disabled)

CONFIG_I2C_SKIP_LEGACY_CONFLICT_CHECK

Skip legacy conflict check

Found in: Component config > Legacy Driver Configurations > Legacy I2C Driver Configurations

This configuration option allows the user to bypass the conflict check mechanism with legacy code.

Default value:

- No (disabled)

Legacy Touch Sensor Driver Configurations

Contains:

- CONFIG_TOUCH_SKIP_LEGACY_CONFLICT_CHECK
- CONFIG_TOUCH_SUPPRESS_DEPRECATE_WARN

CONFIG_TOUCH_SUPPRESS_DEPRECATE_WARN

Suppress legacy driver deprecated warning

Found in: Component config > Legacy Driver Configurations > Legacy Touch Sensor Driver Configurations

whether to suppress the deprecation warnings when using legacy touch sensor driver (driver/touch_sensor.h). If you want to continue using the legacy driver, and don't want to see related deprecation warnings, you can enable this option.

Default value:

- No (disabled)

CONFIG_TOUCH_SKIP_LEGACY_CONFLICT_CHECK

Skip legacy conflict check

Found in: Component config > Legacy Driver Configurations > Legacy Touch Sensor Driver Configurations

This configuration option allows the user to bypass the conflict check mechanism with legacy code.

Default value:

- No (disabled)

eFuse Bit Manager

Contains:

- CONFIG_EFUSE_CODE_SCHEME_SELECTOR
- CONFIG_EFUSE_VIRTUAL
- CONFIG_EFUSE_CUSTOM_TABLE

CONFIG_EFUSE_CUSTOM_TABLE

Use custom eFuse table

Found in: Component config > eFuse Bit Manager

Allows to generate a structure for eFuse from the CSV file.

Default value:

- No (disabled)

CONFIG_EFUSE_CUSTOM_TABLE_FILENAME

Custom eFuse CSV file

Found in: Component config > eFuse Bit Manager > CONFIG_EFUSE_CUSTOM_TABLE

Name of the custom eFuse CSV filename. This path is evaluated relative to the project root directory.

Default value:

- "main/esp_efuse_custom_table.csv" if CONFIG_EFUSE_CUSTOM_TABLE

CONFIG_EFUSE_VIRTUAL

Simulate eFuse operations in RAM

Found in: Component config > eFuse Bit Manager

If "n" - No virtual mode. All eFuse operations are real and use eFuse registers. If "y" - The virtual mode is enabled and all eFuse operations (read and write) are redirected to RAM instead of eFuse registers, all permanent changes (via eFuse) are disabled. Log output will state changes that would be applied, but they will not be.

If it is "y", then SECURE_FLASH_ENCRYPTION_MODE_RELEASE cannot be used. Because the EFUSE VIRT mode is for testing only.

During startup, the eFuses are copied into RAM. This mode is useful for fast tests.

Default value:

- No (disabled)

CONFIG_EFUSE_VIRTUAL_KEEP_IN_FLASH

Keep eFuses in flash

Found in: Component config > eFuse Bit Manager > CONFIG_EFUSE_VIRTUAL

In addition to the "Simulate eFuse operations in RAM" option, this option just adds a feature to keep eFuses after reboots in flash memory. To use this mode the partition_table should have the *efuse* partition. partition.csv: "efuse_em, data, efuse, , 0x2000,"

During startup, the eFuses are copied from flash or, in case if flash is empty, from real eFuse to RAM and then update flash. This mode is useful when need to keep changes after reboot (testing secure_boot and flash_encryption).

CONFIG_EFUSE_VIRTUAL_LOG_ALL_WRITES

Log all virtual writes

Found in: Component config > eFuse Bit Manager > CONFIG_EFUSE_VIRTUAL

If enabled, log efuse burns. This shows changes that would be made.

CONFIG_EFUSE_CODE_SCHEME_SELECTOR

Coding Scheme Compatibility

Found in: Component config > eFuse Bit Manager

Selector eFuse code scheme.

Available options:

- None Only (CONFIG_EFUSE_CODE_SCHEME_COMPAT_NONE)
- 3/4 and None (CONFIG_EFUSE_CODE_SCHEME_COMPAT_3_4)
- Repeat, 3/4 and None (common table does not support it) (CONFIG_EFUSE_CODE_SCHEME_COMPAT_REPEAT)

ESP-TLS

Contains:

- CONFIG_ESP_TLS_INSECURE
- CONFIG_ESP_TLS_SERVER_CERT_SELECT_HOOK
- CONFIG_ESP_TLS_LIBRARY_CHOOSE
- CONFIG_ESP_TLS_CLIENT_SESSION_TICKETS
- CONFIG_ESP_TLS_PSK_VERIFICATION

- CONFIG_ESP_TLS_SERVER_SESSION_TICKETS
- CONFIG_ESP_TLS_SERVER_MIN_AUTH_MODE_OPTIONAL
- CONFIG_ESP_TLS_USE_DS_PERIPHERAL
- CONFIG_ESP_TLS_USE_SECURE_ELEMENT

CONFIG_ESP_TLS_LIBRARY_CHOOSE

Choose SSL/TLS library for ESP-TLS (See help for more Info)

Found in: Component config > ESP-TLS

The ESP-TLS APIs support multiple backend TLS libraries. mbedTLS is supported by default. Custom TLS stacks can be registered via esp_tls_register_stack() API when CONFIG_ESP_TLS_CUSTOM_STACK is selected. Different TLS libraries may support different features and have different resource usage. Consult the ESP-TLS documentation in ESP-IDF Programming guide for more details.

Available options:

- mbedTLS (CONFIG_ESP_TLS_USING_MBEDTLS)
- Custom TLS stack (register via esp_tls_register_stack()) (CONFIG_ESP_TLS_CUSTOM_STACK)

  When selected, allows external components to register their own TLS stack implementation via esp_tls_register_stack() API. The custom stack must be registered before creating any TLS connections, otherwise TLS operations will fail.

  External components can provide any TLS stack implementation by implementing the esp_tls_stack_ops_t interface.

CONFIG_ESP_TLS_USE_SECURE_ELEMENT

Use Secure Element (ATECC608A) with ESP-TLS

Found in: Component config > ESP-TLS

Enable use of Secure Element for ESP-TLS, this enables internal support for ATECC608A peripheral, which can be used for TLS connection.

CONFIG_ESP_TLS_USE_DS_PERIPHERAL

Use Digital Signature (DS) Peripheral with ESP-TLS

Found in: Component config > ESP-TLS

Enable use of the Digital Signature Peripheral for ESP-TLS.The DS peripheral can only be used when it is appropriately configured for TLS. Consult the ESP-TLS documentation in ESP-IDF Programming Guide for more details.

Default value:

- Yes (enabled) if CONFIG_ESP_TLS_USING_MBEDTLS && SOC_DIG_SIGN_SUPPORTED

CONFIG_ESP_TLS_CLIENT_SESSION_TICKETS

Enable client session tickets

Found in: Component config > ESP-TLS

Enable session ticket support as specified in RFC5077.

CONFIG_ESP_TLS_SERVER_SESSION_TICKETS

Enable server session tickets

Found in: Component config > ESP-TLS

Enable session ticket support as specified in RFC5077

CONFIG_ESP_TLS_SERVER_SESSION_TICKET_TIMEOUT

Server session ticket timeout in seconds

Found in: Component config > ESP-TLS > CONFIG_ESP_TLS_SERVER_SESSION_TICKETS

Sets the session ticket timeout used in the tls server.

Default value:

- 86400 if CONFIG_ESP_TLS_SERVER_SESSION_TICKETS

CONFIG_ESP_TLS_SERVER_CERT_SELECT_HOOK

Certificate selection hook

Found in: Component config > ESP-TLS

Ability to configure and use a certificate selection callback during server handshake, to select a certificate to present to the client based on the TLS extensions supplied in the client hello (alpn, sni, etc).

CONFIG_ESP_TLS_SERVER_MIN_AUTH_MODE_OPTIONAL

ESP-TLS Server: Set minimum Certificate Verification mode to Optional

Found in: Component config > ESP-TLS

When this option is enabled, the ESP-TLS server can be configured to request client certificates optionally. This is done by setting the client_cert_authmode_optional field in the esp_https_server_config_t structure.

mbedtls_ssl_get_verify_result() can be called after the handshake is complete to retrieve status of verification of the client certificate, if presented.

Default value:

- No (disabled)

CONFIG_ESP_TLS_PSK_VERIFICATION

Enable PSK verification

Found in: Component config > ESP-TLS

Enable support for pre shared key ciphers, supported for both mbedTLS as well as custom TLS stacks.

CONFIG_ESP_TLS_INSECURE

Allow potentially insecure options

Found in: Component config > ESP-TLS

You can enable some potentially insecure options. These options should only be used for testing purposes. Only enable these options if you are very sure.

CONFIG_ESP_TLS_SKIP_SERVER_CERT_VERIFY

Skip server certificate verification by default (WARNING: ONLY FOR TESTING PURPOSE, READ HELP)

Found in: Component config > ESP-TLS > CONFIG_ESP_TLS_INSECURE

After enabling this option the esp-tls client will skip the server certificate verification by default. Note that this option will only modify the default behaviour of esp-tls client regarding server cert verification. The default behaviour should only be applicable when no other option regarding the server cert verification is opted in the esp-tls config (e.g. crt_bundle_attach, use_global_ca_store etc.). WARNING : Enabling this option comes with a potential risk of establishing a TLS connection with a server which has a fake identity, provided that the server certificate is not provided either through API or other mechanism like ca_store etc.

ADC and ADC Calibration

Contains:

- ADC Calibration Configurations
- CONFIG_ADC_CONTINUOUS_ISR_IRAM_SAFE
- CONFIG_ADC_DISABLE_DAC_OUTPUT
- CONFIG_ADC_ENABLE_DEBUG_LOG
- CONFIG_ADC_ONESHOT_CTRL_FUNC_IN_IRAM

CONFIG_ADC_ONESHOT_CTRL_FUNC_IN_IRAM

Place ISR version ADC oneshot mode read function into IRAM

Found in: Component config > ADC and ADC Calibration

Place ISR version ADC oneshot mode read function into IRAM.

Default value:

- No (disabled)

CONFIG_ADC_CONTINUOUS_ISR_IRAM_SAFE

ADC continuous mode driver ISR IRAM-Safe

Found in: Component config > ADC and ADC Calibration

Ensure the ADC continuous mode ISR is IRAM-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled)

ADC Calibration Configurations

Contains:

- CONFIG_ADC_CALI_EFUSE_VREF_ENABLE
- CONFIG_ADC_CALI_LUT_ENABLE
- CONFIG_ADC_CALI_EFUSE_TP_ENABLE

CONFIG_ADC_CALI_EFUSE_TP_ENABLE

Use Two Point Values

Found in: Component config > ADC and ADC Calibration > ADC Calibration Configurations

Some ESP32s have Two Point calibration values burned into eFuse BLOCK3. This option will allow the ADC calibration component to characterize the ADC-Voltage curve using Two Point values if they are available.

Default value:

- Yes (enabled)

CONFIG_ADC_CALI_EFUSE_VREF_ENABLE

Use eFuse Vref

Found in: Component config > ADC and ADC Calibration > ADC Calibration Configurations

Some ESP32s have Vref burned into eFuse BLOCK0. This option will allow the ADC calibration component to characterize the ADC-Voltage curve using eFuse Vref if it is available.

Default value:

- Yes (enabled)

CONFIG_ADC_CALI_LUT_ENABLE

Use Lookup Tables

Found in: Component config > ADC and ADC Calibration > ADC Calibration Configurations

This option will allow the ADC calibration component to use Lookup Tables to correct for non-linear behavior in 11db attenuation. Other attenuations do not exhibit non-linear behavior hence will not be affected by this option.

Default value:

- Yes (enabled)

CONFIG_ADC_DISABLE_DAC_OUTPUT

Disable DAC when ADC2 is in use

Found in: Component config > ADC and ADC Calibration

By default, this is set. The ADC oneshot driver will disable the output of the corresponding DAC channels: ESP32: IO25 and IO26 ESP32S2: IO17 and IO18

Disable this option so as to measure the output of DAC by internal ADC, for test usage.

Default value:

- Yes (enabled)

CONFIG_ADC_ENABLE_DEBUG_LOG

Enable ADC debug log

Found in: Component config > ADC and ADC Calibration

whether to enable the debug log message for ADC driver. Note that this option only controls the ADC driver log, will not affect other drivers.

Default value:

- No (disabled)

Wireless Coexistence

Contains:

- CONFIG_ESP_COEX_EXTERNAL_COEXIST_ENABLE
- CONFIG_ESP_COEX_GPIO_DEBUG
- CONFIG_ESP_COEX_SW_COEXIST_ENABLE
- CONFIG_ESP_COEX_POWER_MANAGEMENT

CONFIG_ESP_COEX_SW_COEXIST_ENABLE

Software controls WiFi/Bluetooth coexistence

Found in: Component config > Wireless Coexistence

If enabled, WiFi & Bluetooth coexistence is controlled by software rather than hardware. Recommended for heavy traffic scenarios. Both coexistence configuration options are automatically managed, no user intervention is required. If only Bluetooth is used, it is recommended to disable this option to reduce binary file size.

Default value:

- Yes (enabled) if CONFIG_BT_ENABLED || CONFIG_IEEE802154_ENABLED || (CONFIG_IEEE802154_ENABLED && CONFIG_BT_ENABLED)

CONFIG_ESP_COEX_EXTERNAL_COEXIST_ENABLE

External Coexistence

Found in: Component config > Wireless Coexistence

If enabled, HW External coexistence arbitration is managed by GPIO pins. It can support three types of wired combinations so far which are 1-wired/2-wired/3-wired. User can select GPIO pins in application code with configure interfaces.

Default value:

- No (disabled) if SOC_EXTERNAL_COEX_ADVANCE

CONFIG_ESP_COEX_POWER_MANAGEMENT

Support power management under coexistence

Found in: Component config > Wireless Coexistence

If enabled, coexist power management will be enabled.

Default value:

- No (disabled) if CONFIG_ESP_COEX_SW_COEXIST_ENABLE

CONFIG_ESP_COEX_GPIO_DEBUG

GPIO debugging for coexistence

Found in: Component config > Wireless Coexistence

Support coexistence GPIO debugging

CONFIG_ESP_COEX_GPIO_DEBUG_DIAG

Debugging Diagram

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Select type of debugging diagram

Available options:

- General (CONFIG_ESP_COEX_GPIO_DEBUG_DIAG_GENERAL)
- Wi-Fi (CONFIG_ESP_COEX_GPIO_DEBUG_DIAG_WIFI)

CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT

Max number of debugging GPIOs

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 12 if CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 12 if CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX0

Actual IO num for Debug IO ID0

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 0 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 15 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 0 && CONFIG_ESP_COEX_GPIO_DEBUG
- 1 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 0 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX1

Actual IO num for Debug IO ID1

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 1 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 2 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 1 && CONFIG_ESP_COEX_GPIO_DEBUG
- 2 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 1 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX2

Actual IO num for Debug IO ID2

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 2 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 0 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 2 && CONFIG_ESP_COEX_GPIO_DEBUG
- 3 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 2 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX3

Actual IO num for Debug IO ID3

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 3 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 4 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 3 && CONFIG_ESP_COEX_GPIO_DEBUG
- 4 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 3 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX4

Actual IO num for Debug IO ID4

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 4 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 5 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 4 && CONFIG_ESP_COEX_GPIO_DEBUG
- 5 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 4 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX5

Actual IO num for Debug IO ID5

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 5 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 18 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 5 && CONFIG_ESP_COEX_GPIO_DEBUG
- 6 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 5 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX6

Actual IO num for Debug IO ID6

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 6 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 19 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 6 && CONFIG_ESP_COEX_GPIO_DEBUG
- 7 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 6 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX7

Actual IO num for Debug IO ID7

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 7 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 22 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 7 && CONFIG_ESP_COEX_GPIO_DEBUG
- 8 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 7 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX8

Actual IO num for Debug IO ID8

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 8 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 13 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 8 && CONFIG_ESP_COEX_GPIO_DEBUG
- 9 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 8 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX9

Actual IO num for Debug IO ID9

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 9 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 12 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 9 && CONFIG_ESP_COEX_GPIO_DEBUG
- 10 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 9 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX10

Actual IO num for Debug IO ID10

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 10 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 14 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 10 && CONFIG_ESP_COEX_GPIO_DEBUG
- 11 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 10 && CONFIG_ESP_COEX_GPIO_DEBUG

CONFIG_ESP_COEX_GPIO_DEBUG_IO_IDX11

Actual IO num for Debug IO ID11

Found in: Component config > Wireless Coexistence > CONFIG_ESP_COEX_GPIO_DEBUG

Range:

- from 0 to 33 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 11 && CONFIG_ESP_COEX_GPIO_DEBUG

Default value:

- 27 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 11 && CONFIG_ESP_COEX_GPIO_DEBUG
- 12 if CONFIG_ESP_COEX_GPIO_DEBUG_IO_COUNT > 11 && CONFIG_ESP_COEX_GPIO_DEBUG

Common ESP-related

Contains:

- CONFIG_ESP_ERR_TO_NAME_LOOKUP

CONFIG_ESP_ERR_TO_NAME_LOOKUP

Enable lookup of error code strings

Found in: Component config > Common ESP-related

Functions esp_err_to_name() and esp_err_to_name_r() return string representations of error codes from a pre-generated lookup table. This option can be used to turn off the use of the look-up table in order to save memory but this comes at the price of sacrificing distinguishable (meaningful) output string representations.

Default value:

- Yes (enabled)

ESP-Driver:Analog Comparator Configurations

Contains:

- CONFIG_ANA_CMPR_ISR_CACHE_SAFE
- CONFIG_ANA_CMPR_ENABLE_DEBUG_LOG
- CONFIG_ANA_CMPR_CTRL_FUNC_IN_IRAM
- CONFIG_ANA_CMPR_ISR_HANDLER_IN_IRAM

CONFIG_ANA_CMPR_ISR_HANDLER_IN_IRAM

Place Analog Comparator ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:Analog Comparator Configurations

Place Analog Comparator ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled) if SOC_ANA_CMPR_SUPPORTED

CONFIG_ANA_CMPR_CTRL_FUNC_IN_IRAM

Place Analog Comparator control functions into IRAM

Found in: Component config > ESP-Driver:Analog Comparator Configurations

Place Analog Comparator control functions (like ana_cmpr_set_internal_reference) into IRAM, so that these functions can be IRAM-safe and able to be called in an IRAM interrupt context. Enabling this option can improve driver performance as well.

Default value:

- No (disabled) if SOC_ANA_CMPR_SUPPORTED

CONFIG_ANA_CMPR_ISR_CACHE_SAFE

Allow Analog Comparator ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:Analog Comparator Configurations

Enable this option to allow the Analog Comparator Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the comparator functionality is still required to operate correctly.

Default value:

- No (disabled) if SOC_ANA_CMPR_SUPPORTED

CONFIG_ANA_CMPR_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:Analog Comparator Configurations

If enabled, the driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled) if SOC_ANA_CMPR_SUPPORTED

ESP-Driver:BitScrambler Configurations

Contains:

- CONFIG_BITSCRAMBLER_CTRL_FUNC_IN_IRAM

CONFIG_BITSCRAMBLER_CTRL_FUNC_IN_IRAM

Place BitScrambler control functions in IRAM

Found in: Component config > ESP-Driver:BitScrambler Configurations

Place BitScrambler control functions into IRAM for better performance and fewer cache misses.

Default value:

- No (disabled) if SOC_BITSCRAMBLER_SUPPORTED

ESP-Driver:Camera Controller Configurations

Contains:

- CONFIG_CAM_CTLR_MIPI_CSI_ISR_CACHE_SAFE
- CONFIG_CAM_CTLR_DVP_CAM_ISR_CACHE_SAFE
- CONFIG_CAM_CTLR_ISP_DVP_ISR_CACHE_SAFE

CONFIG_CAM_CTLR_MIPI_CSI_ISR_CACHE_SAFE

CSI ISR Cache-Safe

Found in: Component config > ESP-Driver:Camera Controller Configurations

Ensure the CSI driver ISR is Cache-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled) if SOC_MIPI_CSI_SUPPORTED && (SOC_MIPI_CSI_SUPPORTED || SOC_ISP_DVP_SUPPORTED || SOC_LCDCAM_CAM_SUPPORTED)

CONFIG_CAM_CTLR_ISP_DVP_ISR_CACHE_SAFE

ISP_DVP ISR Cache-Safe

Found in: Component config > ESP-Driver:Camera Controller Configurations

Ensure the ISP_DVP driver ISR is Cache-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled) if SOC_ISP_DVP_SUPPORTED && (SOC_MIPI_CSI_SUPPORTED || SOC_ISP_DVP_SUPPORTED || SOC_LCDCAM_CAM_SUPPORTED)

CONFIG_CAM_CTLR_DVP_CAM_ISR_CACHE_SAFE

DVP ISR Cache-Safe

Found in: Component config > ESP-Driver:Camera Controller Configurations

Ensure the DVP driver ISR is Cache-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled) if SOC_LCDCAM_CAM_SUPPORTED && (SOC_MIPI_CSI_SUPPORTED || SOC_ISP_DVP_SUPPORTED || SOC_LCDCAM_CAM_SUPPORTED)

ESP-Driver:DAC Configurations

Contains:

- CONFIG_DAC_DMA_AUTO_16BIT_ALIGN

- CONFIG_DAC_ISR_IRAM_SAFE
- CONFIG_DAC_ENABLE_DEBUG_LOG
- CONFIG_DAC_CTRL_FUNC_IN_IRAM

CONFIG_DAC_CTRL_FUNC_IN_IRAM

Place DAC control functions into IRAM

Found in: Component config > ESP-Driver:DAC Configurations

Place DAC control functions (e.g. 'dac_oneshot_output_voltage') into IRAM, so that this function can be IRAM-safe and able to be called in the other IRAM interrupt context. Enabling this option can improve driver performance as well.

Default value:

- No (disabled)

CONFIG_DAC_ISR_IRAM_SAFE

DAC ISR IRAM-Safe

Found in: Component config > ESP-Driver:DAC Configurations

Ensure the DAC interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

CONFIG_DAC_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:DAC Configurations

whether to enable the debug log message for DAC driver. Note that, this option only controls the DAC driver log, won't affect other drivers.

Default value:

- No (disabled)

CONFIG_DAC_DMA_AUTO_16BIT_ALIGN

Align the continuous data to 16 bit automatically

Found in: Component config > ESP-Driver:DAC Configurations

Whether to left shift the continuous data to align every bytes to 16 bits in the driver. On ESP32, although the DAC resolution is only 8 bits, the hardware requires 16 bits data in continuous mode. By enabling this option, the driver will left shift 8 bits for the input data automatically. Only disable this option when you decide to do this step by yourself. Note that the driver will allocate a new piece of memory to save the converted data.

Default value:

- Yes (enabled)

GDMA Configurations

Contains:

- CONFIG_GDMA_ENABLE_DEBUG_LOG
- CONFIG_GDMA_ENABLE_WEIGHTED_ARBITRATION
- CONFIG_GDMA_CTRL_FUNC_IN_IRAM
- CONFIG_GDMA_ISR_HANDLER_IN_IRAM

CONFIG_GDMA_CTRL_FUNC_IN_IRAM

Place GDMA control functions in IRAM

Found in: Component config > GDMA Configurations

Place GDMA control functions (like start/stop/append/reset) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context.

Default value:

- No (disabled) if SOC_GDMA_SUPPORTED

CONFIG_GDMA_ISR_HANDLER_IN_IRAM

Place GDMA ISR handler in IRAM to reduce latency

Found in: Component config > GDMA Configurations

Place GDMA ISR handler functions in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled) if SOC_GDMA_SUPPORTED

CONFIG_GDMA_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > GDMA Configurations

If enabled, GDMA driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled) if SOC_GDMA_SUPPORTED

CONFIG_GDMA_ENABLE_WEIGHTED_ARBITRATION

GDMA enable weighted arbitration (Experimental)

Found in: Component config > GDMA Configurations

Whether to enable the weighted arbitration for GDMA driver. The default weight of each channel is 1. You need to set weight for each channel before transmissions. If this option is enabled, the buffer should be aligned to the burst size.

Default value:

- No (disabled) if SOC_GDMA_SUPPORT_WEIGHTED_ARBITRATION && CONFIG_IDF_EXPERIMENTAL_FEATURES && SOC_GDMA_SUPPORTED

DW_GDMA Configurations

Contains:

- CONFIG_DW_GDMA_ENABLE_DEBUG_LOG

CONFIG_DW_GDMA_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > DW_GDMA Configurations

Whether to enable the debug log message for DW_GDMA driver. Note that, this option only controls the DW_GDMA driver log, won't affect other drivers.

Default value:

- No (disabled) if SOC_DW_GDMA_SUPPORTED

2D-DMA Configurations

Contains:

- CONFIG_DMA2D_ISR_IRAM_SAFE
- CONFIG_DMA2D_OPERATION_FUNC_IN_IRAM

CONFIG_DMA2D_OPERATION_FUNC_IN_IRAM

Place 2D-DMA operation functions into IRAM

Found in: Component config > 2D-DMA Configurations

Place 2D-DMA all operation functions, including control functions (e.g. start/stop/append/reset) and setter functions (e.g. connect/strategy/callback registration) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context. It also helps optimizing the performance.

Default value:

- No (disabled) if SOC_DMA2D_SUPPORTED

CONFIG_DMA2D_ISR_IRAM_SAFE

2D-DMA ISR IRAM-Safe

Found in: Component config > 2D-DMA Configurations

This will ensure the 2D-DMA interrupt handler is IRAM-Safe, allow to avoid flash cache misses, and also be able to run whilst the cache is disabled. (e.g. SPI Flash write).

Default value:

- No (disabled) if SOC_DMA2D_SUPPORTED

ESP-Driver:GPIO Configurations

Contains:

- CONFIG_GPIO_CTRL_FUNC_IN_IRAM

CONFIG_GPIO_CTRL_FUNC_IN_IRAM

Place GPIO control functions into IRAM

Found in: Component config > ESP-Driver:GPIO Configurations

Place GPIO control functions (like intr_disable/set_level) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context.

Default value:

- No (disabled)

ESP-Driver:GPTimer Configurations

Contains:

- CONFIG_GPTIMER_ISR_CACHE_SAFE
- CONFIG_GPTIMER_ENABLE_DEBUG_LOG
- CONFIG_GPTIMER_CTRL_FUNC_IN_IRAM
- CONFIG_GPTIMER_ISR_HANDLER_IN_IRAM

CONFIG_GPTIMER_ISR_HANDLER_IN_IRAM

Place GPTimer ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:GPTimer Configurations

Place GPTimer ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled)

CONFIG_GPTIMER_CTRL_FUNC_IN_IRAM

Place GPTimer control functions in IRAM

Found in: Component config > ESP-Driver:GPTimer Configurations

Place GPTimer control functions (like start/stop) in IRAM, to reduce latency caused by cache miss. If enabled, these functions can also be called when cache is disabled.

Default value:

- No (disabled)

CONFIG_GPTIMER_ISR_CACHE_SAFE

Allow GPTimer ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:GPTimer Configurations

Enable this option to allow the GPTimer Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the GPTimer functionality is still required to operate correctly.

Default value:

- No (disabled)

CONFIG_GPTIMER_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:GPTimer Configurations

If enabled, GPTimer component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled)

ESP-Driver:I2C Configurations

Contains:

- CONFIG_I2C_ENABLE_DEBUG_LOG
- CONFIG_I2C_ISR_IRAM_SAFE
- CONFIG_I2C_MASTER_ISR_HANDLER_IN_IRAM

CONFIG_I2C_ISR_IRAM_SAFE

I2C ISR IRAM-Safe

Found in: Component config > ESP-Driver:I2C Configurations

Ensure the I2C interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write). note: This cannot be used in the I2C legacy driver.

Default value:

- No (disabled)

CONFIG_I2C_ENABLE_DEBUG_LOG

Enable I2C debug log

Found in: Component config > ESP-Driver:I2C Configurations

whether to enable the debug log message for I2C driver. Note that this option only controls the I2C driver log, will not affect other drivers.

Default value:

- No (disabled)

CONFIG_I2C_MASTER_ISR_HANDLER_IN_IRAM

Place I2C master ISR handler into IRAM

Found in: Component config > ESP-Driver:I2C Configurations

Place I2C master ISR handler into IRAM for better performance and fewer cache misses.

Default value:

- Yes (enabled)

ESP-Driver:I2S Configurations

Contains:

- CONFIG_I2S_ENABLE_DEBUG_LOG
- CONFIG_I2S_ISR_IRAM_SAFE
- CONFIG_I2S_CTRL_FUNC_IN_IRAM

CONFIG_I2S_ISR_IRAM_SAFE

I2S ISR IRAM-Safe

Found in: Component config > ESP-Driver:I2S Configurations

Ensure the I2S interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

CONFIG_I2S_CTRL_FUNC_IN_IRAM

Place I2S control functions into IRAM

Found in: Component config > ESP-Driver:I2S Configurations

Place I2S control functions into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context.

Default value:

- No (disabled)

CONFIG_I2S_ENABLE_DEBUG_LOG

Enable I2S debug log

Found in: Component config > ESP-Driver:I2S Configurations

whether to enable the debug log message for I2S driver. Note that, this option only controls the I2S driver log, will not affect other drivers.

Default value:

- No (disabled)

ESP-Driver:I3C Master Configurations

Contains:

- CONFIG_I3C_MASTER_ENABLE_DEBUG_LOG
- CONFIG_I3C_MASTER_ISR_CACHE_SAFE
- CONFIG_I3C_MASTER_ISR_HANDLER_IN_IRAM

CONFIG_I3C_MASTER_ISR_CACHE_SAFE

I3C ISR Cache-Safe

Found in: Component config > ESP-Driver:I3C Master Configurations

Ensure the I3C interrupt is Cache-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

CONFIG_I3C_MASTER_ENABLE_DEBUG_LOG

Enable I3C debug log

Found in: Component config > ESP-Driver:I3C Master Configurations

whether to enable the debug log message for I3C driver. Note that this option only controls the I3C driver log, will not affect other drivers.

Default value:

- No (disabled)

CONFIG_I3C_MASTER_ISR_HANDLER_IN_IRAM

Place I3C master ISR handler into IRAM

Found in: Component config > ESP-Driver:I3C Master Configurations

Place I3C master ISR handler into IRAM for better performance and fewer cache misses.

Default value:

- No (disabled)

ESP-Driver:ISP Configurations

Contains:

- CONFIG_ISP_ISR_IRAM_SAFE
- CONFIG_ISP_CTRL_FUNC_IN_IRAM

CONFIG_ISP_ISR_IRAM_SAFE

ISP driver ISR IRAM-Safe

Found in: Component config > ESP-Driver:ISP Configurations

Ensure the ISP driver ISR is IRAM-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled) if SOC_ISP_SUPPORTED

CONFIG_ISP_CTRL_FUNC_IN_IRAM

Place ISP control functions into IRAM

Found in: Component config > ESP-Driver:ISP Configurations

Place ISP control functions into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context. Enabling this option can improve driver performance as well.

Function list: - *esp_isp_sharpen_configure*

Default value:

- No (disabled) if SOC_ISP_SUPPORTED

ESP-Driver:JPEG-Codec Configurations

Contains:

- CONFIG_JPEG_ENABLE_DEBUG_LOG

CONFIG_JPEG_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:JPEG-Codec Configurations

whether to enable the debug log message for JPEG driver. Note that, this option only controls the JPEG driver log, won't affect other drivers. Please also note, enable this option will make jpeg codec process speed much slower.

Default value:

- No (disabled) if SOC_JPEG_CODEC_SUPPORTED

ESP-Driver:LEDC Configurations

Contains:

- CONFIG_LEDC_CTRL_FUNC_IN_IRAM

CONFIG_LEDC_CTRL_FUNC_IN_IRAM

Place LEDC control functions into IRAM

Found in: Component config > ESP-Driver:LEDC Configurations

Place LEDC control functions (ledc_update_duty and ledc_stop) into IRAM, so that these functions can be IRAM-safe and able to be called in an IRAM context. Enabling this option can improve driver performance as well.

Default value:

- No (disabled)

ESP-Driver:MCPWM Configurations

Contains:

- CONFIG_MCPWM_ISR_CACHE_SAFE
- CONFIG_MCPWM_ENABLE_DEBUG_LOG
- CONFIG_MCPWM_CTRL_FUNC_IN_IRAM
- CONFIG_MCPWM_ISR_HANDLER_IN_IRAM

CONFIG_MCPWM_ISR_HANDLER_IN_IRAM

Place MCPWM ISR handler into IRAM to reduce latency

Found in: Component config > ESP-Driver:MCPWM Configurations

Place MCPWM ISR handler(s) in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled)

CONFIG_MCPWM_ISR_CACHE_SAFE

Allow MCPWM ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:MCPWM Configurations

Enable this option to allow the MCPWM Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the MCPWM functionality is still required to operate correctly.

Default value:

- No (disabled)

CONFIG_MCPWM_CTRL_FUNC_IN_IRAM

Place MCPWM control functions in IRAM

Found in: Component config > ESP-Driver:MCPWM Configurations

Place MCPWM control functions in IRAM, to reduce latency caused by cache miss. If enabled, these functions can also be called when cache is disabled.

Default value:

- No (disabled)

CONFIG_MCPWM_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:MCPWM Configurations

If enabled, MCPWM component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option with caution, as it will increase the binary size.

Default value:

- No (disabled)

ESP-Driver:Parallel IO Configurations

Contains:

- CONFIG_PARLIO_RX_ISR_CACHE_SAFE
- CONFIG_PARLIO_TX_ISR_CACHE_SAFE
- CONFIG_PARLIO_ENABLE_DEBUG_LOG

- CONFIG_PARLIO_ISR_IRAM_SAFE
- CONFIG_PARLIO_RX_ISR_HANDLER_IN_IRAM
- CONFIG_PARLIO_TX_ISR_HANDLER_IN_IRAM

CONFIG_PARLIO_TX_ISR_HANDLER_IN_IRAM

Place Parallel IO TX ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:Parallel IO Configurations

Place Parallel IO TX ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled) if SOC_PARLIO_SUPPORTED

CONFIG_PARLIO_RX_ISR_HANDLER_IN_IRAM

Place Parallel IO RX ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:Parallel IO Configurations

Place Parallel IO RX ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled) if SOC_PARLIO_SUPPORTED

CONFIG_PARLIO_TX_ISR_CACHE_SAFE

Allow Parallel IO TX ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:Parallel IO Configurations

Enable this option to allow the Parallel IO TX Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the Parallel IO TX functionality is still required to operate correctly.

Default value:

- No (disabled) if SOC_PARLIO_SUPPORTED

CONFIG_PARLIO_RX_ISR_CACHE_SAFE

Allow Parallel IO RX ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:Parallel IO Configurations

Enable this option to allow the Parallel IO RX Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the Parallel IO RX functionality is still required to operate correctly.

Default value:

- No (disabled) if SOC_PARLIO_SUPPORTED

CONFIG_PARLIO_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:Parallel IO Configurations

If enabled, Parallel IO driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled) if SOC_PARLIO_SUPPORTED

CONFIG_PARLIO_ISR_IRAM_SAFE

Parallel IO ISR IRAM-Safe (Deprecated)

Found in: Component config > ESP-Driver:Parallel IO Configurations

Ensure the Parallel IO interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled) if SOC_PARLIO_SUPPORTED

ESP-Driver:PCNT Configurations

Contains:

- CONFIG_PCNT_ENABLE_DEBUG_LOG
- CONFIG_PCNT_ISR_IRAM_SAFE
- CONFIG_PCNT_CTRL_FUNC_IN_IRAM

CONFIG_PCNT_CTRL_FUNC_IN_IRAM

Place PCNT control functions into IRAM

Found in: Component config > ESP-Driver:PCNT Configurations

Place PCNT control functions (like start/stop) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context. Enabling this option can improve driver performance as well.

Default value:

- No (disabled)

CONFIG_PCNT_ISR_IRAM_SAFE

PCNT ISR IRAM-Safe

Found in: Component config > ESP-Driver:PCNT Configurations

Ensure the PCNT interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

CONFIG_PCNT_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:PCNT Configurations

whether to enable the debug log message for PCNT driver. Note that, this option only controls the PCNT driver log, won't affect other drivers.

Default value:

- No (disabled)

ESP-Driver:RMT Configurations

Contains:

- CONFIG_RMT_RX_ISR_CACHE_SAFE
- CONFIG_RMT_TX_ISR_CACHE_SAFE

- CONFIG_RMT_ENABLE_DEBUG_LOG
- CONFIG_RMT_ENCODER_FUNC_IN_IRAM
- CONFIG_RMT_RECV_FUNC_IN_IRAM
- CONFIG_RMT_RX_ISR_HANDLER_IN_IRAM
- CONFIG_RMT_TX_ISR_HANDLER_IN_IRAM
- CONFIG_RMT_ISR_IRAM_SAFE

CONFIG_RMT_ENCODER_FUNC_IN_IRAM

Place RMT encoder function in IRAM

Found in: Component config > ESP-Driver:RMT Configurations

Place RMT encoder function into IRAM for better performance and fewer cache misses.

Default value:

- Yes (enabled)

CONFIG_RMT_TX_ISR_HANDLER_IN_IRAM

Place RMT TX ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:RMT Configurations

Place RMT TX ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled)

CONFIG_RMT_RX_ISR_HANDLER_IN_IRAM

Place RMT RX ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:RMT Configurations

Place RMT RX ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled)

CONFIG_RMT_RECV_FUNC_IN_IRAM

Place RMT receive function in IRAM

Found in: Component config > ESP-Driver:RMT Configurations

Place RMT receive function into IRAM for better performance and fewer cache misses.

Default value:

- No (disabled)

CONFIG_RMT_TX_ISR_CACHE_SAFE

Allow RMT TX ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:RMT Configurations

Enable this option to allow the RMT TX Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the RMT TX functionality is still required to operate correctly.

Default value:

- No (disabled)

CONFIG_RMT_RX_ISR_CACHE_SAFE

Allow RMT RX ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:RMT Configurations

Enable this option to allow the RMT RX Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the RMT RX functionality is still required to operate correctly.

Default value:

- No (disabled)

CONFIG_RMT_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:RMT Configurations

If enabled, RMT driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled)

CONFIG_RMT_ISR_IRAM_SAFE

RMT ISR IRAM-Safe (Deprecated)

Found in: Component config > ESP-Driver:RMT Configurations

Ensure the RMT interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

ESP-Driver:Sigma Delta Modulator Configurations

Contains:

- CONFIG_SDM_ENABLE_DEBUG_LOG
- CONFIG_SDM_CTRL_FUNC_IN_IRAM

CONFIG_SDM_CTRL_FUNC_IN_IRAM

Place SDM control functions into IRAM

Found in: Component config > ESP-Driver:Sigma Delta Modulator Configurations

Place SDM control functions (like set_pulse_density) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context. Enabling this option can improve driver performance as well.

Default value:

- No (disabled)

CONFIG_SDM_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:Sigma Delta Modulator Configurations

If enabled, SDM driver will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled)

ESP-Driver:SD Host SDMMC Controller Configurations

Contains:

- CONFIG_SD_HOST_SDMMC_ISR_CACHE_SAFE

CONFIG_SD_HOST_SDMMC_ISR_CACHE_SAFE

SD Host ISR Cache-Safe

Found in: Component config > ESP-Driver:SD Host SDMMC Controller Configurations

Ensure the SD Host SDMMC driver ISR is Cache-Safe. When enabled, the ISR handler will be available when the cache is disabled.

Default value:

- No (disabled)

ESP-Driver:SPI Configurations

Contains:

- CONFIG_SPI_MASTER_ISR_IN_IRAM
- CONFIG_SPI_SLAVE_ISR_IN_IRAM
- CONFIG_SPI_MASTER_IN_IRAM
- CONFIG_SPI_SLAVE_IN_IRAM

CONFIG_SPI_MASTER_IN_IRAM

Place transmitting functions of SPI master into IRAM

Found in: Component config > ESP-Driver:SPI Configurations

Normally only the ISR of SPI master is placed in the IRAM, so that it can work without the flash when interrupt is triggered. For other functions, there's some possibility that the flash cache miss when running inside and out of SPI functions, which may increase the interval of SPI transactions. Enable this to put `queue\_trans`, `get\_trans\_result` and `transmit` functions into the IRAM to avoid possible cache miss.

This configuration won't be available if *CONFIG_FREERTOS_IN_IRAM* is disabled.

During unit test, this is enabled to measure the ideal case of api.

Default value:

- No (disabled) if CONFIG_FREERTOS_IN_IRAM

CONFIG_SPI_MASTER_ISR_IN_IRAM

Place SPI master ISR function into IRAM

Found in: Component config > ESP-Driver:SPI Configurations

Place the SPI master ISR in to IRAM to avoid possible cache miss.

Enabling this configuration is possible only when HEAP_PLACE_FUNCTION_INTO_FLASH is disabled since the spi master uses can allocate transactions buffers into DMA memory section using the heap component API that ipso facto has to be placed in IRAM.

Also you can forbid the ISR being disabled during flash writing access, by add ESP_INTR_FLAG_IRAM when initializing the driver.

CONFIG_SPI_SLAVE_IN_IRAM

Place transmitting functions of SPI slave into IRAM

Found in: Component config > ESP-Driver:SPI Configurations

Normally only the ISR of SPI slave is placed in the IRAM, so that it can work without the flash when interrupt is triggered. For other functions, there is some possibility that the flash cache miss when running inside and out of SPI functions, which may increase the interval of SPI transactions. Enable this to put `queue\_trans`, `get\_trans\_result` and `transmit` functions into the IRAM to avoid possible cache miss.

Default value:

- No (disabled)

CONFIG_SPI_SLAVE_ISR_IN_IRAM

Place SPI slave ISR function into IRAM

Found in: Component config > ESP-Driver:SPI Configurations

Place the SPI slave ISR in to IRAM to avoid possible cache miss.

Also you can forbid the ISR being disabled during flash writing access, by add ESP_INTR_FLAG_IRAM when initializing the driver.

Default value:

- Yes (enabled)

ESP-Driver:Touch Sensor Configurations

Contains:

- CONFIG_TOUCH_ENABLE_DEBUG_LOG
- CONFIG_TOUCH_CTRL_FUNC_IN_IRAM
- CONFIG_TOUCH_SKIP_FSM_CHECK
- CONFIG_TOUCH_ISR_IRAM_SAFE

CONFIG_TOUCH_CTRL_FUNC_IN_IRAM

Place touch sensor control functions into IRAM

Found in: Component config > ESP-Driver:Touch Sensor Configurations

Place touch sensor oneshot scanning and continuous scanning functions into IRAM, so that these function can be IRAM-safe and able to be called when the flash cache is disabled. Enabling this option can improve driver performance as well.

Default value:

- No (disabled)

CONFIG_TOUCH_ISR_IRAM_SAFE

Touch sensor ISR IRAM-Safe

Found in: Component config > ESP-Driver:Touch Sensor Configurations

Ensure the touch sensor interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled)

CONFIG_TOUCH_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:Touch Sensor Configurations

Whether to enable the debug log message for touch driver. Note that, this option only controls the touch driver log, won't affect other drivers.

Default value:

- No (disabled)

CONFIG_TOUCH_SKIP_FSM_CHECK

Skip the FSM check

Found in: Component config > ESP-Driver:Touch Sensor Configurations

If skipped the FSM check, the driver will allow to re-configure the touch sensor during the runtime. It is mainly used to tune the parameters in the runtime to find a proper set of touch parameters. CAUTION: Not suggest to enable this option in the final product, it might cause false triggering which is not safe in the actual environment.

Default value:

- No (disabled)

ESP-Driver:Temperature Sensor Configurations

Contains:

- CONFIG_TEMP_SENSOR_ENABLE_DEBUG_LOG
- CONFIG_TEMP_SENSOR_ISR_IRAM_SAFE

CONFIG_TEMP_SENSOR_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:Temperature Sensor Configurations

whether to enable the debug log message for temperature sensor driver. Note that, this option only controls the temperature sensor driver log, won't affect other drivers.

Default value:

- No (disabled) if SOC_TEMP_SENSOR_SUPPORTED

CONFIG_TEMP_SENSOR_ISR_IRAM_SAFE

Temperature sensor ISR IRAM-Safe

Found in: Component config > ESP-Driver:Temperature Sensor Configurations

Ensure the Temperature Sensor interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write).

Default value:

- No (disabled) if SOC_TEMPERATURE_SENSOR_INTR_SUPPORT && SOC_TEMP_SENSOR_SUPPORTED

ESP-Driver:TWAI Configurations

Contains:

- CONFIG_TWAI_ISR_CACHE_SAFE
- CONFIG_TWAI_ENABLE_DEBUG_LOG
- CONFIG_TWAI_IO_FUNC_IN_IRAM
- CONFIG_TWAI_ISR_IN_IRAM

CONFIG_TWAI_ISR_IN_IRAM

Place TWAI ISR function in IRAM

Found in: Component config > ESP-Driver:TWAI Configurations

Place the TWAI ISR in to IRAM to reduce latency and increase performance

Default value:

- No (disabled)

CONFIG_TWAI_IO_FUNC_IN_IRAM

Place TWAI I/O functions in IRAM

Found in: Component config > ESP-Driver:TWAI Configurations

Place certain TWAI I/O functions (like twai_transmit) in IRAM to reduce latency

Default value:

- No (disabled)

CONFIG_TWAI_ISR_CACHE_SAFE

Allow TWAI ISR execute when cache disabled

Found in: Component config > ESP-Driver:TWAI Configurations

Allow TWAI works under Cache disabled (such as when writing to SPI Flash), to enabled this config, all callbacks and user_ctx should also place in IRAM

Default value:

- No (disabled)

CONFIG_TWAI_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:TWAI Configurations

If enabled, TWAI driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled)

ESP-Driver:UART Configurations

Contains:

- CONFIG_UART_ISR_IN_IRAM

CONFIG_UART_ISR_IN_IRAM

Place UART ISR function into IRAM

Found in: Component config > ESP-Driver:UART Configurations

If this option is not selected, UART interrupt will be disabled for a long time and may cause data lost when doing spi flash operation.

ESP-Driver:UHCI Configurations

Contains:

- CONFIG_UHCI_ISR_CACHE_SAFE
- CONFIG_UHCI_ENABLE_DEBUG_LOG
- CONFIG_UHCI_ISR_HANDLER_IN_IRAM

CONFIG_UHCI_ISR_HANDLER_IN_IRAM

Place UHCI ISR function into IRAM

Found in: Component config > ESP-Driver:UHCI Configurations

If this option is not selected, UHCI interrupt will be disabled for a long time and may cause data lost when doing spi flash operation.

Default value:

- No (disabled)

CONFIG_UHCI_ISR_CACHE_SAFE

Allow UHCI ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:UHCI Configurations

Enable this option to allow the ISR for UHCI to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the UHCI functionality is still required to operate correctly.

Default value:

- No (disabled)

CONFIG_UHCI_ENABLE_DEBUG_LOG

Enable debug log

Found in: Component config > ESP-Driver:UHCI Configurations

whether to enable the debug log message for UHCI driver. Note that, this option only controls the UHCI driver log, won't affect other drivers.

Default value:

- No (disabled)

ESP-Driver:USB Serial/JTAG Configuration

Contains:

- CONFIG_USJ_ENABLE_USB_SERIAL_JTAG

CONFIG_USJ_ENABLE_USB_SERIAL_JTAG

Enable USB-Serial-JTAG Module

Found in: Component config > ESP-Driver:USB Serial/JTAG Configuration

The USB-Serial-JTAG module on ESP chips is turned on by default after power-on. If your application does not need it and not rely on it to be used as system console or use the built-in JTAG for debugging, you can disable this option, then the clock of this module will be disabled at startup, which will save some power consumption. Note: This will disable USB-Serial-JTAG only in the application; it will still be enabled during the bootloader stage, so you can continue to use it to flash the chip in download mode. To disable it completely, you must burn the EFUSE_DIS_USB_SERIAL_JTAG eFuse.

Default value:

- Yes (enabled) if SOC_USB_SERIAL_JTAG_SUPPORTED

CONFIG_USJ_NO_AUTO_LS_ON_CONNECTION

Don't enter the automatic light sleep when USB Serial/JTAG port is connected

Found in: Component config > ESP-Driver:USB Serial/JTAG Configuration > CONFIG_USJ_ENABLE_USB_SERIAL_JTAG

If enabled, the chip will constantly monitor the connection status of the USB Serial/JTAG port. As long as the USB Serial/JTAG is connected, a ESP_PM_NO_LIGHT_SLEEP power management lock will be acquired to prevent the system from entering light sleep. This option can be useful if serial monitoring is needed via USB Serial/JTAG while power management is enabled, as the USB Serial/JTAG cannot work under light sleep and after waking up from light sleep. Note. This option can only control the automatic Light-Sleep behavior. If esp_light_sleep_start() is called manually from the program, enabling this option will not prevent light sleep entry even if the USB Serial/JTAG is in use.

Ethernet

Contains:

- CONFIG_ETH_TRANSMIT_MUTEX
- CONFIG_ETH_USE_ESP32_EMAC
- CONFIG_ETH_USE_OPENETH
- CONFIG_ETH_USE_SPI_ETHERNET

CONFIG_ETH_USE_ESP32_EMAC

Support ESP32 internal EMAC controller

Found in: Component config > Ethernet

ESP32 integrates a 10/100M Ethernet MAC controller.

Default value:

- Yes (enabled)

Contains:

- CONFIG_ETH_DMA_RX_BUFFER_NUM
- CONFIG_ETH_DMA_TX_BUFFER_NUM
- CONFIG_ETH_IRAM_OPTIMIZATION
- CONFIG_ETH_SOFT_FLOW_CONTROL
- CONFIG_ETH_DMA_BUFFER_SIZE

CONFIG_ETH_DMA_BUFFER_SIZE

Ethernet DMA buffer size (Byte)

Found in: Component config > Ethernet > CONFIG_ETH_USE_ESP32_EMAC

Set the size of each buffer used by Ethernet MAC DMA. !! Important !! Make sure it is 64B aligned for ESP32P4!

Range:

- from 256 to 1600

Default value:

- 512

CONFIG_ETH_DMA_RX_BUFFER_NUM

Amount of Ethernet DMA Rx buffers

Found in: Component config > Ethernet > CONFIG_ETH_USE_ESP32_EMAC

Number of DMA receive buffers. Each buffer's size is ETH_DMA_BUFFER_SIZE. Larger number of buffers could increase throughput somehow.

Range:

- from 3 to 30

Default value:

- 10

CONFIG_ETH_DMA_TX_BUFFER_NUM

Amount of Ethernet DMA Tx buffers

Found in: Component config > Ethernet > CONFIG_ETH_USE_ESP32_EMAC

Number of DMA transmit buffers. Each buffer's size is ETH_DMA_BUFFER_SIZE. Larger number of buffers could increase throughput somehow.

Range:

- from 3 to 30

Default value:

- 10

CONFIG_ETH_SOFT_FLOW_CONTROL

Enable software flow control

Found in: Component config > Ethernet > CONFIG_ETH_USE_ESP32_EMAC

Ethernet MAC engine on ESP32 doesn't feature a flow control logic. The MAC driver can perform a software flow control if you enable this option. Note that, if the RX buffer number is small, enabling software flow control will cause obvious performance loss.

Default value:

- No (disabled) if CONFIG_ETH_DMA_RX_BUFFER_NUM > 15 && CONFIG_ETH_USE_ESP32_EMAC

CONFIG_ETH_IRAM_OPTIMIZATION

Enable IRAM optimization

Found in: Component config > Ethernet > CONFIG_ETH_USE_ESP32_EMAC

If enabled, functions related to RX/TX are placed into IRAM. It can improve Ethernet throughput. If disabled, all functions are placed into FLASH.

Default value:

- No (disabled)

CONFIG_ETH_USE_SPI_ETHERNET

Support SPI to Ethernet Module

Found in: Component config > Ethernet

ESP-IDF can also support SPI-Ethernet. Actual chip drivers are available as components in Component Registry.

Default value:

- Yes (enabled)

CONFIG_ETH_USE_OPENETH

Support OpenCores Ethernet MAC (for use with QEMU)

Found in: Component config > Ethernet

OpenCores Ethernet MAC driver can be used when an ESP-IDF application is executed in QEMU. This driver is not supported when running on a real chip.

Default value:

- No (disabled)

Contains:

- CONFIG_ETH_OPENETH_DMA_RX_BUFFER_NUM
- CONFIG_ETH_OPENETH_DMA_TX_BUFFER_NUM

CONFIG_ETH_OPENETH_DMA_RX_BUFFER_NUM

Number of Ethernet DMA Rx buffers

Found in: Component config > Ethernet > CONFIG_ETH_USE_OPENETH

Number of DMA receive buffers, each buffer is 1600 bytes.

Range:

- from 1 to 64 if CONFIG_ETH_USE_OPENETH

Default value:

- 4 if CONFIG_ETH_USE_OPENETH

CONFIG_ETH_OPENETH_DMA_TX_BUFFER_NUM

Number of Ethernet DMA Tx buffers

Found in: Component config > Ethernet > CONFIG_ETH_USE_OPENETH

Number of DMA transmit buffers, each buffer is 1600 bytes.

Range:

- from 1 to 64 if CONFIG_ETH_USE_OPENETH

Default value:

- 1 if CONFIG_ETH_USE_OPENETH

CONFIG_ETH_TRANSMIT_MUTEX

Enable Transmit Mutex

Found in: Component config > Ethernet

Prevents multiple accesses when Ethernet interface is used as shared resource and multiple functionalities might try to access it at a time.

Default value:

- No (disabled)

Event Loop Library

Contains:

- CONFIG_ESP_EVENT_LOOP_PROFILING

- CONFIG_ESP_EVENT_POST_FROM_ISR

CONFIG_ESP_EVENT_LOOP_PROFILING

Enable event loop profiling

Found in: Component config > Event Loop Library

Enables collections of statistics in the event loop library such as the number of events posted to/recieved by an event loop, number of callbacks involved, number of events dropped to to a full event loop queue, run time of event handlers, and number of times/run time of each event handler.

Default value:

- No (disabled)

CONFIG_ESP_EVENT_POST_FROM_ISR

Support posting events from ISRs

Found in: Component config > Event Loop Library

Enable posting events from interrupt handlers.

Default value:

- Yes (enabled)

CONFIG_ESP_EVENT_POST_FROM_IRAM_ISR

Support posting events from ISRs placed in IRAM

Found in: Component config > Event Loop Library > CONFIG_ESP_EVENT_POST_FROM_ISR

Enable posting events from interrupt handlers placed in IRAM. Enabling this option places API functions esp_event_post and esp_event_post_to in IRAM.

Default value:

- Yes (enabled)

GDB Stub

Contains:

- CONFIG_ESP_GDBSTUB_SUPPORT_TASKS
- CONFIG_ESP_SYSTEM_GDBSTUB_RUNTIME

CONFIG_ESP_SYSTEM_GDBSTUB_RUNTIME

GDBStub at runtime

Found in: Component config > GDB Stub

Enable builtin GDBStub. This allows to debug the target device using serial port: - Run 'idf.py monitor'. - Wait for the device to initialize. - Press Ctrl+C to interrupt the execution and enter GDB attached to your device for debugging. NOTE: all UART input will be handled by GDBStub.

CONFIG_ESP_GDBSTUB_SUPPORT_TASKS

Enable listing FreeRTOS tasks through GDB Stub

Found in: Component config > GDB Stub

If enabled, GDBStub can supply the list of FreeRTOS tasks to GDB. Thread list can be queried from GDB using 'info threads' command. Note that if GDB task lists were corrupted, this feature may not work. If GDBStub fails, try disabling this feature.

Default value:

- Yes (enabled)

CONFIG_ESP_GDBSTUB_MAX_TASKS

Maximum number of tasks supported by GDB Stub

Found in: Component config > GDB Stub > CONFIG_ESP_GDBSTUB_SUPPORT_TASKS

Set the number of tasks which GDB Stub will support.

Default value:

- 32

ESP HID

Contains:

- CONFIG_ESPHID_TASK_SIZE_BLE
- CONFIG_ESPHID_TASK_SIZE_BT

CONFIG_ESPHID_TASK_SIZE_BT

Task stack size for ESP HID BR/EDR

Found in: Component config > ESP HID

This is the stack size for the BT HID task. Default is 2048 bytes.

Range:

- from 2048 to 10240

Default value:

- 2048

CONFIG_ESPHID_TASK_SIZE_BLE

Task stack size for ESP HID BLE

Found in: Component config > ESP HID

This is the stack size for the BLE HID task. Default is 4096 bytes.

Range:

- from 2048 to 10240

Default value:

- 4096

ESP HTTP client

Contains:

- CONFIG_ESP_HTTP_CLIENT_ENABLE_GET_CONTENT_RANGE
- CONFIG_ESP_HTTP_CLIENT_ENABLE_CUSTOM_TRANSPORT
- CONFIG_ESP_HTTP_CLIENT_ENABLE_BASIC_AUTH
- CONFIG_ESP_HTTP_CLIENT_ENABLE_DIGEST_AUTH
- CONFIG_ESP_HTTP_CLIENT_ENABLE_HTTPS
- CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS
- CONFIG_ESP_HTTP_CLIENT_EVENT_POST_TIMEOUT

CONFIG_ESP_HTTP_CLIENT_ENABLE_HTTPS

Enable https

Found in: Component config > ESP HTTP client

This option will enable https protocol by linking esp-tls library and initializing SSL transport

Default value:

- Yes (enabled)

CONFIG_ESP_HTTP_CLIENT_ENABLE_BASIC_AUTH

Enable HTTP Basic Authentication

Found in: Component config > ESP HTTP client

This option will enable HTTP Basic Authentication. It is disabled by default as Basic auth uses unencrypted encoding, so it introduces a vulnerability when not using TLS

Default value:

- No (disabled)

CONFIG_ESP_HTTP_CLIENT_ENABLE_DIGEST_AUTH

Enable HTTP Digest Authentication

Found in: Component config > ESP HTTP client

This option will enable HTTP Digest Authentication. It is enabled by default, but use of this configuration is not recommended as the password can be derived from the exchange, so it introduces a vulnerability when not using TLS

Default value:

- No (disabled)

CONFIG_ESP_HTTP_CLIENT_ENABLE_CUSTOM_TRANSPORT

Enable custom transport

Found in: Component config > ESP HTTP client

This option will enable injection of a custom tcp_transport handle, so the http operation will be performed on top of the user defined transport abstraction (if configured)

Default value:

- No (disabled)

CONFIG_ESP_HTTP_CLIENT_ENABLE_GET_CONTENT_RANGE

Enable content range functionality

Found in: Component config > ESP HTTP client

This enables use of content range header in esp_http_client component.

Default value:

- No (disabled)

CONFIG_ESP_HTTP_CLIENT_EVENT_POST_TIMEOUT

Time in millisecond to wait for posting event

Found in: Component config > ESP HTTP client

This config option helps in setting the time in millisecond to wait for event to be posted to the system default event loop. Set it to -1 if you need to set timeout to portMAX_DELAY.

Default value:

- 2000

CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS

Save response headers

Found in: Component config > ESP HTTP client

This option will enable saving of response headers in esp_http_client component. When enabled, upto ESP_HTTP_CLIENT_MAX_SAVED_RESPONSE_HEADERS headers are saved. Enabling this option will increase the memory footprint of the component.

Default value:

- No (disabled)

CONFIG_ESP_HTTP_CLIENT_MAX_SAVED_RESPONSE_HEADERS

Maximum number of response headers

Found in: Component config > ESP HTTP client > CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS

This config option helps in setting the maximum number of response headers that can be saved in esp_http_client component.

Default value:

- 10 if CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS

CONFIG_ESP_HTTP_CLIENT_MAX_RESPONSE_HEADER_SIZE

Maximum size of response header

Found in: Component config > ESP HTTP client > CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS

This config option helps in setting the maximum size of response header that can be saved in esp_http_client component. Note that the same size is used for key and value in the response headers.

Default value:

- 128 if CONFIG_ESP_HTTP_CLIENT_SAVE_RESPONSE_HEADERS

HTTP Server

Contains:

- CONFIG_HTTPD_MAX_REQ_HDR_LEN

- CONFIG_HTTPD_QUEUE_WORK_BLOCKING
- CONFIG_HTTPD_PURGE_BUF_LEN
- CONFIG_HTTPD_LOG_PURGE_DATA
- CONFIG_HTTPD_MAX_URI_LEN
- CONFIG_HTTPD_SERVER_EVENT_POST_TIMEOUT
- CONFIG_HTTPD_ERR_RESP_NO_DELAY
- CONFIG_HTTPD_WS_POST_HANDSHAKE_CB_SUPPORT
- CONFIG_HTTPD_WS_PRE_HANDSHAKE_CB_SUPPORT
- CONFIG_HTTPD_WS_SUPPORT

CONFIG_HTTPD_MAX_REQ_HDR_LEN

HTTP Request Header Length limit

Found in: Component config > HTTP Server

This sets the default limit for the HTTP request header length. The limit can be configured at run time by setting max_req_hdr_len member of httpd_config_t structure. The memory allocated will depend on the actual header length. Hence keeping a sufficiently large max header length is recommended.

Range:

- from 128 to 65536

Default value:

- 1024

CONFIG_HTTPD_MAX_URI_LEN

Max HTTP URI Length

Found in: Component config > HTTP Server

This sets the maximum supported size of HTTP request URI to be processed by the server

Default value:

- 512

CONFIG_HTTPD_ERR_RESP_NO_DELAY

Use TCP_NODELAY socket option when sending HTTP error responses

Found in: Component config > HTTP Server

Using TCP_NODEALY socket option ensures that HTTP error response reaches the client before the underlying socket is closed. Please note that turning this off may cause multiple test failures

Default value:

- Yes (enabled)

CONFIG_HTTPD_PURGE_BUF_LEN

Length of temporary buffer for purging data

Found in: Component config > HTTP Server

This sets the size of the temporary buffer used to receive and discard any remaining data that is received from the HTTP client in the request, but not processed as part of the server HTTP request handler.

If the remaining data is larger than the available buffer size, the buffer will be filled in multiple iterations. The buffer should be small enough to fit on the stack, but large enough to avoid excessive iterations.

Default value:

- 32

CONFIG_HTTPD_LOG_PURGE_DATA

Log purged content data at Debug level

Found in: Component config > HTTP Server

Enabling this will log discarded binary HTTP request data at Debug level. For large content data this may not be desirable as it will clutter the log.

Default value:

- No (disabled)

CONFIG_HTTPD_WS_SUPPORT

WebSocket server support

Found in: Component config > HTTP Server

This sets the WebSocket server support.

Default value:

- No (disabled)

CONFIG_HTTPD_QUEUE_WORK_BLOCKING

httpd_queue_work as blocking API

Found in: Component config > HTTP Server

This makes httpd_queue_work() API to wait until a message space is available on UDP control socket. It internally uses a counting semaphore with count set to *LWIP_UDP_RECVMBOX_SIZE* to achieve this. This config will slightly change API behavior to block until message gets delivered on control socket.

CONFIG_HTTPD_SERVER_EVENT_POST_TIMEOUT

Time in millisecond to wait for posting event

Found in: Component config > HTTP Server

This config option helps in setting the time in millisecond to wait for event to be posted to the system default event loop. Set it to -1 if you need to set timeout to portMAX_DELAY.

Default value:

- 2000

CONFIG_HTTPD_WS_PRE_HANDSHAKE_CB_SUPPORT

WebSocket pre-handshake callback support

Found in: Component config > HTTP Server

Enable this option to use WebSocket pre-handshake callback. This will allow the server to register a callback function that will be called before the WebSocket handshake is processed i.e. before switching to the WebSocket protocol.

Default value:

- No (disabled) if CONFIG_HTTPD_WS_SUPPORT

CONFIG_HTTPD_WS_POST_HANDSHAKE_CB_SUPPORT

WebSocket post-handshake callback support

Found in: Component config > HTTP Server

Enable this option to use WebSocket post-handshake callback. This will allow the server to register a callback function that will be called after the WebSocket handshake is processed i.e. after switching to the WebSocket protocol.

Default value:

- No (disabled) if CONFIG_HTTPD_WS_SUPPORT

ESP HTTPS OTA

Contains:

- CONFIG_ESP_HTTPS_OTA_ALLOW_HTTP
- CONFIG_ESP_HTTPS_OTA_ENABLE_PARTIAL_DOWNLOAD
- CONFIG_ESP_HTTPS_OTA_DECRYPT_CB
- CONFIG_ESP_HTTPS_OTA_EVENT_POST_TIMEOUT

CONFIG_ESP_HTTPS_OTA_DECRYPT_CB

Provide decryption callback

Found in: Component config > ESP HTTPS OTA

Exposes an additional callback whereby firmware data could be decrypted before being processed by OTA update component. This can help to integrate external encryption related format and removal of such encapsulation layer from firmware image.

Default value:

- No (disabled)

CONFIG_ESP_HTTPS_OTA_ALLOW_HTTP

Allow HTTP for OTA (WARNING: ONLY FOR TESTING PURPOSE, READ HELP)

Found in: Component config > ESP HTTPS OTA

It is highly recommended to keep HTTPS (along with server certificate validation) enabled. Enabling this option comes with potential risk of: - Non-encrypted communication channel with server - Accepting firmware upgrade image from server with fake identity

Default value:

- No (disabled)

CONFIG_ESP_HTTPS_OTA_EVENT_POST_TIMEOUT

Time in millisecond to wait for posting event

Found in: Component config > ESP HTTPS OTA

This config option helps in setting the time in millisecond to wait for event to be posted to the system default event loop. Set it to -1 if you need to set timeout to portMAX_DELAY.

Default value:

- 2000

CONFIG_ESP_HTTPS_OTA_ENABLE_PARTIAL_DOWNLOAD

Enable partial HTTP download for OTA

Found in: Component config > ESP HTTPS OTA

This enables use of range header in esp_https_ota component. The firmware image will be downloaded over multiple HTTP requests.

Default value:

- No (disabled)

ESP HTTPS server

Contains:

- CONFIG_ESP_HTTPS_SERVER_CERT_SELECT_HOOK
- CONFIG_ESP_HTTPS_SERVER_ENABLE
- CONFIG_ESP_HTTPS_SERVER_EVENT_POST_TIMEOUT

CONFIG_ESP_HTTPS_SERVER_ENABLE

Enable ESP_HTTPS_SERVER component

Found in: Component config > ESP HTTPS server

Enable ESP HTTPS server component

CONFIG_ESP_HTTPS_SERVER_EVENT_POST_TIMEOUT

Time in millisecond to wait for posting event

Found in: Component config > ESP HTTPS server

This config option helps in setting the time in millisecond to wait for event to be posted to the system default event loop. Set it to -1 if you need to set timeout to portMAX_DELAY.

Default value:

- 2000

CONFIG_ESP_HTTPS_SERVER_CERT_SELECT_HOOK

Enable certificate selection hook

Found in: Component config > ESP HTTPS server

Enable certificate selection hook for ESP HTTPS Server. When enabled, this allows the server to dynamically select the appropriate certificate based on the client's Server Name Indication (SNI). This is useful for hosting multiple domains on a single server with different SSL certificates.

Default value:

- No (disabled)

Hardware Settings

Contains:

- CONFIG_ESP_ENABLE_PVT
- Chip revision
- Event Task Matrix (ETM) Configurations
- MAC Config
- Main XTAL Config
- Peripheral Control
- CONFIG_ESP_INTR_IN_IRAM
- Power Supplier
- RTC Clock Config
- Sleep Config

Chip revision

Contains:

- CONFIG_ESP_REV_NEW_CHIP_TEST
- CONFIG_ESP_EFUSE_BLOCK_REV_MIN_FULL
- CONFIG_ESP32_REV_MIN

CONFIG_ESP32_REV_MIN

Minimum Supported ESP32 Revision

Found in: Component config > Hardware Settings > Chip revision

Required minimum chip revision. ESP-IDF will check for it and reject to boot if the chip revision fails the check. This ensures the chip used will have some modifications (features, or bugfixes).

The complied binary will only support chips above this revision, this will also help to reduce binary size.

Available options:

- Rev v0.0 (ECO0) (CONFIG_ESP32_REV_MIN_0)
- Rev v1.0 (ECO1) (CONFIG_ESP32_REV_MIN_1)
- Rev v1.1 (ECO1.1) (CONFIG_ESP32_REV_MIN_1_1)
- Rev v2.0 (ECO2) (CONFIG_ESP32_REV_MIN_2)
- Rev v3.0 (ECO3) (CONFIG_ESP32_REV_MIN_3)
- Rev v3.1 (ECO4) (CONFIG_ESP32_REV_MIN_3_1)

CONFIG_ESP_EFUSE_BLOCK_REV_MIN_FULL

Minimum Supported ESP32 eFuse Block Revision

Found in: Component config > Hardware Settings > Chip revision

Required minimum eFuse Block revision. ESP-IDF will check it at the 2nd bootloader stage whether the current image can work correctly for this eFuse Block revision. So that to avoid running an incompatible image on a SoC that contains breaking change in the eFuse Block. If you want to update this value to run the image that not compatible with the current eFuse Block revision, please contact to Espressif's business team for details: https://www.espressif.com.cn/en/contact-us/sales-questions

Default value:

- 0

CONFIG_ESP_REV_NEW_CHIP_TEST

Internal test mode

Found in: Component config > Hardware Settings > Chip revision

For internal chip testing, a small number of new versions chips didn't update the version field in eFuse, you can enable this option to force the software recognize the chip version based on the rev selected in menuconfig.

Default value:

- No (disabled)

MAC Config

Contains:

- CONFIG_ESP_MAC_USE_CUSTOM_MAC_AS_BASE_MAC
- CONFIG_ESP_MAC_IGNORE_MAC_CRC_ERROR
- CONFIG_ESP32_UNIVERSAL_MAC_ADDRESSES

CONFIG_ESP32_UNIVERSAL_MAC_ADDRESSES

Number of universally administered (by IEEE) MAC address

Found in: Component config > Hardware Settings > MAC Config

Configure the number of universally administered (by IEEE) MAC addresses. During initialization, MAC addresses for each network interface are generated or derived from a single base MAC address. If the number of universal MAC addresses is four, all four interfaces (WiFi station, WiFi softap, Bluetooth and Ethernet) receive a universally administered MAC address. These are generated sequentially by adding 0, 1, 2 and 3 (respectively) to the final octet of the base MAC address. If the number of universal MAC addresses is two, only two interfaces (WiFi station and Bluetooth) receive a universally administered MAC address. These are generated sequentially by adding 0 and 1 (respectively) to the base MAC address. The remaining two interfaces (WiFi softap and Ethernet) receive local MAC addresses. These are derived from the universal WiFi station and Bluetooth MAC addresses, respectively. When using the default (Espressif-assigned) base MAC address, either setting can be used. When using a custom universal MAC address range, the correct setting will depend on the allocation of MAC addresses in this range (either 2 or 4 per device.)

Available options:

- Two (CONFIG_ESP32_UNIVERSAL_MAC_ADDRESSES_TWO)
- Four (CONFIG_ESP32_UNIVERSAL_MAC_ADDRESSES_FOUR)

CONFIG_ESP_MAC_IGNORE_MAC_CRC_ERROR

Ignore MAC CRC error (not recommended)

Found in: Component config > Hardware Settings > MAC Config

If you have an invalid MAC CRC (ESP_ERR_INVALID_CRC) problem and you still want to use this chip, you can enable this option to bypass such an error. This applies to both MAC_FACTORY and CUSTOM_MAC efuses.

Default value:

- No (disabled)

CONFIG_ESP_MAC_USE_CUSTOM_MAC_AS_BASE_MAC

Enable using custom mac as base mac

Found in: Component config > Hardware Settings > MAC Config

When this configuration is enabled, the user can invoke *esp_read_mac* to obtain the desired type of MAC using a custom MAC as the base MAC.

Default value:

- No (disabled)

Sleep Config

Contains:

- CONFIG_ESP_SLEEP_GPIO_ENABLE_INTERNAL_RESISTORS
- CONFIG_ESP_SLEEP_CACHE_SAFE_ASSERTION
- CONFIG_ESP_SLEEP_EVENT_CALLBACKS
- CONFIG_ESP_SLEEP_DEBUG
- CONFIG_ESP_SLEEP_WAIT_FLASH_READY_EXTRA_DELAY
- CONFIG_ESP_SLEEP_GPIO_RESET_WORKAROUND
- CONFIG_ESP_SLEEP_POWER_DOWN_FLASH
- CONFIG_ESP_SLEEP_MSPI_NEED_ALL_IO_PU
- CONFIG_ESP_SLEEP_FLASH_LEAKAGE_WORKAROUND
- CONFIG_ESP_SLEEP_PSRAM_LEAKAGE_WORKAROUND

CONFIG_ESP_SLEEP_POWER_DOWN_FLASH

Power down flash in light sleep when there is no SPIRAM or SPIRAM has independent power supply

Found in: Component config > Hardware Settings > Sleep Config

If enabled, chip will try to power down flash as part of esp_light_sleep_start(), which costs more time when chip wakes up. Can only be enabled if there is no SPIRAM configured.

This option will power down flash under a strict but relatively safe condition. Also, it is possible to power down flash under a relaxed condition by using esp_sleep_pd_config() to set ESP_PD_DOMAIN_VDDSDIO to ESP_PD_OPTION_OFF. It should be noted that there is a risk in powering down flash, you can refer *ESP-IDF Programming Guide/API Reference/System API/Sleep Modes/Power-down of Flash* for more details.

Default value:

- No (disabled) if ESP_LDO_RESERVE_PSRAM

CONFIG_ESP_SLEEP_FLASH_LEAKAGE_WORKAROUND

Pull-up Flash CS pin in light sleep

Found in: Component config > Hardware Settings > Sleep Config

All IOs will be set to isolate(floating) state by default during sleep. Since the power supply of SPI Flash is not lost during lightsleep, if its CS pin is recognized as low level(selected state) in the floating state, there will be a large current leakage, and the data in Flash may be corrupted by random signals on other SPI pins. Select this option will set the CS pin of Flash to PULL-UP state during sleep, but this will increase the sleep current about 10 uA. If you are developing with esp32xx modules, you must select this option, but if you are developing with chips, you can also pull up the CS pin of SPI Flash in the external circuit to save power consumption caused by internal pull-up during sleep. (!!! Don't deselect this option if you don't have external SPI Flash CS pin pullups.)

CONFIG_ESP_SLEEP_PSRAM_LEAKAGE_WORKAROUND

Pull-up PSRAM CS pin in light sleep

Found in: Component config > Hardware Settings > Sleep Config

All IOs will be set to isolate(floating) state by default during sleep. Since the power supply of PSRAM is not lost during lightsleep, if its CS pin is recognized as low level(selected state) in the floating state, there will be a large current leakage, and the data in PSRAM may be corrupted by random signals on other SPI pins. Select this option will set the CS pin of PSRAM to PULL-UP state during sleep, but this will increase the sleep current about 10 uA. If you are developing with esp32xx modules, you must select this option, but if you are developing with chips, you can also pull up the CS pin of PSRAM in the external circuit to save power consumption caused by internal pull-up during sleep. (!!! Don't deselect this option if you don't have external PSRAM CS pin pullups.)

Default value:

- Yes (enabled) if CONFIG_SPIRAM

CONFIG_ESP_SLEEP_MSPI_NEED_ALL_IO_PU

Pull-up all SPI pins in light sleep

Found in: Component config > Hardware Settings > Sleep Config

To reduce leakage current, some types of SPI Flash/RAM only need to pull up the CS pin during light sleep. But there are also some kinds of SPI Flash/RAM that need to pull up all pins. It depends on the SPI Flash/RAM chip used.

CONFIG_ESP_SLEEP_GPIO_RESET_WORKAROUND

light sleep GPIO reset workaround

Found in: Component config > Hardware Settings > Sleep Config

All existing chips except esp32 and esp32s2 will reset on wake-up if a GPIO receives a small electrostatic pulse during light sleep, with specific conditions.

- GPIO needs to be configured as input-mode only
- The pin receives a small electrostatic pulse, and reset occurs when the pulse voltage is higher than 6 V

For GPIO set to input mode only, it is not a good practice to leave it open/floating, The hardware design needs to controlled it with determined supply or ground voltage is necessary.

This option provides a software workaround for this issue. Configure to isolate all GPIO pins in sleep state.

CONFIG_ESP_SLEEP_WAIT_FLASH_READY_EXTRA_DELAY

Extra delay (in us) after flash powerdown sleep wakeup to wait flash ready

Found in: Component config > Hardware Settings > Sleep Config

When the chip exits sleep, the CPU and the flash chip are powered on at the same time. CPU will run rom code (deepsleep) or ram code (lightsleep) first, and then load or execute code from flash.

Some flash chips need sufficient time to pass between power on and first read operation. By default, without any extra delay, this time is approximately 900us, although some flash chip types need more than that.

(!!! Please adjust this value according to the Data Sheet of SPI Flash used in your project.) In Flash Data Sheet, the parameters that define the Flash ready timing after power-up (minimum time from Vcc(min) to CS active) are usually named tVSL in ELECTRICAL CHARACTERISTICS chapter, and the configuration value here should be: ESP_SLEEP_WAIT_FLASH_READY_EXTRA_DELAY = tVSL - 900

For esp32 and esp32s3, the default extra delay is set to 2000us. When optimizing startup time for applications which require it, this value may be reduced.

If you are seeing "flash read err, 1000" message printed to the console after deep sleep reset on esp32, or triggered RTC_WDT/LP_WDT after lightsleep wakeup, try increasing this value. (For esp32, the delay will be executed in both deep sleep and light sleep wake up flow. For chips after esp32, the delay will be executed only in light sleep flow, the delay controlled by the EFUSE_FLASH_TPUW in ROM will be executed in deepsleep wake up flow.)

Range:

- from 0 to 5000

Default value:

- 2000
- 0

CONFIG_ESP_SLEEP_CACHE_SAFE_ASSERTION

Check the cache safety of the sleep wakeup code in sleep process

Found in: Component config > Hardware Settings > Sleep Config

Enabling it will check the cache safety of the code before the flash power is ready after light sleep wakeup, and check PM_SLP_IRAM_OPT related code cache safety. This option is only for code quality inspection. Enabling it will increase the time overhead of entering and exiting sleep. It is not recommended to enable it in the release version.

Default value:

- No (disabled)

CONFIG_ESP_SLEEP_DEBUG

esp sleep debug

Found in: Component config > Hardware Settings > Sleep Config

Enable esp sleep debug.

Default value:

- No (disabled)

CONFIG_ESP_SLEEP_GPIO_ENABLE_INTERNAL_RESISTORS

Allow to enable internal pull-up/downs for the Deep-Sleep wakeup IOs

Found in: Component config > Hardware Settings > Sleep Config

When using rtc gpio wakeup source during deepsleep without external pull-up/downs, you may want to make use of the internal ones.

Default value:

- Yes (enabled)

CONFIG_ESP_SLEEP_EVENT_CALLBACKS

Enable registration of sleep event callbacks

Found in: Component config > Hardware Settings > Sleep Config

If enabled, it allows user to register sleep event callbacks. It is primarily designed for internal developers and customers can use PM_LIGHT_SLEEP_CALLBACKS as an alternative.

NOTE: These callbacks are executed from the IDLE task context hence you cannot have any blocking calls in your callbacks.

NOTE: Enabling these callbacks may change sleep duration calculations based on time spent in callback and hence it is highly recommended to keep them as short as possible.

Default value:

- No (disabled) if CONFIG_FREERTOS_USE_TICKLESS_IDLE

RTC Clock Config

Contains:

- CONFIG_RTC_EXT_CRYST_ADDIT_CURRENT_METHOD
- CONFIG_RTC_XTAL_CAL_RETRY
- CONFIG_RTC_CLK_CAL_CYCLES
- CONFIG_RTC_CLK_FUNC_IN_IRAM
- CONFIG_RTC_TIME_FUNC_IN_IRAM
- CONFIG_RTC_CLK_SRC

CONFIG_RTC_CLK_SRC

RTC clock source

Found in: Component config > Hardware Settings > RTC Clock Config

Choose which clock is used as RTC clock source.

- **"Internal 150kHz oscillator" option provides lowest deep sleep current**

  consumption, and does not require extra external components. However frequency stability with respect to temperature is poor, so time may drift in deep/light sleep modes.

- **"External 32kHz crystal" provides better frequency stability, at the**

  expense of slightly higher (1uA) deep sleep current consumption.

- **"External 32kHz oscillator" allows using 32kHz clock generated by an**

  external circuit. In this case, external clock signal must be connected to 32K_XN pin. Additionally, 1nF capacitor must be connected between 32K_XP pin and ground. 32K_XP pin can not be used as a GPIO in this case.

- **"Internal 8.5MHz oscillator divided by 256" option results in higher**

  deep sleep current (by 5uA) but has better frequency stability than the internal 150kHz oscillator. It does not require external components.

Available options:

- Internal 150 kHz RC oscillator (CONFIG_RTC_CLK_SRC_INT_RC)
- External 32kHz crystal (CONFIG_RTC_CLK_SRC_EXT_CRYS)
- External 32kHz oscillator at 32K_XN pin (CONFIG_RTC_CLK_SRC_EXT_OSC)
- Internal 8.5MHz oscillator, divided by 256 (~33kHz) (CONFIG_RTC_CLK_SRC_INT_8MD256)

CONFIG_RTC_EXT_CRYST_ADDIT_CURRENT_METHOD

Additional current for external 32kHz crystal

Found in: Component config > Hardware Settings > RTC Clock Config

With some 32kHz crystal configurations, the X32N and X32P pins may not have enough drive strength to keep the crystal oscillating. Choose the method to provide additional current from touchpad 9 to the external 32kHz crystal. Note that the deep sleep current is slightly high (4-5uA) and the touchpad and the wakeup sources of both touchpad and ULP are not available in method 1 and method 2.

This problem is fixed in ESP32 ECO 3, so this workaround is not needed. Setting the project configuration to minimum revision ECO3 will disable this option, , allow all wakeup sources, and save some code size.

- "None" option will not provide additional current to external crystal
- **"Method 1" option can't ensure 100% to solve the external 32k crystal start failed**

  issue, but the touchpad can work in this method.
- **"Method 2" option can solve the external 32k issue, but the touchpad can't work**

  in this method.

Available options:

- None (CONFIG_RTC_EXT_CRYST_ADDIT_CURRENT_NONE)
- Method 1 (CONFIG_RTC_EXT_CRYST_ADDIT_CURRENT)
- Method 2 (CONFIG_RTC_EXT_CRYST_ADDIT_CURRENT_V2)

CONFIG_RTC_CLK_CAL_CYCLES

Number of cycles for RTC_SLOW_CLK calibration

Found in: Component config > Hardware Settings > RTC Clock Config

When the startup code initializes RTC_SLOW_CLK, it can perform calibration by comparing the RTC_SLOW_CLK frequency with main XTAL frequency. This option sets the number of RTC_SLOW_CLK cycles measured by the calibration routine. Higher numbers increase calibration precision, which may be important for applications which spend a lot of time in deep sleep. Lower numbers reduce startup time.

When this option is set to 0, clock calibration will not be performed at startup, and approximate clock frequencies will be assumed:

- 150000 Hz if internal RC oscillator is used as clock source. For this use value 1024.
- **32768 Hz if the 32k crystal oscillator is used. For this use value 3000 or more.**

  In case more value will help improve the definition of the launch of the crystal. If the crystal could not start, it will be switched to internal RC.

Range:

- from 0 to 27000 if CONFIG_RTC_CLK_SRC_EXT_CRYS || CONFIG_RTC_CLK_SRC_EXT_OSC || CONFIG_RTC_CLK_SRC_INT_8MD256
- from 0 to 32766

Default value:

- 3000 if CONFIG_RTC_CLK_SRC_EXT_CRYS || CONFIG_RTC_CLK_SRC_EXT_OSC || CONFIG_RTC_CLK_SRC_INT_8MD256
- 1024

CONFIG_RTC_XTAL_CAL_RETRY

Number of attempts to repeat 32k XTAL calibration

Found in: Component config > Hardware Settings > RTC Clock Config

Number of attempts to repeat 32k XTAL calibration before giving up and switching to the internal RC. Increase this option if the 32k crystal oscillator does not start and switches to internal RC.

Default value:

- 1 if CONFIG_RTC_CLK_SRC_EXT_CRYS

CONFIG_RTC_CLK_FUNC_IN_IRAM

Place RTC clock module functions into IRAM

Found in: Component config > Hardware Settings > RTC Clock Config

Place RTC clock module (all rtc clock functions and const data) into IRAM.

Default value:

- Yes (enabled)

CONFIG_RTC_TIME_FUNC_IN_IRAM

Place RTC time module functions into IRAM

Found in: Component config > Hardware Settings > RTC Clock Config

Place RTC time module (all rtc clock functions) into IRAM.

Default value:

- Yes (enabled)

Peripheral Control

Contains:

- CONFIG_ESP_MODEM_CLOCK_ENABLE_CHECKING
- CONFIG_ESP_PERIPH_CTRL_FUNC_IN_IRAM
- CONFIG_ESP_REGI2C_CTRL_FUNC_IN_IRAM

CONFIG_ESP_PERIPH_CTRL_FUNC_IN_IRAM

Place peripheral control functions into IRAM

Found in: Component config > Hardware Settings > Peripheral Control

Place peripheral control functions (e.g. periph_module_reset) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context.

Default value:

- Yes (enabled)

CONFIG_ESP_REGI2C_CTRL_FUNC_IN_IRAM

Place regi2c control functions into IRAM

Found in: Component config > Hardware Settings > Peripheral Control

Place analog i2c master control functions (e.g. regi2c_ctrl_read_reg, regi2c_ctrl_write_reg) into IRAM, so that these functions can be IRAM-safe and able to be called in the other IRAM interrupt context.

Default value:

- Yes (enabled)

CONFIG_ESP_MODEM_CLOCK_ENABLE_CHECKING

Enable modem clock configuration checking

Found in: Component config > Hardware Settings > Peripheral Control

If enabled, this option will perform additional checks on modem clock configuration to ensure proper operation and detect potential configuration issues.

Default value:

- No (disabled) if SOC_MODEM_CLOCK_SUPPORTED

Event Task Matrix (ETM) Configurations

Contains:

- CONFIG_ETM_ENABLE_DEBUG_LOG

CONFIG_ETM_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > Hardware Settings > Event Task Matrix (ETM) Configurations

If enabled, ETM driver will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled) if SOC_ETM_SUPPORTED

Main XTAL Config

Contains:

- CONFIG_XTAL_FREQ

CONFIG_XTAL_FREQ

Main XTAL frequency

Found in: Component config > Hardware Settings > Main XTAL Config

This option selects the operating frequency of the XTAL (crystal) clock used to drive the ESP target. The selected value MUST reflect the frequency of the given hardware.

Note: On ESP32, the XTAL_FREQ_AUTO option allows the ESP target to automatically estimating XTAL clock's operating frequency. The ESP32 uses the internal 8MHZ as a reference when estimating. Due to the internal oscillator's frequency being temperature dependent, usage of the XTAL_FREQ_AUTO is not recommended in applications that operate in high ambient temperatures or use high-temperature qualified chips and modules.

Available options:

- 26 MHz (CONFIG_XTAL_FREQ_26)
- 32 MHz (CONFIG_XTAL_FREQ_32)
- 40 MHz (CONFIG_XTAL_FREQ_40)
- Autodetect (CONFIG_XTAL_FREQ_AUTO)

Power Supplier

Contains:

- Brownout Detector

Brownout Detector

Contains:

- CONFIG_ESP_BROWNOUT_DET

CONFIG_ESP_BROWNOUT_DET

Hardware brownout detect & reset

Found in: Component config > Hardware Settings > Power Supplier > Brownout Detector

The ESP has a built-in brownout detector which can detect if the voltage is lower than a specific value. If this happens, it will reset the chip in order to prevent unintended behaviour.

Default value:

- Yes (enabled)

CONFIG_ESP_BROWNOUT_DET_LVL_SEL

Brownout voltage level

Found in: Component config > Hardware Settings > Power Supplier > Brownout Detector > CONFIG_ESP_BROWNOUT_DET

The brownout detector will reset the chip when the supply voltage is approximately below this level. Note that there may be some variation of brownout voltage level between each ESP chip.

#The voltage levels here are estimates, more work needs to be done to figure out the exact voltages #of the brownout threshold levels.

Available options:

- 2.43V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_0)
- 2.48V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_1)
- 2.58V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_2)
- 2.62V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_3)
- 2.67V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_4)
- 2.70V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_5)
- 2.77V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_6)
- 2.80V +/- 0.05 (CONFIG_ESP_BROWNOUT_DET_LVL_SEL_7)

CONFIG_ESP_ENABLE_PVT

Auto adjust hp & lp voltage using pvt function (MUST ENABLE FOR MP)

Found in: Component config > Hardware Settings

If enabled, hp & lp voltage can be auto adjust by PVT characteristic. Otherwise, internal voltage will be set to fix dbias. This is a must for stable mass production. Disable for debugging only.

Default value:

- Yes (enabled) if SOC_PMU_PVT_SUPPORTED

CONFIG_ESP_INTR_IN_IRAM

Place esp_intr_alloc functions in IRAM

Found in: Component config > Hardware Settings

Default value:

- Yes (enabled)

ESP-Driver:LCD Controller Configurations

Contains:

- CONFIG_LCD_DSI_ISR_CACHE_SAFE
- CONFIG_LCD_RGB_RESTART_IN_VSYNC
- CONFIG_LCD_ENABLE_DEBUG_LOG
- CONFIG_LCD_DSI_ISR_HANDLER_IN_IRAM
- CONFIG_LCD_RGB_ISR_IRAM_SAFE

CONFIG_LCD_RGB_ISR_IRAM_SAFE

RGB LCD ISR IRAM-Safe

Found in: Component config > ESP-Driver:LCD Controller Configurations

Ensure the LCD interrupt is IRAM-Safe by allowing the interrupt handler to be executable when the cache is disabled (e.g. SPI Flash write). If you want the LCD driver to keep flushing the screen even when cache ops disabled, you can enable this option. Note, this will also increase the IRAM usage.

Default value:

- No (disabled) if SOC_LCD_RGB_SUPPORTED

CONFIG_LCD_RGB_RESTART_IN_VSYNC

Always restart RGB LCD transmission in VSYNC

Found in: Component config > ESP-Driver:LCD Controller Configurations

Reset the GDMA channel every VBlank to stop permanent desyncs from happening. Only need to enable it when in your application, the DMA can't deliver data as fast as the LCD consumes it.

Default value:

- No (disabled) if SOC_LCD_RGB_SUPPORTED

CONFIG_LCD_DSI_ISR_HANDLER_IN_IRAM

Place DSI ISR handler in IRAM to reduce latency

Found in: Component config > ESP-Driver:LCD Controller Configurations

Place DSI ISR handler in IRAM to reduce latency caused by cache miss.

Default value:

- Yes (enabled) if SOC_MIPI_DSI_SUPPORTED

CONFIG_LCD_DSI_ISR_CACHE_SAFE

Allow DSI ISR to execute when cache is disabled

Found in: Component config > ESP-Driver:LCD Controller Configurations

Enable this option to allow the DSI Interrupt Service Routine (ISR) to execute even when the cache is disabled. This can be useful in scenarios where the cache might be turned off, but the DSI functionality is still required to operate correctly.

Default value:

- No (disabled) if SOC_MIPI_DSI_SUPPORTED

CONFIG_LCD_ENABLE_DEBUG_LOG

Force enable debug log

Found in: Component config > ESP-Driver:LCD Controller Configurations

If enabled, LCD driver component will: 1. ignore the global logging settings 2. compile all log messages into the binary 3. set the runtime log level to VERBOSE Please enable this option by caution, as it will increase the binary size.

Default value:

- No (disabled)

LibC

Contains:

- CONFIG_LIBC_ASSERT_BUFFER_SIZE
- CONFIG_LIBC_NEWLIB_NANO_FORMAT
- CONFIG_LIBC
- CONFIG_LIBC_STDIN_LINE_ENDING
- CONFIG_LIBC_STDOUT_LINE_ENDING
- CONFIG_LIBC_LOCKS_PLACE_IN_IRAM
- CONFIG_LIBC_MISC_IN_IRAM
- CONFIG_LIBC_PICOLIBC_NEWLIB_COMPATIBILITY
- CONFIG_LIBC_TIME_SYSCALL
- CONFIG_LIBC_OPTIMIZED_MISALIGNED_ACCESS

CONFIG_LIBC

LibC to build application with

Found in: Component config > LibC

Available options:

- NewLib (CONFIG_LIBC_NEWLIB)
- Picolibc (CONFIG_LIBC_PICOLIBC)

CONFIG_LIBC_PICOLIBC_NEWLIB_COMPATIBILITY

Provides limited interoperability with libraries built using Newlib headers

Found in: Component config > LibC

This option provides limited compatibility with libraries built using Newlib headers by:

Enabling hidden system-header inclusions that exist in Newlib. Adding tls_stdio, tls_stdout, and tls_stderr variables to Thread Local Storage to allow prebuilt libraries to access them via getreent(). Providing an implementation of getreent() that returns the value of the thread-pointer register.

Limitations:

1. If your application or an external prebuilt library accesses Newlib "struct _reent" implicitly, this may cause memory corruption on the task stack. You have two options:

- Use libc API calls instead of directly accessing "struct _reent" fields.
- Switch ESP-IDF to use the Newlib implementation by setting CONFIG_LIBC_NEWLIB=y in sdkconfig.

2. If your application uses a prebuilt library built with Newlib headers, you may encounter unexpected behavior when overriding stdin, stdout, and stderr. External prebuilt libraries can only read these streams, so overriding them will not affect as it was for Newlib. You have two options to fix this:

- Rebuild the library with Picolibc headers that follow POSIX-standardized stdin, stdout, and stderr declarations.
- Switch ESP-IDF to use the Newlib implementation by setting CONFIG_LIBC_NEWLIB=y in sdkconfig.

Default value:

- Yes (enabled)

CONFIG_LIBC_MISC_IN_IRAM

Place misc libc functions (abort/assert/stdatomics) in IRAM

Found in: Component config > LibC

Default value:

- Yes (enabled)

CONFIG_LIBC_LOCKS_PLACE_IN_IRAM

Place lock API in IRAM

Found in: Component config > LibC

Enable this option to include be able to call the lock API from code that runs while cache is disabled, e.g. IRAM interrupts.

Default value:

- Yes (enabled)

CONFIG_LIBC_STDOUT_LINE_ENDING

Line ending for console output

Found in: Component config > LibC

This option allows configuring the desired line endings sent to console when a newline ('n', LF) appears on stdout. Three options are possible:

CRLF: whenever LF is encountered, prepend it with CR

LF: no modification is applied, stdout is sent as is

CR: each occurrence of LF is replaced with CR

This option doesn't affect behavior of the UART driver (drivers/uart.h).

Available options:

- CRLF (CONFIG_LIBC_STDOUT_LINE_ENDING_CRLF)
- LF (CONFIG_LIBC_STDOUT_LINE_ENDING_LF)
- CR (CONFIG_LIBC_STDOUT_LINE_ENDING_CR)

CONFIG_LIBC_STDIN_LINE_ENDING

Line ending for console input

Found in: Component config > LibC

This option allows configuring which input sequence on console produces a newline ('n', LF) on stdin. Three options are possible:

CRLF: CRLF is converted to LF

LF: no modification is applied, input is sent to stdin as is

CR: each occurrence of CR is replaced with LF

This option doesn't affect behavior of the UART driver (drivers/uart.h).

Available options:

- CRLF (CONFIG_LIBC_STDIN_LINE_ENDING_CRLF)
- LF (CONFIG_LIBC_STDIN_LINE_ENDING_LF)
- CR (CONFIG_LIBC_STDIN_LINE_ENDING_CR)

CONFIG_LIBC_NEWLIB_NANO_FORMAT

Enable 'nano' formatting options for printf/scanf family

Found in: Component config > LibC

In most chips the ROM contains parts of newlib C library, including printf/scanf family of functions. These functions have been compiled with so-called "nano" formatting option. This option doesn't support 64-bit integer formats and C99 features, such as positional arguments.

For more details about "nano" formatting option, please see newlib readme file, search for '--enable-newlib-nano-formatted-io': https://sourceware.org/git/?p=newlib-cygwin.git;a=blob_plain;f=newlib/README;hb=HEAD

If this option is enabled and the ROM contains functions from newlib-nano, the build system will use functions available in ROM, reducing the application binary size. Functions available in ROM run faster than functions which run from flash. Functions available in ROM can also run when flash instruction cache is disabled.

Some chips (e.g. ESP32-C6) has the full formatting versions of printf/scanf in ROM instead of the nano versions and in this building with newlib nano might actually increase the size of the binary. Which functions are present in ROM can be seen from ROM caps: ESP_ROM_HAS_NEWLIB_NANO_FORMAT and ESP_ROM_HAS_NEWLIB_NORMAL_FORMAT.

If you need 64-bit integer formatting support or C99 features, keep this option disabled.

CONFIG_LIBC_TIME_SYSCALL

Timers used for gettimeofday function

Found in: Component config > LibC

This setting defines which hardware timers are used to implement 'gettimeofday' and 'time' functions in C library.

- **If both high-resolution (systimer for all targets except ESP32)**

  and RTC timers are used, timekeeping will continue in deep sleep. Time will be reported at 1 microsecond resolution. This is the default, and the recommended option.

- **If only high-resolution timer (systimer) is used, gettimeofday will**

  provide time at microsecond resolution. Time will not be preserved when going into deep sleep mode.

- **If only RTC timer is used, timekeeping will continue in**

  deep sleep, but time will be measured at 6.(6) microsecond resolution. Also the gettimeofday function itself may take longer to run.

- **If no timers are used, gettimeofday and time functions**

  return -1 and set errno to ENOSYS; they are defined as weak, so they could be overridden. If you want to customize gettimeofday() and other time functions, please choose this option and refer to the 'time.c' source file for the exact prototypes of these functions.

- **When RTC is used for timekeeping, two RTC_STORE registers are**

  used to keep time in deep sleep mode.

Available options:

- RTC and high-resolution timer (CONFIG_LIBC_TIME_SYSCALL_USE_RTC_HRT)
- RTC (CONFIG_LIBC_TIME_SYSCALL_USE_RTC)
- High-resolution timer (CONFIG_LIBC_TIME_SYSCALL_USE_HRT)
- None (CONFIG_LIBC_TIME_SYSCALL_USE_NONE)

CONFIG_LIBC_OPTIMIZED_MISALIGNED_ACCESS

Use performance-optimized memXXX/strXXX functions on misaligned memory access

Found in: Component config > LibC

Enables performance-optimized implementations of memory and string functions when handling misaligned memory.

Require approximately 800–1000 bytes of IRAM.

Optimized functions include:

- memcpy
- memset
- memmove
- str[n]cpy
- str[n]cmp

Default value:

- Yes (enabled) if ESP_ROM_HAS_SUBOPTIMAL_NEWLIB_ON_MISALIGNED_MEMORY

CONFIG_LIBC_ASSERT_BUFFER_SIZE

Assert message buffer size

Found in: Component config > LibC

Size of the buffer used to format assert failure messages.

When assertions fail, the system formats a message containing the function name, file name, line number, and the failed expression. This option controls the maximum length of this message.

If you encounter truncated assert messages (especially with C++ templates or long function names), increase this value. The default value of 200 bytes should be sufficient for most cases, but complex template expressions may require larger buffers.

Range:

- from 100 to 2048

Default value:

- 200

ESP-MM: Memory Management Configurations

Contains:

- CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS

CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS

Enable esp_cache_msync C2M chunked operation

Found in: Component config > ESP-MM: Memory Management Configurations

esp_cache_msync C2M direction takes critical sections, which means during the operation, the interrupts are disabled. Whereas Cache writebacks for large buffers could be especially time intensive, and might cause interrupts to be disabled for a significant amount of time.

Sometimes you want other ISRs to be responded during this C2M process. This option is to slice one C2M operation into multiple chunks, with CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS_MAX_LEN max len. This will give you a breath during the C2M process as sometimes the C2M process is quite long.

Note if the buffer processed by the *esp_cache_msync* (C2M sliced) is interrupted by an ISR, and this ISR also accesses this buffer, this may lead to data coherence issue.

CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS_MAX_LEN

Max len in bytes per C2M chunk

Found in: Component config > ESP-MM: Memory Management Configurations > CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS

Max len in bytes per C2M chunk, operations with size over the max len will be sliced into multiple chunks.

Range:

- from 0 to 0x80000 if CONFIG_ESP_MM_CACHE_MSYNC_C2M_CHUNKED_OPS

ESP NETIF Adapter

Contains:

- CONFIG_ESP_NETIF_SET_DNS_PER_DEFAULT_NETIF
- CONFIG_ESP_NETIF_LOST_IP_TIMER_ENABLE

- CONFIG_ESP_NETIF_BRIDGE_EN
- CONFIG_ESP_NETIF_L2_TAP
- CONFIG_ESP_NETIF_IP_LOST_TIMER_INTERVAL
- CONFIG_ESP_NETIF_REPORT_DATA_TRAFFIC
- CONFIG_ESP_NETIF_USE_TCPIP_STACK_LIB
- CONFIG_ESP_NETIF_PROVIDE_CUSTOM_IMPLEMENTATION

CONFIG_ESP_NETIF_LOST_IP_TIMER_ENABLE

Enable IPv4 lost IP event timer

Found in: Component config > ESP NETIF Adapter

Enable raising LOST_IP events using a timer when the IPv4 address becomes invalid. When disabled, the timer is not started and LOST_IP events are not posted. The delay is configured by ESP_NETIF_IP_LOST_TIMER_INTERVAL.

Default value:

- Yes (enabled)

CONFIG_ESP_NETIF_IP_LOST_TIMER_INTERVAL

IP Address lost timer interval (seconds)

Found in: Component config > ESP NETIF Adapter

The IP address may be lost because of some reasons, e.g. when the station disconnects from soft-AP, or when DHCP IP renew fails etc. If the IP lost timer is enabled, it will be started every time the IP is lost. Event SYSTEM_EVENT_STA_LOST_IP will be raised if the timer expires. The IP lost timer is stopped if the station gets the IP again before the timer expires. For backward compatibility, setting the interval to 0 disables the timer as well.

Range:

- from 0 to 65535

Default value:

- 120

CONFIG_ESP_NETIF_PROVIDE_CUSTOM_IMPLEMENTATION

Use only ESP-NETIF headers

Found in: Component config > ESP NETIF Adapter

No implementation of ESP-NETIF functions is provided. This option is used for adding a custom TCP/IP stack and defining related esp_netif functionality

Default value:

- No (disabled)

CONFIG_ESP_NETIF_USE_TCPIP_STACK_LIB

TCP/IP Stack Library

Found in: Component config > ESP NETIF Adapter

Choose the TCP/IP Stack to work, for example, LwIP, uIP, etc.

Available options:

- LwIP (CONFIG_ESP_NETIF_TCPIP_LWIP)

  lwIP is a small independent implementation of the TCP/IP protocol suite.

- Loopback (CONFIG_ESP_NETIF_LOOPBACK)

  Dummy implementation of esp-netif functionality which connects driver transmit to receive function. This option is for testing purpose only

CONFIG_ESP_NETIF_REPORT_DATA_TRAFFIC

Report data traffic via events

Found in: Component config > ESP NETIF Adapter

Enable if esp_netif_transmit() and esp_netif_receive() should generate events. This can be useful to blink data traffic indication lights.

Default value:

- Yes (enabled)

CONFIG_ESP_NETIF_L2_TAP

Enable netif L2 TAP support

Found in: Component config > ESP NETIF Adapter

A user program can read/write link layer (L2) frames from/to ESP TAP device. The ESP TAP device can be currently associated only with Ethernet physical interfaces.

CONFIG_ESP_NETIF_L2_TAP_MAX_FDS

Maximum number of opened L2 TAP File descriptors

Found in: Component config > ESP NETIF Adapter > CONFIG_ESP_NETIF_L2_TAP

Maximum number of opened File descriptors (FD's) associated with ESP TAP device. ESP TAP FD's take up a certain amount of memory, and allowing fewer FD's to be opened at the same time conserves memory.

Range:

- from 1 to 10 if CONFIG_ESP_NETIF_L2_TAP

Default value:

- 5 if CONFIG_ESP_NETIF_L2_TAP

CONFIG_ESP_NETIF_L2_TAP_RX_QUEUE_SIZE

Size of L2 TAP Rx queue

Found in: Component config > ESP NETIF Adapter > CONFIG_ESP_NETIF_L2_TAP

Maximum number of frames queued in opened File descriptor. Once the queue is full, the newly arriving frames are dropped until the queue has enough room to accept incoming traffic (Tail Drop queue management).

Range:

- from 1 to 100 if CONFIG_ESP_NETIF_L2_TAP

Default value:

- 20 if CONFIG_ESP_NETIF_L2_TAP

CONFIG_ESP_NETIF_BRIDGE_EN

Enable LwIP IEEE 802.1D bridge

Found in: Component config > ESP NETIF Adapter

Enable LwIP IEEE 802.1D bridge support in ESP-NETIF. Note that "Number of clients store data in netif" (LWIP_NUM_NETIF_CLIENT_DATA) option needs to be properly configured to be LwIP bridge available!

Default value:

- No (disabled)

CONFIG_ESP_NETIF_SET_DNS_PER_DEFAULT_NETIF

Enable DNS server per interface

Found in: Component config > ESP NETIF Adapter

Enable this option to use the DNS server which belongs to the selected default network interface. This feature collects DNS server and netif information from LWIP core modules. Whenever a new default netif is selected, global DNS servers in LWIP are updated with the netif related servers.

Default value:

- No (disabled)

Partition API Configuration

PHY

Contains:

- CONFIG_ESP_PHY_CALIBRATION_MODE
- CONFIG_ESP_PHY_DEBUG
- CONFIG_ESP_PHY_PLL_TRACK_DEBUG
- CONFIG_ESP_PHY_ENABLE_CERT_TEST
- CONFIG_ESP_PHY_IMPROVE_RX_11B
- CONFIG_ESP_PHY_ENABLE_USB
- CONFIG_ESP_PHY_MAX_WIFI_TX_POWER
- CONFIG_ESP_PHY_IRAM_OPT
- CONFIG_ESP_PHY_INIT_IRAM
- CONFIG_ESP_PHY_MAC_BB_PD
- CONFIG_ESP_PHY_ENABLE_VERSION_PRINT
- CONFIG_ESP_PHY_RECORD_USED_TIME
- CONFIG_ESP_PHY_REDUCE_TX_POWER

- CONFIG_ESP_PHY_PLL_TRACK_PERIOD_MS
- CONFIG_ESP_PHY_CALIBRATION_AND_DATA_STORAGE
- CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_CALIBRATION_AND_DATA_STORAGE

Store phy calibration data in NVS

Found in: Component config > PHY

If this option is enabled, NVS will be initialized and calibration data will be loaded from there. PHY calibration will be skipped on deep sleep wakeup. If calibration data is not found, full calibration will be performed and stored in NVS. Normally, only partial calibration will be performed. If this option is disabled, full calibration will be performed.

If it's easy that your board calibrate bad data, choose 'n'. Two cases for example, you should choose 'n': 1.If your board is easy to be booted up with antenna disconnected. 2.Because of your board design, each time when you do calibration, the result are too unstable. If unsure, choose 'y'.

Default value:

- Yes (enabled)

CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

Use a partition to store PHY init data

Found in: Component config > PHY

If enabled, PHY init data will be loaded from a partition. When using a custom partition table, make sure that PHY data partition is included (type: 'data', subtype: 'phy'). With default partition tables, this is done automatically. If PHY init data is stored in a partition, it has to be flashed there, otherwise runtime error will occur.

If this option is not enabled, PHY init data will be embedded into the application binary.

If unsure, choose 'n'.

Default value:

- No (disabled)

Contains:

- CONFIG_ESP_PHY_DEFAULT_INIT_IF_INVALID

- CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN

CONFIG_ESP_PHY_DEFAULT_INIT_IF_INVALID

Reset default PHY init data if invalid

Found in: Component config > PHY > CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

If enabled, PHY init data will be restored to default if it cannot be verified successfully to avoid endless bootloops.

If unsure, choose 'n'.

Default value:

- No (disabled) if CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN

Support multiple PHY init data bin

Found in: Component config > PHY > CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

If enabled, the corresponding PHY init data type can be automatically switched according to the country code. China's PHY init data bin is used by default. Can be modified by country information in API esp_wifi_set_country(). The priority of switching the PHY init data type is: 1. Country configured by API esp_wifi_set_country() and the parameter policy is WIFI_COUNTRY_POLICY_MANUAL. 2. Country notified by the connected AP. 3. Country configured by API esp_wifi_set_country() and the parameter policy is WIFI_COUNTRY_POLICY_AUTO.

Default value:

- No (disabled) if CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION && CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN_CUSTOM_PATH

Custom path to phy_multiple_init_data.bin

Found in: Component config > PHY > CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION > CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN

Absolute path, or path relative to the project directory, to the file phy_multiple_init_data.bin, or to a directory containing that file.

If this field is left empty, or the file cannot be found at the resolved path, the default binary located at components/esp_phy/<chip>/phy_multiple_init_data.bin will be used.

Default value:

- if CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN && CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN_EMBED

Support embedded multiple phy init data bin to app bin

Found in: Component config > PHY > CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION > CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN

If enabled, multiple phy init data bin will embedded into app bin If not enabled, multiple phy init data bin will still leave alone, and need to be flashed by users.

Default value:

- No (disabled) if CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN && CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_INIT_DATA_ERROR

Terminate operation when PHY init data error

Found in: Component config > PHY > CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION > CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN

If enabled, when an error occurs while the PHY init data is updated, the program will terminate and restart. If not enabled, the PHY init data will not be updated when an error occurs.

Default value:

- No (disabled) if CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN && CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION

CONFIG_ESP_PHY_MAX_WIFI_TX_POWER

Max WiFi TX power (dBm)

Found in: Component config > PHY

Set maximum transmit power for WiFi radio. Actual transmit power for high data rates may be lower than this setting.

Range:

- from 10 to 20

Default value:

- 20

CONFIG_ESP_PHY_MAC_BB_PD

Power down MAC and baseband of Wi-Fi and Bluetooth when PHY is disabled

Found in: Component config > PHY

If enabled, the MAC and baseband of Wi-Fi and Bluetooth will be powered down when PHY is disabled. Enabling this setting reduces power consumption by a small amount but increases RAM use by approximately 4 KB(Wi-Fi only), 2 KB(Bluetooth only) or 5.3 KB(Wi-Fi + Bluetooth).

Default value:

- No (disabled) if SOC_PM_SUPPORT_MAC_BB_PD && CONFIG_FREERTOS_USE_TICKLESS_IDLE

CONFIG_ESP_PHY_REDUCE_TX_POWER

Reduce PHY TX power when brownout reset

Found in: Component config > PHY

When brownout reset occurs, reduce PHY TX power to keep the code running.

Default value:

- No (disabled)

CONFIG_ESP_PHY_ENABLE_USB

Keep the USB PHY enabled when initializing WiFi

Found in: Component config > PHY

On some ESP targets, the USB PHY can interfere with WiFi thus lowering WiFi performance. As a result, on those affected ESP targets, the ESP PHY library's initialization will automatically disable the USB PHY to get best WiFi performance. This option controls whether or not the ESP PHY library will keep the USB PHY enabled on initialization.

Note: This option can be disabled to increase WiFi performance. However, disabling this option will also mean that the USB PHY cannot be used while WiFi is enabled.

Default value:

- Yes (enabled) if (CONFIG_ESP_CONSOLE_USB_SERIAL_JTAG || CONFIG_ESP_CONSOLE_SECONDARY_USB_SERIAL_JTAG) && SOC_WIFI_PHY_NEEDS_USB_WORKAROUND
- No (disabled) if SOC_WIFI_PHY_NEEDS_USB_WORKAROUND

CONFIG_ESP_PHY_ENABLE_CERT_TEST

Enable RF certification test functions

Found in: Component config > PHY

If enabled, you can use RF certification test APIs.

Default value:

- No (disabled)

CONFIG_ESP_PHY_CALIBRATION_MODE

Calibration mode

Found in: Component config > PHY

Select PHY calibration mode. During RF initialization, the partial calibration method is used by default for RF calibration. Full calibration takes about 100ms more than partial calibration. If boot duration is not critical, it is suggested to use the full calibration method. No calibration method is only used when the device wakes up from deep sleep.

Available options:

- Calibration partial (CONFIG_ESP_PHY_RF_CAL_PARTIAL)
- Calibration none (CONFIG_ESP_PHY_RF_CAL_NONE)
- Calibration full (CONFIG_ESP_PHY_RF_CAL_FULL)

CONFIG_ESP_PHY_IMPROVE_RX_11B

Improve Wi-Fi receive 11b pkts

Found in: Component config > PHY

This is a workaround to improve Wi-Fi receive 11b pkts for some modules using AC-DC power supply with high interference, enable this option will sacrifice Wi-Fi OFDM receive performance. But to guarantee 11b receive performance serves as a bottom line in this case.

Default value:

- No (disabled) if SOC_PHY_IMPROVE_RX_11B

CONFIG_ESP_PHY_PLL_TRACK_PERIOD_MS

Set the period of the pll track

Found in: Component config > PHY

Set the period of the pll track.

Default value:

- 1000

CONFIG_ESP_PHY_PLL_TRACK_DEBUG

Enable pll track logging

Found in: Component config > PHY

If enabled, there will be some logs while pll tracking

Default value:

- No (disabled)

CONFIG_ESP_PHY_RECORD_USED_TIME

Record PHY used time

Found in: Component config > PHY

Select to support record and query phy used time.

Default value:

- No (disabled)

CONFIG_ESP_PHY_IRAM_OPT

PHY IRAM speed optimization

Found in: Component config > PHY

Select this option to place frequently called PHY library functions in IRAM. When this option is disabled, more than 1.1Kbytes of IRAM memory will be saved, but PHY performance will be reduced. This config only affect esp32c2 now.

Default value:

- Yes (enabled)

CONFIG_ESP_PHY_DEBUG

Enable PHY Debug

Found in: Component config > PHY

Enabling this option allows different kinds of phy debugging features.

Default value:

- No (disabled)

Contains:

- CONFIG_ESP_PHY_DISABLE_PLL_TRACK
- CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

CONFIG_ESP_PHY_DISABLE_PLL_TRACK

Disable phy pll track(only for experimental)

Found in: Component config > PHY > CONFIG_ESP_PHY_DEBUG

Disable pll track. This configuration option is used for experimental. PLL track helps the PHY module adapt to temperature changes, ensuring stable performance. When pll enabled, the ESP PHY module will periodically track and adjust PLL parameters.

Default value:

- No (disabled) if CONFIG_ESP_PHY_DEBUG

CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

PHY temperature tracking debug mode

Found in: Component config > PHY > CONFIG_ESP_PHY_DEBUG

If enabled, allows customization of temperature delta for PHY track pll.

Default value:

- No (disabled) if CONFIG_ESP_PHY_DEBUG

CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG_FLAG

Debug flag for PHY temperature tracking

Found in: Component config > PHY > CONFIG_ESP_PHY_DEBUG > CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

Bitmask debug flag passed to the PHY temperature tracking routine. Each bit enables a specific tracking mode.

- bit0: PLL tracking
- bit1: Calibration parameter tracking

Range:

- from 1 to 3 if CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

Default value:

- 1 if CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

CONFIG_ESP_PHY_PLL_TRACK_TEMP_DELTA

Temperature delta for PHY track pll

Found in: Component config > PHY > CONFIG_ESP_PHY_DEBUG > CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

Set the temperature delta for PHY track pll.

Range:

- from 0 to 255 if CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

Default value:

- 10 if CONFIG_ESP_PHY_PLL_TRACK_TEMP_DEBUG

CONFIG_ESP_PHY_INIT_IRAM

Place PHY (de)init in IRAM

Found in: Component config > PHY

Select this option to place esp_phy and PHY library functions related to phy_init in IRAM. This config applies to ESP32-H2, ESP32-H21, and ESP32-H4.

Default value:

- No (disabled) if SOC_IEEE802154_BLE_ONLY

CONFIG_ESP_PHY_ENABLE_VERSION_PRINT

Print PHY version

Found in: Component config > PHY

Select to print PHY version in esp_phy_enable. This config only applies to esp32hxx for now.

Default value:

- Yes (enabled)

Power Management

Contains:

- CONFIG_PM_LIGHTSLEEP_RTC_OSC_CAL_INTERVAL
- CONFIG_PM_SLP_DISABLE_GPIO
- CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_PROTECTION
- CONFIG_PM_LIGHT_SLEEP_CALLBACKS
- CONFIG_PM_SLEEP_FUNC_IN_IRAM
- CONFIG_PM_POWER_DOWN_CPU_IN_LIGHT_SLEEP
- CONFIG_PM_POWER_DOWN_PERIPHERAL_IN_LIGHT_SLEEP
- CONFIG_PM_SLP_IRAM_OPT
- CONFIG_PM_RTOS_IDLE_OPT
- Retentive memory alloaction strategy for light sleep
- CONFIG_PM_ENABLE

CONFIG_PM_SLEEP_FUNC_IN_IRAM

Place Power Management module functions in IRAM

Found in: Component config > Power Management

Default value:

- No (disabled)

CONFIG_PM_ENABLE

Support for power management

Found in: Component config > Power Management

If enabled, application is compiled with support for power management. This option has run-time overhead (increased interrupt latency, longer time to enter idle state), and it also reduces accuracy of RTOS ticks and timers used for timekeeping. Enable this option if application uses power management APIs.

Default value:

- No (disabled) if __DOXYGEN__

CONFIG_PM_DFS_INIT_AUTO

Enable dynamic frequency scaling (DFS) at startup

Found in: Component config > Power Management > CONFIG_PM_ENABLE

If enabled, startup code configures dynamic frequency scaling. Max CPU frequency is set to DEFAULT_CPU_FREQ_MHZ setting, min frequency is set to XTAL frequency. If disabled, DFS will not be active until the application configures it using esp_pm_configure function.

Default value:

- No (disabled) if CONFIG_PM_ENABLE

CONFIG_PM_PROFILING

Enable profiling counters for PM locks

Found in: Component config > Power Management > CONFIG_PM_ENABLE

If enabled, esp_pm_* functions will keep track of the amount of time each of the power management locks has been held, and esp_pm_dump_locks function will print this information. This feature can be used to analyze which locks are preventing the chip from going into a lower power state, and see what time the chip spends in each power saving mode. This feature does incur some run-time overhead, so should typically be disabled in production builds.

Default value:

- No (disabled) if CONFIG_PM_ENABLE

CONFIG_PM_TRACE

Enable debug tracing of PM using GPIOs

Found in: Component config > Power Management > CONFIG_PM_ENABLE

If enabled, some GPIOs will be used to signal events such as RTOS ticks, frequency switching, entry/exit from idle state. Refer to pm_trace.c file for the list of GPIOs. This feature is intended to be used when analyzing/debugging behavior of power management implementation, and should be kept disabled in applications.

Default value:

- No (disabled) if CONFIG_PM_ENABLE

CONFIG_PM_SLP_IRAM_OPT

Put lightsleep related codes in internal RAM

Found in: Component config > Power Management

If enabled, about 2.1KB of lightsleep related source code would be in IRAM and chip would sleep longer for 310us at 160MHz CPU frequency most each time. This feature is intended to be used when lower power consumption is needed while there is enough place in IRAM to place source code.

CONFIG_PM_RTOS_IDLE_OPT

Put RTOS IDLE related codes in internal RAM

Found in: Component config > Power Management

If enabled, about 180Bytes of RTOS_IDLE related source code would be in IRAM and chip would sleep longer for 20us at 160MHz CPU frequency most each time. This feature is intended to be used when lower power consumption is needed while there is enough place in IRAM to place source code.

CONFIG_PM_SLP_DISABLE_GPIO

Disable all GPIO when chip at sleep

Found in: Component config > Power Management

This feature is intended to disable all GPIO pins at automantic sleep to get a lower power mode. If enabled, chips will disable all GPIO pins at automantic sleep to reduce about 200~300 uA current. If you want to specifically use some pins normally as chip wakes when chip sleeps, you can call 'gpio_sleep_sel_dis' to disable this feature on those pins. You can also keep this feature on and call 'gpio_sleep_set_direction' and 'gpio_sleep_set_pull_mode' to have a different GPIO configuration at sleep.

CONFIG_PM_LIGHTSLEEP_RTC_OSC_CAL_INTERVAL

Calibrate the RTC_FAST/SLOW clock every N times of light sleep

Found in: Component config > Power Management

The value of this option determines the calibration interval of the RTC_FAST/SLOW clock during sleep when power management is enabled. When it is configured as N, the RTC_FAST/SLOW clock will be calibrated every N times of lightsleep. Decreasing this value will increase the time the chip is in the active state, thereby increasing the average power consumption of the chip. Increasing this value can reduce the average power consumption, but when the external environment changes drastically and the chip RTC_FAST/SLOW oscillator frequency drifts, it may cause system instability.

Range:

- from 1 to 128 if CONFIG_PM_ENABLE

Default value:

- 1 if CONFIG_PM_ENABLE

CONFIG_PM_POWER_DOWN_CPU_IN_LIGHT_SLEEP

Power down CPU in light sleep

Found in: Component config > Power Management

If enabled, the CPU will be powered down in light sleep, ESP chips supports saving and restoring CPU's running context before and after light sleep, the feature provides applications with seamless CPU powerdowned lightsleep without user awareness. But this will takes up some internal memory. On esp32c3 soc, enabling this option will consume 1.68 KB of internal RAM and will reduce sleep current consumption by about 100 uA. On esp32s3 soc, enabling this option will consume 8.58 KB of internal RAM and will reduce sleep current consumption by about 650 uA.

Retentive memory alloaction strategy for light sleep

In light sleep mode, certain CPU states and data need to be preserved in memory while cpu domain is powered down. This configuration determines how the required memory resources are allocated. The PM_CPU_RETENTION_DYNAMIC option allows more flexible runtime memory management by supporting access to larger memory. However, it carries the risk of insufficient memory when the chip enters the light-sleep mode, even resulting in retention failure. In contrast, the PM_CPU_RETENTION_STATIC option ensures reliable sleep CPU retention during the system initialization, but reduces the memory available during runtime.

Available options:

- Dynamically allocate memory (CONFIG_PM_CPU_RETENTION_DYNAMIC)

  Allocate memory for retention data only when preparing to enter light sleep.

- Statically allocate memory (CONFIG_PM_CPU_RETENTION_STATIC)

  Pre-allocate all required retention memory during system initialization.

CONFIG_PM_POWER_DOWN_PERIPHERAL_IN_LIGHT_SLEEP

Power down Digital Peripheral in light sleep (EXPERIMENTAL)

Found in: Component config > Power Management

If enabled, digital peripherals will try to powered down in light sleep, then all related peripherals will not be available during sleep, including wake-up sources from the peripherals (For detailed availability information, see the note of the corresponding wakeup source enable function). The chip will automatically save/restore register context during sleep/wakeup to make the upper layer user unaware of the peripheral powerdown during sleep. Enabling this option will increase static RAM and heap usage but will also significantly reduce power. consumption during lightsleep, the actual memory cost depends on the peripherals you have initialized, for specific power consumption data in this mode, please refer to Electrical Characteristics section in the chip datasheet. (In order to save/restore the context of the necessary hardware for FreeRTOS to run, it will need at least 4.55 KB free heap at sleep time. Otherwise sleep will not power down the peripherals.)

Note1: Please use this option with caution, the current IDF does not support the retention of all peripherals. When the digital peripherals are powered off and a sleep and wake-up is completed, the peripherals that have not saved the running context are equivalent to performing a reset. !!! Please confirm the peripherals used in your application and their sleep retention support status before enabling this option, peripherals sleep retention driver support status is tracked in power_management.rst

Note2: When this option is enabled simultaneously with FREERTOS_USE_TICKLESS_IDLE, since the UART will be powered down, the uart FIFO will be flushed before sleep to avoid data loss, however, this has the potential to block the sleep process and cause the wakeup time to be skipped, which will cause the tick of freertos to not be compensated correctly when returning from sleep and cause the system to crash. To avoid this, you can increase FREERTOS_IDLE_TIME_BEFORE_SLEEP threshold in menuconfig.

Note3: Enabling this option does not necessarily mean that the peripheral power domain will be turned down during sleep. The control priority of *esp_sleep_pd_config* is higher than this option, user code can still prevent the peripheral power domain from powering down during sleep by *esp_sleep_pd_config(ESP_PD_DOMAIN_TOP, ESP_PD_OPTION_ON)*. In addition, whether the peripheral power domain is powered down during sleep also depends on the sleep working strategy selected by the driver. If any module belonging to the peripheral power domain chooses not to be powered down during sleep, then the peripheral power domain will not be powered off either.

Default value:

- No (disabled) if SOC_PM_SUPPORT_TOP_PD && SOC_PAU_SUPPORTED

CONFIG_PM_LIGHT_SLEEP_CALLBACKS

Enable registration of pm light sleep callbacks

Found in: Component config > Power Management

If enabled, it allows user to register entry and exit callbacks which are called before and after entering auto light sleep.

NOTE: These callbacks are executed from the IDLE task context hence you cannot have any blocking calls in your callbacks.

NOTE: Enabling these callbacks may change sleep duration calculations based on time spent in callback and hence it is highly recommended to keep them as short as possible

Default value:

- No (disabled) if CONFIG_FREERTOS_USE_TICKLESS_IDLE

CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_PROTECTION

Enable light sleep tick overflow protection

Found in: Component config > Power Management

Limits tick compensation after light sleep wakeup to prevent vTaskStepTick() assertion failure when actual wakeup overhead exceeds estimation (due to cache misses, CPU frequency changes, or flash latency variations).

When enabled:

- Silently limits slept_ticks to xExpectedIdleTime when oversleep is within tolerance (configured by PM_LIGHTSLEEP_TICK_OVERFLOW_TOLERANCE), preventing assertion failure
- Does not limit when oversleep exceeds tolerance (may indicate a bug), assertion failure may occur
- May lose ticks in rare cases, causing xTickCount to lag behind esp_timer

When disabled (default):

- Accurate tick compensation, better precision
- May trigger assertion failure and system crash if overslept

Keep disabled by default to maintain RTOS tick accuracy. Enable only when you experience assertion failures related to vTaskStepTick() and can accept slight inaccuracy of RTOS tick time compared to real time.

Default value:

- No (disabled) if CONFIG_FREERTOS_USE_TICKLESS_IDLE

CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_TOLERANCE

Light sleep tick overflow tolerance

Found in: Component config > Power Management > CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_PROTECTION

Maximum number of ticks that can be overslept before triggering. When oversleep is within this tolerance, the system silently limits slept_ticks to prevent assertion failure. When oversleep exceeds this tolerance, assertion failure may occur and system may crash.

Higher values provide more tolerance for estimation variance but may hide real issues and cause more tick loss, leading to greater RTOS tick time inaccuracy compared to real time. Lower values provide better RTOS tick accuracy but may cause false positives.

Range:

- from 1 to 10 if CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_PROTECTION

Default value:

- 2 if CONFIG_PM_LIGHTSLEEP_TICK_OVERFLOW_PROTECTION

ESP PSRAM

Contains:

- CONFIG_SPIRAM

CONFIG_SPIRAM

Support for external, SPI-connected RAM

Found in: Component config > ESP PSRAM

This enables support for an external SPI RAM chip, connected in parallel with the main SPI flash chip.

SPI RAM config

Contains:

- CONFIG_SPIRAM_ALLOW_BSS_SEG_EXTERNAL_MEMORY
- CONFIG_SPIRAM_ALLOW_NOINIT_SEG_EXTERNAL_MEMORY
- CONFIG_SPIRAM_SPIWP_SD3_PIN
- CONFIG_SPIRAM_BANKSWITCH_ENABLE
- CONFIG_SPIRAM_2T_MODE

- CONFIG_SPIRAM_CACHE_WORKAROUND
- CONFIG_SPIRAM_IGNORE_NOTFOUND
- CONFIG_SPIRAM_BOOT_HW_INIT
- CONFIG_SPIRAM_MALLOC_ALWAYSINTERNAL
- CONFIG_SPIRAM_PRE_CONFIGURE_MEMORY_PROTECTION
- PSRAM clock and cs IO for ESP32-D2WD
- PSRAM clock and cs IO for ESP32-DOWD
- PSRAM clock and cs IO for ESP32-PICO-D4
- CONFIG_SPIRAM_MALLOC_RESERVE_INTERNAL
- CONFIG_SPIRAM_MEMTEST
- CONFIG_SPIRAM_SPEED
- CONFIG_SPIRAM_OCCUPY_SPI_HOST
- CONFIG_SPIRAM_USE
- SPIRAM cache workaround debugging
- SPIRAM workaround libraries placement
- CONFIG_SPIRAM_TRY_ALLOCATE_WIFI_LWIP
- CONFIG_SPIRAM_TYPE
- CONFIG_SPIRAM_CUSTOM_SPIWP_SD3_PIN

CONFIG_SPIRAM_TYPE

Type of SPI RAM chip in use

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Available options:

- Auto-detect (CONFIG_SPIRAM_TYPE_AUTO)
- ESP-PSRAM16 or APS1604 (CONFIG_SPIRAM_TYPE_ESPPSRAM16)
- ESP-PSRAM32 (CONFIG_SPIRAM_TYPE_ESPPSRAM32)
- ESP-PSRAM64 or LY68L6400 (CONFIG_SPIRAM_TYPE_ESPPSRAM64)

CONFIG_SPIRAM_SPEED

Set RAM clock speed

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Select the speed for the SPI RAM chip. If SPI RAM is enabled, we only support three combinations of SPI speed mode we supported now:

1. Flash SPI running at 40Mhz and RAM SPI running at 40Mhz
2. Flash SPI running at 80Mhz and RAM SPI running at 40Mhz
3. Flash SPI running at 80Mhz and RAM SPI running at 80Mhz

Note: If the third mode(80Mhz+80Mhz) is enabled for SPI RAM of type 32MBit, one of the HSPI/VSPI host will be occupied by the system. Which SPI host to use can be selected by the config item SPIRAM_OCCUPY_SPI_HOST. Application code should never touch HSPI/VSPI hardware in this case. The option to select 80MHz will only be visible if the flash SPI speed is also 80MHz. (ESPTOOLPY_FLASHFREQ_80M is true)

Available options:

- 40MHz clock speed (CONFIG_SPIRAM_SPEED_40M)
- 80MHz clock speed (CONFIG_SPIRAM_SPEED_80M)

CONFIG_SPIRAM_BOOT_HW_INIT

Initialise the PSRAM related hardware

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

If this is enabled, the PSRAM hardware will be initialized during startup. Enabling this config does not ensure make sure the availability of the PSRAM for usage, but just initialises the PSRAM hardware. This is necessary to configure PSRAM memory protection during the boot up.

CONFIG_SPIRAM_BOOT_INIT

Initialize PSRAM during startup, including the hardware and memory related configurations

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > CONFIG_SPIRAM_BOOT_HW_INIT

If this is enabled, the PSRAM will be enabled during initial boot. Unless you have specific requirements, you'll want to leave this enabled so memory allocated during boot-up can also be placed in PSRAM.

CONFIG_SPIRAM_PRE_CONFIGURE_MEMORY_PROTECTION

Pre-configure memory protection for PSRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

If this is enabled, the PSRAM will be pre-configured for memory protection during initial boot. This configuration takes into consideration the PSRAM memory configurations that are performed by ESP-IDF's default PSRAM initialization function, esp_psram_init(). Thus, the config is enabled by default when SPIRAM_BOOT_INIT is enabled, because the function esp_psram_init() would be called in the startup code.

In case you wish to disable SPIRAM_BOOT_INIT just for delaying the PSRAM initialization and plan to use the ESP-IDF's default PSRAM initialization function, esp_psram_init() in the application code, you should still enable this config to enable memory protection for the PSRAM.

Note that enabling this config also considers that the rest of the PSRAM memory that is left after the memory configurations are performed by esp_psram_init(), can be allocated to the heap using the function esp_psram_extram_add_to_heap_allocator(), thus configures this region with heap memory protection (RW).

As an advanced usage, if you plan to initialize the PSRAM memory regions manually by yourself without using the function esp_psram_init(), you should disable this config to avoid any memory protection and usage conflicts.

CONFIG_SPIRAM_IGNORE_NOTFOUND

Ignore PSRAM when not found

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Normally, if psram initialization is enabled during compile time but not found at runtime, it is seen as an error making the CPU panic. If this is enabled, booting will complete but no PSRAM will be available. In particular, it is important to note that when SPIRAM_IGNORE_NOTFOUND is enabled, some WIFI related configs will be set to the default value used when SPIRAM is disabled. It can be assumed that by enabling this config, WIFI and LWIP will assume that the current chip does not have SPIRAM. SPIRAM_TRY_ALLOCATE_WIFI_LWIP will not work, buffers in WIFI and LWIP will be set to smaller ranges and default values. WIFI_CACHE_TX_BUFFER_NUM and ESP_WIFI_AMSDU_TX_ENABLED will be disabled, ESP_WIFI_TX_BUFFER will be set to ESP_WIFI_DYNAMIC_TX_BUFFER.

CONFIG_SPIRAM_USE

SPI RAM access method

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

The SPI RAM can be accessed in multiple methods: by just having it available as an unmanaged memory region in the CPU's memory map, by integrating it in the heap as 'special' memory needing heap_caps_malloc to allocate, or by fully integrating it making malloc() also able to return SPI RAM pointers.

Available options:

- Integrate RAM into memory map (CONFIG_SPIRAM_USE_MEMMAP)
- Add RAM to heap_caps allocator (malloc() stays internal by default) (CONFIG_SPIRAM_USE_CAPS_ALLOC)

  Add SPI RAM to the capability allocator.

  Use heap_caps_malloc(..., MALLOC_CAP_SPIRAM) to explicitly allocate from SPI RAM. SPI RAM may also be returned by heap_caps_malloc(..., MALLOC_CAP_DEFAULT), since this heap is also registered with MALLOC_CAP_DEFAULT.

  Standard malloc() does not allocate from SPI RAM by default in this mode.
- Make RAM allocatable using malloc() as well (CONFIG_SPIRAM_USE_MALLOC)

CONFIG_SPIRAM_MEMTEST

Run memory test on SPI RAM initialization

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Runs a rudimentary memory test on initialization. Aborts when memory test fails. Disable this for slightly faster startup.

CONFIG_SPIRAM_MALLOC_ALWAYSINTERNAL

Maximum malloc() size, in bytes, to always put in internal memory

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

If malloc() is capable of also allocating SPI-connected ram, its allocation strategy will prefer to allocate chunks less than this size in internal memory, while allocations larger than this will be done from external RAM. If allocation from the preferred region fails, an attempt is made to allocate from the non-preferred region instead, so malloc() will not suddenly fail when either internal or external memory is full.

CONFIG_SPIRAM_TRY_ALLOCATE_WIFI_LWIP

Try to allocate memories of WiFi and LWIP in SPIRAM firstly. If failed, allocate internal memory

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Try to allocate memories of WiFi and LWIP in SPIRAM firstly. If failed, try to allocate internal memory then.

CONFIG_SPIRAM_MALLOC_RESERVE_INTERNAL

Reserve this amount of bytes for data that specifically needs to be in DMA or internal memory

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Because the external/internal RAM allocation strategy is not always perfect, it sometimes may happen that the internal memory is entirely filled up. This causes allocations that are specifically done in internal memory, for example the stack for new tasks or memory to service DMA or have memory that's also available when SPI cache is down, to fail. This option reserves a pool specifically for requests like that; the memory in this pool is not given out when a normal malloc() is called.

Set this to 0 to disable this feature.

Note that because FreeRTOS stacks are forced to internal memory, they will also use this memory pool; be sure to keep this in mind when adjusting this value.

Note also that the DMA reserved pool may not be one single contiguous memory region, depending on the configured size and the static memory usage of the app.

CONFIG_SPIRAM_ALLOW_BSS_SEG_EXTERNAL_MEMORY

Allow .bss segment placed in external memory

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

If enabled, variables with EXT_RAM_BSS_ATTR attribute will be placed in SPIRAM instead of internal DRAM. BSS section of *lwip*, *net80211*, *pp*, *bt* libraries will be automatically placed in SPIRAM. BSS sections from other object files and libraries can also be placed in SPIRAM through linker fragment scheme *extram_bss*.

Note that the variables placed in SPIRAM using EXT_RAM_BSS_ATTR will be zero initialized.

CONFIG_SPIRAM_ALLOW_NOINIT_SEG_EXTERNAL_MEMORY

Allow .noinit segment placed in external memory

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

If enabled, noinit variables can be placed in PSRAM using EXT_RAM_NOINIT_ATTR.

Note the values placed into this section will not be initialized at startup and should keep its value after software restart.

CONFIG_SPIRAM_CACHE_WORKAROUND

Enable workaround for bug in SPI RAM cache for Rev1 ESP32s

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Revision 1 of the ESP32 has a bug that can cause a write to PSRAM not to take place in some situations when the cache line needs to be fetched from external RAM and an interrupt occurs. This enables a fix in the compiler (-mfix-esp32-psram-cache-issue) that makes sure the specific code that is vulnerable to this will not be emitted.

This will also not use any bits of newlib that are located in ROM, opting for a version that is compiled with the workaround and located in flash instead.

The workaround is not required for ESP32 revision 3 and above.

SPIRAM cache workaround debugging

Contains:

- CONFIG_SPIRAM_CACHE_WORKAROUND_STRATEGY

CONFIG_SPIRAM_CACHE_WORKAROUND_STRATEGY

Workaround strategy

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM cache workaround debugging

Select the workaround strategy. Note that the strategy for precompiled libraries (libgcc, libc, bt, wifi) is not affected by this selection.

Unless you know you need a different strategy, it's suggested you stay with the default MEMW strategy. Note that DUPLDST can interfere with hardware encryption and this will be automatically disabled if this workaround is selected. 'Insert nops' is the workaround that was used in older esp-idf versions. This workaround still can cause faulty data transfers from/to SPI RAM in some situation.

Available options:

- Insert memw after vulnerable instructions (default) (CONFIG_SPIRAM_CACHE_WORKAROUND_STRATEGY_MEMW)
- Duplicate LD/ST for 32-bit, memw for 8/16 bit (CONFIG_SPIRAM_CACHE_WORKAROUND_STRATEGY_DUPLDST)
- Insert nops between vulnerable loads/stores (old strategy, obsolete) (CONFIG_SPIRAM_CACHE_WORKAROUND_STRATEGY_NOPS)

SPIRAM workaround libraries placement

Contains:

- CONFIG_SPIRAM_CACHE_LIBCHAR_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBENV_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBFILE_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBIO_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBJMP_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBMATH_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBMEM_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBMISC_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBNUMPARSER_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBRAND_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBSTR_IN_IRAM
- CONFIG_SPIRAM_CACHE_LIBTIME_IN_IRAM

CONFIG_SPIRAM_CACHE_LIBJMP_IN_IRAM

Put libc's jump related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: longjmp and setjmp. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBMATH_IN_IRAM

Put libc's math related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: abs, div, labs, ldiv, quorem, fpclassify, and nan. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBNUMPARSER_IN_IRAM

Put libc's number parsing related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: utoa, itoa, atoi, atol, strtol, and strtoul. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBIO_IN_IRAM

Put libc's I/O related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: wcrtomb, fvwrite, wbuf, wsetup, fputwc, wctomb_r, ungetc, makebuf, fflush, refill, and sccl. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBTIME_IN_IRAM

Put libc's time related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: asctime, asctime_r, ctime, ctime_r, lcltime, lcltime_r, gmtime, gmtime_r, strftime, mktime, tzset_r, tzset, time, gettzinfo, systimes, month_lengths, timelocal, tzvars, tzlock, tzcalc_limits, and strptime. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBCHAR_IN_IRAM

Put libc's characters related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: ctype_, toupper, tolower, toascii, strupr, bzero, isalnum, isalpha, isascii, isblank, iscntrl, isdigit, isgraph, islower, isprint, ispunct, isspace, and isupper. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBMEM_IN_IRAM

Put libc's memory related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: memccpy, memchr memmove, and memrchr. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBSTR_IN_IRAM

Put libc's string related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: strcasecmp, strcasestr, strchr, strcoll, strcpy, strcspn, strdup, strdup_r, strlcat, strlcpy, strlen, strlwr, strncasecmp, strncat, strncmp, strncpy, strndup, strndup_r, strrchr, strsep, strspn, strstr, strtok_r, and strupr. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBRAND_IN_IRAM

Put libc's random related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: srand, rand, and rand_r. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBENV_IN_IRAM

Put libc's environment related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: environ, envlock, and getenv_r. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBFILE_IN_IRAM

Put libc's file related functions in IRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: lock, isatty, fclose, open, close, creat, read, rshift, sbrk, stdio, syssbrk, sysclose, sysopen, creat, sysread, syswrite, impure, fwalk, and findfp. Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_CACHE_LIBMISC_IN_IRAM

Put libc's miscellaneous functions in IRAM, see help

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > SPIRAM workaround libraries placement

The functions affected by this option are: raise and system Putting these function in IRAM will allow them to be called when flash cache is disabled but it will also reduce the available size of free IRAM for the user application.

CONFIG_SPIRAM_BANKSWITCH_ENABLE

Enable bank switching for >4MiB external RAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

The ESP32 only supports 4MiB of external RAM in its address space. The hardware does support larger memories, but these have to be bank-switched in and out of this address space. Enabling this allows you to reserve some MMU pages for this, which allows the use of the esp_himem api to manage these banks.

#Note that this is limited to 62 banks, as esp_psram_extram_writeback_cache needs some kind of mapping of #some banks below that mark to work. We cannot at this moment guarantee this to exist when himem is #enabled.

If spiram 2T mode is enabled, the size of 64Mbit psram will be changed as 32Mbit, so himem will be unusable.

CONFIG_SPIRAM_BANKSWITCH_RESERVE

Amount of 32K pages to reserve for bank switching

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > CONFIG_SPIRAM_BANKSWITCH_ENABLE

Select the amount of banks reserved for bank switching. Note that the amount of RAM allocatable with malloc/esp_heap_alloc_caps will decrease by 32K for each page reserved here.

Note that this reservation is only actually done if your program actually uses the himem API. Without any himem calls, the reservation is not done and the original amount of memory will be available to malloc/esp_heap_alloc_caps.

CONFIG_SPIRAM_OCCUPY_SPI_HOST

SPI host to use for 32MBit PSRAM

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

When both flash and PSRAM is working under 80MHz, and the PSRAM is of type 32MBit, one of the HSPI/VSPI host will be used to output the clock. Select which one to use here.

Available options:

- HSPI host (SPI2) (CONFIG_SPIRAM_OCCUPY_HSPI_HOST)
- VSPI host (SPI3) (CONFIG_SPIRAM_OCCUPY_VSPI_HOST)
- Will not try to use any host, will abort if not able to use the PSRAM (CONFIG_SPIRAM_OCCUPY_NO_HOST)

PSRAM clock and cs IO for ESP32-DOWD

Contains:

- CONFIG_D0WD_PSRAM_CLK_IO
- CONFIG_D0WD_PSRAM_CS_IO

CONFIG_D0WD_PSRAM_CLK_IO

PSRAM CLK IO number

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > PSRAM clock and cs IO for ESP32-DOWD

The PSRAM CLOCK IO can be any unused GPIO, user can config it based on hardware design. If user use 1.8V flash and 1.8V psram, this value can only be one of 6, 7, 8, 9, 10, 11, 16, 17.

If configured to the same pin as Flash, PSRAM shouldn't be rev0. Contact Espressif for more information.

CONFIG_D0WD_PSRAM_CS_IO

PSRAM CS IO number

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > PSRAM clock and cs IO for ESP32-DOWD

The PSRAM CS IO can be any unused GPIO, user can config it based on hardware design. If user use 1.8V flash and 1.8V psram, this value can only be one of 6, 7, 8, 9, 10, 11, 16, 17.

PSRAM clock and cs IO for ESP32-D2WD

Contains:

- CONFIG_D2WD_PSRAM_CLK_IO
- CONFIG_D2WD_PSRAM_CS_IO

CONFIG_D2WD_PSRAM_CLK_IO

PSRAM CLK IO number

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > PSRAM clock and cs IO for ESP32-D2WD

User can config it based on hardware design. For ESP32-D2WD chip, the psram can only be 1.8V psram, so this value can only be one of 6, 7, 8, 9, 10, 11, 16, 17.

If configured to the same pin (GPIO6) as Flash, PSRAM shouldn't be rev0. Contact Espressif for more information.

CONFIG_D2WD_PSRAM_CS_IO

PSRAM CS IO number

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > PSRAM clock and cs IO for ESP32-D2WD

User can config it based on hardware design. For ESP32-D2WD chip, the psram can only be 1.8V psram, so this value can only be one of 6, 7, 8, 9, 10, 11, 16, 17.

PSRAM clock and cs IO for ESP32-PICO-D4

Contains:

- CONFIG_PICO_PSRAM_CS_IO

CONFIG_PICO_PSRAM_CS_IO

PSRAM CS IO number

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config > PSRAM clock and cs IO for ESP32-PICO-D4

The PSRAM CS IO can be any unused GPIO, user can config it based on hardware design.

For ESP32-PICO chip, the psram share clock with flash, so user do not need to configure the clock IO. For the reference hardware design, please refer to https://www.espressif.com/sites/default/files/documentation/esp32-pico-d4_datasheet_en.pdf

CONFIG_SPIRAM_CUSTOM_SPIWP_SD3_PIN

Use custom SPI PSRAM WP(SD3) Pin when flash pins set in eFuse (read help)

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

This setting is only used if the SPI flash pins have been overridden by setting the eFuses SPI_PAD_CONFIG_xxx, and the SPI flash mode is DIO or DOUT.

When this is the case, the eFuse config only defines 3 of the 4 Quad I/O data pins. The WP pin (aka ESP32 pin "SD_DATA_3" or SPI flash pin "IO2") is not specified in eFuse. The psram only has QPI mode, so a WP pin setting is necessary.

If this config item is set to N (default), the correct WP pin will be automatically used for any Espressif chip or module with integrated flash. If a custom setting is needed, set this config item to Y and specify the GPIO number connected to the WP pin.

When flash mode is set to QIO or QOUT, the PSRAM WP pin will be set the same as the SPI Flash WP pin configured in the bootloader.

CONFIG_SPIRAM_SPIWP_SD3_PIN

Custom SPI PSRAM WP(SD3) Pin

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

The option "Use custom SPI PSRAM WP(SD3) pin" must be set or this value is ignored

If burning a customized set of SPI flash pins in eFuse and using DIO or DOUT mode for flash, set this value to the GPIO number of the SPIRAM WP pin.

CONFIG_SPIRAM_2T_MODE

Enable SPI PSRAM 2T mode

Found in: Component config > ESP PSRAM > CONFIG_SPIRAM > SPI RAM config

Enable this option to fix single bit errors inside 64Mbit PSRAM.

Some 64Mbit PSRAM chips have a hardware issue in the RAM which causes bit errors at multiple fixed bit positions.

Note: If this option is enabled, the 64Mbit PSRAM chip will appear to be 32Mbit in size. Applications will not be affected unless the use the esp_himem APIs, which are not supported in 2T mode.

ESP Ringbuf

Contains:

- CONFIG_RINGBUF_PLACE_ISR_FUNCTIONS_INTO_FLASH
- CONFIG_RINGBUF_IN_IRAM

CONFIG_RINGBUF_IN_IRAM

Place ring buffer functions in IRAM

Found in: Component config > ESP Ringbuf

Place ring buffer functions in IRAM for better performance. By default, ring buffer functions are placed in Flash to save IRAM. Enable this option if you need maximum performance for ring buffer operations.

Note: This option increases IRAM usage.

Default value:

- No (disabled)

CONFIG_RINGBUF_PLACE_ISR_FUNCTIONS_INTO_FLASH

Place ISR ringbuf functions into flash

Found in: Component config > ESP Ringbuf

Place ISR ringbuf functions (like xRingbufferSendFromISR/xRingbufferReceiveFromISR) into flash. This frees up IRAM, but the functions can no longer be called when the cache is disabled or from an IRAM interrupt context.

This option is not compatible with ESP-IDF drivers which are configured to run the ISR from an IRAM context, e.g. CONFIG_UART_ISR_IN_IRAM.

ESP-ROM

Contains:

- CONFIG_ESP_ROM_PRINT_IN_IRAM

CONFIG_ESP_ROM_PRINT_IN_IRAM

Place print functions in IRAM

Found in: Component config > ESP-ROM

Default value:

- Yes (enabled)

ESP Security Specific

Contains:

- Crypto DPA Protection
- CONFIG_ESP_ECDSA_ENABLE_P192_CURVE
- CONFIG_ESP_CRYPTO_FORCE_ECC_CONSTANT_TIME_POINT_MUL

Crypto DPA Protection

Contains:

- CONFIG_ESP_CRYPTO_DPA_PROTECTION_AT_STARTUP

CONFIG_ESP_CRYPTO_DPA_PROTECTION_AT_STARTUP

Enable crypto DPA protection at startup

Found in: Component config > ESP Security Specific > Crypto DPA Protection

This config controls the DPA (Differential Power Analysis) protection knob for the crypto peripherals. DPA protection dynamically adjusts clock frequency of the crypto peripheral. DPA protection helps to make it difficult to perform SCA attacks on the crypto peripherals. However, there is also associated performance impact based on the security level set. Please refer to the TRM for more details.

Default value:

- Yes (enabled) if SOC_CRYPTO_DPA_PROTECTION_SUPPORTED

CONFIG_ESP_CRYPTO_DPA_PROTECTION_LEVEL

DPA protection level

Found in: Component config > ESP Security Specific > Crypto DPA Protection > CONFIG_ESP_CRYPTO_DPA_PROTECTION_AT_STARTUP

Configure the DPA protection security level

Available options:

- Security level low (CONFIG_ESP_CRYPTO_DPA_PROTECTION_LEVEL_LOW)
- Security level medium (CONFIG_ESP_CRYPTO_DPA_PROTECTION_LEVEL_MEDIUM)
- Security level high (CONFIG_ESP_CRYPTO_DPA_PROTECTION_LEVEL_HIGH)

CONFIG_ESP_CRYPTO_FORCE_ECC_CONSTANT_TIME_POINT_MUL

Forcefully enable ECC constant time point multiplication operations

Found in: Component config > ESP Security Specific

If enabled, the app startup code will burn the ECC_FORCE_CONST_TIME efuse bit to force the ECC peripheral to always perform constant time point multiplication operations, irrespective of the ECC_MULT_SECURITY_MODE status bit that is present in the ECC_MULT_CONF_REG register. By default, ESP-IDF configures the ECC peripheral to perform constant time point multiplication operations, so enabling this config would provide security enhancement only in the cases when trusted boot is not enabled and the attacker tries carrying out non-constant time point multiplication operations by changing the default ESP-IDF configurations. Performing constant time operations protect the ECC multiplication operations from timing attacks.

For targets that support Secure Boot using ECDSA-P384, the write-protection bit of the efuse bit could be shared by multiple other efuse bits and can be programmed by the application when Secure Boot is enabled. Thus, you could select CONFIG_SECURE_BOOT_SKIP_WRITE_PROTECTION_SCA, in case you would like to skip the write-protection of the efuse bit.

Default value:

- No (disabled) if SOC_ECC_CONSTANT_TIME_POINT_MUL

CONFIG_ESP_ECDSA_ENABLE_P192_CURVE

Enable ECDSA 192-curve operations

Found in: Component config > ESP Security Specific

By default, only the 256-bit curve operations are allowed. If this configuration is enabled, it will set the eFuse to allow ECDSA operations using both the 192-bit and 256-bit curves.

Default value:

- No (disabled) if SOC_ECDSA_P192_CURVE_DEFAULT_DISABLED

ESP-STDIO

Contains:

- CONFIG_ESP_CONSOLE_UART
- CONFIG_ESP_CONSOLE_SECONDARY
- CONFIG_ESP_CONSOLE_UART_BAUDRATE
- CONFIG_ESP_CONSOLE_UART_NUM

- CONFIG_ESP_CONSOLE_UART_RX_GPIO
- CONFIG_ESP_CONSOLE_UART_TX_GPIO

CONFIG_ESP_CONSOLE_UART

Channel for console output

Found in: Component config > ESP-STDIO

Select where to send console output (through stdout and stderr).

- Default is to use UART0 on pre-defined GPIOs.
- If "Custom" is selected, UART0 or UART1 can be chosen, and any pins can be selected.
- If "None" is selected, there will be no console output on any UART, except for initial output from ROM bootloader. This ROM output can be suppressed by GPIO strapping or EFUSE, refer to chip datasheet for details.
- On chips with USB OTG peripheral, "USB CDC" option redirects output to the CDC port. This option uses the CDC driver in the chip ROM. This option is incompatible with TinyUSB stack.
- On chips with an USB serial/JTAG debug controller, selecting the option for that redirects output to the CDC/ACM (serial port emulation) component of that device.

Available options:

- Default: UART0 (CONFIG_ESP_CONSOLE_UART_DEFAULT)
- USB CDC (CONFIG_ESP_CONSOLE_USB_CDC)
- USB Serial/JTAG Controller (CONFIG_ESP_CONSOLE_USB_SERIAL_JTAG)
- Custom UART (CONFIG_ESP_CONSOLE_UART_CUSTOM)
- None (CONFIG_ESP_CONSOLE_NONE)

CONFIG_ESP_CONSOLE_SECONDARY

Channel for console secondary output

Found in: Component config > ESP-STDIO

This secondary option supports output through other specific port like USB_SERIAL_JTAG when UART0 port as a primary is selected but not connected. This secondary output currently only supports non-blocking mode without using REPL. If you want to output in blocking mode with REPL or input through this secondary port, please change the primary config to this port in *Channel for console output* menu.

Available options:

- No secondary console (CONFIG_ESP_CONSOLE_SECONDARY_NONE)
- USB_SERIAL_JTAG PORT (CONFIG_ESP_CONSOLE_SECONDARY_USB_SERIAL_JTAG)

  This option supports output through USB_SERIAL_JTAG port when the UART0 port is not connected. The output currently only supports non-blocking mode without using the console. If you want to output in blocking mode with REPL or input through USB_SERIAL_JTAG port, please change the primary config to ESP_CONSOLE_USB_SERIAL_JTAG above.

CONFIG_ESP_CONSOLE_UART_NUM

UART peripheral to use for console output (0-1)

Found in: Component config > ESP-STDIO

This UART peripheral is used for console output from the ESP-IDF Bootloader and the app.

If the configuration is different in the Bootloader binary compared to the app binary, UART is reconfigured after the bootloader exits and the app starts.

Due to an ESP32 ROM bug, UART2 is not supported for console output via esp_rom_printf.

Available options:

- UART0 (CONFIG_ESP_CONSOLE_UART_CUSTOM_NUM_0)
- UART1 (CONFIG_ESP_CONSOLE_UART_CUSTOM_NUM_1)

CONFIG_ESP_CONSOLE_UART_TX_GPIO

UART TX on GPIO<num>

Found in: Component config > ESP-STDIO

This GPIO is used for console UART TX output in the ESP-IDF Bootloader and the app (including boot log output and default standard output and standard error of the app). Value -1 means to continue using the default console UART TX pin.

If the configuration is different in the Bootloader binary compared to the app binary, UART is reconfigured after the bootloader exits and the app starts.

Range:

- from -1 to 33 if CONFIG_ESP_CONSOLE_UART_CUSTOM

Default value:

- 1 if CONFIG_ESP_CONSOLE_UART_CUSTOM
- "-1" if CONFIG_ESP_CONSOLE_UART_CUSTOM

CONFIG_ESP_CONSOLE_UART_RX_GPIO

UART RX on GPIO<num>

Found in: Component config > ESP-STDIO

This GPIO is used for console UART RX input in the ESP-IDF Bootloader and the app (including default standard input of the app). Value -1 means to continue using the default console UART RX pin.

Note: The default ESP-IDF Bootloader configures this pin but doesn't read anything from the UART.

If the configuration is different in the Bootloader binary compared to the app binary, UART is reconfigured after the bootloader exits and the app starts.

Range:

- from -1 to 39 if CONFIG_ESP_CONSOLE_UART_CUSTOM

Default value:

- 3 if CONFIG_ESP_CONSOLE_UART_CUSTOM
- "-1" if CONFIG_ESP_CONSOLE_UART_CUSTOM

CONFIG_ESP_CONSOLE_UART_BAUDRATE

UART console baud rate

Found in: Component config > ESP-STDIO

This baud rate is used by both the ESP-IDF Bootloader and the app (including boot log output and default standard input/output/error of the app).

The app's maximum baud rate depends on the UART clock source. If Power Management is disabled, the UART clock source is the APB clock and all baud rates in the available range will be sufficiently accurate. If Power Management is enabled, REF_TICK clock source is used so the baud rate is divided from 1MHz. Baud rates above 1Mbps are not possible and values between 500Kbps and 1Mbps may not be accurate.

If the configuration is different in the Bootloader binary compared to the app binary, UART is reconfigured after the bootloader exits and the app starts.

Range:

- from 1200 to 1000000 if CONFIG_PM_ENABLE

Default value:

- 115200

ESP System Settings

Contains:

- CONFIG_ESP_SYSTEM_RTC_EXT_XTAL_BOOTSTRAP_CYCLES
- CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ
- CONFIG_ESP_SYSTEM_MEMPROT
- CONFIG_ESP_SYSTEM_ALLOW_RTC_FAST_MEM_AS_HEAP
- CONFIG_ESP_TASK_WDT_EN
- CONFIG_ESP_SYSTEM_EVENT_TASK_STACK_SIZE
- CONFIG_ESP_SYSTEM_HW_PC_RECORD
- CONFIG_ESP_SYSTEM_HW_STACK_GUARD
- CONFIG_ESP_XT_WDT
- CONFIG_ESP_SYSTEM_CHECK_INT_LEVEL
- CONFIG_ESP_INT_WDT
- CONFIG_ESP_MAIN_TASK_AFFINITY
- CONFIG_ESP_MAIN_TASK_STACK_SIZE
- CONFIG_ESP_DEBUG_OCDAWARE
- Memory
- CONFIG_ESP_MINIMAL_SHARED_STACK_SIZE
- CONFIG_ESP_DEBUG_STUBS_ENABLE
- CONFIG_ESP_SYSTEM_PANIC

- CONFIG_ESP_SYSTEM_PANIC_REBOOT_DELAY_SECONDS
- CONFIG_ESP32_DISABLE_BASIC_ROM_CONSOLE
- CONFIG_ESP_PANIC_HANDLER_IRAM
- CONFIG_ESP_SYSTEM_IN_IRAM
- CONFIG_ESP_DEBUG_INCLUDE_OCD_STUB_BINS
- CONFIG_ESP_SYSTEM_EVENT_QUEUE_SIZE
- Trace memory

CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ

CPU frequency

Found in: Component config > ESP System Settings

CPU frequency to be set on application startup.

Available options:

- 40 MHz (CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ_40)
- 80 MHz (CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ_80)
- 160 MHz (CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ_160)
- 240 MHz (CONFIG_ESP_DEFAULT_CPU_FREQ_MHZ_240)

Memory

Contains:

- CONFIG_ESP32_IRAM_AS_8BIT_ACCESSIBLE_MEMORY
- Non-backward compatible options
- CONFIG_ESP32_RTCDATA_IN_FAST_MEM
- CONFIG_ESP32_USE_FIXED_STATIC_RAM_SIZE

CONFIG_ESP32_RTCDATA_IN_FAST_MEM

Place RTC_DATA_ATTR and RTC_RODATA_ATTR variables into RTC fast memory segment

Found in: Component config > ESP System Settings > Memory

This option allows to place .rtc_data and .rtc_rodata sections into RTC fast memory segment to free the slow memory region for ULP programs. This option depends on the CONFIG_ESP_SYSTEM_SINGLE_CORE_MODE option because RTC fast memory can be accessed only by PRO_CPU core.

CONFIG_ESP32_USE_FIXED_STATIC_RAM_SIZE

Use fixed static RAM size

Found in: Component config > ESP System Settings > Memory

If this option is disabled, the DRAM part of the heap starts right after the .bss section, within the dram0_0 region. As a result, adding or removing some static variables will change the available heap size.

If this option is enabled, the DRAM part of the heap starts right after the dram0_0 region, where its length is set with ESP32_FIXED_STATIC_RAM_SIZE

Default value:

- No (disabled)

CONFIG_ESP32_FIXED_STATIC_RAM_SIZE

Fixed Static RAM size

Found in: Component config > ESP System Settings > Memory > CONFIG_ESP32_USE_FIXED_STATIC_RAM_SIZE

RAM size dedicated for static variables (.data & .bss sections). Please note that the actual length will be reduced by BTDM_RESERVE_DRAM if Bluetooth controller is enabled.

Range:

- from 0 to 0x2c200 if CONFIG_ESP32_USE_FIXED_STATIC_RAM_SIZE

Default value:

- "0x1E000" if CONFIG_ESP32_USE_FIXED_STATIC_RAM_SIZE

CONFIG_ESP32_IRAM_AS_8BIT_ACCESSIBLE_MEMORY

Enable IRAM as 8 bit accessible memory

Found in: Component config > ESP System Settings > Memory

If enabled, application can use IRAM as byte accessible region for storing data (Note: IRAM region cannot be used as task stack)

This is possible due to handling of exceptions *LoadStoreError (3)* and *LoadStoreAlignmentError (9)* Each unaligned read/write access will incur a penalty of maximum of 167 CPU cycles.

Non-backward compatible options

Contains:

- CONFIG_ESP_SYSTEM_ESP32_SRAM1_REGION_AS_IRAM

CONFIG_ESP_SYSTEM_ESP32_SRAM1_REGION_AS_IRAM

Reserve parts of SRAM1 for app IRAM (WARNING, read help before enabling)

Found in: Component config > ESP System Settings > Memory > Non-backward compatible options

Reserve parts of SRAM1 for app IRAM which was previously reserved for bootloader DRAM. If booting an app on an older bootloader from before this option was introduced, the app will fail to boot due to not recognizing the new IRAM memory area.

If this is the case please test carefully before pushing out any OTA updates.

Trace memory

Contains:

- CONFIG_ESP32_TRAX

CONFIG_ESP32_TRAX

Use TRAX tracing feature

Found in: Component config > ESP System Settings > Trace memory

The ESP32 contains a feature which allows you to trace the execution path the processor has taken through the program. This is stored in a chunk of 32K (16K for single-processor) of memory that can't be used for general purposes anymore. Disable this if you do not know what this is.

Default value:

- No (disabled)

CONFIG_ESP32_TRAX_TWOBANKS

Reserve memory for tracing both pro as well as app cpu execution

Found in: Component config > ESP System Settings > Trace memory > CONFIG_ESP32_TRAX

The ESP32 contains a feature which allows you to trace the execution path the processor has taken through the program. This is stored in a chunk of 32K (16K for single-processor) of memory that can't be used for general purposes anymore. Disable this if you do not know what this is.

Memory to reverse for trace, used in linker script

Default value:

- No (disabled) if CONFIG_ESP32_TRAX

CONFIG_ESP_SYSTEM_IN_IRAM

Place system functions in IRAM

Found in: Component config > ESP System Settings

The following system functions will be placed in IRAM if this option is enabled: - system startup - system time - system error - system restart - system crosscore - system debug - system APB backup DMA lock - system application tick hook - Unified Behavior Sanitizer (UBSAN) hook - Interrupt watchdog handler - XTAL32K watchdog timer - IPC and IPC ISR

Default value:

- Yes (enabled)

CONFIG_ESP_SYSTEM_PANIC

Panic handler behaviour

Found in: Component config > ESP System Settings

If FreeRTOS detects unexpected behaviour or an unhandled exception, the panic handler is invoked. Configure the panic handler's action here.

Available options:

- Print registers and halt (CONFIG_ESP_SYSTEM_PANIC_PRINT_HALT)

  Outputs the relevant registers over the serial port and halt the processor. Needs a manual reset to restart.

- Print registers and reboot (CONFIG_ESP_SYSTEM_PANIC_PRINT_REBOOT)

  Outputs the relevant registers over the serial port and immediately reset the processor.

- Silent reboot (CONFIG_ESP_SYSTEM_PANIC_SILENT_REBOOT)

  Just resets the processor without outputting anything

- GDBStub on panic (CONFIG_ESP_SYSTEM_PANIC_GDBSTUB)

  Invoke gdbstub on the serial port, allowing for gdb to attach to it to do a postmortem of the crash.

CONFIG_ESP_SYSTEM_PANIC_REBOOT_DELAY_SECONDS

Panic reboot delay (Seconds)

Found in: Component config > ESP System Settings

After the panic handler executes, you can specify a number of seconds to wait before the device reboots.

Range:

- from 0 to 99

Default value:

- 0

CONFIG_ESP_SYSTEM_RTC_EXT_XTAL_BOOTSTRAP_CYCLES

Bootstrap cycles for external 32kHz crystal

Found in: Component config > ESP System Settings

To reduce the startup time of an external RTC crystal, we bootstrap it with a 32kHz square wave for a fixed number of cycles. Setting 0 will disable bootstrapping (if disabled, the crystal may take longer to start up or fail to oscillate under some conditions).

If this value is too high, a faulty crystal may initially start and then fail. If this value is too low, an otherwise good crystal may not start.

To accurately determine if the crystal has started, set a larger "Number of cycles for RTC_SLOW_CLK calibration" (about 3000).

CONFIG_ESP_SYSTEM_ALLOW_RTC_FAST_MEM_AS_HEAP

Enable RTC fast memory for dynamic allocations

Found in: Component config > ESP System Settings

This config option allows to add RTC fast memory region to system heap with capability similar to that of DRAM region but without DMA. Speed wise RTC fast memory operates on APB clock and hence does not have much performance impact.

CONFIG_ESP_SYSTEM_MEMPROT

Enable memory protection

Found in: Component config > ESP System Settings

This option enables memory protection for the valid memory regions. This feature also automatically splits the ROM, RAM and flash memory into data and instruction segments and sets Read/Execute permissions for the instruction part (below given splitting address) and Read/Write permissions for the data part (above the splitting address). The memory protection is effective on all access through the IRAM0 and DRAM0 buses.

Note: allocating memory with MALLOC_CAP_EXEC capability is not possible when this config is enabled.

Default value:

- Yes (enabled) if SOC_CPU_IDRAM_SPLIT_USING_PMP || SECURE_ENABLE_TEE || SOC_MEMPROT_SUPPORTED

CONFIG_ESP_SYSTEM_MEMPROT_MODE

Memory Protection configurations

Found in: Component config > ESP System Settings > CONFIG_ESP_SYSTEM_MEMPROT

Available options:

- Enable Permission Control Module (PMS) configurations (CONFIG_ESP_SYSTEM_MEMPROT_PMS)

  This option enables memory protection using the Permission Control Module (PMS).

- Enable CPU's Physical Memory Protection (PMP) configurations (CONFIG_ESP_SYSTEM_MEMPROT_PMP)

  This option enables memory protection using CPU PMP.

- Enable Trusted Execution Environment (TEE) configurations (CONFIG_ESP_SYSTEM_MEMPROT_TEE)

  This option enables the default memory protection provided by TEE.

CONFIG_ESP_SYSTEM_MEMPROT_PMS_LOCK

Lock memory protection settings

Found in: Component config > ESP System Settings > CONFIG_ESP_SYSTEM_MEMPROT

Once locked, memory protection settings cannot be changed anymore. The lock is reset only on the chip startup.

Default value:

- Yes (enabled) if CONFIG_ESP_SYSTEM_MEMPROT && CONFIG_ESP_SYSTEM_MEMPROT_PMS

CONFIG_ESP_SYSTEM_EVENT_QUEUE_SIZE

System event queue size

Found in: Component config > ESP System Settings

Config system event queue size in different application.

Default value:

- 32

CONFIG_ESP_SYSTEM_EVENT_TASK_STACK_SIZE

Event loop task stack size

Found in: Component config > ESP System Settings

Config system event task stack size in different application.

Default value:

- 2304

CONFIG_ESP_MAIN_TASK_STACK_SIZE

Main task stack size

Found in: Component config > ESP System Settings

Configure the "main task" stack size. This is the stack of the task which calls app_main(). If app_main() returns then this task is deleted and its stack memory is freed.

Default value:

- 3584

CONFIG_ESP_MAIN_TASK_AFFINITY

Main task core affinity

Found in: Component config > ESP System Settings

Configure the "main task" core affinity. This is the used core of the task which calls app_main(). If app_main() returns then this task is deleted.

Available options:

- CPU0 (CONFIG_ESP_MAIN_TASK_AFFINITY_CPU0)
- CPU1 (CONFIG_ESP_MAIN_TASK_AFFINITY_CPU1)
- No affinity (CONFIG_ESP_MAIN_TASK_AFFINITY_NO_AFFINITY)

CONFIG_ESP_MINIMAL_SHARED_STACK_SIZE

Minimal allowed size for shared stack

Found in: Component config > ESP System Settings

Minimal value of size, in bytes, accepted to execute a expression with shared stack.

Default value:

- 2048

CONFIG_ESP_INT_WDT

Interrupt watchdog

Found in: Component config > ESP System Settings

This watchdog timer can detect if the FreeRTOS tick interrupt has not been called for a certain time, either because a task turned off interrupts and did not turn them on for a long time, or because an interrupt handler did not return. It will try to invoke the panic handler first and failing that reset the SoC.

Default value:

- Yes (enabled)

CONFIG_ESP_INT_WDT_TIMEOUT_MS

Interrupt watchdog timeout (ms)

Found in: Component config > ESP System Settings > CONFIG_ESP_INT_WDT

The timeout of the watchdog, in milliseconds. Make this higher than the FreeRTOS tick rate.

Range:

- from 10 to 10000

Default value:

- 800 if CONFIG_SPIRAM && CONFIG_ESP_INT_WDT

CONFIG_ESP_INT_WDT_CHECK_CPU1

Also watch CPU1 tick interrupt

Found in: Component config > ESP System Settings > CONFIG_ESP_INT_WDT

Also detect if interrupts on CPU 1 are disabled for too long.

CONFIG_ESP_TASK_WDT_EN

Enable Task Watchdog Timer

Found in: Component config > ESP System Settings

The Task Watchdog Timer can be used to make sure individual tasks are still running. Enabling this option will enable the Task Watchdog Timer. It can be either initialized automatically at startup or initialized after startup (see Task Watchdog Timer API Reference)

Default value:

- Yes (enabled)

CONFIG_ESP_TASK_WDT_INIT

Initialize Task Watchdog Timer on startup

Found in: Component config > ESP System Settings > CONFIG_ESP_TASK_WDT_EN

Enabling this option will cause the Task Watchdog Timer to be initialized automatically at startup.

Default value:

- Yes (enabled)

CONFIG_ESP_TASK_WDT_PANIC

Invoke panic handler on Task Watchdog timeout

Found in: Component config > ESP System Settings > CONFIG_ESP_TASK_WDT_EN > CONFIG_ESP_TASK_WDT_INIT

If this option is enabled, the Task Watchdog Timer will be configured to trigger the panic handler when it times out. This can also be configured at run time (see Task Watchdog Timer API Reference)

Default value:

- No (disabled)

CONFIG_ESP_TASK_WDT_TIMEOUT_S

Task Watchdog timeout period (seconds)

Found in: Component config > ESP System Settings > CONFIG_ESP_TASK_WDT_EN > CONFIG_ESP_TASK_WDT_INIT

Timeout period configuration for the Task Watchdog Timer in seconds. This is also configurable at run time (see Task Watchdog Timer API Reference)

Range:

- from 1 to 60

Default value:

- 5

CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU0

Watch CPU0 Idle Task

Found in: Component config > ESP System Settings > CONFIG_ESP_TASK_WDT_EN > CONFIG_ESP_TASK_WDT_INIT

If this option is enabled, the Task Watchdog Timer will watch the CPU0 Idle Task. Having the Task Watchdog watch the Idle Task allows for detection of CPU starvation as the Idle Task not being called is usually a symptom of CPU starvation. Starvation of the Idle Task is detrimental as FreeRTOS household tasks depend on the Idle Task getting some runtime every now and then.

Default value:

- Yes (enabled)

CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU1

Watch CPU1 Idle Task

Found in: Component config > ESP System Settings > CONFIG_ESP_TASK_WDT_EN > CONFIG_ESP_TASK_WDT_INIT

If this option is enabled, the Task Watchdog Timer will wach the CPU1 Idle Task.

CONFIG_ESP_XT_WDT

Initialize XTAL32K watchdog timer on startup

Found in: Component config > ESP System Settings

This watchdog timer can detect oscillation failure of the XTAL32K_CLK. When such a failure is detected the hardware can be set up to automatically switch to BACKUP32K_CLK and generate an interrupt.

CONFIG_ESP_XT_WDT_TIMEOUT

XTAL32K watchdog timeout period

Found in: Component config > ESP System Settings > CONFIG_ESP_XT_WDT

Timeout period configuration for the XTAL32K watchdog timer based on RTC_CLK.

Range:

- from 1 to 255 if CONFIG_ESP_XT_WDT

Default value:

- 200 if CONFIG_ESP_XT_WDT

CONFIG_ESP_XT_WDT_BACKUP_CLK_ENABLE

Automatically switch to BACKUP32K_CLK when timer expires

Found in: Component config > ESP System Settings > CONFIG_ESP_XT_WDT

Enable this to automatically switch to BACKUP32K_CLK as the source of RTC_SLOW_CLK when the watchdog timer expires.

Default value:

- Yes (enabled) if CONFIG_ESP_XT_WDT

CONFIG_ESP_PANIC_HANDLER_IRAM

Place panic handler code in IRAM

Found in: Component config > ESP System Settings

If this option is disabled (default), the panic handler code is placed in flash not IRAM. This means that if ESP-IDF crashes while flash cache is disabled, the panic handler will automatically re-enable flash cache before running GDB Stub or Core Dump. This adds some minor risk, if the flash cache status is also corrupted during the crash.

If this option is enabled, the panic handler code (including required UART functions) is placed in IRAM. This may be necessary to debug some complex issues with crashes while flash cache is disabled (for example, when writing to SPI flash) or when flash cache is corrupted when an exception is triggered.

Default value:

- No (disabled)

CONFIG_ESP_DEBUG_STUBS_ENABLE

OpenOCD debug stubs

Found in: Component config > ESP System Settings

Debug stubs are used by OpenOCD to execute pre-compiled onboard code which does some useful debugging stuff, e.g. GCOV data dump.

CONFIG_ESP_DEBUG_INCLUDE_OCD_STUB_BINS

Preload OpenOCD stub binaries to speed up debugging. 8K memory will be reserved

Found in: Component config > ESP System Settings

OpenOCD uses stub code to access flash during programming or when inserting and removing SW flash breakpoints. To execute stub code, OpenOCD allocates memory on the target device, backs up the existing memory, loads the stub binary, runs the binary, and then restores the original memory. This process can be time-consuming, especially when using USB serial JTAG. By enabling this option, 8K of memory in RAM will be preallocated with the stub code, eliminating the need to back up and restore the memory region.

Default value:

- No (disabled) if SOC_DEBUG_HAVE_OCD_STUB_BINS

CONFIG_ESP_DEBUG_OCDAWARE

Make exception and panic handlers JTAG/OCD aware

Found in: Component config > ESP System Settings

The FreeRTOS panic and unhandled exception handlers can detect a JTAG OCD debugger and instead of panicking, have the debugger stop on the offending instruction.

Default value:

- Yes (enabled)

CONFIG_ESP_SYSTEM_CHECK_INT_LEVEL

Interrupt level to use for Interrupt Watchdog and other system checks

Found in: Component config > ESP System Settings

Interrupt level to use for Interrupt Watchdog, IPC_ISR and other system checks.

Available options:

- Level 5 interrupt (CONFIG_ESP_SYSTEM_CHECK_INT_LEVEL_5)

  Using level 5 interrupt for Interrupt Watchdog, IPC_ISR and other system checks.
- Level 4 interrupt (CONFIG_ESP_SYSTEM_CHECK_INT_LEVEL_4)

  Using level 4 interrupt for Interrupt Watchdog, IPC_ISR and other system checks.

CONFIG_ESP32_DISABLE_BASIC_ROM_CONSOLE

Permanently disable BASIC ROM Console

Found in: Component config > ESP System Settings

If set, the first time the app boots it will disable the BASIC ROM Console permanently (by burning an eFuse).

Otherwise, the BASIC ROM Console starts on reset if no valid bootloader is read from the flash.

(Enabling secure boot also disables the BASIC ROM Console by default.)

Default value:

- No (disabled)

CONFIG_ESP_SYSTEM_HW_STACK_GUARD

Hardware stack guard

Found in: Component config > ESP System Settings

This config allows to trigger a panic interrupt when Stack Pointer register goes out of allocated stack memory bounds.

Default value:

- Yes (enabled) if SOC_ASSIST_DEBUG_SUPPORTED

CONFIG_ESP_SYSTEM_HW_PC_RECORD

Hardware PC recording

Found in: Component config > ESP System Settings

This option will enable the PC recording function of assist_debug module. The PC value of the CPU will be recorded to PC record register in assist_debug module in real time. When an exception occurs and the CPU is reset, this register will be kept, then we can use the recorded PC to debug the causes of the reset.

Default value:

- Yes (enabled) if SOC_ASSIST_DEBUG_SUPPORTED

IPC (Inter-Processor Call)

Contains:

- CONFIG_ESP_IPC_TASK_STACK_SIZE
- CONFIG_ESP_IPC_USES_CALLERS_PRIORITY

CONFIG_ESP_IPC_TASK_STACK_SIZE

Inter-Processor Call (IPC) task stack size

Found in: Component config > IPC (Inter-Processor Call)

Configure the IPC tasks stack size. An IPC task runs on each core (in dual core mode), and allows for cross-core function calls. See IPC documentation for more details. The default IPC stack size should be enough for most common simple use cases. However, users can increase/decrease the stack size to their needs.

Range:

- from 2048 to 65536 if APPTRACE_ENABLE

Default value:

- 2048 if APPTRACE_ENABLE
- 1024

CONFIG_ESP_IPC_USES_CALLERS_PRIORITY

IPC runs at caller's priority

Found in: Component config > IPC (Inter-Processor Call)

If this option is not enabled then the IPC task will keep behavior same as prior to that of ESP-IDF v4.0, hence IPC task will run at (configMAX_PRIORITIES - 1) priority.

Default value:

- Yes (enabled)

ESP Timer (High Resolution Timer)

Contains:

- CONFIG_ESP_TIMER_PROFILING
- CONFIG_ESP_TIMER_TASK_AFFINITY
- CONFIG_ESP_TIMER_TASK_STACK_SIZE

- CONFIG_ESP_TIMER_INTERRUPT_LEVEL
- CONFIG_ESP_TIMER_IN_IRAM
- CONFIG_ESP_TIMER_SHOW_EXPERIMENTAL
- CONFIG_ESP_TIMER_SUPPORTS_ISR_DISPATCH_METHOD
- CONFIG_ESP_TIMER_ISR_AFFINITY

CONFIG_ESP_TIMER_IN_IRAM

Place esp_timer functions in IRAM

Found in: Component config > ESP Timer (High Resolution Timer)

Default value:

- Yes (enabled)

CONFIG_ESP_TIMER_PROFILING

Enable esp_timer profiling features

Found in: Component config > ESP Timer (High Resolution Timer)

If enabled, esp_timer_dump will dump information such as number of times the timer was started, number of times the timer has triggered, and the total time it took for the callback to run. This option has some effect on timer performance and the amount of memory used for timer storage, and should only be used for debugging/testing purposes.

Default value:

- No (disabled)

CONFIG_ESP_TIMER_TASK_STACK_SIZE

High-resolution timer task stack size

Found in: Component config > ESP Timer (High Resolution Timer)

Configure the stack size of "timer_task" task. This task is used to dispatch callbacks of timers created using ets_timer and esp_timer APIs. If you are seeing stack overflow errors in timer task, increase this value.

Note that this is not the same as FreeRTOS timer task. To configure FreeRTOS timer task size, see "FreeRTOS timer task stack size" option in "FreeRTOS".

Range:

- from 2048 to 65536

Default value:

- 3584

CONFIG_ESP_TIMER_INTERRUPT_LEVEL

Interrupt level

Found in: Component config > ESP Timer (High Resolution Timer)

This sets the interrupt priority level for esp_timer ISR. A higher value reduces interrupt latency by minimizing the timer processing delay.

Range:

- from 1 to 3

Default value:

- 1

CONFIG_ESP_TIMER_SHOW_EXPERIMENTAL

show esp_timer's experimental features

Found in: Component config > ESP Timer (High Resolution Timer)

This shows some hidden features of esp_timer. Note that they may break other features, use them with care.

CONFIG_ESP_TIMER_TASK_AFFINITY

esp_timer task core affinity

Found in: Component config > ESP Timer (High Resolution Timer)

The default settings: timer TASK on CPU0 and timer ISR on CPU0. Other settings may help in certain cases, but note that they may break other features, use them with care. - "CPU0": (default) esp_timer task is processed by CPU0. - "CPU1": esp_timer task is processed by CPU1. - "No affinity": esp_timer task can be processed by any CPU.

Available options:

- CPU0 (CONFIG_ESP_TIMER_TASK_AFFINITY_CPU0)
- CPU1 (CONFIG_ESP_TIMER_TASK_AFFINITY_CPU1)
- No affinity (CONFIG_ESP_TIMER_TASK_AFFINITY_NO_AFFINITY)

CONFIG_ESP_TIMER_ISR_AFFINITY

timer interrupt core affinity

Found in: Component config > ESP Timer (High Resolution Timer)

The default settings: timer TASK on CPU0 and timer ISR on CPU0. Other settings may help in certain cases, but note that they may break other features, use them with care. - "CPU0": (default) timer interrupt is processed by CPU0. - "CPU1": timer interrupt is processed by CPU1. - "No affinity": timer interrupt can be processed by any CPU. It helps to reduce latency but there is a disadvantage it leads to the timer ISR running on every core. It increases the CPU time usage for timer ISRs by N on an N-core system.

Available options:

- CPU0 (CONFIG_ESP_TIMER_ISR_AFFINITY_CPU0)
- CPU1 (CONFIG_ESP_TIMER_ISR_AFFINITY_CPU1)
- No affinity (CONFIG_ESP_TIMER_ISR_AFFINITY_NO_AFFINITY)

CONFIG_ESP_TIMER_SUPPORTS_ISR_DISPATCH_METHOD

Support ISR dispatch method

Found in: Component config > ESP Timer (High Resolution Timer)

Allows using ESP_TIMER_ISR dispatch method (ESP_TIMER_TASK dispatch method is also available). - ESP_TIMER_TASK - Timer callbacks are dispatched from a high-priority esp_timer task. - ESP_TIMER_ISR - Timer callbacks are dispatched directly from the timer interrupt handler. The ISR dispatch can be used, in some cases, when a callback is very simple or need a lower-latency.

Default value:

- No (disabled)

ESP Trace Configuration

Contains:

- Application Level Tracing
- CONFIG_ESP_TRACE_LIBRARY
- CONFIG_ESP_TRACE_TIMESTAMP_SOURCE
- CONFIG_ESP_TRACE_TRANSPORT

CONFIG_ESP_TRACE_LIBRARY

Trace library

Found in: Component config > ESP Trace Configuration

Select the trace library to use.

Available options:

- External library from component registry (CONFIG_ESP_TRACE_LIB_EXTERNAL)

  Use a custom encoder provided by an external component. The trace core system will be enabled but no built-in encoder is included.

  The external component MUST provide a header file named "esp_trace_freertos_impl.h" in its include directories that defines all required FreeRTOS trace macros.

- Disabled (CONFIG_ESP_TRACE_LIB_NONE)

CONFIG_ESP_TRACE_TRANSPORT

Trace transport

Found in: Component config > ESP Trace Configuration

Select the trace backend to use.

Available options:

- ESP-IDF apptrace (CONFIG_ESP_TRACE_TRANSPORT_APPTRACE)
- External transport from component registry (CONFIG_ESP_TRACE_TRANSPORT_EXTERNAL)

  Use a transport registered by an external component. The trace core system will be enabled but no built-in transport is included.

- None (CONFIG_ESP_TRACE_TRANSPORT_NONE)

Application Level Tracing

Contains:

- CONFIG_APPTRACE_DESTINATION
- CONFIG_APPTRACE_LOCK_ENABLE
- CONFIG_APPTRACE_BUF_SIZE
- CONFIG_APPTRACE_POSTMORTEM_FLUSH_THRESH
- CONFIG_APPTRACE_ONPANIC_HOST_FLUSH_TMO
- CONFIG_APPTRACE_UART_BAUDRATE
- CONFIG_APPTRACE_DEST_UART_NUM
- CONFIG_APPTRACE_UART_RX_GPIO
- CONFIG_APPTRACE_UART_TX_MSG_SIZE
- CONFIG_APPTRACE_UART_TX_GPIO
- CONFIG_APPTRACE_UART_TX_BUFF_SIZE

CONFIG_APPTRACE_DESTINATION

Data Destination

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Select destination for application trace: JTAG, UART, or both.

Available options:

- JTAG (CONFIG_APPTRACE_DEST_JTAG)
- UART (CONFIG_APPTRACE_DEST_UART)
- All (runtime selection) (CONFIG_APPTRACE_DEST_ALL)

  Compile both JTAG and UART interfaces in advance (higher IRAM usage). Allows runtime switching between JTAG and UART via esp_apptrace_get_user_params().

  If esp_apptrace_get_user_params() is not provided by the application, JTAG is used by default with the default configuration defined in components/app_trace/include/esp_app_trace_config.h.

CONFIG_APPTRACE_BUF_SIZE

Size of the apptrace buffer

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Size of the memory buffer for trace data in bytes.

Default value:

- 16384 if CONFIG_APPTRACE_DEST_JTAG && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_DEST_UART_NUM

UART port number

Found in: Component config > ESP Trace Configuration > Application Level Tracing

UART communication port number for the apptrace destination. See UART documentation for available port numbers.

Range:

- from 0 to 2 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE
- from 0 to 4 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 1 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_UART_TX_GPIO

UART TX on GPIO<num>

Found in: Component config > ESP Trace Configuration > Application Level Tracing

This GPIO is used for UART TX pin.

Range:

- from 0 to 46 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 12 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_UART_RX_GPIO

UART RX on GPIO<num>

Found in: Component config > ESP Trace Configuration > Application Level Tracing

This GPIO is used for UART RX pin.

Range:

- from 0 to 46 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 13 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_UART_BAUDRATE

UART baud rate

Found in: Component config > ESP Trace Configuration > Application Level Tracing

This baud rate is used for UART.

The app's maximum baud rate depends on the UART clock source. If Power Management is disabled, the UART clock source is the APB clock and all baud rates in the available range will be sufficiently accurate. If Power Management is enabled, REF_TICK clock source is used so the baud rate is divided from 1MHz. Baud rates above 1Mbps are not possible and values between 500Kbps and 1Mbps may not be accurate.

Range:

- from 1200 to 8000000 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE
- from 1200 to 1000000 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 1000000 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_UART_TX_BUFF_SIZE

UART TX ring buffer size

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Size of the UART output ring buffer. Must be power of two. This size related to the baudrate, system tick frequency and amount of data to transfer.

Range:

- from 2048 to 32768 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 4096 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_UART_TX_MSG_SIZE

UART TX message size

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Maximum size of the single message to transfer.

Range:

- from 64 to 32768 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 128 if CONFIG_APPTRACE_DEST_UART && CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_LOCK_ENABLE

Internal Sync Lock Enable

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Enables/disable application tracing module internal sync lock to prevent data corruption when multiple tasks are writing to the same trace buffer. Keep in mind this will slow down the trace data transfer to the host.

Default value:

- No (disabled) if CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_ONPANIC_HOST_FLUSH_TMO

Timeout for flushing last trace data to host on panic

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Timeout for flushing last trace data to host in case of panic. In ms. Use -1 to disable timeout and wait forever.

Range:

- from -1 to 5000 if CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- "-1" if CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_APPTRACE_POSTMORTEM_FLUSH_THRESH

Threshold for flushing last trace data to host on panic

Found in: Component config > ESP Trace Configuration > Application Level Tracing

Threshold for flushing last trace data to host on panic in post-mortem mode. This is minimal amount of data needed to perform flush. In bytes.

Range:

- from 0 to 16384 if CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

Default value:

- 0 if CONFIG_ESP_TRACE_TRANSPORT_APPTRACE

CONFIG_ESP_TRACE_TIMESTAMP_SOURCE

Trace timestamp source

Found in: Component config > ESP Trace Configuration

Select the timestamp source for tracing.

Available options:

- CPU cycle counter (CCOUNT) (CONFIG_ESP_TRACE_TS_SOURCE_CCOUNT)
- General Purpose Timer (Timer Group) (CONFIG_ESP_TRACE_TS_SOURCE_GPTIMER)
- esp_timer high resolution timer (CONFIG_ESP_TRACE_TS_SOURCE_ESP_TIMER)

Wi-Fi

Contains:

- CONFIG_ESP_WIFI_WPS_SOFTAP_REGISTRAR
- CONFIG_ESP_WIFI_ENABLE_ROAMING_APP
- Beacon Sample Configuration Options
- CONFIG_ESP_WIFI_11KV_SUPPORT
- CONFIG_ESP_WIFI_11R_SUPPORT
- CONFIG_ESP_WIFI_BSS_MAX_IDLE_SUPPORT
- CONFIG_ESP_WIFI_MODEM_RF_FLAG_UPDATE_DEBUG
- CONFIG_ESP_WIFI_DPP_SUPPORT
- CONFIG_ESP_WIFI_ENTERPRISE_SUPPORT
- CONFIG_ESP_WIFI_MBO_SUPPORT
- CONFIG_ESP_WIFI_SUITE_B_192
- CONFIG_ESP_WIFI_ENABLE_WPA3_OWE_STA
- CONFIG_ESP_WIFI_WAPI_PSK
- CONFIG_ESP_WIFI_NAN_SYNC_ENABLE
- CONFIG_ESP_WIFI_NAN_USD_ENABLE
- CONFIG_ESP_WIFI_ENABLE_DUMP_CTRL_BFRP
- CONFIG_ESP_WIFI_ENABLE_DUMP_HESIGB
- CONFIG_ESP_WIFI_ENABLE_DUMP_MU_CFO
- CONFIG_ESP_WIFI_ENABLE_DUMP_CTRL_NDPA
- CONFIG_ESP_WIFI_ENABLE_WIFI_RX_STATS
- CONFIG_ESP_WIFI_ENABLE_WIFI_TX_STATS
- CONFIG_ESP_WIFI_WPA3_COMPATIBLE_SUPPORT
- CONFIG_ESP_WIFI_ENABLE_WPA3_SAE
- CONFIG_ESP_HOST_WIFI_ENABLED
- CONFIG_ESP_WIFI_SCAN_CACHE
- CONFIG_ESP_WIFI_SOFTAP_BEACON_MAX_LEN
- CONFIG_ESP_WIFI_CACHE_TX_BUFFER_NUM
- CONFIG_ESP_WIFI_DYNAMIC_RX_BUFFER_NUM
- CONFIG_ESP_WIFI_DYNAMIC_TX_BUFFER_NUM
- CONFIG_ESP_WIFI_RX_MGMT_BUF_NUM_DEF
- CONFIG_ESP_WIFI_STATIC_RX_BUFFER_NUM
- CONFIG_ESP_WIFI_STATIC_TX_BUFFER_NUM
- CONFIG_ESP_WIFI_ESPNOW_MAX_ENCRYPT_NUM
- CONFIG_ESP_WIFI_SLP_DEFAULT_MAX_ACTIVE_TIME
- CONFIG_ESP_WIFI_SLP_DEFAULT_MIN_ACTIVE_TIME
- CONFIG_ESP_WIFI_SLP_DEFAULT_WAIT_BROADCAST_DATA_TIME
- CONFIG_ESP_WIFI_STA_DISCONNECTED_PM_ENABLE
- CONFIG_ESP_WIFI_DEBUG_PRINT
- CONFIG_ESP_WIFI_PASSIVE_HIDDEN_AP_SUPPORT
- CONFIG_ESP_WIFI_MGMT_RX_BUFFER
- CONFIG_ESP_WIFI_TX_BUFFER

- CONFIG_ESP_WIFI_MBEDTLS_CRYPTO
- CONFIG_ESP_WIFI_AMPDU_RX_ENABLED
- CONFIG_ESP_WIFI_AMPDU_TX_ENABLED
- CONFIG_ESP_WIFI_AMSDU_TX_ENABLED
- CONFIG_ESP_WIFI_CSI_ENABLED
- CONFIG_ESP_WIFI_EXTRA_IRAM_OPT
- CONFIG_ESP_WIFI_FTM_ENABLE
- CONFIG_ESP_WIFI_GCMP_SUPPORT
- CONFIG_ESP_WIFI_GMAC_SUPPORT
- CONFIG_ESP_WIFI_IRAM_OPT
- CONFIG_ESP_WIFI_MGMT_SBUF_NUM
- CONFIG_ESP_WIFI_ENHANCED_LIGHT_SLEEP
- CONFIG_ESP_WIFI_NVS_ENABLED
- CONFIG_ESP_WIFI_RX_IRAM_OPT
- CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT
- CONFIG_ESP_WIFI_SLP_IRAM_OPT
- CONFIG_ESP_WIFI_SOFTAP_SUPPORT
- CONFIG_ESP_WIFI_TASK_CORE_ID
- CONFIG_ESP_WIFI_TX_HETB_QUEUE_NUM
- WPS Configuration Options

CONFIG_ESP_HOST_WIFI_ENABLED

Host WiFi Enable

Found in: Component config > Wi-Fi

Default value:

- No (disabled) if SOC_WIRELESS_HOST_SUPPORTED

CONFIG_ESP_WIFI_STATIC_RX_BUFFER_NUM

Max number of WiFi static RX buffers

Found in: Component config > Wi-Fi

Set the number of WiFi static RX buffers. Each buffer takes approximately 1.6KB of RAM. The static rx buffers are allocated when esp_wifi_init is called, they are not freed until esp_wifi_deinit is called.

WiFi hardware use these buffers to receive all 802.11 frames. A higher number may allow higher throughput but increases memory use. If ESP_WIFI_AMPDU_RX_ENABLED is enabled, this value is recommended to set equal or bigger than ESP_WIFI_RX_BA_WIN in order to achieve better throughput and compatibility with both stations and APs.

Range:

- from 2 to 128 if SOC_WIFI_HE_SUPPORT

Default value:

- 10

CONFIG_ESP_WIFI_DYNAMIC_RX_BUFFER_NUM

Max number of WiFi dynamic RX buffers

Found in: Component config > Wi-Fi

Set the number of WiFi dynamic RX buffers, 0 means unlimited RX buffers will be allocated (provided sufficient free RAM). The size of each dynamic RX buffer depends on the size of the received data frame.

For each received data frame, the WiFi driver makes a copy to an RX buffer and then delivers it to the high layer TCP/IP stack. The dynamic RX buffer is freed after the higher layer has successfully received the data frame.

For some applications, WiFi data frames may be received faster than the application can process them. In these cases we may run out of memory if RX buffer number is unlimited (0).

If a dynamic RX buffer limit is set, it should be at least the number of static RX buffers.

Range:

- from 0 to 1024 if CONFIG_LWIP_WND_SCALE

Default value:

- 32

CONFIG_ESP_WIFI_TX_BUFFER

Type of WiFi TX buffers

Found in: Component config > Wi-Fi

Select type of WiFi TX buffers:

If "Static" is selected, WiFi TX buffers are allocated when WiFi is initialized and released when WiFi is de-initialized. The size of each static TX buffer is fixed to about 1.6KB.

If "Dynamic" is selected, each WiFi TX buffer is allocated as needed when a data frame is delivered to the Wifi driver from the TCP/IP stack. The buffer is freed after the data frame has been sent by the WiFi driver. The size of each dynamic TX buffer depends on the length of each data frame sent by the TCP/IP layer.

If PSRAM is enabled, "Static" should be selected to guarantee enough WiFi TX buffers. If PSRAM is disabled, "Dynamic" should be selected to improve the utilization of RAM.

Available options:

- Static (CONFIG_ESP_WIFI_STATIC_TX_BUFFER)
- Dynamic (CONFIG_ESP_WIFI_DYNAMIC_TX_BUFFER)

CONFIG_ESP_WIFI_STATIC_TX_BUFFER_NUM

Max number of WiFi static TX buffers

Found in: Component config > Wi-Fi

Set the number of WiFi static TX buffers. Each buffer takes approximately 1.6KB of RAM. The static RX buffers are allocated when esp_wifi_init() is called, they are not released until esp_wifi_deinit() is called.

For each transmitted data frame from the higher layer TCP/IP stack, the WiFi driver makes a copy of it in a TX buffer. For some applications especially UDP applications, the upper layer can deliver frames faster than WiFi layer can transmit. In these cases, we may run out of TX buffers.

Range:

- from 1 to 64 if CONFIG_ESP_WIFI_STATIC_TX_BUFFER

Default value:

- 16 if CONFIG_ESP_WIFI_STATIC_TX_BUFFER

CONFIG_ESP_WIFI_CACHE_TX_BUFFER_NUM

Max number of WiFi cache TX buffers

Found in: Component config > Wi-Fi

Set the number of WiFi cache TX buffer number.

For each TX packet from uplayer, such as LWIP etc, WiFi driver needs to allocate a static TX buffer and makes a copy of uplayer packet. If WiFi driver fails to allocate the static TX buffer, it caches the uplayer packets to a dedicated buffer queue, this option is used to configure the size of the cached TX queue.

CONFIG_ESP_WIFI_DYNAMIC_TX_BUFFER_NUM

Max number of WiFi dynamic TX buffers

Found in: Component config > Wi-Fi

Set the number of WiFi dynamic TX buffers. The size of each dynamic TX buffer is not fixed, it depends on the size of each transmitted data frame.

For each transmitted frame from the higher layer TCP/IP stack, the WiFi driver makes a copy of it in a TX buffer. For some applications, especially UDP applications, the upper layer can deliver frames faster than WiFi layer can transmit. In these cases, we may run out of TX buffers.

Range:

- from 1 to 128

Default value:

- 32

CONFIG_ESP_WIFI_MGMT_RX_BUFFER

Type of WiFi RX MGMT buffers

Found in: Component config > Wi-Fi

Select type of WiFi RX MGMT buffers:

If "Static" is selected, WiFi RX MGMT buffers are allocated when WiFi is initialized and released when WiFi is de-initialized. The size of each static RX MGMT buffer is fixed to about 500 Bytes.

If "Dynamic" is selected, each WiFi RX MGMT buffer is allocated as needed when a MGMT data frame is received. The MGMT buffer is freed after the MGMT data frame has been processed by the WiFi driver.

Available options:

- Static (CONFIG_ESP_WIFI_STATIC_RX_MGMT_BUFFER)
- Dynamic (CONFIG_ESP_WIFI_DYNAMIC_RX_MGMT_BUFFER)

CONFIG_ESP_WIFI_RX_MGMT_BUF_NUM_DEF

Max number of WiFi RX MGMT buffers

Found in: Component config > Wi-Fi

Set the number of WiFi RX_MGMT buffers.

For Management buffers, the number of dynamic and static management buffers is the same. In order to prevent memory fragmentation, the management buffer type should be set to static first.

Range:

- from 1 to 10

Default value:

- 5

CONFIG_ESP_WIFI_CSI_ENABLED

WiFi CSI(Channel State Information)

Found in: Component config > Wi-Fi

Select this option to enable CSI(Channel State Information) feature. CSI takes about CONFIG_ESP_WIFI_STATIC_RX_BUFFER_NUM KB of RAM. If CSI is not used, it is better to disable this feature in order to save memory.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_AMPDU_TX_ENABLED

WiFi AMPDU TX

Found in: Component config > Wi-Fi

Select this option to enable AMPDU TX feature

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_TX_BA_WIN

WiFi AMPDU TX BA window size

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_AMPDU_TX_ENABLED

Set the size of WiFi Block Ack TX window. Generally a bigger value means higher throughput but more memory. Most of time we should NOT change the default value unless special reason, e.g. test the maximum UDP TX throughput with iperf etc. For iperf test in shieldbox, the recommended value is 9~12.

Range:

- from 2 to 64 if SOC_WIFI_HE_SUPPORT && CONFIG_ESP_WIFI_AMPDU_TX_ENABLED

Default value:

- 6

CONFIG_ESP_WIFI_AMPDU_RX_ENABLED

WiFi AMPDU RX

Found in: Component config > Wi-Fi

Select this option to enable AMPDU RX feature

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_RX_BA_WIN

WiFi AMPDU RX BA window size

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_AMPDU_RX_ENABLED

Set the size of WiFi Block Ack RX window. Generally a bigger value means higher throughput and better compatibility but more memory. Most of time we should NOT change the default value unless special reason, e.g. test the maximum UDP RX throughput with iperf etc. For iperf test in shieldbox, the recommended value is 9~12. If PSRAM is used and WiFi memory is preferred to allocate in PSRAM first, the default and minimum value should be 16 to achieve better throughput and compatibility with both stations and APs.

Range:

- from 2 to 64 if SOC_WIFI_HE_SUPPORT && CONFIG_ESP_WIFI_AMPDU_RX_ENABLED

Default value:

- 6

CONFIG_ESP_WIFI_AMSDU_TX_ENABLED

WiFi AMSDU TX

Found in: Component config > Wi-Fi

Select this option to enable AMSDU TX feature

Default value:

- No (disabled) if CONFIG_ESP_WIFI_CACHE_TX_BUFFER_NUM >= 2

CONFIG_ESP_WIFI_NVS_ENABLED

WiFi NVS flash

Found in: Component config > Wi-Fi

Select this option to enable WiFi NVS flash

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_TASK_CORE_ID

WiFi Task Core ID

Found in: Component config > Wi-Fi

Pinned WiFi task to core 0 or core 1.

Available options:

- Core 0 (CONFIG_ESP_WIFI_TASK_PINNED_TO_CORE_0)
- Core 1 (CONFIG_ESP_WIFI_TASK_PINNED_TO_CORE_1)

CONFIG_ESP_WIFI_SOFTAP_BEACON_MAX_LEN

Max length of WiFi SoftAP Beacon

Found in: Component config > Wi-Fi

ESP-MESH utilizes beacon frames to detect and resolve root node conflicts (see documentation). However the default length of a beacon frame can simultaneously hold only five root node identifier structures, meaning that a root node conflict of up to five nodes can be detected at one time. In the occurrence of more root nodes conflict involving more than five root nodes, the conflict resolution process will detect five of the root nodes, resolve the conflict, and re-detect more root nodes. This process will repeat until all root node conflicts are resolved. However this process can generally take a very long time.

To counter this situation, the beacon frame length can be increased such that more root nodes can be detected simultaneously. Each additional root node will require 36 bytes and should be added on top of the default beacon frame length of 752 bytes. For example, if you want to detect 10 root nodes simultaneously, you need to set the beacon frame length as 932 (752+36*5).

Setting a longer beacon length also assists with debugging as the conflicting root nodes can be identified more quickly.

Range:

- from 752 to 1256

Default value:

- 752

CONFIG_ESP_WIFI_MGMT_SBUF_NUM

WiFi mgmt short buffer number

Found in: Component config > Wi-Fi

Set the maximum number of Wi-Fi management short buffers. These buffers are dynamically allocated, with their size determined by the length of the management packet to be sent. When a management packet is less than 64 bytes, the Wi-Fi driver classifies it as a short management packet and assigns it to one of these buffers.

Range:

- from 6 to 32

Default value:

- 32

CONFIG_ESP_WIFI_IRAM_OPT

WiFi IRAM speed optimization

Found in: Component config > Wi-Fi

Select this option to place frequently called Wi-Fi library functions in IRAM. When this option is disabled, more than 10Kbytes of IRAM memory will be saved but Wi-Fi throughput will be reduced.

Default value:

- No (disabled) if CONFIG_BT_ENABLED && CONFIG_SPIRAM
- Yes (enabled)

CONFIG_ESP_WIFI_EXTRA_IRAM_OPT

WiFi EXTRA IRAM speed optimization

Found in: Component config > Wi-Fi

Select this option to place additional frequently called Wi-Fi library functions in IRAM. When this option is disabled, more than 5Kbytes of IRAM memory will be saved but Wi-Fi throughput will be reduced.

Default value:

- Yes (enabled) if SOC_WIFI_HE_SUPPORT
- No (disabled)

CONFIG_ESP_WIFI_RX_IRAM_OPT

WiFi RX IRAM speed optimization

Found in: Component config > Wi-Fi

Select this option to place frequently called Wi-Fi library RX functions in IRAM. When this option is disabled, more than 17Kbytes of IRAM memory will be saved but Wi-Fi performance will be reduced.

Default value:

- No (disabled) if CONFIG_BT_ENABLED && CONFIG_SPIRAM
- Yes (enabled)

CONFIG_ESP_WIFI_ENABLE_WPA3_SAE

Enable WPA3-Personal

Found in: Component config > Wi-Fi

Select this option to allow the device to establish a WPA3-Personal connection with eligible AP's. PMF (Protected Management Frames) is a prerequisite feature for a WPA3 connection, it needs to be explicitly configured before attempting connection. Please refer to the Wi-Fi Driver API Guide for details.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_ENABLE_SAE_H2E

Enable SAE-H2E

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_WPA3_SAE

Select this option to enable SAE-H2E

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_ENABLE_SAE_PK

Enable SAE-PK

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_WPA3_SAE > CONFIG_ESP_WIFI_ENABLE_SAE_H2E

Select this option to enable SAE-PK

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_SOFTAP_SAE_SUPPORT

Enable WPA3 Personal(SAE) SoftAP

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_WPA3_SAE

Select this option to enable SAE support in softAP mode.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_ENABLE_WPA3_OWE_STA

Enable OWE STA

Found in: Component config > Wi-Fi

Select this option to allow the device to establish OWE connection with eligible AP's. PMF (Protected Management Frames) is a prerequisite feature for a WPA3 connection, it needs to be explicitly configured before attempting connection. Please refer to the Wi-Fi Driver API Guide for details.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_WPA3_COMPATIBLE_SUPPORT

Enable WPA3 Compatible support

Found in: Component config > Wi-Fi

Select this option to support wpa3_compatible mode for station and AP

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_SLP_IRAM_OPT

WiFi SLP IRAM speed optimization

Found in: Component config > Wi-Fi

Select this option to place called Wi-Fi library TBTT process and receive beacon functions in IRAM. Some functions can be put in IRAM either by ESP_WIFI_IRAM_OPT and ESP_WIFI_RX_IRAM_OPT, or this one. If already enabled ESP_WIFI_IRAM_OPT, the other 7.3KB IRAM memory would be taken by this option. If already enabled ESP_WIFI_RX_IRAM_OPT, the other 1.3KB IRAM memory would be taken by this option. If neither of them are enabled, the other 7.4KB IRAM memory would be taken by this option. Wi-Fi power-save mode average current would be reduced if this option is enabled.

Default value:

- Yes (enabled) if SOC_WIFI_HE_SUPPORT

CONFIG_ESP_WIFI_SLP_DEFAULT_MIN_ACTIVE_TIME

Minimum active time

Found in: Component config > Wi-Fi

Only for station in WIFI_PS_MIN_MODEM or WIFI_PS_MAX_MODEM. When the station enters the active state, it will work for at least ESP_WIFI_SLP_DEFAULT_MIN_ACTIVE_TIME. If a data packet is received or sent during this period, the time will be refreshed. If the time is up, but the station still has packets to receive or send, the time will also be refreshed. unit: milliseconds.

Range:

- from 8 to 60

Default value:

- 50

CONFIG_ESP_WIFI_BSS_MAX_IDLE_SUPPORT

Enable bss max idle support

Found in: Component config > Wi-Fi

Enables bss max idle support for Station and SoftAP. BSS max idle period enables an SoftAP to indicate a time period during which the AP does not disassociate a STA due to nonreceipt of frames from the STA. For station, max idle period is default 10 (1000TUs) and can be set through ESP_WIFI_SLP_DEFAULT_MAX_ACTIVE_TIME. For softap, bss max idle parameters will be set through bss_max_idle_cfg in wifi_ap_config_t.

Default value:

- Yes (enabled) if SOC_WIFI_HE_SUPPORT || CONFIG_ESP_WIFI_WNM_SUPPORT

CONFIG_ESP_WIFI_SLP_DEFAULT_MAX_ACTIVE_TIME

Maximum keep alive time

Found in: Component config > Wi-Fi

Only for station. If no packet has been sent within ESP_WIFI_SLP_DEFAULT_MAX_ACTIVE_TIME, a null data packet will be sent to maintain the connection with the AP. If ESP_WIFI_BSS_MAX_IDLE_SUPPORT is set, it will be sent as bss max idle period in association request. unit: seconds.

Range:

- from 10 to 60

Default value:

- 10

CONFIG_ESP_WIFI_SLP_DEFAULT_WAIT_BROADCAST_DATA_TIME

Minimum wait broadcast data time

Found in: Component config > Wi-Fi

Only for station in WIFI_PS_MIN_MODEM or WIFI_PS_MAX_MODEM. When the station knows through the beacon that AP will send broadcast packet, it will wait for ESP_WIFI_SLP_DEFAULT_WAIT_BROADCAST_DATA_TIME before entering the sleep process. If a broadcast packet is received with more data bits, the time will refreshed. unit: milliseconds.

Range:

- from 10 to 30

Default value:

- 15

CONFIG_ESP_WIFI_FTM_ENABLE

WiFi FTM

Found in: Component config > Wi-Fi

Enable feature Fine Timing Measurement for calculating WiFi Round-Trip-Time (RTT).

Default value:

- No (disabled) if SOC_WIFI_FTM_SUPPORT

CONFIG_ESP_WIFI_FTM_INITIATOR_SUPPORT

FTM Initiator support

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_FTM_ENABLE

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_FTM_ENABLE

CONFIG_ESP_WIFI_FTM_RESPONDER_SUPPORT

FTM Responder support

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_FTM_ENABLE

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_FTM_ENABLE

CONFIG_ESP_WIFI_STA_DISCONNECTED_PM_ENABLE

Power Management for station at disconnected

Found in: Component config > Wi-Fi

Select this option to enable power_management for station when disconnected. Chip will do modem-sleep when rf module is not in use any more.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_GCMP_SUPPORT

WiFi GCMP Support(GCMP128 and GCMP256)

Found in: Component config > Wi-Fi

Select this option to enable GCMP support. GCMP support is compulsory for WiFi Suite-B support.

Default value:

- No (disabled) if SOC_WIFI_GCMP_SUPPORT

CONFIG_ESP_WIFI_GMAC_SUPPORT

WiFi GMAC Support(GMAC128 and GMAC256)

Found in: Component config > Wi-Fi

Select this option to enable GMAC support. GMAC support is compulsory for WiFi 192 bit certification.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_SOFTAP_SUPPORT

WiFi SoftAP Support

Found in: Component config > Wi-Fi

WiFi module can be compiled without SoftAP to save code size.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_ENHANCED_LIGHT_SLEEP

WiFi modem automatically receives the beacon

Found in: Component config > Wi-Fi

The wifi modem automatically receives the beacon frame during light sleep.

Default value:

- No (disabled) if CONFIG_ESP_PHY_MAC_BB_PD && SOC_PM_SUPPORT_BEACON_WAKEUP

CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Wifi sleep optimize when beacon lost

Found in: Component config > Wi-Fi

Enable wifi sleep optimization when beacon loss occurs and immediately enter sleep mode when the WiFi module detects beacon loss.

CONFIG_ESP_WIFI_SLP_BEACON_LOST_TIMEOUT

Beacon loss timeout

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Timeout time for close rf phy when beacon loss occurs, Unit: 1024 microsecond.

Range:

- from 5 to 100 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Default value:

- 25 if CONFIG_ESP_COEX_SW_COEXIST_ENABLE && CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

CONFIG_ESP_WIFI_SLP_BEACON_LOST_THRESHOLD

Maximum number of consecutive lost beacons allowed

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Maximum number of consecutive lost beacons allowed, the WiFi Rx state behaviour will be determined by ESP_WIFI_SLP_BEACON_LOST_OVER_THRESHOLD_POLICY when the number of consecutive beacons lost is greater than the given threshold.

Range:

- from 0 to 8 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Default value:

- 3 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

CONFIG_ESP_WIFI_SLP_PHY_ON_DELTA_EARLY_TIME

Delta early time for RF PHY on

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Delta early time for rf phy on, When the beacon is lost, the next rf phy on will be earlier the time specified by the configuration item, Unit: 32 microsecond.

Range:

- from 0 to 100 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Default value:

- 2 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

CONFIG_ESP_WIFI_SLP_PHY_OFF_DELTA_TIMEOUT_TIME

Delta timeout time for RF PHY off

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Delta timeout time for rf phy off, When the beacon is lost, the next rf phy off will be delayed for the time specified by the configuration item. Unit: 1024 microsecond.

Range:

- from 0 to 8 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Default value:

- 2 if CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

CONFIG_ESP_WIFI_SLP_BEACON_LOST_OVER_THRESHOLD_POLICY

Beacon strategy when beacon loss exceeds threshold

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Select the strategy to apply when the number of lost beacons exceeds the threshold.

- "Receive beacon": Keep RF on until a beacon is successfully received.
- "Drop beacon": Turn off RF and skip beacon reception during this period.
- **"Auto": Beacon will been dropped only if ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE enabled**

  and expected rx beacon probability stays below the standard.

Available options:

- Receive beacon (CONFIG_ESP_WIFI_SLP_BEACON_LOST_OVER_THRESHOLD_RECEIVE)
- Drop beacon (CONFIG_ESP_WIFI_SLP_BEACON_LOST_OVER_THRESHOLD_DROP)
- Auto (CONFIG_ESP_WIFI_SLP_BEACON_LOST_OVER_THRESHOLD_AUTO)

CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

Sample beacon to calculate beacon offset

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_SLP_BEACON_LOST_OPT

Select to enable feature sampling beacons to calculate beacon offset.

Beacon Sample Configuration Options

Contains:

- CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_DIFFERENCE_PERCENT
- CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_RESAMPLE_PERIOD
- CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_COUNT
- CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_STANDARD_PERCENT

CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_COUNT

Sample beacons at wifi connected to adjust beacon parameters

Found in: Component config > Wi-Fi > Beacon Sample Configuration Options

Numble of sampled beacons at wifi connected to adjust beacon parameters.

Range:

- from 100 to 1200 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

Default value:

- 300 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_DIFFERENCE_PERCENT

Difference percentage triggers unstable event

Found in: Component config > Wi-Fi > Beacon Sample Configuration Options

Difference triggers event WIFI_EVENT_STA_BEACON_OFFSET_UNSTABLE when the actual rx beacon probability continuously falls below the expected probability by this value. unit: percentage

Range:

- from 0 to 100 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

Default value:

- 20 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_STANDARD_PERCENT

Standard percentage triggers beacon drop

Found in: Component config > Wi-Fi > Beacon Sample Configuration Options

Standard triggers beacon drop when the expected rx beacon probability falls below this value under ESP_WIFI_SLP_BEACON_LOST_DROP_BEACON_AUTO mode. unit: percentage

Range:

- from 0 to 100 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

Default value:

- 55 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_RESAMPLE_PERIOD

Resample period

Found in: Component config > Wi-Fi > Beacon Sample Configuration Options

Resample period if beacon drop is active under ESP_WIFI_SLP_BEACON_LOST_DROP_BEACON_AUTO mode. It means never resample if setting this value to 0. unit: hours

Range:

- from 0 to 255 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

Default value:

- 3 if CONFIG_ESP_WIFI_SLP_SAMPLE_BEACON_FEATURE

CONFIG_ESP_WIFI_ESPNOW_MAX_ENCRYPT_NUM

Maximum espnow encrypt peers number

Found in: Component config > Wi-Fi

Maximum number of encrypted peers supported by espnow. The number of hardware keys for encryption is fixed. And the espnow and SoftAP share the same hardware keys. So this configuration will affect the maximum connection number of SoftAP. Maximum espnow encrypted peers number + maximum number of connections of SoftAP = Max hardware keys number. When using ESP mesh, this value should be set to a maximum of 6.

Range:

- from 0 to 17

Default value:

- 7

CONFIG_ESP_WIFI_NAN_SYNC_ENABLE

Enable Wi-Fi Aware: Synchronization (NAN-Sync)

Found in: Component config > Wi-Fi

Enable Wi-Fi Aware: Synchronization feature (NAN-Sync).

Default value:

- No (disabled)

CONFIG_ESP_WIFI_NAN_USD_ENABLE

Enable Wi-Fi Aware: Unsynchronized service discovery (NAN-USD)

Found in: Component config > Wi-Fi

Enable Wi-Fi Aware: Unsynchronized service discovery (NAN-USD)

Default value:

- No (disabled) if CONFIG_IDF_EXPERIMENTAL_FEATURES

CONFIG_ESP_WIFI_MBEDTLS_CRYPTO

Use MbedTLS crypto APIs

Found in: Component config > Wi-Fi

Select this option to enable the use of MbedTLS crypto APIs. The internal crypto support within the supplicant is limited and may not suffice for all new security features, including WPA3.

It is recommended to always keep this option enabled. Additionally, note that MbedTLS can leverage hardware acceleration if available, resulting in significantly faster cryptographic operations.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_MBEDTLS_TLS_CLIENT

Use MbedTLS TLS client for WiFi Enterprise connection

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_MBEDTLS_CRYPTO

Select this option to use MbedTLS TLS client for WPA2 enterprise connection. Please note that from MbedTLS-3.0 onwards, MbedTLS does not support SSL-3.0 TLS-v1.0, TLS-v1.1 versions. In case your server is using one of these version, it is advisable to update your server. Please disable this option for compatibility with older TLS versions.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_EAP_TLS1_3

Enable EAP-TLS v1.3 Support for WiFi Enterprise connection

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_MBEDTLS_CRYPTO > CONFIG_ESP_WIFI_MBEDTLS_TLS_CLIENT

Select this option to support EAP with TLS v1.3. This configuration still supports compatibility with EAP-TLS v1.2. Please note that enabling this configuration will cause every application which uses TLS go for TLS1.3 if server supports that. TLS1.3 is still in development in mbedtls and there may be interoperability issues with this. Please modify your application to set max version as TLS1.2 if you want to enable TLS1.3 only for WiFi connection.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_MBEDTLS_TLS_CLIENT && CONFIG_IDF_EXPERIMENTAL_FEATURES && CONFIG_ESP_WIFI_MBEDTLS_CRYPTO

CONFIG_ESP_WIFI_WAPI_PSK

Enable WAPI PSK support

Found in: Component config > Wi-Fi

Select this option to enable WAPI-PSK which is a Chinese National Standard Encryption for Wireless LANs (GB 15629.11-2003).

Default value:

- No (disabled)

CONFIG_ESP_WIFI_SUITE_B_192

Enable NSA suite B support with 192 bit key

Found in: Component config > Wi-Fi

Select this option to enable 192 bit NSA suite-B. This is necessary to support WPA3 192 bit security.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_ENTERPRISE_SUPPORT && SOC_WIFI_GCMP_SUPPORT

CONFIG_ESP_WIFI_11KV_SUPPORT

Enable 802.11k, 802.11v APIs Support

Found in: Component config > Wi-Fi

Select this option to enable 802.11k 802.11v APIs(RRM and BTM support).

Default value:

- No (disabled)

CONFIG_ESP_WIFI_RRM_SUPPORT

Enable 802.11k APIs Support

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_11KV_SUPPORT

Select this option to enable 802.11k APIs(RRM support). Only APIs which are helpful for network assisted roaming are supported for now. Enable this option with RRM enabled in sta config to make device ready for network assisted roaming. RRM: Radio measurements enable STAs to understand the radio environment, it enables STAs to observe and gather data on radio link performance and on the radio environment. Current implementation adds beacon report, link measurement, neighbor report.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_11KV_SUPPORT

CONFIG_ESP_WIFI_WNM_SUPPORT

Enable 802.11v APIs Support

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_11KV_SUPPORT

Select this option to enable 802.11v APIs(BTM support). Only APIs which are helpful for network assisted roaming are supported for now. Enable this option with BTM enabled in sta config to make device ready for network assisted roaming. BTM: BSS transition management enables an AP to request a station to transition to a specific AP, or to indicate to a station a set of preferred APs.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_11KV_SUPPORT

CONFIG_ESP_WIFI_SCAN_CACHE

Keep scan results in cache

Found in: Component config > Wi-Fi

Keep scan results in cache, if not enabled, those will be flushed immediately.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_RRM_SUPPORT

CONFIG_ESP_WIFI_MBO_SUPPORT

Enable Multi Band Operation Certification Support

Found in: Component config > Wi-Fi

Select this option to enable WiFi Multiband operation certification support.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Advanced support for Wi-Fi Roaming (Experimental)

Found in: Component config > Wi-Fi

Enable Espressif's roaming app to allow for efficient Wi-Fi roaming. This includes configurable periodic environment scans, maintaining a cache of the best APs, handling low rssi events etc.

Risk Warning Please note that this feature is still experimental and enabling this potentially can lead to unpredictable scanning, connection and roaming attempts. We are still working on tuning and optimising this feature to ensure reliable and stable use.

Default value:

- No (disabled) if CONFIG_IDF_EXPERIMENTAL_FEATURES

Configure roaming App

Contains:

- Blacklist Configuration
- CONFIG_ESP_WIFI_ROAMING_BACKOFF_TIME

- Roaming Methods
- Roaming triggers
- Scan Configuration
- CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING

Roaming triggers

Contains:

- CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR
- CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING

CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING

Use Low RSSI to trigger roaming.

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers

Enable to use a RSSI threshold to trigger roaming.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_THRESHOLD

WiFi RSSI threshold to trigger roaming

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers > CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING

WiFi RSSI threshold to trigger roaming value in dBm (-99 to -1). Values under -30 dbm might lead to a flood of low rssi events. This interferes with normal functioning and TX/Rx performance.

Range:

- from -99 to -30 if CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- "-60" if CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_OFFSET

Offset by which to reset the RSSI Threshold after attempt to roam.

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers > CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING

Decide the offset by which to decrease the Low RSSI threshold set by ESP_WIFI_ROAMING_LOW_RSSI_THRESHOLD after each failed attempt to roam. This allows for the station to keep scanning for better AP's after the Low RSSI threshold is reached in a stepped manner, rather than only attempting to roam the first time the current AP's RSSI breaches the set RSSI threshold. Setting 0 here may cause station to be flooded with low rssi events, therefore that's not recommended to be kept.

Range:

- from 0 to 99 if CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 5 if CONFIG_ESP_WIFI_ROAMING_LOW_RSSI_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR

Conduct periodic scans to check if a better AP is available

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers

Conduct periodic scans periodically to check if a better AP is available.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_THRESHOLD

Threshold at which to begin periodic scanning for a better AP.

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers > CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR

Threshold at which the station will begin scanning to find an AP with better RSSI.

Range:

- from -99 to -1 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- "-50" if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_SCAN_MONITOR_INTERVAL

Time intervals (in seconds) at which station will initiate a scan

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers > CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR

Intervals at which station will periodically scan to check if better AP is available

Range:

- from 1 to 1500 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 30 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_SCAN_ROAM_RSSI_DIFF

RSSI difference b/w current AP and candidate AP to initiate connection

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming triggers > CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR

Minimum RSSI difference b/w current AP and a potential roaming candidate AP to trigger a roaming attempt.

Range:

- from 0 to 99 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 15 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_SCAN_MONITOR && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Roaming Methods

Contains:

- CONFIG_ESP_WIFI_ROAMING_IP_RENEW_SKIP
- CONFIG_ESP_WIFI_ROAMING_LEGACY_ROAMING
- CONFIG_ESP_WIFI_ROAMING_NETWORK_ASSISTED_ROAM

CONFIG_ESP_WIFI_ROAMING_LEGACY_ROAMING

Support Legacy roaming approach

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming Methods

Roaming between APs that do not support 802.11v. This will allow station to roam even when connection is not BTM supported, by forcefully disconnecting from current AP and connecting to better AP.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_NETWORK_ASSISTED_ROAM

Support Network Assisted roaming using 802.11v

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming Methods

Roaming between APs using network assisted Roaming. This involves BSS Transition Management mechanisms outlined in 802.11v. Note that this moves the responsibility to the AP's network, and hence isn't guaranteed to cause the station to attempt to roam each time.

Default value:

- Yes (enabled) if CONFIG_ESP_WIFI_WNM_SUPPORT && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_NETWORK_ASSISTED_ROAMING_RETRY_COUNT

Retry count after which to switch to legacy roaming

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming Methods > CONFIG_ESP_WIFI_ROAMING_NETWORK_ASSISTED_ROAM

Retry threshold after which the station should stop using Network Assisted roaming methods and start using legacy roaming instead.

Range:

- from 1 to 5 if CONFIG_ESP_WIFI_ROAMING_NETWORK_ASSISTED_ROAM && CONFIG_ESP_WIFI_ROAMING_LEGACY_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 2 if CONFIG_ESP_WIFI_ROAMING_NETWORK_ASSISTED_ROAM && CONFIG_ESP_WIFI_ROAMING_LEGACY_ROAMING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_IP_RENEW_SKIP

Skip IP renew after roaming

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Roaming Methods

Skip the default IP renew path when the Wi-Fi stack reports a roaming event. Enable this only if your network preserves IP state across a roam, for example during 802.11r or BTM-based roaming.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Scan Configuration

Contains:

- CONFIG_ESP_WIFI_ROAMING_HOME_CHANNEL_DWELL_TIME
- CONFIG_ESP_WIFI_ROAMING_SCAN_MAX_SCAN_TIME
- CONFIG_ESP_WIFI_ROAMING_SCAN_MIN_SCAN_TIME
- CONFIG_ESP_WIFI_ROAMING_SCAN_CHAN_LIST
- CONFIG_ESP_WIFI_ROAMING_SCAN_EXPIRY_WINDOW

CONFIG_ESP_WIFI_ROAMING_SCAN_MIN_SCAN_TIME

Minimum duration (in milliseconds) of station's per channel active scan

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Scan Configuration

Minimum duration of active scanning per channel in milliseconds.

Range:

- from 0 to 120 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 30 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_SCAN_MAX_SCAN_TIME

Maximum duration (in milliseconds) of station's per channel active scan time

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Scan Configuration

Maximum duration of active scanning per channel in milliseconds.

Range:

- from 30 to 120 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 70 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_HOME_CHANNEL_DWELL_TIME

Home channel dwell time scanning between consecutive channels

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Scan Configuration

If connected, duration for which the station will return to it's home channel for Tx/Rx of frames stored in buffers between scanning on consecutive channels.

Range:

- from 30 to 150 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 30 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_SCAN_CHAN_LIST

Preferred channel list for scanning

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Scan Configuration

Channels your wireless network operates on to allow for faster scanning. Specify channels in a comma-separated manner. The list is used as a channel filter, not as an ordered scan sequence. When 5 GHz support is enabled, valid 5 GHz channels are also accepted.

Default value:

- "None" if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_SCAN_EXPIRY_WINDOW

Scan results expiry window (in seconds)

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Scan Configuration

Duration for which the results from the most recent scans can be used by the roaming app for determining the roaming candidates.

Range:

- from 5 to 20 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 10 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_BACKOFF_TIME

Default time to wait between subsequent roaming attempts.

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App

Time to wait (in seconds) by station before registering for the RSSI event again or start continuous montoring to find better AP.

Range:

- from 0 to 120 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 15 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING

Send periodic neighbor report request to AP for internal list updation

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App

This option will enable station to keep sending RRM neighbor list request to AP and update its internal list.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_RRM_SUPPORT && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_RRM_MONITOR_TIME

Time interval (in seconds) between neighbor report requests to an AP

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING

Enable this to send periodic neighbor report requests to the AP. These neighbor report requests provide information about other APs in the same managed network. This information is used for more intelligent roaming.

Range:

- from 0 to 1500 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 60 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_RRM_MONITOR_THRESHOLD

Threshold for sending periodic neighbor report requests

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING

The RSSI threshold beyond which we start sending periodic neighbor report requests.

Range:

- from -99 to 0 if CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- "-20" if CONFIG_ESP_WIFI_ROAMING_PERIODIC_RRM_MONITORING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Blacklist Configuration

Contains:

- CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST
- CONFIG_ESP_WIFI_ROAMING_MAX_CANDIDATES

CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST

Enable BSSID blacklisting

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Blacklist Configuration

Enable this to blacklist BSSIDs.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_AUTO_BLACKLISTING

Enable automatic BSSID blacklisting

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Blacklist Configuration > CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST

Enable this to automatically blacklist BSSIDs after multiple failed connection attempts.

CONFIG_ESP_WIFI_ROAMING_MAX_CONN_FAILURES

Maximum connection failures

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Blacklist Configuration > CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST > CONFIG_ESP_WIFI_ROAMING_AUTO_BLACKLISTING

Maximum number of connection failures before a BSSID is blacklisted.

Range:

- from 1 to 10 if CONFIG_ESP_WIFI_ROAMING_AUTO_BLACKLISTING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 3 if CONFIG_ESP_WIFI_ROAMING_AUTO_BLACKLISTING && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_BLACKLIST_TIMEOUT

Blacklist timeout (in seconds)

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Blacklist Configuration > CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST

Time in seconds for which a BSSID remains in the blacklist. This applies to both automatically and manually blacklisted BSSIDs.

Range:

- from 10 to 3600 if CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 300 if CONFIG_ESP_WIFI_ROAMING_BSSID_BLACKLIST && CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_ROAMING_MAX_CANDIDATES

Maximum number of roaming candidates

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_ROAMING_APP > Configure roaming App > Blacklist Configuration

Maximum number of roaming candidates to consider. This also defines the size of the blacklist.

Range:

- from 1 to 10 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

Default value:

- 5 if CONFIG_ESP_WIFI_ENABLE_ROAMING_APP

CONFIG_ESP_WIFI_DPP_SUPPORT

Enable DPP support

Found in: Component config > Wi-Fi

Select this option to enable WiFi Easy Connect Support.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_11R_SUPPORT

Enable 802.11R (Fast Transition) Support

Found in: Component config > Wi-Fi

Select this option to enable WiFi Fast Transition Support.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_WPS_SOFTAP_REGISTRAR

Add WPS Registrar support in SoftAP mode

Found in: Component config > Wi-Fi

Select this option to enable WPS registrar support in softAP mode.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_ENABLE_WIFI_TX_STATS

Enable Wi-Fi transmission statistics

Found in: Component config > Wi-Fi

Enable Wi-Fi transmission statistics. Total support 4 access category. Each access category will use 346 bytes memory.

Default value:

- No (disabled) if SOC_WIFI_HE_SUPPORT

CONFIG_ESP_WIFI_ENABLE_WIFI_RX_STATS

Enable Wi-Fi reception statistics

Found in: Component config > Wi-Fi

Enable Wi-Fi reception statistics. Total support 2 access category. Each access category will use 190 bytes memory.

Default value:

- No (disabled) if SOC_WIFI_HE_SUPPORT

CONFIG_ESP_WIFI_ENABLE_WIFI_RX_MU_STATS

Enable Wi-Fi DL MU-MIMO and DL OFDMA reception statistics

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENABLE_WIFI_RX_STATS

Enable Wi-Fi DL MU-MIMO and DL OFDMA reception statistics. Will use 10932 bytes memory.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_ENABLE_WIFI_RX_STATS

CONFIG_ESP_WIFI_TX_HETB_QUEUE_NUM

WiFi TX HE TB QUEUE number for STA HE TB PPDU transmission

Found in: Component config > Wi-Fi

Set the maximum number of queue that can be aggregated by the STA in the A-MPDU carried in the HE TB PPDU.

Range:

- from 1 to 4 if SOC_WIFI_HE_SUPPORT

Default value:

- 3 if SOC_WIFI_HE_SUPPORT

CONFIG_ESP_WIFI_ENABLE_DUMP_HESIGB

Enable Wi-Fi dump HE-SIGB which is contained in DL HE MU PPDUs

Found in: Component config > Wi-Fi

Enable Wi-Fi dump HE-SIGB which is contained in DL HE MU PPDUs.

Default value:

- No (disabled) if SOC_WIFI_SUPPORT_5G

CONFIG_ESP_WIFI_ENABLE_DUMP_MU_CFO

Enable Wi-Fi dump MU CFO

Found in: Component config > Wi-Fi

Enable Wi-Fi dump MU CFO.

Default value:

- No (disabled) if SOC_WIFI_SUPPORT_5G

CONFIG_ESP_WIFI_ENABLE_DUMP_CTRL_NDPA

Enable Wi-Fi dump NDPA frames

Found in: Component config > Wi-Fi

Enable Wi-Fi dump NDPA frames.

Default value:

- No (disabled) if SOC_WIFI_SUPPORT_5G

CONFIG_ESP_WIFI_ENABLE_DUMP_CTRL_BFRP

Enable Wi-Fi dump BFRP frames

Found in: Component config > Wi-Fi

Enable Wi-Fi dump BFRP frames.

Default value:

- No (disabled) if SOC_WIFI_SUPPORT_5G

WPS Configuration Options

Contains:

- CONFIG_ESP_WIFI_WPS_PASSPHRASE
- CONFIG_ESP_WIFI_WPS_RECONNECT_ON_FAIL
- CONFIG_ESP_WIFI_WPS_STRICT

CONFIG_ESP_WIFI_WPS_STRICT

Strictly validate all WPS attributes

Found in: Component config > Wi-Fi > WPS Configuration Options

Select this option to enable validate each WPS attribute rigorously. Disabling this add the workarounds with various APs. Enabling this may cause inter operability issues with some APs.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_WPS_PASSPHRASE

Get WPA2 passphrase in WPS config

Found in: Component config > Wi-Fi > WPS Configuration Options

Select this option to get passphrase during WPS configuration. This option fakes the virtual display capabilities to get the configuration in passphrase mode. Not recommended to be used since WPS credentials should not be shared to other devices, making it in readable format increases that risk, also passphrase requires pbkdf2 to convert in psk.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_WPS_RECONNECT_ON_FAIL

Reconnect to previous SSID if WPS failed

Found in: Component config > Wi-Fi > WPS Configuration Options

Select this option to enable reconnection to previous SSID if WPS fails. This option will only work if station was connected to a network when WPS was started.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_DEBUG_PRINT

Print debug messages from WPA Supplicant

Found in: Component config > Wi-Fi

Select this option to print logging information from WPA supplicant, this includes handshake information and key hex dumps depending on the project logging level.

Enabling this could increase the build size ~60kb depending on the project logging level.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_ENTERPRISE_SUPPORT

Enable enterprise option

Found in: Component config > Wi-Fi

Select this to enable/disable enterprise connection support.

disabling this will reduce binary size. disabling this will disable the use of any esp_wifi_sta_wpa2_ent_* (as APIs will be meaningless)

Note that when using bigger certificates on low-power chips without crypto hardware acceleration, it is recommended to adjust the task watchdog timer (TWDT) if it is enabled. For precise information on timing requirements, you can check performance numbers at https://github.com/espressif/mbedtls/wiki/Performance-Numbers.

Default value:

- Yes (enabled)

CONFIG_ESP_WIFI_ENT_FREE_DYNAMIC_BUFFER

Free dynamic buffers during WiFi enterprise connection

Found in: Component config > Wi-Fi > CONFIG_ESP_WIFI_ENTERPRISE_SUPPORT

Select this configuration to free dynamic buffers during WiFi enterprise connection. This will enable chip to reduce heap consumption during WiFi enterprise connection.

Default value:

- No (disabled)

CONFIG_ESP_WIFI_MODEM_RF_FLAG_UPDATE_DEBUG

Enable debug assertions for modem RF flag update

Found in: Component config > Wi-Fi

Enable debug assertions to verify modem RF flag update operations. This option enables assert checks to verify that modem RF power state is correctly cleared before pmu sleep.

Default value:

- No (disabled) if CONFIG_ESP_WIFI_ENHANCED_LIGHT_SLEEP

CONFIG_ESP_WIFI_PASSIVE_HIDDEN_AP_SUPPORT

Support station connection to hidden APs using passive scan when country policy is auto

Found in: Component config > Wi-Fi

Select this option to enable/disable support for station connections to hidden APs using passive scan when the country policy is set to auto.

Default value:

- No (disabled)

Core dump

Contains:

- CONFIG_ESP_COREDUMP_CHECK_BOOT
- CONFIG_ESP_COREDUMP_TO_FLASH_OR_UART
- CONFIG_ESP_COREDUMP_UART_DELAY
- CONFIG_ESP_COREDUMP_FLASH_NO_OVERWRITE
- CONFIG_ESP_COREDUMP_LOGS
- CONFIG_ESP_COREDUMP_DECODE
- CONFIG_ESP_COREDUMP_CAPTURE_DRAM
- CONFIG_ESP_COREDUMP_MAX_TASKS_NUM
- CONFIG_ESP_COREDUMP_STACK_SIZE

CONFIG_ESP_COREDUMP_TO_FLASH_OR_UART

Data destination

Found in: Component config > Core dump

Select place to store core dump: flash, uart or none (to disable core dumps generation).

Core dumps to Flash are not available if PSRAM is used for task stacks.

If core dump is configured to be stored in flash and custom partition table is used add corresponding entry to your CSV. For examples, please see predefined partition table CSV descriptions in the components/partition_table directory.

Available options:

- Flash (CONFIG_ESP_COREDUMP_ENABLE_TO_FLASH)
- UART (CONFIG_ESP_COREDUMP_ENABLE_TO_UART)
- None (CONFIG_ESP_COREDUMP_ENABLE_TO_NONE)

CONFIG_ESP_COREDUMP_CAPTURE_DRAM

Include whole .bss and .data sections and heap data into core dump file

Found in: Component config > Core dump

Storing these sections can help with easier debugging and troubleshooting. However, additional storage space will be required in the core dump partition. At least 128KB should be reserved, but the actual amount required may vary based on the application's DRAM usage. Note that sections located in external RAM will not be stored.

CONFIG_ESP_COREDUMP_CHECK_BOOT

Check core dump data integrity on boot

Found in: Component config > Core dump

When enabled, if any data are found on the flash core dump partition, they will be checked by calculating their checksum.

Default value:

- Yes (enabled) if CONFIG_ESP_COREDUMP_ENABLE_TO_FLASH

CONFIG_ESP_COREDUMP_LOGS

Enable coredump logs for debugging

Found in: Component config > Core dump

Enable/disable coredump logs. Logs strings from espcoredump component are placed in DRAM. Disabling these helps to save ~5KB of internal memory.

CONFIG_ESP_COREDUMP_MAX_TASKS_NUM

Maximum number of tasks

Found in: Component config > Core dump

Maximum number of tasks that will be included in the core dump. Crashed task registers and stacks are always included. Other tasks are included in order of their priority. (Highest priority ready task first)

CONFIG_ESP_COREDUMP_UART_DELAY

Delay before print to UART

Found in: Component config > Core dump

Config delay (in ms) before printing core dump to UART. Delay can be interrupted by pressing Enter key.

Default value:

- 0 if CONFIG_ESP_COREDUMP_ENABLE_TO_UART

CONFIG_ESP_COREDUMP_FLASH_NO_OVERWRITE

Don't overwrite existing core dump

Found in: Component config > Core dump

Don't overwrite an existing core dump already present in flash. Enable this option to only keep the first of multiple core dumps.

If enabled, the core dump partition must be erased before the first core dump can be written.

Default value:

- No (disabled) if CONFIG_ESP_COREDUMP_ENABLE_TO_FLASH

CONFIG_ESP_COREDUMP_STACK_SIZE

Reserved stack size

Found in: Component config > Core dump

Size of the memory to be reserved for core dump stack. If 0 core dump process will run on the stack of crashed task/ISR, otherwise special stack will be allocated. To ensure that core dump itself will not overflow task/ISR stack set this to the value around 1300-1800 depending on the chosen checksum calculation method. SHA256 method needs more stack space than CRC32. NOTE: It eats DRAM.

CONFIG_ESP_COREDUMP_DECODE

Handling of UART core dumps in IDF Monitor

Found in: Component config > Core dump

Available options:

- Decode and show summary (info_corefile) (CONFIG_ESP_COREDUMP_DECODE_INFO)
- Don't decode (CONFIG_ESP_COREDUMP_DECODE_DISABLE)

FAT Filesystem support

Contains:

- CONFIG_FATFS_API_ENCODING
- CONFIG_FATFS_VFS_FSTAT_BLKSIZE
- CONFIG_FATFS_IMMEDIATE_FSYNC
- CONFIG_FATFS_USE_FASTSEEK
- CONFIG_FATFS_USE_STRFUNC_CHOICE
- CONFIG_FATFS_FAST_SEEK_BUFFER_SIZE
- CONFIG_FATFS_STRF_ENCODE_CHOICE
- File system free space calculation behavior
- CONFIG_FATFS_LONG_FILENAMES
- CONFIG_FATFS_PRINT_FLOAT
- CONFIG_FATFS_PRINT_LLI
- CONFIG_FATFS_MAX_LFN
- CONFIG_FATFS_FS_LOCK
- CONFIG_FATFS_VOLUME_COUNT
- CONFIG_FATFS_CHOOSE_CODEPAGE
- CONFIG_FATFS_LINK_LOCK
- CONFIG_FATFS_ALLOC_PREFER_EXTRAM

- CONFIG_FATFS_SECTOR_SIZE
- CONFIG_FATFS_TIMEOUT_MS
- CONFIG_FATFS_USE_DYN_BUFFERS
- CONFIG_FATFS_USE_LABEL
- CONFIG_FATFS_PER_FILE_CACHE

CONFIG_FATFS_VOLUME_COUNT

Number of volumes

Found in: Component config > FAT Filesystem support

Number of volumes (logical drives) to use.

Range:

- from 1 to 10

Default value:

- 2

CONFIG_FATFS_LONG_FILENAMES

Long filename support

Found in: Component config > FAT Filesystem support

Support long filenames in FAT. Long filename data increases memory usage. FATFS can be configured to store the buffer for long filename data in stack or heap.

Available options:

- No long filenames (CONFIG_FATFS_LFN_NONE)
- Long filename buffer in heap (CONFIG_FATFS_LFN_HEAP)
- Long filename buffer on stack (CONFIG_FATFS_LFN_STACK)

CONFIG_FATFS_SECTOR_SIZE

Sector size

Found in: Component config > FAT Filesystem support

Specify the size of the sector in bytes for FATFS partition generator.

Available options:

- 512 (CONFIG_FATFS_SECTOR_512)
- 4096 (CONFIG_FATFS_SECTOR_4096)

CONFIG_FATFS_CHOOSE_CODEPAGE

OEM Code Page

Found in: Component config > FAT Filesystem support

OEM code page used for file name encodings.

If "Dynamic" is selected, code page can be chosen at runtime using f_setcp function. Note that choosing this option will increase application size by ~480kB.

Available options:

- Dynamic (all code pages supported) (CONFIG_FATFS_CODEPAGE_DYNAMIC)
- US (CP437) (CONFIG_FATFS_CODEPAGE_437)
- Arabic (CP720) (CONFIG_FATFS_CODEPAGE_720)
- Greek (CP737) (CONFIG_FATFS_CODEPAGE_737)
- KBL (CP771) (CONFIG_FATFS_CODEPAGE_771)
- Baltic (CP775) (CONFIG_FATFS_CODEPAGE_775)
- Latin 1 (CP850) (CONFIG_FATFS_CODEPAGE_850)
- Latin 2 (CP852) (CONFIG_FATFS_CODEPAGE_852)
- Cyrillic (CP855) (CONFIG_FATFS_CODEPAGE_855)
- Turkish (CP857) (CONFIG_FATFS_CODEPAGE_857)
- Portuguese (CP860) (CONFIG_FATFS_CODEPAGE_860)
- Icelandic (CP861) (CONFIG_FATFS_CODEPAGE_861)
- Hebrew (CP862) (CONFIG_FATFS_CODEPAGE_862)
- Canadian French (CP863) (CONFIG_FATFS_CODEPAGE_863)
- Arabic (CP864) (CONFIG_FATFS_CODEPAGE_864)
- Nordic (CP865) (CONFIG_FATFS_CODEPAGE_865)
- Russian (CP866) (CONFIG_FATFS_CODEPAGE_866)
- Greek 2 (CP869) (CONFIG_FATFS_CODEPAGE_869)
- Japanese (DBCS) (CP932) (CONFIG_FATFS_CODEPAGE_932)
- Simplified Chinese (DBCS) (CP936) (CONFIG_FATFS_CODEPAGE_936)
- Korean (DBCS) (CP949) (CONFIG_FATFS_CODEPAGE_949)
- Traditional Chinese (DBCS) (CP950) (CONFIG_FATFS_CODEPAGE_950)

CONFIG_FATFS_MAX_LFN

Max long filename length

Found in: Component config > FAT Filesystem support

Maximum long filename length. Can be reduced to save RAM.

CONFIG_FATFS_API_ENCODING

API character encoding

Found in: Component config > FAT Filesystem support

Choose encoding for character and string arguments/returns when using FATFS APIs. The encoding of arguments will usually depend on text editor settings.

Available options:

- API uses ANSI/OEM encoding (CONFIG_FATFS_API_ENCODING_ANSI_OEM)
- API uses UTF-8 encoding (CONFIG_FATFS_API_ENCODING_UTF_8)

CONFIG_FATFS_FS_LOCK

Number of simultaneously open files protected by lock function

Found in: Component config > FAT Filesystem support

This option sets the FATFS configuration value _FS_LOCK. The option _FS_LOCK switches file lock function to control duplicated file open and illegal operation to open objects.

* 0: Disable file lock function. To avoid volume corruption, application should avoid illegal open, remove and rename to the open objects.

* >0: Enable file lock function. The value defines how many files/sub-directories can be opened simultaneously under file lock control.

Note that the file lock control is independent of re-entrancy.

Range:

- from 0 to 65535

Default value:

- 0

CONFIG_FATFS_TIMEOUT_MS

Timeout for acquiring a file lock, ms

Found in: Component config > FAT Filesystem support

This option sets FATFS configuration value _FS_TIMEOUT, scaled to milliseconds. Sets the number of milliseconds FATFS will wait to acquire a mutex when operating on an open file. For example, if one task is performing a lengthy operation, another task will wait for the first task to release the lock, and time out after amount of time set by this option.

Default value:

- 10000

CONFIG_FATFS_PER_FILE_CACHE

Use separate cache for each file

Found in: Component config > FAT Filesystem support

This option affects FATFS configuration value _FS_TINY.

If this option is set, _FS_TINY is 0, and each open file has its own cache, size of the cache is equal to the _MAX_SS variable (512 or 4096 bytes). This option uses more RAM if more than 1 file is open, but needs less reads and writes to the storage for some operations.

If this option is not set, _FS_TINY is 1, and single cache is used for all open files, size is also equal to _MAX_SS variable. This reduces the amount of heap used when multiple files are open, but increases the number of read and write operations which FATFS needs to make.

Default value:

- Yes (enabled)

CONFIG_FATFS_ALLOC_PREFER_EXTRAM

Prefer external RAM when allocating FATFS buffers

Found in: Component config > FAT Filesystem support

When the option is enabled, internal buffers used by FATFS will be allocated from external RAM. If the allocation from external RAM fails, the buffer will be allocated from the internal RAM. Disable this option if optimizing for performance. Enable this option if optimizing for internal memory size.

Default value:

- Yes (enabled) if CONFIG_SPIRAM_USE_CAPS_ALLOC || CONFIG_SPIRAM_USE_MALLOC

CONFIG_FATFS_USE_FASTSEEK

Enable fast seek algorithm when using lseek function through VFS FAT

Found in: Component config > FAT Filesystem support

The fast seek feature enables fast backward/long seek operations without FAT access by using an in-memory CLMT (cluster link map table). Please note, fast-seek is only allowed for read-mode files, if a file is opened in write-mode, the seek mechanism will automatically fallback to the default implementation.

Default value:

- No (disabled)

CONFIG_FATFS_USE_STRFUNC_CHOICE

Enable string functions, f_gets(), f_putc(), f_puts() and f_printf()

Found in: Component config > FAT Filesystem support

These are specialized alternatives to stdio functions for working directly with FATFS without VFS. Legacy code may need functions, but for new development, it is advised to use stdio under VFS.

0: Disable. FF_PRINT_LLI, FF_PRINT_FLOAT and FF_STRF_ENCODE have no effect. 1: Enable without LF-CRLF conversion. 2: Enable with LF-CRLF conversion.

Available options:

- 0:Disable (CONFIG_FATFS_USE_STRFUNC_NONE)
- 1:Enable without LF-CRLF conversion (CONFIG_FATFS_USE_STRFUNC_WITHOUT_CRLF_CONV)
- 2:Enable with LF-CRLF conversion (CONFIG_FATFS_USE_STRFUNC_WITH_CRLF_CONV)

CONFIG_FATFS_PRINT_LLI

Make fatfs f_printf() support long long argument

Found in: Component config > FAT Filesystem support

CONFIG_FATFS_PRINT_FLOAT

Make fatfs f_printf() support floating point argument

Found in: Component config > FAT Filesystem support

CONFIG_FATFS_STRF_ENCODE_CHOICE

FatFS string functions: convert character encoding

Found in: Component config > FAT Filesystem support

When FF_LFN_UNICODE >= 1 with LFN enabled, string functions convert the character encoding in it. FF_STRF_ENCODE selects assumption of character encoding ON THE FILE to be read/written via those functions. 0: ANSI/OEM in current CP 1: Unicode in UTF-16LE 2: Unicode in UTF-16BE 3: Unicode in UTF-8

Available options:

- 0:ANSI/OEM in current CP (CONFIG_FATFS_STRF_ENCODE_ANSI)
- 1:Unicode in UTF-16LE (CONFIG_FATFS_STRF_ENCODE_UTF16LE)
- 2:Unicode in UTF-16BE (CONFIG_FATFS_STRF_ENCODE_UTF16BE)
- 3:Unicode in UTF-8 (CONFIG_FATFS_STRF_ENCODE_UTF8)

CONFIG_FATFS_FAST_SEEK_BUFFER_SIZE

Fast seek CLMT buffer size

Found in: Component config > FAT Filesystem support

If fast seek algorithm is enabled, this defines the size of CLMT buffer used by this algorithm in 32-bit word units. This value should be chosen based on prior knowledge of maximum elements of each file entry would store.

Default value:

- 64 if CONFIG_FATFS_USE_FASTSEEK

CONFIG_FATFS_VFS_FSTAT_BLKSIZE

Default block size

Found in: Component config > FAT Filesystem support

If set to 0, the 'newlib' library's default size (BLKSIZ) is used (128 B). If set to a non-zero value, the value is used as the block size. Default file buffer size is set to this value and the buffer is allocated when first attempt of reading/writing to a file is made. Increasing this value improves fread() speed, however the heap usage is increased as well.

NOTE: The block size value is shared by all the filesystem functions accessing target media for given file descriptor! See 'Improving I/O performance' section of 'Maximizing Execution Speed' documentation page for more details.

Default value:

- 0

CONFIG_FATFS_IMMEDIATE_FSYNC

Enable automatic f_sync

Found in: Component config > FAT Filesystem support

Enables automatic calling of f_sync() to flush recent file changes after each call of vfs_fat_write(), vfs_fat_pwrite(), vfs_fat_link(), vfs_fat_truncate() and vfs_fat_ftruncate() functions. This feature improves file-consistency and size reporting accuracy for the FatFS, at a price on decreased performance due to frequent disk operations

Default value:

- No (disabled)

CONFIG_FATFS_USE_LABEL

Use FATFS volume label

Found in: Component config > FAT Filesystem support

Allows FATFS volume label to be specified using f_setlabel

Default value:

- No (disabled)

CONFIG_FATFS_LINK_LOCK

Perform the whole link operation under lock

Found in: Component config > FAT Filesystem support

If enabled, the whole link operation (including file copying) is performed under lock. This ensures that the link operation is atomic, but may cause performance for large files. It may create less fragmented file copy.

Default value:

- Yes (enabled)

CONFIG_FATFS_USE_DYN_BUFFERS

Use dynamic buffers

Found in: Component config > FAT Filesystem support

If enabled, the buffers used by FATFS will be allocated separately from the rest of the structure. This option is useful when using multiple FATFS instances with different sector sizes, as the buffers will be allocated according to the sector size. If disabled, the greatest sector size will be used for all FATFS instances. (In most cases, this would be the sector size of Wear Levelling library) This might cause more memory to be used than necessary.

Default value:

- Yes (enabled)

File system free space calculation behavior

Controls if the file system does or does not trust cached data like free cluster count and allocated cluster number. Setting these to do not trust the data may result of more accurate output from *f_getfree()* function but increased overhead (forces a full FAT scan, etc.).

Contains:

- CONFIG_FATFS_DONT_TRUST_LAST_ALLOC
- CONFIG_FATFS_DONT_TRUST_FREE_CLUSTER_CNT

CONFIG_FATFS_DONT_TRUST_FREE_CLUSTER_CNT

Don't trust free cluster count

Found in: Component config > FAT Filesystem support > File system free space calculation behavior

If 1, the file system will not trust the free cluster count in the FSINFO (in FATFS struct). This may result in more accurate output from *f_getfree()* function but increased overhead.

Range:

- from 0 to 1

Default value:

- 0

CONFIG_FATFS_DONT_TRUST_LAST_ALLOC

Don't trust allocated cluster number

Found in: Component config > FAT Filesystem support > File system free space calculation behavior

If 1, the file system will not trust the last allocated cluster number in the FSINFO (in FATFS struct). This may result in more accurate output from *f_getfree()* function but increased overhead.

Range:

- from 0 to 1

Default value:

- 0

FreeRTOS

Contains:

- Extra
- Kernel
- Port

Kernel

Contains:

- CONFIG_FREERTOS_CHECK_STACKOVERFLOW

- CONFIG_FREERTOS_ENABLE_BACKWARD_COMPATIBILITY
- CONFIG_FREERTOS_GENERATE_RUN_TIME_STATS
- CONFIG_FREERTOS_MAX_TASK_NAME_LEN
- CONFIG_FREERTOS_IDLE_TASK_STACKSIZE
- CONFIG_FREERTOS_THREAD_LOCAL_STORAGE_POINTERS
- CONFIG_FREERTOS_QUEUE_REGISTRY_SIZE
- CONFIG_FREERTOS_TASK_NOTIFICATION_ARRAY_ENTRIES
- CONFIG_FREERTOS_HZ
- CONFIG_FREERTOS_USE_APPLICATION_TASK_TAG
- CONFIG_FREERTOS_USE_IDLE_HOOK
- CONFIG_FREERTOS_USE_LIST_DATA_INTEGRITY_CHECK_BYTES
- CONFIG_FREERTOS_OPTIMIZED_SCHEDULER
- CONFIG_FREERTOS_USE_TICK_HOOK
- CONFIG_FREERTOS_USE_TICKLESS_IDLE
- CONFIG_FREERTOS_USE_TIMERS
- CONFIG_FREERTOS_USE_TRACE_FACILITY
- CONFIG_FREERTOS_VTASKLIST_INCLUDE_COREID
- CONFIG_FREERTOS_UNICORE
- CONFIG_FREERTOS_SMP
- CONFIG_FREERTOS_USE_PASSIVE_IDLE_HOOK

CONFIG_FREERTOS_SMP

Run the Amazon SMP FreeRTOS kernel instead (FEATURE UNDER DEVELOPMENT)

Found in: Component config > FreeRTOS > Kernel

Amazon has released an SMP version of the FreeRTOS Kernel which can be found via the following link: https://github.com/FreeRTOS/FreeRTOS-Kernel/tree/smp

IDF has added an experimental port of this SMP kernel located in components/freertos/FreeRTOS-Kernel-SMP. Enabling this option will cause IDF to use the Amazon SMP kernel. Note that THIS FEATURE IS UNDER ACTIVE DEVELOPMENT, users use this at their own risk.

Leaving this option disabled will mean the IDF FreeRTOS kernel is used instead, which is located in: components/freertos/FreeRTOS-Kernel. Both kernel versions are SMP capable, but differ in their implementation and features.

Default value:

- No (disabled)

CONFIG_FREERTOS_UNICORE

Run FreeRTOS only on first core

Found in: Component config > FreeRTOS > Kernel

This version of FreeRTOS normally takes control of all cores of the CPU. Select this if you only want to start it on the first core. This is needed when e.g. another process needs complete control over the second core.

CONFIG_FREERTOS_HZ

configTICK_RATE_HZ

Found in: Component config > FreeRTOS > Kernel

Sets the FreeRTOS tick interrupt frequency in Hz (see configTICK_RATE_HZ documentation for more details).

Range:

- from 1 to 1000

Default value:

- 100

CONFIG_FREERTOS_OPTIMIZED_SCHEDULER

configUSE_PORT_OPTIMISED_TASK_SELECTION

Found in: Component config > FreeRTOS > Kernel

Enables port specific task selection method. This option can speed up the search of ready tasks when scheduling (see configUSE_PORT_OPTIMISED_TASK_SELECTION documentation for more details).

CONFIG_FREERTOS_CHECK_STACKOVERFLOW

configCHECK_FOR_STACK_OVERFLOW

Found in: Component config > FreeRTOS > Kernel

Enables FreeRTOS to check for stack overflows (see configCHECK_FOR_STACK_OVERFLOW documentation for more details).

Note: If users do not provide their own `vApplicationStackOverflowHook()` function, a default function will be provided by ESP-IDF.

Available options:

- No checking (CONFIG_FREERTOS_CHECK_STACKOVERFLOW_NONE)

  Do not check for stack overflows (configCHECK_FOR_STACK_OVERFLOW = 0)

- Check by stack pointer value (Method 1) (CONFIG_FREERTOS_CHECK_STACKOVERFLOW_PTRVAL)

  Check for stack overflows on each context switch by checking if the stack pointer is in a valid range. Quick but does not detect stack overflows that happened between context switches (configCHECK_FOR_STACK_OVERFLOW = 1)

- Check using canary bytes (Method 2) (CONFIG_FREERTOS_CHECK_STACKOVERFLOW_CANARY)

  Places some magic bytes at the end of the stack area and on each context switch, check if these bytes are still intact. More thorough than just checking the pointer, but also slightly slower. (configCHECK_FOR_STACK_OVERFLOW = 2)

CONFIG_FREERTOS_THREAD_LOCAL_STORAGE_POINTERS

configNUM_THREAD_LOCAL_STORAGE_POINTERS

Found in: Component config > FreeRTOS > Kernel

Set the number of thread local storage pointers in each task (see configNUM_THREAD_LOCAL_STORAGE_POINTERS documentation for more details).

Note: In ESP-IDF, this value must be at least 1. Index 0 is reserved for use by the pthreads API thread-local-storage. Other indexes can be used for any desired purpose.

Range:

- from 1 to 256

Default value:

- 1

CONFIG_FREERTOS_IDLE_TASK_STACKSIZE

configMINIMAL_STACK_SIZE (Idle task stack size)

Found in: Component config > FreeRTOS > Kernel

Sets the idle task stack size in bytes (see configMINIMAL_STACK_SIZE documentation for more details).

Note:

- ESP-IDF specifies stack sizes in bytes instead of words.
- The default size is enough for most use cases.
- The stack size may need to be increased above the default if the app installs idle or thread local storage cleanup hooks that use a lot of stack memory.
- Conversely, the stack size can be reduced to the minimum if non of the idle features are used.

Range:

- from 768 to 32768

Default value:

- 1536

CONFIG_FREERTOS_USE_IDLE_HOOK

configUSE_IDLE_HOOK

Found in: Component config > FreeRTOS > Kernel

Enables the idle task application hook (see configUSE_IDLE_HOOK documentation for more details).

Note:

- The application must provide the hook function `void vApplicationIdleHook( void );`
- `vApplicationIdleHook()` is called from FreeRTOS idle task(s)
- The FreeRTOS idle hook is NOT the same as the ESP-IDF Idle Hook, but both can be enabled simultaneously.

Default value:

- No (disabled)

CONFIG_FREERTOS_USE_PASSIVE_IDLE_HOOK

Use FreeRTOS minimal idle hook

Found in: Component config > FreeRTOS > Kernel

Enables the minimal idle task application hook (see configUSE_IDLE_HOOK documentation for more details).

Note:

- The application must provide the hook function `void vApplicationPassiveIdleHook( void );`
- `vApplicationPassiveIdleHook()` is called from FreeRTOS minimal idle task(s)

Default value:

- No (disabled) if CONFIG_FREERTOS_SMP

CONFIG_FREERTOS_USE_TICK_HOOK

configUSE_TICK_HOOK

Found in: Component config > FreeRTOS > Kernel

Enables the tick hook (see configUSE_TICK_HOOK documentation for more details).

Note:

- The application must provide the hook function `void vApplicationTickHook( void );`
- `vApplicationTickHook()` is called from FreeRTOS's tick handling function `xTaskIncrementTick()`
- The FreeRTOS tick hook is NOT the same as the ESP-IDF Tick Interrupt Hook, but both can be enabled simultaneously.

Default value:

- No (disabled)

CONFIG_FREERTOS_MAX_TASK_NAME_LEN

configMAX_TASK_NAME_LEN

Found in: Component config > FreeRTOS > Kernel

Sets the maximum number of characters for task names (see configMAX_TASK_NAME_LEN documentation for more details).

Note: For most uses, the default of 16 characters is sufficient.

Range:

- from 1 to 256

Default value:

- 16

CONFIG_FREERTOS_ENABLE_BACKWARD_COMPATIBILITY

configENABLE_BACKWARD_COMPATIBILITY

Found in: Component config > FreeRTOS > Kernel

Enable backward compatibility with APIs prior to FreeRTOS v8.0.0. (see configENABLE_BACKWARD_COMPATIBILITY documentation for more details).

Default value:

- No (disabled)

CONFIG_FREERTOS_USE_TIMERS

configUSE_TIMERS

Found in: Component config > FreeRTOS > Kernel

Enable FreeRTOS Software Timers. Normally the timer task will only get pulled into the build and created if any software timer related functions are used. This is achieved through IDF defining a weak empty function for xTimerCreateTimerTask, which should take effect if timers.c is not pulled into the build.

In certain special cases (if you use configUSE_TRACE_FACILITY=y and event groups) the linker will still pull in the xTimerCreateTimerTask from timers.c even if the function that utilized it gets discarded due to not being used.

In these cases you can use this option to force the timer task to be disabled.

Default value:

- Yes (enabled)

CONFIG_FREERTOS_TIMER_SERVICE_TASK_NAME

configTIMER_SERVICE_TASK_NAME

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TIMERS

Sets the timer task's name (see configTIMER_SERVICE_TASK_NAME documentation for more details).

Default value:

- "Tmr Svc"

CONFIG_FREERTOS_TIMER_SERVICE_TASK_CORE_AFFINITY

configTIMER_SERVICE_TASK_CORE_AFFINITY

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TIMERS

Sets the timer task's core affinity (see configTIMER_SERVICE_TASK_CORE_AFFINITY documentation for more details).

Available options:

- CPU0 (CONFIG_FREERTOS_TIMER_TASK_AFFINITY_CPU0)
- CPU1 (CONFIG_FREERTOS_TIMER_TASK_AFFINITY_CPU1)
- No affinity (CONFIG_FREERTOS_TIMER_TASK_NO_AFFINITY)

CONFIG_FREERTOS_TIMER_TASK_PRIORITY

configTIMER_TASK_PRIORITY

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TIMERS

Sets the timer task's priority (see configTIMER_TASK_PRIORITY documentation for more details).

Range:

- from 1 to 25

Default value:

- 1

CONFIG_FREERTOS_TIMER_TASK_STACK_DEPTH

configTIMER_TASK_STACK_DEPTH

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TIMERS

Set the timer task's stack size (see configTIMER_TASK_STACK_DEPTH documentation for more details).

Range:

- from 1536 to 32768

Default value:

- 2048

CONFIG_FREERTOS_TIMER_QUEUE_LENGTH

configTIMER_QUEUE_LENGTH

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TIMERS

Set the timer task's command queue length (see configTIMER_QUEUE_LENGTH documentation for more details).

Range:

- from 5 to 20

Default value:

- 10

CONFIG_FREERTOS_QUEUE_REGISTRY_SIZE

configQUEUE_REGISTRY_SIZE

Found in: Component config > FreeRTOS > Kernel

Set the size of the queue registry (see configQUEUE_REGISTRY_SIZE documentation for more details).

Note: A value of 0 will disable queue registry functionality

Range:

- from 0 to 20

Default value:

- 0

CONFIG_FREERTOS_TASK_NOTIFICATION_ARRAY_ENTRIES

configTASK_NOTIFICATION_ARRAY_ENTRIES

Found in: Component config > FreeRTOS > Kernel

Set the size of the task notification array of each task. When increasing this value, keep in mind that this means additional memory for each and every task on the system. However, task notifications in general are more light weight compared to alternatives such as semaphores.

Range:

- from 1 to 32

Default value:

- 1

CONFIG_FREERTOS_USE_TRACE_FACILITY

configUSE_TRACE_FACILITY

Found in: Component config > FreeRTOS > Kernel

Enables additional structure members and functions to assist with execution visualization and tracing (see configUSE_TRACE_FACILITY documentation for more details).

Default value:

- No (disabled)

CONFIG_FREERTOS_USE_STATS_FORMATTING_FUNCTIONS

configUSE_STATS_FORMATTING_FUNCTIONS

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TRACE_FACILITY

Set configUSE_TRACE_FACILITY and configUSE_STATS_FORMATTING_FUNCTIONS to 1 to include the `vTaskList()` and `vTaskGetRunTimeStats()` functions in the build (see configUSE_STATS_FORMATTING_FUNCTIONS documentation for more details).

Default value:

- No (disabled) if CONFIG_FREERTOS_USE_TRACE_FACILITY

CONFIG_FREERTOS_USE_LIST_DATA_INTEGRITY_CHECK_BYTES

configUSE_LIST_DATA_INTEGRITY_CHECK_BYTES

Found in: Component config > FreeRTOS > Kernel

Enable list integrity checker (see configUSE_LIST_DATA_INTEGRITY_CHECK_BYTES documentation for more details).

Default value:

- No (disabled)

CONFIG_FREERTOS_VTASKLIST_INCLUDE_COREID

Enable display of xCoreID in vTaskList

Found in: Component config > FreeRTOS > Kernel

If enabled, this will include an extra column when vTaskList is called to display the CoreID the task is pinned to (0,1) or -1 if not pinned.

CONFIG_FREERTOS_GENERATE_RUN_TIME_STATS

configGENERATE_RUN_TIME_STATS

Found in: Component config > FreeRTOS > Kernel

Enables collection of run time statistics for each task (see configGENERATE_RUN_TIME_STATS documentation for more details).

Note: The clock used for run time statistics can be configured in FREERTOS_RUN_TIME_STATS_CLK.

Default value:

- No (disabled)

CONFIG_FREERTOS_RUN_TIME_COUNTER_TYPE

configRUN_TIME_COUNTER_TYPE

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_GENERATE_RUN_TIME_STATS

Sets the data type used for the FreeRTOS run time stats. A larger data type can be used to reduce the frequency of the counter overflowing.

Available options:

- uint32_t (CONFIG_FREERTOS_RUN_TIME_COUNTER_TYPE_U32)

  configRUN_TIME_COUNTER_TYPE is set to uint32_t
- uint64_t (CONFIG_FREERTOS_RUN_TIME_COUNTER_TYPE_U64)

  configRUN_TIME_COUNTER_TYPE is set to uint64_t

CONFIG_FREERTOS_USE_TICKLESS_IDLE

configUSE_TICKLESS_IDLE

Found in: Component config > FreeRTOS > Kernel

If power management support is enabled, FreeRTOS will be able to put the system into light sleep mode when no tasks need to run for a number of ticks. This number can be set using FREERTOS_IDLE_TIME_BEFORE_SLEEP option. This feature is also known as "automatic light sleep".

Note that timers created using esp_timer APIs may prevent the system from entering sleep mode, even when no tasks need to run. To skip unnecessary wake-up initialize a timer with the "skip_unhandled_events" option as true.

If disabled, automatic light sleep support will be disabled.

Default value:

- No (disabled) if CONFIG_PM_ENABLE

CONFIG_FREERTOS_IDLE_TIME_BEFORE_SLEEP

configEXPECTED_IDLE_TIME_BEFORE_SLEEP

Found in: Component config > FreeRTOS > Kernel > CONFIG_FREERTOS_USE_TICKLESS_IDLE

FreeRTOS will enter light sleep mode if no tasks need to run for this number of ticks. You can enable PM_PROFILING feature in esp_pm components and dump the sleep status with esp_pm_dump_locks, if the proportion of rejected sleeps is too high, please increase this value to improve scheduling efficiency

Range:

- from 2 to 4294967295 if CONFIG_FREERTOS_USE_TICKLESS_IDLE

Default value:

- 3 if CONFIG_FREERTOS_USE_TICKLESS_IDLE

CONFIG_FREERTOS_USE_APPLICATION_TASK_TAG

configUSE_APPLICATION_TASK_TAG

Found in: Component config > FreeRTOS > Kernel

Enables task tagging functionality and its associated API (see configUSE_APPLICATION_TASK_TAG documentation for more details).

Default value:

- No (disabled)

Port

Contains:

- CONFIG_FREERTOS_RUN_TIME_STATS_CLK
- CONFIG_FREERTOS_INTERRUPT_BACKTRACE
- CONFIG_FREERTOS_WATCHPOINT_END_OF_STACK
- CONFIG_FREERTOS_TASK_PRE_DELETION_HOOK
- CONFIG_FREERTOS_TLSP_DELETION_CALLBACKS
- CONFIG_FREERTOS_ISR_STACKSIZE
- CONFIG_FREERTOS_PLACE_ISR_FUNCTIONS_INTO_FLASH
- CONFIG_FREERTOS_IN_IRAM
- CONFIG_FREERTOS_CHECK_PORT_CRITICAL_COMPLIANCE
- CONFIG_FREERTOS_CORETIMER
- CONFIG_FREERTOS_FPU_IN_ISR
- CONFIG_FREERTOS_TASK_FUNCTION_WRAPPER

CONFIG_FREERTOS_TASK_FUNCTION_WRAPPER

Wrap task functions

Found in: Component config > FreeRTOS > Port

If enabled, all FreeRTOS task functions will be enclosed in a wrapper function. If a task function mistakenly returns (i.e. does not delete), the call flow will return to the wrapper function. The wrapper function will then log an error and abort the application. This option is also required for GDB backtraces and C++ exceptions to work correctly inside top-level task functions.

Default value:

- Yes (enabled)

CONFIG_FREERTOS_WATCHPOINT_END_OF_STACK

Enable stack overflow debug watchpoint

Found in: Component config > FreeRTOS > Port

FreeRTOS can check if a stack has overflown its bounds by checking either the value of the stack pointer or by checking the integrity of canary bytes. (See FREERTOS_CHECK_STACKOVERFLOW for more information.) These checks only happen on a context switch, and the situation that caused the stack overflow may already be long gone by then. This option will use the last debug memory watchpoint to allow breaking into the debugger (or panic'ing) as soon as any of the last 32 bytes on the stack of a task are overwritten. The side effect is that using gdb, you effectively have one hardware watchpoint less because the last one is overwritten as soon as a task switch happens.

Another consequence is that due to alignment requirements of the watchpoint, the usable stack size decreases by up to 60 bytes. This is because the watchpoint region has to be aligned to its size and the size for the stack watchpoint in IDF is 32 bytes.

This check only triggers if the stack overflow writes within 32 bytes near the end of the stack, rather than overshooting further, so it is worth combining this approach with one of the other stack overflow check methods.

When this watchpoint is hit, gdb will stop with a SIGTRAP message. When no JTAG OCD is attached, esp-idf will panic on an unhandled debug exception.

Default value:

- No (disabled)

CONFIG_FREERTOS_TLSP_DELETION_CALLBACKS

Enable thread local storage pointers deletion callbacks

Found in: Component config > FreeRTOS > Port

ESP-IDF provides users with the ability to free TLSP memory by registering TLSP deletion callbacks. These callbacks are automatically called by FreeRTOS when a task is deleted. When this option is turned on, the memory reserved for TLSPs in the TCB is doubled to make space for storing the deletion callbacks. If the user does not wish to use TLSP deletion callbacks then this option could be turned off to save space in the TCB memory.

Default value:

- Yes (enabled)

CONFIG_FREERTOS_TASK_PRE_DELETION_HOOK

Enable task pre-deletion hook

Found in: Component config > FreeRTOS > Port

Enable this option to make FreeRTOS call a user provided hook function right before it deletes a task (i.e., frees/releases a dynamically/statically allocated task's memory). This is useful if users want to know when a task is actually deleted (in case the task's deletion is delegated to the IDLE task).

If this config option is enabled, users must define a `void vTaskPreDeletionHook( void \* pxTCB )` hook function in their application.

Default value:

- No (disabled)

CONFIG_FREERTOS_ISR_STACKSIZE

ISR stack size

Found in: Component config > FreeRTOS > Port

The interrupt handlers have their own stack. The size of the stack can be defined here. Each processor has its own stack, so the total size occupied will be twice this.

Range:

- from 1536 to 32768

Default value:

- 1536

CONFIG_FREERTOS_INTERRUPT_BACKTRACE

Enable backtrace from interrupt to task context

Found in: Component config > FreeRTOS > Port

If this option is enabled, interrupt stack frame will be modified to point to the code of the interrupted task as its return address. This helps the debugger (or the panic handler) show a backtrace from the interrupt to the task which was interrupted. This also works for nested interrupts: higher level interrupt stack can be traced back to the lower level interrupt. This option adds 4 instructions to the interrupt dispatching code.

Default value:

- Yes (enabled)

CONFIG_FREERTOS_FPU_IN_ISR

Use float in Level 1 ISR

Found in: Component config > FreeRTOS > Port

When enabled, the usage of float type is allowed inside Level 1 ISRs. Note that usage of float types in higher level interrupts is still not permitted.

Default value:

- No (disabled)

CONFIG_FREERTOS_CORETIMER

Tick timer source (Xtensa Only)

Found in: Component config > FreeRTOS > Port

FreeRTOS needs a timer with an associated interrupt to use as the main tick source to increase counters, run timers and do pre-emptive multitasking with. There are multiple timers available to do this, with different interrupt priorities.

Available options:

- Timer 0 (int 6, level 1) (CONFIG_FREERTOS_CORETIMER_0)

  Select this to use timer 0

- Timer 1 (int 15, level 3) (CONFIG_FREERTOS_CORETIMER_1)

  Select this to use timer 1

- SYSTIMER 0 (level 1) (CONFIG_FREERTOS_CORETIMER_SYSTIMER_LVL1)

  Select this to use systimer with the 1 interrupt priority.

- SYSTIMER 0 (level 3) (CONFIG_FREERTOS_CORETIMER_SYSTIMER_LVL3)

  Select this to use systimer with the 3 interrupt priority.

CONFIG_FREERTOS_RUN_TIME_STATS_CLK

Choose the clock source for run time stats

Found in: Component config > FreeRTOS > Port

Choose the clock source for FreeRTOS run time stats. Options are CPU0's CPU Clock or the ESP Timer. Both clock sources are 32 bits. The CPU Clock can run at a higher frequency hence provide a finer resolution but will overflow much quicker. Note that run time stats are only valid until the clock source overflows.

Available options:

- Use ESP TIMER for run time stats (CONFIG_FREERTOS_RUN_TIME_STATS_USING_ESP_TIMER)

  ESP Timer will be used as the clock source for FreeRTOS run time stats. The ESP Timer runs at a frequency of 1MHz regardless of Dynamic Frequency Scaling. Therefore the ESP Timer will overflow in approximately 4290 seconds.

- Use CPU Clock for run time stats (CONFIG_FREERTOS_RUN_TIME_STATS_USING_CPU_CLK)

  CPU Clock will be used as the clock source for the generation of run time stats. The CPU Clock has a frequency dependent on ESP_DEFAULT_CPU_FREQ_MHZ and Dynamic Frequency Scaling (DFS). Therefore the CPU Clock frequency can fluctuate between 80 to 240MHz. Run time stats generated using the CPU Clock represents the number of CPU cycles each task is allocated and DOES NOT reflect the amount of time each task runs for (as CPU clock frequency can change). If the CPU clock consistently runs at the maximum frequency of 240MHz, it will overflow in approximately 17 seconds.

CONFIG_FREERTOS_IN_IRAM

Place FreeRTOS functions in IRAM

Found in: Component config > FreeRTOS > Port

Place FreeRTOS functions in IRAM for better performance. By default, FreeRTOS functions are placed in Flash to save IRAM. Enable this option if you need maximum performance for FreeRTOS operations.

Note: This option increases IRAM usage.

Default value:

- No (disabled)

CONFIG_FREERTOS_PLACE_ISR_FUNCTIONS_INTO_FLASH

Place FreeRTOS functions called from ISR context into Flash

Found in: Component config > FreeRTOS > Port

When enabled additional FreeRTOS functions which maybe called from an ISR context are also placed in Flash, thus freeing up more IRAM. This option is only applicable when the SPI_FLASH_AUTO_SUSPEND feature is supported.

Default value:

- No (disabled) if CONFIG_SPI_FLASH_AUTO_SUSPEND

CONFIG_FREERTOS_CHECK_PORT_CRITICAL_COMPLIANCE

Tests compliance with Vanilla FreeRTOS port*_CRITICAL calls

Found in: Component config > FreeRTOS > Port

If enabled, context of port*_CRITICAL calls (ISR or Non-ISR) would be checked to be in compliance with Vanilla FreeRTOS. e.g Calling port*_CRITICAL from ISR context would cause assert failure

Default value:

- No (disabled)

Extra

Contains:

- CONFIG_FREERTOS_TASK_CREATE_ALLOW_EXT_MEM

CONFIG_FREERTOS_TASK_CREATE_ALLOW_EXT_MEM

Allow external memory as an argument to xTaskCreateStatic (READ HELP)

Found in: Component config > FreeRTOS > Extra

Accessing memory in PSRAM has certain restrictions, so task stacks allocated by xTaskCreate are by default allocated from internal RAM.

This option allows for passing memory allocated from SPIRAM to be passed to xTaskCreateStatic. This should only be used for tasks where the stack is never accessed while the cache is disabled.

Extra notes for ESP32:

Because some bits of the ESP32 code environment cannot be recompiled with the cache workaround, normally tasks cannot be safely run with their stack residing in external memory; for this reason xTaskCreate (and related task creation functions) always allocate stack in internal memory and xTaskCreateStatic will check if the memory passed to it is in internal memory. If you have a task that needs a large amount of stack and does not call on ROM code in any way (no direct calls, but also no Bluetooth/WiFi), you can try enable this to cause xTaskCreateStatic to allow tasks stack in external memory.

Default value:

- No (disabled) if CONFIG_SPIRAM
- Yes (enabled) if CONFIG_SPIRAM

Hardware Abstraction Layer (HAL) and Low Level (LL)

Contains:

- CONFIG_HAL_DEFAULT_ASSERTION_LEVEL
- CONFIG_HAL_LOG_LEVEL
- CONFIG_HAL_GPIO_USE_ROM_IMPL
- CONFIG_HAL_SYSTIMER_USE_ROM_IMPL
- CONFIG_HAL_WDT_USE_ROM_IMPL

CONFIG_HAL_DEFAULT_ASSERTION_LEVEL

Default HAL assertion level

Found in: Component config > Hardware Abstraction Layer (HAL) and Low Level (LL)

Set the assert behavior / level for HAL component. HAL component assert level can be set separately, but the level can't exceed the system assertion level. e.g. If the system assertion is disabled, then the HAL assertion can't be enabled either. If the system assertion is enable, then the HAL assertion can still be disabled by this Kconfig option.

Available options:

- Same as system assertion level (CONFIG_HAL_ASSERTION_EQUALS_SYSTEM)
- Disabled (CONFIG_HAL_ASSERTION_DISABLE)
- Silent (CONFIG_HAL_ASSERTION_SILENT)
- Enabled (CONFIG_HAL_ASSERTION_ENABLE)

CONFIG_HAL_LOG_LEVEL

HAL layer log verbosity

Found in: Component config > Hardware Abstraction Layer (HAL) and Low Level (LL)

Specify how much output to see in HAL logs.

Available options:

- No output (CONFIG_HAL_LOG_LEVEL_NONE)
- Error (CONFIG_HAL_LOG_LEVEL_ERROR)
- Warning (CONFIG_HAL_LOG_LEVEL_WARN)
- Info (CONFIG_HAL_LOG_LEVEL_INFO)
- Debug (CONFIG_HAL_LOG_LEVEL_DEBUG)
- Verbose (CONFIG_HAL_LOG_LEVEL_VERBOSE)

CONFIG_HAL_SYSTIMER_USE_ROM_IMPL

Use ROM implementation of SysTimer HAL driver

Found in: Component config > Hardware Abstraction Layer (HAL) and Low Level (LL)

Enable this flag to use HAL functions from ROM instead of ESP-IDF.

If keeping this as "n" in your project, you will have less free IRAM. If making this as "y" in your project, you will increase free IRAM, but you will lose the possibility to debug this module, and some new features will be added and bugs will be fixed in the IDF source but cannot be synced to ROM.

Default value:

- Yes (enabled) if ESP_ROM_HAS_HAL_SYSTIMER

CONFIG_HAL_WDT_USE_ROM_IMPL

Use ROM implementation of WDT HAL driver

Found in: Component config > Hardware Abstraction Layer (HAL) and Low Level (LL)

Enable this flag to use HAL functions from ROM instead of ESP-IDF.

If keeping this as "n" in your project, you will have less free IRAM. If making this as "y" in your project, you will increase free IRAM, but you will lose the possibility to debug this module, and some new features will be added and bugs will be fixed in the IDF source but cannot be synced to ROM.

Default value:

- Yes (enabled) if ESP_ROM_HAS_HAL_WDT

CONFIG_HAL_GPIO_USE_ROM_IMPL

Use ROM implementation of GPIO HAL driver

Found in: Component config > Hardware Abstraction Layer (HAL) and Low Level (LL)

Enable this flag to use HAL functions from ROM when applicable instead of ESP-IDF.

If keeping this as "n" in your project, you will have less free IRAM. When compiling an application for a CPU that cannot access to the ROM memory, this option should be disabled.

Default value:

- Yes (enabled)

Heap memory debugging

Contains:

- CONFIG_HEAP_ABORT_WHEN_ALLOCATION_FAILS
- CONFIG_HEAP_TASK_TRACKING
- CONFIG_HEAP_PLACE_FUNCTION_INTO_FLASH
- CONFIG_HEAP_CORRUPTION_DETECTION
- CONFIG_HEAP_TRACING_DEST
- CONFIG_HEAP_TRACING_STACK_DEPTH
- CONFIG_HEAP_USE_HOOKS
- CONFIG_HEAP_TRACE_HASH_MAP
- CONFIG_HEAP_TLSF_USE_ROM_IMPL

CONFIG_HEAP_CORRUPTION_DETECTION

Heap corruption detection

Found in: Component config > Heap memory debugging

Enable heap poisoning features to detect heap corruption caused by out-of-bounds access to heap memory.

See the "Heap Memory Debugging" page of the IDF documentation for a description of each level of heap corruption detection.

Available options:

- Basic (no poisoning) (CONFIG_HEAP_POISONING_DISABLED)
- Light impact (CONFIG_HEAP_POISONING_LIGHT)
- Comprehensive (CONFIG_HEAP_POISONING_COMPREHENSIVE)

CONFIG_HEAP_TRACING_DEST

Heap tracing

Found in: Component config > Heap memory debugging

Enables the heap tracing API defined in esp_heap_trace.h.

This function causes a moderate increase in IRAM code side and a minor increase in heap function (malloc/free/realloc) CPU overhead, even when the tracing feature is not used. So it's best to keep it disabled unless tracing is being used.

Available options:

- Disabled (CONFIG_HEAP_TRACING_OFF)
- Standalone (CONFIG_HEAP_TRACING_STANDALONE)
- Host-based (CONFIG_HEAP_TRACING_TOHOST)

CONFIG_HEAP_TRACE_HASH_MAP

Use hash map mechanism to access heap trace records

Found in: Component config > Heap memory debugging

Enable this flag to use a hash map to increase performance in handling heap trace records.

Heap trace standalone supports storing records as a list, or a list + hash map.

Using only a list takes less memory, but calls to 'free' will get slower as the list grows. This is particularly affected when using HEAP_TRACE_ALL mode.

By using a list + hash map, calls to 'free' remain fast, at the cost of additional memory to store the hash map.

Default value:

- No (disabled) if CONFIG_HEAP_TRACING_STANDALONE

CONFIG_HEAP_TRACE_HASH_MAP_IN_EXT_RAM

Place hash map in external RAM

Found in: Component config > Heap memory debugging > CONFIG_HEAP_TRACE_HASH_MAP

When enabled this configuration forces the hash map to be placed in external RAM.

Default value:

- No (disabled) if CONFIG_SPIRAM && CONFIG_HEAP_TRACE_HASH_MAP

CONFIG_HEAP_TRACE_HASH_MAP_SIZE

The number of entries in the hash map

Found in: Component config > Heap memory debugging > CONFIG_HEAP_TRACE_HASH_MAP

Defines the number of entries in the heap trace hashmap. Each entry takes 8 bytes. The bigger this number is, the better the performance. Recommended range: 200 - 2000.

Default value:

- 512 if CONFIG_HEAP_TRACE_HASH_MAP

CONFIG_HEAP_TRACING_STACK_DEPTH

Heap tracing stack depth

Found in: Component config > Heap memory debugging

Number of stack frames to save when tracing heap operation callers.

More stack frames uses more memory in the heap trace buffer (and slows down allocation), but can provide useful information.

CONFIG_HEAP_USE_HOOKS

Use allocation and free hooks

Found in: Component config > Heap memory debugging

Enable the user to implement function hooks triggered for each successful allocation and free.

CONFIG_HEAP_TASK_TRACKING

Enable heap task tracking

Found in: Component config > Heap memory debugging

Enables tracking the task responsible for each heap allocation.

Note: Allocating or freeing memory or using the task tracking API will lead to a crash when the scheduler is not working (e.g, after calling vTaskSuspendAll).

CONFIG_HEAP_TRACK_DELETED_TASKS

Keep information about the memory usage of deleted tasks

Found in: Component config > Heap memory debugging > CONFIG_HEAP_TASK_TRACKING

When enabled, this configuration allows the user to keep trace of the memory usage of a task that has been deleted.

This allows the user to verify that no memory allocated within a task remains unfreed before terminating the task

Note that this feature cannot keep track of a task deletion if the task is allocated statically

Default value:

- No (disabled) if CONFIG_HEAP_TASK_TRACKING

CONFIG_HEAP_ABORT_WHEN_ALLOCATION_FAILS

Abort if memory allocation fails

Found in: Component config > Heap memory debugging

When enabled, if a memory allocation operation fails it will cause a system abort.

Default value:

- No (disabled)

CONFIG_HEAP_TLSF_USE_ROM_IMPL

Use ROM implementation of heap tlsf library

Found in: Component config > Heap memory debugging

Enable this flag to use heap functions from ROM instead of ESP-IDF.

If keeping this as "n" in your project, you will have less free IRAM. If making this as "y" in your project, you will increase free IRAM, but you will lose the possibility to debug this module, and some new features will be added and bugs will be fixed in the IDF source but cannot be synced to ROM.

Default value:

- Yes (enabled) if ESP_ROM_HAS_HEAP_TLSF

CONFIG_HEAP_PLACE_FUNCTION_INTO_FLASH

Force the entire heap component to be placed in flash memory

Found in: Component config > Heap memory debugging

Enable this flag to save up RAM space by placing the heap component in the flash memory

Note that it is only safe to enable this configuration if no functions from esp_heap_caps.h or esp_heap_trace.h are called from IRAM ISR which runs when cache is disabled.

Default value:

- No (disabled)

IEEE 802.15.4

Contains:

- CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_ENABLED

IEEE802154 Enable

Found in: Component config > IEEE 802.15.4

Default value:

- Yes (enabled) if SOC_IEEE802154_SUPPORTED

CONFIG_IEEE802154_RX_BUFFER_SIZE

The number of 802.15.4 receive buffers

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Number of receive buffers allocated for IEEE 802.15.4 frames. Increasing this value allows more frames to be buffered before processing, but consumes more memory.

Range:

- from 2 to 100 if CONFIG_IEEE802154_ENABLED

Default value:

- 20 if CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_CCA_MODE

Clear Channel Assessment (CCA) mode

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Select the Clear Channel Assessment (CCA) mode used to determine if the channel is clear before transmission. CCA is required by the IEEE 802.15.4 standard to avoid collisions.

Available options:

- Carrier sense only (CONFIG_IEEE802154_CCA_CARRIER)

  CCA reports channel busy only if a valid IEEE 802.15.4 signal is detected. This mode is less sensitive to non-802.15.4 interference.

- Energy above threshold (CONFIG_IEEE802154_CCA_ED)

  CCA reports channel busy if the energy level exceeds the configured threshold. This is the default mode and provides good balance between sensitivity and reliability.

- Carrier sense OR energy above threshold (CONFIG_IEEE802154_CCA_CARRIER_OR_ED)

  CCA reports channel busy if either a valid IEEE 802.15.4 signal is detected OR the energy level exceeds the threshold. This is the most sensitive mode.

- Carrier sense AND energy above threshold (CONFIG_IEEE802154_CCA_CARRIER_AND_ED)

  CCA reports channel busy only if both a valid IEEE 802.15.4 signal is detected AND the energy level exceeds the threshold. This is the least sensitive mode.

CONFIG_IEEE802154_CCA_THRESHOLD

CCA detection threshold

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Energy detection threshold for CCA, in dBm. The channel is considered busy if the detected energy exceeds this threshold. Lower values make the radio more sensitive to interference but may cause false positives.

Range:

- from -120 to 0 if CONFIG_IEEE802154_ENABLED

Default value:

- "-75" if CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_PENDING_TABLE_SIZE

Pending table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Size of the pending data table used to track frames waiting to be sent to sleeping devices. Each entry consumes memory, so adjust based on the number of sleeping child devices in your network.

Range:

- from 1 to 100 if CONFIG_IEEE802154_ENABLED

Default value:

- 20 if CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_MULTI_PAN_ENABLE

Enable multi-pan feature for frame filter

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Enable IEEE 802.15.4 multi-PAN (Personal Area Network) support. This allows the device to participate in multiple PANs simultaneously by filtering frames based on PAN ID.

Default value:

- No (disabled) if CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_INTERFACE_NUM

Number of IEEE802154 interfaces

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Number of IEEE802154 interfaces

Range:

- from 1 to 4 if CONFIG_IEEE802154_MULTI_PAN_ENABLE && CONFIG_IEEE802154_ENABLED

Default value:

- 2 if CONFIG_IEEE802154_MULTI_PAN_ENABLE && CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_TIMING_OPTIMIZATION

Enable throughput optimization

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Enabling this option increases throughput by ~5% at the expense of ~2.1k IRAM code size increase.

Default value:

- Yes (enabled) if CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_SLEEP_ENABLE

Enable IEEE802154 light sleep

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Enabling this option allows the IEEE802.15.4 module to be powered down during automatic light sleep, which reduces current consumption.

Default value:

- No (disabled) if CONFIG_PM_ENABLE && CONFIG_IEEE802154_ENABLED

CONFIG_IEEE802154_DEBUG

Enable IEEE802154 Debug

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED

Enabling this option allows different kinds of IEEE802154 debug output. All IEEE802154 debug features increase the size of the final binary.

Default value:

- No (disabled) if CONFIG_IEEE802154_ENABLED

IEEE 802.15.4 Debug Parameters

Contains:

- CONFIG_IEEE802154_DEBUG_ASSERT_MONITOR
- CONFIG_IEEE802154_PRINT_PHY_REG
- CONFIG_IEEE802154_TXRX_STATISTIC

- CONFIG_IEEE802154_RECORD_TXRX_FRAME
- CONFIG_IEEE802154_ASSERT
- CONFIG_IEEE802154_RECORD
- CONFIG_IEEE802154_RX_BUFFER_STATISTIC

CONFIG_IEEE802154_DEBUG_ASSERT_MONITOR

Enable IEEE802154 assert monitor

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable monitoring and detection of abnormal or unexpected states during IEEE 802.15.4 operation. When enabled, performs additional runtime checks and assertions to catch potential driver issues early.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RX_BUFFER_STATISTIC

Rx buffer statistic

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable tracking of receive buffer allocation and deallocation. This provides statistics on buffer usage patterns, which can help optimize buffer sizing for your application.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_ASSERT

Enrich the assert information

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable enhanced assert information. When an assertion fails, additional context including state, events, and commands will be printed to help diagnose the issue. Automatically enables IEEE802154_RECORD.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD

Record the information with IEEE802154 state and event

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable recording of IEEE 802.15.4 driver state, events, and commands. Probe code is added throughout the driver to capture runtime information for debugging purposes. This is a prerequisite for other RECORD_* options.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_EVENT

Enable record event information for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD

Record IEEE 802.15.4 driver events in a circular buffer. When an assertion occurs, the recorded event history will be printed to help trace the sequence of events leading to the failure.

Default value:

- No (disabled) if CONFIG_IEEE802154_RECORD && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_EVENT_SIZE

Record event table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD > CONFIG_IEEE802154_RECORD_EVENT

Size of the circular buffer for recording events. Larger values provide more history but consume more memory.

Range:

- from 1 to 50 if CONFIG_IEEE802154_RECORD_EVENT && CONFIG_IEEE802154_DEBUG

Default value:

- 30 if CONFIG_IEEE802154_RECORD_EVENT && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_STATE

Enable record state information for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD

Record IEEE 802.15.4 driver state transitions in a circular buffer. When an assertion occurs, the recorded state history will be printed to show the sequence of state changes.

Default value:

- No (disabled) if CONFIG_IEEE802154_RECORD && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_STATE_SIZE

Record state table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD > CONFIG_IEEE802154_RECORD_STATE

Size of the circular buffer for recording state transitions. Larger values provide more history but consume more memory.

Range:

- from 1 to 50 if CONFIG_IEEE802154_RECORD_STATE && CONFIG_IEEE802154_DEBUG

Default value:

- 10 if CONFIG_IEEE802154_RECORD_STATE && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_CMD

Enable record command information for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD

Record IEEE 802.15.4 driver commands in a circular buffer. When an assertion occurs, the recorded command history will be printed to show what commands were issued before the failure.

Default value:

- No (disabled) if CONFIG_IEEE802154_RECORD && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_CMD_SIZE

Record command table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD > CONFIG_IEEE802154_RECORD_CMD

Size of the circular buffer for recording commands. Larger values provide more history but consume more memory.

Range:

- from 1 to 50 if CONFIG_IEEE802154_RECORD_CMD && CONFIG_IEEE802154_DEBUG

Default value:

- 10 if CONFIG_IEEE802154_RECORD_CMD && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_ABORT

Enable record abort information for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD

Record IEEE 802.15.4 operation aborts in a circular buffer. When an assertion occurs, the recorded abort history will be printed to show what operations were aborted before the failure.

Default value:

- No (disabled) if CONFIG_IEEE802154_RECORD && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_ABORT_SIZE

Record abort table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD > CONFIG_IEEE802154_RECORD_ABORT

Size of the circular buffer for recording aborts. Larger values provide more history but consume more memory.

Range:

- from 1 to 50 if CONFIG_IEEE802154_RECORD_ABORT && CONFIG_IEEE802154_DEBUG

Default value:

- 10 if CONFIG_IEEE802154_RECORD_ABORT && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_TXRX_FRAME

Enable record txrx packets for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Record transmitted and received IEEE 802.15.4 frames in a circular buffer. When an assertion occurs, the recorded frame history will be printed to help diagnose frame-level issues. Note: This can consume significant memory depending on frame size and buffer size.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_RECORD_TXRX_FRAME_SIZE

Record frame table size

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters > CONFIG_IEEE802154_RECORD_TXRX_FRAME

Size of the circular buffer for recording frames. Larger values provide more frame history but consume significantly more memory (each frame can be up to 127 bytes plus metadata).

Range:

- from 1 to 50 if CONFIG_IEEE802154_RECORD_TXRX_FRAME && CONFIG_IEEE802154_DEBUG

Default value:

- 15 if CONFIG_IEEE802154_RECORD_TXRX_FRAME && CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_TXRX_STATISTIC

Enable record tx/rx packets information for debugging

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable collection of transmit and receive packet statistics. This tracks counts, errors, and other metrics for transmitted and received frames. Statistics can be accessed via debug APIs for performance analysis.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

CONFIG_IEEE802154_PRINT_PHY_REG

Enable print IEEE802154 PHY register command

Found in: Component config > IEEE 802.15.4 > CONFIG_IEEE802154_ENABLED > CONFIG_IEEE802154_DEBUG > IEEE 802.15.4 Debug Parameters

Enable the 'phyreg' console command to print IEEE 802.15.4 PHY register information for debugging purposes. This command displays register values and configuration settings of the IEEE 802.15.4 hardware.

Default value:

- No (disabled) if CONFIG_IEEE802154_DEBUG

Log

Contains:

- Format
- Log Level
- CONFIG_LOG_VERSION
- CONFIG_LOG_IN_IRAM
- Settings

CONFIG_LOG_VERSION

Log version

Found in: Component config > Log

Select the log version to be used by the ESP log component.

- **"V1": This version integrates log formatting into the format string provided by the user.**

  Logs are processed and formatted during compile time, leading to a larger binary file. Example: ESP_LOGI("boot", "chip revision: v%d.%d", major, minor); Output: I (56) boot: chip revision: v3.0 Note: Log strings are stored in Flash with added formatting characters. Format string on flash: "[0;32mI (%lu) %s: chip revision: v%d.%d [0m"

- **"V2": This version centralizes log formatting within the esp_log() function.**

  User-supplied format strings are stored without added formatting, reducing binary size. Example: ESP_LOGI("boot", "chip revision: v%d.%d", major, minor); Output: I (56) boot: chip revision: v3.0 Note: This version supports runtime configuration of formatting and is more flexible, logging from constrained environments (ex.: ISR, Startup, Cache disabled). It may consumes a bit more stack and affect performance. Format string on flash: "chip revision: v%d.%d"

Use V1 for minimal stack usage and simpler implementation. Use V2 for smaller binary sizes, more flexible log formatting, and advanced features like disabling colors or timestamps.

Available options:

- V1 (CONFIG_LOG_VERSION_1)

  Select this option to use Log V1. Recommended for projects with strict stack constraints or that prioritize performance over flexibility.

- V2 (CONFIG_LOG_VERSION_2)

  Select this option to use Log V2. Recommended for projects that require smaller binaries, runtime log formatting configuration, or advanced logging features.

Log Level

Contains:

- CONFIG_LOG_DEFAULT_LEVEL
- Level Settings
- CONFIG_LOG_MAXIMUM_LEVEL

CONFIG_LOG_DEFAULT_LEVEL

Default log verbosity

Found in: Component config > Log > Log Level

Specify how much output to see in logs by default. You can set lower verbosity level at runtime using esp_log_level_set() function if LOG_DYNAMIC_LEVEL_CONTROL is enabled.

By default, this setting limits which log statements are compiled into the program. For example, selecting "Warning" would mean that changing log level to "Debug" at runtime will not be possible. To allow increasing log level above the default at runtime, see the next option.

Available options:

- No output (CONFIG_LOG_DEFAULT_LEVEL_NONE)
- Error (CONFIG_LOG_DEFAULT_LEVEL_ERROR)
- Warning (CONFIG_LOG_DEFAULT_LEVEL_WARN)
- Info (CONFIG_LOG_DEFAULT_LEVEL_INFO)
- Debug (CONFIG_LOG_DEFAULT_LEVEL_DEBUG)
- Verbose (CONFIG_LOG_DEFAULT_LEVEL_VERBOSE)

CONFIG_LOG_MAXIMUM_LEVEL

Maximum log verbosity

Found in: Component config > Log > Log Level

This config option sets the highest log verbosity that it's possible to select at runtime by calling esp_log_level_set(). This level may be higher than the default verbosity level which is set when the app starts up.

This can be used enable debugging output only at a critical point, for a particular tag, or to minimize startup time but then enable more logs once the firmware has loaded.

Note that increasing the maximum available log level will increase the firmware binary size.

This option only applies to logging from the app, the bootloader log level is fixed at compile time to the separate "Bootloader log verbosity" setting.

Available options:

- Same as default (CONFIG_LOG_MAXIMUM_EQUALS_DEFAULT)
- Error (CONFIG_LOG_MAXIMUM_LEVEL_ERROR)
- Warning (CONFIG_LOG_MAXIMUM_LEVEL_WARN)
- Info (CONFIG_LOG_MAXIMUM_LEVEL_INFO)
- Debug (CONFIG_LOG_MAXIMUM_LEVEL_DEBUG)
- Verbose (CONFIG_LOG_MAXIMUM_LEVEL_VERBOSE)

Level Settings

Contains:

- CONFIG_LOG_TAG_LEVEL_CACHE_IMPL
- CONFIG_LOG_DYNAMIC_LEVEL_CONTROL
- CONFIG_LOG_MASTER_LEVEL
- CONFIG_LOG_TAG_LEVEL_IMPL_CACHE_SIZE
- CONFIG_LOG_TAG_LEVEL_IMPL

CONFIG_LOG_MASTER_LEVEL

Enable global master log level

Found in: Component config > Log > Log Level > Level Settings

Enables an additional global "master" log level check that occurs before a log tag cache lookup. This is useful if you want to compile in a lot of logs that are selectable at runtime, but avoid the performance hit during periods where you don't want log output.

Examples include remote log forwarding, or disabling logs during a time-critical or CPU-intensive section and re-enabling them later. Results in larger program size depending on number of logs compiled in.

If enabled, defaults to LOG_DEFAULT_LEVEL and can be set using esp_log_set_level_master(). This check takes precedence over ESP_LOG_LEVEL_LOCAL.

Default value:

- No (disabled)

CONFIG_LOG_DYNAMIC_LEVEL_CONTROL

Enable dynamic log level changes at runtime

Found in: Component config > Log > Log Level > Level Settings

Enabling this option allows dynamic changes to the log level at runtime (using esp_log_level_set()), providing the ability to increase or decrease the log level during program execution. If disabled, the log level remains static once set at compile-time and calling esp_log_level_set() will have no effect. If binary size is a critical consideration and dynamic log level changes are not needed, consider disabling this option when LOG_TAG_LEVEL_IMPL_NONE=y to minimize program size.

Default value:

- Yes (enabled)

CONFIG_LOG_TAG_LEVEL_IMPL

Method of tag level checks

Found in: Component config > Log > Log Level > Level Settings

Choose the per-tag log level implementation for the log library. This functionality is used to enable/disable logs for a particular tag at run time. Applicable only for application logs (i.e., not bootloader logs).

Available options:

- None (CONFIG_LOG_TAG_LEVEL_IMPL_NONE)

  This option disables the ability to set the log level per tag. The ability to change the log level at runtime depends on LOG_DYNAMIC_LEVEL_CONTROL. If LOG_DYNAMIC_LEVEL_CONTROL is disabled, then changing the log level at runtime using *esp_log_level_set()* is not possible. This implementation is suitable for highly constrained environments.

- Linked List (CONFIG_LOG_TAG_LEVEL_IMPL_LINKED_LIST)

  Select this option to use the linked list-only implementation (no cache) for log level retrieval. This approach searches the linked list of all tags for the log level, which may be slower for a large number of tags but may have lower memory requirements than the CACHE approach. The linked list approach compares the whole strings of log tags for finding the log level.

- Cache + Linked List (CONFIG_LOG_TAG_LEVEL_IMPL_CACHE_AND_LINKED_LIST)

  Select this option to use a hybrid mode: cache in combination with the linked list for log tag level checks. This hybrid approach offers a balance between speed and memory usage.

  The cache stores recently accessed log tags and their corresponding log levels, providing faster lookups for frequently used tags. The cache approach compares the tag pointers, which is faster than comparing the whole strings.

  For less frequently used tags, the linked list is used to search for the log level, which may be slower for a large number of tags but has lower memory requirements compared to a full cache.

  This hybrid approach aims to improve the efficiency of log level retrieval by combining the benefits of both cache and linked list implementations.

CONFIG_LOG_TAG_LEVEL_CACHE_IMPL

Cache implementation

Found in: Component config > Log > Log Level > Level Settings

The cache stores recently accessed log tags (address of tag) and their corresponding log levels, providing faster lookups for frequently used tags. Cache size can be configured using the LOG_TAG_LEVEL_IMPL_CACHE_SIZE option. The cache approach compares the tag pointers, which is faster than comparing the whole strings.

Available options:

- Array (CONFIG_LOG_TAG_LEVEL_CACHE_ARRAY)

  This option enables the use of a simple array-based cache implementation for storing and retrieving log tag levels. There is no additional code that reorders the cache for fast lookups. Suitable for projects where memory usage optimization is crucial and the simplicity of implementation is preferred.

- Binary Min-Heap (CONFIG_LOG_TAG_LEVEL_CACHE_BINARY_MIN_HEAP)

  This option enables the use of a binary min-heap-based cache implementation for efficient storage and retrieval of log tag levels. It does automatically optimizing cache for fast lookups. Suitable for projects where speed of lookup is critical and memory usage can accommodate the overhead of maintaining a binary min-heap structure.

CONFIG_LOG_TAG_LEVEL_IMPL_CACHE_SIZE

Log Tag Cache Size

Found in: Component config > Log > Log Level > Level Settings

This option sets the size of the cache used for log tag entries. The cache stores recently accessed log tags and their corresponding log levels, which helps improve the efficiency of log level retrieval. The value must be a power of 2 minus 1 (e.g., 1, 3, 7, 15, 31, 63, 127, 255, ...) to ensure proper cache behavior. For LOG_TAG_LEVEL_CACHE_ARRAY option the value can be any, without restrictions.

Note: A larger cache size can improve lookup performance for frequently used log tags but may consume more memory. Conversely, a smaller cache size reduces memory usage but may lead to more frequent cache evictions for less frequently used log tags.

Default value:

- 31

Format

Contains:

- CONFIG_LOG_COLORS_SUPPORT
- CONFIG_LOG_TIMESTAMP_SUPPORT
- CONFIG_LOG_COLORS
- CONFIG_LOG_TIMESTAMP_SOURCE

CONFIG_LOG_COLORS

Color

Found in: Component config > Log > Format

Enable ANSI terminal color codes. Logs (info, errors, warnings) will contain color codes. In order to view these, your terminal program must support ANSI color codes.

This is disabled by default, as colors are added in IDF Monitor. If you want to use new lines in the messages or you are using a different terminal program, you may want to enable this option.

Default value:

- No (disabled)

CONFIG_LOG_COLORS_SUPPORT

Allow enabling color output at run time

Found in: Component config > Log > Format

Enables support for color codes in the esp_log() function. If CONFIG_LOG_COLORS is enabled, this option is always active. If CONFIG_LOG_COLORS is disabled, this option allows you to still handle color codes in specific files by defining ESP_LOG_COLOR_DISABLED as 0 before including esp_log.h.

Note that enabling this option may slightly increase IRAM usage due to additional color handling functionality. It provides flexibility to manage color output even when CONFIG_LOG_COLORS is turned off.

Default value:

- No (disabled) if CONFIG_LOG_VERSION_2

CONFIG_LOG_TIMESTAMP_SOURCE

Timestamp

Found in: Component config > Log > Format

Choose what sort of timestamp is displayed in the log output:

- "None" - The log will only contain the actual log messages themselves without any time-related information. Avoiding timestamps can help conserve processing power and memory. It might useful when you perform log analysis or debugging, sometimes it's more straightforward to work with logs that lack timestamps, especially if the time of occurrence is not critical for understanding the issues.
- "Milliseconds since boot" is calculated from the RTOS tick count multiplied by the tick period. This time will reset after a software reboot. e.g. (90000)
- "System time (HH:MM:SS.sss)" is taken from POSIX time functions which use the chip's RTC and high resolution timers to maintain an accurate time. The system time is initialized to 0 on startup, it can be set with an SNTP sync, or with POSIX time functions. This time will not reset after a software reboot. e.g. (00:01:30.000)
- "System time (YY-MM-DD HH:MM:SS.sss)" it is the same as the above, but also prints the date as well.
- "Unix time in milliseconds" is the same as the two above, but in Unix time format and in milliseconds. e.g. (1718795571035).
- NOTE: Currently this will not get used in logging from binary blobs (i.e WiFi & Bluetooth libraries), these will always print milliseconds since boot.

Available options:

- None (CONFIG_LOG_TIMESTAMP_SOURCE_NONE)
- Milliseconds Since Boot (CONFIG_LOG_TIMESTAMP_SOURCE_RTOS)
- System Time (HH:MM:SS.sss) (CONFIG_LOG_TIMESTAMP_SOURCE_SYSTEM)
- System Time (YY-MM-DD HH:MM:SS.sss) (CONFIG_LOG_TIMESTAMP_SOURCE_SYSTEM_FULL)
- Unix time in milliseconds (CONFIG_LOG_TIMESTAMP_SOURCE_UNIX)

CONFIG_LOG_TIMESTAMP_SUPPORT

Allow enabling timestamp output at run time

Found in: Component config > Log > Format

Enables support for timestamp in the esp_log() function. If CONFIG_LOG_TIMESTAMP_SOURCE_NONE, this option allows you to still handle timestamp in specific files by defining ESP_LOG_TIMESTAMP_DISABLED as 0 before including esp_log.h.

Note that enabling this option may slightly increase IRAM usage due to additional timestamp handling functionality. It provides flexibility to manage timestamp output even when CONFIG_LOG_TIMESTAMP_SOURCE_NONE.

Default value:

- Yes (enabled) if CONFIG_LOG_VERSION_2

Settings

Contains:

- CONFIG_LOG_MODE

CONFIG_LOG_MODE

Log Mode

Found in: Component config > Log > Settings

Available options:

- Text Log Mode (CONFIG_LOG_MODE_TEXT)

  Enables text-based logging, where log messages are stored in a human-readable format. This mode is useful for development and debugging, as it allows logs to be easily read and interpreted without additional processing.

- Binary Log Mode (CONFIG_LOG_MODE_BINARY)

  Enables binary logging with host-side format string expansion. In this mode, the format argument of ESP_LOGx, ESP_EARLY_LOG, and ESP_DRAM_LOG macros is stored in a NOLOAD section, not included in the final binary file. This reduces flash usage by approximately 10% - 35%. The esp_log() function uses the binary log handler to output messages. Instead of sending the full log string, the chip transmits only the addresses of these strings (if present in the ELF file). If the format string cannot be found in the ELF file, the chip sends the entire string. The host-side monitor tool, which has access to the ELF file, reconstructs the log message using the format string. This reduces firmware size by eliminating format strings from flash memory and removing the usage of printf-like functions, potentially freeing up a few kilobytes of space. To further reduce firmware size, wrap string data with ESP_LOG_ATTR_STR.

CONFIG_LOG_IN_IRAM

Place logging functions in IRAM

Found in: Component config > Log

Default value:

- Yes (enabled)

LWIP

Contains:

- CONFIG_LWIP_CHECK_THREAD_SAFETY
- Checksums
- CONFIG_LWIP_DHCP_CHECKS_OFFERED_ADDRESS
- CONFIG_LWIP_DHCP_COARSE_TIMER_SECS
- DHCP server
- CONFIG_LWIP_DHCP_OPTIONS_LEN

- CONFIG_LWIP_DHCP_DISABLE_CLIENT_ID
- CONFIG_LWIP_DHCP_DISABLE_VENDOR_CLASS_ID
- CONFIG_LWIP_DHCP_RESTORE_LAST_IP
- DNS
- CONFIG_LWIP_L2_TO_L3_COPY
- CONFIG_LWIP_IPV6_DHCP6
- CONFIG_LWIP_IP4_FRAG
- CONFIG_LWIP_IP6_FRAG
- CONFIG_LWIP_IP_FORWARD
- CONFIG_LWIP_NETBUF_RECVINFO
- CONFIG_LWIP_IPV4
- CONFIG_LWIP_AUTOIP
- CONFIG_LWIP_IPV6
- CONFIG_LWIP_IPV6_ND6_ROUTE_INFO_OPTION_SUPPORT
- CONFIG_LWIP_NETIF_LINK_CALLBACK
- CONFIG_LWIP_ESP_LWIP_ASSERT
- CONFIG_LWIP_DEBUG
- CONFIG_LWIP_IRAM_OPTIMIZATION
- CONFIG_LWIP_EXTRA_IRAM_OPTIMIZATION
- CONFIG_LWIP_ENABLE
- CONFIG_LWIP_STATS
- CONFIG_LWIP_TIMERS_ONDEMAND
- CONFIG_LWIP_DNS_SUPPORT_MDNS_QUERIES
- CONFIG_LWIP_PPP_SUPPORT
- CONFIG_LWIP_IP4_REASSEMBLY
- CONFIG_LWIP_IP6_REASSEMBLY
- CONFIG_LWIP_SLIP_SUPPORT
- CONFIG_LWIP_SO_LINGER
- CONFIG_LWIP_SO_RCVBUF
- CONFIG_LWIP_SO_REUSE
- CONFIG_LWIP_NETIF_STATUS_CALLBACK
- CONFIG_LWIP_TCPIP_CORE_LOCKING
- Hooks
- ICMP
- CONFIG_LWIP_LOCAL_HOSTNAME
- CONFIG_LWIP_ND6
- LWIP RAW API
- CONFIG_LWIP_TCPIP_TASK_PRIO
- CONFIG_LWIP_IPV6_ND6_NUM_ROUTERS
- CONFIG_LWIP_IPV6_ND6_NUM_DESTINATIONS
- CONFIG_LWIP_IPV6_ND6_NUM_NEIGHBORS
- CONFIG_LWIP_IPV6_ND6_NUM_PREFIXES
- CONFIG_LWIP_IPV6_MEMP_NUM_ND6_QUEUE
- CONFIG_LWIP_MAX_SOCKETS
- CONFIG_LWIP_BRIDGEIF_MAX_PORTS

- CONFIG_LWIP_NUM_NETIF_CLIENT_DATA
- CONFIG_LWIP_IPV6_DUP_DETECT_ATTEMPTS
- CONFIG_LWIP_ESP_GRATUITOUS_ARP
- CONFIG_LWIP_ESP_MLDV6_REPORT
- SNTP
- CONFIG_LWIP_USE_ONLY_LWIP_SELECT
- CONFIG_LWIP_NETIF_LOOPBACK
- TCP
- CONFIG_LWIP_TCPIP_TASK_AFFINITY
- CONFIG_LWIP_TCPIP_TASK_STACK_SIZE
- CONFIG_LWIP_TCPIP_RECVMBOX_SIZE
- CONFIG_LWIP_IP_REASS_MAX_PBUFS
- CONFIG_LWIP_IP_DEFAULT_TTL
- UDP
- CONFIG_LWIP_IPV6_RDNSS_MAX_DNS_SERVERS

CONFIG_LWIP_ENABLE

Enable LwIP stack

Found in: Component config > LWIP

Builds normally if selected. Excludes LwIP from build if unselected, even if it is a dependency of a component or application. Some applications can switch their IP stacks, e.g., when switching between chip and Linux targets (LwIP stack vs. Linux IP stack). Since the LwIP dependency cannot easily be excluded based on a Kconfig option, it has to be a dependency in all cases. This switch allows the LwIP stack to be built selectively, even if it is a dependency.

Default value:

- Yes (enabled)

CONFIG_LWIP_LOCAL_HOSTNAME

Local netif hostname

Found in: Component config > LWIP

The default name this device will report to other devices on the network. Could be updated at runtime with esp_netif_set_hostname()

Default value:

- "espressif"

CONFIG_LWIP_TCPIP_TASK_PRIO

LWIP TCP/IP Task Priority

Found in: Component config > LWIP

LWIP tcpip task priority. In case of high throughput, this parameter could be changed up to (configMAX_PRIORITIES-1).

Range:

- from 1 to 24

Default value:

- 18

CONFIG_LWIP_TCPIP_CORE_LOCKING

Enable tcpip core locking

Found in: Component config > LWIP

If Enable tcpip core locking,Creates a global mutex that is held during TCPIP thread operations.Can be locked by client code to perform lwIP operations without changing into TCPIP thread using callbacks. See LOCK_TCPIP_CORE() and UNLOCK_TCPIP_CORE().

If disable tcpip core locking,TCP IP will perform tasks through context switching

Default value:

- No (disabled)

CONFIG_LWIP_TCPIP_CORE_LOCKING_INPUT

Enable tcpip core locking input

Found in: Component config > LWIP > CONFIG_LWIP_TCPIP_CORE_LOCKING

when LWIP_TCPIP_CORE_LOCKING is enabled, this lets tcpip_input() grab the mutex for input packets as well, instead of allocating a message and passing it to tcpip_thread.

Default value:

- No (disabled) if CONFIG_LWIP_TCPIP_CORE_LOCKING

CONFIG_LWIP_CHECK_THREAD_SAFETY

Checks that lwip API runs in expected context

Found in: Component config > LWIP

Enable to check that the project does not violate lwip thread safety. If enabled, all lwip functions that require thread awareness run an assertion to verify that the TCP/IP core functionality is either locked or accessed from the correct thread.

Default value:

- No (disabled)

CONFIG_LWIP_DNS_SUPPORT_MDNS_QUERIES

Enable mDNS queries in resolving host name

Found in: Component config > LWIP

If this feature is enabled, standard API such as gethostbyname support .local addresses by sending one shot multicast mDNS query

Default value:

- Yes (enabled)

CONFIG_LWIP_L2_TO_L3_COPY

Enable copy between Layer2 and Layer3 packets

Found in: Component config > LWIP

If this feature is enabled, all traffic from layer2(WIFI Driver) will be copied to a new buffer before sending it to layer3(LWIP stack), freeing the layer2 buffer. Please be notified that the total layer2 receiving buffer is fixed and ESP32 currently supports 25 layer2 receiving buffer, when layer2 buffer runs out of memory, then the incoming packets will be dropped in hardware. The layer3 buffer is allocated from the heap, so the total layer3 receiving buffer depends on the available heap size, when heap runs out of memory, no copy will be sent to layer3 and packet will be dropped in layer2. Please make sure you fully understand the impact of this feature before enabling it.

Default value:

- No (disabled)

CONFIG_LWIP_IRAM_OPTIMIZATION

Enable LWIP IRAM optimization

Found in: Component config > LWIP

If this feature is enabled, some functions relating to RX/TX in LWIP will be put into IRAM, it can improve UDP/TCP throughput by >10% for single core mode, it doesn't help too much for dual core mode. On the other hand, it needs about 10KB IRAM for these optimizations.

If this feature is disabled, all lwip functions will be put into FLASH.

Default value:

- No (disabled)

CONFIG_LWIP_EXTRA_IRAM_OPTIMIZATION

Enable LWIP IRAM optimization for TCP part

Found in: Component config > LWIP

If this feature is enabled, some tcp part functions relating to RX/TX in LWIP will be put into IRAM, it can improve TCP throughput. On the other hand, it needs about 17KB IRAM for these optimizations.

Default value:

- No (disabled)

CONFIG_LWIP_TIMERS_ONDEMAND

Enable LWIP Timers on demand

Found in: Component config > LWIP

If this feature is enabled, IGMP and MLD6 timers will be activated only when joining groups or receiving QUERY packets.

This feature will reduce the power consumption for applications which do not use IGMP and MLD6.

Default value:

- Yes (enabled)

CONFIG_LWIP_ND6

LWIP NDP6 Enable/Disable

Found in: Component config > LWIP

This option is used to disable the Network Discovery Protocol (NDP) if it is not required. Please use this option with caution, as the NDP is essential for IPv6 functionality within a local network.

Default value:

- Yes (enabled)

CONFIG_LWIP_FORCE_ROUTER_FORWARDING

LWIP Force Router Forwarding Enable/Disable

Found in: Component config > LWIP > CONFIG_LWIP_ND6

This option is used to set the the router flag for the NA packets. When enabled, the router flag in NA packet will always set to 1, otherwise, never set router flag for NA packets.

Default value:

- No (disabled)

CONFIG_LWIP_MAX_SOCKETS

Max number of open sockets

Found in: Component config > LWIP

The practical maximum limit is determined by available heap memory at runtime.

Sockets take up a certain amount of memory, and allowing fewer sockets to be open at the same time conserves memory. Specify the maximum amount of sockets here. The valid value is from 1 to 253. If using value above 61, update CMakeLists defining FD_SETSIZE to the number of sockets used plus the expected open files (minimum of +3 for stdout, stderr and stdin).

Range:

- from 1 to 253

Default value:

- 10

CONFIG_LWIP_USE_ONLY_LWIP_SELECT

Support LWIP socket select() only (DEPRECATED)

Found in: Component config > LWIP

This option is deprecated. Do not use this option, use VFS_SUPPORT_SELECT instead.

Default value:

- No (disabled)

CONFIG_LWIP_SO_LINGER

Enable SO_LINGER processing

Found in: Component config > LWIP

Enabling this option allows SO_LINGER processing. l_onoff = 1,l_linger can set the timeout.

If l_linger=0, When a connection is closed, TCP will terminate the connection. This means that TCP will discard any data packets stored in the socket send buffer and send an RST to the peer.

If l_linger!=0,Then closesocket() calls to block the process until the remaining data packets has been sent or timed out.

Default value:

- No (disabled)

CONFIG_LWIP_SO_REUSE

Enable SO_REUSEADDR option

Found in: Component config > LWIP

Enabling this option allows binding to a port which remains in TIME_WAIT.

Default value:

- Yes (enabled)

CONFIG_LWIP_SO_REUSE_RXTOALL

SO_REUSEADDR copies broadcast/multicast to all matches

Found in: Component config > LWIP > CONFIG_LWIP_SO_REUSE

Enabling this option means that any incoming broadcast or multicast packet will be copied to all of the local sockets that it matches (may be more than one if SO_REUSEADDR is set on the socket.)

This increases memory overhead as the packets need to be copied, however they are only copied per matching socket. You can safely disable it if you don't plan to receive broadcast or multicast traffic on more than one socket at a time.

Default value:

- Yes (enabled)

CONFIG_LWIP_SO_RCVBUF

Enable SO_RCVBUF option

Found in: Component config > LWIP

Enabling this option allows checking for available data on a netconn.

Default value:

- No (disabled)

CONFIG_LWIP_NETBUF_RECVINFO

Enable IP_PKTINFO option

Found in: Component config > LWIP

Enabling this option allows checking for the destination address of a received IPv4 Packet.

Default value:

- No (disabled)

CONFIG_LWIP_IP_DEFAULT_TTL

The value for Time-To-Live used by transport layers

Found in: Component config > LWIP

Set value for Time-To-Live used by transport layers.

Range:

- from 1 to 255

Default value:

- 64

CONFIG_LWIP_IP4_FRAG

Enable fragment outgoing IP4 packets

Found in: Component config > LWIP

Enabling this option allows fragmenting outgoing IP4 packets if their size exceeds MTU.

Default value:

- Yes (enabled)

CONFIG_LWIP_IP6_FRAG

Enable fragment outgoing IP6 packets

Found in: Component config > LWIP

Enabling this option allows fragmenting outgoing IP6 packets if their size exceeds MTU.

Default value:

- Yes (enabled)

CONFIG_LWIP_IP4_REASSEMBLY

Enable reassembly incoming fragmented IP4 packets

Found in: Component config > LWIP

Enabling this option allows reassemblying incoming fragmented IP4 packets.

Default value:

- No (disabled)

CONFIG_LWIP_IP6_REASSEMBLY

Enable reassembly incoming fragmented IP6 packets

Found in: Component config > LWIP

Enabling this option allows reassemblying incoming fragmented IP6 packets.

Default value:

- No (disabled)

CONFIG_LWIP_IP_REASS_MAX_PBUFS

The maximum amount of pbufs waiting to be reassembled

Found in: Component config > LWIP

Set the maximum amount of pbufs waiting to be reassembled.

Range:

- from 10 to 100

Default value:

- 10

CONFIG_LWIP_IPV6_DUP_DETECT_ATTEMPTS

Number of duplicate address detection attempts

Found in: Component config > LWIP

Set the number of duplicate address detection attempts.

Range:

- from 0 to 7

Default value:

- 1

CONFIG_LWIP_IP_FORWARD

Enable IP forwarding

Found in: Component config > LWIP

Enabling this option allows packets forwarding across multiple interfaces.

Default value:

- No (disabled)

CONFIG_LWIP_IPV4_NAPT

Enable NAT

Found in: Component config > LWIP > CONFIG_LWIP_IP_FORWARD

Enabling this option allows Network Address and Port Translation.

Default value:

- No (disabled) if CONFIG_LWIP_IP_FORWARD

CONFIG_LWIP_IPV4_NAPT_PORTMAP

Enable NAT Port Mapping

Found in: Component config > LWIP > CONFIG_LWIP_IP_FORWARD > CONFIG_LWIP_IPV4_NAPT

Enabling this option allows Port Forwarding or Port mapping.

Default value:

- Yes (enabled) if CONFIG_LWIP_IPV4_NAPT

CONFIG_LWIP_STATS

Enable LWIP statistics

Found in: Component config > LWIP

Enabling this option allows LWIP statistics

Default value:

- No (disabled)

CONFIG_LWIP_ESP_GRATUITOUS_ARP

Send gratuitous ARP periodically

Found in: Component config > LWIP

Enable this option allows to send gratuitous ARP periodically.

This option solve the compatibility issues.If the ARP table of the AP is old, and the AP doesn't send ARP request to update it's ARP table, this will lead to the STA sending IP packet fail. Thus we send gratuitous ARP periodically to let AP update it's ARP table.

Default value:

- Yes (enabled)

CONFIG_LWIP_GARP_TMR_INTERVAL

GARP timer interval(seconds)

Found in: Component config > LWIP > CONFIG_LWIP_ESP_GRATUITOUS_ARP

Set the timer interval for gratuitous ARP. The default value is 60s

Default value:

- 60

CONFIG_LWIP_ESP_MLDV6_REPORT

Send mldv6 report periodically

Found in: Component config > LWIP

Enable this option allows to send mldv6 report periodically.

This option solve the issue that failed to receive multicast data. Some routers fail to forward multicast packets. To solve this problem, send multicast mdlv6 report to routers regularly.

Default value:

- Yes (enabled)

CONFIG_LWIP_MLDV6_TMR_INTERVAL

mldv6 report timer interval(seconds)

Found in: Component config > LWIP > CONFIG_LWIP_ESP_MLDV6_REPORT

Set the timer interval for mldv6 report. The default value is 30s

Default value:

- 40

CONFIG_LWIP_TCPIP_RECVMBOX_SIZE

TCPIP task receive mail box size

Found in: Component config > LWIP

Set TCPIP task receive mail box size. Generally bigger value means higher throughput but more memory. The value should be bigger than UDP/TCP mail box size.

Range:

- from 6 to 1024 if CONFIG_LWIP_WND_SCALE

Default value:

- 32

CONFIG_LWIP_DHCP_CHECKS_OFFERED_ADDRESS

Choose how DHCP validates offered IP

Found in: Component config > LWIP

Choose the preferred way of DHCP client to check if the offered address is available: * Using Address Conflict Detection (ACD) module assures that the offered IP address is properly probed and announced before binding in DHCP. This conforms to RFC5227, but takes several seconds. * Using ARP check, we only send two ARP requests to check for replies. This process lasts 1 - 2 seconds. * No conflict detection: We directly bind the offered address.

Available options:

- DHCP provides simple ARP check (CONFIG_LWIP_DHCP_DOES_ARP_CHECK)
- DHCP provides Address Conflict Detection (ACD) (CONFIG_LWIP_DHCP_DOES_ACD_CHECK)
- DHCP does not detect conflict on the offered IP (CONFIG_LWIP_DHCP_DOES_NOT_CHECK_OFFERED_IP)

CONFIG_LWIP_DHCP_DISABLE_CLIENT_ID

DHCP: Disable Use of HW address as client identification

Found in: Component config > LWIP

This option could be used to disable DHCP client identification with its MAC address. (Client id is used by DHCP servers to uniquely identify clients and are included in the DHCP packets as an option 61) Set this option to "y" in order to exclude option 61 from DHCP packets.

Default value:

- No (disabled)

CONFIG_LWIP_DHCP_DISABLE_VENDOR_CLASS_ID

DHCP: Disable Use of vendor class identification

Found in: Component config > LWIP

This option could be used to disable DHCP client vendor class identification. Set this option to "y" in order to exclude option 60 from DHCP packets.

Default value:

- Yes (enabled)

CONFIG_LWIP_DHCP_RESTORE_LAST_IP

DHCP: Restore last IP obtained from DHCP server

Found in: Component config > LWIP

When this option is enabled, DHCP client tries to re-obtain last valid IP address obtained from DHCP server. Last valid DHCP configuration is stored in nvs and restored after reset/power-up. If IP is still available, there is no need for sending discovery message to DHCP server and save some time.

Default value:

- No (disabled)

CONFIG_LWIP_DHCP_OPTIONS_LEN

DHCP total option length

Found in: Component config > LWIP

Set total length of outgoing DHCP option msg. Generally bigger value means it can carry more options and values. If your code meets LWIP_ASSERT due to option value is too long. Please increase the LWIP_DHCP_OPTIONS_LEN value.

Range:

- from 69 to 255

Default value:

- 69

CONFIG_LWIP_NUM_NETIF_CLIENT_DATA

Number of clients store data in netif

Found in: Component config > LWIP

Number of clients that may store data in client_data member array of struct netif.

Range:

- from 0 to 256

Default value:

- 0

CONFIG_LWIP_DHCP_COARSE_TIMER_SECS

DHCP coarse timer interval(s)

Found in: Component config > LWIP

Set DHCP coarse interval in seconds. A higher value will be less precise but cost less power consumption.

Range:

- from 1 to 10

Default value:

- 1

DHCP server

Contains:

- CONFIG_LWIP_DHCPS
- CONFIG_LWIP_DHCPS_STATIC_ENTRIES
- CONFIG_LWIP_DHCPS_MAX_HOSTNAME_LEN

CONFIG_LWIP_DHCPS

DHCPS: Enable IPv4 Dynamic Host Configuration Protocol Server (DHCPS)

Found in: Component config > LWIP > DHCP server

Enabling this option allows the device to run the DHCP server (to dynamically assign IPv4 addresses to clients).

Default value:

- Yes (enabled)

CONFIG_LWIP_DHCPS_REPORT_CLIENT_HOSTNAME

DHCPS: Report client hostname in assigned-IP event

Found in: Component config > LWIP > DHCP server > CONFIG_LWIP_DHCPS

When enabled, the DHCP server parses client hostname (DHCP option 12) and makes it available via IP_EVENT_ASSIGNED_IP_TO_CLIENT. This adds a small amount of RAM usage per lease to store the hostname.

Default value:

- Yes (enabled)

CONFIG_LWIP_DHCPS_LEASE_UNIT

Multiplier for lease time, in seconds

Found in: Component config > LWIP > DHCP server > CONFIG_LWIP_DHCPS

The DHCP server is calculating lease time multiplying the sent and received times by this number of seconds per unit. The default is 60, that equals one minute.

Range:

- from 1 to 3600

Default value:

- 60

CONFIG_LWIP_DHCPS_MAX_STATION_NUM

Maximum number of stations

Found in: Component config > LWIP > DHCP server > CONFIG_LWIP_DHCPS

The maximum number of DHCP clients that are connected to the server. After this number is exceeded, DHCP server removes of the oldest device from it's address pool, without notification.

Range:

- from 1 to 64

Default value:

- 8

CONFIG_LWIP_DHCPS_MAX_HOSTNAME_LEN

Maximum client hostname length stored by DHCPS

Found in: Component config > LWIP > DHCP server

Maximum number of bytes stored per-client for DHCP option 12 (hostname), including the terminating null when used in esp-netif events. Longer hostnames will be truncated.

Range:

- from 1 to 255

Default value:

- 64

CONFIG_LWIP_DHCPS_STATIC_ENTRIES

Enable ARP static entries

Found in: Component config > LWIP > DHCP server

Enabling this option allows DHCP server to support temporary static ARP entries for DHCP Client. This will help the DHCP server to send the DHCP OFFER and DHCP ACK using IP unicast.

Default value:

- Yes (enabled)

CONFIG_LWIP_AUTOIP

Enable IPV4 Link-Local Addressing (AUTOIP)

Found in: Component config > LWIP

Enabling this option allows the device to self-assign an address in the 169.256/16 range if none is assigned statically or via DHCP.

See RFC 3927.

Default value:

- No (disabled)

Contains:

- CONFIG_LWIP_AUTOIP_TRIES
- CONFIG_LWIP_AUTOIP_MAX_CONFLICTS
- CONFIG_LWIP_AUTOIP_RATE_LIMIT_INTERVAL

CONFIG_LWIP_AUTOIP_TRIES

DHCP Probes before self-assigning IPv4 LL address

Found in: Component config > LWIP > CONFIG_LWIP_AUTOIP

DHCP client will send this many probes before self-assigning a link local address.

From LWIP help: "This can be set as low as 1 to get an AutoIP address very quickly, but you should be prepared to handle a changing IP address when DHCP overrides AutoIP." (In the case of ESP-IDF, this means multiple SYSTEM_EVENT_STA_GOT_IP events.)

Range:

- from 1 to 100 if CONFIG_LWIP_AUTOIP

Default value:

- 2 if CONFIG_LWIP_AUTOIP

CONFIG_LWIP_AUTOIP_MAX_CONFLICTS

Max IP conflicts before rate limiting

Found in: Component config > LWIP > CONFIG_LWIP_AUTOIP

If the AUTOIP functionality detects this many IP conflicts while self-assigning an address, it will go into a rate limited mode.

Range:

- from 1 to 100 if CONFIG_LWIP_AUTOIP

Default value:

- 9 if CONFIG_LWIP_AUTOIP

CONFIG_LWIP_AUTOIP_RATE_LIMIT_INTERVAL

Rate limited interval (seconds)

Found in: Component config > LWIP > CONFIG_LWIP_AUTOIP

If rate limiting self-assignment requests, wait this long between each request.

Range:

- from 5 to 120 if CONFIG_LWIP_AUTOIP

Default value:

- 20 if CONFIG_LWIP_AUTOIP

CONFIG_LWIP_IPV4

Enable IPv4

Found in: Component config > LWIP

Enable IPv4 stack. If you want to use IPv6 only TCP/IP stack, disable this.

Default value:

- Yes (enabled)

CONFIG_LWIP_IPV6

Enable IPv6

Found in: Component config > LWIP

Enable IPv6 function. If not use IPv6 function, set this option to n. If disabling LWIP_IPV6 then some other components (asio) will no longer be available.

Default value:

- Yes (enabled)

CONFIG_LWIP_IPV6_AUTOCONFIG

Enable IPV6 stateless address autoconfiguration (SLAAC)

Found in: Component config > LWIP > CONFIG_LWIP_IPV6

Enabling this option allows the devices to IPV6 stateless address autoconfiguration (SLAAC).

See RFC 4862.

Default value:

- No (disabled)

CONFIG_LWIP_IPV6_NUM_ADDRESSES

Number of IPv6 addresses on each network interface

Found in: Component config > LWIP > CONFIG_LWIP_IPV6

The maximum number of IPv6 addresses on each interface. Any additional addresses will be discarded.

Default value:

- 3

CONFIG_LWIP_IPV6_FORWARD

Enable IPv6 forwarding between interfaces

Found in: Component config > LWIP > CONFIG_LWIP_IPV6

Forwarding IPv6 packets between interfaces is only required when acting as a router.

Default value:

- No (disabled)

CONFIG_LWIP_IPV6_RDNSS_MAX_DNS_SERVERS

Use IPv6 Router Advertisement Recursive DNS Server Option

Found in: Component config > LWIP

Use IPv6 Router Advertisement Recursive DNS Server Option (as per RFC 6106) to copy a defined maximum number of DNS servers to the DNS module. Set this option to a number of desired DNS servers advertised in the RA protocol. This feature is disabled when set to 0.

Default value:

- 0 if CONFIG_LWIP_IPV6_AUTOCONFIG

CONFIG_LWIP_IPV6_DHCP6

Enable DHCPv6 stateless address autoconfiguration

Found in: Component config > LWIP

Enable DHCPv6 for IPv6 stateless address autoconfiguration. Note that the dhcpv6 client has to be started using dhcp6_enable_stateless(netif); Note that the stateful address autoconfiguration is not supported.

Default value:

- No (disabled) if CONFIG_LWIP_IPV6_AUTOCONFIG

CONFIG_LWIP_NETIF_STATUS_CALLBACK

Enable status callback for network interfaces

Found in: Component config > LWIP

Enable callbacks when the network interface is up/down and addresses are changed.

Default value:

- No (disabled)

CONFIG_LWIP_NETIF_LINK_CALLBACK

Enable link callback for network interfaces

Found in: Component config > LWIP

Enable callbacks when the physical network link is up/down.

Default value:

- No (disabled)

CONFIG_LWIP_NETIF_LOOPBACK

Support per-interface loopback

Found in: Component config > LWIP

Enabling this option means that if a packet is sent with a destination address equal to the interface's own IP address, it will "loop back" and be received by this interface. Disabling this option disables support of loopback interface in lwIP

Default value:

- Yes (enabled)

Contains:

- CONFIG_LWIP_LOOPBACK_MAX_PBUFS

CONFIG_LWIP_LOOPBACK_MAX_PBUFS

Max queued loopback packets per interface

Found in: Component config > LWIP > CONFIG_LWIP_NETIF_LOOPBACK

Configure the maximum number of packets which can be queued for loopback on a given interface. Reducing this number may cause packets to be dropped, but will avoid filling memory with queued packet data.

Range:

- from 0 to 16

Default value:

- 8

TCP

Contains:

- CONFIG_LWIP_TCP_WND_DEFAULT
- CONFIG_LWIP_TCP_SND_BUF_DEFAULT
- CONFIG_LWIP_TCP_ACCEPTMBOX_SIZE
- CONFIG_LWIP_TCP_RECVMBOX_SIZE
- CONFIG_LWIP_TCP_RTO_TIME
- CONFIG_LWIP_MAX_ACTIVE_TCP
- CONFIG_LWIP_TCP_FIN_WAIT_TIMEOUT
- CONFIG_LWIP_MAX_LISTENING_TCP
- CONFIG_LWIP_TCP_MAXRTX
- CONFIG_LWIP_TCP_SYNMAXRTX
- CONFIG_LWIP_TCP_MSL
- CONFIG_LWIP_TCP_MSS
- CONFIG_LWIP_TCP_OVERSIZE
- CONFIG_LWIP_TCP_QUEUE_OOSEQ
- CONFIG_LWIP_WND_SCALE
- CONFIG_LWIP_TCP_HIGH_SPEED_RETRANSMISSION
- CONFIG_LWIP_TCP_TMR_INTERVAL

CONFIG_LWIP_MAX_ACTIVE_TCP

Maximum active TCP Connections

Found in: Component config > LWIP > TCP

The maximum number of simultaneously active TCP connections. The practical maximum limit is determined by available heap memory at runtime.

Changing this value by itself does not substantially change the memory usage of LWIP, except for preventing new TCP connections after the limit is reached.

Range:

- from 1 to 1024

Default value:

- 16

CONFIG_LWIP_MAX_LISTENING_TCP

Maximum listening TCP Connections

Found in: Component config > LWIP > TCP

The maximum number of simultaneously listening TCP connections. The practical maximum limit is determined by available heap memory at runtime.

Changing this value by itself does not substantially change the memory usage of LWIP, except for preventing new listening TCP connections after the limit is reached.

Range:

- from 1 to 1024

Default value:

- 16

CONFIG_LWIP_TCP_HIGH_SPEED_RETRANSMISSION

TCP high speed retransmissions

Found in: Component config > LWIP > TCP

Speed up the TCP retransmission interval. If disabled, it is recommended to change the number of SYN retransmissions to 6, and TCP initial rto time to 3000.

Default value:

- Yes (enabled)

CONFIG_LWIP_TCP_MAXRTX

Maximum number of retransmissions of data segments

Found in: Component config > LWIP > TCP

Set maximum number of retransmissions of data segments.

Range:

- from 3 to 12

Default value:

- 12

CONFIG_LWIP_TCP_SYNMAXRTX

Maximum number of retransmissions of SYN segments

Found in: Component config > LWIP > TCP

Set maximum number of retransmissions of SYN segments.

Range:

- from 3 to 12

Default value:

- 12

CONFIG_LWIP_TCP_MSS

Maximum Segment Size (MSS)

Found in: Component config > LWIP > TCP

Set maximum segment size for TCP transmission.

Can be set lower to save RAM, the default value 1460(ipv4)/1440(ipv6) will give best throughput. IPv4 TCP_MSS Range: 576 <= TCP_MSS <= 1460 IPv6 TCP_MSS Range: 1220<= TCP_MSS <= 1440

Range:

- from 536 to 1460

Default value:

- 1440

CONFIG_LWIP_TCP_TMR_INTERVAL

TCP timer interval(ms)

Found in: Component config > LWIP > TCP

Set TCP timer interval in milliseconds.

Can be used to speed connections on bad networks. A lower value will redeliver unacked packets faster.

Default value:

- 250

CONFIG_LWIP_TCP_MSL

Maximum segment lifetime (MSL)

Found in: Component config > LWIP > TCP

Set maximum segment lifetime in milliseconds.

Default value:

- 60000

CONFIG_LWIP_TCP_FIN_WAIT_TIMEOUT

Maximum FIN segment lifetime

Found in: Component config > LWIP > TCP

Set maximum segment lifetime in milliseconds.

Default value:

- 20000

CONFIG_LWIP_TCP_SND_BUF_DEFAULT

Default send buffer size

Found in: Component config > LWIP > TCP

Set default send buffer size for new TCP sockets.

Per-socket send buffer size can be changed at runtime with lwip_setsockopt(s, TCP_SNDBUF, ...).

This value must be at least 2x the MSS size, and the default is 4x the default MSS size.

Setting a smaller default SNDBUF size can save some RAM, but will decrease performance.

Range:

- from 2440 to 1024000 if CONFIG_LWIP_WND_SCALE

Default value:

- 5760

CONFIG_LWIP_TCP_WND_DEFAULT

Default receive window size

Found in: Component config > LWIP > TCP

Set default TCP receive window size for new TCP sockets.

Per-socket receive window size can be changed at runtime with lwip_setsockopt(s, TCP_WINDOW, ...).

Setting a smaller default receive window size can save some RAM, but will significantly decrease performance.

Range:

- from 2440 to 1024000 if CONFIG_LWIP_WND_SCALE

Default value:

- 5760

CONFIG_LWIP_TCP_RECVMBOX_SIZE

Default TCP receive mail box size

Found in: Component config > LWIP > TCP

Set TCP receive mail box size. Generally bigger value means higher throughput but more memory. The recommended value is: LWIP_TCP_WND_DEFAULT/TCP_MSS + 2, e.g. if LWIP_TCP_WND_DEFAULT=14360, TCP_MSS=1436, then the recommended receive mail box size is (14360/1436 + 2) = 12.

TCP receive mail box is a per socket mail box, when the application receives packets from TCP socket, LWIP core firstly posts the packets to TCP receive mail box and the application then fetches the packets from mail box. It means LWIP can caches maximum LWIP_TCP_RECCVMBOX_SIZE packets for each TCP socket, so the maximum possible cached TCP packets for all TCP sockets is LWIP_TCP_RECCVMBOX_SIZE multiples the maximum TCP socket number. In other words, the bigger LWIP_TCP_RECVMBOX_SIZE means more memory. On the other hand, if the receive mail box is too small, the mail box may be full. If the mail box is full, the LWIP drops the packets. So generally we need to make sure the TCP receive mail box is big enough to avoid packet drop between LWIP core and application.

Range:

- from 6 to 1024 if CONFIG_LWIP_WND_SCALE

Default value:

- 6

CONFIG_LWIP_TCP_ACCEPTMBOX_SIZE

Default TCP accept mail box size

Found in: Component config > LWIP > TCP

Set TCP accept mail box size. Generally bigger value means supporting larger backlogs but more memory. The recommended value is 6, but applications can set it to a lower value if listening servers are meant to have a smaller backlog.

TCP accept mail box is a per socket mail box, when the application listens for connections with a given listening TCP socket. If the mailbox is full, LWIP will send a RST packet and the client will fail to connect.

Range:

- from 1 to 255 if CONFIG_LWIP_WND_SCALE

Default value:

- 6

CONFIG_LWIP_TCP_QUEUE_OOSEQ

Queue incoming out-of-order segments

Found in: Component config > LWIP > TCP

Queue incoming out-of-order segments for later use.

Disable this option to save some RAM during TCP sessions, at the expense of increased retransmissions if segments arrive out of order.

Default value:

- Yes (enabled)

CONFIG_LWIP_TCP_OOSEQ_TIMEOUT

Timeout for each pbuf queued in TCP OOSEQ, in RTOs.

Found in: Component config > LWIP > TCP > CONFIG_LWIP_TCP_QUEUE_OOSEQ

The timeout value is TCP_OOSEQ_TIMEOUT * RTO.

Range:

- from 1 to 30

Default value:

- 6

CONFIG_LWIP_TCP_OOSEQ_MAX_PBUFS

The maximum number of pbufs queued on OOSEQ per pcb

Found in: Component config > LWIP > TCP > CONFIG_LWIP_TCP_QUEUE_OOSEQ

If LWIP_TCP_OOSEQ_MAX_PBUFS = 0, TCP will not control the number of OOSEQ pbufs.

In a poor network environment, many out-of-order tcp pbufs will be received. These out-of-order pbufs will be cached in the TCP out-of-order queue which will cause Wi-Fi/Ethernet fail to release RX buffer in time. It is possible that all RX buffers for MAC layer are used by OOSEQ.

Control the number of out-of-order pbufs to ensure that the MAC layer has enough RX buffer to receive packets.

In the Wi-Fi scenario, recommended OOSEQ PBUFS Range: 0 <= TCP_OOSEQ_MAX_PBUFS <= CONFIG_ESP_WIFI_DYNAMIC_RX_BUFFER_NUM/(MAX_TCP_NUMBER + 1)

In the Ethernet scenario,recommended Ethernet OOSEQ PBUFS Range: 0 <= TCP_OOSEQ_MAX_PBUFS <= CONFIG_ETH_DMA_RX_BUFFER_NUM/(MAX_TCP_NUMBER + 1)

Within the recommended value range, the larger the value, the better the performance.

MAX_TCP_NUMBER represent Maximum number of TCP connections in Wi-Fi(STA+SoftAP) and Ethernet scenario.

Range:

- from 0 to 12

Default value:

- 4

CONFIG_LWIP_TCP_SACK_OUT

Support sending selective acknowledgements

Found in: Component config > LWIP > TCP > CONFIG_LWIP_TCP_QUEUE_OOSEQ

TCP will support sending selective acknowledgements (SACKs).

Default value:

- No (disabled)

CONFIG_LWIP_TCP_OVERSIZE

Pre-allocate transmit PBUF size

Found in: Component config > LWIP > TCP

Allows enabling "oversize" allocation of TCP transmission pbufs ahead of time, which can reduce the length of pbuf chains used for transmission.

This will not make a difference to sockets where Nagle's algorithm is disabled.

Default value of MSS is fine for most applications, 25% MSS may save some RAM when only transmitting small amounts of data. Disabled will have worst performance and fragmentation characteristics, but uses least RAM overall.

Available options:

- MSS (CONFIG_LWIP_TCP_OVERSIZE_MSS)
- 25% MSS (CONFIG_LWIP_TCP_OVERSIZE_QUARTER_MSS)
- Disabled (CONFIG_LWIP_TCP_OVERSIZE_DISABLE)

CONFIG_LWIP_WND_SCALE

Support TCP window scale

Found in: Component config > LWIP > TCP

Enable this feature to support TCP window scaling.

CONFIG_LWIP_TCP_RCV_SCALE

Set TCP receiving window scaling factor

Found in: Component config > LWIP > TCP > CONFIG_LWIP_WND_SCALE

Enable this feature to support TCP window scaling.

Range:

- from 0 to 14 if CONFIG_LWIP_WND_SCALE

Default value:

- 0 if CONFIG_LWIP_WND_SCALE

CONFIG_LWIP_TCP_RTO_TIME

Default TCP rto time

Found in: Component config > LWIP > TCP

Set default TCP rto time for a reasonable initial rto. In bad network environment, recommend set value of rto time to 1500.

Default value:

- 1500

UDP

Contains:

- CONFIG_LWIP_UDP_RECVMBOX_SIZE
- CONFIG_LWIP_MAX_UDP_PCBS

CONFIG_LWIP_MAX_UDP_PCBS

Maximum active UDP control blocks

Found in: Component config > LWIP > UDP

The maximum number of active UDP "connections" (ie UDP sockets sending/receiving data). The practical maximum limit is determined by available heap memory at runtime.

Range:

- from 1 to 1024

Default value:

- 16

CONFIG_LWIP_UDP_RECVMBOX_SIZE

Default UDP receive mail box size

Found in: Component config > LWIP > UDP

Set UDP receive mail box size. The recommended value is 6.

UDP receive mail box is a per socket mail box, when the application receives packets from UDP socket, LWIP core firstly posts the packets to UDP receive mail box and the application then fetches the packets from mail box. It means LWIP can caches maximum UDP_RECCVMBOX_SIZE packets for each UDP socket, so the maximum possible cached UDP packets for all UDP sockets is UDP_RECCVMBOX_SIZE multiples the maximum UDP socket number. In other words, the bigger UDP_RECVMBOX_SIZE means more memory. On the other hand, if the receive mail box is too small, the mail box may be full. If the mail box is full, the LWIP drops the packets. So generally we need to make sure the UDP receive mail box is big enough to avoid packet drop between LWIP core and application.

Range:

- from 6 to 64

Default value:

- 6

Checksums

Contains:

- CONFIG_LWIP_CHECKSUM_CHECK_ICMP
- CONFIG_LWIP_CHECKSUM_CHECK_IP
- CONFIG_LWIP_CHECKSUM_CHECK_UDP

CONFIG_LWIP_CHECKSUM_CHECK_IP

Enable LWIP IP checksums

Found in: Component config > LWIP > Checksums

Enable checksum checking for received IP messages

Default value:

- No (disabled)

CONFIG_LWIP_CHECKSUM_CHECK_UDP

Enable LWIP UDP checksums

Found in: Component config > LWIP > Checksums

Enable checksum checking for received UDP messages

Default value:

- No (disabled)

CONFIG_LWIP_CHECKSUM_CHECK_ICMP

Enable LWIP ICMP checksums

Found in: Component config > LWIP > Checksums

Enable checksum checking for received ICMP messages

Default value:

- Yes (enabled)

CONFIG_LWIP_TCPIP_TASK_STACK_SIZE

TCP/IP Task Stack Size

Found in: Component config > LWIP

Configure TCP/IP task stack size, used by LWIP to process multi-threaded TCP/IP operations. Setting this stack too small will result in stack overflow crashes.

Range:

- from 2048 to 65536

Default value:

- 3072

CONFIG_LWIP_TCPIP_TASK_AFFINITY

TCP/IP task affinity

Found in: Component config > LWIP

Allows setting LwIP tasks affinity, i.e. whether the task is pinned to CPU0, pinned to CPU1, or allowed to run on any CPU. Currently this applies to "TCP/IP" task and "Ping" task.

Available options:

- No affinity (CONFIG_LWIP_TCPIP_TASK_AFFINITY_NO_AFFINITY)
- CPU0 (CONFIG_LWIP_TCPIP_TASK_AFFINITY_CPU0)
- CPU1 (CONFIG_LWIP_TCPIP_TASK_AFFINITY_CPU1)

CONFIG_LWIP_IPV6_MEMP_NUM_ND6_QUEUE

Max number of IPv6 packets to queue during MAC resolution

Found in: Component config > LWIP

Config max number of IPv6 packets to queue during MAC resolution.

Range:

- from 3 to 20

Default value:

- 3

CONFIG_LWIP_IPV6_ND6_NUM_NEIGHBORS

Max number of entries in IPv6 neighbor cache

Found in: Component config > LWIP

Config max number of entries in IPv6 neighbor cache

Range:

- from 3 to 10

Default value:

- 5

CONFIG_LWIP_IPV6_ND6_NUM_PREFIXES

Max number of entries in IPv6 on-link prefixes cache

Found in: Component config > LWIP

Maximum number of entries in IPv6 on-link prefixes cache

Default value:

- 5

CONFIG_LWIP_IPV6_ND6_NUM_ROUTERS

Max number of entries in IPv6 default routers cache

Found in: Component config > LWIP

Maximum number of entries in IPv6 default routers cache

Default value:

- 3

CONFIG_LWIP_IPV6_ND6_NUM_DESTINATIONS

Max number of entries in IPv6 destinations cache

Found in: Component config > LWIP

Maximum number of entries in IPv6 destinations cache

Default value:

- 10

CONFIG_LWIP_IPV6_ND6_ROUTE_INFO_OPTION_SUPPORT

Enable IPV6 ND6 route information option support

Found in: Component config > LWIP

Enable route information option (RIO) support in IPV6 ND6

Default value:

- No (disabled)

CONFIG_LWIP_IPV6_ND6_NUM_ROUTE_INFO

Max number of entries in IPv6 route information options cache

Found in: Component config > LWIP > CONFIG_LWIP_IPV6_ND6_ROUTE_INFO_OPTION_SUPPORT

Maximum number of entries in IPv6 route information options cache

Default value:

- 10 if CONFIG_LWIP_IPV6_ND6_ROUTE_INFO_OPTION_SUPPORT

CONFIG_LWIP_PPP_SUPPORT

Enable PPP support

Found in: Component config > LWIP

Enable PPP stack. Now only PPP over serial is possible.

Default value:

- No (disabled)

Contains:

- CONFIG_LWIP_PPP_CHAP_SUPPORT
- CONFIG_LWIP_PPP_ENABLE_IPV4
- CONFIG_LWIP_PPP_ENABLE_IPV6
- CONFIG_LWIP_ENABLE_LCP_ECHO
- CONFIG_LWIP_PPP_MPPE_SUPPORT
- CONFIG_LWIP_PPP_MSCHAP_SUPPORT
- CONFIG_LWIP_PPP_NOTIFY_PHASE_SUPPORT
- CONFIG_LWIP_PPP_PAP_SUPPORT
- CONFIG_LWIP_PPP_DEBUG_ON
- CONFIG_LWIP_PPP_SERVER_SUPPORT
- CONFIG_LWIP_PPP_VJ_HEADER_COMPRESSION
- CONFIG_LWIP_USE_EXTERNAL_MBEDTLS

CONFIG_LWIP_PPP_ENABLE_IPV4

Enable IPV4 support for PPP connections (IPCP)

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable IPCP protocol in PPP negotiations, which assigns IPv4 addresses to the PPP client, as well as IPv4 DNS servers. You can disable this if your modem supports IPv6 only.

Default value:

- Yes (enabled) if CONFIG_LWIP_PPP_SUPPORT && CONFIG_LWIP_IPV4

CONFIG_LWIP_PPP_ENABLE_IPV6

Enable IPV6 support for PPP connections (IPV6CP)

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable IPV6 support in PPP for the local link between the DTE (processor) and DCE (modem). There are some modems which do not support the IPV6 addressing in the local link. If they are requested for IPV6CP negotiation, they may time out. This would in turn fail the configuration for the whole link. If your modem is not responding correctly to PPP Phase Network, try to disable IPV6 support.

Default value:

- Yes (enabled) if CONFIG_LWIP_PPP_SUPPORT && CONFIG_LWIP_IPV6

CONFIG_LWIP_PPP_NOTIFY_PHASE_SUPPORT

Enable Notify Phase Callback

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable to set a callback which is called on change of the internal PPP state machine.

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_PAP_SUPPORT

Enable PAP support

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable Password Authentication Protocol (PAP) support

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_CHAP_SUPPORT

Enable CHAP support

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable Challenge Handshake Authentication Protocol (CHAP) support

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_MSCHAP_SUPPORT

Enable MSCHAP support

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable Microsoft version of the Challenge-Handshake Authentication Protocol (MSCHAP) support

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_MPPE_SUPPORT

Enable MPPE support

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable Microsoft Point-to-Point Encryption (MPPE) support

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_SERVER_SUPPORT

Enable PPP server support

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable to use PPP server

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_PPP_VJ_HEADER_COMPRESSION

Enable VJ IP Header compression

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable support for VJ header compression. Please disable this if you're using NAPT on PPP interface, since the compressed IP header might not be correctly interpreted in NAT causing the compressed packet to be dropped.

Default value:

- Yes (enabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_ENABLE_LCP_ECHO

Enable LCP ECHO

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable LCP echo keepalive requests

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_LCP_ECHOINTERVAL

Echo interval (s)

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT > CONFIG_LWIP_ENABLE_LCP_ECHO

Interval in seconds between keepalive LCP echo requests, 0 to disable.

Range:

- from 0 to 1000000 if CONFIG_LWIP_ENABLE_LCP_ECHO

Default value:

- 3 if CONFIG_LWIP_ENABLE_LCP_ECHO

CONFIG_LWIP_LCP_MAXECHOFAILS

Maximum echo failures

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT > CONFIG_LWIP_ENABLE_LCP_ECHO

Number of consecutive unanswered echo requests before failure is indicated.

Range:

- from 0 to 100000 if CONFIG_LWIP_ENABLE_LCP_ECHO

Default value:

- 3 if CONFIG_LWIP_ENABLE_LCP_ECHO

CONFIG_LWIP_PPP_DEBUG_ON

Enable PPP debug log output

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

Enable PPP debug log output

Default value:

- No (disabled) if CONFIG_LWIP_PPP_SUPPORT

CONFIG_LWIP_USE_EXTERNAL_MBEDTLS

Use mbedTLS instead of internal polarSSL

Found in: Component config > LWIP > CONFIG_LWIP_PPP_SUPPORT

This option uses mbedTLS crypto functions (instead of internal PolarSSL implementation) for PPP authentication modes (PAP, CHAP, etc.). You can use this option to address symbol duplication issues, since the internal functions are not namespaced (e.g. md5_init()).

CONFIG_LWIP_SLIP_SUPPORT

Enable SLIP support (new/experimental)

Found in: Component config > LWIP

Enable SLIP stack. Now only SLIP over serial is possible.

SLIP over serial support is experimental and unsupported.

Default value:

- No (disabled)

Contains:

- CONFIG_LWIP_SLIP_DEBUG_ON

CONFIG_LWIP_SLIP_DEBUG_ON

Enable SLIP debug log output

Found in: Component config > LWIP > CONFIG_LWIP_SLIP_SUPPORT

Enable SLIP debug log output

Default value:

- No (disabled) if CONFIG_LWIP_SLIP_SUPPORT

ICMP

Contains:

- CONFIG_LWIP_ICMP
- CONFIG_LWIP_BROADCAST_PING
- CONFIG_LWIP_MULTICAST_PING

CONFIG_LWIP_ICMP

ICMP: Enable ICMP

Found in: Component config > LWIP > ICMP

Enable ICMP module for check network stability

Default value:

- Yes (enabled)

CONFIG_LWIP_MULTICAST_PING

Respond to multicast pings

Found in: Component config > LWIP > ICMP

Default value:

- No (disabled)

CONFIG_LWIP_BROADCAST_PING

Respond to broadcast pings

Found in: Component config > LWIP > ICMP

Default value:

- No (disabled)

LWIP RAW API

Contains:

- CONFIG_LWIP_MAX_RAW_PCBS

CONFIG_LWIP_MAX_RAW_PCBS

Maximum LWIP RAW PCBs

Found in: Component config > LWIP > LWIP RAW API

The maximum number of simultaneously active LWIP RAW protocol control blocks. The practical maximum limit is determined by available heap memory at runtime.

Range:

- from 1 to 1024

Default value:

- 16

SNTP

Contains:

- CONFIG_LWIP_SNTP_STARTUP_DELAY
- CONFIG_LWIP_SNTP_MAX_SERVERS
- CONFIG_LWIP_SNTP_UPDATE_DELAY
- CONFIG_LWIP_DHCP_GET_NTP_SRV

CONFIG_LWIP_SNTP_MAX_SERVERS

Maximum number of NTP servers

Found in: Component config > LWIP > SNTP

Set maximum number of NTP servers used by LwIP SNTP module. First argument of sntp_setserver/sntp_setservername functions is limited to this value.

Range:

- from 1 to 16

Default value:

- 1

CONFIG_LWIP_DHCP_GET_NTP_SRV

Request NTP servers from DHCP

Found in: Component config > LWIP > SNTP

If enabled, LWIP will add 'NTP' to Parameter-Request Option sent via DHCP-request. DHCP server might reply with an NTP server address in option 42. SNTP callback for such replies should be set accordingly (see sntp_servermode_dhcp() func.)

Default value:

- No (disabled)

CONFIG_LWIP_DHCP_MAX_NTP_SERVERS

Maximum number of NTP servers acquired via DHCP

Found in: Component config > LWIP > SNTP > CONFIG_LWIP_DHCP_GET_NTP_SRV

Set maximum number of NTP servers acquired via DHCP-offer. Should be less or equal to "Maximum number of NTP servers", any extra servers would be just ignored.

Range:

- from 1 to 16 if CONFIG_LWIP_DHCP_GET_NTP_SRV

Default value:

- 1 if CONFIG_LWIP_DHCP_GET_NTP_SRV

CONFIG_LWIP_SNTP_UPDATE_DELAY

Request interval to update time (ms)

Found in: Component config > LWIP > SNTP

This option allows you to set the time update period via SNTP. Default is 1 hour. Must not be below 15 seconds by specification. (SNTPv4 RFC 4330 enforces a minimum update time of 15 seconds).

Range:

- from 15000 to 4294967295

Default value:

- 3600000

CONFIG_LWIP_SNTP_STARTUP_DELAY

Enable SNTP startup delay

Found in: Component config > LWIP > SNTP

It is recommended (RFC 4330) to delay the initial request after by a random timeout from 1 to 5 minutes to reduce potential load of NTP servers after simultaneous power-up of many devices. This option disables this initial delay. Please use this option with care, it could improve a single device responsiveness but might cause peaks on the network after reset. Another option to address responsiveness of devices while using the initial random delay is to adjust LWIP_SNTP_MAXIMUM_STARTUP_DELAY.

Default value:

- Yes (enabled)

CONFIG_LWIP_SNTP_MAXIMUM_STARTUP_DELAY

Maximum startup delay (ms)

Found in: Component config > LWIP > SNTP > CONFIG_LWIP_SNTP_STARTUP_DELAY

RFC 4330 recommends a startup delay before sending the initial request. LWIP calculates this delay to a random number of milliseconds between 0 and this value.

Range:

- from 100 to 300000

Default value:

- 5000

DNS

Contains:

- CONFIG_LWIP_FALLBACK_DNS_SERVER_SUPPORT
- CONFIG_LWIP_DNS_SETSERVER_WITH_NETIF
- CONFIG_LWIP_USE_ESP_GETADDRINFO
- CONFIG_LWIP_DNS_MAX_SERVERS
- CONFIG_LWIP_DNS_MAX_HOST_IP

CONFIG_LWIP_DNS_MAX_HOST_IP

Maximum number of IP addresses per host

Found in: Component config > LWIP > DNS

Maximum number of IP addresses that can be returned by DNS queries for a single host.

Default value:

- 1

CONFIG_LWIP_DNS_MAX_SERVERS

Maximum number of DNS servers

Found in: Component config > LWIP > DNS

Set maximum number of DNS servers. If fallback DNS servers are supported, the number of DNS servers needs to be greater than or equal to 3.

Range:

- from 1 to 4

Default value:

- 3

CONFIG_LWIP_FALLBACK_DNS_SERVER_SUPPORT

Enable DNS fallback server support

Found in: Component config > LWIP > DNS

Enable this feature to support DNS fallback server.

Default value:

- No (disabled)

CONFIG_LWIP_FALLBACK_DNS_SERVER_ADDRESS

DNS fallback server address

Found in: Component config > LWIP > DNS > CONFIG_LWIP_FALLBACK_DNS_SERVER_SUPPORT

This option allows you to config dns fallback server address.

Default value:

- "114.114.114.114" if CONFIG_LWIP_FALLBACK_DNS_SERVER_SUPPORT

CONFIG_LWIP_DNS_SETSERVER_WITH_NETIF

Enable DNS server settings with netif

Found in: Component config > LWIP > DNS

This option allows collecting DNS server settings per netif using configurable callback function. It's typically used with CONFIG_ESP_NETIF_SET_DNS_PER_DEFAULT_NETIF which configures a callback to collect the DNS info on esp_netif layer.

Default value:

- No (disabled)

CONFIG_LWIP_USE_ESP_GETADDRINFO

Enable esp_getaddrinfo() instead of lwip_getaddrinfo()

Found in: Component config > LWIP > DNS

Use esp_getaddrinfo() for DNS lookups instead of lwip_getaddrinfo(). This function correctly handles the AF_UNSPEC flag for resolving both IPv4 and IPv6 addresses. Available only when both IPv4 and IPv6 are enabled.

Default value:

- No (disabled)

CONFIG_LWIP_BRIDGEIF_MAX_PORTS

Maximum number of bridge ports

Found in: Component config > LWIP

Set maximum number of ports a bridge can consists of.

Range:

- from 1 to 63

Default value:

- 7

CONFIG_LWIP_ESP_LWIP_ASSERT

Enable LWIP ASSERT checks

Found in: Component config > LWIP

Enable this option keeps LWIP assertion checks enabled. It is recommended to keep this option enabled.

If asserts are disabled for the entire project, they are also disabled for LWIP and this option is ignored.

Hooks

Contains:

- CONFIG_LWIP_HOOK_DHCP_EXTRA_OPTION
- CONFIG_LWIP_HOOK_DNS_EXTERNAL_RESOLVE
- CONFIG_LWIP_HOOK_ND6_GET_GW
- CONFIG_LWIP_HOOK_IP6_INPUT
- CONFIG_LWIP_HOOK_IP6_ROUTE
- CONFIG_LWIP_HOOK_IP6_SELECT_SRC_ADDR
- CONFIG_LWIP_HOOK_NETCONN_EXTERNAL_RESOLVE
- CONFIG_LWIP_HOOK_TCP_ISN

CONFIG_LWIP_HOOK_TCP_ISN

TCP ISN Hook

Found in: Component config > LWIP > Hooks

Enables to define a TCP ISN hook to randomize initial sequence number in TCP connection. The default TCP ISN algorithm used in IDF (standardized in RFC 6528) produces ISN by combining an MD5 of the new TCP id and a stable secret with the current time. This is because the lwIP implementation (*tcp_next_iss*) is not very strong, as it does not take into consideration any platform specific entropy source.

Set to LWIP_HOOK_TCP_ISN_CUSTOM to provide custom implementation. Set to LWIP_HOOK_TCP_ISN_NONE to use lwIP implementation.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_TCP_ISN_NONE)
- Default implementation (CONFIG_LWIP_HOOK_TCP_ISN_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_TCP_ISN_CUSTOM)

CONFIG_LWIP_HOOK_IP6_ROUTE

IPv6 route Hook

Found in: Component config > LWIP > Hooks

Enables custom IPv6 route hook. Setting this to "default" provides weak implementation stub that could be overwritten in application code. Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_IP6_ROUTE_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_IP6_ROUTE_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_IP6_ROUTE_CUSTOM)

CONFIG_LWIP_HOOK_ND6_GET_GW

IPv6 get gateway Hook

Found in: Component config > LWIP > Hooks

Enables custom IPv6 route hook. Setting this to "default" provides weak implementation stub that could be overwritten in application code. Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_ND6_GET_GW_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_ND6_GET_GW_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_ND6_GET_GW_CUSTOM)

CONFIG_LWIP_HOOK_IP6_SELECT_SRC_ADDR

IPv6 source address selection Hook

Found in: Component config > LWIP > Hooks

Enables custom IPv6 source address selection. Setting this to "default" provides weak implementation stub that could be overwritten in application code. Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_IP6_SELECT_SRC_ADDR_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_IP6_SELECT_SRC_ADDR_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_IP6_SELECT_SRC_ADDR_CUSTOM)

CONFIG_LWIP_HOOK_DHCP_EXTRA_OPTION

DHCP extra option hook

Found in: Component config > LWIP > Hooks

Enables custom hook to receive and parse extra DHCP options. Setting this to "default" provides weak implementation stub that could be overwritten in application code. Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_DHCP_EXTRA_OPTION_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_DHCP_EXTRA_OPTION_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_DHCP_EXTRA_OPTION_CUSTOM)

CONFIG_LWIP_HOOK_NETCONN_EXTERNAL_RESOLVE

Netconn external resolve Hook

Found in: Component config > LWIP > Hooks

Enables custom DNS resolve hook (without callback). Setting this to "default" provides weak implementation stub that could be overwritten in application code. Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_NETCONN_EXT_RESOLVE_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_NETCONN_EXT_RESOLVE_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_NETCONN_EXT_RESOLVE_CUSTOM)

CONFIG_LWIP_HOOK_DNS_EXTERNAL_RESOLVE

DNS external resolve Hook

Found in: Component config > LWIP > Hooks

Enables custom DNS resolve hook (with callback). Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_DNS_EXT_RESOLVE_NONE)
- Custom implementation (CONFIG_LWIP_HOOK_DNS_EXT_RESOLVE_CUSTOM)

CONFIG_LWIP_HOOK_IP6_INPUT

IPv6 packet input

Found in: Component config > LWIP > Hooks

Enables custom IPv6 packet input. Setting this to "default" provides weak IDF implementation, which drops all incoming IPv6 traffic if the interface has no link local address. (this default implementation is "weak" and could be still overwritten in the application if some additional IPv6 input packet filtering is needed) Setting this to "none" removes this default filter and conforms to the lwIP implementation (which accepts multicasts even if the interface has no link local address) Setting this to "custom" provides hook's declaration only and expects the application to implement it.

Available options:

- No hook declared (CONFIG_LWIP_HOOK_IP6_INPUT_NONE)
- Default (weak) implementation (CONFIG_LWIP_HOOK_IP6_INPUT_DEFAULT)
- Custom implementation (CONFIG_LWIP_HOOK_IP6_INPUT_CUSTOM)

CONFIG_LWIP_DEBUG

Enable LWIP Debug

Found in: Component config > LWIP

Enabling this option allows different kinds of lwIP debug output.

All lwIP debug features increase the size of the final binary.

Default value:

- No (disabled)

Contains:

- CONFIG_LWIP_API_LIB_DEBUG
- CONFIG_LWIP_BRIDGEIF_FDB_DEBUG
- CONFIG_LWIP_BRIDGEIF_FW_DEBUG
- CONFIG_LWIP_BRIDGEIF_DEBUG
- CONFIG_LWIP_DHCP_DEBUG
- CONFIG_LWIP_DHCP_STATE_DEBUG
- CONFIG_LWIP_DNS_DEBUG
- CONFIG_LWIP_ETHARP_DEBUG
- CONFIG_LWIP_ICMP_DEBUG
- CONFIG_LWIP_ICMP6_DEBUG
- CONFIG_LWIP_IP_DEBUG
- CONFIG_LWIP_IP6_DEBUG
- CONFIG_LWIP_NAPT_DEBUG
- CONFIG_LWIP_NETIF_DEBUG
- CONFIG_LWIP_PBUF_DEBUG
- CONFIG_LWIP_SNTP_DEBUG
- CONFIG_LWIP_SOCKETS_DEBUG
- CONFIG_LWIP_TCP_DEBUG
- CONFIG_LWIP_UDP_DEBUG
- CONFIG_LWIP_DEBUG_ESP_LOG

CONFIG_LWIP_DEBUG_ESP_LOG

Route LWIP debugs through ESP_LOG interface

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Enabling this option routes all enabled LWIP debugs through ESP_LOGD.

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_NETIF_DEBUG

Enable netif debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_PBUF_DEBUG

Enable pbuf debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_ETHARP_DEBUG

Enable etharp debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_API_LIB_DEBUG

Enable api lib debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_SOCKETS_DEBUG

Enable socket debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_IP_DEBUG

Enable IP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_ICMP_DEBUG

Enable ICMP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG && CONFIG_LWIP_ICMP

CONFIG_LWIP_DHCP_STATE_DEBUG

Enable DHCP state tracking

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_DHCP_DEBUG

Enable DHCP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_IP6_DEBUG

Enable IP6 debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_ICMP6_DEBUG

Enable ICMP6 debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_TCP_DEBUG

Enable TCP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_UDP_DEBUG

Enable UDP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_SNTP_DEBUG

Enable SNTP debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_DNS_DEBUG

Enable DNS debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_NAPT_DEBUG

Enable NAPT debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG && CONFIG_LWIP_IPV4_NAPT

CONFIG_LWIP_BRIDGEIF_DEBUG

Enable bridge generic debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_BRIDGEIF_FDB_DEBUG

Enable bridge FDB debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

CONFIG_LWIP_BRIDGEIF_FW_DEBUG

Enable bridge forwarding debug messages

Found in: Component config > LWIP > CONFIG_LWIP_DEBUG

Default value:

- No (disabled) if CONFIG_LWIP_DEBUG

mbedTLS

Contains:

- Asymmetric Ciphers
- Certificates
- Core Configuration

- CONFIG_MBEDTLS_TLS_ENABLED
- Encoding/Decoding
- Entropy and Random Number Generation
- Hardware Acceleration
- Hash functions
- Stream Cipher
- CONFIG_MBEDTLS_SSL_PROTO_DTLS
- Symmetric Ciphers
- CONFIG_MBEDTLS_USE_CRYPTO_ROM_IMPL
- CONFIG_MBEDTLS_USE_CRYPTO_ROM_IMPL_BOOTLOADER

Core Configuration

Contains:

- CONFIG_MBEDTLS_ASYMMETRIC_CONTENT_LEN
- CONFIG_MBEDTLS_COMPILER_OPTIMIZATION
- CONFIG_MBEDTLS_ERROR_STRERROR_DUMMY
- CONFIG_MBEDTLS_ERROR_STRINGS
- CONFIG_MBEDTLS_FS_IO
- CONFIG_MBEDTLS_DEBUG
- CONFIG_MBEDTLS_SELF_TEST
- CONFIG_MBEDTLS_HAVE_TIME
- CONFIG_MBEDTLS_VERSION_FEATURES
- CONFIG_MBEDTLS_THREADING_C
- CONFIG_MBEDTLS_VERSION_C
- CONFIG_MBEDTLS_MEM_ALLOC_MODE
- CONFIG_MBEDTLS_SSL_MAX_CONTENT_LEN
- CONFIG_MBEDTLS_DYNAMIC_BUFFER

CONFIG_MBEDTLS_COMPILER_OPTIMIZATION

Compiler optimization level

Found in: Component config > mbedTLS > Core Configuration

This option allows you to select the compiler optimization level for mbedTLS. The default is set to the optimization level used by the rest of the ESP-IDF project.

Available options:

- No optimization (CONFIG_MBEDTLS_COMPILER_OPTIMIZATION_NONE)
- Optimize for size (-Os) (CONFIG_MBEDTLS_COMPILER_OPTIMIZATION_SIZE)
- Optimize for performance (-O2) (CONFIG_MBEDTLS_COMPILER_OPTIMIZATION_PERF)

CONFIG_MBEDTLS_FS_IO

Enable functions that use the filesystem

Found in: Component config > mbedTLS > Core Configuration

This option enables functions in mbedTLS that use the filesystem. It uses the default filesystem support for the target, which is added through vfs component for ESP32 based targets or by the host system when the target is Linux.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_THREADING_C

Enable the threading abstraction layer

Found in: Component config > mbedTLS > Core Configuration

If you do intend to use contexts between threads, you will need to enable this layer to prevent race conditions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_THREADING_IMPLEMENTATION

Threading implementation

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_THREADING_C

Available options:

- Enable threading alternate implementation (CONFIG_MBEDTLS_THREADING_ALT)

  Enable threading alt to allow your own alternate threading implementation.

- Enable threading pthread implementation (CONFIG_MBEDTLS_THREADING_PTHREAD)

  Enable the pthread wrapper layer for the threading layer.

CONFIG_MBEDTLS_ERROR_STRINGS

Enable error code to error string conversion

Found in: Component config > mbedTLS > Core Configuration

Enables mbedtls_strerror() for converting error codes to error strings. Disabling this config can save some code/rodata size as the error string conversion implementation is replaced with an empty stub.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ERROR_STRERROR_DUMMY

Enable a dummy error function to make use of mbedtls_strerror()

Found in: Component config > mbedTLS > Core Configuration

This option enables a dummy error function to make use of mbedtls_strerror() when MBEDTLS_ERROR_STRINGS is disabled. This is useful for applications that use mbedtls_strerror() but do not need the actual error strings. This option can be used to save code size when MBEDTLS_ERROR_STRINGS is disabled.

CONFIG_MBEDTLS_VERSION_C

Enable version information

Found in: Component config > mbedTLS > Core Configuration

Enable version information functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_HAVE_TIME

Enable mbedtls time support

Found in: Component config > mbedTLS > Core Configuration

Enable use of time.h functions (time() and gmtime()) by mbedTLS.

This option doesn't require the system time to be correct, but enables functionality that requires relative timekeeping - for example periodic expiry of TLS session tickets or session cache entries.

Disabling this option will save some firmware size, particularly if the rest of the firmware doesn't call any standard timekeeping functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PLATFORM_TIME_ALT

Enable mbedtls time support: platform-specific

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_HAVE_TIME

Enabling this config will provide users with a function "mbedtls_platform_set_time()" that allows to set an alternative time function pointer.

Default value:

- No (disabled)

CONFIG_MBEDTLS_HAVE_TIME_DATE

Enable mbedtls certificate expiry check

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_HAVE_TIME

Enables X.509 certificate expiry checks in mbedTLS.

If this option is disabled (default) then X.509 certificate "valid from" and "valid to" timestamp fields are ignored.

If this option is enabled, these fields are compared with the current system date and time. The time is retrieved using the standard time() and gmtime() functions. If the certificate is not valid for the current system time then verification will fail with code MBEDTLS_X509_BADCERT_FUTURE or MBEDTLS_X509_BADCERT_EXPIRED.

Enabling this option requires adding functionality in the firmware to set the system clock to a valid timestamp before using TLS. The recommended way to do this is via ESP-IDF's SNTP functionality, but any method can be used.

In the case where only a small number of certificates are trusted by the device, please carefully consider the tradeoffs of enabling this option. There may be undesired consequences, for example if all trusted certificates expire while the device is offline and a TLS connection is required to update. Or if an issue with the SNTP server means that the system time is invalid for an extended period after a reset.

Default value:

- No (disabled)

CONFIG_MBEDTLS_MEM_ALLOC_MODE

Memory allocation strategy

Found in: Component config > mbedTLS > Core Configuration

Allocation strategy for mbedTLS, essentially provides ability to allocate all required dynamic allocations from,

- Internal DRAM memory only
- External SPIRAM memory only
- Either internal or external memory based on default malloc() behavior in ESP-IDF
- **Custom allocation mode, by overwriting calloc()/free()**

  using mbedtls_platform_set_calloc_free() function
- Internal IRAM memory wherever applicable else internal DRAM

Recommended mode here is always internal (*), since that is most preferred from security perspective. But if application requirement does not allow sufficient free internal memory then alternate mode can be selected.

(*) In case of ESP32-S2/ESP32-S3, hardware allows encryption of external SPIRAM contents provided hardware flash encryption feature is enabled. In that case, using external SPIRAM allocation strategy is also safe choice from security perspective.

Available options:

- Internal memory (CONFIG_MBEDTLS_INTERNAL_MEM_ALLOC)
- External SPIRAM (CONFIG_MBEDTLS_EXTERNAL_MEM_ALLOC)
- Default alloc mode (CONFIG_MBEDTLS_DEFAULT_MEM_ALLOC)
- Custom alloc mode (CONFIG_MBEDTLS_CUSTOM_MEM_ALLOC)
- Internal IRAM (CONFIG_MBEDTLS_IRAM_8BIT_MEM_ALLOC)

  Allows to use IRAM memory region as 8bit accessible region.

  TLS input and output buffers will be allocated in IRAM section which is 32bit aligned memory. Every unaligned (8bit or 16bit) access will result in an exception and incur penalty of certain clock cycles per unaligned read/write.

CONFIG_MBEDTLS_SSL_MAX_CONTENT_LEN

TLS maximum message content length

Found in: Component config > mbedTLS > Core Configuration

Maximum TLS message length (in bytes) supported by mbedTLS.

16384 is the default and this value is required to comply fully with TLS standards.

However you can set a lower value in order to save RAM. This is safe if the other end of the connection supports Maximum Fragment Length Negotiation Extension (max_fragment_length, see RFC6066) or you know for certain that it will never send a message longer than a certain number of bytes.

If the value is set too low, symptoms are a failed TLS handshake or a return value of MBEDTLS_ERR_SSL_INVALID_RECORD (-0x7200).

CONFIG_MBEDTLS_ASYMMETRIC_CONTENT_LEN

Asymmetric in/out fragment length

Found in: Component config > mbedTLS > Core Configuration

If enabled, this option allows customizing TLS in/out fragment length in asymmetric way. Please note that enabling this with default values saves 12KB of dynamic memory per TLS connection.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_IN_CONTENT_LEN

TLS maximum incoming fragment length

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_ASYMMETRIC_CONTENT_LEN

This defines maximum incoming fragment length, overriding default maximum content length (MBEDTLS_SSL_MAX_CONTENT_LEN).

Range:

- from 512 to 16384

Default value:

- 16384

CONFIG_MBEDTLS_SSL_OUT_CONTENT_LEN

TLS maximum outgoing fragment length

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_ASYMMETRIC_CONTENT_LEN

This defines maximum outgoing fragment length, overriding default maximum content length (MBEDTLS_SSL_MAX_CONTENT_LEN).

Range:

- from 512 to 16384

Default value:

- 4096

CONFIG_MBEDTLS_DYNAMIC_BUFFER

Using dynamic TX/RX buffer

Found in: Component config > mbedTLS > Core Configuration

Using dynamic TX/RX buffer. After enabling this option, mbedTLS will allocate TX buffer when need to send data and then free it if all data is sent, allocate RX buffer when need to receive data and then free it when all data is used or read by upper layer.

By default, when SSL is initialized, mbedTLS also allocate TX and RX buffer with the default value of "MBEDTLS_SSL_OUT_CONTENT_LEN" or "MBEDTLS_SSL_IN_CONTENT_LEN", so to save more heap, users can set the options to be an appropriate value.

CONFIG_MBEDTLS_DYNAMIC_FREE_CONFIG_DATA

Free private key and DHM data after its usage

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_DYNAMIC_BUFFER

Free private key and DHM data after its usage in handshake process.

The option will decrease heap cost when handshake, but also lead to problem:

Because all certificate, private key and DHM data are freed so users should register certificate and private key to ssl config object again.

Default value:

- No (disabled) if CONFIG_MBEDTLS_DYNAMIC_BUFFER

CONFIG_MBEDTLS_DYNAMIC_FREE_CA_CERT

Free SSL CA certificate after its usage

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_DYNAMIC_BUFFER > CONFIG_MBEDTLS_DYNAMIC_FREE_CONFIG_DATA

Free CA certificate after its usage in the handshake process. This option will decrease the heap footprint for the TLS handshake, but may lead to a problem: If the respective ssl object needs to perform the TLS handshake again, the CA certificate should once again be registered to the ssl object.

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_DYNAMIC_FREE_CONFIG_DATA

CONFIG_MBEDTLS_VERSION_FEATURES

Enable mbedTLS version features

Found in: Component config > mbedTLS > Core Configuration

Enable mbedTLS version features. This option allows Allow run-time checking of compile-time enabled features. Disabling this option will save some code size.

Default value:

- No (disabled)

CONFIG_MBEDTLS_DEBUG

Enable mbedTLS debugging

Found in: Component config > mbedTLS > Core Configuration

Enable mbedTLS debugging functions at compile time.

If this option is enabled, you can include "mbedtls/esp_debug.h" and call mbedtls_esp_enable_debug_log() at runtime in order to enable mbedTLS debug output via the ESP log mechanism.

Default value:

- No (disabled)

CONFIG_MBEDTLS_DEBUG_LEVEL

Set mbedTLS debugging level

Found in: Component config > mbedTLS > Core Configuration > CONFIG_MBEDTLS_DEBUG

Set mbedTLS debugging level

Available options:

- Warning (CONFIG_MBEDTLS_DEBUG_LEVEL_WARN)
- Info (CONFIG_MBEDTLS_DEBUG_LEVEL_INFO)
- Debug (CONFIG_MBEDTLS_DEBUG_LEVEL_DEBUG)
- Verbose (CONFIG_MBEDTLS_DEBUG_LEVEL_VERBOSE)

CONFIG_MBEDTLS_SELF_TEST

Enable mbedTLS self-test

Found in: Component config > mbedTLS > Core Configuration

Enable mbedTLS self-test functions.

Default value:

- Yes (enabled)

Certificates

Contains:

- CONFIG_MBEDTLS_ALLOW_WEAK_CERTIFICATE_VERIFICATION
- CONFIG_MBEDTLS_ASN1_PARSE_C
- CONFIG_MBEDTLS_ASN1_WRITE_C
- CONFIG_MBEDTLS_PK_C
- CONFIG_MBEDTLS_X509_TRUSTED_CERT_CALLBACK
- CONFIG_MBEDTLS_CERTIFICATE_BUNDLE
- CONFIG_MBEDTLS_X509_CRT_PARSE_C
- CONFIG_MBEDTLS_X509_USE_C
- CONFIG_MBEDTLS_PEM_PARSE_C
- CONFIG_MBEDTLS_X509_REMOVE_INFO
- CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_CROSS_SIGNED_VERIFY
- CONFIG_MBEDTLS_PEM_WRITE_C
- CONFIG_MBEDTLS_X509_CREATE_C
- CONFIG_MBEDTLS_X509_CRL_PARSE_C
- CONFIG_MBEDTLS_X509_CSR_PARSE_C
- CONFIG_MBEDTLS_X509_RSASSA_PSS_SUPPORT

CONFIG_MBEDTLS_X509_USE_C

Enable X.509 certificate support

Found in: Component config > mbedTLS > Certificates

Enable X.509 certificate support.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PEM_PARSE_C

Read & Parse PEM formatted certificates

Found in: Component config > mbedTLS > Certificates

Enable decoding/parsing of PEM formatted certificates.

If your certificates are all in the simpler DER format, disabling this option will save some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PEM_WRITE_C

Write PEM formatted certificates

Found in: Component config > mbedTLS > Certificates

Enable writing of PEM formatted certificates.

If writing certificate data only in DER format, disabling this option will save some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PK_C

Enable generic public key layer

Found in: Component config > mbedTLS > Certificates

Enable support for generic public key layer.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PK_PARSE_C

Enables generic public key parsing functions

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_PK_C

Enable generic public key parsing functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PK_WRITE_C

Enables generic public key writing functions

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_PK_C

Enable generic public key writing functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_X509_REMOVE_INFO

Remove X.509 debug info

Found in: Component config > mbedTLS > Certificates

Removes mbedtls_x509_*_info(), as well as mbedtls_debug_print_crt() and other functions/constants only used by these functions. This will save some code size.

Default value:

- No (disabled)

CONFIG_MBEDTLS_X509_CRL_PARSE_C

X.509 CRL parsing

Found in: Component config > mbedTLS > Certificates

Support for parsing X.509 Certificate Revocation Lists.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_X509_CRT_PARSE_C

Enable X.509 certificate parsing

Found in: Component config > mbedTLS > Certificates

Enable X.509 certificate parsing. This is required for TLS and DTLS.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_X509_CSR_PARSE_C

X.509 CSR parsing

Found in: Component config > mbedTLS > Certificates

Support for parsing X.509 Certificate Signing Requests

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_X509_CREATE_C

X.509 certificate creation

Found in: Component config > mbedTLS > Certificates

Support for creating X.509 certificates and CSRs.

Default value:

- No (disabled)

CONFIG_MBEDTLS_X509_CRT_WRITE_C

X.509 certificate writing

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_X509_CREATE_C

Support for writing X.509 certificates

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_X509_CREATE_C

CONFIG_MBEDTLS_X509_CSR_WRITE_C

X.509 CSR writing

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_X509_CREATE_C

Support for writing X.509 CSRs

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_X509_CREATE_C

CONFIG_MBEDTLS_X509_RSASSA_PSS_SUPPORT

X.509 PSS support

Found in: Component config > mbedTLS > Certificates

Support for parsing X.509 certificates with RSASSA-PSS signatures.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_X509_TRUSTED_CERT_CALLBACK

Enable trusted certificate callbacks

Found in: Component config > mbedTLS > Certificates

Enables users to configure the set of trusted certificates through a callback instead of a linked list.

See mbedTLS documentation for required API and more details.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ASN1_PARSE_C

Enable ASN.1 parsing

Found in: Component config > mbedTLS > Certificates

Enable ASN.1 parsing functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ASN1_WRITE_C

Enable ASN.1 writing

Found in: Component config > mbedTLS > Certificates

Enable ASN.1 writing functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CERTIFICATE_BUNDLE

Enable trusted root certificate bundle

Found in: Component config > mbedTLS > Certificates

Enable support for large number of default root certificates

When enabled this option allows user to store default as well as customer specific root certificates in compressed format rather than storing full certificate. For the root certificates the public key and the subject name will be stored.

Default value:

- Yes (enabled)

Certificate Bundle Configuration

Contains:

- CONFIG_MBEDTLS_CUSTOM_CERTIFICATE_BUNDLE
- CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEPRECATED_LIST
- CONFIG_MBEDTLS_DEFAULT_CERTIFICATE_BUNDLE
- CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_MAX_CERTS

CONFIG_MBEDTLS_DEFAULT_CERTIFICATE_BUNDLE

Default certificate bundle options

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_CERTIFICATE_BUNDLE > Certificate Bundle Configuration

Available options:

- Use the full default certificate bundle (CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEFAULT_FULL)
- Use only the most common certificates from the default bundles (CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEFAULT_CMN)

  Use only the most common certificates from the default bundles, reducing the size with 50%, while still having around 99% coverage.

- Do not use the default certificate bundle (CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEFAULT_NONE)

CONFIG_MBEDTLS_CUSTOM_CERTIFICATE_BUNDLE

Add custom certificates to the default bundle

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_CERTIFICATE_BUNDLE > Certificate Bundle Configuration

Default value:

- No (disabled)

CONFIG_MBEDTLS_CUSTOM_CERTIFICATE_BUNDLE_PATH

Custom certificate bundle path

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_CERTIFICATE_BUNDLE > Certificate Bundle Configuration > CONFIG_MBEDTLS_CUSTOM_CERTIFICATE_BUNDLE

Name of the custom certificate directory or file. This path is evaluated relative to the project root directory.

CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEPRECATED_LIST

Add deprecated root certificates

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_CERTIFICATE_BUNDLE > Certificate Bundle Configuration

Include the deprecated list of root certificates in the bundle. This list gets updated when a certificate is removed from the Mozilla's NSS root certificate store. This config can be enabled if you would like to ensure that none of the certificates that were deployed in the product are affected because of the update to bundle. In turn, enabling this setting keeps expired, retracted certificates in the bundle and it may pose a security risk. - Deprecated cert list may grow based based on sync with upstream bundle - Deprecated certs would be removed in ESP-IDF (next) major release

CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_MAX_CERTS

Maximum no of certificates allowed in certificate bundle

Found in: Component config > mbedTLS > Certificates > CONFIG_MBEDTLS_CERTIFICATE_BUNDLE > Certificate Bundle Configuration

Default value:

- 200

CONFIG_MBEDTLS_ALLOW_WEAK_CERTIFICATE_VERIFICATION

Allow weak certificate verification

Found in: Component config > mbedTLS > Certificates

This options allows weak certificate verification by skipping the hostname verification. It is not recommended to use this option.

Default value:

- No (disabled)

CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_CROSS_SIGNED_VERIFY

Support cross-signed certificate verification in certificate bundle

Found in: Component config > mbedTLS > Certificates

Enable support for cross-signed certificate verification in the certificate bundle. This feature uses an internal callback to verify the cross-signed certificates. This feature is kept disabled by default as enabling this feature increases heap usage by approximately 700 bytes.

Default value:

- No (disabled)

CONFIG_MBEDTLS_TLS_ENABLED

Enable TLS protocol support

Found in: Component config > mbedTLS

Enable support for the TLS protocol, which is used for secure communication over networks. This option is required for most secure network protocols, including HTTPS, FTPS, and others.

If you do not need TLS support, you can disable this option to save code size.

Default value:

- Yes (enabled)

TLS Protocol Configuration

Contains:

- CONFIG_MBEDTLS_SSL_ALL_ALERT_MESSAGES
- CONFIG_MBEDTLS_SSL_KEYING_MATERIAL_EXPORT
- CONFIG_MBEDTLS_SSL_CONTEXT_SERIALIZATION
- CONFIG_MBEDTLS_SSL_SERVER_NAME_INDICATION
- CONFIG_MBEDTLS_SSL_CACHE_C
- CONFIG_MBEDTLS_SSL_RECORD_SIZE_LIMIT
- CONFIG_MBEDTLS_SSL_MAX_FRAGMENT_LENGTH
- CONFIG_MBEDTLS_SSL_KEEP_PEER_CERTIFICATE
- CONFIG_MBEDTLS_SSL_CID_PADDING_GRANULARITY
- CONFIG_MBEDTLS_SSL_ALPN
- CONFIG_MBEDTLS_SSL_PROTO_GMTSSL1_1
- CONFIG_MBEDTLS_SSL_PROTO_TLS1_2
- CONFIG_MBEDTLS_SSL_PROTO_TLS1_3

- CONFIG_MBEDTLS_SSL_RENEGOTIATION
- TLS Key Exchange Configuration
- CONFIG_MBEDTLS_TLS_MODE
- CONFIG_MBEDTLS_CLIENT_SSL_SESSION_TICKETS
- CONFIG_MBEDTLS_SERVER_SSL_SESSION_TICKETS
- CONFIG_MBEDTLS_SSL_VARIABLE_BUFFER_LENGTH

CONFIG_MBEDTLS_SSL_PROTO_TLS1_2

Support TLS 1.2 protocol

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_PROTO_TLS1_3

Support TLS 1.3 protocol

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Default value:

- No (disabled)

TLS 1.3 Configuration

Contains:

- CONFIG_MBEDTLS_SSL_EARLY_DATA
- CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_EPHEMERAL
- CONFIG_MBEDTLS_SSL_TLS1_3_COMPATIBILITY_MODE
- CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_PSK_EPHEMERAL
- CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_PSK

CONFIG_MBEDTLS_SSL_TLS1_3_COMPATIBILITY_MODE

TLS 1.3 middlebox compatibility mode

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 > TLS 1.3 Configuration

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_PSK

TLS 1.3 PSK key exchange mode

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 > TLS 1.3 Configuration

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_EPHEMERAL

TLS 1.3 ephemeral key exchange mode

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 > TLS 1.3 Configuration

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_PSK_EPHEMERAL

TLS 1.3 PSK ephemeral key exchange mode

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 > TLS 1.3 Configuration

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_EARLY_DATA

TLS 1.3 early data

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 > TLS 1.3 Configuration

Enable support for TLS 1.3 early data (0-RTT).

Default value:

- No (disabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_CLIENT_SSL_SESSION_TICKETS && (CONFIG_MBEDTLS_SSL_TLS1_3_KEXM_PSK || MBEDTLS_SSL_TLS1_3_KEXM_EPHEMER) && CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_PROTO_GMTSSL1_1

Support GM/T SSL 1.1 protocol

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Provisions for GM/T SSL 1.1 support

Default value:

- No (disabled)

CONFIG_MBEDTLS_TLS_MODE

TLS Protocol Role

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

mbedTLS can be compiled with protocol support for the TLS server, TLS client, or both server and client.

Reducing the number of TLS roles supported saves code size.

Available options:

- Server & Client (CONFIG_MBEDTLS_TLS_SERVER_AND_CLIENT)
- Server (CONFIG_MBEDTLS_TLS_SERVER_ONLY)
- Client (CONFIG_MBEDTLS_TLS_CLIENT_ONLY)

CONFIG_MBEDTLS_SSL_CID_PADDING_GRANULARITY

Record plaintext padding

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Controls the use of record plaintext padding in TLS 1.3 and when using the Connection ID extension in DTLS 1.2.

The padding will always be chosen so that the length of the padded plaintext is a multiple of the value of this option.

Notes: A value of 1 means that no padding will be used for outgoing records. On systems lacking division instructions, a power of two should be preferred.

Range:

- from 0 to 32 if (CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 || CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID) && CONFIG_MBEDTLS_TLS_ENABLED

Default value:

- 16 if (CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 || CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID) && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_KEEP_PEER_CERTIFICATE

Keep peer certificate after handshake completion

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Keep the peer's certificate after completion of the handshake. Disabling this option will save about 4kB of heap and some code size.

See mbedTLS documentation for required API and more details.

Default value:

- No (disabled)

CONFIG_MBEDTLS_SSL_CONTEXT_SERIALIZATION

Enable serialization of the TLS context structures

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable serialization of the TLS context structures This is a local optimization in handling a single, potentially long-lived connection.

See mbedTLS documentation for required API and more details. Disabling this option will save some code and RAM size.

Default value:

- No (disabled)

CONFIG_MBEDTLS_SSL_CACHE_C

Enable SSL session cache

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable simple SSL session cache implementation.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_ALL_ALERT_MESSAGES

Enable all TLS alert messages

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable all TLS alert messages in case of encountered errors as per RFC. If disabled, Mbed TLS can still communicate with other servers, only debugging of failures is harder. The advantage of not sending alert messages, is that no information is given about reasons for failures thus preventing adversaries of gaining intel.

Default value:

- Yes (enabled)

TLS Key Exchange Configuration

Contains:

- CONFIG_MBEDTLS_KEY_EXCHANGE_ECJPAKE
- CONFIG_MBEDTLS_PSK_MODES
- CONFIG_MBEDTLS_KEY_EXCHANGE_RSA
- CONFIG_MBEDTLS_KEY_EXCHANGE_ELLIPTIC_CURVE

CONFIG_MBEDTLS_PSK_MODES

Enable pre-shared-key ciphersuites

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration

Enable to show configuration for different types of pre-shared-key TLS authentatication methods.

Leaving this options disabled will save code size if they are not used.

Default value:

- No (disabled)

CONFIG_MBEDTLS_KEY_EXCHANGE_PSK

Enable PSK based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration > CONFIG_MBEDTLS_PSK_MODES

Enable to support symmetric key PSK (pre-shared-key) TLS key exchange modes.

Default value:

- No (disabled) if CONFIG_MBEDTLS_PSK_MODES && CONFIG_MBEDTLS_TLS_ENABLED && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_KEY_EXCHANGE_ECDHE_PSK

Enable ECDHE-PSK based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration > CONFIG_MBEDTLS_PSK_MODES

Enable to support Elliptic-Curve-Diffie-Hellman PSK (pre-shared-key) TLS authentication modes.

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_PSK_MODES && CONFIG_MBEDTLS_ECDH_C && CONFIG_MBEDTLS_TLS_ENABLED && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_KEY_EXCHANGE_RSA_PSK

Enable RSA-PSK based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration > CONFIG_MBEDTLS_PSK_MODES

Enable to support RSA PSK (pre-shared-key) TLS authentication modes.

Default value:

- No (disabled) if CONFIG_MBEDTLS_PSK_MODES && CONFIG_MBEDTLS_TLS_ENABLED && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_KEY_EXCHANGE_RSA

Enable RSA-only based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration

Enable to support ciphersuites with prefix TLS-RSA-WITH-

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_KEY_EXCHANGE_ELLIPTIC_CURVE

Support Elliptic Curve based ciphersuites

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration

Enable to show Elliptic Curve based ciphersuite mode options. Disabling all Elliptic Curve ciphersuites saves code size and can give slightly faster TLS handshakes, provided the server supports RSA-only ciphersuite modes.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_KEY_EXCHANGE_ECDHE_RSA

Enable ECDHE-RSA based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration > CONFIG_MBEDTLS_KEY_EXCHANGE_ELLIPTIC_CURVE

Enable to support ciphersuites with prefix TLS-ECDHE-RSA-WITH-

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_KEY_EXCHANGE_ECDHE_ECDSA

Enable ECDHE-ECDSA based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration > CONFIG_MBEDTLS_KEY_EXCHANGE_ELLIPTIC_CURVE

Enable to support ciphersuites with prefix TLS-ECDHE-ECDSA-WITH-

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_KEY_EXCHANGE_ECJPAKE

Enable ECJPAKE based ciphersuite modes

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration > TLS Key Exchange Configuration

Enable to support ciphersuites with prefix TLS-ECJPAKE-WITH-

Default value:

- No (disabled) if CONFIG_MBEDTLS_ECJPAKE_C && CONFIG_MBEDTLS_ECP_DP_SECP256R1_ENABLED && CONFIG_MBEDTLS_TLS_ENABLED && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_SERVER_NAME_INDICATION

Enable server name indication

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable support for RFC 6066 server name indication (SNI).

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_ALPN

Support ALPN (Application Layer Protocol Negotiation)

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Disabling this option will save some code size if it is not needed.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_MAX_FRAGMENT_LENGTH

Enable support for TLS max fragment length extension

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable support for the TLS max fragment length extension.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_RECORD_SIZE_LIMIT

Enable support for record size limit

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable support for record size limit in TLS 1.3.

Default value:

- No (disabled) if CONFIG_MBEDTLS_SSL_PROTO_TLS1_3 && CONFIG_MBEDTLS_TLS_ENABLED

CONFIG_MBEDTLS_SSL_VARIABLE_BUFFER_LENGTH

Variable SSL buffer length

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

This enables the SSL buffer to be resized automatically based on the negotiated maximum fragment length in each direction.

Default value:

- No (disabled)

CONFIG_MBEDTLS_SSL_RENEGOTIATION

Support TLS renegotiation

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

The two main uses of renegotiation are (1) refresh keys on long-lived connections and (2) client authentication after the initial handshake. If you don't need renegotiation, disabling it will save code size and reduce the possibility of abuse/vulnerability.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CLIENT_SSL_SESSION_TICKETS

TLS: Client Support for RFC 5077 SSL session tickets

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Client support for RFC 5077 session tickets. See mbedTLS documentation for more details. Disabling this option will save some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SERVER_SSL_SESSION_TICKETS

TLS: Server Support for RFC 5077 SSL session tickets

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Server support for RFC 5077 session tickets. See mbedTLS documentation for more details. Disabling this option will save some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SSL_KEYING_MATERIAL_EXPORT

Enable keying material export

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_TLS_ENABLED > TLS Protocol Configuration

Enable shared symmetric keys export for TLS sessions using mbedtls_ssl_export_keying_material() after SSL handshake. The process for deriving the keys is specified in RFC 5705 for TLS 1.2 and in RFC 8446, Section 7.5, for TLS 1.3.

Default value:

- No (disabled)

CONFIG_MBEDTLS_SSL_PROTO_DTLS

Support DTLS protocol (all versions)

Found in: Component config > mbedTLS

Requires TLS 1.2 to be enabled for DTLS 1.2

Default value:

- No (disabled)

DTLS-based configurations

Contains:

- CONFIG_MBEDTLS_SSL_COOKIE_C
- CONFIG_MBEDTLS_SSL_DTLS_SRTP
- CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID

CONFIG_MBEDTLS_SSL_COOKIE_C

Enable SSL session cookie

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_SSL_PROTO_DTLS > DTLS-based configurations

Enable basic DTLS cookie implementation for hello verification.

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_SSL_PROTO_DTLS

CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID

Support for the DTLS Connection ID extension

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_SSL_PROTO_DTLS > DTLS-based configurations

Enable support for the DTLS Connection ID extension which allows to identify DTLS connections across changes in the underlying transport.

Default value:

- No (disabled) if CONFIG_MBEDTLS_SSL_PROTO_DTLS

CONFIG_MBEDTLS_SSL_CID_IN_LEN_MAX

Maximum length of CIDs used for incoming DTLS messages

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_SSL_PROTO_DTLS > DTLS-based configurations > CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID

Maximum length of CIDs used for incoming DTLS messages

Range:

- from 0 to 32 if CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID && CONFIG_MBEDTLS_SSL_PROTO_DTLS

Default value:

- 32 if CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID && CONFIG_MBEDTLS_SSL_PROTO_DTLS

CONFIG_MBEDTLS_SSL_CID_OUT_LEN_MAX

Maximum length of CIDs used for outgoing DTLS messages

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_SSL_PROTO_DTLS > DTLS-based configurations > CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID

Maximum length of CIDs used for outgoing DTLS messages

Range:

- from 0 to 32 if CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID && CONFIG_MBEDTLS_SSL_PROTO_DTLS

Default value:

- 32 if CONFIG_MBEDTLS_SSL_DTLS_CONNECTION_ID && CONFIG_MBEDTLS_SSL_PROTO_DTLS

CONFIG_MBEDTLS_SSL_DTLS_SRTP

Enable support for negotiation of DTLS-SRTP (RFC 5764)

Found in: Component config > mbedTLS > CONFIG_MBEDTLS_SSL_PROTO_DTLS > DTLS-based configurations

Enable support for negotiation of DTLS-SRTP (RFC 5764) through the use_srtp extension.

See mbedTLS documentation for required API and more details. Disabling this option will save some code size.

Default value:

- No (disabled) if CONFIG_MBEDTLS_SSL_PROTO_DTLS

Symmetric Ciphers

Contains:

- CONFIG_MBEDTLS_AES_C
- CONFIG_MBEDTLS_ARIA_C
- CONFIG_MBEDTLS_CAMELLIA_C
- CONFIG_MBEDTLS_CIPHER_MODE_CBC
- CONFIG_MBEDTLS_CCM_C
- CONFIG_MBEDTLS_CIPHER_MODE_CFB
- CONFIG_MBEDTLS_CIPHER_MODE_CTR
- CONFIG_MBEDTLS_DES_C
- CONFIG_MBEDTLS_CMAC_C
- CONFIG_MBEDTLS_GCM_C

- CONFIG_MBEDTLS_NIST_KW_C
- CONFIG_MBEDTLS_CIPHER_MODE_OFB
- CONFIG_MBEDTLS_AES_ONLY_128_BIT_KEY_LENGTH
- CONFIG_MBEDTLS_AES_ROM_TABLES
- CONFIG_MBEDTLS_AES_FEWER_TABLES
- CONFIG_MBEDTLS_CAMELLIA_SMALL_MEMORY
- CONFIG_MBEDTLS_CIPHER_MODE_XTS

CONFIG_MBEDTLS_AES_C

AES block cipher

Found in: Component config > mbedTLS > Symmetric Ciphers

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CAMELLIA_C

Camellia block cipher

Found in: Component config > mbedTLS > Symmetric Ciphers

Default value:

- No (disabled)

CONFIG_MBEDTLS_ARIA_C

ARIA block cipher

Found in: Component config > mbedTLS > Symmetric Ciphers

Default value:

- No (disabled)

CONFIG_MBEDTLS_CAMELLIA_SMALL_MEMORY

Use small memory implementation of Camellia

Found in: Component config > mbedTLS > Symmetric Ciphers

Reduces ROM usage of the Camellia implementation

Default value:

- No (disabled) if CONFIG_MBEDTLS_CAMELLIA_C

CONFIG_MBEDTLS_DES_C

DES block cipher (legacy, insecure)

Found in: Component config > mbedTLS > Symmetric Ciphers

Enables the DES block cipher to support 3DES-based TLS ciphersuites.

3DES is vulnerable to the Sweet32 attack and should only be enabled if absolutely necessary.

Default value:

- No (disabled)

CONFIG_MBEDTLS_CCM_C

CCM (Counter with CBC-MAC) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Counter with CBC-MAC (CCM) modes for AES and/or Camellia ciphers.

Disabling this option saves some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CIPHER_MODE_CBC

CBC (Cipher Block Chaining) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Cipher Block Chaining (CBC) modes for AES and/or Camellia ciphers.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CIPHER_MODE_CFB

CFB (Cipher Feedback) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Cipher Feedback (CFB) modes for AES and/or Camellia ciphers.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CIPHER_MODE_CTR

CTR (Counter) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Counter (CTR) modes for AES and/or Camellia ciphers.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CIPHER_MODE_OFB

OFB (Output Feedback) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Output Feedback (OFB) modes for AES and/or Camellia ciphers.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_CIPHER_MODE_XTS

XTS (XEX Tweakable Block Cipher with Ciphertext Stealing) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable XEX Tweakable Block Cipher with Ciphertext Stealing (XTS) modes for AES and/or Camellia ciphers.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_GCM_C

GCM (Galois/Counter) block cipher modes

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable Galois/Counter Mode for AES and/or Camellia ciphers.

This option is generally faster than CCM.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_NIST_KW_C

NIST key wrapping (KW) and KW padding (KWP)

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable NIST key wrapping and key wrapping padding.

Default value:

- No (disabled)

CONFIG_MBEDTLS_AES_ROM_TABLES

Store AES tables in ROM

Found in: Component config > mbedTLS > Symmetric Ciphers

Store the AES tables in ROM instead of generating them at runtime. Using precomputed ROM tables reduces RAM usage, but increases flash usage.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_AES_FEWER_TABLES

Use fewer AES tables

Found in: Component config > mbedTLS > Symmetric Ciphers

Use fewer AES tables to reduce ROM/RAM usage. Using fewer tables increases the time taken to generate the tables at runtime, but reduces ROM/RAM usage.

Default value:

- No (disabled)

CONFIG_MBEDTLS_AES_ONLY_128_BIT_KEY_LENGTH

Only support 128-bit AES keys

Found in: Component config > mbedTLS > Symmetric Ciphers

Only support 128-bit AES keys. This reduces code size, but disables support for 192-bit and 256-bit AES keys.

Default value:

- No (disabled)

CONFIG_MBEDTLS_CMAC_C

Enable CMAC mode for block ciphers

Found in: Component config > mbedTLS > Symmetric Ciphers

Enable the CMAC (Cipher-based Message Authentication Code) mode for block ciphers.

Default value:

- Yes (enabled)

Asymmetric Ciphers

Contains:

- Elliptic Curve Ciphers Configuration
- CONFIG_MBEDTLS_ECP_C
- CONFIG_MBEDTLS_RSA_C
- Supported Curves

CONFIG_MBEDTLS_RSA_C

RSA public key cryptosystem

Found in: Component config > mbedTLS > Asymmetric Ciphers

Enable RSA. Needed to use RSA-xxx TLS ciphersuites.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECP_C

Enable Elliptic Curve Ciphers(ECC) support

Found in: Component config > mbedTLS > Asymmetric Ciphers

Default value:

- Yes (enabled)

Supported Curves

Contains:

- CONFIG_MBEDTLS_ECP_DP_BP256R1_ENABLED

- CONFIG_MBEDTLS_ECP_DP_BP384R1_ENABLED
- CONFIG_MBEDTLS_ECP_DP_BP512R1_ENABLED
- CONFIG_MBEDTLS_ECP_DP_CURVE25519_ENABLED
- CONFIG_MBEDTLS_ECP_DP_SECP256K1_ENABLED
- CONFIG_MBEDTLS_ECP_DP_SECP256R1_ENABLED
- CONFIG_MBEDTLS_ECP_DP_SECP384R1_ENABLED
- CONFIG_MBEDTLS_ECP_DP_SECP521R1_ENABLED

CONFIG_MBEDTLS_ECP_DP_SECP256R1_ENABLED

Enable SECP256R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

Enable support for SECP256R1 Elliptic Curve.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECP_DP_SECP384R1_ENABLED

Enable SECP384R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

Enable support for SECP384R1 Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_SECP521R1_ENABLED

Enable SECP521R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

Enable support for SECP521R1 Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_SECP256K1_ENABLED

Enable SECP256K1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

Enable support for SECP256K1 Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_BP256R1_ENABLED

Enable BP256R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

support for DP Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_BP384R1_ENABLED

Enable BP384R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

support for DP Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_BP512R1_ENABLED

Enable BP512R1 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

support for DP Elliptic Curve.

CONFIG_MBEDTLS_ECP_DP_CURVE25519_ENABLED

Enable CURVE25519 curve

Found in: Component config > mbedTLS > Asymmetric Ciphers > Supported Curves

Enable support for CURVE25519 Elliptic Curve.

Elliptic Curve Ciphers Configuration

Contains:

- CONFIG_MBEDTLS_DHM_C
- CONFIG_MBEDTLS_ECDH_C
- CONFIG_MBEDTLS_ECDSA_C
- CONFIG_MBEDTLS_ECJPAKE_C
- CONFIG_MBEDTLS_ECDSA_DETERMINISTIC
- CONFIG_MBEDTLS_ECP_FIXED_POINT_OPTIM
- CONFIG_MBEDTLS_ECP_RESTARTABLE
- CONFIG_MBEDTLS_PK_PARSE_EC_COMPRESSED
- CONFIG_MBEDTLS_PK_PARSE_EC_EXTENDED
- CONFIG_MBEDTLS_ECP_NIST_OPTIM
- CONFIG_MBEDTLS_ECDH_LEGACY_CONTEXT

CONFIG_MBEDTLS_ECP_NIST_OPTIM

NIST 'modulo p' optimisations

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

NIST 'modulo p' optimisations increase Elliptic Curve operation performance.

Disabling this option saves some code size.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECP_FIXED_POINT_OPTIM

Enable fixed-point multiplication optimisations

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

This configuration option enables optimizations to speedup (about 3 ~ 4 times) the ECP fixed point multiplication using pre-computed tables in the flash memory. Enabling this configuration option increases the flash footprint (about 29KB if all Elliptic Curve selected) in the application binary.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ECDH_LEGACY_CONTEXT

Use a backward compatible ECDH context (Experimental)

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Use the legacy ECDH context format. Define this option only if you enable MBEDTLS_ECP_RESTARTABLE or if you want to access ECDH context fields directly.

Default value:

- No (disabled) if CONFIG_MBEDTLS_ECDH_C && CONFIG_MBEDTLS_ECP_RESTARTABLE && CONFIG_MBEDTLS_ECP_C

CONFIG_MBEDTLS_DHM_C

Diffie-Hellman-Merkle key exchange (DHM)

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable DHM. Needed to use DHE-xxx TLS ciphersuites.

Note that the security of Diffie-Hellman key exchanges depends on a suitable prime being used for the exchange. Please see detailed warning text about this in file *mbedtls/dhm.h* file.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ECDH_C

Elliptic Curve Diffie-Hellman (ECDH)

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable ECDH. Needed to use ECDHE-xxx TLS ciphersuites.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECJPAKE_C

Elliptic curve J-PAKE

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable ECJPAKE. Needed to use ECJPAKE-xxx TLS ciphersuites.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ECDSA_C

Elliptic Curve DSA

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable ECDSA. Needed to use ECDSA-xxx TLS ciphersuites.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PK_PARSE_EC_EXTENDED

Enhance support for reading EC keys

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enhance support for reading EC keys using variants of SEC1 not allowed by RFC 5915 and RFC 5480.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PK_PARSE_EC_COMPRESSED

Enable the support for parsing public keys of type Short Weierstrass

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable the support for parsing public keys of type Short Weierstrass (MBEDTLS_ECP_DP_SECP_XXX and MBEDTLS_ECP_DP_BP_XXX) which are using the compressed point format. This parsing is done through ECP module's functions. depends on MBEDTLS_ECP_C

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECDSA_DETERMINISTIC

Enable deterministic ECDSA

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Standard ECDSA is "fragile" in the sense that lack of entropy when signing may result in a compromise of the long-term signing key.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_ECP_RESTARTABLE

Enable mbedTLS ecp restartable

Found in: Component config > mbedTLS > Asymmetric Ciphers > Elliptic Curve Ciphers Configuration

Enable "non-blocking" ECC operations that can return early and be resumed.

Default value:

- No (disabled)

Hash functions

Contains:

- CONFIG_MBEDTLS_MD_C
- CONFIG_MBEDTLS_RIPEMD160_C
- CONFIG_MBEDTLS_SHA256_SMALLER
- CONFIG_MBEDTLS_SHA512_SMALLER
- CONFIG_MBEDTLS_MD5_C
- CONFIG_MBEDTLS_SHA1_C
- CONFIG_MBEDTLS_SHA224_C
- CONFIG_MBEDTLS_SHA256_C
- CONFIG_MBEDTLS_SHA512_C
- CONFIG_MBEDTLS_SHA384_C
- CONFIG_MBEDTLS_SHA3_C
- CONFIG_MBEDTLS_ROM_MD5

CONFIG_MBEDTLS_RIPEMD160_C

Enable RIPEMD-160 hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enable the RIPEMD-160 hash algorithm.

Default value:

- No (disabled)

CONFIG_MBEDTLS_MD_C

Enable message digest support

Found in: Component config > mbedTLS > Hash functions

Enable generic layer for message digest algorithms.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_MD5_C

Enable the MD5 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enables support for MD5. This module is required for TLS 1.2 depending on the handshake parameters. Further, it is used for checking MD5-signed certificates, and for PBKDF1 when decrypting PEM-encoded encrypted keys. MD5 is considered a weak message digest and its use constitutes a security risk. If possible, consider stronger message digests such as SHA-256 (part of the SHA-2 family).

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA1_C

Enable the SHA-1 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enabling MBEDTLS_SHA1_C adds support for SHA-1. SHA-1 is considered a weak message digest and its use constitutes a security risk. Disabling this configuration option could impact TLS 1.2 / Wi-Fi Enterprise compatibility with certain older certificates that rely on SHA-1 for digital signatures. Before proceeding, ensure that all your certificates are using stronger hash algorithms, such as SHA-256 (part of the SHA-2 family). If you're using older certificates or if you're unsure about the impact on your product, please consider testing the changes in a controlled environment for individual features like OTA updates, cloud connectivity, secure local control, etc.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA224_C

Enable the SHA-224 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enable MBEDTLS_SHA224_C adds support for SHA-224.

Default value:

- No (disabled)

CONFIG_MBEDTLS_SHA256_C

Enable the SHA-256 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enable MBEDTLS_SHA256_C adds support for SHA-256.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA384_C

Enable the SHA-384 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enable MBEDTLS_SHA384_C adds support for SHA-384.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA512_C

Enable the SHA-384 and SHA-512 cryptographic hash algorithms

Found in: Component config > mbedTLS > Hash functions

Enable MBEDTLS_SHA512_C adds support for SHA-512.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA3_C

Enable the SHA3 cryptographic hash algorithm

Found in: Component config > mbedTLS > Hash functions

Enabling MBEDTLS_SHA3_C adds support for SHA3. Enabling this configuration option increases the flash footprint by almost 4KB.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ROM_MD5

Use MD5 implementation in ROM

Found in: Component config > mbedTLS > Hash functions

Use ROM MD5 in mbedTLS.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_SHA256_SMALLER

Enable SHA-256 smaller implementation

Found in: Component config > mbedTLS > Hash functions

Enable a smaller implementation of SHA-256 that has lower ROM footprint but is slower than the default implementation.

CONFIG_MBEDTLS_SHA512_SMALLER

Enable SHA-512 smaller implementation

Found in: Component config > mbedTLS > Hash functions

Enable a smaller implementation of SHA-512 that has lower ROM footprint but is slower than the default implementation.

Hardware Acceleration

Contains:

- CONFIG_MBEDTLS_HARDWARE_ECDSA_VERIFY
- CONFIG_MBEDTLS_HARDWARE_ECDSA_SIGN
- CONFIG_MBEDTLS_TEE_SEC_STG_ECDSA_SIGN
- CONFIG_MBEDTLS_HARDWARE_AES
- CONFIG_MBEDTLS_HARDWARE_ECC
- CONFIG_MBEDTLS_ATCA_HW_ECDSA_SIGN
- CONFIG_MBEDTLS_ATCA_HW_ECDSA_VERIFY
- CONFIG_MBEDTLS_HARDWARE_MPI
- CONFIG_MBEDTLS_HARDWARE_RSA_DS_PERIPHERAL
- CONFIG_MBEDTLS_HARDWARE_SHA
- CONFIG_MBEDTLS_AES_HW_SMALL_DATA_LEN_OPTIM
- Enable Software Countermeasure for ECDSA signing using on-chip ECDSA peripheral

CONFIG_MBEDTLS_HARDWARE_ECDSA_VERIFY

Enable ECDSA signature verification using on-chip ECDSA peripheral

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated ECDSA peripheral to verify signature on curve SECP192R1 and SECP256R1 in mbedTLS. Note that SECP192R1 support is disabled by default.

Default value:

- Yes (enabled) if SOC_ECDSA_SUPPORTED

Enable Software Countermeasure for ECDSA signing using on-chip ECDSA peripheral

CONFIG_MBEDTLS_HARDWARE_ECDSA_SIGN

Enable ECDSA signing using on-chip ECDSA peripheral

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated ECDSA peripheral to sign data on curve SECP192R1 and SECP256R1 in mbedTLS. Note that SECP192R1 support is disabled by default.

Note that for signing, the private key has to be burnt in an efuse key block with key purpose set to ECDSA_KEY. If no key is burnt, it will report an error

The key should be burnt in little endian format. espefuse utility handles it internally but care needs to be taken while burning using esp_efuse APIs

Default value:

- No (disabled) if SOC_ECDSA_SUPPORTED

CONFIG_MBEDTLS_TEE_SEC_STG_ECDSA_SIGN

Enable ECDSA signing using TEE secure storage

Found in: Component config > mbedTLS > Hardware Acceleration

Default value:

- Yes (enabled) if SECURE_ENABLE_TEE

CONFIG_MBEDTLS_HARDWARE_ECC

Enable hardware ECC acceleration

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated ECC point multiplication and point verification for points on curve SECP192R1 and SECP256R1 in mbedTLS Note that SECP192R1 support is disabled by default.

Default value:

- Yes (enabled) if SOC_ECC_SUPPORTED

CONFIG_MBEDTLS_ECC_OTHER_CURVES_SOFT_FALLBACK

Fallback to software implementation for curves not supported in hardware

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_ECC

Fallback to software implementation of ECC point multiplication and point verification for curves not supported in hardware.

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_HARDWARE_ECC

CONFIG_MBEDTLS_HARDWARE_SHA

Enable hardware SHA acceleration

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated SHA1, SHA256, SHA384 & SHA512 in mbedTLS.

Due to a hardware limitation, on the ESP32 hardware acceleration is only guaranteed if SHA digests are calculated one at a time. If more than one SHA digest is calculated at the same time, one will be calculated fully in hardware and the rest will be calculated (at least partially calculated) in software. This happens automatically.

SHA hardware acceleration is faster than software in some situations but slower in others. You should benchmark to find the best setting for you.

CONFIG_MBEDTLS_HARDWARE_MPI

Enable hardware MPI (bignum) acceleration

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated multiple precision integer operations.

Hardware accelerated multiplication, modulo multiplication, and modular exponentiation for up to SOC_RSA_MAX_BIT_LEN bit results.

These operations are used by RSA.

CONFIG_MBEDTLS_LARGE_KEY_SOFTWARE_MPI

Fallback to software implementation for larger MPI values

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_MPI

Fallback to software implementation for RSA key lengths larger than SOC_RSA_MAX_BIT_LEN. If this is not active then the ESP will be unable to process keys greater than SOC_RSA_MAX_BIT_LEN.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_HARDWARE_AES

Enable hardware AES acceleration

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated AES encryption & decryption.

Note that if the ESP32 CPU is running at 240MHz, hardware AES does not offer any speed boost over software AES.

CONFIG_MBEDTLS_AES_SOFT_FALLBACK

Enable software fallback for AES

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_AES

Enable software fallback for AES. This option is used to fallback to software implementation of AES if the hardware AES does not support the requested operation.

Default value:

- No (disabled)

CONFIG_MBEDTLS_HARDWARE_GCM

Enable partially hardware accelerated GCM

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_AES

Enable partially hardware accelerated GCM. GHASH calculation is still done in software.

If MBEDTLS_HARDWARE_GCM is disabled and MBEDTLS_HARDWARE_AES is enabled then mbedTLS will still use the hardware accelerated AES block operation, but on a single block at a time.

Default value:

- Yes (enabled) if SOC_AES_SUPPORT_GCM && CONFIG_MBEDTLS_HARDWARE_AES

CONFIG_MBEDTLS_GCM_SUPPORT_NON_AES_CIPHER

Enable support for non-AES ciphers in GCM operation

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_AES

Enable this config to support fallback to software definitions for a non-AES cipher GCM operation as we support hardware acceleration only for AES cipher. Some of the non-AES ciphers used in a GCM operation are DES, ARIA, CAMELLIA, CHACHA20.

If this config is disabled, performing a non-AES cipher GCM operation with the config MBEDTLS_HARDWARE_AES enabled will result in calculation of an AES-GCM operation instead for the given input values and thus could lead to failure in certificate validation which would ultimately lead to a SSL handshake failure.

This config being by-default enabled leads to an increase in binary size footprint of ~2.5KB. In case you are sure that your use case (for example, client and server settings in case of a TLS handshake) would not involve any GCM operations using a non-AES cipher, you can safely disable this config, leading to reduction in binary size footprint.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC

Enable AES hardware's pseudo round function

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_AES

Enables the pseudo round function of the AES peripheral. Enabling this would impact the performance of the AES operations. For more info regarding the performance impact, please checkout the pseudo round function section of the security guide.

Default value:

- No (disabled) if SOC_AES_SUPPORT_PSEUDO_ROUND_FUNCTION && CONFIG_MBEDTLS_HARDWARE_AES

CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC_STRENGTH

Strength of the pseudo rounds function

Found in: Component config > mbedTLS > Hardware Acceleration > CONFIG_MBEDTLS_HARDWARE_AES > CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC

The strength of the pseudo rounds functions can be configured to low, medium and high. You can configure the strength of the pseudo rounds functions according to your use cases, for example, increasing the strength would provide higher security but would slow down the hardware AES encryption/decryption operations.

Available options:

- Low (CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC_STRENGTH_LOW)
- Medium (CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC_STRENGTH_MEDIUM)
- High (CONFIG_MBEDTLS_AES_USE_PSEUDO_ROUND_FUNC_STRENGTH_HIGH)

CONFIG_MBEDTLS_AES_HW_SMALL_DATA_LEN_OPTIM

Enable performance optimisation for the small data length hardware AES operations

Found in: Component config > mbedTLS > Hardware Acceleration

This option enables dynamically switching between the hardware AES peripheral's block and DMA modes based on the length of the input data, thus, significantly speeding up the AES operations with shorter data lengths. For example, NVS encryption/decryption operations, TLS communication, etc. with smaller data lengths.

It is enabled by default due to the significant performance impact but note that it also increases the binary size by ~1.2 KB as it pulls in the peripheral's block mode code as well.

Default value:

- Yes (enabled) if CONFIG_MBEDTLS_HARDWARE_AES && SOC_AES_SUPPORT_DMA

CONFIG_MBEDTLS_ATCA_HW_ECDSA_SIGN

Enable hardware ECDSA sign acceleration when using ATECC608A

Found in: Component config > mbedTLS > Hardware Acceleration

This option enables hardware acceleration for ECDSA sign function, only when using ATECC608A cryptoauth chip.

Default value:

- No (disabled)

CONFIG_MBEDTLS_ATCA_HW_ECDSA_VERIFY

Enable hardware ECDSA verify acceleration when using ATECC608A

Found in: Component config > mbedTLS > Hardware Acceleration

This option enables hardware acceleration for ECDSA sign function, only when using ATECC608A cryptoauth chip.

Default value:

- No (disabled)

CONFIG_MBEDTLS_HARDWARE_RSA_DS_PERIPHERAL

Enable hardware RSA digital signature peripheral acceleration

Found in: Component config > mbedTLS > Hardware Acceleration

Enable hardware accelerated digital signature peripheral for RSA digital signature generation.

Default value:

- No (disabled) if SOC_DIG_SIGN_SUPPORTED

Entropy and Random Number Generation

Contains:

- CONFIG_MBEDTLS_CTR_DRBG_C
- CONFIG_MBEDTLS_HMAC_DRBG_C
- CONFIG_MBEDTLS_ENTROPY_FORCE_SHA256

CONFIG_MBEDTLS_ENTROPY_FORCE_SHA256

Force SHA-256 for entropy

Found in: Component config > mbedTLS > Entropy and Random Number Generation

Force SHA-256 to be used for the entropy pool if both SHA-256 and SHA-512 are enabled. On 32-bit architectures, SHA-256 can be faster than SHA-512

Default value:

- No (disabled)

CONFIG_MBEDTLS_CTR_DRBG_C

Enable CTR_DRBG

Found in: Component config > mbedTLS > Entropy and Random Number Generation

Enable CTR_DRBG (CTR mode Deterministic Random Bit Generator). The CTR_DRBG generator uses AES-256 by default.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_HMAC_DRBG_C

Enable HMAC_DRBG

Found in: Component config > mbedTLS > Entropy and Random Number Generation

Enable HMAC_DRBG (HMAC mode Deterministic Random Bit Generator).

Default value:

- Yes (enabled)

Encoding/Decoding

Contains:

- CONFIG_MBEDTLS_BASE64_C
- CONFIG_MBEDTLS_PKCS7_C
- CONFIG_MBEDTLS_PKCS1_V15
- CONFIG_MBEDTLS_PKCS1_V21
- CONFIG_MBEDTLS_PKCS5_C

CONFIG_MBEDTLS_BASE64_C

Enable Base64 encoding/decoding

Found in: Component config > mbedTLS > Encoding/Decoding

Enable Base64 encoding and decoding functions. This is required for PEM support.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PKCS5_C

Enable PKCS#5 functions

Found in: Component config > mbedTLS > Encoding/Decoding

Enable support for PKCS#5 functions.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PKCS7_C

Enable PKCS number 7

Found in: Component config > mbedTLS > Encoding/Decoding

Enable PKCS number 7 core for using PKCS number 7-formatted signatures.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PKCS1_V15

Enable PKCS#1 v1.5 padding

Found in: Component config > mbedTLS > Encoding/Decoding

Enable support for PKCS#1 v1.5 operations.

Default value:

- Yes (enabled)

CONFIG_MBEDTLS_PKCS1_V21

Enable PKCS#1 v2.1 padding

Found in: Component config > mbedTLS > Encoding/Decoding

Enable support for PKCS#1 v2.1 operations.

Default value:

- Yes (enabled)

Stream Cipher

Contains:

- CONFIG_MBEDTLS_CHACHA20_C
- CONFIG_MBEDTLS_CHACHAPOLY_C

CONFIG_MBEDTLS_CHACHAPOLY_C

ChaCha20-Poly1305 AEAD algorithm

Found in: Component config > mbedTLS > Stream Cipher

Enable support for ChaCha20-Poly1305 AEAD algorithm.

Default value:

- No (disabled) if CONFIG_MBEDTLS_CHACHA20_C

CONFIG_MBEDTLS_CHACHA20_C

Chacha20 stream cipher

Found in: Component config > mbedTLS > Stream Cipher

Enable support for Chacha20 stream cipher.

Default value:

- No (disabled)

CONFIG_MBEDTLS_USE_CRYPTO_ROM_IMPL_BOOTLOADER

Use ROM implementation of the crypto algorithm in the bootloader

Found in: Component config > mbedTLS

Enable this flag to use mbedtls crypto algorithm from ROM instead of ESP-IDF in case of a bootloader build. Similar to the MBEDTLS_USE_CRYPTO_ROM_IMPL config but enables usage of the mbedtls crypto algorithm from ROM for the bootloader build.

CONFIG_MBEDTLS_USE_CRYPTO_ROM_IMPL

Use ROM implementation of the crypto algorithm

Found in: Component config > mbedTLS

Enable this flag to use mbedtls crypto algorithm from ROM instead of ESP-IDF.

This configuration option saves flash footprint in the application binary. Note that the version of mbedtls crypto algorithm library in ROM(ECO1~ECO3) is v2.16.12, and the version of mbedtls crypto algorithm library in ROM(ECO4) is v3.6.0. We have done the security analysis of the mbedtls revision in ROM (ECO1~ECO4) and ensured that affected symbols have been patched (removed). If in the future mbedtls revisions there are security issues that also affects the version in ROM (ECO1~ECO4) then we shall patch the relevant symbols. This would increase the flash footprint and hence care must be taken to keep some reserved space for the application binary in flash layout.

NVS

Contains:

- CONFIG_NVS_LEGACY_DUP_KEYS_COMPATIBILITY
- CONFIG_NVS_ENCRYPTION
- CONFIG_NVS_FLASH_VERIFY_ERASE
- CONFIG_NVS_COMPATIBLE_PRE_V4_3_ENCRYPTION_FLAG
- CONFIG_NVS_ALLOCATE_CACHE_IN_SPIRAM
- CONFIG_NVS_BDL_STACK
- CONFIG_NVS_ASSERT_ERROR_CHECK

CONFIG_NVS_ENCRYPTION

Enable NVS encryption

Found in: Component config > NVS

This option enables encryption for NVS. When enabled, XTS-AES is used to encrypt the complete NVS data, except the page headers. It requires XTS encryption keys to be stored in an encrypted partition (enabling flash encryption is mandatory here) or to be derived from an HMAC key burnt in eFuse.

Default value:

- Yes (enabled) if CONFIG_SECURE_FLASH_ENC_ENABLED && (CONFIG_SECURE_FLASH_ENC_ENABLED || SOC_HMAC_SUPPORTED)

CONFIG_NVS_COMPATIBLE_PRE_V4_3_ENCRYPTION_FLAG

NVS partition encrypted flag compatible with ESP-IDF before v4.3

Found in: Component config > NVS

Enabling this will ignore "encrypted" flag for NVS partitions. NVS encryption scheme is different than hardware flash encryption and hence it is not recommended to have "encrypted" flag for NVS partitions. This was not being checked in pre v4.3 IDF. Hence, if you have any devices where this flag is kept enabled in partition table then enabling this config will allow to have same behavior as pre v4.3 IDF.

CONFIG_NVS_ASSERT_ERROR_CHECK

Use assertions for error checking

Found in: Component config > NVS

This option switches error checking type between assertions (y) or return codes (n).

Default value:

- No (disabled)

CONFIG_NVS_LEGACY_DUP_KEYS_COMPATIBILITY

Enable legacy nvs_set function behavior when same key is reused with different data types

Found in: Component config > NVS

Enabling this option will switch the nvs_set() family of functions to the legacy mode: when called repeatedly with the same key but different data type, the existing value in the NVS remains active and the new value is just stored, actually not accessible through corresponding nvs_get() call for the key given. Use this option only when your application relies on such NVS API behaviour.

Default value:

- No (disabled)

CONFIG_NVS_ALLOCATE_CACHE_IN_SPIRAM

Prefers allocation of in-memory cache structures in SPI connected PSRAM

Found in: Component config > NVS

Enabling this option lets NVS library try to allocate page cache and key hash list in SPIRAM instead of internal RAM. It can help applications using large nvs partitions or large number of keys to save heap space in internal RAM. SPIRAM heap allocation negatively impacts speed of NVS operations as the CPU accesses NVS cache via SPI instead of direct access to the internal RAM.

Default value:

- No (disabled) if CONFIG_SPIRAM && (CONFIG_SPIRAM_USE_CAPS_ALLOC || CONFIG_SPIRAM_USE_MALLOC)

CONFIG_NVS_BDL_STACK

Run NVS on BDL instead of ESP_Partition

Found in: Component config > NVS

This option enforces internal use of Block Device Layer instead ESP_Partition.

Default value:

- No (disabled)

CONFIG_NVS_FLASH_VERIFY_ERASE

Enable verification of erase operations by reading back the erased data

Found in: Component config > NVS

This option ensures that erase operations are verified by reading back the flash memory.

Default value:

- No (disabled)

CONFIG_NVS_FLASH_ERASE_ATTEMPTS

Total number of flash erase attempts in case the erase verification fails

Found in: Component config > NVS > CONFIG_NVS_FLASH_VERIFY_ERASE

Configures the total number of flash erase attempts. The flash erase operation will be attempted up to this number of times before returning an error. Minimum value is 1, maximum value is 10.

Range:

- from 1 to 10 if CONFIG_NVS_FLASH_VERIFY_ERASE

Default value:

- 2 if CONFIG_NVS_FLASH_VERIFY_ERASE

NVS Security Provider

Contains:

- CONFIG_NVS_SEC_HMAC_EFUSE_KEY_ID
- CONFIG_NVS_SEC_KEY_PROTECTION_SCHEME

CONFIG_NVS_SEC_KEY_PROTECTION_SCHEME

NVS Encryption: Key Protection Scheme

Found in: Component config > NVS Security Provider

This choice defines the default NVS encryption keys protection scheme; which will be used for the default NVS partition. Users can use the corresponding scheme registration APIs to register other schemes for the default as well as other NVS partitions.

Available options:

- Using Flash Encryption (CONFIG_NVS_SEC_KEY_PROTECT_USING_FLASH_ENC)

  Protect the NVS Encryption Keys using Flash Encryption Requires a separate 'nvs_keys' partition (which will be encrypted by flash encryption) for storing the NVS encryption keys

- Using HMAC peripheral (CONFIG_NVS_SEC_KEY_PROTECT_USING_HMAC)

  Derive and protect the NVS Encryption Keys using the HMAC peripheral Requires the specified eFuse block (NVS_SEC_HMAC_EFUSE_KEY_ID or the v2 API argument) to be empty or pre-written with a key with the purpose ESP_EFUSE_KEY_PURPOSE_HMAC_UP

- None (CONFIG_NVS_SEC_KEY_PROTECT_NONE)

  Select this option if key derivation/protection is handled by a custom implementation, and not by the nvs_sec_provider component.

CONFIG_NVS_SEC_HMAC_EFUSE_KEY_ID

eFuse key ID storing the HMAC key

Found in: Component config > NVS Security Provider

eFuse block key ID storing the HMAC key for deriving the NVS encryption keys

Note: The eFuse block key ID required by the HMAC scheme (CONFIG_NVS_SEC_KEY_PROTECT_USING_HMAC) is set using this config when the default NVS partition is initialized with nvs_flash_init(). The eFuse block key ID can also be set at runtime by passing the appropriate value to the NVS security scheme registration APIs.

Range:

- from -1 to 5 if CONFIG_NVS_SEC_KEY_PROTECT_USING_HMAC

Default value:

- "-1" if CONFIG_NVS_SEC_KEY_PROTECT_USING_HMAC

OpenThread

Contains:

- CONFIG_OPENTHREAD_DEBUG
- CONFIG_OPENTHREAD_ENABLED
- OpenThread Spinel

CONFIG_OPENTHREAD_ENABLED

OpenThread

Found in: Component config > OpenThread

Select this option to enable OpenThread and show the submenu with OpenThread configuration choices.

Default value:

- No (disabled)

Thread Task Parameters

Contains:

- CONFIG_OPENTHREAD_TASK_NAME
- CONFIG_OPENTHREAD_TASK_PRIORITY
- CONFIG_OPENTHREAD_TASK_SIZE

CONFIG_OPENTHREAD_TASK_NAME

OpenThread task name

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Task Parameters

The OpenThread task name.

Default value:

- "ot_main" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TASK_SIZE

Size of OpenThread task

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Task Parameters

The size in bytes of OpenThread task.

Default value:

- 8192 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TASK_PRIORITY

Priority of OpenThread task

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Task Parameters

The priority of OpenThread task.

Default value:

- 5 if CONFIG_OPENTHREAD_ENABLED

Thread Version Message

Contains:

- CONFIG_OPENTHREAD_PACKAGE_NAME
- CONFIG_OPENTHREAD_PLATFORM_INFO

CONFIG_OPENTHREAD_PACKAGE_NAME

OpenThread package name

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Version Message

The OpenThread package name.

Default value:

- "openthread-esp32" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PLATFORM_INFO

platform information

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Version Message

The OpenThread platform information.

Default value:

- "esp32" if CONFIG_OPENTHREAD_ENABLED

Thread Console

Contains:

- CONFIG_OPENTHREAD_CLI
- CONFIG_OPENTHREAD_CONSOLE_ENABLE
- CONFIG_OPENTHREAD_CONSOLE_COMMAND_PREFIX

CONFIG_OPENTHREAD_CONSOLE_ENABLE

Enable OpenThread console

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Console

Enable the OpenThread-specific console provided by the SDK. This only controls whether the SDK sets up a dedicated console for OpenThread. Even if disabled, the default ESP-IDF console (if initialized elsewhere) can still be used independently.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CONSOLE_TYPE

OpenThread console type

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Console > CONFIG_OPENTHREAD_CONSOLE_ENABLE

Select OpenThread console type

Available options:

- OpenThread console type UART (CONFIG_OPENTHREAD_CONSOLE_TYPE_UART)
- OpenThread console type USB Serial/JTAG Controller (CONFIG_OPENTHREAD_CONSOLE_TYPE_USB_SERIAL_JTAG)

CONFIG_OPENTHREAD_CLI

Enable Openthread Command-Line Interface

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Console

Select this option to enable Command-Line Interface in OpenThread.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CONSOLE_COMMAND_PREFIX

The prefix of the openthread CLI command registered on the esp console

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Console

A prefix string used before a Thread CLI command, allowing the ESP console to identify it and delegate the remaining command to the OpenThread callback for processing.

Default value:

- "ot" if CONFIG_OPENTHREAD_ENABLED

Thread Core Features

Contains:

- CSL Configurations

- CONFIG_OPENTHREAD_BORDER_AGENT_ENABLE
- CONFIG_OPENTHREAD_BORDER_ROUTER
- CONFIG_OPENTHREAD_COMMISSIONER
- CONFIG_OPENTHREAD_CSL_ENABLE
- CONFIG_OPENTHREAD_DIAG
- CONFIG_OPENTHREAD_DNS_CLIENT
- CONFIG_OPENTHREAD_DNS64_CLIENT
- CONFIG_OPENTHREAD_PLATFORM_NETIF
- CONFIG_OPENTHREAD_JOINER
- CONFIG_OPENTHREAD_LINK_METRICS
- CONFIG_OPENTHREAD_MACFILTER_ENABLE
- CONFIG_OPENTHREAD_RX_ON_WHEN_IDLE
- CONFIG_OPENTHREAD_PARENT_SEARCH_MTD
- CONFIG_OPENTHREAD_RADIO_STATS_ENABLE
- CONFIG_OPENTHREAD_SRP_CLIENT
- CONFIG_OPENTHREAD_TIME_SYNC
- CONFIG_OPENTHREAD_TIMING_OPTIMIZATION
- OpenThread Stack Parameters
- Parent Search Configurations
- Thread 15.4 Radio Link
- CONFIG_OPENTHREAD_DEVICE_TYPE
- Thread Memory Allocation
- Thread Operational Dataset
- Thread Radio Co-Processor Feature
- Thread Trel Radio Link

Thread Operational Dataset

Contains:

- CONFIG_OPENTHREAD_NETWORK_EXTPANID
- CONFIG_OPENTHREAD_MESH_LOCAL_PREFIX
- CONFIG_OPENTHREAD_NETWORK_CHANNEL
- CONFIG_OPENTHREAD_NETWORK_MASTERKEY
- CONFIG_OPENTHREAD_NETWORK_NAME
- CONFIG_OPENTHREAD_NETWORK_PANID
- CONFIG_OPENTHREAD_NETWORK_PSKC

CONFIG_OPENTHREAD_NETWORK_NAME

OpenThread network name

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

Default value:

- "OpenThread-ESP" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_MESH_LOCAL_PREFIX

OpenThread mesh local prefix, format <address>/<plen>

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

A string in the format "<address>/<plen>", where *<address>* is an IPv6 address and *<plen>* is a prefix length. For example "fd00:db8:a0:0::/64"

Default value:

- "fd00:db8:a0:0::/64" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NETWORK_CHANNEL

OpenThread network channel

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

Range:

- from 11 to 26 if CONFIG_OPENTHREAD_ENABLED

Default value:

- 15 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NETWORK_PANID

OpenThread network pan id

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

Range:

- from 0 to 0xFFFE if CONFIG_OPENTHREAD_ENABLED

Default value:

- "0x1234" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NETWORK_EXTPANID

OpenThread extended pan id

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

The OpenThread network extended pan id in hex string format

Default value:

- dead00beef00cafe if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NETWORK_MASTERKEY

OpenThread network key

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

The OpenThread network network key in hex string format

Default value:

- 00112233445566778899aabbccddeeff if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NETWORK_PSKC

OpenThread pre-shared commissioner key

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Operational Dataset

The OpenThread pre-shared commissioner key in hex string format

Default value:

- 104810e2315100afd6bc9215a6bfac53 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_DEVICE_TYPE

Thread device type

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

OpenThread can be configured to different device types (FTD, MTD, Radio)

Available options:

- Full Thread Device (CONFIG_OPENTHREAD_FTD)

  Select this to enable Full Thread Device which can act as router and leader in a Thread network.

- Minimal Thread Device (CONFIG_OPENTHREAD_MTD)

  Select this to enable Minimal Thread Device which can only act as end device in a Thread network. This will reduce the code size of the OpenThread stack.

- Radio Only Device (CONFIG_OPENTHREAD_RADIO)

  Select this to enable Radio Only Device which can only forward 15.4 packets to the host. The OpenThread stack will be run on the host and OpenThread will have minimal footprint on the radio only device.

Thread Trel Radio Link

Contains:

- CONFIG_OPENTHREAD_RADIO_TREL

CONFIG_OPENTHREAD_RADIO_TREL

Enable Thread Radio Encapsulation Link (TREL)

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Trel Radio Link

Select this option to enable Thread Radio Encapsulation Link.

Default value:

- No (disabled) if (EXAMPLE_CONNECT_WIFI || EXAMPLE_CONNECT_ETHERNET) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TREL_PORT

The port of openthread trel service

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Trel Radio Link > CONFIG_OPENTHREAD_RADIO_TREL

Configure the port number of TREL service.

Default value:

- 12390 if CONFIG_OPENTHREAD_RADIO_TREL && (EXAMPLE_CONNECT_WIFI || EXAMPLE_CONNECT_ETHERNET) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TREL_BUFFER_SIZE

The receive buffer size of openthread trel

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Trel Radio Link > CONFIG_OPENTHREAD_RADIO_TREL

Configure the receive buffer size of TREL service.

Range:

- from 10 to 255 if CONFIG_OPENTHREAD_RADIO_TREL && (EXAMPLE_CONNECT_WIFI || EXAMPLE_CONNECT_ETHERNET) && CONFIG_OPENTHREAD_ENABLED

Default value:

- 50 if CONFIG_OPENTHREAD_RADIO_TREL && (EXAMPLE_CONNECT_WIFI || EXAMPLE_CONNECT_ETHERNET) && CONFIG_OPENTHREAD_ENABLED

Thread 15.4 Radio Link

Contains:

- CONFIG_OPENTHREAD_RADIO_TYPE

CONFIG_OPENTHREAD_RADIO_TYPE

Config the Thread radio type with 15.4 link

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread 15.4 Radio Link

Configure how OpenThread connects to the 15.4 radio

Available options:

- Native 15.4 radio (CONFIG_OPENTHREAD_RADIO_NATIVE)

  Select this to use the native 15.4 radio.

- Connect via UART (CONFIG_OPENTHREAD_RADIO_SPINEL_UART)

  Select this to connect to a Radio Co-Processor via UART.

- Connect via SPI (CONFIG_OPENTHREAD_RADIO_SPINEL_SPI)

  Select this to connect to a Radio Co-Processor via SPI.

- Disable the Thread radio based on 15.4 link (CONFIG_OPENTHREAD_RADIO_154_NONE)

  Select this to disable the Thread radio based on 15.4 link.

Thread Radio Co-Processor Feature

Contains:

- CONFIG_OPENTHREAD_RCP_SPINEL_CONSOLE
- CONFIG_OPENTHREAD_NCP_VENDOR_HOOK
- CONFIG_OPENTHREAD_RCP_TRANSPORT

CONFIG_OPENTHREAD_RCP_TRANSPORT

The RCP transport type

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Radio Co-Processor Feature

Available options:

- UART RCP (CONFIG_OPENTHREAD_RCP_UART)

  Select this to enable UART connection to host.

- SPI RCP (CONFIG_OPENTHREAD_RCP_SPI)

  Select this to enable SPI connection to host.

- USB RCP (CONFIG_OPENTHREAD_RCP_USB_SERIAL_JTAG)

  Select this to enable connection to host over USB JTAG serial.

CONFIG_OPENTHREAD_NCP_VENDOR_HOOK

Enable vendor command for RCP

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Radio Co-Processor Feature

Select this to enable OpenThread NCP vendor commands.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_RADIO && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_RCP_SPINEL_CONSOLE

Enable RCP console via Spinel

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Radio Co-Processor Feature

Select this to enable sending console commands to OpenThread RCP via Spinel.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_RADIO && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_BORDER_ROUTER

Enable Border Router

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable border router features in OpenThread.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_FTD && CONFIG_OPENTHREAD_PLATFORM_NETIF && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_COMMISSIONER

Enable Commissioner

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable commissioner in OpenThread. This will enable the device to act as a commissioner in the Thread network. A commissioner checks the pre-shared key from a joining device with the Thread commissioning protocol and shares the network parameter with the joining device upon success.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

Commissioner Configurations

Contains:

- CONFIG_OPENTHREAD_COMM_MAX_JOINER_ENTRIES

CONFIG_OPENTHREAD_COMM_MAX_JOINER_ENTRIES

The size of max commissioning joiner entries

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CONFIG_OPENTHREAD_COMMISSIONER > Commissioner Configurations

Default value:

- 2 if CONFIG_OPENTHREAD_COMMISSIONER && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_JOINER

Enable Joiner

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable Joiner in OpenThread. This allows a device to join the Thread network with a pre-shared key using the Thread commissioning protocol.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_SRP_CLIENT

Enable SRP Client

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable SRP Client in OpenThread. This allows a device to register SRP services to SRP Server.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

SRP Client Configurations

Contains:

- CONFIG_OPENTHREAD_SRP_CLIENT_MAX_SERVICES

CONFIG_OPENTHREAD_SRP_CLIENT_MAX_SERVICES

Specifies number of service entries in the SRP client service pool

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CONFIG_OPENTHREAD_SRP_CLIENT > SRP Client Configurations

Set the max buffer size of service entries in the SRP client service pool.

Default value:

- 5 if CONFIG_OPENTHREAD_SRP_CLIENT && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_DNS_CLIENT

Enable DNS Client

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable DNS Client in OpenThread.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_DNS64_CLIENT

Enable DNS64 Client

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to acquire NAT64 address from dns servers.

Default value:

- No (disabled) if CONFIG_LWIP_IPV4 && CONFIG_OPENTHREAD_PLATFORM_NETIF && CONFIG_OPENTHREAD_ENABLED

DNS64 Client Configurations

Contains:

- CONFIG_OPENTHREAD_DNS_SERVER_ADDR

CONFIG_OPENTHREAD_DNS_SERVER_ADDR

DNS server address (IPv4)

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CONFIG_OPENTHREAD_DNS64_CLIENT > DNS64 Client Configurations

Set the DNS server IPv4 address.

Default value:

- "8.8.8.8" if CONFIG_OPENTHREAD_DNS64_CLIENT && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TIMING_OPTIMIZATION

Enable timing optimization

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable timing optimization for link metrics / CSL features.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_LINK_METRICS

Enable link metrics feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable link metrics feature

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_BORDER_AGENT_ENABLE

Enable border agent feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable border agent feature

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_BORDER_ROUTER && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_MACFILTER_ENABLE

Enable mac filter feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable mac filter feature

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CSL_ENABLE

Enable CSL feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable CSL feature

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CSL Configurations

Contains:

- CONFIG_OPENTHREAD_CSL_DEBUG_ENABLE
- CONFIG_OPENTHREAD_CSL_UNCERTAIN
- CONFIG_OPENTHREAD_CSL_ACCURACY

CONFIG_OPENTHREAD_CSL_ACCURACY

The current CSL rx/tx scheduling drift, in units of ± ppm

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CSL Configurations

The current accuracy of the clock used for scheduling CSL operations

Default value:

- 50 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CSL_UNCERTAIN

The CSL Uncertainty in units of 10 us.

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CSL Configurations

The fixed uncertainty of the Device for scheduling CSL Transmissions in units of 10 microseconds.

Default value:

- 50 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CSL_DEBUG_ENABLE

Enable CSL debug

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > CSL Configurations

Select this option to set rx on when sleep in CSL feature, only for debug

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TIME_SYNC

Enable the time synchronization service feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable time synchronization feature, the devices in the same Thread network could sync to the same network time.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_RADIO_STATS_ENABLE

Enable Radio Statistics feature

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable the radio statistics feature, you can use radio command to print some radio Statistics information.

Default value:

- No (disabled) if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_MTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_RX_ON_WHEN_IDLE

Enable OpenThread radio capability rx on when idle

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable OpenThread radio capability rx on when idle. Do not support this feature when SW coexistence is enabled.

Default value:

- No (disabled) if (CONFIG_ESP_COEX_SW_COEXIST_ENABLE || CONFIG_ESP_COEX_EXTERNAL_COEXIST_ENABLE) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_DIAG

Enable diag

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable Diag in OpenThread. This will enable diag mode and a series of diag commands in the OpenThread command line. These commands allow users to manipulate low-level features of the storage and 15.4 radio.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PARENT_SEARCH_MTD

Enable Parent Search

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Select this option to enable "Periodic Parent Search" function for MTD. This checks the average RSS to its current parent every periodically and starts a parent search process if the average RSS is below OPENTHREAD_PARENT_SEARCH_RSS_THRESHOLD. This feature is always enabled for FTDs.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_MTD && CONFIG_OPENTHREAD_ENABLED

Parent Search Configurations

Contains:

- CONFIG_OPENTHREAD_PARENT_SEARCH_BACKOFF_INTERVAL_MINS
- CONFIG_OPENTHREAD_PARENT_SEARCH_CHECK_INTERVAL_MINS
- CONFIG_OPENTHREAD_PARENT_SEARCH_RESELECT_TIMEOUT_MINS
- CONFIG_OPENTHREAD_PARENT_SEARCH_RSS_MARGIN
- CONFIG_OPENTHREAD_PARENT_SEARCH_RSS_THRESHOLD

CONFIG_OPENTHREAD_PARENT_SEARCH_CHECK_INTERVAL_MINS

The interval in minutes for a child to check the trigger condition to perform a parent search

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Parent Search Configurations

Default value:

- 9 if (CONFIG_OPENTHREAD_PARENT_SEARCH_MTD || CONFIG_OPENTHREAD_FTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PARENT_SEARCH_BACKOFF_INTERVAL_MINS

The backoff interval in minutes for a child to not perform a parent search after triggering one

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Parent Search Configurations

Default value:

- 600 if (CONFIG_OPENTHREAD_PARENT_SEARCH_MTD || CONFIG_OPENTHREAD_FTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PARENT_SEARCH_RSS_THRESHOLD

The RSS threshold used to trigger a parent search

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Parent Search Configurations

Default value:

- "-65" if (CONFIG_OPENTHREAD_PARENT_SEARCH_MTD || CONFIG_OPENTHREAD_FTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PARENT_SEARCH_RESELECT_TIMEOUT_MINS

The parent reselect timeout duration in minutes used on FTD child devices

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Parent Search Configurations

Default value:

- 90 if (CONFIG_OPENTHREAD_PARENT_SEARCH_MTD || CONFIG_OPENTHREAD_FTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PARENT_SEARCH_RSS_MARGIN

The RSS margin over the current parent RSS used on FTD child devices

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Parent Search Configurations

Default value:

- 7 if (CONFIG_OPENTHREAD_PARENT_SEARCH_MTD || CONFIG_OPENTHREAD_FTD) && CONFIG_OPENTHREAD_ENABLED

Thread Memory Allocation

Contains:

- CONFIG_OPENTHREAD_PLATFORM_MALLOC_CAP_SPIRAM

CONFIG_OPENTHREAD_PLATFORM_MALLOC_CAP_SPIRAM

Allocate memory from PSRAM

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Memory Allocation

Select this option to allocate buffer from PSRAM for Thread

Default value:

- Yes (enabled) if (CONFIG_SPIRAM_USE_CAPS_ALLOC || CONFIG_SPIRAM_USE_MALLOC) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PLATFORM_MSGPOOL_MANAGEMENT

Allocate message pool buffer from PSRAM

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > Thread Memory Allocation > CONFIG_OPENTHREAD_PLATFORM_MALLOC_CAP_SPIRAM

If enabled, the message pool is managed by platform defined logic.

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_PLATFORM_MALLOC_CAP_SPIRAM && (CONFIG_SPIRAM_USE_CAPS_ALLOC || CONFIG_SPIRAM_USE_MALLOC) && CONFIG_OPENTHREAD_ENABLED

OpenThread Stack Parameters

Contains:

- CONFIG_OPENTHREAD_MAC_MAX_CSMA_BACKOFFS_DIRECT
- CONFIG_OPENTHREAD_PREFERRED_CHANNEL_MASK
- CONFIG_OPENTHREAD_SUPPORTED_CHANNEL_MASK
- CONFIG_OPENTHREAD_XTAL_ACCURACY
- CONFIG_OPENTHREAD_BUS_LATENCY
- CONFIG_OPENTHREAD_NUM_MESSAGE_BUFFERS
- CONFIG_OPENTHREAD_MLE_MAX_CHILDREN
- CONFIG_OPENTHREAD_TMF_ADDR_CACHE_ENTRIES
- CONFIG_OPENTHREAD_UART_BUFFER_SIZE
- Thread Address Query Config

Thread Address Query Config

Contains:

- CONFIG_OPENTHREAD_ADDRESS_QUERY_RETRY_DELAY
- CONFIG_OPENTHREAD_ADDRESS_QUERY_MAX_RETRY_DELAY
- CONFIG_OPENTHREAD_ADDRESS_QUERY_TIMEOUT

CONFIG_OPENTHREAD_ADDRESS_QUERY_TIMEOUT

Timeout (in seconds) for a address notification response after sending an address query.

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters > Thread Address Query Config

Default value:

- 3 if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_MTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_ADDRESS_QUERY_RETRY_DELAY

Initial retry delay for address query (in seconds).

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters > Thread Address Query Config

Default value:

- 15 if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_MTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_ADDRESS_QUERY_MAX_RETRY_DELAY

Maximum retry delay for address query (in seconds).

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters > Thread Address Query Config

Default value:

- 120 if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_MTD) && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PREFERRED_CHANNEL_MASK

Preferred channel mask

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Default value:

- "0x7fff800" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_SUPPORTED_CHANNEL_MASK

Supported channel mask

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Default value:

- "0x7fff800" if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_NUM_MESSAGE_BUFFERS

The number of openthread message buffers

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Default value:

- 1024 if CONFIG_OPENTHREAD_PLATFORM_MSGPOOL_MANAGEMENT && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_XTAL_ACCURACY

The accuracy of the XTAL

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

The device's XTAL accuracy, in ppm.

Default value:

- 10 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_BUS_LATENCY

The bus latency between host and radio chip

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

The device's bus latency, in us.

Default value:

- 4000 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_MLE_MAX_CHILDREN

The size of max MLE children entries

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Default value:

- 10 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_TMF_ADDR_CACHE_ENTRIES

The size of max TMF address cache entries

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Default value:

- 20 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_UART_BUFFER_SIZE

The uart received buffer size of openthread

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

Set the OpenThread UART buffer size.

Default value:

- 2048 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_MAC_MAX_CSMA_BACKOFFS_DIRECT

Maximum backoffs times before declaring a channel access failure.

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features > OpenThread Stack Parameters

The maximum number of backoffs the CSMA-CA algorithm will attempt before declaring a channel access failure.

Default value:

- 4 if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_PLATFORM_NETIF

Enable esp_netif/lwIP glue

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Core Features

Enable the default ESP-IDF OpenThread integration which creates an OpenThread esp_netif, attaches the esp_netif glue, and enables lwIP-specific DNS/netif handling.

Default value:

- Yes (enabled) if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_MTD) && CONFIG_OPENTHREAD_ENABLED

Thread Log

Contains:

- CONFIG_OPENTHREAD_LOG_LEVEL_DYNAMIC
- CONFIG_OPENTHREAD_LOG_LEVEL

CONFIG_OPENTHREAD_LOG_LEVEL_DYNAMIC

Enable dynamic log level control

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Log

Select this option to enable dynamic log level control for OpenThread

Default value:

- Yes (enabled) if CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_LOG_LEVEL

OpenThread log verbosity

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Log

Select OpenThread log level.

Available options:

- No logs (CONFIG_OPENTHREAD_LOG_LEVEL_NONE)
- Error logs (CONFIG_OPENTHREAD_LOG_LEVEL_CRIT)
- Warning logs (CONFIG_OPENTHREAD_LOG_LEVEL_WARN)
- Notice logs (CONFIG_OPENTHREAD_LOG_LEVEL_NOTE)
- Info logs (CONFIG_OPENTHREAD_LOG_LEVEL_INFO)
- Debug logs (CONFIG_OPENTHREAD_LOG_LEVEL_DEBG)

Thread Extensioned Features

Contains:

- CONFIG_OPENTHREAD_HEADER_CUSTOM

CONFIG_OPENTHREAD_HEADER_CUSTOM

Use a header file defined by customer

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Extensioned Features

This option allows users to tailor the values of openthread macros according to their requirements. The openthread submodule contains numerous macros, each with a default value set. In the Kconfig of ESP openthread, users can set specific openthread parameters, which will be applied for certain openthread macros in the openthread-core-esp32x-xxx-config.h file. During compilation, the values specified in openthread-core-esp32x-xxx-config.h will replace the default settings in the openthread submodule. However, Kconfig does not cover all openthread macros, particularly those typically using default values. For such macros, users can enable the OPENTHREAD_HEADER_CUSTOM option in the Kconfig and provide a custom header file. Macros defined in the custom header file will have the highest priority.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_ENABLED

OpenThread Custom Header Config

Contains:

- CONFIG_OPENTHREAD_CUSTOM_HEADER_FILE_NAME
- CONFIG_OPENTHREAD_CUSTOM_HEADER_PATH

CONFIG_OPENTHREAD_CUSTOM_HEADER_PATH

Path of custom header file

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Extensioned Features > CONFIG_OPENTHREAD_HEADER_CUSTOM > OpenThread Custom Header Config

Please use relative paths with respect to the project folder.

Default value:

- if CONFIG_OPENTHREAD_HEADER_CUSTOM && CONFIG_OPENTHREAD_HEADER_CUSTOM && CONFIG_OPENTHREAD_ENABLED

CONFIG_OPENTHREAD_CUSTOM_HEADER_FILE_NAME

Name of custom header file

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_ENABLED > Thread Extensioned Features > CONFIG_OPENTHREAD_HEADER_CUSTOM > OpenThread Custom Header Config

Name of custom header file.

Default value:

- "esp_ot_custom_config.h" if CONFIG_OPENTHREAD_HEADER_CUSTOM && CONFIG_OPENTHREAD_HEADER_CUSTOM && CONFIG_OPENTHREAD_ENABLED

OpenThread Spinel

Contains:

- CONFIG_OPENTHREAD_SPINEL_ONLY
- CONFIG_OPENTHREAD_SPINEL_MAC_MAX_CSMA_BACKOFFS_DIRECT
- CONFIG_OPENTHREAD_SPINEL_RX_FRAME_BUFFER_SIZE

- CONFIG_OPENTHREAD_SPINEL_UART_DRIVER_BUFFER_SIZE

CONFIG_OPENTHREAD_SPINEL_ONLY

Enable OpenThread External Radio Spinel feature

Found in: Component config > OpenThread > OpenThread Spinel

Select this option to enable the OpenThread Radio Spinel for external protocol stack, such as Zigbee.

Default value:

- No (disabled)

CONFIG_OPENTHREAD_SPINEL_RX_FRAME_BUFFER_SIZE

The size of openthread spinel rx frame buffer

Found in: Component config > OpenThread > OpenThread Spinel

Default value:

- 1024 if (CONFIG_OPENTHREAD_MTD || CONFIG_OPENTHREAD_RADIO) && (CONFIG_OPENTHREAD_ENABLED || CONFIG_OPENTHREAD_SPINEL_ONLY)
- 2048 if (CONFIG_OPENTHREAD_FTD || CONFIG_OPENTHREAD_SPINEL_ONLY) && (CONFIG_OPENTHREAD_ENABLED || CONFIG_OPENTHREAD_SPINEL_ONLY)

CONFIG_OPENTHREAD_SPINEL_MAC_MAX_CSMA_BACKOFFS_DIRECT

Maximum backoffs times before declaring a channel access failure.

Found in: Component config > OpenThread > OpenThread Spinel

The maximum number of backoffs the CSMA-CA algorithm will attempt before declaring a channel access failure.

Default value:

- 4 if CONFIG_OPENTHREAD_SPINEL_ONLY

CONFIG_OPENTHREAD_SPINEL_UART_DRIVER_BUFFER_SIZE

UART driver buffer size for radio spinel interface

Found in: Component config > OpenThread > OpenThread Spinel

RX buffer size passed to uart_driver_install() for the radio spinel UART interface.

Default value:

- 1024 if CONFIG_OPENTHREAD_SPINEL_ONLY || CONFIG_OPENTHREAD_RADIO_SPINEL_UART

CONFIG_OPENTHREAD_DEBUG

Enable ESP OpenThread Debug

Found in: Component config > OpenThread

Enable additional debugging support for ESP OpenThread integration. This includes various diagnostic tools and logs to help track down issues in OpenThread networking or system integration

Default value:

- No (disabled)

Contains:

- CONFIG_OPENTHREAD_DUMP_MAC_ON_ASSERT
- CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR

CONFIG_OPENTHREAD_DUMP_MAC_ON_ASSERT

Dump 802.15.4 MAC debug info on OpenThread assert

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_DEBUG

When enabled, this option triggers the printing of 802.15.4 MAC layer debug information whenever an OpenThread assert occurs. This can help developers analyze unexpected failures by providing additional MAC layer context.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_DEBUG && CONFIG_OPENTHREAD_RADIO_NATIVE && CONFIG_IEEE802154_DEBUG

CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR

Enable OpenThread Task Block Monitor

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_DEBUG

Enable monitoring of OpenThread tasks to detect if a task is blocked and unable to continue its normal execution loop. Useful for debugging issues where tasks stop progressing due to deadlocks or resource waits. On RISC-V targets, in order to get meaningful backtraces when a task is detected as blocked, ESP_SYSTEM_USE_FRAME_POINTER must be enabled in ESP System Settings -> Backtracing method.

Default value:

- No (disabled) if CONFIG_OPENTHREAD_DEBUG

CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR_TIMEOUT

Task Block Monitor Timeout (seconds)

Found in: Component config > OpenThread > CONFIG_OPENTHREAD_DEBUG > CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR

Set the timeout (in seconds) for detecting a blocked task. The timer is started at the beginning of each task main loop and stopped at the end. If the timer expires, the task is considered blocked.

Range:

- from 1 to 60 if CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR

Default value:

- 20 if CONFIG_OPENTHREAD_TASK_BLOCK_MONITOR

Protocomm

Contains:

- CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_0
- CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_1
- CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_2

CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_0

Support protocomm security version 0 (no security)

Found in: Component config > Protocomm

Enable support of security version 0. This version provides no encryption or authentication and should not be used in production. Use only for development and testing. Disabling this option saves some code size. Consult the Enabling protocomm security version section of the Protocomm documentation in ESP-IDF Programming guide for more details.

Default value:

- No (disabled)

CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_1

Support protocomm security version 1 (Curve25519 key exchange + AES-CTR encryption/decryption)

Found in: Component config > Protocomm

Enable support of security version 1. Security version 2 (SRP6a + AES-GCM) is recommended over this version for new designs. Disabling this option saves some code size. Consult the Enabling protocomm security version section of the Protocomm documentation in ESP-IDF Programming guide for more details.

Default value:

- No (disabled)

CONFIG_ESP_PROTOCOMM_SUPPORT_SECURITY_VERSION_2

Support protocomm security version 2 (SRP6a-based key exchange + AES-GCM encryption/decryption)

Found in: Component config > Protocomm

Enable support of security version 2. Disabling this option saves some code size. Consult the Enabling protocomm security version section of the Protocomm documentation in ESP-IDF Programming guide for more details.

Default value:

- Yes (enabled)

PThreads

Contains:

- CONFIG_PTHREAD_TASK_NAME_DEFAULT
- CONFIG_PTHREAD_TASK_CORE_DEFAULT
- CONFIG_PTHREAD_TASK_PRIO_DEFAULT
- CONFIG_PTHREAD_TASK_STACK_SIZE_DEFAULT
- CONFIG_PTHREAD_STACK_MIN

CONFIG_PTHREAD_TASK_PRIO_DEFAULT

Default task priority

Found in: Component config > PThreads

Priority used to create new tasks with default pthread parameters.

Range:

- from 0 to 255

Default value:

- 5

CONFIG_PTHREAD_TASK_STACK_SIZE_DEFAULT

Default task stack size

Found in: Component config > PThreads

Stack size used to create new tasks with default pthread parameters.

Default value:

- 3072

CONFIG_PTHREAD_STACK_MIN

Minimum allowed pthread stack size

Found in: Component config > PThreads

Minimum allowed pthread stack size set in attributes passed to pthread_create

Default value:

- 768

CONFIG_PTHREAD_TASK_CORE_DEFAULT

Default pthread core affinity

Found in: Component config > PThreads

The default core to which pthreads are pinned.

Available options:

- No affinity (CONFIG_PTHREAD_DEFAULT_CORE_NO_AFFINITY)
- Core 0 (CONFIG_PTHREAD_DEFAULT_CORE_0)
- Core 1 (CONFIG_PTHREAD_DEFAULT_CORE_1)

CONFIG_PTHREAD_TASK_NAME_DEFAULT

Default name of pthreads

Found in: Component config > PThreads

The default name of pthreads.

Default value:

- "pthread"

SD Protocol Layer Configuration

Contains:

- CONFIG_SD_ENABLE_SDIO_SUPPORT

CONFIG_SD_ENABLE_SDIO_SUPPORT

Enable SDIO support

Found in: Component config > SD Protocol Layer Configuration

Enable SDIO support. Disabling this will skip SDIO-specific initialization steps

Default value:

- Yes (enabled)

SoC Settings

Contains:

- MMU Config

MMU Config

Main Flash configuration

Contains:

- Optional and Experimental Features (READ DOCS FIRST)
- SPI Flash behavior when brownout

SPI Flash behavior when brownout

Contains:

- CONFIG_SPI_FLASH_BROWNOUT_RESET_XMC

CONFIG_SPI_FLASH_BROWNOUT_RESET_XMC

Enable sending reset when brownout for XMC flash chips

Found in: Component config > Main Flash configuration > SPI Flash behavior when brownout

When this option is selected, the patch will be enabled for XMC. Follow the recommended flow by XMC for better stability.

DO NOT DISABLE UNLESS YOU KNOW WHAT YOU ARE DOING.

Optional and Experimental Features (READ DOCS FIRST)

Contains:

- CONFIG_SPI_FLASH_AUTO_SUSPEND
- CONFIG_SPI_FLASH_AUTO_CHECK_SUSPEND_STATUS
- CONFIG_SPI_FLASH_DISABLE_SCHEDULER_IN_SUSPEND
- CONFIG_SPI_FLASH_FORCE_ENABLE_C6_H2_SUSPEND
- CONFIG_SPI_FLASH_FORCE_ENABLE_XMC_C_SUSPEND
- CONFIG_SPI_FLASH_HPM
- CONFIG_SPI_FLASH_PLACE_FUNCTIONS_IN_IRAM
- CONFIG_SPI_FLASH_SOFTWARE_RESUME
- CONFIG_SPI_FLASH_SUSPEND_TRS_VAL_US
- CONFIG_SPI_FLASH_SUSPEND_TSUS_VAL_US
- CONFIG_SPI_FLASH_HPM_DC

CONFIG_SPI_FLASH_HPM

High Performance Mode (READ DOCS FIRST, > 80MHz)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

Whether the High Performance Mode of Flash is enabled. As an optional feature, user needs to manually enable this option as a confirmation. To be back-compatible with earlier IDF version, this option is automatically enabled with warning when Flash running > 80Mhz.

Available options:

- Enable (CONFIG_SPI_FLASH_HPM_ENA)
- Auto (Not recommended) (CONFIG_SPI_FLASH_HPM_AUTO)
- Disabled (CONFIG_SPI_FLASH_HPM_DIS)

CONFIG_SPI_FLASH_HPM_DC

Support HPM using DC (READ DOCS FIRST)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

This feature needs your bootloader to be compiled DC-aware (BOOTLOADER_FLASH_DC_AWARE=y). Otherwise the chip will not be able to boot after a reset.

Available options:

- Auto (Enable when bootloader support enabled (BOOTLOADER_FLASH_DC_AWARE)) (CONFIG_SPI_FLASH_HPM_DC_AUTO)
- Disable (READ DOCS FIRST) (CONFIG_SPI_FLASH_HPM_DC_DISABLE)

CONFIG_SPI_FLASH_AUTO_SUSPEND

Auto suspend long erase/write operations (READ DOCS FIRST)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

This option is disabled by default because it is supported only for specific flash chips and for specific Espressif chips. To evaluate if you can use this feature refer to *Optional Features for Flash > Auto Suspend & Resume* of the *ESP-IDF Programming Guide*.

CAUTION: If you want to OTA to an app with this feature turned on, please make sure the bootloader has the support for it. (later than IDF v4.3)

If you are using an official Espressif module, please contact Espressif Business support to check if the module has the flash that support this feature installed. Also refer to *Concurrency Constraints for Flash on SPI1 > Flash Auto Suspend Feature* before enabling this option.

CONFIG_SPI_FLASH_SUSPEND_TSUS_VAL_US

SPI flash tSUS value (refer to chapter AC CHARACTERISTICS)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

This config is used for setting Tsus parameter. Tsus means CS# high to next command after suspend. You can refer to the chapter of AC CHARACTERISTICS of flash datasheet.

CONFIG_SPI_FLASH_SUSPEND_TRS_VAL_US

SPI flash tRS value (refer to chapter AC CHARACTERISTICS)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

This config is used for setting Trs parameter. Trs means CS Latency Between Resume And Next Suspend. You can refer to the chapter of AC CHARACTERISTICS of flash datasheet. For high-performance scenarios, some flash chips allow this set value to be smaller than the given value in the datasheet without causing errors in the flash state machine. When you have any related needs, please contact espressif business team.

CONFIG_SPI_FLASH_FORCE_ENABLE_XMC_C_SUSPEND

Enable XMC-C series flash chip suspend feature anyway

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

XMC-C series is regarded as not qualified for the Suspend feature, since its specification has a tRS >= 1ms restriction. We strongly do not suggest using it for the Suspend feature. However, if your product in field has enabled this feature, you may still enable this config option to keep the legacy behavior.

For new users, DO NOT enable this config.

CONFIG_SPI_FLASH_FORCE_ENABLE_C6_H2_SUSPEND

Enable chip suspend feature on c6 or h2 anyway (DO NOT ENABLE FOR NEW USERS OR APPLICATIONS)

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

Flash suspend has a defect on ESP32C6 until v0.2 and ESP32H2 until v1.2. If you already use suspend feature for mass production, you can enable this for bypassing check after knowing the risk. But if you are new users, or developing new applications, or producing a new batch, please DO NOT enable this config option.

For more information, please refer to errata or connect to Espressif business support team.

CONFIG_SPI_FLASH_SOFTWARE_RESUME

Resume flash program/erase form suspend state by software control

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

Enable this config will disable auto-resume from hardware. Thus the software will resume the chip after any higher priority task/interrupt which suspend the chip. The benefit is that the suspend-resume will not disturb the higher priority task and interrupt.

This currently is only valid on single core chip.

CONFIG_SPI_FLASH_DISABLE_SCHEDULER_IN_SUSPEND

Disable task scheduler when suspend is enabled when SPI1 operation is ongoing

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

Disable freertos task scheduler when CONFIG_SPI_FLASH_AUTO_SUSPEND is enabled. Thus only interrupt can trigger a suspend. When SPI_FLASH_AUTO_SUSPEND is enabled, default behavior is not disable the task scheduler, so both interrupt and high priority task can suspend the erase/program operation. When this option is enabled, task scheduler is disabled, only interrupt can suspend erase/program operation.

CONFIG_SPI_FLASH_AUTO_CHECK_SUSPEND_STATUS

Check flash status automatically after flash suspend

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

Majority flash supports to use flash register to judge if flash suspend status is done or not. So enable this config, the behavior would use flash register WIP bit to judge whether suspend is valid instead of waiting for a specific long time, which can save a lot of time and benefit for performance improvement.

CONFIG_SPI_FLASH_PLACE_FUNCTIONS_IN_IRAM

Place spi_flash operation functions into IRAM

Found in: Component config > Main Flash configuration > Optional and Experimental Features (READ DOCS FIRST)

When disabled, certain functions in *spi_flash* component will be placed into Flash memory instead of IRAM. Disabling this option will save almost 10KB of IRAM depending on which functions are used.

When enabled, these functions will be placed in internal RAM, with better performance.

For more information please refer to programming guide.

SPI Flash driver

Contains:

- Auto-detect flash chips
- CONFIG_SPI_FLASH_BYPASS_BLOCK_ERASE
- CONFIG_SPI_FLASH_ENABLE_ENCRYPTED_READ_WRITE
- CONFIG_SPI_FLASH_ENABLE_COUNTERS
- CONFIG_SPI_FLASH_YIELD_DURING_ERASE
- CONFIG_SPI_FLASH_CHECK_ERASE_TIMEOUT_DISABLED
- CONFIG_SPI_FLASH_WRITE_CHUNK_SIZE
- CONFIG_SPI_FLASH_OVERRIDE_CHIP_DRIVER_LIST
- CONFIG_SPI_FLASH_SIZE_OVERRIDE
- CONFIG_SPI_FLASH_SHARE_SPI1_BUS
- CONFIG_SPI_FLASH_ROM_IMPL
- CONFIG_SPI_FLASH_VERIFY_WRITE
- CONFIG_SPI_FLASH_DANGEROUS_WRITE

CONFIG_SPI_FLASH_VERIFY_WRITE

Verify SPI flash writes

Found in: Component config > SPI Flash driver

If this option is enabled, any time SPI flash is written then the data will be read back and verified. This can catch hardware problems with SPI flash, or flash which was not erased before verification.

This will slightly influence the write performance.

CONFIG_SPI_FLASH_LOG_FAILED_WRITE

Log errors if verification fails

Found in: Component config > SPI Flash driver > CONFIG_SPI_FLASH_VERIFY_WRITE

If this option is enabled, if SPI flash write verification fails then a log error line will be written with the address, expected & actual values. This can be useful when debugging hardware SPI flash problems.

CONFIG_SPI_FLASH_WARN_SETTING_ZERO_TO_ONE

Log warning if writing zero bits to ones

Found in: Component config > SPI Flash driver > CONFIG_SPI_FLASH_VERIFY_WRITE

If this option is enabled, any SPI flash write which tries to set zero bits in the flash to ones will log a warning. Such writes will not result in the requested data appearing identically in flash once written, as SPI NOR flash can only set bits to one when an entire sector is erased. After erasing, individual bits can only be written from one to zero.

Note that some software (such as SPIFFS) which is aware of SPI NOR flash may write one bits as an optimisation, relying on the data in flash becoming a bitwise AND of the new data and any existing data. Such software will log spurious warnings if this option is enabled.

CONFIG_SPI_FLASH_ENABLE_COUNTERS

Enable operation counters

Found in: Component config > SPI Flash driver

This option enables the following APIs:

- esp_flash_reset_counters
- esp_flash_dump_counters
- esp_flash_get_counters

These APIs may be used to collect performance data for spi_flash APIs and to help understand behaviour of libraries which use SPI flash.

CONFIG_SPI_FLASH_ROM_IMPL

Use esp_flash implementation in ROM

Found in: Component config > SPI Flash driver

Enable this flag to use new SPI flash driver functions from ROM instead of ESP-IDF.

If keeping this as "n" in your project, you will have less free IRAM. But you can use all of our flash features.

If making this as "y" in your project, you will increase free IRAM. But you may miss out on some flash features and support for new flash chips.

Currently the ROM cannot support the following features:

- SPI_FLASH_AUTO_SUSPEND (C3, S3)

CONFIG_SPI_FLASH_DANGEROUS_WRITE

Writing to dangerous flash regions

Found in: Component config > SPI Flash driver

SPI flash APIs can optionally abort or return a failure code if erasing or writing addresses that fall at the beginning of flash (covering the bootloader and partition table) or that overlap the app partition that contains the running app.

It is not recommended to ever write to these regions from an IDF app, and this check prevents logic errors or corrupted firmware memory from damaging these regions.

Note that this feature *does not* check calls to the esp_rom_xxx SPI flash ROM functions. These functions should not be called directly from IDF applications.

Available options:

- Aborts (CONFIG_SPI_FLASH_DANGEROUS_WRITE_ABORTS)
- Fails (CONFIG_SPI_FLASH_DANGEROUS_WRITE_FAILS)
- Allowed (CONFIG_SPI_FLASH_DANGEROUS_WRITE_ALLOWED)

CONFIG_SPI_FLASH_SHARE_SPI1_BUS

Support other devices attached to SPI1 bus

Found in: Component config > SPI Flash driver

Each SPI bus needs a lock for arbitration among devices. This allows multiple devices on a same bus, but may reduce the speed of esp_flash driver access to the main flash chip.

If you only need to use esp_flash driver to access the main flash chip, disable this option, and the lock will be bypassed on SPI1 bus. Otherwise if extra devices are needed to attach to SPI1 bus, enable this option.

CONFIG_SPI_FLASH_BYPASS_BLOCK_ERASE

Bypass a block erase and always do sector erase

Found in: Component config > SPI Flash driver

Some flash chips can have very high "max" erase times, especially for block erase (32KB or 64KB). This option allows to bypass "block erase" and always do sector erase commands. This will be much slower overall in most cases, but improves latency for other code to run.

CONFIG_SPI_FLASH_YIELD_DURING_ERASE

Enables yield operation during flash erase

Found in: Component config > SPI Flash driver

This allows to yield the CPUs between erase commands. Prevents starvation of other tasks. Please use this configuration together with `SPI\_FLASH\_ERASE\_YIELD\_DURATION\_MS` and `SPI\_FLASH\_ERASE\_YIELD\_TICKS` after carefully checking flash datasheet to avoid a watchdog timeout. For more information, please check *SPI Flash API* reference documentation under section *OS Function*.

CONFIG_SPI_FLASH_ERASE_YIELD_DURATION_MS

Duration of erasing to yield CPUs (ms)

Found in: Component config > SPI Flash driver > CONFIG_SPI_FLASH_YIELD_DURING_ERASE

If a duration of one erase command is large then it will yield CPUs after finishing a current command.

CONFIG_SPI_FLASH_ERASE_YIELD_TICKS

CPU release time (tick) for an erase operation

Found in: Component config > SPI Flash driver > CONFIG_SPI_FLASH_YIELD_DURING_ERASE

Defines how many ticks will be before returning to continue a erasing.

CONFIG_SPI_FLASH_WRITE_CHUNK_SIZE

Flash write chunk size

Found in: Component config > SPI Flash driver

Flash write is broken down in terms of multiple (smaller) write operations. This configuration options helps to set individual write chunk size, smaller value here ensures that cache (and non-IRAM resident interrupts) remains disabled for shorter duration.

CONFIG_SPI_FLASH_SIZE_OVERRIDE

Override flash size in bootloader header by ESPTOOLPY_FLASHSIZE

Found in: Component config > SPI Flash driver

SPI Flash driver uses the flash size configured in bootloader header by default. Enable this option to override flash size with latest ESPTOOLPY_FLASHSIZE value from the app header if the size in the bootloader header is incorrect.

CONFIG_SPI_FLASH_CHECK_ERASE_TIMEOUT_DISABLED

Flash timeout checkout disabled

Found in: Component config > SPI Flash driver

This option is helpful if you are using a flash chip whose timeout is quite large or unpredictable.

CONFIG_SPI_FLASH_OVERRIDE_CHIP_DRIVER_LIST

Override default chip driver list

Found in: Component config > SPI Flash driver

This option allows the chip driver list to be customized, instead of using the default list provided by ESP-IDF.

When this option is enabled, the default list is no longer compiled or linked. Instead, the *default_registered_chips* structure must be provided by the user.

See example: custom_chip_driver under examples/storage for more details.

Auto-detect flash chips

Contains:

- CONFIG_SPI_FLASH_SUPPORT_BOYA_CHIP
- CONFIG_SPI_FLASH_SUPPORT_GD_CHIP
- CONFIG_SPI_FLASH_SUPPORT_ISSI_CHIP
- CONFIG_SPI_FLASH_SUPPORT_MXIC_CHIP
- CONFIG_SPI_FLASH_SUPPORT_TH_CHIP
- CONFIG_SPI_FLASH_SUPPORT_WINBOND_CHIP

CONFIG_SPI_FLASH_SUPPORT_ISSI_CHIP

ISSI

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of ISSI chips if chip vendor not directly given by `chip\_drv` member of the chip struct. This adds support for variant chips, however will extend detecting time.

CONFIG_SPI_FLASH_SUPPORT_MXIC_CHIP

MXIC

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of MXIC chips if chip vendor not directly given by `chip\_drv` member of the chip struct. This adds support for variant chips, however will extend detecting time.

CONFIG_SPI_FLASH_SUPPORT_GD_CHIP

GigaDevice

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of GD (GigaDevice) chips if chip vendor not directly given by `chip\_drv` member of the chip struct. If you are using Wrover modules, please don't disable this, otherwise your flash may not work in 4-bit mode.

This adds support for variant chips, however will extend detecting time and image size. Note that the default chip driver supports the GD chips with product ID 60H.

CONFIG_SPI_FLASH_SUPPORT_WINBOND_CHIP

Winbond

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of Winbond chips if chip vendor not directly given by `chip\_drv` member of the chip struct. This adds support for variant chips, however will extend detecting time.

CONFIG_SPI_FLASH_SUPPORT_BOYA_CHIP

BOYA

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of BOYA chips if chip vendor not directly given by `chip\_drv` member of the chip struct. This adds support for variant chips, however will extend detecting time.

CONFIG_SPI_FLASH_SUPPORT_TH_CHIP

TH

Found in: Component config > SPI Flash driver > Auto-detect flash chips

Enable this to support auto detection of TH chips if chip vendor not directly given by `chip\_drv` member of the chip struct. This adds support for variant chips, however will extend detecting time.

CONFIG_SPI_FLASH_ENABLE_ENCRYPTED_READ_WRITE

Enable encrypted partition read/write operations

Found in: Component config > SPI Flash driver

This option enables flash read/write operations to encrypted partition/s. This option is kept enabled irrespective of state of flash encryption feature. However, in case application is not using flash encryption feature and is in need of some additional memory from IRAM region (~1KB) then this config can be disabled.

SPIFFS Configuration

Contains:

- Debug Configuration
- CONFIG_SPIFFS_USE_MAGIC
- CONFIG_SPIFFS_GC_STATS
- CONFIG_SPIFFS_PAGE_CHECK
- CONFIG_SPIFFS_FOLLOW_SYMLINKS
- CONFIG_SPIFFS_MAX_PARTITIONS
- CONFIG_SPIFFS_USE_MTIME
- CONFIG_SPIFFS_GC_MAX_RUNS
- CONFIG_SPIFFS_OBJ_NAME_LEN
- CONFIG_SPIFFS_META_LENGTH
- SPIFFS Cache Configuration
- CONFIG_SPIFFS_PAGE_SIZE
- CONFIG_SPIFFS_MTIME_WIDE_64_BITS

CONFIG_SPIFFS_MAX_PARTITIONS

Maximum Number of Partitions

Found in: Component config > SPIFFS Configuration

Define maximum number of partitions that can be mounted.

Range:

- from 1 to 10

Default value:

- 3

SPIFFS Cache Configuration

Contains:

- CONFIG_SPIFFS_CACHE

CONFIG_SPIFFS_CACHE

Enable SPIFFS Cache

Found in: Component config > SPIFFS Configuration > SPIFFS Cache Configuration

Enables/disable memory read caching of nucleus file system operations.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_CACHE_WR

Enable SPIFFS Write Caching

Found in: Component config > SPIFFS Configuration > SPIFFS Cache Configuration > CONFIG_SPIFFS_CACHE

Enables memory write caching for file descriptors in hydrogen.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_CACHE_STATS

Enable SPIFFS Cache Statistics

Found in: Component config > SPIFFS Configuration > SPIFFS Cache Configuration > CONFIG_SPIFFS_CACHE

Enable/disable statistics on caching. Debug/test purpose only.

Default value:

- No (disabled)

CONFIG_SPIFFS_PAGE_CHECK

Enable SPIFFS Page Check

Found in: Component config > SPIFFS Configuration

Always check header of each accessed page to ensure consistent state. If enabled it will increase number of reads from flash, especially if cache is disabled.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_GC_MAX_RUNS

Set Maximum GC Runs

Found in: Component config > SPIFFS Configuration

Define maximum number of GC runs to perform to reach desired free pages.

Range:

- from 1 to 10000

Default value:

- 10

CONFIG_SPIFFS_GC_STATS

Enable SPIFFS GC Statistics

Found in: Component config > SPIFFS Configuration

Enable/disable statistics on gc. Debug/test purpose only.

Default value:

- No (disabled)

CONFIG_SPIFFS_PAGE_SIZE

SPIFFS logical page size

Found in: Component config > SPIFFS Configuration

Logical page size of SPIFFS partition, in bytes. Must be multiple of flash page size (which is usually 256 bytes). Larger page sizes reduce overhead when storing large files, and improve filesystem performance when reading large files. Smaller page sizes reduce overhead when storing small (< page size) files.

Range:

- from 256 to 1024

Default value:

- 256

CONFIG_SPIFFS_OBJ_NAME_LEN

Set SPIFFS Maximum Name Length

Found in: Component config > SPIFFS Configuration

Object name maximum length. Note that this length include the zero-termination character, meaning maximum string of characters can at most be SPIFFS_OBJ_NAME_LEN - 1.

SPIFFS_OBJ_NAME_LEN + SPIFFS_META_LENGTH should not exceed SPIFFS_PAGE_SIZE - 64.

Range:

- from 1 to 256

Default value:

- 32

CONFIG_SPIFFS_FOLLOW_SYMLINKS

Enable symbolic links for image creation

Found in: Component config > SPIFFS Configuration

If this option is enabled, symbolic links are taken into account during partition image creation.

Default value:

- No (disabled)

CONFIG_SPIFFS_USE_MAGIC

Enable SPIFFS Filesystem Magic

Found in: Component config > SPIFFS Configuration

Enable this to have an identifiable spiffs filesystem. This will look for a magic in all sectors to determine if this is a valid spiffs system or not at mount time.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_USE_MAGIC_LENGTH

Enable SPIFFS Filesystem Length Magic

Found in: Component config > SPIFFS Configuration > CONFIG_SPIFFS_USE_MAGIC

If this option is enabled, the magic will also be dependent on the length of the filesystem. For example, a filesystem configured and formatted for 4 megabytes will not be accepted for mounting with a configuration defining the filesystem as 2 megabytes.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_META_LENGTH

Size of per-file metadata field

Found in: Component config > SPIFFS Configuration

This option sets the number of extra bytes stored in the file header. These bytes can be used in an application-specific manner. Set this to at least 4 bytes to enable support for saving file modification time.

SPIFFS_OBJ_NAME_LEN + SPIFFS_META_LENGTH should not exceed SPIFFS_PAGE_SIZE - 64.

Default value:

- 4

CONFIG_SPIFFS_USE_MTIME

Save file modification time

Found in: Component config > SPIFFS Configuration

If enabled, then the first 4 bytes of per-file metadata will be used to store file modification time (mtime), accessible through stat/fstat functions. Modification time is updated when the file is opened.

Default value:

- Yes (enabled)

CONFIG_SPIFFS_MTIME_WIDE_64_BITS

The time field occupies 64 bits in the image instead of 32 bits

Found in: Component config > SPIFFS Configuration

If this option is not set, the time field is 32 bits (up to 2106 year), otherwise it is 64 bits and make sure it matches SPIFFS_META_LENGTH. If the chip already has the spiffs image with the time field = 32 bits then this option cannot be applied in this case. Erase it first before using this option. To resolve the Y2K38 problem for the spiffs, use a toolchain with 64-bit time_t support.

Default value:

- No (disabled) if CONFIG_SPIFFS_META_LENGTH >= 8

Debug Configuration

Contains:

- CONFIG_SPIFFS_DBG
- CONFIG_SPIFFS_API_DBG
- CONFIG_SPIFFS_CACHE_DBG
- CONFIG_SPIFFS_CHECK_DBG
- CONFIG_SPIFFS_TEST_VISUALISATION
- CONFIG_SPIFFS_GC_DBG

CONFIG_SPIFFS_DBG

Enable general SPIFFS debug

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enabling this option will print general debug messages to the console.

Default value:

- No (disabled)

CONFIG_SPIFFS_API_DBG

Enable SPIFFS API debug

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enabling this option will print API debug messages to the console.

Default value:

- No (disabled)

CONFIG_SPIFFS_GC_DBG

Enable SPIFFS Garbage Cleaner debug

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enabling this option will print GC debug messages to the console.

Default value:

- No (disabled)

CONFIG_SPIFFS_CACHE_DBG

Enable SPIFFS Cache debug

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enabling this option will print cache debug messages to the console.

Default value:

- No (disabled)

CONFIG_SPIFFS_CHECK_DBG

Enable SPIFFS Filesystem Check debug

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enabling this option will print Filesystem Check debug messages to the console.

Default value:

- No (disabled)

CONFIG_SPIFFS_TEST_VISUALISATION

Enable SPIFFS Filesystem Visualization

Found in: Component config > SPIFFS Configuration > Debug Configuration

Enable this option to enable SPIFFS_vis function in the API.

Default value:

- No (disabled)

TCP Transport

Contains:

- Websocket

Websocket

Contains:

- CONFIG_WS_TRANSPORT

CONFIG_WS_TRANSPORT

Enable Websocket Transport

Found in: Component config > TCP Transport > Websocket

Enable support for creating websocket transport.

Default value:

- Yes (enabled)

CONFIG_WS_BUFFER_SIZE

Websocket transport buffer size

Found in: Component config > TCP Transport > Websocket > CONFIG_WS_TRANSPORT

Size of the buffer used for constructing the HTTP Upgrade request during connect

Default value:

- 1024

CONFIG_WS_DYNAMIC_BUFFER

Using dynamic websocket transport buffer

Found in: Component config > TCP Transport > Websocket > CONFIG_WS_TRANSPORT

If enable this option, websocket transport buffer will be freed after connection succeed to save more heap.

Default value:

- No (disabled)

Ultra Low Power (ULP) Co-processor

Contains:

- CONFIG_ULP_ROM_PRINT_ENABLE
- CONFIG_ULP_COPROC_ENABLED
- CONFIG_ULP_TRAP_WAKEUP
- ULP Debugging Options
- ULP RISC-V Settings

CONFIG_ULP_COPROC_ENABLED

Enable Ultra Low Power (ULP) Co-processor

Found in: Component config > Ultra Low Power (ULP) Co-processor

Enable this feature if you plan to use the ULP Co-processor. Once this option is enabled, further ULP co-processor configuration will appear in the menu.

Default value:

- No (disabled)

CONFIG_ULP_COPROC_TYPE

ULP Co-processor type

Found in: Component config > Ultra Low Power (ULP) Co-processor > CONFIG_ULP_COPROC_ENABLED

Choose the ULP Coprocessor type: ULP FSM (Finite State Machine) or ULP RISC-V.

Available options:

- ULP FSM (Finite State Machine) (CONFIG_ULP_COPROC_TYPE_FSM)
- ULP RISC-V (CONFIG_ULP_COPROC_TYPE_RISCV)
- LP core RISC-V (CONFIG_ULP_COPROC_TYPE_LP_CORE)

CONFIG_ULP_COPROC_RESERVE_MEM

RTC slow memory reserved for coprocessor

Found in: Component config > Ultra Low Power (ULP) Co-processor > CONFIG_ULP_COPROC_ENABLED

Bytes of memory to reserve for ULP Co-processor firmware & data. Data is reserved at the beginning of RTC slow memory.

Range:

- from 32 to 8176 if CONFIG_ULP_COPROC_ENABLED

Default value:

- 512 if CONFIG_ULP_COPROC_ENABLED

ULP RISC-V Settings

Contains:

- CONFIG_ULP_RISCV_UART_BAUDRATE
- CONFIG_ULP_RISCV_INTERRUPT_ENABLE
- CONFIG_ULP_RISCV_I2C_RW_TIMEOUT

CONFIG_ULP_RISCV_INTERRUPT_ENABLE

Enable ULP RISC-V interrupts

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP RISC-V Settings

Turn on this setting to enabled interrupts on the ULP RISC-V core.

Default value:

- No (disabled) if CONFIG_ULP_COPROC_TYPE_RISCV

CONFIG_ULP_RISCV_UART_BAUDRATE

Baudrate used by the bitbanged ULP RISC-V UART driver

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP RISC-V Settings

The accuracy of the bitbanged UART driver is limited, it is not recommend to increase the value above 19200.

Default value:

- 9600 if CONFIG_ULP_COPROC_TYPE_RISCV

CONFIG_ULP_RISCV_I2C_RW_TIMEOUT

Set timeout for ULP RISC-V I2C transaction timeout in ticks.

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP RISC-V Settings

Set the ULP RISC-V I2C read/write timeout. Set this value to -1 if the ULP RISC-V I2C read and write APIs should wait forever. Please note that the tick rate of the ULP co-processor would be different than the OS tick rate of the main core and therefore can have different timeout value depending on which core the API is invoked on.

Range:

- from -1 to 4294967295 if CONFIG_ULP_COPROC_TYPE_RISCV

Default value:

- 500 if CONFIG_ULP_COPROC_TYPE_RISCV

CONFIG_ULP_ROM_PRINT_ENABLE

Enable print utilities from LP ROM

Found in: Component config > Ultra Low Power (ULP) Co-processor

Set this option to enable printf functionality from LP ROM. This option can help reduce the LP core binary size by not linking printf functionality from RAM code. Note: For LP ROM prints to work properly, make sure that the LP core boots from the LP ROM.

Default value:

- Yes (enabled) if CONFIG_ULP_COPROC_TYPE_LP_CORE && ESP_ROM_HAS_LP_ROM

CONFIG_ULP_TRAP_WAKEUP

Enable wakeup of HP-CPU when LP-core encounters an exception

Found in: Component config > Ultra Low Power (ULP) Co-processor

Set this option to also trigger a wakeup signal to the HP-CPU when the LP-core encounters an exception.

Default value:

- Yes (enabled) if CONFIG_ULP_COPROC_TYPE_LP_CORE

ULP Debugging Options

Contains:

- CONFIG_ULP_NORESET_UNDER_DEBUG
- CONFIG_ULP_PANIC_OUTPUT_ENABLE
- CONFIG_ULP_HP_UART_CONSOLE_PRINT

CONFIG_ULP_PANIC_OUTPUT_ENABLE

Enable panic handler which outputs over LP UART

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP Debugging Options

Set this option to enable panic handler functionality. If this option is enabled then the LP Core will output a panic dump over LP UART, similar to what the main core does. Output depends on LP UART already being initialized and configured. Disabling this option will reduce the LP core binary size by not linking in panic handler functionality.

Default value:

- No (disabled) if CONFIG_ULP_COPROC_TYPE_LP_CORE && SOC_ULP_LP_UART_SUPPORTED

CONFIG_ULP_HP_UART_CONSOLE_PRINT

Route lp_core_printf to the console HP-UART

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP Debugging Options

Set this option to route lp_core_printf to the console HP-UART. This allows you to easily view print outputs from the LP core, without having to connect to the LP-UART. This option comes with the following limitations:

1. There is no mutual exclusion between the HP-Core and the LP-Core accessing the HP-UART, which means that if both cores are logging heavily the output strings might get mangled together. 2. The HP-UART can only work while the HP-Core is running, which means that if the HP-Core is in deep sleep, the LP-Core will not be able to print to the console HP-UART.

Due to these limitations it is only recommended to use this option for easy debugging. For more serious use-cases you should use the LP-UART.

CONFIG_ULP_NORESET_UNDER_DEBUG

Avoid resetting LP core when debugger is attached

Found in: Component config > Ultra Low Power (ULP) Co-processor > ULP Debugging Options

Enable this feature to avoid resetting LP core in sleep mode when debugger is attached, otherwise configured HW breakpoints and dcsr.ebreak* bits will be missed. This is a workaround until it will be fixed in HW.

Default value:

- Yes (enabled) if CONFIG_ULP_COPROC_TYPE_LP_CORE

Unity unit testing library

Contains:

- CONFIG_UNITY_ENABLE_COLOR
- CONFIG_UNITY_ENABLE_IDF_TEST_RUNNER
- CONFIG_UNITY_ENABLE_FIXTURE
- CONFIG_UNITY_ENABLE_BACKTRACE_ON_FAIL
- CONFIG_UNITY_ENABLE_64BIT
- CONFIG_UNITY_ENABLE_DOUBLE
- CONFIG_UNITY_ENABLE_FLOAT

CONFIG_UNITY_ENABLE_FLOAT

Support for float type

Found in: Component config > Unity unit testing library

If not set, assertions on float arguments will not be available.

Default value:

- Yes (enabled)

CONFIG_UNITY_ENABLE_DOUBLE

Support for double type

Found in: Component config > Unity unit testing library

If not set, assertions on double arguments will not be available.

Default value:

- Yes (enabled)

CONFIG_UNITY_ENABLE_64BIT

Support for 64-bit integer types

Found in: Component config > Unity unit testing library

If not set, assertions on 64-bit integer types will always fail. If this feature is enabled, take care not to pass pointers (which are 32 bit) to UNITY_ASSERT_EQUAL, as that will cause pointer-to-int-cast warnings.

Default value:

- No (disabled)

CONFIG_UNITY_ENABLE_COLOR

Colorize test output

Found in: Component config > Unity unit testing library

If set, Unity will colorize test results using console escape sequences.

Default value:

- No (disabled)

CONFIG_UNITY_ENABLE_IDF_TEST_RUNNER

Include ESP-IDF test registration/running helpers

Found in: Component config > Unity unit testing library

If set, then the following features will be available:

- TEST_CASE macro which performs automatic registration of test functions
- Functions to run registered test functions: unity_run_all_tests, unity_run_tests_with_filter, unity_run_single_test_by_name.
- Interactive menu which lists test cases and allows choosing the tests to be run, available via unity_run_menu function.

Disable if a different test registration mechanism is used.

Default value:

- Yes (enabled)

CONFIG_UNITY_ENABLE_FIXTURE

Include Unity test fixture

Found in: Component config > Unity unit testing library

If set, unity_fixture.h header file and associated source files are part of the build. These provide an optional set of macros and functions to implement test groups.

Default value:

- No (disabled)

CONFIG_UNITY_ENABLE_BACKTRACE_ON_FAIL

Print a backtrace when a unit test fails

Found in: Component config > Unity unit testing library

If set, the unity framework will print the backtrace information before jumping back to the test menu. The jumping is usually occurs in assert functions such as TEST_ASSERT, TEST_FAIL etc.

Default value:

- No (disabled)

Virtual file system

Contains:

- Host File System I/O (Semihosting)
- CONFIG_VFS_INITIALIZE_DEV_NULL
- CONFIG_VFS_SUPPORT_IO
- CONFIG_VFS_SUPPRESS_CTX_DEPRECATION

CONFIG_VFS_SUPPORT_IO

Provide basic I/O functions

Found in: Component config > Virtual file system

If enabled, the following functions are provided by the VFS component.

open, close, read, write, pread, pwrite, lseek, fstat, fsync, ioctl, fcntl

Filesystem drivers can then be registered to handle these functions for specific paths.

Disabling this option can save memory when the support for these functions is not required.

Note that the following functions can still be used with socket file descriptors when this option is disabled:

close, read, write, ioctl, fcntl.

Default value:

- Yes (enabled)

CONFIG_VFS_SUPPORT_DIR

Provide directory related functions

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO

If enabled, the following functions are provided by the VFS component.

stat, link, unlink, rename, utime, access, truncate, rmdir, mkdir, opendir, closedir, readdir, readdir_r, seekdir, telldir, rewinddir

Filesystem drivers can then be registered to handle these functions for specific paths.

Disabling this option can save memory when the support for these functions is not required.

Default value:

- Yes (enabled)

CONFIG_VFS_SUPPORT_SELECT

Provide select function

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO

If enabled, select function is provided by the VFS component, and can be used on peripheral file descriptors (such as UART) and sockets at the same time.

If disabled, the default select implementation will be provided by LWIP for sockets only.

Disabling this option can reduce code size if support for "select" on UART file descriptors is not required.

CONFIG_VFS_SUPPRESS_SELECT_DEBUG_OUTPUT

Suppress select() related debug outputs

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO > CONFIG_VFS_SUPPORT_SELECT

Select() related functions might produce an inconveniently lot of debug outputs when one sets the default log level to DEBUG or higher. It is possible to suppress these debug outputs by enabling this option.

Default value:

- Yes (enabled)

CONFIG_VFS_SELECT_IN_RAM

Make VFS driver select() callbacks IRAM-safe

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO > CONFIG_VFS_SUPPORT_SELECT

If enabled, VFS driver select() callback function will be placed in IRAM.

Default value:

- No (disabled)

CONFIG_VFS_SUPPORT_TERMIOS

Provide termios.h functions

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO

Disabling this option can save memory when the support for termios.h is not required.

Default value:

- No (disabled)

CONFIG_VFS_MAX_COUNT

Maximum Number of Virtual Filesystems

Found in: Component config > Virtual file system > CONFIG_VFS_SUPPORT_IO

Define maximum number of virtual filesystems that can be registered.

Range:

- from 1 to 20

Default value:

- 8

CONFIG_VFS_SUPPRESS_CTX_DEPRECATION

Suppress context-less API deprecation warning

Found in: Component config > Virtual file system

The VFS APIs without context pointer are deprecated and will be removed in the next major release. Please switch to the APIs with context pointer, or alternatively enable this option to silence the warning.

Default value:

- No (disabled)

Host File System I/O (Semihosting)

Contains:

- CONFIG_VFS_SEMIHOSTFS_MAX_MOUNT_POINTS

CONFIG_VFS_SEMIHOSTFS_MAX_MOUNT_POINTS

Host FS: Maximum number of the host filesystem mount points

Found in: Component config > Virtual file system > Host File System I/O (Semihosting)

Define maximum number of host filesystem mount points.

Default value:

- 1

CONFIG_VFS_INITIALIZE_DEV_NULL

Initialize /dev/null VFS

Found in: Component config > Virtual file system

If enabled, /dev/null VFS will be automatically initialized at startup.

Default value:

- Yes (enabled)

Wear Levelling

Contains:

- CONFIG_WL_SECTOR_MODE
- CONFIG_WL_SECTOR_SIZE

CONFIG_WL_SECTOR_SIZE

Wear Levelling library sector size

Found in: Component config > Wear Levelling

Sector size used by wear levelling library. You can set default sector size or size that will fit to the flash device sector size.

With sector size set to 4096 bytes, wear levelling library is more efficient. However if FAT filesystem is used on top of wear levelling library, it will need more temporary storage: 4096 bytes for each mounted filesystem and 4096 bytes for each opened file.

With sector size set to 512 bytes, wear levelling library will perform more operations with flash memory, but less RAM will be used by FAT filesystem library (512 bytes for the filesystem and 512 bytes for each file opened).

Available options:

- 512 (CONFIG_WL_SECTOR_SIZE_512)
- 4096 (CONFIG_WL_SECTOR_SIZE_4096)

CONFIG_WL_SECTOR_MODE

Sector store mode

Found in: Component config > Wear Levelling

Specify the mode to store data into flash:

- In Performance mode a data will be stored to the RAM and then stored back to the flash. Compared to the Safety mode, this operation is faster, but if power will be lost when erase sector operation is in progress, then the data from complete flash device sector will be lost.
- In Safety mode data from complete flash device sector will be read from flash, modified, and then stored back to flash. Compared to the Performance mode, this operation is slower, but if power is lost during erase sector operation, then the data from full flash device sector will not be lost.

Available options:

- Performance (CONFIG_WL_SECTOR_MODE_PERF)
- Safety (CONFIG_WL_SECTOR_MODE_SAFE)

CONFIG_IDF_EXPERIMENTAL_FEATURES

Make experimental features visible

Found in:

By enabling this option, ESP-IDF experimental feature options will be visible.

Note you should still enable a certain experimental feature option to use it, and you should read the corresponding risk warning and known issue list carefully.

Current experimental feature list:

- CONFIG_ESPTOOLPY_FLASHFREQ_120M && CONFIG_ESPTOOLPY_FLASH_SAMPLE_MODE_DTR
- CONFIG_SPIRAM_SPEED_120M && CONFIG_SPIRAM_MODE_OCT
- CONFIG_BOOTLOADER_CACHE_32BIT_ADDR_QUAD_FLASH
- CONFIG_ESP_WIFI_EAP_TLS1_3
- CONFIG_ESP_WIFI_ENABLE_ROAMING_APP
- CONFIG_USB_HOST_EXT_PORT_RESET_ATTEMPTS
- CONFIG_GDMA_ENABLE_WEIGHTED_ARBITRATION
- CONFIG_I3C_MASTER_ENABLED

Default value:

- No (disabled)

Deprecated options and their replacements

- CONFIG_A2DP_ENABLE (CONFIG_BT_A2DP_ENABLE)
- **CONFIG_A2D_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_A2D_TRACE_LEVEL)**
  - CONFIG_A2D_TRACE_LEVEL_NONE
  - CONFIG_A2D_TRACE_LEVEL_ERROR
  - CONFIG_A2D_TRACE_LEVEL_WARNING
  - CONFIG_A2D_TRACE_LEVEL_API
  - CONFIG_A2D_TRACE_LEVEL_EVENT
  - CONFIG_A2D_TRACE_LEVEL_DEBUG
  - CONFIG_A2D_TRACE_LEVEL_VERBOSE
- CONFIG_ADC_CAL_EFUSE_TP_ENABLE (CONFIG_ADC_CALI_EFUSE_TP_ENABLE)
- CONFIG_ADC_CAL_EFUSE_VREF_ENABLE (CONFIG_ADC_CALI_EFUSE_VREF_ENABLE)
- CONFIG_ADC_CAL_LUT_ENABLE (CONFIG_ADC_CALI_LUT_ENABLE)
- CONFIG_ADC_DISABLE_DAC (CONFIG_ADC_DISABLE_DAC_OUTPUT)
- CONFIG_ANA_CMPR_ISR_IRAM_SAFE (CONFIG_ANA_CMPR_ISR_CACHE_SAFE)

- **CONFIG_APPL_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_APPL_TRACE_LEVEL)**
    - CONFIG_APPL_TRACE_LEVEL_NONE
    - CONFIG_APPL_TRACE_LEVEL_ERROR
    - CONFIG_APPL_TRACE_LEVEL_WARNING
    - CONFIG_APPL_TRACE_LEVEL_API
    - CONFIG_APPL_TRACE_LEVEL_EVENT
    - CONFIG_APPL_TRACE_LEVEL_DEBUG
    - CONFIG_APPL_TRACE_LEVEL_VERBOSE
- CONFIG_APP_ANTI_ROLLBACK (CONFIG_BOOTLOADER_APP_ANTI_ROLLBACK)
- CONFIG_APP_ROLLBACK_ENABLE (CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE)
- CONFIG_APP_SECURE_VERSION (CONFIG_BOOTLOADER_APP_SECURE_VERSION)
- CONFIG_APP_SECURE_VERSION_SIZE_EFUSE_FIELD (CONFIG_BOOTLOADER_APP_SEC_VER_SIZE_EFUSE_FIELD)
- **CONFIG_AVCT_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_AVCT_TRACE_LEVEL)**
    - CONFIG_AVCT_TRACE_LEVEL_NONE
    - CONFIG_AVCT_TRACE_LEVEL_ERROR
    - CONFIG_AVCT_TRACE_LEVEL_WARNING
    - CONFIG_AVCT_TRACE_LEVEL_API
    - CONFIG_AVCT_TRACE_LEVEL_EVENT
    - CONFIG_AVCT_TRACE_LEVEL_DEBUG
    - CONFIG_AVCT_TRACE_LEVEL_VERBOSE
- **CONFIG_AVDT_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_AVDT_TRACE_LEVEL)**
    - CONFIG_AVDT_TRACE_LEVEL_NONE
    - CONFIG_AVDT_TRACE_LEVEL_ERROR
    - CONFIG_AVDT_TRACE_LEVEL_WARNING
    - CONFIG_AVDT_TRACE_LEVEL_API
    - CONFIG_AVDT_TRACE_LEVEL_EVENT
    - CONFIG_AVDT_TRACE_LEVEL_DEBUG
    - CONFIG_AVDT_TRACE_LEVEL_VERBOSE
- **CONFIG_AVRC_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_AVRC_TRACE_LEVEL)**
    - CONFIG_AVRC_TRACE_LEVEL_NONE
    - CONFIG_AVRC_TRACE_LEVEL_ERROR
    - CONFIG_AVRC_TRACE_LEVEL_WARNING
    - CONFIG_AVRC_TRACE_LEVEL_API
    - CONFIG_AVRC_TRACE_LEVEL_EVENT
    - CONFIG_AVRC_TRACE_LEVEL_DEBUG
    - CONFIG_AVRC_TRACE_LEVEL_VERBOSE
- CONFIG_BLE_ACTIVE_SCAN_REPORT_ADV_SCAN_RSP_INDIVIDUALLY (CONFIG_BT_BLE_ACT_SCAN_REP_ADV_SCAN)

- CONFIG_BLE_ADV_REPORT_DISCARD_THRSHOLD (CONFIG_BTDM_BLE_ADV_REPORT_DISCARD_THRSHOLD)
- CONFIG_BLE_ADV_REPORT_FLOW_CONTROL_NUM (CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_NUM)
- CONFIG_BLE_ADV_REPORT_FLOW_CONTROL_SUPPORTED (CONFIG_BTDM_BLE_ADV_REPORT_FLOW_CTRL_SUPP)
- CONFIG_BLE_ESTABLISH_LINK_CONNECTION_TIMEOUT (CONFIG_BT_BLE_ESTAB_LINK_CONN_TOUT)
- CONFIG_BLE_MESH_GATT_PROXY (CONFIG_BLE_MESH_GATT_PROXY_SERVER)
- CONFIG_BLE_MESH_SCAN_DUPLICATE_EN (CONFIG_BTDM_BLE_MESH_SCAN_DUPL_EN)
- CONFIG_BLE_SCAN_DUPLICATE (CONFIG_BTDM_BLE_SCAN_DUPL)
- CONFIG_BLE_SMP_ENABLE (CONFIG_BT_BLE_SMP_ENABLE)
- CONFIG_BLUEDROID_MEM_DEBUG (CONFIG_BT_BLUEDROID_MEM_DEBUG)
- **CONFIG_BLUEDROID_PINNED_TO_CORE_CHOICE (CONFIG_BT_BLUEDROID_PINNED_TO_CORE_CHOICE)**
    - CONFIG_BLUEDROID_PINNED_TO_CORE_0
    - CONFIG_BLUEDROID_PINNED_TO_CORE_1
- **CONFIG_BLUFI_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_BLUFI_TRACE_LEVEL)**
    - CONFIG_BLUFI_TRACE_LEVEL_NONE
    - CONFIG_BLUFI_TRACE_LEVEL_ERROR
    - CONFIG_BLUFI_TRACE_LEVEL_WARNING
    - CONFIG_BLUFI_TRACE_LEVEL_API
    - CONFIG_BLUFI_TRACE_LEVEL_EVENT
    - CONFIG_BLUFI_TRACE_LEVEL_DEBUG
    - CONFIG_BLUFI_TRACE_LEVEL_VERBOSE
- CONFIG_BNEP_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_BNEP_TRACE_LEVEL)
- CONFIG_BROWNOUT_DET (CONFIG_ESP_BROWNOUT_DET)
- **CONFIG_BROWNOUT_DET_LVL_SEL (CONFIG_ESP_BROWNOUT_DET_LVL_SEL)**
    - CONFIG_BROWNOUT_DET_LVL_SEL_0
    - CONFIG_BROWNOUT_DET_LVL_SEL_1
    - CONFIG_BROWNOUT_DET_LVL_SEL_2
    - CONFIG_BROWNOUT_DET_LVL_SEL_3
    - CONFIG_BROWNOUT_DET_LVL_SEL_4
    - CONFIG_BROWNOUT_DET_LVL_SEL_5
    - CONFIG_BROWNOUT_DET_LVL_SEL_6
    - CONFIG_BROWNOUT_DET_LVL_SEL_7
- **CONFIG_BTC_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_BTC_TRACE_LEVEL)**
    - CONFIG_BTC_TRACE_LEVEL_NONE
    - CONFIG_BTC_TRACE_LEVEL_ERROR
    - CONFIG_BTC_TRACE_LEVEL_WARNING

    - CONFIG_BTC_TRACE_LEVEL_API
    - CONFIG_BTC_TRACE_LEVEL_EVENT
    - CONFIG_BTC_TRACE_LEVEL_DEBUG
    - CONFIG_BTC_TRACE_LEVEL_VERBOSE

- CONFIG_BTC_TASK_STACK_SIZE (CONFIG_BT_BTC_TASK_STACK_SIZE)
- CONFIG_BTDM_CONTROLLER_BLE_MAX_CONN (CONFIG_BTDM_CTRL_BLE_MAX_CONN)
- CONFIG_BTDM_CONTROLLER_BR_EDR_MAX_ACL_CONN (CONFIG_BTDM_CTRL_BR_EDR_MAX_ACL_CONN)
- CONFIG_BTDM_CONTROLLER_BR_EDR_MAX_SYNC_CONN (CONFIG_BTDM_CTRL_BR_EDR_MAX_SYNC_CONN)
- CONFIG_BTDM_CONTROLLER_FULL_SCAN_SUPPORTED (CONFIG_BTDM_CTRL_FULL_SCAN_SUPPORTED)
- **CONFIG_BTDM_CONTROLLER_HCI_MODE_CHOICE (CONFIG_BTDM_CTRL_HCI_MODE_CHOICE)**
    - CONFIG_BTDM_CONTROLLER_HCI_MODE_VHCI
    - CONFIG_BTDM_CONTROLLER_HCI_MODE_UART_H4
- **CONFIG_BTDM_CONTROLLER_MODE (CONFIG_BTDM_CTRL_MODE)**
    - CONFIG_BTDM_CONTROLLER_MODE_BLE_ONLY
    - CONFIG_BTDM_CONTROLLER_MODE_BR_EDR_ONLY
    - CONFIG_BTDM_CONTROLLER_MODE_BTDM
- CONFIG_BTDM_CONTROLLER_MODEM_SLEEP (CONFIG_BTDM_CTRL_MODEM_SLEEP)
- CONFIG_BTDM_CONTROLLER_PINNED_TO_CORE_CHOICE (CONFIG_BTDM_CTRL_PINNED_TO_CORE_CHOICE)
- **CONFIG_BTH_LOG_SDP_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_SDP_TRACE_LEVEL)**
    - CONFIG_SDP_TRACE_LEVEL_NONE
    - CONFIG_SDP_TRACE_LEVEL_ERROR
    - CONFIG_SDP_TRACE_LEVEL_WARNING
    - CONFIG_SDP_TRACE_LEVEL_API
    - CONFIG_SDP_TRACE_LEVEL_EVENT
    - CONFIG_SDP_TRACE_LEVEL_DEBUG
    - CONFIG_SDP_TRACE_LEVEL_VERBOSE
- **CONFIG_BTIF_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_BTIF_TRACE_LEVEL)**
    - CONFIG_BTIF_TRACE_LEVEL_NONE
    - CONFIG_BTIF_TRACE_LEVEL_ERROR
    - CONFIG_BTIF_TRACE_LEVEL_WARNING
    - CONFIG_BTIF_TRACE_LEVEL_API
    - CONFIG_BTIF_TRACE_LEVEL_EVENT
    - CONFIG_BTIF_TRACE_LEVEL_DEBUG

    - CONFIG_BTIF_TRACE_LEVEL_VERBOSE
- **CONFIG_BTM_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_BTM_TRACE_LEVEL)**
    - CONFIG_BTM_TRACE_LEVEL_NONE
    - CONFIG_BTM_TRACE_LEVEL_ERROR
    - CONFIG_BTM_TRACE_LEVEL_WARNING
    - CONFIG_BTM_TRACE_LEVEL_API
    - CONFIG_BTM_TRACE_LEVEL_EVENT
    - CONFIG_BTM_TRACE_LEVEL_DEBUG
    - CONFIG_BTM_TRACE_LEVEL_VERBOSE
- CONFIG_BTU_TASK_STACK_SIZE (CONFIG_BT_BTU_TASK_STACK_SIZE)
- CONFIG_BT_NIMBLE_ACL_BUF_COUNT (CONFIG_BT_NIMBLE_TRANSPORT_ACL_FROM_LL_COUNT)
- CONFIG_BT_NIMBLE_ACL_BUF_SIZE (CONFIG_BT_NIMBLE_TRANSPORT_ACL_SIZE)
- CONFIG_BT_NIMBLE_HCI_EVT_BUF_SIZE (CONFIG_BT_NIMBLE_TRANSPORT_EVT_SIZE)
- CONFIG_BT_NIMBLE_HCI_EVT_HI_BUF_COUNT (CONFIG_BT_NIMBLE_TRANSPORT_EVT_COUNT)
- CONFIG_BT_NIMBLE_HCI_EVT_LO_BUF_COUNT (CONFIG_BT_NIMBLE_TRANSPORT_EVT_DISCARD_COUNT)
- CONFIG_BT_NIMBLE_MSYS1_BLOCK_COUNT (CONFIG_BT_NIMBLE_MSYS_1_BLOCK_COUNT)
- CONFIG_BT_NIMBLE_SM_SC_LVL (CONFIG_BT_NIMBLE_SM_LVL)
- CONFIG_BT_NIMBLE_TASK_STACK_SIZE (CONFIG_BT_NIMBLE_HOST_TASK_STACK_SIZE)
- CONFIG_CAM_CTLR_DVP_CAM_ISR_IRAM_SAFE (CONFIG_CAM_CTLR_DVP_CAM_ISR_CACHE_SAFE)
- CONFIG_CAM_CTLR_ISP_DVP_ISR_IRAM_SAFE (CONFIG_CAM_CTLR_ISP_DVP_ISR_CACHE_SAFE)
- CONFIG_CAM_CTLR_MIPI_CSI_ISR_IRAM_SAFE (CONFIG_CAM_CTLR_MIPI_CSI_ISR_CACHE_SAFE)
- CONFIG_CLASSIC_BT_ENABLED (CONFIG_BT_CLASSIC_ENABLED)
- **CONFIG_CONSOLE_UART (CONFIG_ESP_CONSOLE_UART)**
    - CONFIG_CONSOLE_UART_DEFAULT
    - CONFIG_CONSOLE_UART_CUSTOM
    - CONFIG_CONSOLE_UART_NONE, CONFIG_ESP_CONSOLE_UART_NONE
- CONFIG_CONSOLE_UART_BAUDRATE (CONFIG_ESP_CONSOLE_UART_BAUDRATE)
- **CONFIG_CONSOLE_UART_NUM (CONFIG_ESP_CONSOLE_UART_NUM)**
    - CONFIG_CONSOLE_UART_CUSTOM_NUM_0
    - CONFIG_CONSOLE_UART_CUSTOM_NUM_1
- CONFIG_CONSOLE_UART_RX_GPIO (CONFIG_ESP_CONSOLE_UART_RX_GPIO)
- CONFIG_CONSOLE_UART_TX_GPIO (CONFIG_ESP_CONSOLE_UART_TX_GPIO)

- CONFIG_CXX_EXCEPTIONS (CONFIG_COMPILER_CXX_EXCEPTIONS)
- CONFIG_CXX_EXCEPTIONS_EMG_POOL_SIZE (CONFIG_COMPILER_CXX_EXCEPTIONS_EMG_POOL_SIZE)
- CONFIG_DUPLICATE_SCAN_CACHE_SIZE (CONFIG_BTDM_SCAN_DUPL_CACHE_SIZE)
- CONFIG_EFUSE_SECURE_VERSION_EMULATE (CONFIG_BOOTLOADER_EFUSE_SECURE_VERSION_EMULATE)
- CONFIG_ENABLE_STATIC_TASK_CLEAN_UP_HOOK (CONFIG_FREERTOS_TASK_PRE_DELETION_HOOK)
- CONFIG_ESP32_COMPATIBLE_PRE_V2_1_BOOTLOADERS (CONFIG_APP_COMPATIBLE_PRE_V2_1_BOOTLOADERS)
- CONFIG_ESP32_COMPATIBLE_PRE_V3_1_BOOTLOADERS (CONFIG_APP_COMPATIBLE_PRE_V3_1_BOOTLOADERS)
- **CONFIG_ESP32_CORE_DUMP_DECODE (CONFIG_ESP_COREDUMP_DECODE)**
    - CONFIG_ESP32_CORE_DUMP_DECODE_INFO
    - CONFIG_ESP32_CORE_DUMP_DECODE_DISABLE
- CONFIG_ESP32_CORE_DUMP_MAX_TASKS_NUM (CONFIG_ESP_COREDUMP_MAX_TASKS_NUM)
- CONFIG_ESP32_CORE_DUMP_STACK_SIZE (CONFIG_ESP_COREDUMP_STACK_SIZE)
- CONFIG_ESP32_CORE_DUMP_UART_DELAY (CONFIG_ESP_COREDUMP_UART_DELAY)
- CONFIG_ESP32_DEBUG_STUBS_ENABLE (CONFIG_ESP_DEBUG_STUBS_ENABLE)
- CONFIG_ESP32_PHY_CALIBRATION_AND_DATA_STORAGE (CONFIG_ESP_PHY_CALIBRATION_AND_DATA_STORAGE)
- CONFIG_ESP32_PHY_DEFAULT_INIT_IF_INVALID (CONFIG_ESP_PHY_DEFAULT_INIT_IF_INVALID)
- CONFIG_ESP32_PHY_INIT_DATA_ERROR (CONFIG_ESP_PHY_INIT_DATA_ERROR)
- CONFIG_ESP32_PHY_INIT_DATA_IN_PARTITION (CONFIG_ESP_PHY_INIT_DATA_IN_PARTITION)
- CONFIG_ESP32_PHY_MAC_BB_PD (CONFIG_ESP_PHY_MAC_BB_PD)
- CONFIG_ESP32_PHY_MAX_WIFI_TX_POWER (CONFIG_ESP_PHY_MAX_WIFI_TX_POWER)
- CONFIG_ESP32_PTHREAD_STACK_MIN (CONFIG_PTHREAD_STACK_MIN)
- **CONFIG_ESP32_PTHREAD_TASK_CORE_DEFAULT (CONFIG_PTHREAD_TASK_CORE_DEFAULT)**
    - CONFIG_ESP32_DEFAULT_PTHREAD_CORE_NO_AFFINITY
    - CONFIG_ESP32_DEFAULT_PTHREAD_CORE_0
    - CONFIG_ESP32_DEFAULT_PTHREAD_CORE_1
- CONFIG_ESP32_PTHREAD_TASK_NAME_DEFAULT (CONFIG_PTHREAD_TASK_NAME_DEFAULT)
- CONFIG_ESP32_PTHREAD_TASK_PRIO_DEFAULT (CONFIG_PTHREAD_TASK_PRIO_DEFAULT)

- CONFIG_ESP32_PTHREAD_TASK_STACK_SIZE_DEFAULT (CONFIG_PTHREAD_TASK_STACK_SIZE_DEFAULT)
- CONFIG_ESP32_REDUCE_PHY_TX_POWER (CONFIG_ESP_PHY_REDUCE_TX_POWER)
- CONFIG_ESP32_RTC_XTAL_BOOTSTRAP_CYCLES (CONFIG_ESP_SYSTEM_RTC_EXT_XTAL_BOOTSTRAP_CYCLES)
- CONFIG_ESP32_SUPPORT_MULTIPLE_PHY_INIT_DATA_BIN (CONFIG_ESP_PHY_MULTIPLE_INIT_DATA_BIN)
- CONFIG_ESP32_WIFI_AMPDU_RX_ENABLED (CONFIG_ESP_WIFI_AMPDU_RX_ENABLED)
- CONFIG_ESP32_WIFI_AMPDU_TX_ENABLED (CONFIG_ESP_WIFI_AMPDU_TX_ENABLED)
- CONFIG_ESP32_WIFI_AMSDU_TX_ENABLED (CONFIG_ESP_WIFI_AMSDU_TX_ENABLED)
- CONFIG_ESP32_WIFI_CACHE_TX_BUFFER_NUM (CONFIG_ESP_WIFI_CACHE_TX_BUFFER_NUM)
- CONFIG_ESP32_WIFI_CSI_ENABLED (CONFIG_ESP_WIFI_CSI_ENABLED)
- CONFIG_ESP32_WIFI_DYNAMIC_RX_BUFFER_NUM (CONFIG_ESP_WIFI_DYNAMIC_RX_BUFFER_NUM)
- CONFIG_ESP32_WIFI_DYNAMIC_TX_BUFFER_NUM (CONFIG_ESP_WIFI_DYNAMIC_TX_BUFFER_NUM)
- CONFIG_ESP32_WIFI_ENABLE_WPA3_OWE_STA (CONFIG_ESP_WIFI_ENABLE_WPA3_OWE_STA)
- CONFIG_ESP32_WIFI_ENABLE_WPA3_SAE (CONFIG_ESP_WIFI_ENABLE_WPA3_SAE)
- CONFIG_ESP32_WIFI_IRAM_OPT (CONFIG_ESP_WIFI_IRAM_OPT)
- CONFIG_ESP32_WIFI_MGMT_SBUF_NUM (CONFIG_ESP_WIFI_MGMT_SBUF_NUM)
- CONFIG_ESP32_WIFI_NVS_ENABLED (CONFIG_ESP_WIFI_NVS_ENABLED)
- CONFIG_ESP32_WIFI_RX_BA_WIN (CONFIG_ESP_WIFI_RX_BA_WIN)
- CONFIG_ESP32_WIFI_RX_IRAM_OPT (CONFIG_ESP_WIFI_RX_IRAM_OPT)
- CONFIG_ESP32_WIFI_SOFTAP_BEACON_MAX_LEN (CONFIG_ESP_WIFI_SOFTAP_BEACON_MAX_LEN)
- CONFIG_ESP32_WIFI_STATIC_RX_BUFFER_NUM (CONFIG_ESP_WIFI_STATIC_RX_BUFFER_NUM)
- CONFIG_ESP32_WIFI_STATIC_TX_BUFFER_NUM (CONFIG_ESP_WIFI_STATIC_TX_BUFFER_NUM)
- CONFIG_ESP32_WIFI_SW_COEXIST_ENABLE (CONFIG_ESP_COEX_SW_COEXIST_ENABLE)
- **CONFIG_ESP32_WIFI_TASK_CORE_ID (CONFIG_ESP_WIFI_TASK_CORE_ID)**
    - CONFIG_ESP32_WIFI_TASK_PINNED_TO_CORE_0
    - CONFIG_ESP32_WIFI_TASK_PINNED_TO_CORE_1
- CONFIG_ESP32_WIFI_TX_BA_WIN (CONFIG_ESP_WIFI_TX_BA_WIN)
- **CONFIG_ESP32_WIFI_TX_BUFFER (CONFIG_ESP_WIFI_TX_BUFFER)**
    - CONFIG_ESP32_WIFI_STATIC_TX_BUFFER
    - CONFIG_ESP32_WIFI_DYNAMIC_TX_BUFFER

- CONFIG_ESP_GRATUITOUS_ARP (CONFIG_LWIP_ESP_GRATUITOUS_ARP)
- CONFIG_ESP_SYSTEM_MEMPROT_FEATURE (CONFIG_ESP_SYSTEM_MEMPROT)
- CONFIG_ESP_SYSTEM_MEMPROT_FEATURE_LOCK (CONFIG_ESP_SYSTEM_MEMPROT_PMS_LOCK)
- CONFIG_ESP_SYSTEM_MEMPROT_FEATURE_VIA_TEE (CONFIG_ESP_SYSTEM_MEMPROT)
- CONFIG_ESP_SYSTEM_PD_FLASH (CONFIG_ESP_SLEEP_POWER_DOWN_FLASH)
- CONFIG_ESP_SYSTEM_PMP_IDRAM_SPLIT (CONFIG_ESP_SYSTEM_MEMPROT)
- CONFIG_ESP_SYSTEM_PM_POWER_DOWN_CPU (CONFIG_PM_POWER_DOWN_CPU_IN_LIGHT_SLEEP)
- CONFIG_ESP_TASK_WDT (CONFIG_ESP_TASK_WDT_INIT)
- CONFIG_ESP_WIFI_EXTERNAL_COEXIST_ENABLE (CONFIG_ESP_COEX_EXTERNAL_COEXIST_ENABLE)
- CONFIG_ESP_WIFI_NAN_ENABLE (CONFIG_ESP_WIFI_NAN_SYNC_ENABLE)
- CONFIG_ESP_WIFI_NETWORK_ASSISTED_ROAMING_IP_RENEW_SKIP (CONFIG_ESP_WIFI_ROAMING_IP_RENEW_SKIP)
- CONFIG_ESP_WIFI_SW_COEXIST_ENABLE (CONFIG_ESP_COEX_SW_COEXIST_ENABLE)
- CONFIG_EVENT_LOOP_PROFILING (CONFIG_ESP_EVENT_LOOP_PROFILING)
- CONFIG_EXTERNAL_COEX_ENABLE (CONFIG_ESP_COEX_EXTERNAL_COEXIST_ENABLE)
- CONFIG_FLASH_ENCRYPTION_ENABLED (CONFIG_SECURE_FLASH_ENC_ENABLED)
- CONFIG_FLASH_ENCRYPTION_UART_BOOTLOADER_ALLOW_CACHE (CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_CACHE)
- CONFIG_FLASH_ENCRYPTION_UART_BOOTLOADER_ALLOW_DECRYPT (CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_DEC)
- CONFIG_FLASH_ENCRYPTION_UART_BOOTLOADER_ALLOW_ENCRYPT (CONFIG_SECURE_FLASH_UART_BOOTLOADER_ALLOW_ENC)
- **CONFIG_GAP_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_GAP_TRACE_LEVEL)**
    - CONFIG_GAP_TRACE_LEVEL_NONE
    - CONFIG_GAP_TRACE_LEVEL_ERROR
    - CONFIG_GAP_TRACE_LEVEL_WARNING
    - CONFIG_GAP_TRACE_LEVEL_API
    - CONFIG_GAP_TRACE_LEVEL_EVENT
    - CONFIG_GAP_TRACE_LEVEL_DEBUG
    - CONFIG_GAP_TRACE_LEVEL_VERBOSE
- CONFIG_GARP_TMR_INTERVAL (CONFIG_LWIP_GARP_TMR_INTERVAL)
- CONFIG_GATTC_CACHE_NVS_FLASH (CONFIG_BT_GATTC_CACHE_NVS_FLASH)
- CONFIG_GATTC_ENABLE (CONFIG_BT_GATTC_ENABLE)
- CONFIG_GATTS_ENABLE (CONFIG_BT_GATTS_ENABLE)
- **CONFIG_GATTS_SEND_SERVICE_CHANGE_MODE (CONFIG_BT_GATTS_SEND_SERVICE_CHANGE_MODE)**
    - CONFIG_GATTS_SEND_SERVICE_CHANGE_MANUAL

    - CONFIG_GATTS_SEND_SERVICE_CHANGE_AUTO
- **CONFIG_GATT_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_GATT_TRACE_LEVEL)**
    - CONFIG_GATT_TRACE_LEVEL_NONE
    - CONFIG_GATT_TRACE_LEVEL_ERROR
    - CONFIG_GATT_TRACE_LEVEL_WARNING
    - CONFIG_GATT_TRACE_LEVEL_API
    - CONFIG_GATT_TRACE_LEVEL_EVENT
    - CONFIG_GATT_TRACE_LEVEL_DEBUG
    - CONFIG_GATT_TRACE_LEVEL_VERBOSE
- CONFIG_GDBSTUB_MAX_TASKS (CONFIG_ESP_GDBSTUB_MAX_TASKS)
- CONFIG_GDBSTUB_SUPPORT_TASKS (CONFIG_ESP_GDBSTUB_SUPPORT_TASKS)
- CONFIG_GPTIMER_ISR_IRAM_SAFE (CONFIG_GPTIMER_ISR_CACHE_SAFE)
- **CONFIG_HCI_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_HCI_TRACE_LEVEL)**
    - CONFIG_HCI_TRACE_LEVEL_NONE
    - CONFIG_HCI_TRACE_LEVEL_ERROR
    - CONFIG_HCI_TRACE_LEVEL_WARNING
    - CONFIG_HCI_TRACE_LEVEL_API
    - CONFIG_HCI_TRACE_LEVEL_EVENT
    - CONFIG_HCI_TRACE_LEVEL_DEBUG
    - CONFIG_HCI_TRACE_LEVEL_VERBOSE
- CONFIG_HFP_AG_ENABLE (CONFIG_BT_HFP_AG_ENABLE)
- **CONFIG_HFP_AUDIO_DATA_PATH (CONFIG_BT_HFP_AUDIO_DATA_PATH)**
    - CONFIG_HFP_AUDIO_DATA_PATH_PCM
    - CONFIG_HFP_AUDIO_DATA_PATH_HCI
- CONFIG_HFP_CLIENT_ENABLE (CONFIG_BT_HFP_CLIENT_ENABLE)
- CONFIG_HFP_ENABLE (CONFIG_BT_HFP_ENABLE)
- **CONFIG_HID_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_HID_TRACE_LEVEL)**
    - CONFIG_HID_TRACE_LEVEL_NONE
    - CONFIG_HID_TRACE_LEVEL_ERROR
    - CONFIG_HID_TRACE_LEVEL_WARNING
    - CONFIG_HID_TRACE_LEVEL_API
    - CONFIG_HID_TRACE_LEVEL_EVENT
    - CONFIG_HID_TRACE_LEVEL_DEBUG
    - CONFIG_HID_TRACE_LEVEL_VERBOSE
- CONFIG_INT_WDT (CONFIG_ESP_INT_WDT)
- CONFIG_INT_WDT_CHECK_CPU1 (CONFIG_ESP_INT_WDT_CHECK_CPU1)
- CONFIG_INT_WDT_TIMEOUT_MS (CONFIG_ESP_INT_WDT_TIMEOUT_MS)
- CONFIG_IPC_TASK_STACK_SIZE (CONFIG_ESP_IPC_TASK_STACK_SIZE)
- **CONFIG_L2CAP_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_L2CAP_TRACE_LEVEL)**
    - CONFIG_L2CAP_TRACE_LEVEL_NONE

    - CONFIG_L2CAP_TRACE_LEVEL_ERROR
    - CONFIG_L2CAP_TRACE_LEVEL_WARNING
    - CONFIG_L2CAP_TRACE_LEVEL_API
    - CONFIG_L2CAP_TRACE_LEVEL_EVENT
    - CONFIG_L2CAP_TRACE_LEVEL_DEBUG
    - CONFIG_L2CAP_TRACE_LEVEL_VERBOSE

- CONFIG_L2_TO_L3_COPY (CONFIG_LWIP_L2_TO_L3_COPY)
- CONFIG_LCD_DSI_ISR_IRAM_SAFE (CONFIG_LCD_DSI_ISR_CACHE_SAFE)
- **CONFIG_LOG_BOOTLOADER_LEVEL (CONFIG_BOOTLOADER_LOG_LEVEL)**
    - CONFIG_LOG_BOOTLOADER_LEVEL_NONE
    - CONFIG_LOG_BOOTLOADER_LEVEL_ERROR
    - CONFIG_LOG_BOOTLOADER_LEVEL_WARN
    - CONFIG_LOG_BOOTLOADER_LEVEL_INFO
    - CONFIG_LOG_BOOTLOADER_LEVEL_DEBUG
    - CONFIG_LOG_BOOTLOADER_LEVEL_VERBOSE
- CONFIG_MAC_BB_PD (CONFIG_ESP_PHY_MAC_BB_PD)
- CONFIG_MAIN_TASK_STACK_SIZE (CONFIG_ESP_MAIN_TASK_STACK_SIZE)
- **CONFIG_MCA_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_MCA_TRACE_LEVEL)**
    - CONFIG_MCA_TRACE_LEVEL_NONE
    - CONFIG_MCA_TRACE_LEVEL_ERROR
    - CONFIG_MCA_TRACE_LEVEL_WARNING
    - CONFIG_MCA_TRACE_LEVEL_API
    - CONFIG_MCA_TRACE_LEVEL_EVENT
    - CONFIG_MCA_TRACE_LEVEL_DEBUG
    - CONFIG_MCA_TRACE_LEVEL_VERBOSE
- CONFIG_MCPWM_ISR_IRAM_SAFE (CONFIG_MCPWM_ISR_CACHE_SAFE)
- CONFIG_MESH_DUPLICATE_SCAN_CACHE_SIZE (CONFIG_BTDM_MESH_DUPL_SCAN_CACHE_SIZE)
- CONFIG_NEWLIB_NANO_FORMAT (CONFIG_LIBC_NEWLIB_NANO_FORMAT)
- **CONFIG_NEWLIB_STDIN_LINE_ENDING (CONFIG_LIBC_STDIN_LINE_ENDING)**
    - CONFIG_NEWLIB_STDIN_LINE_ENDING_CRLF
    - CONFIG_NEWLIB_STDIN_LINE_ENDING_LF
    - CONFIG_NEWLIB_STDIN_LINE_ENDING_CR
- **CONFIG_NEWLIB_STDOUT_LINE_ENDING (CONFIG_LIBC_STDOUT_LINE_ENDING)**
    - CONFIG_NEWLIB_STDOUT_LINE_ENDING_CRLF
    - CONFIG_NEWLIB_STDOUT_LINE_ENDING_LF
    - CONFIG_NEWLIB_STDOUT_LINE_ENDING_CR
- **CONFIG_NEWLIB_TIME_SYSCALL (CONFIG_LIBC_TIME_SYSCALL)**
    - CONFIG_NEWLIB_TIME_SYSCALL_USE_RTC_HRT
    - CONFIG_NEWLIB_TIME_SYSCALL_USE_RTC

    - CONFIG_NEWLIB_TIME_SYSCALL_USE_HRT
    - CONFIG_NEWLIB_TIME_SYSCALL_USE_NONE

- CONFIG_NIMBLE_ATT_PREFERRED_MTU (CONFIG_BT_NIMBLE_ATT_PREFERRED_MTU)
- CONFIG_NIMBLE_CRYPTO_STACK_MBEDTLS (CONFIG_BT_NIMBLE_CRYPTO_STACK_MBEDTLS)
- CONFIG_NIMBLE_DEBUG (CONFIG_BT_NIMBLE_DEBUG)
- CONFIG_NIMBLE_GAP_DEVICE_NAME_MAX_LEN (CONFIG_BT_NIMBLE_GAP_DEVICE_NAME_MAX_LEN)
- CONFIG_NIMBLE_HS_FLOW_CTRL (CONFIG_BT_NIMBLE_HS_FLOW_CTRL)
- CONFIG_NIMBLE_HS_FLOW_CTRL_ITVL (CONFIG_BT_NIMBLE_HS_FLOW_CTRL_ITVL)
- CONFIG_NIMBLE_HS_FLOW_CTRL_THRESH (CONFIG_BT_NIMBLE_HS_FLOW_CTRL_THRESH)
- CONFIG_NIMBLE_HS_FLOW_CTRL_TX_ON_DISCONNECT (CONFIG_BT_NIMBLE_HS_FLOW_CTRL_TX_ON_DISCONNECT)
- CONFIG_NIMBLE_L2CAP_COC_MAX_NUM (CONFIG_BT_NIMBLE_L2CAP_COC_MAX_NUM)
- CONFIG_NIMBLE_MAX_BONDS (CONFIG_BT_NIMBLE_MAX_BONDS)
- CONFIG_NIMBLE_MAX_CCCDS (CONFIG_BT_NIMBLE_MAX_CCCDS)
- CONFIG_NIMBLE_MAX_CONNECTIONS (CONFIG_BT_NIMBLE_MAX_CONNECTIONS)
- **CONFIG_NIMBLE_MEM_ALLOC_MODE (CONFIG_BT_NIMBLE_MEM_ALLOC_MODE)**
    - CONFIG_NIMBLE_MEM_ALLOC_MODE_INTERNAL
    - CONFIG_NIMBLE_MEM_ALLOC_MODE_EXTERNAL
    - CONFIG_NIMBLE_MEM_ALLOC_MODE_DEFAULT
- CONFIG_NIMBLE_MESH (CONFIG_BT_NIMBLE_MESH)
- CONFIG_NIMBLE_MESH_DEVICE_NAME (CONFIG_BT_NIMBLE_MESH_DEVICE_NAME)
- CONFIG_NIMBLE_MESH_FRIEND (CONFIG_BT_NIMBLE_MESH_FRIEND)
- CONFIG_NIMBLE_MESH_GATT_PROXY (CONFIG_BT_NIMBLE_MESH_GATT_PROXY)
- CONFIG_NIMBLE_MESH_LOW_POWER (CONFIG_BT_NIMBLE_MESH_LOW_POWER)
- CONFIG_NIMBLE_MESH_PB_ADV (CONFIG_BT_NIMBLE_MESH_PB_ADV)
- CONFIG_NIMBLE_MESH_PB_GATT (CONFIG_BT_NIMBLE_MESH_PB_GATT)
- CONFIG_NIMBLE_MESH_PROV (CONFIG_BT_NIMBLE_MESH_PROV)
- CONFIG_NIMBLE_MESH_PROXY (CONFIG_BT_NIMBLE_MESH_PROXY)
- CONFIG_NIMBLE_MESH_RELAY (CONFIG_BT_NIMBLE_MESH_RELAY)
- CONFIG_NIMBLE_NVS_PERSIST (CONFIG_BT_NIMBLE_NVS_PERSIST)
- **CONFIG_NIMBLE_PINNED_TO_CORE_CHOICE (CONFIG_BT_NIMBLE_PINNED_TO_CORE_CHOICE)**
    - CONFIG_NIMBLE_PINNED_TO_CORE_0
    - CONFIG_NIMBLE_PINNED_TO_CORE_1

- CONFIG_NIMBLE_ROLE_BROADCASTER (CONFIG_BT_NIMBLE_ROLE_BROADCASTER)
- CONFIG_NIMBLE_ROLE_CENTRAL (CONFIG_BT_NIMBLE_ROLE_CENTRAL)
- CONFIG_NIMBLE_ROLE_OBSERVER (CONFIG_BT_NIMBLE_ROLE_OBSERVER)
- CONFIG_NIMBLE_ROLE_PERIPHERAL (CONFIG_BT_NIMBLE_ROLE_PERIPHERAL)
- CONFIG_NIMBLE_RPA_TIMEOUT (CONFIG_BT_NIMBLE_RPA_TIMEOUT)
- CONFIG_NIMBLE_SM_LEGACY (CONFIG_BT_NIMBLE_SM_LEGACY)
- CONFIG_NIMBLE_SM_SC (CONFIG_BT_NIMBLE_SM_SC)
- CONFIG_NIMBLE_SM_SC_DEBUG_KEYS (CONFIG_BT_NIMBLE_SM_SC_DEBUG_KEYS)
- CONFIG_NIMBLE_SVC_GAP_APPEARANCE (CONFIG_BT_NIMBLE_SVC_GAP_APPEARANCE)
- CONFIG_NIMBLE_SVC_GAP_DEVICE_NAME (CONFIG_BT_NIMBLE_SVC_GAP_DEVICE_NAME)
- CONFIG_NIMBLE_TASK_STACK_SIZE (CONFIG_BT_NIMBLE_HOST_TASK_STACK_SIZE)
- CONFIG_NO_BLOBS (CONFIG_APP_NO_BLOBS)
- **CONFIG_NUMBER_OF_UNIVERSAL_MAC_ADDRESS (CONFIG_ESP32_UNIVERSAL_MAC_ADDRESSES)**
    - CONFIG_TWO_UNIVERSAL_MAC_ADDRESS
    - CONFIG_FOUR_UNIVERSAL_MAC_ADDRESS
- **CONFIG_OPTIMIZATION_ASSERTION_LEVEL (CONFIG_COMPILER_OPTIMIZATION_ASSERTION_LEVEL)**
    - CONFIG_OPTIMIZATION_ASSERTIONS_ENABLED
    - CONFIG_OPTIMIZATION_ASSERTIONS_SILENT
    - CONFIG_OPTIMIZATION_ASSERTIONS_DISABLED
- **CONFIG_OPTIMIZATION_COMPILER (CONFIG_COMPILER_OPTIMIZATION)**
    - CONFIG_OPTIMIZATION_LEVEL_DEBUG, CONFIG_COMPILER_OPTIMIZATION_LEVEL_DEBUG, CONFIG_COMPILER_OPTIMIZATION_DEFAULT
    - CONFIG_OPTIMIZATION_LEVEL_RELEASE, CONFIG_COMPILER_OPTIMIZATION_LEVEL_RELEASE
- **CONFIG_OSI_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_OSI_TRACE_LEVEL)**
    - CONFIG_OSI_TRACE_LEVEL_NONE
    - CONFIG_OSI_TRACE_LEVEL_ERROR
    - CONFIG_OSI_TRACE_LEVEL_WARNING
    - CONFIG_OSI_TRACE_LEVEL_API
    - CONFIG_OSI_TRACE_LEVEL_EVENT
    - CONFIG_OSI_TRACE_LEVEL_DEBUG
    - CONFIG_OSI_TRACE_LEVEL_VERBOSE
- CONFIG_OTA_ALLOW_HTTP (CONFIG_ESP_HTTPS_OTA_ALLOW_HTTP)
- **CONFIG_PAN_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_PAN_TRACE_LEVEL)**

    - CONFIG_PAN_TRACE_LEVEL_NONE
    - CONFIG_PAN_TRACE_LEVEL_ERROR
    - CONFIG_PAN_TRACE_LEVEL_WARNING
    - CONFIG_PAN_TRACE_LEVEL_API
    - CONFIG_PAN_TRACE_LEVEL_EVENT
    - CONFIG_PAN_TRACE_LEVEL_DEBUG
    - CONFIG_PAN_TRACE_LEVEL_VERBOSE

- CONFIG_PERIPH_CTRL_FUNC_IN_IRAM (CONFIG_ESP_PERIPH_CTRL_FUNC_IN_IRAM)
- CONFIG_POST_EVENTS_FROM_IRAM_ISR (CONFIG_ESP_EVENT_POST_FROM_IRAM_ISR)
- CONFIG_POST_EVENTS_FROM_ISR (CONFIG_ESP_EVENT_POST_FROM_ISR)
- CONFIG_PPP_CHAP_SUPPORT (CONFIG_LWIP_PPP_CHAP_SUPPORT)
- CONFIG_PPP_DEBUG_ON (CONFIG_LWIP_PPP_DEBUG_ON)
- CONFIG_PPP_MPPE_SUPPORT (CONFIG_LWIP_PPP_MPPE_SUPPORT)
- CONFIG_PPP_MSCHAP_SUPPORT (CONFIG_LWIP_PPP_MSCHAP_SUPPORT)
- CONFIG_PPP_NOTIFY_PHASE_SUPPORT (CONFIG_LWIP_PPP_NOTIFY_PHASE_SUPPORT)
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- CONFIG_PPP_SUPPORT (CONFIG_LWIP_PPP_SUPPORT)
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    - CONFIG_RFCOMM_TRACE_LEVEL_ERROR
    - CONFIG_RFCOMM_TRACE_LEVEL_WARNING
    - CONFIG_RFCOMM_TRACE_LEVEL_API
    - CONFIG_RFCOMM_TRACE_LEVEL_EVENT
    - CONFIG_RFCOMM_TRACE_LEVEL_DEBUG
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    - CONFIG_SCAN_DUPLICATE_BY_DEVICE_ADDR
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- **CONFIG_SMP_INITIAL_TRACE_LEVEL (CONFIG_BT_LOG_SMP_TRACE_LEVEL)**
    - CONFIG_SMP_TRACE_LEVEL_NONE
    - CONFIG_SMP_TRACE_LEVEL_ERROR
    - CONFIG_SMP_TRACE_LEVEL_WARNING
    - CONFIG_SMP_TRACE_LEVEL_API
    - CONFIG_SMP_TRACE_LEVEL_EVENT

    - CONFIG_SMP_TRACE_LEVEL_DEBUG
    - CONFIG_SMP_TRACE_LEVEL_VERBOSE

- CONFIG_SMP_SLAVE_CON_PARAMS_UPD_ENABLE (CONFIG_BT_SMP_SLAVE_CON_PARAMS_UPD_ENABLE)
- CONFIG_SPIRAM_ALLOW_STACK_EXTERNAL_MEMORY (CONFIG_FREERTOS_TASK_CREATE_ALLOW_EXT_MEM)
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- CONFIG_SPI_FLASH_QUAD_32BIT_ADDR_ENABLE (CONFIG_BOOTLOADER_CACHE_32BIT_ADDR_QUAD_FLASH)
- **CONFIG_SPI_FLASH_WRITING_DANGEROUS_REGIONS (CONFIG_SPI_FLASH_DANGEROUS_WRITE)**
    - CONFIG_SPI_FLASH_WRITING_DANGEROUS_REGIONS_ABORTS
    - CONFIG_SPI_FLASH_WRITING_DANGEROUS_REGIONS_FAILS
    - CONFIG_SPI_FLASH_WRITING_DANGEROUS_REGIONS_ALLOWED
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    - CONFIG_STACK_CHECK_NONE
    - CONFIG_STACK_CHECK_NORM
    - CONFIG_STACK_CHECK_STRONG
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- CONFIG_SYSTEM_EVENT_TASK_STACK_SIZE (CONFIG_ESP_SYSTEM_EVENT_TASK_STACK_SIZE)
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- CONFIG_TASK_WDT_CHECK_IDLE_TASK_CPU1 (CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU1)
- CONFIG_TASK_WDT_PANIC (CONFIG_ESP_TASK_WDT_PANIC)
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- **CONFIG_TCPIP_TASK_AFFINITY (CONFIG_LWIP_TCPIP_TASK_AFFINITY)**
    - CONFIG_TCPIP_TASK_AFFINITY_NO_AFFINITY
    - CONFIG_TCPIP_TASK_AFFINITY_CPU0
    - CONFIG_TCPIP_TASK_AFFINITY_CPU1
- CONFIG_TCPIP_TASK_STACK_SIZE (CONFIG_LWIP_TCPIP_TASK_STACK_SIZE)
- CONFIG_TCP_MAXRTX (CONFIG_LWIP_TCP_MAXRTX)

- CONFIG_TCP_MSL (CONFIG_LWIP_TCP_MSL)
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    - CONFIG_TCP_OVERSIZE_MSS
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- CONFIG_WPA_WPS_SOFTAP_REGISTRAR (CONFIG_ESP_WIFI_WPS_SOFTAP_REGISTRAR)
- CONFIG_WPA_WPS_STRICT (CONFIG_ESP_WIFI_WPS_STRICT)

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